

Local quantitative refinements of Gowers’s proof of Szemerédi’s theorem

Density transfer and phase-flat partitions; energies, restriction and dependent random choice;
cube and progression counts; floor patterns and the Proposition 17.7 phase extraction

Ten research manuscripts prepared for Vladimir Reshetnikov,
merged into one ProveIt research report (sources 01–10)

All sources: 6 October 2026 merged: 6 October 2026

Abstract

Gowers’s 2001 proof of Szemerédi’s theorem passes through many local estimates: phase-flat partitions and density increments (Section 2 and Lemma 5.15 of the paper), the energy and restriction steps of the inverse theorem (Proposition 6.1, Proposition 7.3, Lemma 7.4, Lemmas 9.2–9.4, 12.4, 14.7–14.8), cube and progression counts (Corollaries 3.3–3.6, Lemma 3.10) and floor-pattern counts (Corollaries 17.5–17.6, Proposition 17.7). This report merges five independent research manuscripts, all dated 6 October 2026 and prepared for Vladimir Reshetnikov, that sharpen these estimates against the corrected statements of ProveIt’s transcription and Lean development. The material is arranged in five thematic Parts. *Part I*: a sharp one-sided density–mass bound, proved in four sources; a sharp one-cell phase envelope; the lower constant $1/3$ (optimal) in Lemma 2.3; Corollary 2.4 with 36π and $3\alpha/4$; three density-sensitive Fourier increments, of which the increment $3\delta(1-\delta)\kappa/(4-(4-3\delta)\kappa)$, $\kappa = \eta/(2\delta(1-\delta))$, of one source is at least $3/2$ times those of the other two at their stated parameters and lengths of the same order (proved in the write); simultaneous phase-flat partitions with the optimal exponent $1/(m+1)$. *Part II*: a mixed, weighted extension of Proposition 6.1 on every finite abelian group with coefficient one; a cubic dependent-random-choice certificate giving $7\delta^3 n/(3\sqrt{6})$ in Lemma 7.4; an elementary Balog–Szemerédi–Gowers bound $(c/8, 2^{19}c^{-9})$; exact moment interpolation and a polynomial restriction theorem over prime fields. *Part III*: the exact fourth-order term of the centered cube count on every finite abelian group; the sharp gap $\|f\|_{U^3}^8 \geq 2\|f\|_{U^2}^8$ for real centered functions on odd-order groups, hence the sharp leading constant $6\sqrt{2}$ and, in the write, the answer $c_3^{\text{odd}} = 2^{-1/2}$ to one source’s question; the fractional correction exponent $16/3$; density-weighted progression counting. *Part IV*: the floor-pattern count C_k (two sources) with $|\mathcal{F}(\mathbf{r})| = (\prod_i 2r_i)C_k$, $2^{k(k-1)/2} \leq C_k$, $\log_2 C_k \sim k^2$ (using the classical threshold-function count), certified $C_1, \dots, C_4 = 1, 2, 10, 154$, and a local phase-extraction coefficient $2^{-k^2 - k \log_2 k - O(k)}$ at Proposition 17.7. *Part V*: the crosswalk to the formal ledger, the sources’ formalization plans, and their 46 research questions, deduplicated with credits. No source claims a new global bound for $r_k(N)$. The report is unrefereed. Of its statements only one, a lifting lemma from modular to integer progressions, is formalized, and it was formalized before the sources arrived; the report gains no formal status.

[*Second write, 6 October 2026: sources 06–10.*] Five further manuscripts of the same date are added to the same Parts, so the report now merges ten. *Part I* gains the exact optimum for the largest dense cell with a fixed number of cells (strictly better than the Lemma 5.15 replacement above for $\alpha < 1$), the optimal density cap $\delta + \delta S/(2\delta - S)$ and a density–length frontier; the sharp one-sided bound is now proved in six sources. *Part II* gains a chapter on Lemma 10.9: a transport-sensitive rigidity theorem with the sharp threshold $\delta < 1/(2k)$, which upgrades the

Freiman order from two to five under exactly the hypotheses of the Lean statement and to nine under further hypotheses of the construction; and an energy extraction with $(c/4, 2^{10}c^{-5})$ and a Freiman extraction giving $2^{-209}\gamma^{103}$ in place of $2^{-1882}\gamma^{1164}$ in Corollary 7.6. *Part III* gains the complete classification of the cube count through order five on every finite abelian group, with Stirling multiplicities and order-two and order-three torsion terms, a determinant-five configuration at order six, the refutation of the quartic lower bound with coefficient 12 (recorded with an explicit counterexample), a quadratic progression discrepancy, torsion-sensitive counting, and an exact dimension threshold for polynomial colourings. *Part IV* gains a second certificate of $C_4 = 154$ and carry entropy uniform in the modulus. *Part V* gains iteration potentials and 50 further questions (96 in all). No statement of the new sources is formalized.

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F.1 Guide to this report

This report merges five research manuscripts, numbered 01–05 in arrival order and then in ASCII order of their archive names. All five refine local estimates of W. T. Gowers’s proof of Szemerédi’s theorem [1], and all five compare their statements with ProveIt’s edited transcription of the paper (`Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`) and its Lean development (`Combinatorics/Ramsey/Lean/GowersSzemerédi`). The report lives beside both, in `Combinatorics/Ramsey/Research/GowersSzemerédi/local-quantitative-refinements`, by Vladimir’s direction; that placement confers no formal status (Section F.5).

No.	Delivered title	Pin	Parts
01	Sharp Local Refinements of Gowers’s Szemerédi Argument: density transfer, carry-pattern entropy, and formalization interfaces	3a25020a6	I, IV, V
02	Energy-Sensitive Inverse Steps and Sharp Density Extraction	66c02f8a1*	I, II, V
03	Sharper local estimates in Gowers’s proof of Szemerédi’s theorem: density increments, exact cube expansions, and dependent random choice	3a25020a6	I, II, III, V
04	Sharper Local Bounds in Gowers’s Proof of Szemerédi’s Theorem: polynomial homomorphism restriction, sharp cube estimates, and floor-pattern complexity (the base)	327773149	II, III, IV, V
05	Sharp Finite Phase Partitions and Density-Sensitive Increments	327773149	I, V
<i>Added in the second write, 6 October 2026</i>			
06	Sharp Local Bounds in Gowers’ Proof of Szemerédi’s Theorem: carry patterns, quadratic counting errors, and optimal density extraction	7d99e5dee	I, III, IV, V
07	Transport-sensitive rigidity in Gowers’s approximate-homomorphism argument	ae0e89574	II, V
08	Fourth-order stability of additive cube counts	none [†]	III, V
09	Arithmetic Circuit Expansions for Cubical Counts	7d99e5dee	III, V
10	Quantitative Refinements Around Gowers’ Proof of Szemerédi’s Theorem: torsion, density profiles, additive energy, and polynomial colorings	7d99e5dee	I, II, III, V

* source 02 records no pin; this is the branch reference it observed (its Appendix A). [†] source 08 records blob identifiers only; they are those of 7d99e5dee.

Arrangement. Every manuscript spans two to four themes, so the report is arranged by theme, not by manuscript. Part I treats density transfer and phase-flat partitions; Part II the inverse step (energies, moments, restriction, dependent random choice); Part III cube and progression counts; Part IV floor patterns and the phase extraction of Proposition 17.7; Part V the formal interface, the crosswalk to the ledger, and the research questions. Inside a Part the sources follow in numerical order. Each source’s sections are printed whole, as delivered, with their own subsections; a section heading carries its source tag (for example “[01]”) and a line naming the delivered section. Each manuscript’s abstract and opening section are printed in the front matter (Sections F.6–F.10), where its title block is quoted.

Numbering. Sections restart in each Part and carry its numeral: Section I.9 is the ninth section

of Part I, and the front matter’s sections are F.1–F.10. Statements and equations are numbered within sections, so source 02’s Theorem 7.1 is Theorem I.9.1 here. A statement or equation $k.j$ of a source keeps its index j within its section, except that sources 02 and 03 numbered their equations consecutively through the whole manuscript, whereas here they are numbered within sections. Source 04’s tags (M1)–(M7), (R1)–(R32), (A1)–(A14), (Q1), (Q2) and (*) are kept. *Bare numbers such as “Lemma 5.15” or “Corollary 2.4” always refer to Gowers’s paper* (in the corrected transcription when the text says so), as in all five sources; this report’s own numbers always carry a Part numeral or an F. Plain-text references of a source to its own sections were made live in one place (source 02, Section II.4); everywhere else the sources refer to their own numbers through cross-references, which now print this report’s numbers.

Concordance. Delivered sections and the sections here:

Source	Delivered section → section here
01	1→F.6, 2→I.1, 3→I.2, 4→I.3, 5→I.4, 6→I.5, 7→I.6, 8→I.7, 9→IV.1, 10→IV.2, 11→IV.3, 12→IV.4, 13→IV.5, 14→V.3, 15→V.4, 16→V.5, App. A→V.6, App. B→V.7
02	1→F.7, 2→II.1, 3→II.2, 4→II.3, 5→II.4, 6→I.8, 7→I.9, 8→I.10, 9→I.11, 10→V.8, 11→V.9, 12→V.10, App. A→V.11, App. B→I.12
03	1→F.8, 2→I.13, 3→I.14, 4→I.15, 5→III.1, 6→III.2, 7→II.5, 8→II.6, 9→V.12, 10→V.13, 11→V.14, App. A→V.15
04	1→F.9, 2→II.7, 3→II.8, 4→II.9, 5→II.10, 6→III.3, 7→III.4, 8→IV.6, 9→V.16, 10→V.17, 11→V.18, App. A→V.19, App. B→V.20
05	1→F.10, 2→I.16, 3→I.17, 4→I.18, 5→I.19, 6→I.20, 7→V.21, 8→V.22, 9→V.23, 10→V.24
06	1→F.11, 2→IV.7, 3→III.5, 4→IV.8, 5→I.21, 6→V.25, 7→IV.9, 8→V.26, 9→V.27
07	1→F.12, 2→II.13, 3→II.14, 4→II.15, 5→II.16, 6→II.17, 7→II.18, 8→II.19, 9→V.28, 10→V.29, 11→V.30, App. A→II.12, App. B→V.31
08	1→F.13, 2→III.6, 3→III.7, 4→III.8, 5→III.9, 6→III.10, 7→III.11, 8→III.12, 9→III.13, 10→III.14, 11→V.32, 12→V.33, 13→V.34, 14→V.35, App. A→III.15, App. B→V.36
09	1→F.14, 2→III.16, 3→III.17, 4→III.18, 5→III.19, 6→III.20, 7→III.21, 8→III.22, 9→III.23, 10→III.24, 11→V.37, 12→V.38, 13→III.25, 14→V.39, 15→V.40, App. A→III.26, App. B→III.27, App. C→V.41
10	1→F.15, 2→I.22, 3→I.23, 4→I.24, 5→III.28, 6→III.29, 7→II.20, 8→II.21, 9→III.30, 10→V.42, 11→V.43, 12→V.44, App. A→V.45

Notes and edits. Notes added in the write are set in small type, tagged [write] and dated 6 October 2026. They credit results that several sources prove, mark second routes, correct stale statements about the repository against HEAD, record questions answered by another source, and say what the write checked. The write proves four small statements itself, each inside a note with its proof: the comparison of the three Fourier increments (after Table 3), the identification of the four forms of the one-sided bound (after Theorem I.4.1), the value $c_3^{\text{odd}} = 2^{-1/2}$ (after Theorem III.3.4), and the comparison of the two elementary lower bounds for C_k (after Theorem IV.6.2). No delivered statement or proof was changed or dropped. The write’s interventions in delivered text are all of the following kind:

- labels prefixed (`gsr:part:source:`); a label on every delivered section (`...:sec:dk`) and on two remarks of source 03 (`gsr:dt:03:rem:repo`, `gsr:cc:03:rem:rectify`); citation keys unified into one bibliography;
- section headings tagged with the source; theorem styles unified (remarks of sources 04 and 05 were set in the definition style); the sources’ page layout (title pages, running heads, colours, fonts) not reproduced;
- source 02’s plain “Section 3” made a live reference (above), the two figure paths renamed to

the shipped files, and plain-text bookmark forms (`\texorpdfstring`) given to two headings with mathematics, invisible in print;

- source 04’s missing backslash in `_{\epsilon\in\Omega_k}` restored, twice, with a dated note (Section IV.6).

The final section of Part V that the sources’ own questions feed, Section V.2, applies Vladimir’s standing rule of 4 October 2026.

Five further sources. Five more manuscripts on the same subject (sources 06–10, arrival 62e21161f) were placed in this directory by ad37962c4; their prefixed code, data and figures are present. They are not used in this text. A later write will add them to these Parts as dated additions, without renumbering anything printed here.

[write] **6 October 2026; sources 06–10.** *Done in the second write.* Sources 06–10 are now part of the report (rows above). Every delivered section of each of them is printed whole in exactly one place: its abstract and Section 1 in the front matter (Sections F.11–F.15, so the front matter now runs to F.15), the other sections appended at the end of Parts I–V after the sections of sources 01–05, in source order, with source 07 as its own chapter of Part II (Section II.11). Nothing printed for sources 01–05 was renumbered or relabelled; the concordance above has gained the rows 06–10. Each Part head and Sections F.2–F.5 and V.1–V.2 have dated additions; results proved by several sources carry dated credit notes at both places. So this paragraph and the opening sentence of this section (“five research manuscripts”) describe the first write only; the title page now says ten.

Further sources. While this write was in progress, sources 11–14, 15–17 and 18–20 were placed in this directory by 6f530c29c, eed2ab863 and 58aa0c493 respectively. Their prefixed files are present (README). They are not used in this text. The placement of sources 11–14 records source 14 as settling source 04’s Question 1 (item III.3); that finding is for the write that adds it.

Edits of the second write in delivered text of sources 06–10 (in addition to the kinds listed above): source 06’s macro `\snapshot` (its own pin) renamed `\snapshotsix`; source 09’s ‘question’ environment (printed “Research question”) set as this report’s ‘Research question’; source 10’s three damaged subscripts M_{rest} restored (note in Section III.28); the two figure paths renamed to the shipped files; source 07’s Appendix A (its notation table) placed at the start of its chapter. Tables and figures of sources 06–10 are numbered within their sections (“Table III.13.1”), and the write’s remark on the false quartic bound is numbered W1, so that no number printed for sources 01–05 moved. Source 06 numbered its equations consecutively through the manuscript; here, as for sources 02 and 03, they are numbered within sections, while every statement of sources 06–10 keeps its index within its section. Source 10’s invisible fragment comments are kept.

Statements proved in the second write, each inside a dated note with its proof: the comparison of source 01’s increment with source 10’s frontier (after Table 3); source 06’s fixed- M bound against the $M - 1$ corollary, strict exactly for $\alpha < 1$ (after Corollary I.9.3); the order of the leading step-count coefficients of sources 06, 10 and 03 (after Proposition V.12.1); the comparison of the four progression-existence criteria (after Corollary III.4.4); and the implications from source 10’s Lemma 7.5 and Corollary 7.6 replacements to the formal statements (after Corollary II.21.2). The counterexample to the quartic lower bound (Remark W1) and source 09’s determinant-five configuration were recomputed independently.

F.2 What is proved, and where

“Proved” means proved in the cited source by an ordinary written proof, unrefereed. “External” marks a published theorem used and not reproved. Nothing here is checked by a proof assistant, except the one formalized statement marked **formal**.

Statement	Where	Sources
<i>Part I. Density transfer and phase-flat partitions</i>		

Statement	Where	Sources
Sharp one-sided mass bound	Thm. I.9.1, Thm. I.14.1; case	02, 03; 01, 05
$\Pr(X > t) \geq ((a+t)T - at)/(a(b-t))$ for mean-zero $X \in [-a, b]$, $T = \mathbb{E}X_+$; forced maximum $aT/(a-T)$; extremizers	$a = \delta$: Thm. I.4.1, Lem. I.17.3	
Short-arc density tail with phase parameter s , occupied-mass exceptions, optimal forced maximum a_* , the trade-off $s_{\max}(t)$; sharp in the finite weighted model, not claimed for arithmetic characters	Thms. I.2.1, I.2.3, I.3.1	01
One-cell phase envelope $C^2 \leq (1-\rho^2)F^2 + \rho^2L^2$ and the linear law $C \leq \rho L + (1-\rho)S$ (real f only)	Lem. I.1.2, I.1.1; Thm. I.8.1, Cor. I.8.2	01, 02
Finite-knot linear-programming reduction; multiphase score	Thm. I.4.2, Cor. I.4.3	01
Lemma 5.15 with $ P \geq \alpha(2+\alpha)N/(2(4-\alpha)(M-1))$ (needs $\alpha > 0$, nonempty cells)	Cor. I.9.3, (I.14.4)	02, 03
Simultaneous phase-flat partition, $L = \lfloor (N \prod_i \varepsilon_i / 2^m)^{1/(m+1)} \rfloor$, every cell in $[L, 2L-1]$	Thm. I.5.1, Thm. I.18.2	01, 05
Fourier increment $\delta + 3\delta b\kappa/(4 - (4-3\delta)\kappa) \geq \delta + 3\eta/8$, $b = 1 - \delta$, length $\lfloor \sqrt{(N/2\pi) \arcsin(\eta/4\delta b)} \rfloor$; real frequency, bounded weights, any N	Thm. I.6.1	01
Fourier increment $\delta + \delta\alpha/(4\delta - \alpha)$, length $\geq \sqrt{\alpha N/(16\pi\delta(1-\delta))}$, exact integer length, popularity, trade-off	Thm. I.17.4	05
Fourier increment with adjustable loss λ	Thm. I.15.2	03
Increment comparison: source 01's is at least 3/2 times source 05's (and source 03's at $\lambda = 1/2$) at stated parameters	[write] note after Table 3	write
Lemma 2.3 with lower constant 1/3 (optimal, also under the exact Lean hypotheses)	Thm. I.16.4	05
Corollary 2.4 with 36π and $3\alpha/4$, same upper length	Cor. I.16.6	05
Optimal simultaneous localization order $1 + (\varepsilon^m N)^{1/(m+1)}$; one-phase barrier $\ell \leq 1 + \sqrt{3r(N-1)}$	Thm. I.18.5; Thm. I.7.2	05; 01
Spectral extraction $E \leq \sqrt{EV} + 2R(v-V)$, Gram correction, optimal width allocation, short-cell discard	Thm. I.19.2, Cor. I.19.4, Prop. I.19.3, Prop. I.20.1	05
Robust phase-to-density transfer; refinement with one global phase budget and its optimum, <i>conditional on</i> $\text{PP}(K, R_0)$ (Corollary 5.6, now formal for intervals) <i>Part I, sources 06–10 (second write)</i>	Thms. I.10.1, I.11.1, Cor. I.11.2, Prop. I.11.3	02
Exact fixed- M optimum $\min\{P_+/(b(M-1)), S/(M-2)\}$ for the largest high cell, with extremizers; Lemma 5.15 with both branches (strictly better than the $M-1$ corollary for $\alpha < 1$); simultaneous $\tau(\alpha)$; signed drift and exceptional sets	Thm. I.21.1, Cors. I.21.2, I.21.3, Prop. I.21.4	06
Sharp one-sided mass bound, now in six sources	Thm. I.21.1; Prop. I.23.2; note after Thm. I.4.1	06, 10 (with 01, 02, 03, 05)
Optimal cap $M \geq \delta + \delta S/(2\delta - S)$ and variance cap $M \geq \delta + V/\delta$; correlation transfer with an exceptional set	Thm. I.23.1, Thms. I.23.3, I.23.4	10
Explicit phase partition and the density-length frontier D_γ, L_γ ; variance form of the U^2 inverse step	Lem. I.24.1, Thm. I.24.2, Cor. I.24.3, Prop. I.24.4	10
Source 01's increment is at least 3/2 times source 10's for $\gamma \leq 1/2$, at least equal for $\gamma \leq 3/(4-\kappa)$	[write] note after Table 3	write

Statement	Where	Sources
<i>Part II. The inverse step</i>		
$\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4 \mathcal{U}(g), m(g)^4 \mathcal{U}(f)\} \mathcal{E}(w)$ on every finite abelian group, coefficient 1 optimal; $\ f\ _{U^3} \leq \ f\ _2$; graph and balanced forms; spectral certificate; deficits; alignment; rigidity	Thm. II.2.1 and Sections II.2–II.4	02
Lemma 7.4 with $ K \geq 7\delta^3 n / (3\sqrt{6})$, same threshold and 90%; optimal scalar tangent κ_* ; weighted tuples; one-witness trade-off	Thm. II.5.2, Section II.5	03
Elementary Proposition 7.3: $ A' \geq c A /8$, $ A' - A \leq 2^{19} c^{-9} A $ (weaker than Reiher–Schoen)	Thm. II.6.3	03
Exact energy interpolation (M1)–(M7); Lemma 9.2 strengthened to $\gamma^7 \beta^{15} p^{15}$	Thm. II.8.1, Cor. II.8.2, Prop. II.8.3	04
Polynomial restriction over $\mathbb{F}_p$; order eight: $2^{-95} \gamma^{112} \eta^{15} B ^{15}$ under $p \geq 2^{102} \gamma^{-112} \eta^{-16} \beta^{-1}$; method limit (prime fields only)	Thm. II.9.1, Cor. II.9.5, Prop. II.9.6	04
Arrangement interpolation with profile σ ; Lemmas 14.8, 12.4 with $\beta^{7.4^{k+1}-6}, \beta^{106}$	Thm. II.10.1	04
<i>Part II, sources 06–10 (second write)</i>		
Coupled-path rigidity with transport profiles; Freiman order k when $2[1 + (k-1)\kappa]\delta < 1$; sharp threshold $\delta < 1/(2k)$ for constant fibres	Thm. II.15.1, Cor. II.15.3, Thm. II.16.1	07
Lemma 10.9 with Freiman order five under exactly the Lean hypotheses; order nine with the ambient package (not among the Lean hypotheses); allowances 120^{-5} , 75^{-5}	Cors. II.17.3, II.17.6, Props. II.17.4, II.17.7	07
Root-law linear program; collision-to-mode exactification; weighted laws; nonabelian groups	Thm. II.18.3, Props. II.19.1, II.19.2, Thm. II.19.3	07
Proposition 7.3 with $(c/4, 2^{10} c^{-5})$ on any abelian group	Thm. II.20.1	10
Lemma 7.5 with $ B /(k(h_k - 1) + 1)$; Corollary 7.6 with $2^{-209} \gamma^{103}$; general finite codomains	Thm. II.21.1, Cor. II.21.2, Thm. II.21.3	10
<i>Part III. Cube and progression counts</i>		
No centered terms of degree 1–3; error $\sum_{j \geq 4} \binom{m}{j} \delta^{m-j} \eta^j$	Lem. III.2.1, Cor. III.2.2; Lem. III.3.1	03, 04
Exact quartic term $\delta^{m-4} (R_d \mathcal{E}(f) + T_d \mathcal{E}_{(2)}(f))$ on every finite abelian group; K_d optimal over all groups; power 4 optimal for sets; fifth-order remainder necessary; an order-three correlation	Thm. III.2.6, Section III.2	03 (odd order also 04)
$\ f\ _{U^3}^8 \geq 2\ f\ _{U^2}^8$ for real centered f on odd-order groups, equality, stability	Thm. III.3.4, Cor. III.3.5	04
Three-cube bound with sharp leading term $6\sqrt{2} \delta^4 u^4$; correction $u^{16/3}$ with optimal exponent for $\gcd(G , 6) = 1$; sharpness for sets	Thms. III.3.6, III.3.8, Props. III.3.7, III.3.9, III.3.10	04
$c_3^{\text{odd}} = 2^{-1/2}$, attained by real parts of characters (source 03's question at $d = 3$)	[write] note after Thm. III.3.4	write, from 04
Generalized von Neumann bound for invertible (admissible) slopes; telescoping error $(k-2)\alpha$ in place of $2^k \alpha$; optimal elimination order	Lem. III.1.1, Thm. III.1.2; Lem. III.4.2, Thm. III.4.3	03, 04
$T_k^*(A) \geq N^2(\delta^k - u S_{k-2}(\delta)) - \delta N$ and its thresholds	Cor. III.1.3, Cor. III.4.4	03, 04
Lifting a modular progression from a short interval	Lem. III.4.5, Rem. III.1.4	04, 03; formal
<i>Part III, sources 06–10 (second write)</i>		

Statement	Where	Sources
Generalized von Neumann bound with two distinguished functions; progression discrepancy $\sqrt{\delta}B_k(\delta)\varepsilon^2$; sharp three-term variance factor; modular criterion	Thm. III.5.1, Cor. III.5.5, Prop. III.5.6, Cor. III.5.7	06
Cancellation through degree three; exact quartic term with order-two correction (also 03)	Thms. III.7.3, III.8.3; Thms. III.17.2, III.21.1	08, 09
Complete odd-order three-cube polynomial; Fourier lower bound $(\delta^4 + E_4)^{p/4}$; inhomogeneous weights; character-kernel survival criterion	Thm. III.9.2, Thms. III.12.1, III.13.1, Prop. III.14.1	08
Complete four- and five-vertex classifications on every finite abelian group; exact fifth order with Stirling multiplicities; sharpness through order six; transfer to sets	Thms. III.19.1, III.20.1, III.21.1, Props. III.22.1, III.22.3, Cor. III.22.5	09
The quartic lower bound with coefficient 12 is false	Rem. W1; Thm. III.10.1, Prop. III.22.2	03, 08, 09
Determinant-five configuration on six vertices; isolated ℓ -torsion for every prime ℓ	Thm. III.23.1, Prop. III.23.2	09
Torsion-sensitive mixed- L^p generalized von Neumann bound, sharp on $(\mathbb{Z}/2\mathbb{Z})^n$; density-weighted telescope with error $\delta S_k(\delta)\eta$	Thm. III.28.2, Prop. III.28.4, Thm. III.29.1, Cor. III.29.2	10
Exact dimension threshold $d \leq \sum_{\ell} \lceil D_{\ell}/2 \rceil^2$ for polynomial colourings without symmetrically coloured even progressions ($D_{\ell} \leq k < p$)	Thm. III.30.3	10
<i>Part IV. Floor patterns</i>		
$ \mathcal{F}(\mathbf{r}) = (\prod_i 2r_i)C_k$; arbitrary integer intervals and the modular count	Thm. IV.1.2; Thm. IV.6.3	01, 04
$C_k \leq \mathcal{R}(H_k, k)$, $\log_2 C_k \leq k^2 + (\log_2 e - 1)k$; $A_k \leq \mathcal{R}(k2^{k-1}, k+1)$	Thms. IV.1.4, IV.3.1; Thm. IV.6.2	01, 04
$C_k \geq 2^{\lfloor k^2/4 \rfloor}$ (01); $C_k \geq 2^{k(k-1)/2}$, $A_k \geq \prod_{j < k} (2^j + 1)$ (04; stronger for $k \geq 3$)	Thm. IV.2.1; Thm. IV.6.2	01; 04
$\log_2 C_k \sim k^2$, $\log_2 A_k \sim k^2$ (external: the classical threshold-function count)	Thms. IV.2.3, IV.3.1	01
Translation-class budgets independent of the modulus	Thm. IV.3.2; Cor. IV.6.4	01, 04
Polynomial floor entropy $(1 + o(1))n^{d+1}/d!$ (lower half external: Baldi–Vershynin)	Thm. IV.4.1	01
$C_1, \dots, C_4 = 1, 2, 10, 154$; $A_1, A_2, A_3 = 2, 6, 38$ (exact rational certificates, Python verifiers)	Prop. IV.5.1; Section IV.6	01; 04
Local phase extraction at Proposition 17.7 with coefficient $2^{-k^2 - k \log_2 k - O(k)}$ in place of $2^{-2(k+1)^3}$ (not propagated to Section 18)	Thm. IV.6.7, Cor. IV.6.8; Prop. IV.3.3	04; 01
<i>Part IV, sources 06–10 (second write)</i>		
Range-carry separation for every $V \subset \mathbb{Z}_{\geq 0}^k$; modular count $\min(2r, M)^k C(V; M)$; translation classes; $\log_2 C_k \sim k^2$ uniformly in M (external: Zuev)	Thms. IV.7.3, IV.7.4, IV.7.6	06
$C_4 = 154$ certified a second time; $D_1, D_2, D_3 = 2, 6, 38$	Section IV.9	06 (with 01, 04)
Proposition 17.7 with coefficient $1/(8K_k^2) = 2^{-2k^2 - O(k \log k)}$	Thm. IV.8.1, Cor. IV.8.2	06
<i>Part V</i>		
Density and length bookkeeping for an abstract iteration	Prop. V.12.1	03
Discrete potential with optimal leading cost; explicit double-logarithmic budget	Lem. V.25.1, Thm. V.25.2	06

Statement	Where	Sources
Step count $321(\delta_0^{-1} - 1)$; exact propagation of length losses	Props. V.42.1, V.42.2	10
<i>Open, or not certified</i> (Section V.2)		
$C_5 = 8410$ and $A_4 = 770$ (intake recount, floating-point infeasibility; <i>not certified</i>)	Item IV.2	intake
The second coefficient of the $u^{16/3}$ term (between $(9/4)^{1/3}$ and 6); c_d^{odd} for $d \geq 4$; the $k \log k$ term of $\log_2 C_k$; 46 delivered questions in all	Section V.2	all
Sources 06–10: 50 further questions and two question paragraphs (96 questions in all), merged with credits	Section V.2	06–10

What no source claims. No source claims a new bound for $r_k(N)$; all five cite Leng, Sah and Sawhney’s $r_k(N) \ll N \exp(-(\log \log N)^{c_k})$ for $k \geq 5$ [2] as the benchmark, and say their results are local. No source claims literature priority for its formulations or an exhaustive priority search. No source compiled Lean code or claims kernel verification. Each source’s further limitations are printed in place and collected in Section V.2.

[write] **6 October 2026; sources 06–10.** The same holds for sources 06–10: none claims a new bound for $r_k(N)$ (all five cite Leng–Sah–Sawhney; source 10 also Raghavan’s 2026 bound for r_3 and Green–Tao’s for r_4), none claims priority, none compiled Lean code. Sources 08 and 09 state AI assistance (provenance, Section F.4). The table rows marked “06–10” were added in the second write.

F.3 Notation across the five sources

No delivered symbol was renamed; the same letter often means different things in different sources. Each Part opens with a short table for its own sections; this is the whole dictionary. The tempting false readings are in italics.

Symbol	Meanings, by source and Part
$\mathbb{E}, \mathcal{E}, E$	$\mathbb{E}$ is always an average (expectation). $\mathcal{E}(w)$: 02, normalized additive energy of a weight on $G \times \widehat{G}$ (factor N^{-3} ; Part II). $\mathcal{E}_j(w)$: 04, $M^{-1} \sum \widehat{w} ^{2j}$ with the <i>unnormalized</i> transform, counting $2j$ -tuples (Part II). $\mathcal{E}(f) = \ f\ _{U^2}^4$ and $\mathcal{E}_{(2)}(f)$, the order-two energy: 03 (Part III). $\mathcal{E}(\lambda)$: 04, the additive energy of $\lambda_r = a_r ^2$ (Part III). E : 05, captured Fourier energy $\sum c_i ^2$ (Part I); 01, an exceptional set (Part I); 02 and 03, the additive energy $E(\Gamma)$, $E(A)$ of a graph or a set (Part II); 04, E and E_j in Lemma IV.6.6 (Part IV). E_{osc} : 03, an oscillation error. <i>None of these is a normalization of another.</i>
Q	03, 04: Q_d the normalized cube count (the same quantity, Part III). 01: Q_i box counts (Part I). 05: $Q_i(L)$ box counts and Q in Lemma I.16.1 (Part I). 02: Q and $Q_{a,b}(p, t)$, good-mass certificates; Q_i input progressions (Part I). 04: Q the kernel ratio sum (R14), Q_K (R32), $Q_k = k^{2k-1}$, intervals Q_i (Parts II, IV); questions Q1–Q12 and displays (Q1), (Q2) (Part V). 03: Q in Lemma I.15.1.
$R, \mathcal{R}$	03: R_d the number of ordinary parallelograms of $\{0, 1\}^d$, $\mathcal{R}_d(F)$ a remainder (Part III). 04: P_d the same number; R_d, R_6, R_7 remainders; R_j counts of respected arrangements (Part II); $R_d(H) = \sum_{j \leq d} \binom{H}{j}$ (Part IV); R_i cells. 01: $\mathcal{R}(h, d) = \sum_{j \leq d} \binom{h}{j}$ (Part IV), <i>so 01’s $\mathcal{R}(H, d)$ is 04’s $R_d(H)$, arguments reversed.</i> 02: $R = \mathbb{E} K_u(s) ^2$ (Part II), R_0 the partition threshold, R_j refined cells (Part I). 05: R an oscillation budget (Part I). 03: R the number of steps (Part V). <i>04’s remainder R_d is not 03’s count R_d.</i>

Symbol	Meanings, by source and Part
$T, \mathcal{T}$	02: $\mathcal{T}_w(f, g)$ and T , weighted derivative correlations (Part II). 03: T_d binary tetrahedra (Part III); $T = \mathbb{E}X_+$, T_0 (Part I). 04: $T_d(B)$, $T_d^\phi(B)$ tuple counts (Part II); $T(s, t)$ Fourier cube coefficients, $T_k^*(A)$, $T_i(a)$ (Parts III, IV). 01: T_n , $T(n, d)$ threshold-function counts (Part IV); $T = \prod_i Q_i$ (Part I).
κ	01: $\eta/(2\delta(1-\delta))$ (Part I). 02: $\mathcal{U}(f)/m(f)^4$ (Part II). 03: $7/(3\sqrt{6})$ and κ_* (Part II). 04: $\kappa_d = 1/(2(400d)^d)$ (Part II). 05: a Gram-matrix bound (Part I).
ρ	02: half the chordal diameter of a phase set (Part I). 03: a normalized Fourier coefficient (Part I); $c/(2-c)$ (Part II). 04: m/M (Part II); the largest Fourier modulus (Part III). 01: $\max\{0, \lambda_* - \beta_E\}$ (Corollary I.2.4).
s	01: $\sin \theta$ for an arc of half-angle θ . 05: a positive mass, and $s(\varepsilon)$. 02: a weighted absolute mean. 03: $\sqrt{1-\gamma/\varepsilon}$ (Part II). 01's s is 02's ρ , not 05's s .
C_k, C_d, C	01: C_k fractional carry patterns (= 04's F_k , Part IV). 04: $C_d = 1 + \binom{2d}{2}$ (Part II). 02: C a local correlation, C_0 (Part I).
B, B_k	01: $B_k = (2(k+1))^k \mathcal{R}(k2^{k-1}, k+1)$ and $B_{\delta,s}(a)$. 04: $B_k = (2k)^k R_{k+1}(k2^{k-1})$, $B_5(f)$, the set $B \subset \mathbb{F}_p$. 02: B_0 , B_ρ , a set B . 05: $B = \mathbb{E} f $, a width budget B . 03: $B(S)$ (= 04's $D(S)$). 01's and 04's B_k are different numbers.
L	02: the total absolute mass $\sum f $. 03: $L = D $ (Part II), a length (Part I). 04: kernel bandwidth (Part II), number of labels (Part IV), cube forms L_v (Part III, also 03). 01, 05: cell lengths.
M, m, q	M : 01 $M_\delta = 2\delta(1-\delta)$ and a modulus; 02, 03 a number of cells; 04 $ H $, $\sum \lambda^4$, $M_\delta(u)$, a modulus; 05 the modulus of Lemma 2.3. m : 03 2^d (cubes), $ V $ (DRC), $ A $ (energy); 04 $ B $, $ A $, $2h+1$; 01 anchor count, monomial count; 05 the number of phases. q : 04 2^d (Part III), kernel degree (Part II), $\lceil h/k \rceil$ (Part IV); 01, 03, 05 a common difference; 01 q_j cell densities; 02 q_t good mass.
α, η, u	Gowers's "α-uniform of degree $d-1$ " means $\ f\ _{U^d}^{2^d} \leq \alpha$: α is a norm power, not a norm (03, 04). Elsewhere α is a correlation (02, 05, Gowers's Corollaries 2.4–2.5) or a retained proportion (04, Part II). η : 01 a normalized correlation; 03 a U^d norm (Part III) or a cell discrepancy bound; 04 the unrespected proportion (Part II). u : 04 a U^d norm.
arcs	01: an arc of r turns, half-angle $\theta = \pi r$, $s = \sin \theta$. 05: an arc of width ε turns, disk radius $\sin(\pi\varepsilon)$. 02: chordal diameter 2ρ , so $\rho = \sin \theta$. 03: chordal error ε . <i>A width in turns is not an angle.</i>
Fourier	02, 03, 04 (outside its Section II.8), 05: normalized, $\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}$. 04 Section II.8: unnormalized. 01: normalized correlations $\mathbb{E}(h-\delta)e(\xi n)$, and the unnormalized cyclic transform in Corollary I.6.2. The Lean development: unnormalized (<code>ZMod.dft</code>), so its hypothesis $ \widehat{1_A}(r) \geq \alpha N$ is a normalized coefficient at least α .
Gowers norms	02: $\mathcal{U}(f) = \ f\ _{U^3}^8$, normalized; its repository counterpart is $N^4 \mathcal{U}(f)$, and the degree parameter of <code>UniformOfDegree</code> is 2. 03, 04: $\ f\ _{U^d}$ normalized; Lean's <code>gowersNorm</code> $d f = N^{(d+1)/2^d} \ f\ _{U^d}$.
numbering	"Lemma 5.15", "Corollary 2.4", "Section 17" and other bare numbers are Gowers's. This report's numbers carry a Part numeral (I–V) or F. Source 04's display tags (Mj), (Rj), (Aj) are not this report's equation numbers; (A1) is not Appendix A, and (R3) is not a remainder.

Sources 06–10 (added in the second write; the Part heads have the full rows)

η, δ, γ	06: η a density threshold (Part I). 07: δ, δ_d failure probabilities of directions, <i>not densities</i> (Part II). 10: η a correlation (Part I) or a U^{k-1} norm (Part III); γ a retained correlation fraction (Part I) or a graph-energy proportion (Part II).
$\mathcal{C}, \mathcal{C}_d, C_d$	07: $\mathcal{C}(\mathbf{d})$ the cost of a word (Part II). 08: $\mathcal{C}_d(F)$, 09: $C_d(F)$, the normalized cube count, = 03's and 04's Q_d (Part III).
$Q_d, R_d, P_d, T_d, F_d, A_d, K_d, S_d$	08's $R_d = 09$'s $P_d = 03$'s $R_d = 04$'s P_d ; T_d the same in 03, 08, 09; 08's $F_d = 09$'s $A_d = 03$'s K_d . 08's $Q_d(f)$ (the quartic coefficient), 09's Q_d (star count) and 08's $K_d = \sum_{r \geq 4} \binom{p}{r}$ are not 03's or 04's Q_d , K_d ; 09's S_d counts ternary supports.

Symbol	Meanings, by source and Part
κ	07: a fibre-size ratio. 10: $\kappa_G(c) = \{x : cx = 0\} $. 08: $\kappa_d^{\text{odd}} = 03$'s c_d^{odd} .
$B_k, c_k, h_k,$	06: $B_k(\delta) = \sum_t (t+1)\delta^t$ (Part III), $K_k = 01$'s B_k and $c_k = 1/(8K_k^2)$ (Part IV),
H_k	$h_k = (k-2)2^{k-1} + 1 = H_k$ of 01, 04. 10: H_k, h_k a vertical obstruction fibre and its size (Part II). 06's c_k is not 04's c_k .
M, v, S	06: M the number of cells, v the cell variation. 10: M the largest cell density, $v = \delta(1-\delta)$, $S_{\mathcal{P}}$ the first moment (= 06's v for indicators).

F.4 Provenance and merge decisions

Archives. The five archives arrived in commit 66f24b0d0 (“New research reports”, 6 October 2026) and were placed in this directory by aefb0b44a (“Place batch 115 (1/2)”, batch 115 of ProveIt’s incoming-reports intake), which also retired them from docs/incoming/. They survive in the arrival commit (git show 66f24b0d0:docs/incoming/<archive>).

No.	Archive	Bytes	Files	Pin	Manuscript
01	Gowers_Quantitative_Refinements(1).zip	541,437	15	3a25020a6	1,980 lines, 31 pp.
02	Gowers_Szemeredi_Refinements.zip	492,008	7	(66c02f8a1)	823 lines, 20 pp.
03	gowers_local_refinements.zip	408,961	10	3a25020a6	2,392 lines, 36 pp.
04	gowers_quantitative_refinements.zip	788,239	14	327773149	2,987 lines, 40 pp.
05	gowers_strengthening.zip	399,349	8	327773149	847 lines, 18 pp.

The pins are three commits of 6 October 2026, in the order 3a25020a6 $\rightarrow$ 327773149 $\rightarrow$ 66c02f8a1 $\rightarrow \dots \rightarrow$ HEAD. Every repository blob that a source cites was checked at intake against its pin and found correct.

Authorship and disclosures. Sources 02, 03 and 04 are “prepared for Vladimir Reshetnikov” (title block and PDF author field), source 05 is a “research note prepared for Vladimir Reshetnikov” with an empty PDF author field. *Source 01 credits ChatGPT*: its author line reads “Research report prepared for Vladimir Reshetnikov / Mathematical development and computational checks with ChatGPT”, and its PDF author field reads “Research report prepared with ChatGPT”. No source names a human author. Source 03 mentions “independent mathematical reviews” that are not part of its delivery (note in Section V.13).

What each source contributed. 01: the sharp weighted density-tail theory with a phase parameter, simultaneous partitions, the strongest single-coefficient Fourier increment, the scale barrier (Part I), and the carry-pattern count with $\log_2 C_k \sim k^2$ and $C_4 = 154$ (Part IV). 02: the mixed weighted extension of Proposition 6.1 (Part II), sharp real phase freezing, the general sharp tail bound, the Lemma 5.15 replacement and the refinement theorem under $\text{PP}(K, R_0)$ (Part I). 03: the selection theorem and a Fourier increment (Part I), dependent random choice and an elementary Proposition 7.3 (Part II), progression counting and the exact quartic cube term on every group (Part III), iteration bookkeeping (Part V). 04: moments and polynomial restriction (Part II), the sharp odd-order cube constants (Part III), floor patterns with the stronger lower bound and the Proposition 17.7 coefficient (Part IV). 05: the sharp constant $1/3$, Corollary 2.4 with 36π , a density-sensitive increment with popularity, simultaneous localization with its optimal exponent, spectral extraction (Part I).

Overlaps. The five are independent manuscripts: no two share more than 0.8% of their text (intake measurement, 8-word shingles), and sources 01 and 04 are not editions of one another (different titles, pins and files; 0.23% shared text). The results proved by several sources are listed in Section F.2 with all their sources.

Where the merge had to choose.

- *Base.* Source 04 is the base: it was the largest and broadest at intake; its manuscript was staged as `article.tex` and its delivery README as `README.md`, both now replaced by this report and its README.
- *Thematic Parts,* by the intake’s recommendation and Vladimir’s direction, since every source spans two to four themes. Each delivered section is kept whole in exactly one Part; each source’s abstract and Section 1 are in the front matter. Sources’ conventions sections go with their first use: source 03’s Section 2 opens its Part I chapter, source 02’s Section 2 and source 04’s Section 2 open their Part II chapters. Source 02’s Appendix B, a rational example for its tail theorem, is in Part I with that theorem; its Appendix A is in Part V.
- *Results proved by several sources* are printed in each source’s own form, because the forms and hypotheses differ (general $[-a, b]$ against $a = \delta$; strict against non-strict events; all groups against odd-order groups), and because the proofs are independent, mostly by different routes. A dated note at each later print names the first print and credits every source; where two proofs follow the same argument, the note says so. No proof was replaced by a pointer.
- *The sources’ question sections* are printed as delivered, and the write’s Section V.2 deduplicates them with credits.

[write] **6 October 2026; sources 06–10.** *Sources 06–10.* The five archives arrived in commit 62e21161f (6 October 2026) and were placed in this directory by ad37962c4 (“Place batch 115 (2/2)”), which retired them from `docs/incoming/`; they survive in the arrival commit (`git show 62e21161f:docs/incoming/<archive>`). Not staged: the five PDFs and the five manuscripts and delivery READMEs, whose text is printed here.

No.	Archive	Bytes	Files	Pin	Manuscript
06	Gowers_Refinements (1).zip	647,760	19	7d99e5dee	1,958 lines, 27 pp.
07	Gowers_Transport_Rigidity.zip	359,160	9	ae0e89574	1,472 lines, 21 pp.
08	gowers_cube_refinements.zip	541,854	8	(blobs)	1,677 lines, 25 pp.
09	gowers_cubical_expansions.zip	512,022	10	7d99e5dee	1,794 lines, 27 pp.
10	gowers_refinements.zip	664,416	15	7d99e5dee	2,443 lines, 34 pp.

The pins precede HEAD; ae0e89574 is a descendant of 7d99e5dee. Source 08 records blob identifiers, not a commit; they are those of 7d99e5dee. All blob identifiers the sources cite were checked at intake.

Authorship and disclosures. Source 06: “Research report prepared for Vladimir Reshetnikov” (title block and PDF author field). Source 07: “Research draft” (title page, running head, PDF author field). *Source 08* is “Prepared for Vladimir Reshetnikov”, PDF author field “Research memorandum prepared for Vladimir Reshetnikov”, and its title page calls it “This AI-assisted research draft”. *Source 09* credits ChatGPT: its author line reads “Research manuscript prepared with ChatGPT”, and so does its PDF author field. Source 10: “A mathematical research report”, PDF author field “Research report”. No source names a human author. Source 10 mentions an “independent review” that is not part of its delivery (note in Section V.43).

What each contributed. 06: the exact fixed- M extraction (Part I), a two-target generalized von Neumann bound and quadratic progression discrepancy (Part III), carry counts for general families with entropy uniform in the modulus, a second certificate of $C_4 = 154$, and a Proposition 17.7 coefficient (Part IV), an iteration potential with optimal leading cost (Part V). 07: transport-sensitive rigidity at Lemma 10.9, with orders five and nine and a sharp threshold (Part II). 08: the fourth-order cube theory with the complete odd-order three-cube polynomial and the refutation of the quartic lower bound (Part III). 09: the complete classification

through order five on every finite abelian group, the sixth-order and determinant-five phenomena (Part III). 10: the optimal density cap and the frontier (Part I), a weighted energy extraction and the vertical-fibre Freiman extraction with $2^{-209}\gamma^{103}$ (Part II), torsion-sensitive counting and the polynomial-colouring threshold (Part III), iteration bookkeeping (Part V).

Overlaps. The ten are independent manuscripts: no two of them share more than 1.14% of their text (intake measurement, 8-word shingles). Sources 06 (`Gowers_Refinements (1).zip`) and 10 (`gowers_refinements.zip`) are not editions of one another (1.09%; different titles and theorem sets).

Where the second write had to choose. Each delivered section of 06–10 is printed whole in exactly one Part, by its main subject; a section that serves several themes goes where its main result is used: source 06’s computation section (Section 7) is in Part IV with its certified carry counts, source 09’s Section 13 (relation to progression bounds) in Part III with its cube results, sources 08’s and 09’s certificate appendices beside the theorems they certify, their provenance appendices in Part V. Source 07 is a chapter of Part II with its notation table first, by the intake’s recommendation; source 09 is printed after source 08 in Part III as the fullest cube treatment, and sources 03 and 04 are not moved. Results proved by several sources are printed in each source’s form with dated credit notes at both places, as in the first write.

F.5 Relation to the formal project

The formal project consists of the corrected transcription (`Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`), the Lean development (`Combinatorics/Ramsey/Lean/GowersSzemerédi`, namespace `LeanProofs.GowersSzemerédi`) and the ledger `Combinatorics/Ramsey/gowers-proof-status.json`, which records each numbered statement of the paper as a Lean proposition and its exact companion theorem if there is one. The sources compare their results with *corrected* statements of this project: Lemma 2.3 with denominator 16, Corollary 2.4 with 128π and $\alpha/2$, Corollary 2.5’s exported pair and the stronger pair inside its proof, and so on.

Ledger at the write. At HEAD 25df8755d (6 October 2026) the ledger records 98 statements with exact companions and 22 open, against 88 and 32 at all three pins. The ten statements closed since the pins are `corollary_5_6`, `corollary_5_7`, `lemma_5_9`, `lemma_5_10`, `corollary_5_11`, `lemma_5_14`, `lemma_16_1`, `lemma_16_3`, `lemma_16_5` and `lemma_16_6`. The development is being extended by another session as this report is written, so these counts may already have moved; the crosswalk (Section V.1) records every declaration it cites with file and line at 25df8755d. Two stale statements of the sources are corrected in dated notes: source 02’s description of Corollary 5.6 as an unverified import (it is now `corollary_5_6_holds`), and source 03’s (and implicitly source 01’s) count of 88 exact and 32 open statements.

What is formalized. One statement of the five sources is formalized: source 04’s Lemma III.4.5 (lifting a modular progression from a short interval), with source 03’s rectification remark, as `hasNatAP_of_short_modular_sequence` and `hasNatAP_of_hasModAP_image` (`Proofs18IntervalTransfer.lean`, lines 22 and 92). Those declarations were added in 327773149, source 04’s own pin, before the sources arrived. Some baselines the sources improve are proof-level facts of the development: the stronger Corollary 2.5 pair (`cor25_large_scale`) and the exact second moment of Lemma 9.2 (inside `lemma_9_2_holds`). The Gowers–Cauchy–Schwarz inequality (`gowers_cauchy_schwarz`) and the triangle inequality (`gowersNorm_add_le`) are formal tools that sources 03 and 04 use. The simultaneous partition of sources 01 and 05 is not the formal degree-one partition `section5_strong_linear_partition` (`Proofs05LinearPartition.lean`, line 86), which is a one-phase statement with target lengths up to $r^{1/16}$ and diameter $r^{-1/8}N$. *Nothing else in this report is formalized, and the report gains no formal status* by its placement beside the Lean development. Each source’s formalization plan is printed in Part V.

[write] 6 October 2026; sources 06–10. *Sources 06–10.* At HEAD 764740f07 the ledger records 99 exact companions and 21 open statements (crosswalk, Section V.1). No statement of sources 06–10 is formalized. Two of them strengthen the conclusion of a formal statement under exactly its hypotheses: source 07’s Corollary II.17.3 (order five in place of order two in lemma_10_9) and, by a short argument recorded at the write, source 10’s Corollary II.21.2 and Theorem II.21.1 (which imply the conclusions of corollary_7_6 and lemma_7_5); none of these implications is formalized. Source 07’s order nine needs hypotheses that lemma_10_9 does not have. The report still gains no formal status by its placement.

F.6 Source 01. The two interfaces and the main conclusions

Source 01, delivered Section 1.

Delivered title block. “Sharp Local Refinements of Gowers’s Szemerédi Argument / Density transfer, carry-pattern entropy, and formalization interfaces / Research report prepared for Vladimir Reshetnikov / Mathematical development and computational checks with ChatGPT / 6 October 2026”; PDF author field “Research report prepared with ChatGPT”. Archive `Gowers_Quantitative_Refinements(1).zip`, manuscript `gowers_refinements.tex` (1,980 lines, 31 pages), pin 3a25020a6. Printed below: the abstract, the status paragraph and Section 1; Sections 2–8 are in Part I, 9–13 in Part IV, 14–16 and Appendices A–B in Part V.

Abstract of source 01 (as delivered). We develop two quantitative refinements of components of Gowers’s 2001 proof. First, a convex-envelope calculation solves the finite weighted extremal problem of converting correlation with a cellwise short-arc phase into a density tail. The resulting bound is sharp in the stated information model, keeps track of occupied mass in exceptional cells, and yields an explicit Fourier density increment on a progression of length on the scale $\sqrt{\eta N / [\delta(1 - \delta)]}$, in place of the $\sqrt{\eta^3 N}$ length in the printed Corollary 2.5. Second, the number of Boolean subset-sum floor patterns with coefficients in $(-r, r)$ is exactly $(2r)^k C_k$, where C_k counts fractional carry patterns. We prove a quadratic-exponent upper bound, an elementary $2^{\lfloor k^2/4 \rfloor}$ lower bound, and, using the classical enumeration of linear threshold functions, $\log_2 C_k \sim k^2$. A polynomial-feature extension has matching leading entropy for fixed degree. An affine version bounds translation classes independently of the modulus, providing a smaller combinatorial loss at the phase-removal interface in Section 17. Exact rational certificates give $C_1 = 1$, $C_2 = 2$, $C_3 = 10$, and $C_4 = 154$. All local results are proved here; the threshold-function input is explicitly external. These refinements are not asserted to improve the best global bounds for Szemerédi’s theorem, and no Lean kernel verification is claimed.

Status and scope. This is a research manuscript, not a refereed publication. “Sharp” below always refers to a precisely specified extremal problem, not to every arithmetic realization of that problem. The work establishes strengthened statements relative to the cited components of Gowers’s paper; an exhaustive priority search for every formulation has not been completed. The exact factorization, extremal transfer formulas, and their proofs can be assessed independently of any novelty claim.

Two recurrent losses in a density-increment proof are logically distinct. One converts phase correlation into a local increase of density. The other counts the discrete floor patterns created when a continuous phase is cut into progression cells. Improving one does not automatically improve every subsequent loss. We keep these interfaces separate.

F.6.1 An explicit density-increment replacement

Write $e(t) = \exp(2\pi it)$ and $[N]_0 = \{0, \dots, N-1\}$. Suppose $A \subseteq [N]_0$ has density $0 < \delta < 1$ and, for some real ξ ,

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (\mathbf{1}_A(n) - \delta) e(\xi n) \right| \geq \eta > 0. \quad (\text{F.6.1})$$

Put $\kappa = \eta/[2\delta(1-\delta)]$. Necessarily $0 < \kappa \leq 1$. Theorem I.6.1 proves that a proper arithmetic progression $P \subseteq [N]_0$ exists with

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{N}{2\pi}} \arcsin \frac{\eta}{4\delta(1-\delta)} \right\rfloor \right\}, \quad (\text{F.6.2})$$

$$\frac{|A \cap P|}{|P|} \geq \delta + \frac{3\delta(1-\delta)\kappa}{4 - (4-3\delta)\kappa} \geq \delta + \frac{3\eta}{8}. \quad (\text{F.6.3})$$

No primality hypothesis is needed. A nonzero cyclic Fourier coefficient is a special case of (F.6.1). The printed Corollary 2.5 of Gowers gives length $\sqrt{\eta^3 N/(128\pi)}$ and increment $\eta/8$ in that special case [1, Corollary 2.5].

The ingredients here are elementary. In particular, the fact that a minimum-cell-size partition avoids some subsequent length loss is already implicit in the preceding Corollary 2.4. Our point is the density-sensitive, integer-exact formulation and the optimal finite weighted transfer theorem behind it, not a claim that the Fourier density-increment strategy itself is new.

F.6.2 A quadratic rather than cubic floor-pattern exponent

For $\alpha \in \mathbb{R}^k$, define

$$F_\alpha(S) = \left\lfloor \sum_{i \in S} \alpha_i \right\rfloor, \quad S \subseteq [k].$$

For integers $r_i \geq 1$, let $\mathcal{F}(\mathbf{r})$ be the set of these functions when $-r_i < \alpha_i < r_i$. Define

$$C_k = \# \left\{ S \mapsto \left\lfloor \sum_{i \in S} x_i \right\rfloor : x \in [0, 1)^k \right\}.$$

Theorem IV.1.2 gives the exact identity

$$|\mathcal{F}(\mathbf{r})| = \left(\prod_{i=1}^k 2r_i \right) C_k. \quad (\text{F.6.4})$$

Let

$$H_k = (k-2)2^{k-1} + 1, \quad \mathcal{R}(h, d) = \sum_{j=0}^d \binom{h}{j},$$

where $\binom{h}{j} = 0$ for $j > h$. We prove

$$2^{\lfloor k^2/4 \rfloor} \leq C_k \leq \mathcal{R}(H_k, k). \quad (\text{F.6.5})$$

In the isotropic case this replaces the 2^{2rk^3} upper bound in Gowers's Corollary 17.5 by $(2r)^k \mathcal{R}(H_k, k)$ [1, Corollaries 17.5–17.6]. The threshold-function theorem discussed in Section IV.2 then yields

$$\log_2 C_k = k^2 + o(k^2). \quad (\text{F.6.6})$$

Thus a subquadratic exponent cannot bound the full family, even when the coefficient range is fixed. This optimal leading exponent is obtained by combining a reduction proved here with a classical external theorem.

F.6.3 What is, and is not, concluded

The proofs below solve two specific auxiliary problems: the optimal correlation-to-density-tail problem with freely placed short-arc phases, and the exact dependence of floor-pattern counts on integral coefficient ranges. They also locate a smaller translation-class budget for a phase-removal step. They do not supply a new inverse theorem for higher Gowers norms or a new globally optimized iteration.

For comparison, Leng, Sah, and Sawhney prove, for $k \geq 5$ and some $c_k > 0$,

$$r_k(N) \ll N \exp(-(\log \log N)^{c_k}),$$

where $r_k(N)$ is the largest size of a k -term-progression-free subset of $[N]$ [2]. That checked benchmark is already much stronger than what is obtained by merely adjusting constants in the 2001 argument.

F.7 Source 02. Scope, sources, and the precise improvements

Source 02, delivered Section 1.

Delivered title block. “Energy-Sensitive Inverse Steps and Sharp Density Extraction / Quantitative refinements of Gowers’s Szemerédi argument / Research article prepared for Vladimir Reshetnikov / October 6, 2026”. Archive `Gowers_Szemerédi_Refinements.zip`, manuscript `gowers_refinements.tex` (823 lines, 20 pages; not the same file as source 01’s `gowers_refinements.tex`). No pin in its README; its Appendix A records the observed branch reference `66c02f8a1`. Printed below: the abstract and Section 1; Sections 2–5 are in Part II, 6–9 and Appendix B in Part I, 10–12 and Appendix A in Part V.

Abstract of source 02 (as delivered). We strengthen two interfaces in the proof of Szemerédi’s theorem: the passage from derivative Fourier correlations to additive energy, and the passage from local phase correlations to a density increment. The first result is a sharp, mixed-function, weighted extension of Gowers’s Proposition 6.1, valid on every finite abelian group without a pointwise boundedness assumption. If $m(f) = \mathbb{E}|f|^2$, if $\mathcal{U}(f) = \|f\|_{U^3}^8$, and if $\mathcal{T}_w(f, g)$ is a weighted derivative-correlation statistic, then

$$\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4 \mathcal{U}(g), m(g)^4 \mathcal{U}(f)\} \mathcal{E}(w).$$

Here $\mathcal{E}(w)$ is explicitly normalized additive energy on $G \times \widehat{G}$. We prove the identities underlying this inequality, give equality examples, and quantify its two Cauchy–Schwarz deficits. For balanced indicators, the resulting graph-energy estimate retains the variance $\delta(1 - \delta)$ rather than discarding it.

The second result is a sharp real-valued phase-freezing inequality. If the phases in every cell have chordal diameter at most 2ρ , local correlation C , signed cell mass S , and total absolute mass L satisfy $C \leq \rho L + (1 - \rho)S$. A sharp one-sided moment inequality then gives the total mass of cells with any prescribed positive density increment. This strengthens Lemma 5.15, yields an explicit refinement theorem using a single global phase budget, and permits closed-form optimization of that budget. The elementary results are proved in full; the polynomial partition specialization explicitly imports the repository’s corrected Corollary 5.6. These are local quantitative refinements, not a claimed improvement of the best global bound for Szemerédi’s theorem. Literature priority for the exact combined formulations is not established.

F.7.1 The two proof interfaces

Gowers’s proof [1] repeatedly turns a failure of uniformity into structure and then into increased density on an arithmetic progression. The supplied ProveIt transcription and Lean development [20] make the intermediate interfaces particularly useful targets: they have concrete finite statements, explicit normalizations, and independently reusable conclusions.

The starting point for our first improvement is Proposition 6.1. In the paper’s unnormalized Fourier convention, if $f : \mathbb{Z}_N \rightarrow \mathbb{C}$ satisfies $|f| \leq 1$, $B \subseteq \mathbb{Z}_N$, and $\phi : B \rightarrow \mathbb{Z}_N$ satisfies

$$\sum_{h \in B} |\widehat{\Delta_h f}(\phi(h))|^2 \geq \alpha N^3, \quad (\text{F.7.1})$$

then the graph of ϕ contains at least $\alpha^4 N^3$ additive quadruples. The proof uses Cauchy–Schwarz and Fourier orthogonality, but bounds two quantities depending on f by their worst-case values. We retain both quantities and allow a weighted, multivalued spectrum and two different input functions.

For our second improvement, Lemma 5.15 assumes a real, mean-zero, 1-bounded function and a partition $P_1, \dots, P_M$ such that

$$\sum_j \left| \sum_{x \in P_j} f(x) \right| \geq \alpha N.$$

It produces a cell with mean at least $\alpha/4$ and size at least $\alpha N/(4M)$. We obtain a full tradeoff between the requested cell mean and the mass of all cells attaining it. The estimate uses the asymmetric range of a balanced indicator, $[-\delta, 1 - \delta]$, instead of only $[-1, 1]$.

Between these two interfaces lie the phase partitions of Section 5. We sharpen the geometric step that removes a nearly constant phase and then track one global error budget through an arbitrary collection of input progressions. No lower bound on each individual local correlation is needed.

F.7.2 What is and is not claimed

The weighted Fourier-energy inequality, the real-phase inequalities, the sharp moment bound, and their stated consequences have complete proofs below. The coefficient in the main energy inequality and the constants in the abstract phase and moment inequalities are optimal in the senses explicitly proved. “Sharp” does not mean that every later progression-length bound, or the complete density-increment iteration, is optimal.

The estimate $\|f\|_{U^3} \leq \|f\|_2$ and the general subject of extremizers are established mathematics; the broader near-extremizer theory is treated by Eisner and Tao [7]. Our elementary proof of that estimate is included to make the constants and the treatment of torsion transparent. Similarly, we do not claim that retaining a Cauchy–Schwarz denominator is a new general technique.

These results do not supply a new best asymptotic estimate for $r_k(N)$. For comparison, Leng, Sah, and Sawhney prove, for $k \geq 5$, a bound of the form

$$r_k(N) \ll N \exp(-(\log \log N)^{c_k}), \quad c_k > 0,$$

using quasipolynomial inverse-theorem bounds and a different quantitative organization of the density-increment argument [2, 3]. We use that published preprint statement as a benchmark, not as a claim to have exhaustively determined the latest record in every progression length.

The exact combined formulations here are independently derived refinements of the supplied proof. The literature check does not establish publication priority, and no previously documented

major open problem is declared solved. Section V.9 separates useful next research problems from results proved here.

F.8 Source 03. Scope, starting point, and principal results

Source 03, delivered Section 1.

Delivered title block. “Additive Combinatorics / Sharper local estimates in Gowers’s proof of Szemerédi’s theorem / Density increments, exact cube expansions, and dependent random choice / Research report prepared for Vladimir Reshetnikov / 6 October 2026”. Archive `gowers_local_refinements.zip`, manuscript `gowers_local_refinements.tex` (2,392 lines, 36 pages), pin 3a25020a6. Its title page also says: “Source basis: Gowers (2001) and the two requested `ProveIt` directories, pinned to the revision documented in Section V.13. The accompanying code checks finite instances; the proofs in the text establish the general statements.” Printed below: the abstract and Section 1; Sections 2–4 are in Part I, 5–6 in Part III, 7–8 in Part II, 9–11 and Appendix A in Part V.

Abstract of source 03 (as delivered). We develop explicit local strengthenings of several estimates used in Gowers’s proof of Szemerédi’s theorem and relate them to the accompanying `ProveIt` Lean development. First, an exact asymmetric selection inequality converts partition discrepancy into a sharp lower bound for the mass of cells above a chosen density threshold. Combined with an elementary progression partition, it gives a Fourier density increment whose length retains a factor $[\delta(1 - \delta)]^{-1/2}$. Second, a telescoping progression count replaces an exponential error coefficient by a density-weighted sum. Third, we compute the entire fourth-order term of the centered cube count on every finite abelian group. Its two contributions are ordinary Fourier energy and an explicit correction from the quotient by doubling. The resulting fourth-order error and its universal leading constant are sharp, including for sequences of actual sets. Finally, a cubic dependent-random-choice certificate improves the exact-threshold subset size in Lemma 7.4 from order $\delta^5 n$ to order $\delta^3 n$; a separate one-witness argument greatly improves the elementary constants in Proposition 7.3. All claimed refinements have complete proofs. We identify their relation to established methods, distinguish them from stronger modern theorems, and give a concrete formalization and research agenda. No improvement to the global record for Szemerédi’s theorem or solution of a major outstanding conjecture is asserted.

The aim of this report is to strengthen concrete parts of the proof in [1], with statements precise enough to be independently checked and incorporated into the supplied formal development [21]. The two supplied directories have different roles. The paper directory contains an edited transcription of the original article, including finite-scale repairs and explanatory comments. The Lean directory contains a catalogue of numbered propositions together with many companion proofs. We compare both with the published article rather than silently identifying their statements.

The work concentrates on three places where numerical losses can be removed without inventing a new inverse theorem: selection from a partition, expansion of a multilinear count, and extraction of a structured subfamily by random sampling. These are useful mathematical components in their own right. Their improvements are also easier to audit than a claim to have sharpened the final tower of parameters after an unexamined substitution into a long proof.

F.8.1 What is established

The following are the central conclusions. The conventions and all finite-scale assumptions appear in the cited statements within this report.

1. **Optimal selection from asymmetric discrepancy data.** For a mean-zero random variable $X \in [-a, b]$, its positive first moment T determines the exact best universal lower bound

$$\Pr(X > t) \geq \max \left\{ 0, \frac{(a+t)T - at}{a(b-t)} \right\}.$$

Theorem I.14.1 includes extremizers. Specializing to cell averages strengthens Lemma 5.15 and allows the density and size requirements to be chosen simultaneously.

2. **A Fourier increment with the density factor preserved.** For a normalized nonzero Fourier coefficient of size ρ , an explicit proper integer progression has the bounds in (I.15.3). The construction works for composite moduli and handles singletons and integer rounding directly. It improves on the stronger estimate already present inside the repository's proof, not merely its weaker exported conclusion.
3. **A smaller progression-count error.** Theorem III.1.2 gives a density-weighted sum with $k-2$ terms. Under the hypotheses of Corollary 3.3, in its original index order, this replaces $2^k \alpha$ by at most $(k-2)\alpha$. Theorem III.1.2 also permits any family of slopes whose pairwise differences act invertibly on the group.
4. **An exact fourth-order cube expansion.** Theorem III.2.6 computes the leading possible error rather than only bounding it. There are no terms of degrees one, two, or three. Among degree-four supports, only ordinary parallelograms and a second family visible to order-two characters contribute. Corollary III.2.7 changes the displayed error scale in Lemma 3.10 from $\alpha^{1/2^d}$ to $\alpha^{4/2^d}$. The fourth power and the universal leading coefficient are optimal in the precise senses proved below.
5. **A larger common-neighborhood core at the same threshold.** Theorem II.5.2 replaces $2^{-1/2} \delta^5 n$ by $7\delta^3 n / (3\sqrt{6})$, while retaining both the threshold $\delta^2 m / 2$ and the 90% requirement in Lemma 7.4. The certificate also gives weighted higher-tuple versions.
6. **An explicit elementary energy-to-structure improvement.** For $E(A) \geq c|A|^3$, Theorem II.6.3 gives

$$|A'| \geq (c/8)|A|, \quad |A' - A'| \leq 2^{19} c^{-9} |A|.$$

Both exponents improve on Proposition 7.3's displayed constants. The proof works in every abelian group and uses only elementary counting. It is substantially weaker than modern BSG results.

Several ingredients here are established general methods: normalized Gowers norms, telescoping, dependent random choice, and the passage from many additive quadruples to a small difference set. The contribution of this report is the explicitly proved package of refinements, the exact quartic classification with its sharpness statements, the optimized scalar selection problems, and a source-aware route to formalization. We make no historical priority claim for these formulations without a separate, exhaustive literature investigation.

F.8.2 Relation to contemporary quantitative results

Let $r_k(N)$ denote the largest size of a subset of $\{1, \dots, N\}$ containing no nonconstant k -term arithmetic progression. Gowers's quantitative theorem gave a bound of the shape $r_k(N) \ll_k$

$N(\log \log N)^{-c_k}$. Leng, Sah, and Sawhney prove the stronger estimate

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}) \quad (k \geq 5) \quad (\text{F.8.1})$$

for suitable $c_k > 0$ [2]. Their argument uses quasipolynomial inverse-theorem bounds [3] and a density increment strategy related to [5]. None of the estimates established here improves (F.8.1).

The historical questions at the end of Gowers’s paper must also be read in their date. For example, Jaber, Liu, Lovett, Ostuni, and Sawhney prove that a corner-free set in $G \times G$, with G finite abelian, has size at most

$$|G|^2 \exp(-(\log |G|)^{\Omega(1)})$$

[13]. Thus simply restating the 2001 request for a reasonable two-dimensional corners bound would not identify an unsolved problem. Our research questions in Section V.14 instead specify which finer optimization, extension, or formalization remains to be done.

F.9 Source 04. Purpose, results, and scope

Source 04, delivered Section 1.

Delivered title block. “Additive Combinatorics / Sharper Local Bounds in Gowers’s Proof of Szemerédi’s Theorem / Polynomial homomorphism restriction, sharp cube estimates, and floor-pattern complexity / Prepared for Vladimir Reshetnikov / 6 October 2026”. Archive `gowers_quantitative_refinements.zip`, manuscript `gowers_quantitative_refinements.tex` (2,987 lines, 40 pages; the base of the merge, staged as `article.tex`), pin 327773149. Its first lines read “Complete manuscript source. See README.txt for build instructions. Prepared on 2026-10-06. No new Lean kernel verification is claimed.”, and its title page “Companion materials: editable $\text{\LaTeX}$ source, compiled PDF, exact rational pattern certificates, verification programs, and a source manifest.” Printed below: the abstract and Section 1; Sections 2–5 are in Part II, 6–7 in Part III, 8 in Part IV, 9–11 and Appendices A–B in Part V.

Abstract of source 04 (as delivered). We give explicit refinements of local estimates in Gowers’s proof of Szemerédi’s theorem and identify interfaces for their formalization. Exact Fourier second moments sharpen the lifting of respected additive relations. A powered-cosine selection argument, with Fourier-weighted control of exceptional relations and separate treatment of repetitions, gives a polynomial restriction theorem at every fixed tuple order. For real centered functions on odd-order groups, we prove the sharp gap $\|f\|_{U^3}^8 \geq 2\|f\|_{U^2}^8$, classify equality, and obtain stability. The resulting three-cube bound has sharp leading coefficient $6\sqrt{2}$. Without 2- or 3-torsion, its first correction has optimal exponent $16/3$, witnessed by an explicit two-frequency family. Further results retain density factors in progression counting and determine the quadratic exponential scale of fractional floor patterns, improving the associated local phase-extraction loss. The article includes complete proofs, exact examples, supplementary verification programs, a map to the inspected Lean files, and twelve research questions. Improvements are relative to the specified local source estimates; worldwide priority, a new best global Szemerédi bound, and new kernel-checked Lean proofs are not claimed.

This article develops explicit improvements to several local estimates in Gowers’s proof of Szemerédi’s theorem [1]. The starting points are the paper and its companion Lean development in the ProveIt repository [23]. The improvements come from retaining quantities that a convenient coarse estimate discards: the size of a Fourier support measure, cancellation in a centered cube expansion, the simultaneous dependence of floor values on their parameters, and the number of patterns that occur at a fixed parameter.

The most substantial quantitative change concerns the passage from a map respecting many additive quadruples to a restriction respecting almost all additive tuples of a higher order. Here is a concrete consequence, stated at the same order as Gowers's Corollary 9.4. Suppose that $B \subset \mathbb{F}_p$, $|B| = \beta p$, and $\phi : B \rightarrow \mathbb{F}_p$ respects at least $\gamma|B|^3$ additive quadruples. If $0 < \beta, \gamma, \eta \leq 1$ and

$$p \geq 2^{102} \gamma^{-112} \eta^{-16} \beta^{-1},$$

then there is $B' \subset B$ containing at least

$$2^{-95} \gamma^{112} \eta^{15} |B|^{15}$$

ordered additive 16-tuples, at least a proportion $1 - \eta$ of which are respected by ϕ . Both the exponents and the size hypothesis are explicit. The proof uses one powered cosine filter and bounds the exceptional relations with their Fourier weights. It generalizes to every fixed tuple order $2d$, with a retention bound proportional to $\alpha^{2d} \eta^{2d-1}$ when the initial respected-tuple proportion is normalized by $|B|^{2d-1}$.

The second set of results concerns additive cubes. For a real centered function on a finite abelian group of odd order, we prove the sharp inequality

$$\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8.$$

Equality, apart from the zero function, is characterized by support on one conjugate pair of Fourier characters. A quantitative stability statement identifies an almost extremizing Fourier pair. Combined with an exact enumeration of the surviving terms in a centered three-cube expansion, this gives a sharp leading coefficient $6\sqrt{2}$ for the cube excess as a function of $\|f\|_{U^3}$. In groups whose order is coprime to 6, a further refinement replaces a fifth-order remainder by a remainder of order $u^{16/3}$. An explicit two-frequency family shows why that fractional exponent occurs.

The remaining results provide useful quantitative interfaces for the larger proof. A telescoping expansion reduces the exponential counting factor in Corollary 3.3 to a density-weighted sum with $k - 2$ terms. The statement extends to composite cyclic groups and other finite abelian groups under exact invertibility hypotheses. In Section 17, separating integer parts from fractional parts gives polynomial dependence on the slope range. Matching quadratic-order upper and lower bounds determine the exponential scale of the fractional floor-pattern count. A distinction between global patterns and the patterns for a fixed slope improves a subsequent Cauchy–Schwarz loss.

F.9.1 What is established, and what is not claimed

All results asserted as theorems below have proofs. The supplementary programs check finite instances, exact support counts, coefficient inequalities, and rational certificates. These computations are additional evidence and reproducibility aids; the general mathematical statements do not rest on numerical experimentation.

The claim of improvement is relative to the explicitly identified estimates in Gowers's paper and the inspected repository files. Hölder interpolation, generalized von Neumann inequalities, and hyperplane-arrangement estimates are established tools. A targeted literature review did not determine priority for every exact refinement assembled here. Accordingly this article does not present the collection as a certified first discovery in the literature.

There is also a material distinction between improving a local lemma and improving the best bound in Szemerédi's theorem. Leng, Sah, and Sawhney prove, for fixed $k \geq 5$, a bound of the form

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}), \quad c_k > 0,$$

using their quantitative inverse theorem [2, 3]. The local arguments here do not improve that result. Green and Tao’s four-term work [6] provides an additional example of why approximate-homomorphism restriction is a useful component of a density increment proof. Manners’s polynomial true-complexity theorem [14] places the counting arguments in a broader established theory. No unresolved global Szemerédi or inverse-theorem conjecture is declared solved in this article.

F.9.2 The inspected repository

The repository was inspected on 6 October 2026. Its paper transcription already contains editorial corrections, and the Lean directory separates formal statement definitions from companion proof modules. The referenced files agree with the snapshot

327773149bd272239f1f6c0015f76d5097512c88.

In particular, the relevant files include `Proofs09Moments.lean`, `Proofs03CubeUpper.lean`, `Proofs03ProgressionAverage.lean`, and `Proofs03ProgressionCount.lean`. The appendix gives a concrete plan for adding the strengthened results. No new Lean proof is claimed to have been compiled or kernel checked in the present project, and no repository changes are part of this deliverable.

F.10 Source 05. Scope, baseline, and main contributions

Source 05, delivered Section 1.

Delivered title block. “Sharp Finite Phase Partitions and Density-Sensitive Increments / Strengthenings of Gowers’ localization lemmas, simultaneous phases, and spectral-energy extraction / Research note prepared for Vladimir Reshetnikov / 6 October 2026”; empty PDF author field. Archive `gowers_strengthening.zip`, manuscript `article.tex` (847 lines, 18 pages), pin 327773149. Printed below: the abstract and Section 1; Sections 2–6 are in Part I, 7–10 in Part V.

Abstract of source 05 (as delivered). We investigate the finite localization step in Gowers’ proof of Szemerédi’s theorem, using the corrected paper and the corresponding Lean modules in the ProveIt repository as the comparison baseline. Three improvements are established with complete proofs. First, the corrected lower constant $1/4$ in Lemma 2.3 can be replaced by $1/3$ without changing its upper length bound; $1/3$ is the largest possible universal constant in that formulation. Second, a nonzero Fourier coefficient of normalized magnitude at least α for a set of density δ produces an arithmetic progression of length at least

$$\sqrt{\frac{\alpha N}{16\pi\delta(1-\delta)}} \quad \text{and density at least} \quad \delta + \frac{\delta\alpha}{4\delta - \alpha}.$$

An exact integer formula is stronger, and a positive proportion of the partition has a controlled increment. Third, simultaneous localization of m real linear phases has optimal order $1 + (\varepsilon^m N)^{1/(m+1)}$, including a partition with a lower bound on every cell and an explicit algebraic obstruction to a better exponent. A density-sensitive spectral-energy inequality links these partitions to multiplicative increments, and unequal phase tolerances can be optimized explicitly. The methods are elementary refinements of classical arguments; claims of literature priority, a new global Szemerédi bound, and completed Lean verification are not made. Exact-arithmetic partition tests and high-precision Fourier diagnostics accompany the article.

F.10.1 What is being strengthened

Gowers’ proof [1] separates a counting argument from the extraction of structure and then a density increment. Our subject is the last conversion: turning a structured correlation into density on a sufficiently long ordinary arithmetic progression. Even at the linear-phase level, this conversion involves three distinct sources of loss: integer rounding in a partition, the error from replacing an oscillating phase by a constant, and the passage from absolute block sums to a positive block average.

The requested repository contains an edited transcription of the paper and a substantial Lean development [24, 25, 26]. The transcription is not identical to the printed paper. In particular, it explicitly repairs finite rounding issues in Lemma 2.3 and Corollary 2.4. The relevant comparison is therefore with the *corrected repository statements*, not merely with the historical printed constants.

Write $t = rs/M$, where r is the length of the domain interval, M the modulus, and s the allowed image diameter. The repository’s version of Lemma 2.3 asserts a partition into progressions with lengths between $\sqrt{t}/4$ and $\sqrt{t}$. The printed lower constant $1/2$ cannot hold uniformly: when $r = s = M = 5$ and the phase is constant, it would require a partition of five points into sets all of size two. We prove that $1/3$, not $1/4$, is the sharp universal lower constant when the upper constant is fixed at one.

The repository’s Corollary 2.5 states a progression length of at least $\sqrt{\alpha^3 N / (128\pi)}$ and a density increment $\alpha/8$. Its large-scale proof already obtains the stronger pair $\sqrt{\alpha N / (128\pi)}$, $\alpha/4$ before weakening the conclusion [26]. Thus simply noticing this stronger pair would not constitute an advance over the repository. Our density-sensitive length estimate improves beyond that proof-level baseline as well.

F.10.2 A map of the results

The exact finite strengthening is Theorem I.16.4, including the optimality of $1/3$. Corollary I.16.6 simultaneously improves the corrected Corollary 2.4 while keeping its upper length bound. The density-sensitive replacement for Corollary 2.5 is Theorem I.17.4. The simultaneous-phase theorem and its matching exponent obstruction are Theorems I.18.2 and I.18.5. The spectral statement, Theorem I.19.2, depends on the residual-variance estimate

$$E \leq \sqrt{EV} + 2R(v - V), \tag{F.10.1}$$

where E is captured Fourier energy, V is the variance of the block densities, $v = \delta(1 - \delta)$, and R is a weighted phase-oscillation budget.

F.10.3 What is not asserted

The elementary ideas used below—pigeonholing on a torus, grouping residue classes, algebraic norms, and conditional expectation—are established mathematical tools. Our contribution is a set of explicit, proved finite statements and their quantitative combination. We have not established that every displayed refinement is absent from the literature.

Nor do we claim an improvement to the best global bounds for $r_k(N)$. Already in 2024, Leng, Sah, and Sawhney proved, for $k \geq 5$, an estimate of the form $r_k(N) \ll N \exp[-(\log \log N)^{c_k}]$, using substantially deeper inverse and localization results [2]. This is a comparison point, not an assertion of an exhaustive October 2026 literature ranking. Their work also emphasizes simultaneous

localization and multiplicative increments. The results here concern an explicit linear-phase interface and do not replace the higher-order part of either proof.

F.11 Source 06. Scope, conclusions, and mathematical provenance

Source 06, delivered Section 1.

Added in the second write, 6 October 2026. Delivered title block. “Sharp Local Bounds in Gowers’ Proof of Szemerédi’s Theorem / Carry patterns, quadratic counting errors, and optimal density extraction / Research report prepared for Vladimir Reshetnikov / 6 October 2026”; the same in the PDF author field. Archive `Gowers_Refinements (1).zip` (an independent manuscript: not an edition of source 10’s `gowers_refinements.zip` or of source 01’s archive), manuscript `Gowers_Refinements.tex` (1,958 lines, 27 pages), pin 7d99e5dee (its `provenance.json`). Printed below: the abstract and Section 1; Section 5 is in Part I, 3 in Part III, 2, 4 and 7 in Part IV, 6, 8 and 9 in Part V.

Abstract of source 06 (as delivered). We develop explicit refinements of several local arguments in Gowers’ proof of Szemerédi’s theorem, with the accompanying ProveIt development as a precise reference point. For Boolean floor patterns with slopes in $(-r, r)^k$, we prove the exact count $(2r)^k C_k$, where C_k is a unit-cube carry count. An elementary arrangement bound and a reduction from Boolean threshold functions show $\log_2 C_k \sim k^2$; the same leading entropy persists after reduction modulo any fixed $M \geq 2$. A corresponding affine bound changes the pattern loss in a localized phase-removal lemma from $2^{-O(k^3)}$ to $2^{-2k^2 - O(k \log k)}$. For progression counts, a Cauchy–Schwarz induction retaining two distinguished functions gives quadratic, rather than linear, dependence on the balanced uniformity norm, with explicit density factors. We also solve a weighted finite-partition density-extraction problem exactly, including its fixed-cell-count extremizers, and give a density-dependent iteration potential with an asymptotically optimal leading cost for a specified abstract recurrence. Complete proofs, computational certificates, a source-level Lean interface map, and a further-research agenda are included. These results strengthen the indicated auxiliary statements; a new global bound for Szemerédi’s theorem is not established.

The most productive way to improve a long quantitative proof is to isolate the places where information is discarded, determine which losses are necessary, and then propagate the improved statements through the actual dependencies. This report carries out that program at four points of Gowers’ argument [1]: finite floor-pattern counting, uniformity-based progression counting, selection of a dense partition cell, and the final density iteration.

The strongest combinatorial conclusion concerns the maps

$$\varepsilon \mapsto \left\lfloor \sum_{i=1}^k \varepsilon_i \alpha_i \right\rfloor, \quad \varepsilon \in \{0, 1\}^k, \quad -r < \alpha_i < r.$$

Their number is exactly polynomial in the integer slope range r : it is $(2r)^k C_k$. The remaining coefficient satisfies $\log_2 C_k \sim k^2$. Thus the range dependence is determined exactly, and the leading quadratic exponent in the dimension is determined as well. The lower bound uses the known threshold-function asymptotic of Zuev [9]; we give the reduction in full. This is a concrete resolution of the counting subproblem used here, rather than a claim to have settled a major open problem about arithmetic progressions.

The analytical refinement is equally explicit. If A has density δ in a suitable finite abelian group

and $\epsilon = \|\mathbf{1}_A - \delta\|_{U^{k-1}}$, then

$$\left| \Lambda_k(\mathbf{1}_A) - \delta^k \right| \leq \sqrt{\delta} \epsilon^2 \sum_{t=0}^{k-3} (t+1) \delta^t \quad (k \geq 3). \quad (\text{F.11.1})$$

The group need not have prime order: invertibility of the relevant slope differences suffices. For three-term progressions the factor $\sqrt{\delta}$ can be replaced by $\sqrt{\delta(1-\delta)}$, and singletons give equality. The Cauchy–Schwarz and telescoping mechanisms are standard; the contribution here is their explicit, stronger formulation and its careful comparison with the displayed source bounds.

F.11.1 Which source is being strengthened?

We inspected both user-specified paths in the ProveIt repository and pinned the discussion to commit `7d99e5deeb209a4671060990cb1ea3d672d57064` [41]. Its `sz-thm-gowers-proof.tex` is an edited transcription with mathematical and rounding corrections. A reference below to a numbered “source” statement means that inspected version unless the journal paper is specified separately. This distinction matters: the edited source explicitly records that a changed estimate in Corollary 16.11 leaves the displayed quantitative proof of Theorem 18.1 unsupported. That is a statement about this derivation and formalization checkpoint, not a refutation of Gowers’ published theorem.

The repository’s checkpoint reports 95 exact companion proofs among 120 numbered statements, with 25 still open. It also reports that 140 reachable modules compiled and that its axiom audit checked 1,365 public theorems. These are source-reported results, not a fresh Lean execution in this report. In particular, a proposition-valued catalogue definition and a proved conditional implication have different mathematical statuses from an unconditional proof of the catalogue entry. Section V.26 records the interfaces relevant to the present work.

[write] **6 October 2026; sources 06–10.** *Stale since the pin.* These figures are those of the pin `7d99e5dee` (its `FORMALIZATION_STATUS.txt` and ledger). At HEAD `764740f07` (6 October 2026) the ledger records 99 exact companions and 21 open statements, and `FORMALIZATION_STATUS.txt` reports 196 reachable modules and 1,805 audited public theorems. The 4 statements closed since the pin are `lemma_16_1`, `lemma_16_5`, `lemma_16_6` and `lemma_16_9`. `corollary_16_11`, `theorem_18_1` and `theorem_18_2` are still open, so the dependency warning above stands. Crosswalk: Section V.1.

F.11.2 Results and their precise status

Source location	Refinement proved here	What it establishes
Corollaries 17.5–17.6	Exact integer/fractional decomposition; sharp leading carry entropy	Polynomial range dependence, a quadratic dimension exponent, and modular saturation
Proposition 17.7	Carry-count bound propagated through localized Cauchy–Schwarz	An explicit improved local phase-removal loss
Theorem 3.2 and Corollaries 3.3, 3.6	Two distinguished norms, double telescoping, exact diagonal subtraction	Stronger modular counting and existence criteria
Lemma 5.15	Weighted asymmetric extraction with extremizers	Optimal mass and fixed-cell-count guarantees from the stated data
Section 18 iteration	Explicit discrete potential	An optimal leading cost for a stated abstract local recurrence

The results are proved below and are quantitative strengthenings relative to those source formulations. We make no historical priority claim for every inequality or decomposition. A bounded literature search cannot establish such a claim. The comparison with modern work is also important: Leng, Sah, and Sawhney proved, for $k \geq 5$, a bound of the shape

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}), \quad c_k > 0,$$

using quasipolynomial inverse-theorem bounds [2, 3]. Our auxiliary refinements do not improve that result. They are useful as standalone estimates, as replacements in a carefully tracked proof, and as tractable formalization targets.

F.11.3 Conventions

All averages on finite sets are normalized unless a sum is written explicitly. Logarithms are natural except for $\log_2$. We write

$$\mathcal{R}_d(m) = \sum_{j=0}^d \binom{m}{j},$$

where $\binom{m}{j} = 0$ for $j > m$. The standard hyperplane-arrangement bound is that m affine hyperplanes in $\mathbb{R}^d$ have at most $\mathcal{R}_d(m)$ complementary regions. Adding one hyperplane and counting the regions it cuts proves the recurrence $R(m, d) \leq R(m-1, d) + R(m-1, d-1)$, and induction gives the binomial sum. When $m \geq d \geq 1$,

$$\mathcal{R}_d(m) \leq (em/d)^d. \quad (\text{F.11.2})$$

For completeness, take $z = d/m \leq 1$. Since $z^j \geq z^d$ for $j \leq d$, the binomial theorem gives $z^d \mathcal{R}_d(m) \leq (1+z)^m \leq e^d$.

Floor patterns on hyperplanes require more care than patterns in open regions. A one-sided perturbation lemma below supplies the needed justification; counting only open regions without that step would leave a genuine gap.

F.12 Source 07. The problem and the concrete improvements

Source 07, delivered Section 1.

Added in the second write, 6 October 2026. Delivered title block. “Additive combinatorics / Quantitative proof refinement / Transport-sensitive rigidity in Gowers’s approximate-homomorphism argument / Higher-order Freiman structure without square-root losses / Research draft / 6 October 2026”; PDF author field “Research draft”. Archive `Gowers_Transport_Rigidity.zip`, manuscript `transport_rigidity.tex` (1,472 lines, 21 pages), pin `ae0e89574`. Printed below: the abstract, the title-page statements and Section 1; Appendix A and Sections 2–8 form a chapter of Part II, Sections 9–11 and Appendix B are in Part V.

Abstract of source 07 (as delivered). We replace a nested almost-everywhere argument by a coupled-path argument that keeps local errors in their averaged form. The resulting theorem applies to finite fibered domains, nonuniform fiber sizes, missing fibers, and arbitrary abelian target groups. Its exact error budget is expressed by a finite transport profile; coarse versions require $2[1 + (k-1)\kappa]\delta < 1$, or $[2k + \tau k(k-1)]\delta < 1$ under additive fiber-size invariance, to force a Freiman homomorphism of order k . For constant fiber sizes the strict threshold $\delta < 1/(2k)$ is optimal, even with identical cyclic source and target groups.

Applied to the hypotheses recorded for Gowers’s Lemma 10.9, the argument upgrades order two to order five without a stronger smallness assumption. Retaining the ambient fiber-invariance assumptions of the original construction upgrades it further to order nine. We also give explicit local relaxations of the error allowance, a collision-to-mode criterion, a nonabelian word-relation extension, and a rational linear program for optimizing the starting law. Every asserted general result is proved in the text. Exact finite tests are included as supplementary checks, not as replacements for the proofs.

Scope. This is a local strengthening of an exactification step, not a new upper bound for progression-free sets. The sharp threshold problem solved here is explicitly formulated in this note. Priority over the full literature has not been established. The new results have not been compiled in Lean or independently refereed.

Repository baseline. The comparison is pinned to `ProveIt` commit `ae0e895748209f0dd079c6b5ca6df6c6fc065a81`; it does not depend on the subsequent state of `main`.

Package. LaTeX source, compiled PDF, exact-arithmetic Python checks, their output, a source manifest, and a formalization plan.

A recurring task in additive combinatorics is to convert a nearly consistent system of local differences into an exactly additive map. Suppose that a candidate value $\psi(d)$ usually equals $\phi(y) - \phi(x)$ when the index of y is the index of x plus d . A relation

$$a + b = c + d$$

can then be tested by comparing two paths with the same initial and terminal indices. If all four edges are correct, their sums agree. The substantive issues are which probability law to put on the paths, how that law changes under translation, and how to treat a path that leaves the usable domain.

Gowers’s Lemma 10.9 in [1] obtains an order-two Freiman homomorphism from the local model of Lemma 10.6 under

$$\theta = 10\eta^{1/5}, \quad 6\sqrt{\theta} < 1. \quad (\text{F.12.1})$$

The repository records the relevant assumptions explicitly and contains a companion proof [42, 43]. We work at that precise interface. The replacement keeps a single edge-defect budget rather than successively balancing outer and inner exceptional sets.

F.12.1 What is proved

Write $\text{Fr}_k(D, H)$ for the maps from D to H preserving every equality between two sums of k elements of D , repetitions allowed. The main results are as follows; all parameters are defined in detail below.

Input or setting	Conclusion	Location
Nonuniform finite fibers	Every violated word relation has transported error budget at least one.	Theorem II.15.1
Fiber ratio at most κ	$2[1 + (k - 1)\kappa]\delta < 1$ forces Fr_k .	Corollary II.15.3
Constant fiber sizes	The threshold $\delta < 1/(2k)$ is optimal for arbitrary direction sets, for every $k \geq 2$.	Theorem II.16.1
Recorded Lemma 10.9 hypotheses	Order five, rather than merely order two.	Corollary II.17.3
Full ambient fiber invariance	Order nine under the same original parameter range; $\eta \leq (40k)^{-5}$ suffices for order k .	Theorem II.17.5, Corollary II.17.6
Local order-two exactification	Explicit allowances 120^{-5} and 75^{-5} , under two different hypothesis packages.	Propositions II.17.4, II.17.7

The numbers five and nine are certified consequences, not claims that these are the largest possible orders under the respective assumptions. The sharp claim concerns the uniform directional defect δ , not the original parameter η , which has already passed through an earlier construction.

F.12.2 Position relative to other results

Homomorphism testing and correction have an established literature. In particular, the majority-difference method and optimal average-test thresholds are treated in [27, 28]. Our assumptions are different: we control each selected direction in a possibly punctured, fibered domain, and keep explicit track of the law of the source point. Thus the threshold proved below should not be read as a replacement for a sharp *average* homomorphism test.

There is also a substantial distinction between refining one lemma and improving a final density bound. For context, the 2024 theorem of Leng, Sah, and Sawhney gives, for each fixed $\ell \geq 5$, an estimate of the form $r_\ell(N) \ll N \exp[-(\log \log N)^{c_\ell}]$, with $c_\ell > 0$ [2]. We do not improve that estimate or assert that the local gains below survive the entire quantitative iteration. No claim about a most recent record bound is needed for the arguments in this note.

F.13 Source 08. The target lemma and the contribution

Source 08, delivered Section 1.

Added in the second write, 6 October 2026. Delivered title block. “Additive combinatorics / Research memorandum / Fourth-order stability of additive cube counts / Torsion-sensitive refinements of Gowers’ Lemmas 3.7 and 3.10 / Prepared for Vladimir Reshetnikov / 6 October 2026”; PDF author field “Research memorandum prepared for Vladimir Reshetnikov”. Archive `gowers_cube_refinements.zip`, manuscript `fourth_order_cube_counts.tex` (1,677 lines, 25 pages). It records no commit: its `source_provenance.json` gives blob identifiers, which are those of `7d99e5dee` (checked at intake), and its links follow `main`. Printed below: the abstract, the title-page status and Section 1; Sections 2–10 and Appendix A are in Part III, Sections 11–14 and Appendix B in Part V.

Abstract of source 08 (as delivered). For a finite abelian group G , write $p = 2^d$, $d \geq 2$, and let $\mathcal{C}_d(F)$ be the normalized count of d -dimensional additive cubes weighted by a real function $F = \delta + f$, where $\mathbb{E}f = 0$. The first three nonconstant terms of this cube-count polynomial vanish identically. Consequently, with $t = \|f\|_{U^d(G)}$, we obtain

$$0 \leq \mathcal{C}_d(F) - \delta^p \leq \sum_{r=4}^p \binom{p}{r} \delta^{p-r} t^r.$$

This removes the linear, quadratic, and cubic losses in the Minkowski bound used in Gowers’ Lemma 3.10. In the paper’s uniformity parameter $\alpha = t^p$, the resulting exponent is $4/p$, and this exponent cannot be increased uniformly, even for genuine subsets of cyclic groups of odd prime order.

We determine the exact fourth-order coefficient over every finite abelian group. It is $R_d \|f\|_{U^2}^4 + T_d E_{\langle 2 \rangle}(f)$, where R_d counts ordinary parallelograms in the Boolean cube, T_d counts additional binary affine planes, and $E_{\langle 2 \rangle}(f)$ is the fourth Fourier moment on nontrivial order-two characters. We also give the complete three-cube expansion over odd-order groups, identify its signed fifth-order term, and construct a counterexample to a natural global lower bound suggested by the fourth-order coefficient. Further results include a Fourier stability inequality, an inhomogeneous version, and a linear-algebraic criterion for which centered terms can survive.

Status. The main statements below have mathematical proofs; finite identities are additionally checked by the accompanying exact-arithmetic program. These are local refinements of cube-count estimates, not a new global bound for Szemerédi’s theorem. This AI-assisted research draft has not undergone independent peer review. Bibliographic priority and Lean verification are not asserted.

F.13.1 What is being strengthened

Gowers’ *A new proof of Szemerédi’s theorem* [1] contains two complementary estimates for additive cubes. Lemma 3.7 gives a lower count $\delta^{2^d}|G|^{d+1}$. In the cyclic setting of the paper, Lemma 3.10 gives the upper count

$$(\delta + \alpha^{1/2^d})^{2^d}|G|^{d+1} \tag{F.13.1}$$

when the set is α -uniform of degree $d - 1$. The degree shift is important: a d -dimensional cube corresponds to the normalized U^d norm, while the paper calls the relevant uniformity *degree* $d - 1$. The displayed bound follows from Minkowski’s inequality.

The supplied Lean file `Proofs03CubeUpper.lean` proves this statement directly with the unnormalized norm, applying `gowersNorm_add_1e` [46]. We keep its conclusion as our baseline rather than replacing it with a looser, already expanded estimate. The improvement is that centering eliminates the first three terms that Minkowski would retain.

Throughout the article, α may be taken to be the actual normalized balanced cube moment $\|f\|_{U^d}^{2^d}$. If the hypothesis supplies only an upper bound on that moment, every upper estimate remains valid after replacing t by $\alpha^{1/2^d}$.

F.13.2 A guide to the results

The central results are organized by the kind of information they use. Theorem III.7.3 uses only centering and Gowers–Cauchy–Schwarz. Theorem III.8.3 additionally uses the arithmetic of four cube vertices, and distinguishes odd groups from groups with order-two characters. Theorem III.9.2 computes every coefficient of the three-cube polynomial in terms of a small number of configuration averages. Theorem III.11.4 proves that the improved exponent is already optimal in the original indicator-function setting.

Two conclusions are worth keeping separate. The exponent $4/2^d$ is optimal for a uniform estimate in the balanced U^d moment. The exact leading coefficient is a Fourier quantity, not merely a universal multiple of $\|f\|_{U^d}^4$. In particular, the coefficient obtained by replacing the Fourier quantity with $\|f\|_{U^d}^4$ need not be sharp on odd-order groups.

F.13.3 Relation to existing work and limits of the claim

The method is an explicit use of familiar tools: multilinear expansion, character orthogonality, and repeated Cauchy–Schwarz. The broader study of linear configurations includes the true-complexity program of Gowers and Wolf [29], Manners’ polynomial bounds obtained through iterated Cauchy–Schwarz [14], and the study of norming linear systems by Cho, Conlon, Lee, Skokan, and Versteegen [30]. We do not claim to introduce any of these general methods.

The specific contributions here are the sharpened statement relative to (F.13.1), its optimal exponent for sets, the all-dimension quartic formula with its order-two correction, and the explicit

signed three-cube analysis. A targeted literature check did not establish priority for this package of formulas. They should therefore be described as *refinements derived in this memorandum*, not as certified first results.

For comparison, Leng, Sah, and Sawhney prove, for $k \geq 5$, a bound of the form $r_k(N) \ll_k N \exp(-(\log \log N)^{c_k})$ with $c_k > 0$ [2]. Nothing below improves that global result. A stronger cube-count estimate does not by itself improve the inverse theorem or the density-increment iteration required to count long arithmetic progressions.

F.14 Source 09. Scope, source theorem, and principal results

Source 09, delivered Section 1.

Added in the second write, 6 October 2026. Delivered title block. “Arithmetic Circuit Expansions for Cubical Counts / A fourth-order strengthening of Gowers’ Lemma 3.10, exact torsion corrections, and a formalization roadmap / Research manuscript prepared with ChatGPT / 6 October 2026”; the PDF author field reads “Research manuscript prepared with ChatGPT”. Archive `gowers_cubical_expansions.zip`, manuscript `article.tex` (1,794 lines, 27 pages), pin `7d99e5dee` (its source audit, shipped as `09-cubical-expansions-SOURCE_AUDIT.md`). Printed below: the abstract and Section 1; Sections 2–10, 13 and Appendices A–B are in Part III, Sections 11, 12, 14, 15 and Appendix C in Part V. Its ‘question’ environment, printed “Research question” there, is set in this report’s ‘Research question’ style.

Abstract of source 09 (as delivered). Let G be a finite abelian group, $d \geq 2$, $m = 2^d$, and $F = \delta + f$ a real function with $\mathbb{E}f = 0$ and $\varepsilon = \|f\|_{U^d}$. The triangle-inequality estimate behind Gowers’ Lemma 3.10 bounds the normalized d -cube count by $(\delta + \varepsilon)^m$. We retain the balancing condition and prove that all terms of degrees one, two, and three vanish. Consequently the error is fourth order, with a sharp exponent, rather than first order. We then classify every four- and five-vertex contribution over arbitrary finite abelian groups. The exact fourth-order term is $\delta^{m-4}(P_d E_4 + T_d T_2)$, and the fifth-order term is $\delta^{m-5}(Q_d M_5 + S_d T_{3,5})$; the four coefficients are explicit Stirling-number expressions. The additional statistics T_2 and $T_{3,5}$ are supported on characters of orders two and three. The remaining error is bounded by an explicit sixth-order binomial tail. We prove sharpness statements, construct a negative fifth-order correction, transfer weighted examples to actual sets, and exhibit a six-vertex determinant-5 configuration showing the first possible order-five torsion contribution. Exact finite computations and a repository-specific formalization plan accompany the proofs.

These are local counting results proved in this manuscript. Historical originality has not been established, and the manuscript has not been refereed or verified by a proof assistant. No improved final bound for Szemerédi’s theorem is claimed.

F.14.1 The precise target

The target is Lemma 3.10 of Gowers’ *A new proof of Szemerédi’s theorem* [1]. In the notation of that paper, a set of density δ which is α -uniform of degree $d - 1$ has at most

$$(\delta + \alpha^{1/2^d})^{2^d} N^{d+1} \tag{F.14.1}$$

parameterized d -dimensional cubes. The proof applies the norm triangle inequality to the indicator written as its mean plus its balanced part. The present paper asks what is lost when the mean-zero property is forgotten at that step.

The supplied repository contains both a transcription of the paper and a Lean development [47]. At the inspected commit, `Proofs03CubeUpper.lean` proves `lemma_3_10_holds` using exactly this norm decomposition. Our comparison is with that theorem, not with an imagined gap in its proof. The repository's definitions use unnormalized Fourier transforms and unnormalized Gowers norms. Section V.38 gives the precise conversion and the inspected source identifiers.

F.14.2 The first strengthening

Write $m = 2^d$, let $F : G \rightarrow [0, 1]$, and put $\delta = \mathbb{E}F$, $f = F - \delta$, and $\varepsilon = \|f\|_{U^d(G)}$. We prove

$$0 \leq \mathcal{C}_d(F) - \delta^m \leq \sum_{s=4}^m \binom{m}{s} \delta^{m-s} \varepsilon^s. \quad (\text{F.14.2})$$

Thus, at fixed positive density and dimension, the error in (F.14.1) improves from $O(\alpha^{1/m})$ to $O(\alpha^{4/m})$. This comparison concerns the excess above δ^m ; it does not mean that all constants are uniform as d increases or δ decreases. The full polynomial in (F.14.2) is the appropriate statement in those regimes.

The exponent four cannot be increased, even for actual sets in growing finite groups. Moreover, the coefficient of the fourth-order term can be substantially smaller than $\binom{m}{4}$. Set

$$A_d = \frac{m(m-1)(m-2)}{24}. \quad (\text{F.14.3})$$

The exact classification implies a universal upper bound with $A_d \delta^{m-4} \varepsilon^4$ as its fourth-order term. This coefficient is sharp when all finite abelian groups are permitted. Compared with $\binom{m}{4}$, it saves the factor $m-3$.

F.14.3 The complete expansion through degree five

For $d \geq 3$ we prove

$$\begin{aligned} \mathcal{C}_d(\delta + f) &= \delta^m + \delta^{m-4}(P_d \mathbb{E}_4(f) + T_d \mathbb{T}_2(f)) \\ &\quad + \delta^{m-5}(Q_d \mathbb{M}_5(f) + S_d \mathbb{T}_{3,5}(f)) + R_6, \end{aligned} \quad (\text{F.14.4})$$

where the Fourier statistics are defined in Section III.21, and

$$\begin{aligned} P_d &= 2^{d-2} \left\{ \begin{matrix} d+1 \\ 3 \end{matrix} \right\}, & T_d &= 2^{d-2} \left\{ \begin{matrix} d+1 \\ 4 \end{matrix} \right\}, \\ Q_d &= 2^d \left\{ \begin{matrix} d+1 \\ 4 \end{matrix} \right\}, & S_d &= 2^d \left\{ \begin{matrix} d+1 \\ 5 \end{matrix} \right\}. \end{aligned} \quad (\text{F.14.5})$$

Here $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ is a Stirling number of the second kind. The remainder satisfies

$$|R_6| \leq \sum_{s=6}^m \binom{m}{s} \delta^{m-s} \varepsilon^s \leq \binom{m}{6} \varepsilon^6 (\delta + \varepsilon)^{m-6}. \quad (\text{F.14.6})$$

All identities are exact finite identities, before any limiting process. The formula does not require a prime modulus, distinct evaluated cube vertices, or a lower bound on $|G|$.

There are two genuinely different sources of terms. Real affine relations among cube vertices give E_4 and M_5 . Integer lattice indices give the torsion terms T_2 and $T_{3,5}$. Ignoring this arithmetic distinction would give false formulas on groups with small prime torsion.

F.14.4 Boundaries of the claims

The proofs below establish the stated identities, estimates, examples, and finite-configuration classifications. We do not assert that these identities are absent from all previous literature. The broad framework of multilinear Gowers inequalities and norm geometry is established work; see, for example, Gowers [1] and Hatami [31]. The contribution pursued here is a specific, fully explicit local refinement, including its arithmetic corrections and sharpness analysis.

An improved auxiliary cube estimate is not automatically an improved arithmetic-progression theorem. A new global bound would require a complete quantitative argument connecting this refinement to an inverse step and a density-increment iteration. No such argument is supplied. Current results and the status of the paper’s conjectural progression statements are discussed separately in Section III.25.

F.15 Source 10. Scope, results, and the baseline

Source 10, delivered Section 1.

Added in the second write, 6 October 2026. Delivered title block. “Research report / Quantitative Refinements Around Gowers’ Proof of Szemerédi’s Theorem / Torsion, density profiles, additive energy, and sharp limits for polynomial colorings / A mathematical research report / 6 October 2026”; PDF author field “Research report”. Archive `gowers_refinements.zip` (independent of source 06’s `Gowers_Refinements(1).zip`; its manuscript is not the file of the same name of sources 01 and 02), manuscript `gowers_refinements.tex` (2,443 lines, 34 pages), pin `7d99e5dee`. Printed below: the abstract, the title-page note and Section 1; Sections 2–4 are in Part I, 7–8 in Part II, 5, 6 and 9 in Part III, 10–12 and Appendix A in Part V.

Abstract of source 10 (as delivered). We develop explicit strengthenings of several finite steps surrounding Gowers’ 2001 proof of Szemerédi’s theorem, using the supplied paper and Lean development as a precisely identified baseline. The results include an optimal conversion from partition discrepancy to a density increment, a density-sensitive Fourier-to-progression tradeoff with all integer rounding explicit, a mixed- L^p generalized von Neumann inequality on finite abelian groups with quantified torsion losses, and a density-weighted counting estimate replacing an exponential expansion loss. A weighted energy argument and a sharper vertical-fibre extraction improve the explicit order-eight Freiman restriction from $2^{-1882}\gamma^{1164}$ to $2^{-209}\gamma^{103}$ of the original domain. We also determine exactly the feasible dimensions for a field-coordinate polynomial coloring that excludes symmetrically colored even k -term progressions, when every coordinate degree is at most k and the prime characteristic exceeds k . Complete proofs, sharp examples, finite diagnostic checks, a formalization map, and a proposed research program are provided. These are local refinements and a restricted extremal theorem; the report does not claim a new best general bound for Szemerédi’s theorem or priority for methods already present in the literature.

Source snapshot: `7d99e5deeb209a4671060990cb1ea3d672d57064`. All newly stated results are proved in this report. The accompanying computations are diagnostic supplements, not Lean kernel certification.

The most productive response to a request to strengthen a long proof is to identify exactly which hypotheses can be weakened, which losses can be reduced, and which parts of the argument obstruct further progress. This report carries out that program for several stages of Gowers’ proof [1]. Its main outputs are explicit finite theorems, with enough detail to support later insertion into the accompanying formal development.

F.15.1 The source version matters

The two supplied repository paths were read at commit

7d99e5deeb209a4671060990cb1ea3d672d57064.

Its recorded commit time is 6 October 2026, 19:47:40 UTC [48]. The repository’s $\text{\LaTeX}$ is an edited transcription: it includes editorial corrections and qualifications and is not identical to the published 2001 article. For example, the proof of Corollary 2.5 already contains a stronger intermediate density and length estimate than its displayed conclusion. Theorem 3.2 has explicit prime-modulus and range assumptions addressing a noninvertible reindexing step. We compare against these existing repairs where relevant.

The supplied status ledger records 120 numbered catalogue statements: 95 have exact companion proofs and 25 remain open. Its axiom-audit source explicitly distinguishes an axiom check from a check that the hypotheses of conditional results have been discharged. We have inspected these source records; we have not rerun the Lean project or certified the new theorems in Lean. Nothing below treats an unproved catalogue declaration as an established mathematical input.

[write] **6 October 2026; sources 06–10.** *Stale since the pin.* The counts 95 and 25 are those of the pin 7d99e5dee. At HEAD 764740f07 (6 October 2026) the ledger records 99 exact companions and 21 open statements; the 4 statements closed since the pin are `lemma_16_1`, `lemma_16_5`, `lemma_16_6` and `lemma_16_9`. Crosswalk: Section V.1.

F.15.2 A comparison of the main finite guarantees

In this table δ is density, η a normalized correlation or Gowers norm as specified in the relevant theorem, c an energy proportion, and γ a graph-energy proportion. The meanings are not interchangeable.

Step	Baseline in the supplied argument	Refinement proved here
Partition selection	Increment at least $S/2$ from total balanced discrepancy S	The optimal increment $\delta S/(2\delta - S)$; an exact tail bound and an exceptional-set version (Section I.23).
Fourier extraction, Corollary 2.5	The proof already gives increment $\eta/4$ and length $\sqrt{\eta N}/(128\pi)$ in its large-scale range	The density-sensitive formula (I.24.1), increment $\delta\eta/(4\delta - \eta)$, and a complete tradeoff with no size side condition (Section I.24).
Generalized von Neumann, Theorem 3.2	Prime modulus, $k \leq N$, and 1-bounded functions	Arbitrary finite abelian groups, explicit kernel factors, general coefficient systems with an invertible target-neighbor difference, and mixed L^p norms (Theorem III.28.2).

Step	Baseline in the supplied argument	Refinement proved here
Progression counting, Corollary 3.3	Error bounded by $2^k \alpha$ after normalization	A weighted telescoping sum, at most $(k-2)\alpha$ for $k \geq 3$ in the same no-torsion-loss setting, with additional density factors (Section III.29).
Energy extraction, Proposition 7.3	Retain $2^{-20} c^{12} m$; difference set at most $2^{38} c^{-24} m$	Retain $cm/4$; difference set at most $2^{10} c^{-5} m$ (Theorem II.20.1).
Freiman extraction, Lemma 7.5	Retain $ B /(8kC^{4k})$	Retain $ B /[k(h_k-1)+1]$, and hence $ B /(kC^{2k+1})$ (Theorem II.21.1).
Order eight, Corollary 7.6	Retain $2^{-1882} \gamma^{1164} B_0 $	Retain $2^{-209} \gamma^{103} B_0 $ (Corollary II.21.2).
Polynomial coloring restriction	A layered-norm construction and a qualitative Chevalley–Warning limitation in recent work	An exact dimension criterion for prescribed coordinate degrees $1 \leq D_\ell \leq k$ over $\mathbb{F}_p$, $p > k$ (Section III.30).

F.15.3 What sort of contribution these results make

The partition cap and tail bounds are sharp finite optimization statements. The torsion factor is exactly sharp for three-term progressions in elementary abelian 2-groups and has the correct group-size order for a family of longer patterns. The polynomial dimension theorem gives both a universal obstruction in a precisely specified class and a construction attaining it. These are explicit mathematical conclusions of the proofs below.

The underlying tools—Cauchy–Schwarz, Fourier orthogonality, convexity, dependent random choice, sumset inequalities, and Chevalley–Warning—are established. The matching norm-layer coloring is credited to the recent literature. The exact formulations assembled here may be useful new entries in this project, but no assertion of publication-level novelty is made without a broader priority investigation. In particular, an improvement over a 2001 constant is not automatically an improvement over the strongest known theorem in 2026.

F.15.4 Relevant contemporary benchmarks

Let $r_k(N)$ be the largest size of a subset of $\{1, \dots, N\}$ without a nonconstant k -term progression. Gowers’ historical general bound has the shape $N(\log \log N)^{-c_k}$ [1]. Relevant later benchmarks, checked against primary sources for this report, include

$$\begin{aligned}
r_3(N) &\leq N \exp\left(-c \frac{(\log N)^{1/6}}{\log \log N}\right) && \text{(Raghavan, 2026 preprint),} \\
r_4(N) &\ll N(\log N)^{-c} && \text{(Green–Tao),} \\
r_k(N) &\ll_k N \exp(-(\log \log N)^{c_k}), \quad k \geq 5 && \text{(Leng–Sah–Sawhney).}
\end{aligned}$$

See [34, 6, 2] for the precise theorems and their hypotheses. This is a set of comparison benchmarks, not a claim that the present report reproduces their proofs.

The status of the old uniform-set conjectures must also be updated. Gowers’ later construction disproves the random four-term lower bound under Fourier uniformity [35]. The 2026 preprint of Shi and Dong reports counterexamples to the corresponding conjectured lower bound for every even length at least four [33]. Those statements should not be proposed again as untouched open conjectures. Their coloring construction and the unresolved questions around it instead motivate the restricted extremal problem in Section III.30.

Part I

Density transfer and phase-flat partitions

Contents. The last step of a density increment: a correlation with a phase that is nearly constant on the cells of a partition into progressions is turned into a dense cell, and such partitions are built. Sections I.1–I.7: source 01, delivered Sections 2–8 (the finite weighted phase problem, the sharp density-tail theorem with occupied-mass exceptions, the optimal forced maximum, finite convex optimization, integer-exact simultaneous partitions, the Fourier increment, the scale barrier). Sections I.8–I.12: source 02, delivered Sections 6–9 and Appendix B (sharp real phase freezing, the sharp density–mass trade-off, the replacement for Lemma 5.15, the refinement theorem under the partition input $\text{PP}(K, R_0)$). Sections I.13–I.15: source 03, delivered Sections 2–4 (its conventions, the exact selection theorem, a density-sensitive Fourier increment). Sections I.16–I.20: source 05, delivered Sections 2–6 (the sharp constant $1/3$ in Lemma 2.3, Corollary 2.4 with 36π and $3\alpha/4$, the density-sensitive Fourier increment, simultaneous phases with the optimal exponent, spectral energy, short-cell discards).

Printed more than once, credited in dated notes. The sharp one-sided mass bound (sources 01, 02, 03, 05; note after Theorem I.4.1); the one-cell phase envelope (01, 02; note after Lemma I.1.3); the Lemma 5.15 replacement (02, 03); the simultaneous phase-flat partition (01, 05); the scale barrier (01 for one phase, 05 for m phases). The three Fourier increments (01, 03, 05) differ; source 01’s increment is at least $3/2$ times the others at the same length scale (note after Table 3).

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
arc / phase width	01: an arc of r turns has half-angle $\theta = \pi r$ and $s = \sin \theta$; 02: chordal diameter 2ρ , so $\rho = s$; 03: chordal error $ e(\theta x) - z \leq \varepsilon$; 05: arc width ε turns, disk radius $\sin(\pi\varepsilon)$. <i>01’s r and 05’s ε are widths in turns, not angles.</i>
η, α, ρ	01: η = normalized correlation with $e(\xi n)$; 05: α = normalized Fourier coefficient; 03: ρ = normalized Fourier coefficient. For one coefficient 01’s η = 05’s α = 03’s ρ . 02: αN bounds a <i>sum over cells</i> of local correlations, and ρ is a phase half-diameter.
D, p, T, S, s, L	01: $D = \sum_j w_j q_j - \delta $ (twice the positive mass); 02: p positive mass, S sum of absolute cell sums, L total absolute mass; 03: $T = \mathbb{E}X_+$; 05: $S = \mathbb{E}g_+$, $s \leq S$, and $s(\varepsilon)$; 01: $s = \sin \theta$. <i>05’s s is a mass, 01’s s a sine.</i>
κ	01: $\kappa = \eta/(2\delta(1 - \delta))$; 05: a Gram-matrix bound (Corollary I.19.4).
Q, M, B, E	01: Q_i box counts, $M_\delta = 2\delta(1 - \delta)$, $B_{\delta,s}(a)$ largest correlation, E an exceptional set; 02: Q , $Q_{a,b}(p, t)$ good-mass certificates, M number of cells, B_0, B_ρ size constants; 05: $Q_i(L)$ box counts, M the modulus, $B = \mathbb{E} f $ and a width budget B , E captured Fourier energy, R an oscillation budget; 03: E_{osc} an oscillation error.

Added in the second write (6 October 2026): sources 06–10. Contents. Section I.21: source 06, delivered Section 5 (the exact trade-off between the mass of the cells above a threshold and the largest such cell for a fixed number M of cells, with exact extremizers; a strengthening of Lemma 5.15 with both branches of that optimum; the simultaneous improvement $\tau(\alpha)$; the indicator range; exceptional sets and signed drift). Sections I.22–I.24: source 10, delivered Sections 2–4 (its conventions; the optimal conversion of partition discrepancy into a density increment, its variance form and the mass above a threshold; correlation-to-density transfer, also with an exceptional set; an explicit phase partition with all rounding exposed and the one-parameter density–length frontier; the variance form of the U^2 inverse step).

Printed more than once (dated notes at both ends). The sharp one-sided mass bound is now proved in six sources: source 06’s Theorem I.21.1 (mass form, any total mass and signed mean) and source 10’s Proposition I.23.2 join sources 01, 02, 03 and 05 (note after Theorem I.4.1); the forced maximum is also source 10’s (I.23.2). The Lemma 5.15 replacement of sources 02 and 03 is sharpened by source 06’s fixed- M optimum (notes after Corollary I.9.3 and Corollary I.21.2). Exceptional mass appears in three different

models: source 01's Theorem I.2.3, source 06's Proposition I.21.4, source 10's Theorem I.23.4. Source 10's Corollary I.24.3 has source 01's length formula and source 05's increment; its frontier, Theorem I.24.2, is compared with source 01's increment in the note after Table 3.

Reading conventions for the sections of sources 06–10 in this Part.

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
$\eta, \tau, \gamma, \lambda$	06: η a density <i>threshold</i> in Theorem I.21.1, $\tau(\alpha)$ the simultaneous threshold of Corollary I.21.3. 10: η a normalized correlation (= 01's η , 05's α), γ the retained fraction of the correlation, τ a density threshold (= 06's η), λ a phase angle. <i>06's η is not a correlation; 10's λ is not 03's loss parameter.</i>
v, S, V, M, W	06: $v = \sum_j w_j x_j $ the cell variation (for $f = 1_A - \delta$ this is 01's D), m the signed mean, $P_{\pm} = (v \pm m)/2$, S the high-mass bound, W the total mass, M the number of cells, $[-a, b]$ the range (as 02, 03). 10: $S_{\mathcal{P}}$ the same first moment as 06's v , $V_{\mathcal{P}}$ the second moment, $v = \delta(1 - \delta)$, M the <i>largest cell density</i> , D_{γ} an increment. <i>10's v is not 06's v, and 10's M is not a number of cells.</i>
ρ, s, ε	06: ρ the discarded (exceptional) mass, ε a bound on the signed mean. 10: ρ the disc-approximation error of the phase, $s = \eta - 2v\rho$ the residual correlation, ε the exceptional mass. <i>10's s is not 01's $\sin \theta$.</i>

I.1 The finite weighted phase problem

Source 01, delivered Section 2.

I.1.1 Conventions

Let (Ω, μ) be a finite probability space with positive atom weights. Let $h : \Omega \rightarrow [0, 1]$, and put

$$\delta = \mathbb{E}h \in (0, 1), \quad b = 1 - \delta, \quad f = h - \delta, \quad M_{\delta} = 2\delta b. \quad (\text{I.1.1})$$

A partition $\mathcal{P} = (P_j)$ consists of nonempty cells. Set

$$w_j = \mu(P_j), \quad q_j = \mathbb{E}(h \mid P_j).$$

Then $\sum_j w_j = 1$ and $\sum_j w_j q_j = \delta$. Every assertion for this model includes ordinary subsets of a finite set, by taking uniform weights and $h = \mathbf{1}_A$.

A unit-modulus function ψ is *short-arc on a cell* if there is a real γ_j such that

$$\psi(x) \in \{\exp(i(\gamma_j + t)) : -\theta \leq t \leq \theta\} \quad (x \in P_j), \quad 0 \leq \theta < \pi/2. \quad (\text{I.1.2})$$

Write $s = \sin \theta \in [0, 1)$. Arc centers may differ between cells. We assume

$$|\mathbb{E}f\psi| \geq \eta \geq 0. \quad (\text{I.1.3})$$

The elementary chord inequality

$$|u - \delta| \leq \delta + (1 - 2\delta)u \quad (0 \leq u \leq 1) \quad (\text{I.1.4})$$

gives $\mathbb{E}|f| \leq M_{\delta}$, and hence $\eta \leq M_{\delta}$. Equality holds in the first inequality for indicator functions.

I.1.2 A cancellation-sensitive preliminary bound

Lemma I.1.1 (Cancellation and phase diameter). *Let f be any real function on a finite probability space, and suppose $|\psi| \leq 1$ and the diameter of $\psi(P_j)$ is at most $2s$, where $0 \leq s \leq 1$. If*

$$M = \mathbb{E}|f|, \quad D = \sum_j \left| \mathbb{E}(f \mathbf{1}_{P_j}) \right|,$$

then

$$|\mathbb{E}f\psi| \leq sM + (1-s)D. \quad (\text{I.1.5})$$

Proof. In one cell let A and B be the positive and negative masses of f , measured with the original probability weights. Choose the corresponding phase averages u, v (arbitrarily within the phase hull if a mass is zero). They belong to the convex hull of the cell's phase values, so $|u|, |v| \leq 1$ and $|u - v| \leq 2s$. If $A \geq B$, then

$$|Au - Bv| = |(A - B)u + B(u - v)| \leq A - B + 2sB.$$

This equals $s(A + B) + (1-s)|A - B|$. Interchanging the two signs proves the same estimate when $B > A$. Sum over cells and apply the triangle inequality. $\square$

For indicators, the coefficient M in (I.1.5) is $2\delta(1 - \delta)$, not the cruder bound 1. Consequently,

$$D \geq \frac{\eta - 2\delta(1 - \delta)s}{1 - s} \quad (\text{I.1.6})$$

whenever the right-hand side is positive. The next calculation retains more information than D alone.

I.1.3 The exact one-cell envelope

Define, for $0 \leq q \leq 1$,

$$\Phi_{\delta,s}(q) = \sqrt{(q - \delta)^2 + 4\delta bq(1 - q)s^2}. \quad (\text{I.1.7})$$

Lemma I.1.2 (Sharp cell bound). *Under (I.1.2),*

$$|\mathbb{E}((h - \delta)\psi \mid P_j)| \leq \Phi_{\delta,s}(q_j). \quad (\text{I.1.8})$$

For each $q \in [0, 1]$, equality can be realized on a finite weighted cell with mean q and an indicator h .

Proof. First suppose h is an indicator. Relative to the cell probability measure, the positive mass is $A = bq$ and the negative mass is $B = \delta(1 - q)$. Let U, V be the phase averages on the occupied and unoccupied portions. Taking independent samples of their phase values,

$$|AU - BV| \leq \max_{|u|, |v| \leq \theta} |Ae^{iu} - Be^{iv}|.$$

The omitted common rotation has modulus one. Since $|u - v| \leq 2\theta < \pi$,

$$|Ae^{iu} - Be^{iv}|^2 = (A - B)^2 + 4AB \sin^2 \frac{u - v}{2} \leq (A - B)^2 + 4ABs^2.$$

Here $A - B = q - \delta$, giving (I.1.8). Place all occupied mass at one endpoint of the arc and all unoccupied mass at the other to attain it. Cases $q = 0, 1$ use only one phase value.

For general h , replace an atom of weight w and value $h(x)$ by an occupied atom of weight $wh(x)$ and an unoccupied atom of weight $w(1 - h(x))$, with the same phase. Delete zero-weight atoms. This preserves both the cell mean and the correlation, so the indicator argument applies. $\square$

Lemma I.1.3 (Convexity and the endpoint chord). *The function $\Phi_{\delta,s}$ is convex on $[0, 1]$, with $\Phi_{\delta,s}(0) = \delta$ and $\Phi_{\delta,s}(1) = b$. Moreover,*

$$\Phi_{\delta,s}(q) < \delta + (1 - 2\delta)q \quad (0 < q < 1, 0 \leq s < 1). \quad (\text{I.1.9})$$

Proof. For $s = 0$, this is the convex function $|q - \delta|$. For $0 < s < 1$, differentiation gives

$$\Phi''_{\delta,s}(q) = \frac{4\delta^2 b^2 s^2 (1 - s^2)}{\Phi_{\delta,s}(q)^3} > 0. \quad (\text{I.1.10})$$

For completeness, if $G = \Phi_{\delta,s}^2$ then $(\sqrt{G})'' = (2GG'' - (G')^2)/(4G^{3/2})$, and the numerator divided by four is $4\delta^2 b^2 s^2 (1 - s^2)$. The endpoint values are immediate. Finally, the square of the right-hand side of (I.1.9), minus $\Phi_{\delta,s}(q)^2$, is $4\delta b q(1 - q)(1 - s^2) > 0$. The chord is positive, so taking square roots proves the strict inequality. $\square$

[write] **6 October 2026.** *One bound, two parametrizations.* Lemma I.1.2 and source 02's Theorem I.8.1 are the same bound. In a cell with positive mass $A = bq$ and negative mass $B = \delta(1 - q)$ (cell-normalized), source 02's signed and absolute masses are $F = A - B = q - \delta$ and $L = A + B$, and its bound $C^2 \leq (1 - \rho^2)F^2 + \rho^2 L^2 = F^2 + 4\rho^2 AB$ is $\Phi_{\delta,s}(q)^2$ with $\rho = s$. The arc conventions agree: an arc of half-angle θ has chordal diameter $2 \sin \theta$, so source 02's ρ (half the chordal diameter) is this $s = \sin \theta$. Both proofs sample the phases of the positive and the negative mass independently; source 02's statement is for arbitrary real f , and this one reduces bounded weights to indicators by splitting atoms. The linear consequence, Lemma I.1.1, is source 02's (I.8.2) and Corollary I.8.2, proved here directly and there from the quadratic bound. Credited to both sources. Source 05's Proposition I.17.2 (a disk centre inside the arc) and source 03's Lemma I.14.3 (arbitrary unit constants) are different, complex-valued transfer estimates.

I.2 A sharp density-tail theorem

Source 01, delivered Section 3.

Fix a target density $a \in [\delta, 1)$. Define

$$u_a = \frac{\Phi_{\delta,s}(a) - \delta}{a}, \quad (\text{I.2.1})$$

$$B_{\delta,s}(a) = \delta + \delta u_a, \quad (\text{I.2.2})$$

$$\Gamma_{\delta,s}(a) = 1 - 2\delta - u_a = \frac{M_\delta - B_{\delta,s}(a)}{\delta} > 0. \quad (\text{I.2.3})$$

Positivity in (I.2.3) follows from (I.1.9). The quantity $B_{\delta,s}(a)$ is the largest correlation permitted by these constraints if every cell density is at most a .

Theorem I.2.1 (Sharp excess and density-tail transfer). *Under the hypotheses of Section I.1, put*

$$\lambda_*(\delta, s, a, \eta) = \frac{(\eta - B_{\delta,s}(a))_+}{\Gamma_{\delta,s}(a)} = \frac{\delta(\eta - B_{\delta,s}(a))_+}{M_\delta - B_{\delta,s}(a)}. \quad (\text{I.2.4})$$

Then

$$\sum_{q_j > a} w_j \geq \sum_j w_j \frac{(q_j - a)_+}{1 - a} \geq \lambda_*(\delta, s, a, \eta). \quad (\text{I.2.5})$$

For every $\eta \in [0, M_\delta]$, the last bound and the corresponding lower bound on the mass of cells with $q_j > a$ are optimal among all finite weighted spaces and phases satisfying the stated constraints.

Proof. Convexity bounds $\Phi_{\delta,s}$ by the chord joining 0 to a on the first interval, and by the chord joining a to 1 on the second. Written as one formula, this is

$$\Phi_{\delta,s}(q) \leq \delta + u_a q + \Gamma_{\delta,s}(a) \frac{(q - a)_+}{1 - a}. \quad (\text{I.2.6})$$

To check the second chord, its slope minus u_a is

$$\frac{b - \Phi_{\delta,s}(a)}{1 - a} - u_a = \frac{b - \delta - u_a}{1 - a} = \frac{\Gamma_{\delta,s}(a)}{1 - a}.$$

Lemmas I.1.2 and I.1.3, the triangle inequality, and $\sum_j w_j q_j = \delta$ now give

$$\eta \leq \sum_j w_j \Phi_{\delta,s}(q_j) \leq B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \sum_j w_j \frac{(q_j - a)_+}{1 - a}. \quad (\text{I.2.7})$$

Rearrange and use $0 \leq (q - a)_+/(1 - a) \leq \mathbf{1}_{q > a}$.

For sharpness, first take $\eta \geq B_{\delta,s}(a)$ and write $\lambda = (\eta - B_{\delta,s}(a))/\Gamma_{\delta,s}(a)$. Since $\eta \leq M_\delta$, we have $0 \leq \lambda \leq \delta$. Give cells of densities 1, a , 0 respective weights

$$w_1 = \lambda, \quad w_a = \frac{\delta - \lambda}{a}, \quad w_0 = 1 - \lambda - \frac{\delta - \lambda}{a}. \quad (\text{I.2.8})$$

They are nonnegative, sum to one, and have weighted mean δ . Use the equality construction of Lemma I.1.2 in each cell, and rotate the arcs independently so that all cell correlations point along the positive real axis. The resulting correlation is

$$w_0 \delta + w_a \Phi_{\delta,s}(a) + w_1 b = B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \lambda = \eta.$$

Only the density-1 cells exceed a , and their normalized excess is also λ . Thus both bounds in (I.2.5) are equalities.

For $0 \leq \eta \leq B_{\delta,s}(a)$, use the $\lambda = 0$ configuration. If $B_{\delta,s}(a) > 0$, take two disjoint copies with opposite global phase rotations and relative weights $(1 + \eta/B_{\delta,s}(a))/2$ and $(1 - \eta/B_{\delta,s}(a))/2$. They still have mean δ , no cell exceeds a , and their total correlation is η . If $B_{\delta,s}(a) = 0$, only $\eta = 0$ arises in this case. Zero-weight cells are omitted throughout. $\square$

Remark I.2.2 (The scope of sharpness). The extremizers allow arbitrary real atom weights and independent choices of arc centers. With rational weights they can be converted to ordinary finite sets by replacing an atom with equally weighted copies. General weights can be approximated by rational ones. This is not a proof that the same extremizers occur for a fixed character on $\mathbb{Z}/N\mathbb{Z}$, a fixed progression partition, or a prescribed list of cell sizes. Such arithmetic constraints may permit further improvements.

I.2.1 Exceptional cells cost occupied mass

A usual exceptional-set estimate pays for the whole exceptional set. Here its occupied mass is the relevant quantity.

Theorem I.2.3 (Occupied-mass robust transfer). *Let $E \subseteq \Omega$, and partition $\Omega \setminus E$ into nonempty regular cells P_j . Impose the short-arc condition only on regular cells, and set*

$$\beta_E = \mathbb{E}(h\mathbf{1}_E).$$

Retain the global mean δ and correlation lower bound η . For $\delta \leq a < 1$,

$$\sum_{\substack{j \text{ regular} \\ q_j > a}} w_j \geq \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} \geq \max\{0, \lambda_*(\delta, s, a, \eta) - \beta_E\}. \quad (\text{I.2.9})$$

Here $w_j = \mu(P_j)$ uses the original measure, not the measure conditioned on the regular set.

Proof. Write $\varepsilon = \mu(E)$. On regular cells use (I.2.6); their total weight and occupied mass are $1 - \varepsilon$ and $\delta - \beta_E$. On E , use (I.1.4). Thus

$$\begin{aligned} \eta &\leq \delta(1 - \varepsilon) + u_a(\delta - \beta_E) + \Gamma_{\delta,s}(a) \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} + \delta\varepsilon + (1 - 2\delta)\beta_E \\ &= B_{\delta,s}(a) + \Gamma_{\delta,s}(a) \left(\beta_E + \sum_{j \text{ regular}} w_j \frac{(q_j - a)_+}{1 - a} \right). \end{aligned}$$

Rearranging proves the result, including the case $\eta \leq B_{\delta,s}(a)$. $\square$

The estimate is especially useful when the excluded cells have little of A , even if their total length is not particularly small. An exceptional set consisting of unoccupied points has $\beta_E = 0$: it incurs no loss in this formula. There is no contradiction with global cancellation, since the density and constant term were never renormalized after deletion.

[write] **6 October 2026; sources 06–10.** *Exceptional mass in three models.* Source 06's Proposition I.21.4 (extraction from a cell variation after a set of mass ρ is discarded, or with a signed mean of size at most ε) and source 10's Theorem I.23.4 (correlation transfer with an exceptional set of mass $\varepsilon < \delta$, under $s > 2(1 - \delta)\varepsilon$) also allow an exceptional set. The theorem above charges only the occupied mass $\beta_E \leq \mu(E)$; the other two charge the total exceptional mass. The three statements have different hypotheses and conclusions, and none is claimed to imply another.

Corollary I.2.4 (A large good cell). *For a uniform space of N points with m regular cells, let $\rho = \max\{0, \lambda_* - \beta_E\}$. If $\rho > 0$, some regular cell has*

$$q_j > a, \quad |P_j| \geq \frac{\rho N}{m}.$$

If all regular cells have size at least L , that cell also has size at least L . If instead all have size at most U , there are at least $\rho N/U$ good cells.

Proof. Their union has at least ρN points by Theorem I.2.3. Pigeonhole over at most m cells for the first assertion; divide their total size by the upper size bound for the last. $\square$

I.3 The optimal forced maximum and its tradeoff

Source 01, delivered Section 4.

The density-tail theorem contains a closed-form answer to a simpler question: how large must at least one cell density be?

Theorem I.3.1 (Optimal maximum density from a short-arc correlation). *Assume there are no exceptional points, and put $\kappa = \eta/M_\delta$. If $0 \leq s < \kappa \leq 1$, then*

$$\max_j q_j \geq a_* = \frac{\delta(\kappa - s^2)}{\kappa(1 - b\kappa) - \delta s^2}. \quad (\text{I.3.1})$$

Equivalently, the forced increment is

$$t_* = a_* - \delta = \frac{\delta b(\kappa^2 - s^2)}{\kappa(1 - b\kappa) - \delta s^2}. \quad (\text{I.3.2})$$

These bounds are sharp in the finite weighted model. If $\kappa \leq s$, no positive density increment is forced by this information alone.

Proof. The function $B_{\delta,s}(a)$ is continuous and strictly increasing for $\delta \leq a \leq 1$, after defining its endpoint value at 1 by continuity. Indeed, its nonconstant part is the secant slope $(\Phi_{\delta,s}(a) - \Phi_{\delta,s}(0))/a$. Strict convexity proves the assertion for $s > 0$; for $s = 0$ and $a \geq \delta$ it follows directly from $B_{\delta,s}(a) = 2\delta(a - \delta)/a$. Its endpoint values are

$$B_{\delta,s}(\delta) = M_\delta s, \quad B_{\delta,s}(1) = M_\delta.$$

If all $q_j \leq a$, the first chord gives $\eta \leq B_{\delta,s}(a)$. Thus the unique solution of $B_{\delta,s}(a_*) = M_\delta \kappa$ is a lower bound on $\max_j q_j$, including the endpoint $a_* = 1$ when $\kappa = 1$.

The equation for a_* is

$$\Phi_{\delta,s}(a_*) = \delta + a_*(2b\kappa - 1). \quad (\text{I.3.3})$$

Squaring, cancelling the common constant term, and removing the positive factor a_* gives a linear equation in a_* . It yields (I.3.1). There is no extraneous square-root branch. To see this explicitly, let

$$D_0 = \kappa(1 - b\kappa) - \delta s^2.$$

Since $s < \kappa$, $D_0 > \kappa(1 - b\kappa) - \delta \kappa^2 = \kappa(1 - \kappa) \geq 0$. Also

$$D_0 - \delta(\kappa - s^2) = b\kappa(1 - \kappa) \geq 0,$$

so $a_* \leq 1$, and (I.3.2) shows $a_* > \delta$. The right side of (I.3.3), after substitution, equals

$$\frac{\delta b(\kappa^2 + (1 - 2\kappa)s^2)}{D_0} > 0.$$

For $\kappa \leq 1/2$ positivity is immediate; for $\kappa > 1/2$, use $s^2 < \kappa^2$ and $1 - 2\kappa < 0$, with the case $\kappa = 1$ giving $1 - s^2 > 0$. Hence the squared solution satisfies the original root equation.

Sharpness uses densities 0 and a_* with weights $1 - \delta/a_*$ and δ/a_* and the aligned equality phases from Lemma I.1.2. Their correlation is $B_{\delta,s}(a_*) = \eta$. Finally, if $\kappa \leq s$, cells all of density δ can have correlation $M_\delta s$; opposite rotated copies reduce it to any $\eta \leq M_\delta s$ without changing any cell density. $\square$

I.3.1 Solving for the allowed arc width

At $s = 0$, the optimal increment is

$$t_0 = \frac{\delta b \kappa}{1 - b \kappa}.$$

For $0 < \kappa < 1$ and a requested increment $0 < t \leq t_0$, solving $t_*(s) = t$ gives

$$s_{\max}^2(t) = \frac{\kappa[\delta b \kappa - t(1 - b \kappa)]}{\delta(b - t)}. \quad (\text{I.3.4})$$

Here $t_0 < b$, so the denominator is positive. The right side decreases from κ^2 to 0 as t increases from 0 to t_0 . For $0 \leq s \leq s_{\max}(t)$ the forced increment is at least t . This follows either from (I.3.2), or from monotonicity of $\Phi_{\delta,s}$ in s , which makes a fixed correlation easier to obtain when the allowed arc is wider. At $\kappa = 1$, maximal correlation instead forces a density-1 cell for every $s < 1$; formula (I.3.4) need not be used at its removable endpoint.

The tradeoff has a practical interpretation. A wider allowed arc permits longer phase-flat progressions, but weakens the density increase. Formula (I.3.4) separates this exact probabilistic tradeoff from the Diophantine approximation that constructs the progressions.

I.4 Partition discrepancy and finite convex optimization

Source 01, delivered Section 5.

I.4.1 A sharp transfer using only cell discrepancy

For a partition define its normalized balanced discrepancy by

$$D = \sum_j w_j |q_j - \delta|. \quad (\text{I.4.1})$$

The equality $\sum_j w_j (q_j - \delta) = 0$ implies $D = 2 \sum_j w_j (q_j - \delta)_+$, and convexity gives $D \leq M_\delta$.

Theorem I.4.1 (Discrepancy-to-density tail). *Let $0 \leq t < 1 - \delta$. Then*

$$\sum_{q_j > \delta + t} w_j \geq \max \left\{ 0, \frac{D(\delta + t) - 2\delta t}{2\delta(1 - \delta - t)} \right\}. \quad (\text{I.4.2})$$

The lower bound is sharp among all finite weighted partitions of mean δ and discrepancy D . In addition,

$$\max_j q_j \geq \delta + \frac{\delta D}{2\delta - D}. \quad (\text{I.4.3})$$

Proof. Choose a phase constant on each cell, equal to 1 or -1 according to the sign of $q_j - \delta$. Then its global correlation is exactly D . Apply Theorem I.2.1 with $s = 0$, $\eta = D$, and $a = \delta + t$. In this case

$$B_{\delta,s}(a) = \frac{2\delta t}{\delta + t}, \quad \Gamma_{\delta,s}(a) = \frac{2\delta(1 - \delta - t)}{\delta + t},$$

which gives (I.4.2). The three-density equality construction in (I.2.8) has exactly this discrepancy, proving sharpness. For (I.4.3), use Theorem I.3.1 with $s = 0$ and $\kappa = D/M_\delta$; the case $D = 0$ is just the mean bound. The denominator is positive because $D \leq 2\delta(1 - \delta) < 2\delta$. $\square$

Suppose a uniform N -point space has a partition into m cells and a lower bound on its *average* cell size, $\bar{L} = N/m$. Let the positive right-hand side of (I.4.2) be Λ . Then some cell simultaneously has density greater than $\delta + t$ and size at least $\Lambda\bar{L}$. This is a direct local output for any supplied partition; it neither replaces an average by a minimum nor assumes that the good cell is among the largest cells in advance.

[write] 6 October 2026. *The sharp one-sided bound, in four sources.* The bound (I.4.2) is the case $a = \delta$, $b = 1 - \delta$ of a sharp one-sided mass bound that four of the five sources prove independently, each in its own form: source 02's Theorem I.9.1 (cell means in $[-a, b]$, with the nonzero-mean extension (I.9.5)), source 03's Theorem I.14.1 (a random variable in $[-a, b]$, with the bound (I.14.1) on its essential supremum), and source 05's Lemma I.17.3 ($a = \delta$, $b = 1 - \delta$, with the non-strict event $g \geq t$, which makes its bound weaker, and a popularity corollary). The identification (checked at the write): the discrepancy is twice the positive mass, so with $T = D/2$, $a = \delta$ and $b = 1 - \delta$,

$$\frac{(a+t)T - at}{a(b-t)} = \frac{D(\delta+t) - 2\delta t}{2\delta(1-\delta-t)}, \quad \frac{aT}{a-T} = \frac{\delta D}{2\delta - D},$$

which are (I.4.2) and (I.4.3). The proof above is a genuinely different route, the case $s = 0$ of the broken-chord envelope of Theorem I.2.1; sources 02, 03 and 05 count the negative support directly. All four statements and proofs are kept. Only source 01 proves a tail bound that is sharp for a positive arc parameter $s > 0$ (Theorems I.2.1, I.2.3 and I.3.1); source 02's Theorem I.10.1, source 03's Theorem I.15.2 and source 05's Theorem I.17.4 first remove the phase and then apply the $s = 0$ bound.

[write] 6 October 2026; sources 06–10. *Two more sources prove the same bound.* Source 06's Theorem I.21.1 states it in mass form: for cell averages in $[-a, b]$ with total mass W , positive and negative integrals $P_{\pm}$ and threshold η , the cells above η have mass at least $S = (P_+ - \eta(W - P_-/a))/(b - \eta)$. With $W = 1$ and $P_+ = P_- = T$ (mean zero) this is $((a + \eta)T - a\eta)/(a(b - \eta))$, source 02's bound at $t = \eta$; source 06 keeps a nonzero signed mean (its (I.21.11)) and adds the exact optimum for a fixed number of cells (note after Corollary I.9.3). Source 10's Proposition I.23.2 is the case $a = \delta$, $b = 1 - \delta$: with $T = S_{\mathcal{P}}/2$ and $t = \tau - \delta$, the bound $((\delta + t)T - \delta t)/(\delta(1 - \delta - t))$ equals its $(\tau S_{\mathcal{P}} - 2\delta(\tau - \delta))/(2\delta(1 - \tau))$; and its (I.23.2) is the forced maximum $aT/(a - T) = \delta S_{\mathcal{P}}/(2\delta - S_{\mathcal{P}})$, i.e. (I.4.3) with $D = S_{\mathcal{P}}$ (identifications checked at the second write). Source 10's variance form (I.23.3) is not in the other sources. The bound is therefore proved independently in six sources (01, 02, 03, 05, 06, 10); all six statements and proofs are kept.

I.4.2 A master reduction for additional moment information

The preceding broken-chord argument is a small exact linear program. The same mechanism handles more density statistics without repeated pigeonholing.

Theorem I.4.2 (Finite-knot extremal reduction). *Let $0 = t_0 < t_1 < \dots < t_m = 1$, let $F : [0, 1] \rightarrow \mathbb{R}$ be convex, and let $g_1, \dots, g_r$ be continuous functions affine on each interval $[t_i, t_{i+1}]$. Fix feasible values of $\mathbb{E}g_{\ell}(Q)$ for a finitely supported random variable $Q \in [0, 1]$. The maximum possible $\mathbb{E}F(Q)$ is attained by a distribution supported on $\{t_0, \dots, t_m\}$. It is therefore the value of a finite linear program. An optimizer may be chosen with at most the rank of the equality-constraint matrix (including total mass) many positive weights.*

Proof. If $q \in [t_i, t_{i+1}]$, write $q = (1 - \lambda)t_i + \lambda t_{i+1}$. Replace its probability mass by masses in the proportions $1 - \lambda$ and λ at the two endpoints. Every prescribed g_{ℓ} -moment is unchanged, since g_{ℓ} is affine on this interval. The objective does not decrease, by convexity of F . Do this for every atom. The resulting optimization has finitely many nonnegative weights, one mass equation, and the given linear moment equations. Its feasible set is a closed subset of the probability simplex, so an optimum exists.

If an optimum has more positive weights than the equality matrix rank, there is a nonzero vector supported on those weights in the kernel of that matrix. Move in one of its two directions without decreasing the objective, until a weight first becomes zero. The mass equation ensures that the kernel vector has both signs, so such a move is possible. Iterate to reach the claimed support size. $\square$

Suitable moments include the mean, $|Q - \delta|$, and the stop-loss quantities $(Q - a_i)_+$, provided their breakpoints are included among the knots. The discontinuous function $\mathbf{1}_{Q > a_i}$ is not among the permitted continuous moments; its lower bound is obtained afterward from $(Q - a_i)_+ / (1 - a_i) \leq \mathbf{1}_{Q > a_i}$. This distinction matters at equality.

I.4.3 Several phases on one common partition

Corollary I.4.3 (A simultaneous-phase score). *Suppose phases $\psi_1, \dots, \psi_d$ are short-arc on the same partition, with half-angles θ_i and $s_i = \sin \theta_i < 1$. For weights $\omega_i \geq 0$ with $\sum_i \omega_i = 1$, put*

$$F(q) = \sum_i \omega_i \Phi_{\delta, s_i}(q), \quad \eta = \sum_i \omega_i |\mathbb{E}(h - \delta)\psi_i|.$$

Theorem I.2.1 and its occupied-mass version hold with $\Phi_{\delta, s}$ replaced by F in the definitions of u_a , B , and Γ .

Proof. The cell bound and the triangle inequality give $\eta \leq \sum_j w_j F(q_j)$. The function F is convex, has endpoint values δ, b , and lies strictly below their chord in the interior. The proofs of Theorems I.2.1 and I.2.3 use exactly these facts. $\square$

This is an interface, not an unconditional gain over choosing the largest Fourier coefficient. Requiring one partition to control many phases has its own dimension cost. Additional relations between phases could also rule out the independent equality constructions used for the scalar weighted model.

I.5 Integer-exact simultaneous progression partitions

Source 01, delivered Section 6.

We now construct short-arc cells without asymptotic rounding conventions. An arithmetic progression here has the form $\{a, a + q, \dots, a + (\ell - 1)q\}$ with positive integer q and $\ell \geq 1$. All its terms are required to lie in $[N]_0$; no modular wrapping is used.

Theorem I.5.1 (Anisotropic phase-flat partition). *Let $\xi_1, \dots, \xi_d \in \mathbb{R}$, $d \geq 1$, and let $0 < r_i \leq 1/2$. Choose an integer $L \geq 2$ and set*

$$Q_i = \left\lceil \frac{2(L-1)}{r_i} \right\rceil, \quad T = \prod_{i=1}^d Q_i. \quad (\text{I.5.1})$$

If $N \geq LT$, there is a partition of $[N]_0$ into arithmetic progressions, all with the same positive step $q \leq T$, such that every cell has length between L and $2L - 1$ and, on every cell, each $e(\xi_i n)$ lies in an arc of length at most r_i turns.

Proof. Partition $[0, 1)^d$ into T half-open boxes with side lengths $1/Q_i$. Among the $T + 1$ points

$$(\{j\xi_1\}, \dots, \{j\xi_d\}), \quad 0 \leq j \leq T,$$

two belong to the same box. Subtracting their indices gives an integer $1 \leq q \leq T$ and integers p_i with

$$q\xi_i = p_i + \varepsilon_i, \quad |\varepsilon_i| < 1/Q_i. \quad (\text{I.5.2})$$

Partition $[N]_0$ first into residue classes modulo q . Each class has at least $\lfloor N/q \rfloor \geq L$ elements. If a class has $n = vL + w$ elements, where $v \geq 1$ and $0 \leq w < L$, divide its ordered list into $v - 1$ consecutive blocks of length L and one final block of length $L + w$. Thus every block has length in $[L, 2L - 1]$ and retains step q .

On a block of length ℓ , a real lift of the i th phase has successive differences ε_i . Its total span in turns is at most

$$(\ell - 1)|\varepsilon_i| \leq \frac{2L - 2}{Q_i} \leq r_i.$$

Projection of this real interval to the circle gives the required arc. Every point belongs to exactly one block, proving the partition assertion. $\square$

Corollary I.5.2 (A convenient explicit length). *For every positive integer N and the same frequencies and widths, one may take*

$$L = \max \left\{ 1, \left\lfloor \left(\frac{N \prod_i r_i}{2^d} \right)^{1/(d+1)} \right\rfloor \right\} \quad (\text{I.5.3})$$

and obtain cell lengths between L and $2L - 1$ with the stated arc widths. When $L = 1$, use the singleton partition with step 1.

Proof. For $L \geq 2$, the definition of Q_i gives

$$Q_i \leq \frac{2(L - 1)}{r_i} + 1 \leq \frac{2L}{r_i}.$$

Consequently $L \prod_i Q_i \leq 2^d L^{d+1} / \prod_i r_i \leq N$ by (I.5.3). Apply Theorem I.5.1. Singletons satisfy the remaining case directly. $\square$

The exact condition $N \geq L \prod_i \lceil 2(L - 1)/r_i \rceil$ can sometimes permit a larger L than (I.5.3). It is monotone in L and can be checked by integer arithmetic once the widths have certified rational lower bounds. The proof requires no optimal Diophantine approximation algorithm: the pigeonhole construction suffices.

[write] 6 October 2026. Theorem I.5.1 with Corollary I.5.2 and source 05's Theorem I.18.2 are the same construction, a rectangular pigeonhole followed by residue-class blocking into lengths $[L, 2L - 1]$, with the same box counts $\lceil 2(L - 1)/r_i \rceil$ and the same explicit length; credited to both. The hypotheses differ slightly: here the widths satisfy $r_i \leq 1/2$ and the theorem takes $L \geq 2$ (the corollary adds singletons), while source 05 allows widths $\varepsilon_i \leq 1$ and treats $L = 1$ inside the theorem. The one-phase case is also source 05's Proposition I.16.3 and, with a chordal error in place of an arc width, source 03's Lemma I.15.1.

I.6 A strengthened Fourier density-increment lemma

Source 01, delivered Section 7.

Theorem I.6.1 (An explicit replacement for Corollary 2.5). *Let $N \geq 1$, $h : [N]_0 \rightarrow [0, 1]$, and $\delta = N^{-1} \sum_n h(n) \in (0, 1)$. Suppose*

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (h(n) - \delta) e(\xi n) \right| \geq \eta > 0$$

for some real ξ . Put $b = 1 - \delta$ and $\kappa = \eta/(2\delta b)$. There exists a proper arithmetic progression $P \subseteq [N]_0$ such that

$$|P| \geq L := \max \left\{ 1, \left\lfloor \sqrt{\frac{N}{2\pi}} \arcsin \frac{\eta}{4\delta b} \right\rfloor \right\}, \quad (\text{I.6.1})$$

$$\frac{1}{|P|} \sum_{n \in P} h(n) \geq \delta + \frac{3\delta b \kappa}{4 - (4 - 3\delta)\kappa} \geq \delta + \frac{3\eta}{8}. \quad (\text{I.6.2})$$

The progression can be chosen from a partition whose cell lengths all lie between L and $2L - 1$.

Proof. As in (I.1.4), $0 < \kappa \leq 1$. Set

$$\theta = \arcsin(\kappa/2), \quad s = \sin \theta = \kappa/2, \quad r = \theta/\pi.$$

Then $0 < r \leq 1/6$. Corollary I.5.2, with one frequency, supplies the claimed partition and length. Each arc of length r turns has half-angle $\pi r = \theta$, so Theorem I.3.1 applies, also for the singleton partition. Substituting $s = \kappa/2$ into (I.3.2) gives

$$t_* = \frac{\delta b(3\kappa^2/4)}{\kappa(1 - b\kappa) - \delta\kappa^2/4} = \frac{3\delta b \kappa}{4 - (4 - 3\delta)\kappa}.$$

The denominator is positive and at most 4. Since $\eta = 2\delta b \kappa$, this is at least $3\eta/8$. $\square$

Corollary I.6.2 (Nonzero cyclic Fourier coefficients, any modulus). *Let $N \geq 2$, $A \subseteq \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$, and let $r \not\equiv 0 \pmod{N}$. Use the unnormalized transform*

$$\widehat{\mathbf{1}_A}(r) = \sum_{x \in \mathbb{Z}/N\mathbb{Z}} \mathbf{1}_A(x) e(-rx/N).$$

If $|\widehat{\mathbf{1}_A}(r)| \geq \eta N > 0$, the set of standard representatives of A has the density-increment conclusion of Theorem I.6.1 on some ambient progression $P \subseteq [N]_0$. Primality of N is unnecessary.

Proof. The geometric sum $\sum_{x=0}^{N-1} e(-rx/N)$ vanishes for any nonzero residue r , whether or not r is a unit. Thus the Fourier coefficient of $\mathbf{1}_A - \delta$ is the same as that of $\mathbf{1}_A$ at r . Apply Theorem I.6.1 with $h = \mathbf{1}_A$ and $\xi = -r/N$. $\square$

I.6.1 A simpler asymptotic comparison with all rounding exposed

Since $\arcsin u \geq u$ for $0 \leq u \leq 1$, if $N\eta \geq 32\pi\delta(1-\delta)$ then the square root in (I.6.1) is at least 2, and

$$L \geq \sqrt{\frac{N\eta}{32\pi\delta(1-\delta)}}. \quad (\text{I.6.3})$$

Here we used only $\lfloor x \rfloor \geq x/2$ for $x \geq 2$. Compared with the printed lower bound $L_{\text{old}} = \sqrt{\eta^3 N / (128\pi)}$, this yields

$$\frac{L}{L_{\text{old}}} \geq \frac{2}{\eta\sqrt{\delta(1-\delta)}} \quad (\text{I.6.4})$$

in that parameter range. This is a comparison of two valid local conclusions, not a global extremal-set estimate.

δ	η	L_{old}	L from (I.6.1)	$\eta/8$	increment (I.6.2)
0.50	0.100	1576.96	126262	0.01250	0.042857
0.10	0.020	141.05	94055	0.00250	0.008359
0.01	0.002	4.46	89674	0.00025	0.000834
0.01	0.010	49.87	201570	0.00125	0.007519

Table 3: Numerical evaluations of the conditional bounds at $N = 10^{12}$. These are parameter comparisons, not constructions of sets with those Fourier coefficients. The new length is an integer; the old displayed bound is left as a real number.

[write] **6 October 2026.** *Two comparisons the source does not make.* (1) *The repository’s proof-level baseline.* The comparison above is with the printed Corollary 2.5, which is also the statement exported by the formal project (`corollary_2_5`). The repository’s proof of that corollary already obtains the stronger pair $\sqrt{\eta N / (128\pi)}$ and $\eta/4$ before weakening it (the private lemma `cor25_large_scale`, `Proofs02DensityIncrement.lean`, line 375 at HEAD); sources 03 and 05 say so (Remark I.15.3 and Section I.17, Subsection “Comparison with the actual repository baseline”). Against that baseline Theorem I.6.1 still gains: since $\arcsin u \geq u$, its length before the floor is at least $\sqrt{\eta N / (8\pi\delta(1-\delta))}$, a factor $4/\sqrt{\delta(1-\delta)}$ above $\sqrt{\eta N / (128\pi)}$, and its increment $3\eta/8$ exceeds $\eta/4$.

(2) *Cross-source comparison (intake finding, proved at the write).* Under the same hypothesis (this η is source 05’s α and source 03’s ρ), the increment (I.6.2) is at least $3/2$ times source 05’s increment $\delta\alpha/(4\delta-\alpha)$ (Theorem I.17.4), which is also source 03’s increment at $\lambda = 1/2$ (I.15.3). *Proof.* Put $b = 1-\delta$ and $\kappa = \eta/(2\delta b) \in (0, 1]$. Then $\delta\eta/(4\delta-\eta) = 2\delta b\kappa/(4-2b\kappa)$, so the ratio of the two increments is

$$\frac{3\delta b\kappa}{4-(4-3\delta)\kappa} \cdot \frac{4-2b\kappa}{2\delta b\kappa} = \frac{3}{2} \cdot \frac{4-2b\kappa}{4-(4-3\delta)\kappa} = \frac{3}{2} + \frac{3}{2} \cdot \frac{(2-\delta)\kappa}{4-(4-3\delta)\kappa},$$

because $(4-2b\kappa) - (4-(4-3\delta)\kappa) = (2-\delta)\kappa$. The denominator is at least $4-4\kappa+3\delta\kappa \geq 3\delta > 0$, so the ratio exceeds $3/2$. The lengths are on the same scale: for fixed δ, η and $N \rightarrow \infty$, with $u = \eta/(4\delta b)$, the length L here is asymptotic to $\sqrt{N} \arcsin(u)/(2\pi)$, source 05’s exact L to $\sqrt{Nu}/(2\pi)$, and source 03’s length at $\lambda = 1/2$ to $\sqrt{Nu}/(4\pi)$; so this length is not asymptotically shorter than either. Among the three statements at their stated parameters, Theorem I.6.1 is therefore the strongest. The comparison is of these stated pairs, not of trade-off curves: source 03’s parameter $\lambda \rightarrow 0$ and source 05’s width $\varepsilon \rightarrow 0$ in (I.17.9) buy increments approaching $\delta\eta/(2\delta-\eta) = \delta b\kappa/(1-b\kappa)$, which exceeds the increment in (I.6.2) for every $\kappa < 1$ (the difference of $3(1-b\kappa)$ and $4-(4-3\delta)\kappa$ is $1-\kappa$), at the price of much shorter progressions, and source 01’s own trade-off is (I.3.4). Sources 03 and 05 are kept for these and their other features (popularity bound, exact integer length). The intake’s scan of 5,000 random (δ, α) found minimum ratio 1.500014, in agreement.

[write] **6 October 2026; sources 06–10.** *Source 10’s frontier (proved at the second write).* Source 10’s Corollary I.24.3 (its frontier at $\gamma = 1/2$, stated for sets in $\mathbb{Z}/N\mathbb{Z}$, the setting of Corollary I.6.2) has exactly

the length L of (I.6.1), with increment $\delta\eta/(4\delta - \eta)$, which is source 05's; so the increment (I.6.2) is at least $3/2$ times source 10's at the same length (part (2) above). Source 10's Theorem I.24.2 is a one-parameter family: for $0 < \gamma < 1$, length $L_\gamma = \max\{1, \lfloor \sqrt{N\lambda_\gamma/(2\pi)} \rfloor\}$ with $\lambda_\gamma = \arcsin((1 - \gamma)\kappa)$, and increment $D_\gamma = \delta\gamma\eta/(2\delta - \gamma\eta) = \gamma\delta b\kappa/(1 - \gamma b\kappa)$, in the notation $b = 1 - \delta$, $\kappa = \eta/(2\delta b) \in (0, 1]$ used above. *Claim.* Writing I for the increment in (I.6.2), $I \geq D_\gamma$ if and only if $\gamma(4 - \kappa) \leq 3$, and $I \geq \frac{3}{2}D_\gamma$ if and only if $\gamma(4 - (2 - \delta)\kappa) \leq 2$. *Proof.* Both denominators are positive ($4 - (4 - 3\delta)\kappa \geq 3\delta\kappa > 0$ and $1 - \gamma b\kappa > 0$), so $I \geq cD_\gamma$ is equivalent to $3(1 - \gamma b\kappa) \geq c\gamma(4 - (4 - 3\delta)\kappa)$. For $c = 1$ this is $3 - 4\gamma + \gamma\kappa(4 - 3\delta - 3b) \geq 0$, and $4 - 3\delta - 3b = 1$. For $c = 3/2$ it is $2 - 4\gamma + \gamma\kappa(4 - 3\delta - 2b) \geq 0$, and $4 - 3\delta - 2b = 2 - \delta$. Consequences: for every $\gamma \leq 1/2$ the increment here is at least $3/2$ times D_γ , while source 10's length is at least L (because $(1 - \gamma)\kappa \geq \kappa/2$); for $1/2 < \gamma \leq 3/(4 - \kappa)$ the increment here is at least D_γ and $L \geq L_\gamma$; for $\gamma > 3/(4 - \kappa)$ source 10's increment is larger, at a shorter length. So the intake's "at least $3/2$ " comparison holds for source 10 at its stated choice $\gamma = 1/2$ and for every $\gamma \leq 1/2$, not along the whole frontier. As $\gamma \rightarrow 1$, $D_\gamma \rightarrow \delta\eta/(2\delta - \eta)$, the limit described in part (2) for sources 03 and 05. (Exact rational spot checks of both equivalences, 200,000 random parameter triples, found no exception.)

I.6.2 Many increment cells instead of one

On the partition in Theorem I.6.1, fix any target $\delta < a < a_*$. Then $\lambda_*(\delta, \kappa/2, a, \eta) > 0$. Theorem I.2.1 gives a disjoint family of good progressions whose union has at least λ_*N points and whose number is at least

$$\frac{\lambda_*N}{2L - 1}.$$

Each has density greater than a and length at least L . This branching conclusion is often more useful than selecting one progression. Any further count of longer arithmetic progressions inside these cells still requires an appropriate theorem at the new density and length.

I.7 A sharp scale barrier for the phase-flat method

Source 01, delivered Section 8.

The $\sqrt{rN}$ scale of the one-frequency partition is not just an artifact of the proof. A concrete badly approximable frequency gives a matching upper scale for every individual short-arc progression.

Lemma I.7.1 (A quadratic irrational estimate). *For $\xi = (\sqrt{5} - 1)/2$ and every integer $q \geq 1$,*

$$\|q\xi\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{1}{3q}. \quad (\text{I.7.1})$$

Proof. Let p be a nearest integer to $q\xi$, and set $\varphi = (1 + \sqrt{5})/2$. The nonzero integer $p^2 + pq - q^2$ is the product $(p - q\xi)(p + q\varphi)$. It is nonzero because ξ is irrational. Also $p \geq 0$ and

$$0 < p + q\varphi \leq q(\xi + \varphi) + \frac{1}{2} = q\sqrt{5} + \frac{1}{2} < 3q.$$

Taking absolute values of the product proves (I.7.1). $\square$

Theorem I.7.2 (An intrinsic square-root obstruction). *Let $0 < r < 1/2$ and $\xi = (\sqrt{5} - 1)/2$. If a proper progression $P = \{a + jq : 0 \leq j < \ell\} \subseteq [N]_0$ has all phase values $e(\xi n)$ in one arc of length r turns, then*

$$\ell \leq 1 + \sqrt{3r(N - 1)}. \quad (\text{I.7.2})$$

Proof. There is nothing to prove for $\ell = 1$. Otherwise lift every phase to the same real interval of length r . Consecutive differences of the lifts lie in $[-r, r] \subset (-1/2, 1/2)$ and are congruent to $q\xi$ modulo one. There is at most one such representative, so all these differences equal the same number ε , with $|\varepsilon| = \|q\xi\|_{\mathbb{R}/\mathbb{Z}}$. Hence

$$(\ell - 1)\|q\xi\|_{\mathbb{R}/\mathbb{Z}} \leq r.$$

Lemma I.7.1 implies $\ell - 1 \leq 3rq$. Containment in $[N]_0$ gives $q(\ell - 1) \leq N - 1$. Multiplying these two inequalities gives $(\ell - 1)^2 \leq 3r(N - 1)$. $\square$

This proves optimality of the exponent $1/2$ for a uniform theorem about single-frequency phase-flat progressions. It does not rule out longer *density-increment* progressions that are obtained by another mechanism, or longer phase-flat progressions for special frequencies.

[write] **6 October 2026.** Source 05's Theorem I.18.5 proves the same kind of obstruction for m simultaneous phases, with the algebraic vector of Lemma I.18.3; the unwrapping step is its Lemma I.18.4. At $m = 1$ its constant $c_1 = 1/20$ gives $\ell \leq 1 + \sqrt{20\varepsilon(N - 1)}$ for arcs of width ε turns, weaker than $\ell \leq 1 + \sqrt{3r(N - 1)}$ above (here $r = \varepsilon$). Credited to both; the golden-ratio constant is source 01's.

I.8 Sharp phase freezing for real functions

Source 02, delivered Section 6.

The balanced function of a set is real. This allows cancellation between its positive and negative parts to be treated more accurately than by a complex-valued triangle inequality.

Theorem I.8.1 (One-cell phase geometry). *Let P be a finite set, let $f : P \rightarrow \mathbb{R}$, and let $z : P \rightarrow \mathbb{C}$ satisfy $|z(x)| = 1$. Suppose*

$$|z(x) - z(y)| \leq 2\rho \quad (x, y \in P), \quad 0 \leq \rho \leq 1.$$

Writing $F = \sum_P f$, $L = \sum_P |f|$, and $C = |\sum_P fz|$, we have

$$\boxed{C^2 \leq (1 - \rho^2)F^2 + \rho^2L^2.} \tag{I.8.1}$$

Consequently

$$C \leq \rho L + (1 - \rho)|F|. \tag{I.8.2}$$

The quadratic bound is sharp for any prescribed positive and negative masses and any $\rho \in [0, 1]$.

Proof. Let $p = \sum f_+$ and $q = \sum f_-$, so $F = p - q$ and $L = p + q$. If either mass is zero, the ordinary triangle inequality proves (I.8.1). Otherwise, sample a phase Z_+ with probabilities proportional to f_+ and independently sample Z_- with probabilities proportional to f_- . Then

$$\sum_P fz = \mathbb{E}(pZ_+ - qZ_-).$$

For every pair of sampled phases,

$$|pZ_+ - qZ_-|^2 = (p - q)^2 + pq|Z_+ - Z_-|^2 \leq F^2 + 4pq\rho^2 = (1 - \rho^2)F^2 + \rho^2L^2.$$

Take expectations and use the triangle inequality. Equality is attained when all positive mass has one phase and all negative mass has another phase at distance 2ρ .

For (I.8.2), put $s = |F| \leq L$. Squaring the claimed linear majorant and subtracting the right side of (I.8.1) gives

$$[\rho L + (1 - \rho)s]^2 - [(1 - \rho^2)s^2 + \rho^2 L^2] = 2\rho(1 - \rho)s(L - s) \geq 0.$$

Both sides being nonnegative, the claim follows. $\square$

Corollary I.8.2 (Sharp aggregate freezing). *Let $P_1, \dots, P_M$ partition a finite set X , and permit a different unit-modulus phase z_j on each cell, with chordal diameter at most 2ρ . Set*

$$C = \sum_j \left| \sum_{P_j} f z_j \right|, \quad S = \sum_j \left| \sum_{P_j} f \right|, \quad L = \sum_X |f|.$$

For real f ,

$$C \leq \rho L + (1 - \rho)S. \tag{I.8.3}$$

For $\rho < 1$ this gives $S \geq (C - \rho L)/(1 - \rho)$, whenever the latter lower bound is useful.

Proof. Sum (I.8.2). To see optimality of the aggregate linear law from knowledge of L, S, ρ alone, combine cells with $F = 0$ and two endpoint phases, for which $C = \rho L$, with one-sign cells having constant phase, for which $C = L = |F|$. Their contributions realize every ratio $S/L \in [0, 1]$. A zero global mean may also be imposed by using one positive and one negative one-sign cell of equal absolute mass, and rotating their phases if necessary. $\square$

For heterogeneous diameter bounds $2\rho_j$, the proof yields the more informative inequality

$$C \leq S + \sum_j \rho_j \left(\sum_{P_j} |f| - \left| \sum_{P_j} f \right| \right).$$

Thus only the part of a cell capable of positive–negative cancellation incurs a phase-freezing penalty.

Remark I.8.3 (Why the real-valued hypothesis is essential). Take phases $e^{i\theta}, e^{-i\theta}$ with $0 < \theta < \pi/2$ and take f to be their complex conjugates. Then $C = L = 2$ and $S = 2 \cos \theta$, while $\rho = \sin \theta < 1$. The right side of (I.8.3) is strictly below 2. Therefore the improved law must not be exported to arbitrary complex inputs. Gowers’s complex-disc-valued formulations still require a different estimate; the present refinement applies directly to real balanced functions.

[write] **6 October 2026.** Theorem I.8.1 is the bound printed first as source 01’s Lemma I.1.2 (note there: $\rho = s$), here for arbitrary real f and with sharpness for any prescribed masses; Corollary I.8.2 is source 01’s Lemma I.1.1. The heterogeneous inequality above and the complex-input counterexample of the Remark are source 02’s alone.

I.8.1 Balanced indicators and the correct error scale

For $f = \mathbf{1}_A - \delta$ on $|X| = N$,

$$\sum_X f = 0, \quad \frac{1}{N} \sum_X |f| = \lambda := 2\delta(1 - \delta). \tag{I.8.4}$$

If $C \geq \alpha N$ and $\rho < \alpha/\lambda$, Corollary I.8.2 gives

$$\frac{S}{N} \geq \sigma(\rho) := \frac{\alpha - \rho\lambda}{1 - \rho} > 0. \quad (\text{I.8.5})$$

This improves both the use of $L \leq N$ and the loss incurred by freezing at an arbitrary phase representative. For example, at $\delta = 1/10$, $\alpha = 1/50$, and $\rho = 1/20$,

$$\lambda = \frac{9}{50}, \quad \sigma = \frac{11}{950}.$$

The crude representative-phase error bound $2\rho N = N/10$ is larger than the assumed correlation $N/50$ and is useless at these parameters, while (I.8.5) remains positive.

For a modular polynomial phase $z(x) = \exp(2\pi i\phi(x)/N)$, containment of $\phi(P)$ in a modular interval of length at most DN , with $D \leq 1/2$, gives chordal diameter at most $2\sin(\pi D)$. The simpler estimate $\rho \leq \pi D$ will suffice in the refinement theorem.

I.9 A sharp density–mass tradeoff

Source 02, delivered Section 7.

We now isolate the selection step from all arithmetic structure.

Theorem I.9.1 (Sharp positive-tail bound). *Let $n_1, \dots, n_M > 0$ have sum N , and let $a, b > 0$. Suppose the cell means $x_j \in [-a, b]$ satisfy*

$$\sum_j n_j x_j = 0, \quad p = \frac{1}{N} \sum_j n_j \max(x_j, 0) > 0.$$

For $0 \leq t < b$, let $J_t = \{j : x_j > t\}$ and let $q_t = N^{-1} \sum_{j \in J_t} n_j$. Then

$$q_t \geq Q_{a,b}(p, t) := \frac{p(1 + t/a) - t}{b - t}. \quad (\text{I.9.1})$$

The bound is positive exactly when $t < ap/(a - p)$. Necessarily $p \leq ab/(a + b)$, so $a - p > 0$. Whenever the right side is nonnegative, the bound is attained in the weighted model by means in $\{-a, t, b\}$, allowing a zero-weight type to be omitted.

Proof. The negative mass also equals pN . Since the negative means have magnitude at most a , the total weight of negative cells is at least pN/a . Hence the total weight of positive cells is at most $N(1 - p/a)$. The cells in J_t contribute at most $bq_t N$ positive mass, and the other positive cells contribute at most

$$t(N(1 - p/a) - q_t N).$$

Therefore $p \leq bq_t + t(1 - p/a - q_t)$, which rearranges to (I.9.1). The feasibility bound follows from $p \leq b(1 - p/a)$.

For sharpness, assign weight p/a to the mean $-a$, weight $q = Q_{a,b}(p, t)$ to b , and the remaining weight $r = 1 - p/a - q$ to t . Direct calculation gives $bq + tr = p$, so the overall mean is zero. Moreover

$$r = \frac{b(1 - p/a) - p}{b - t} \geq 0.$$

The strict threshold $x > t$ counts precisely the b -type. Thus equality holds. Rational data can be realized by integer cell sizes after scaling; arbitrary real data give a weighted sharpness statement or can be approximated by rational data. $\square$

Corollary I.9.2 (Long good cells). *Under the hypotheses of Theorem I.9.1, if $Q_{a,b}(p, t) > 0$, there is a cell with*

$$x_j > t, \quad n_j \geq \frac{Q_{a,b}(p, t)N}{M-1}.$$

The version with M in the denominator is also valid. If every cell has size at least L_0 , there is a cell of size at least L_0 and mean at least $ap/(a-p)$.

Proof. There is at least one negative cell, and hence at most $M-1$ good cells. Pigeonhole the good mass from (I.9.1). For the final assertion, divide the total positive mass pN by the upper bound $N(1-p/a)$ for the total positive-cell weight. $\square$

The mass bound in Theorem I.9.1 is sharp. The particular $M-1$ length corollary is not asserted to solve the finer extremal problem with a fixed number of cells and prescribed individual sizes.

[write] **6 October 2026.** Theorem I.9.1 is the sharp one-sided bound in its general form $[-a, b]$, proved independently in sources 02 (here), 03 (Theorem I.14.1, the same counting of the negative support in probabilistic language), 01 (the case $a = \delta$, $b = 1 - \delta$, by a convex envelope, Theorem I.4.1) and 05 (the same case with a non-strict event, Lemma I.17.3); see the note after Theorem I.4.1. The final assertion of Corollary I.9.2 (a cell of mean at least $ap/(a-p)$) is source 03's (I.14.1) and source 05's (I.17.3). The rational example of Section I.12 attains the bound.

I.9.1 A direct strengthening of Lemma 5.15

Suppose $f : X \rightarrow [-1, 1]$ has mean zero, and let x_j be its mean on a nonempty partition cell. If $S = \sum_j |\sum_{P_j} f| \geq \alpha N$ with $\alpha > 0$, then $p = S/(2N) \geq \alpha/2$ and necessarily $\alpha \leq 1$. The right side of (I.9.1) increases with p . Taking $a = b = 1$ and $t = \alpha/4$ gives the following explicit replacement.

Corollary I.9.3. *Under the hypotheses of Gowers's Lemma 5.15, after empty cells are discarded, some cell P satisfies*

$$\frac{1}{|P|} \sum_{x \in P} f(x) > \frac{\alpha}{4}, \quad |P| \geq \frac{\alpha(2+\alpha)}{2(4-\alpha)} \frac{N}{M-1}. \quad (\text{I.9.2})$$

The coefficient of N/M , even without using $M-1$, is larger than $\alpha/4$ by the factor

$$\frac{2(2+\alpha)}{4-\alpha} > 1.$$

For $\alpha = 2/5$ it is $2/15$, compared with the original $1/10$. The improvement follows from accounting for the negative support before spending the remaining mass on small positive means.

[write] **6 October 2026.** Source 03 states the same replacement, (I.14.4). *Comparison with the formal statement* (checked at HEAD): `lemma_5_15` (Section05.lean, line 352; exact companion `lemma_5_15_holds`, Proofs05_10.lean, line 735) allows $\alpha \geq 0$ and empty cells among its M cells, and concludes a cell with $\alpha|P_j|/4 \leq \sum_{P_j} f$ and $|P_j| \geq \alpha N/(4M)$. Corollary I.9.3 needs $\alpha > 0$ and counts only nonempty cells (at most M); under these conditions its mean bound is strict and its size bound is the larger one displayed. At $\alpha = 2/5$ the coefficient is $\alpha(2+\alpha)/(2(4-\alpha)) = 0.96/7.2 = 2/15$ (rechecked). Not formalized.

[write] **6 October 2026; sources 06–10.** *The exact optimum for a fixed number of cells (source 06).* Source 06’s Theorem I.21.1 determines the largest guaranteed high cell for fixed data and a fixed number $M \geq 3$ of nonnull cells, $\min\{P_+/(b(M-1)), S/(M-2)\}$, with exact extremizers; its Corollary I.21.2 specializes it, under the hypotheses here ($f \in [-1, 1]$, mean zero, nonempty cells, $0 < \alpha \leq 1$), to a cell with mean $> \alpha/4$ and

$$|P_j| \geq N \min\left\{\frac{\alpha}{2(M-1)}, \frac{S_\alpha}{M-2}\right\}, \quad S_\alpha = \frac{\alpha(2+\alpha)}{2(4-\alpha)}.$$

Claim (second write). This is at least the bound $S_\alpha N/(M-1)$ of Corollary I.9.3, strictly for $0 < \alpha < 1$, with equality at $\alpha = 1$. *Proof.* $S_\alpha/(M-2) > S_\alpha/(M-1)$. And $\alpha/2 \geq S_\alpha$ is equivalent to $4-\alpha \geq 2+\alpha$, i.e. $\alpha \leq 1$, with equality only at $\alpha = 1$; at $\alpha = 1$ the minimum is $1/(2(M-1)) = S_1/(M-1)$. For $M = 2$ source 06 gives $|P_j| \geq \alpha N/2 \geq S_\alpha N$. So the intake’s “strictly beats” holds for $\alpha < 1$ only. Source 06’s Corollary I.21.3 also improves both conclusions at once, with threshold $\tau(\alpha) = \alpha/4 + 3\alpha^2/32 + O(\alpha^3)$. Neither is formalized; lemma_5_15 is unchanged.

I.9.2 The asymmetric range of a balanced indicator

For $f = \mathbf{1}_A - \delta$, put $a = \delta$, $b = 1 - \delta$. If $S \geq \sigma N > 0$, then every

$$0 \leq t < \frac{\delta\sigma}{2\delta - \sigma}$$

satisfies

$$\frac{1}{N} \sum_{j: |A \cap P_j|/|P_j| > \delta + t} |P_j| \geq \frac{(\sigma/2)(1+t/\delta) - t}{1 - \delta - t}. \quad (\text{I.9.3})$$

In particular there is a cell with density at least

$$\delta + \frac{\delta\sigma}{2\delta - \sigma}. \quad (\text{I.9.4})$$

The denominator is positive because $\sigma \leq 2\delta(1 - \delta)$. The expression in (I.9.4) never exceeds 1 under this feasibility condition.

I.9.3 A nonzero-mean extension

For incomplete covers or weighted variants, mean zero can fail. If the weighted mean is μ and the weighted absolute mean is s , define $p_+ = (s + \mu)/2$ and $p_- = (s - \mu)/2$. The same proof gives

$$q_t \geq \frac{p_+ - t(1 - p_-/a)}{b - t}, \quad 0 \leq t < b. \quad (\text{I.9.5})$$

Indeed the negative support has weight at least p_-/a , while the positive mass is p_+ . This formula avoids silently using the zero-mean identity on a proper subset of the original ambient set.

I.10 Combining correlation, phase control, and density extraction

Source 02, delivered Section 8.

Theorem I.10.1 (Robust phase-to-density transfer). *Let $A \subseteq X$, $|X| = N$, $\delta = |A|/N \in (0, 1)$, and $f = \mathbf{1}_A - \delta$. Let $P_1, \dots, P_M$ be a partition into nonempty cells. Suppose each cell has a unit-modulus phase of chordal diameter at most 2ρ , where $0 \leq \rho < 1$, and*

$$\sum_j \left| \sum_{P_j} f z_j \right| \geq \alpha N, \quad \alpha > 2\rho\delta(1 - \delta).$$

Define

$$\sigma = \frac{\alpha - 2\rho\delta(1 - \delta)}{1 - \rho}, \quad Q = \frac{(\sigma/2)(1 + t/\delta) - t}{1 - \delta - t}.$$

For every $0 \leq t < \delta\sigma/(2\delta - \sigma)$, the cells with density greater than $\delta + t$ have total size at least QN , and one of them has size at least $QN/(M - 1)$. If the cells are arithmetic progressions, so is the selected cell.

Proof. Use (I.8.4) and Corollary I.8.2 to obtain $S \geq \sigma N$. Apply (I.9.3) and Corollary I.9.2. No property of a progression is used in selection, so that property is preserved automatically. $\square$

For the numerical parameters at the end of Section I.8, $\sigma = 11/950$. Formula (I.9.4) gives a density increment of at least $11/1790$. Requiring the smaller increment $t = 1/400$ guarantees good-cell mass at least $261/68210$ of the ambient set. These are exact rational consequences, not estimates obtained from a simulation.

This theorem is a replacement interface for phase-to-density arguments, rather than an unconditional restatement of every printed version of Corollary 5.8. In particular, assertions about the length of a progression must be supplied by an actual partition theorem with its thresholds and rounding conditions made explicit.

I.11 A uniform refinement theorem and an optimized phase budget

Source 02, delivered Section 9.

I.11.1 The imported phase-partition input

Here we make the only non-elementary input in our progression-length specialization explicit. Suppose a class of polynomial phases on injectively parameterized modular arithmetic progressions has the following property, for constants $K \geq 1$ and $R_0 \geq 1$:

Partition input $\text{PP}(K, R_0)$. For a progression of size $m > R_0$, and any integer $1 \leq v \leq m^{1/K}$, there is a partition into nonempty arithmetic subprogressions of lengths $v - 1$ or v , such that the phase values in each cell lie in a modular interval of length at most $Nm^{-1/K}$.

The supplied, edited transcription of Corollary 5.6 states this form for

$$K = (k!)^2 2^{(k+1)^2}, \quad R_0 = 2^{2^{40k^3+1}}, \quad k \geq 1, \quad (\text{I.11.1})$$

with the explicit requirement that the progression length is at most N [20]. Composing a polynomial with an injective progression chart $x \mapsto a + dx$ permits its use on each such input progression. We import this stated partition result; the lengthy Weyl-type recurrence proof is not reproved or

independently Lean-verified here. The elementary theorem that follows is valid for any correctly established $\text{PP}(K, R_0)$, including stronger future inputs.

[write] **6 October 2026.** *Stale since the pin: Corollary 5.6 is now proved in Lean.* When source 02 was written (observed branch reference `66c02f8a1`), Corollary 5.6 was an open statement of the formal project, and the two sentences above were accurate. At HEAD (`25df8755d`, 6 October 2026) `corollary_5_6` (`Section05.lean`, line 205) has the exact companion `corollary_5_6_holds` (`Proofs05FullPartition.lean`, line 85). Its statement is $\text{PP}(K, R_0)$ for a polynomial ϕ on $\mathbb{Z}_N$ of degree at most k and the initial interval $\{0, \dots, r-1\}$: $K = \text{polynomialPartitionConstant } k = (k!)^2 2^{(k+1)^2}$; the threshold $\text{polynomialPartitionThreshold } k = \text{weylThreshold } k^2 = 2^{40k^3+1} = R_0$, used with the strict inequality $R_0 < r$; $r \leq N$; $1 \leq v \leq r^{1/K}$; nonempty proper cells of length $v-1$ or v ; image diameter at most $r^{-1/K}N$. So the Weyl-type recurrence behind (I.11.1) is now proved in the Lean development (an exact companion in the ledger). Theorem I.11.1 also composes the polynomial with the chart $x \mapsto a+dx$ of each input progression. The degree half of that step is in Lean (`polynomialOn_affine_pullback`, `Proofs05Downstream.lean`, line 244), but Theorem I.11.1, Corollary I.11.2 and the transfer of cells back to the input progressions are not formalized. They gain no formal status. The related `corollary_5_8` is still open; `corollary_5_8_with_scale_holds` proves only its scale-qualified repair, and Section I.10 rightly declines to restate it.

Theorem I.11.1 (One global budget for many local phases). *Assume $\text{PP}(K, R_0)$. Let $Q_1, \dots, Q_M$ partition $G = \mathbb{Z}_N$ into nonempty injectively parameterized arithmetic progressions, and allow an independent polynomial phase z_i on every Q_i . Let $f : G \rightarrow \mathbb{R}$ and write*

$$C_0 = \sum_i \left| \sum_{Q_i} f z_i \right|, \quad L = \sum_G |f|.$$

For $0 < \rho < 1$, put

$$B_0 = \max\{4, (R_0 + 1)^{1/K}\}, \quad B_\rho = \max\{B_0, \pi/\rho\}. \quad (\text{I.11.2})$$

There is a refinement into J nonempty arithmetic progressions R_j such that

$$J \leq B_\rho M^{1/K} N^{1-1/K}, \quad (\text{I.11.3})$$

$$\sum_j \left| \sum_{R_j} f \right| \geq \frac{C_0 - \rho L}{1 - \rho}. \quad (\text{I.11.4})$$

Proof. Set $T_0 = \max\{4^K, R_0 + 1, (\pi/\rho)^K\} = B_\rho^K$. Treat an input cell of size m in two ways. If $m < T_0$, use singletons. Their count satisfies

$$m = m^{1-1/K} m^{1/K} \leq B_\rho m^{1-1/K},$$

and their phase diameter is zero.

If $m \geq T_0$, take $v = \lfloor m^{1/K} \rfloor$ in PP. Since $m^{1/K} \geq 4$, every resulting cell has length at least

$$v - 1 \geq m^{1/K} - 2 \geq \frac{1}{2} m^{1/K}.$$

The number of cells is at most $2m^{1-1/K}$. On each cell the half chordal diameter is at most $\pi m^{-1/K} \leq \rho$. The conditions $m > R_0$ and $m \leq N$ needed by the imported input are respected.

Summing the counts and applying concavity, or Hölder's inequality, gives

$$J \leq B_\rho \sum_i |Q_i|^{1-1/K} \leq B_\rho M^{1/K} N^{1-1/K}.$$

The sum of absolute local correlations on the refinement is at least C_0 , by the triangle inequality within each original cell. Apply Corollary I.8.2 to the refined cells and their inherited phases to obtain (I.11.4). $\square$

Unlike an argument applying a correlation threshold separately to each Q_i , the cost in (I.11.3) does not depend on a smallest positive local correlation. Cells with tiny or zero correlation cause no singular dependence. The theorem also allows different polynomials in different input cells. The exponent $1/K$ is inherited unchanged from PP.

Corollary I.11.2 (Explicit progression density increment). *In Theorem I.11.1, take $f = \mathbf{1}_A - \delta$, assume $C_0 \geq \alpha N$, and choose ρ, t as in Theorem I.10.1. With its value of Q , there is an arithmetic progression P such that*

$$\frac{|A \cap P|}{|P|} > \delta + t, \quad |P| \geq \frac{Q}{B_\rho} \left(\frac{N}{M} \right)^{1/K}. \quad (\text{I.11.5})$$

Proof. Apply Theorem I.10.1 to the refinement and use $J - 1 \leq J \leq B_\rho M^{1/K} N^{1-1/K}$. The positive lower bound for Q ensures that a negative cell and a good cell exist, so $J \geq 2$. $\square$

The word “progression” in this corollary means an injectively parameterized modular progression when the input cells are modular. Converting a wrapping modular progression to proper integer progressions is a separate cost. It must not be omitted when this interface is inserted into a proof over $[N]$.

I.11.2 Closed-form optimization

The free phase budget can be optimized exactly within the certificate just proved. Fix $\delta \in (0, 1)$, $0 < \alpha < \lambda = 2\delta(1 - \delta)$, and

$$0 \leq t < \frac{\delta\alpha}{2\delta - \alpha}.$$

Define

$$A_t = \frac{1 + t/\delta}{2}, \quad d = A_t\alpha - t, \quad e = A_t\lambda - t. \quad (\text{I.11.6})$$

The identity $e = \delta(1 - \delta - t)$ makes its positivity explicit. Then $0 < d < e$ and the good-mass certificate is

$$Q(\rho) = \frac{d - e\rho}{(1 - \rho)(1 - \delta - t)}, \quad 0 < \rho < d/e.$$

Proposition I.11.3 (Optimal uniform phase budget). *The coefficient $Q(\rho)/B_\rho$ in (I.11.5) is maximized at*

$$\boxed{\rho_{\text{opt}} = \min \left\{ 1 - \sqrt{1 - d/e}, \frac{\pi}{B_0} \right\}.} \quad (\text{I.11.7})$$

If the first entry is no larger than the second, the maximum coefficient is

$$\frac{e(1 - \sqrt{1 - d/e})^2}{\pi(1 - \delta - t)} = \frac{\delta}{\pi}(1 - \sqrt{1 - d/e})^2. \quad (\text{I.11.8})$$

Proof. When $\rho \leq \pi/B_0$, the denominator B_ρ equals π/ρ . Apart from the positive constant $1/[\pi(1 - \delta - t)]$, the objective is

$$H(\rho) = \frac{\rho(d - e\rho)}{1 - \rho}.$$

Its derivative has the sign of $d - 2e\rho + e\rho^2$. The unique critical point in $(0, d/e)$ is $r = 1 - \sqrt{1 - d/e}$; the derivative is positive before r and negative after it. At this point $d = e(2r - r^2)$, so $H(r) = er^2$.

When $\rho \geq \pi/B_0$, the denominator is constant B_0 , while $Q'(\rho) = (d - e)/[(1 - \delta - t)(1 - \rho)^2] < 0$. Hence no larger budget beyond that boundary can improve the certificate. Combining the two regions proves (I.11.7) and (I.11.8). If the boundary is outside the admissible interval, the first critical point is used automatically. $\square$

This optimization improves a particular rigorously derived lower bound; it is not a proof that the underlying polynomial partition itself is optimal. In the regime dominated by the enormous recurrence threshold R_0 , the boundary π/B_0 can control the optimum, so optimizing the elementary interface alone cannot erase the main recurrence loss.

I.12 A rational equality example for the moment bound

Source 02, delivered Appendix B.

Take $a = b = 1$, $p = 1/5$, $t = 1/10$, and assign weights

$$\frac{1}{5}, \quad \frac{2}{3}, \quad \frac{2}{15}$$

to means $-1, 1/10, 1$. Their weighted mean is

$$-\frac{1}{5} + \frac{1}{15} + \frac{2}{15} = 0,$$

the positive mass is $1/5$, and the mass of cells with mean strictly above $1/10$ is $2/15$. Formula (I.9.1) gives exactly

$$\frac{\frac{1}{5}(1 + \frac{1}{10}) - \frac{1}{10}}{1 - \frac{1}{10}} = \frac{2}{15}.$$

For $N = 150$, the three cell sizes are 30, 100, 20. Taking f constant on these cells realizes the abstract example on a finite set with integer cardinalities.

[write] 6 October 2026. The example attains Theorem I.9.1 with $a = b = 1$, $p = 1/5$, $t = 1/10$ (arithmetic rechecked at the write); it is printed here, in Part I, next to the theorem it illustrates, rather than with source 02's other appendix (Section V.11).

I.13 Conventions and analytic background

Source 03, delivered Section 2.

Unless stated otherwise, G is a finite abelian group, written additively, and $\mathbb{E}_{x \in G} = |G|^{-1} \sum_{x \in G}$. Fourier transforms in this report are normalized:

$$\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}, \quad f(x) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(x).$$

Character orthogonality says that the average of a character is zero unless it is trivial. For $f: G \rightarrow \mathbb{C}$ and $d \geq 1$, define

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{v \in \{0,1\}^d} \mathcal{C}^{|v|} f \left(x + \sum_i v_i h_i \right), \quad (\text{I.13.1})$$

where $\mathcal{C}$ denotes complex conjugation. These expressions are nonnegative. One has $\|f\|_{U^1} = |\mathbb{E}f|$, and for $d \geq 2$ they are norms, satisfy the triangle inequality, and are monotone in d . The recursive identity

$$\|f\|_{U^{d+1}}^{2^{d+1}} = \mathbb{E}_h \|f(\cdot) \overline{f(\cdot + h)}\|_{U^d}^{2^d}$$

follows by splitting the last cube coordinate. Iterated Cauchy–Schwarz proves positivity, monotonicity, and the mixed cube inequality used below; these are the standard elementary analytic facts of the source’s Section 3. Fourier orthogonality also gives $\|f\|_{U^2}^4 = \sum_\chi |\hat{f}(\chi)|^4$.

For $A \subset G$, write $\delta = |A|/|G|$ and $f_A = \mathbf{1}_A - \delta$. The source’s statement that A is α -uniform of degree $d - 1$ means

$$\|f_A\|_{U^d}^{2^d} \leq \alpha. \quad (\text{I.13.2})$$

In particular its parameter α is a norm power, not the norm itself. This distinction is essential when comparing exponents.

A cube is a parameter tuple $(x, h_1, \dots, h_d)$ with its 2^d labelled vertices. We count parameter tuples; repeated vertices, zero increments, and coincident cubes under symmetries are retained. Arithmetic progressions in a finite group are similarly counted by their parameters (x, t) unless an ordinary integer progression is specified. A positive integer step and containment in the representative interval are proved explicitly in Section I.15.

The symbols m, N, δ are local to each section. In the cube section $m = 2^d$ is the number of vertices. In the DRC section m is the size of the witness set, while in the energy section it is $|A|$. Every statement gives its own quantifiers. An empty sum is zero and a zeroth power, including 0^0 in finite polynomial identities, is one.

I.14 An exact selection theorem for density increments

Source 03, delivered Section 3.

The last step of a density increment argument is often a finite averaging problem. A partition has appreciable total discrepancy, and one wants a cell which is both large and unusually dense. Keeping the asymmetric bounds of a balanced indicator gives more information than replacing them by $[-1, 1]$. The result below solves the associated one-sided mass problem exactly.

Theorem I.14.1 (Sharp one-sided mass bound). *Let X be a real random variable on a probability space, with $-a \leq X \leq b$, $\mathbb{E}X = 0$, where $a, b > 0$. Set*

$$T = \mathbb{E}X_+ = \mathbb{E}X_- = \frac{1}{2}\mathbb{E}|X|.$$

Then $0 \leq T \leq ab/(a+b)$. If $T > 0$, then

$$\text{ess sup } X \geq \frac{aT}{a - T}. \quad (\text{I.14.1})$$

For every $0 \leq t < b$,

$$\Pr(X > t) \geq \max \left\{ 0, \frac{(a+t)T - at}{a(b-t)} \right\}. \quad (\text{I.14.2})$$

Whenever $0 \leq t < aT/(a-T)$, the positive lower bound in (I.14.2) is attained by a probability distribution supported on $\{-a, t, b\}$. Thus it cannot be increased using only the stated information. Bound (I.14.1) is sharp as well.

Proof. Writing $p_- = \Pr(X < 0)$ gives $p_- \geq T/a$. Likewise $\Pr(X > 0) \geq T/b$, so $T/a + T/b \leq 1$. If h is the essential supremum of X , then

$$T \leq h \Pr(X > 0) \leq h(1 - T/a),$$

which proves (I.14.1). Put $q = \Pr(X > t)$. The mass where $0 < X \leq t$ is at most $1 - p_- - q$; when $t = 0$ the same inequality below still holds. Consequently

$$T \leq t(1 - p_- - q) + bq \leq t(1 - T/a) + (b - t)q.$$

Rearranging proves (I.14.2).

For sharpness set

$$p_{-a} = T/a, \quad p_b = \frac{(a+t)T - at}{a(b-t)}, \quad p_t = 1 - p_{-a} - p_b.$$

The assumed range of t makes p_b positive, and

$$p_t = \frac{ab - (a+b)T}{a(b-t)} \geq 0.$$

These weights sum to one, the negative first moment is T , and the positive first moment is $tp_t + bp_b = T$. Only the atom at b lies above t , giving equality. For (I.14.1), use the two-point distribution at $-a$ and $aT/(a-T)$, with weights T/a and $1 - T/a$. $\square$

The sharpness assertion concerns arbitrary probability weights. Rational weights give literal finite models after clearing denominators; approximation gives the corresponding limiting optimality when prescribed cardinalities or indicator-valued functions impose integrality restrictions.

[write] **6 October 2026.** Theorem I.14.1 is printed first, in cell form, as source 02's Theorem I.9.1; the proof here is the same counting of the negative support, and is kept with the statement it proves (this form adds the essential-supremum bound (I.14.1), which source 02 states as the last assertion of its Corollary I.9.2). Credited to sources 01, 02, 03 and 05 (note after source 01's Theorem I.4.1).

Corollary I.14.2 (A large cell with a specified increment). *Let $a, b > 0$, let $P_1, \dots, P_M$ be a partition of a finite probability space into cells of positive measure, and let f be real, mean zero, and valued in $[-a, b]$. Write $w_j = \mu(P_j)$ and $x_j = \mathbb{E}(f \mid P_j)$. Suppose*

$$\sum_j w_j |x_j| \geq 2T_0 > 0.$$

If $0 \leq t < aT_0/(a - T_0)$, some cell satisfies

$$x_j > t, \quad w_j \geq \frac{(a+t)T_0 - at}{a(b-t)(M-1)}. \quad (\text{I.14.3})$$

In particular $M \geq 2$ under these hypotheses.

Proof. Apply Theorem I.14.1 to the random variable taking value x_j with probability w_j . Its T is at least T_0 , and the bound in (I.14.2) is increasing in T . Since $T > 0$ and the mean is zero, at least one cell has negative average. There are therefore at most $M - 1$ cells above t . One of them carries at least $1/(M - 1)$ of their total mass. Also $T_0 \leq ab/(a + b) < a$, so the displayed denominators are positive. $\square$

For $f = 1_A - \delta$ take $a = \delta$, $b = 1 - \delta$. The resulting formula retains the density throughout the extraction. For the symmetric setting of Gowers's Lemma 5.15, put $a = b = 1$ and $2T_0 = \alpha$. At the same requested increment $t = \alpha/4$, it gives

$$\frac{|P_j|}{N} \geq \frac{\alpha(2 + \alpha)}{2(4 - \alpha)(M - 1)}, \quad \mathbb{E}(f \mid P_j) > \alpha/4, \quad (\text{I.14.4})$$

for $0 < \alpha \leq 1$. This improves the historical $\alpha/(4M)$ guarantee. More usefully, (I.14.3) allows the required increment and the retained mass to be chosen together. The exactness claim applies to the total mass formula; dividing by the number of eligible cells is a separate averaging step.

[write] **6 October 2026.** (I.14.4) is source 02's Corollary I.9.3; the comparison with the formal lemma_5_15 is in the note after it.

I.14.1 Phase removal with the actual error budget

Lemma I.14.3 (Partition phase transfer). *Let f be an integrable complex function and let P_j partition a finite probability space. On each cell let ψ_j be a phase, with $|\psi_j(x)| = 1$, and choose a constant z_j of modulus one. Put*

$$\rho = \sum_j \left| \int_{P_j} f \psi_j d\mu \right|, \quad E_{\text{osc}} = \sum_j \int_{P_j} |f| |\psi_j - z_j| d\mu.$$

Then

$$\sum_j \left| \int_{P_j} f d\mu \right| \geq \rho - E_{\text{osc}}. \quad (\text{I.14.5})$$

If f is real and mean zero, the total positive cell discrepancy is at least $(\rho - E_{\text{osc}})/2$.

Proof. For each j , subtract $z_j \int_{P_j} f$ from $\int_{P_j} f \psi_j$ and use the triangle inequality. Sum the resulting inequalities. For real f of mean zero, the positive and negative cell discrepancies have equal total mass. $\square$

This formulation allows different phases and different oscillations on different cells. Cells on which $|f|$ has small mass may tolerate a much larger phase error. If all oscillations are at most ε , then $E_{\text{osc}} \leq \varepsilon \mathbb{E}|f|$. In particular,

$$\mathbb{E}|1_A - \delta| = 2\delta(1 - \delta). \quad (\text{I.14.6})$$

For a weight $g: \Omega \rightarrow [0, 1]$ of mean δ , the same expression is an upper bound for $\mathbb{E}|g - \delta|$: pointwise, $(g - \delta)_+ \leq (1 - \delta)g$. Discarding this factor and using only $|f| \leq 1$ loses useful information, especially for small δ .

I.15 A density-sensitive Fourier increment on proper progressions

Source 03, delivered Section 4.

We give an independent progression construction, so the constants below do not depend on any unrounded version of the partition lemma in the source. All progressions in this section are ordinary progressions of integers inside $\{0, \dots, N-1\}$, with positive integer step.

Lemma I.15.1 (Elementary phase flattening). *Let $N \geq 1$, $\theta \in \mathbb{R}$, and $0 < \varepsilon \leq 1$. Set*

$$L = \max \left\{ 1, \left\lfloor \sqrt{\frac{\varepsilon N}{4\pi}} \right\rfloor \right\}.$$

There is a partition of $\{0, \dots, N-1\}$ into proper integer arithmetic progressions, each of length between L and $2L-1$, on each of which $e(\theta x) = \exp(2\pi i \theta x)$ differs from a fixed unit complex number by at most ε .

Proof. If $L = 1$, take singletons. Otherwise set $Q = \lfloor N/L \rfloor$. The circular gaps between the $Q+1$ points $0, \theta, \dots, Q\theta$ modulo one give an integer q with $1 \leq q \leq Q$ and

$$\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{Q+1}.$$

This remains true if some points coincide. Split the integer interval into residue classes modulo q . Every class has at least $\lfloor N/q \rfloor \geq L$ elements. A chain of $s \geq L$ elements can be split into consecutive blocks of sizes in $[L, 2L-1]$: write $s = hL + r$, $0 \leq r < L$, and use $h-1$ blocks of size L and one of size $L+r$. Within any such block, the phase difference from its first point is bounded by

$$2\pi(2L-2)\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{4\pi(L-1)}{Q+1} < \frac{4\pi L^2}{N} \leq \varepsilon.$$

Here $|e(u) - e(v)| \leq 2\pi\|u - v\|_{\mathbb{R}/\mathbb{Z}}$ was used together with the triangle inequality on the circle. The blocks inherit the positive step q and remain inside the original interval. $\square$

Theorem I.15.2 (Fourier increment with density retained). *Let $A \subset \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$. Suppose a nontrivial character has normalized Fourier coefficient of magnitude at least $\rho > 0$ on 1_A . For any $0 < \lambda < 1$, there is a proper integer arithmetic progression $P \subset \{0, \dots, N-1\}$ such that*

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{\lambda \rho N}{8\pi\delta(1-\delta)}} \right\rfloor \right\}, \tag{I.15.1}$$

$$\frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta(1-\lambda)\rho}{2\delta - (1-\lambda)\rho}. \tag{I.15.2}$$

No primality assumption on N is needed.

Proof. Let $f = 1_A - \delta$. Orthogonality of the nontrivial character implies $\rho \leq \min\{\delta, 1-\delta\}$ and $|\mathbb{E}_x f(x)e(-rx/N)| \geq \rho$ for the relevant r . Set

$$\varepsilon = \frac{\lambda \rho}{2\delta(1-\delta)}.$$

Since $\max\{\delta, 1 - \delta\} \geq 1/2$, this satisfies $0 < \varepsilon \leq \lambda < 1$. Apply Lemma I.15.1 to $\theta = -r/N$. The triangle inequality over cells gives total cellwise phase correlation at least ρ . Equations (I.14.5) and (I.14.6) therefore give

$$\sum_P \mu(P) |\mathbb{E}(f \mid P)| \geq (1 - \lambda)\rho.$$

The positive discrepancy is at least $T_0 = (1 - \lambda)\rho/2$. Apply (I.14.1) with $a = \delta$, $b = 1 - \delta$ to obtain a cell of average at least $\delta T_0/(\delta - T_0)$. Every cell has the lower length bound (I.15.1), proving both conclusions. The denominator is positive because $\rho \leq \delta$ and $\lambda > 0$. $\square$

The balanced choice $\lambda = 1/2$ gives the particularly simple bounds

$$|P| \geq \max \left\{ 1, \left\lfloor \sqrt{\frac{\rho N}{16\pi\delta(1-\delta)}} \right\rfloor \right\}, \quad \frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta\rho}{4\delta - \rho}. \quad (\text{I.15.3})$$

The increment is at least $\rho/4$. The length keeps the factor $[\delta(1 - \delta)]^{-1/2}$, and all finite rounding is explicit. For example, if $\rho = \kappa\delta$ with $0 < \kappa \leq 1$ and $\delta \leq 1/2$, the lower length scale is $\sqrt{\kappa N/[16\pi(1 - \delta)]}$, while the density grows by the relative factor $1 + \kappa/(4 - \kappa)$. This observation is conditional on a Fourier coefficient of that relative size; it is not an assertion that every sparse set has one.

Remark I.15.3 (What was already in the repository). The pinned implementation of Corollary 2.5 already obtains an increment $\rho/4$ and length $\sqrt{\rho N/(128\pi)}$ when $\rho N \geq 4\pi$, before weakening these to its public conclusion. Thus the $\rho/4$ constant alone is not a contribution of this article. The additional conclusions here are the asymmetric selection formula, the density factor in the progression length, the adjustable loss λ , and an independent construction valid at every finite scale.

[write] **6 October 2026.** Checked at HEAD (6 October 2026): unchanged. The private `cor25_large_scale` is at `Proofs02DensityIncrement.lean`, line 375, and the exported `corollary_2_5_holds` at line 517; the file's blob (85b5c3a) is the one at both pins. *Relation to the other sources.* At $\lambda = 1/2$, (I.15.3) is the simplified form (I.17.8) of source 05's Theorem I.17.4: the same length bound and the same increment, by a different construction (a chordal error bound in place of an arc width and a disk centre). Source 05's exact integer length is asymptotically $\sqrt{2}$ times longer, and source 01's increment (Theorem I.6.1) is at least $3/2$ times this one (note after Table 3).

I.16 Exact finite partitions and the sharp rounding constant

Source 05, delivered Section 2.

I.16.1 Notation and two elementary tools

Set $I_N = \{0, 1, \dots, N - 1\}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. An arithmetic progression is

$$P = \{a, a + q, \dots, a + (\ell - 1)q\} \subset I_N, \quad q \geq 1, \quad \ell \geq 1.$$

Its length is ℓ . All progressions considered here are proper as subsets of the integers, not just as cyclic sequences. An arc of width ε in $\mathbb{T}$ is the image of a real interval of length ε . Arc containment is a little stronger than a bound on pairwise circular distances, and it is the condition used throughout.

Lemma I.16.1 (Circular pigeonholing). *For every $\theta \in \mathbb{R}$ and integer $Q \geq 1$, there is an integer q with*

$$1 \leq q \leq Q, \quad \|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{Q+1}.$$

Proof. Place $0, \theta, \dots, Q\theta$ on the circle. A repeated point gives a zero return. Otherwise, the $Q+1$ cyclic gaps have total length one, so a smallest gap has length at most $1/(Q+1)$. The difference of the two corresponding indices has absolute value between one and Q , and the circular distance of its multiple of θ is no larger than that gap. $\square$

Lemma I.16.2 (Exact residue-class blocking). *Let N, q, L be positive integers with $qL \leq N$. The interval I_N can be partitioned into progressions of common difference q , each of length between L and $2L-1$. The number of cells is at most $\lfloor N/L \rfloor$.*

Proof. Each residue class modulo q has at least L elements. If one class has $n = aL + b$ elements with $a \geq 1$ and $0 \leq b < L$, take $a-1$ consecutive blocks of length L and one final block of length $L+b$. The resulting cells have the asserted lengths. Since the cells are disjoint and each has at least L points, their number is at most N/L . $\square$

Proposition I.16.3 (Flexible one-phase partition). *Let $N \geq 1$, $0 < \varepsilon \leq 1$, and let $L \geq 1$ be an integer such that*

$$2L(L-1) \leq \varepsilon N. \tag{I.16.1}$$

Then I_N has a partition into progressions of one common positive difference, all with lengths in $[L, 2L-1]$, on each of which $\theta x + \beta$ lies in an arc of width at most ε . This holds for every real θ, β .

Proof. Condition (I.16.1), with $\varepsilon \leq 1$, implies $L \leq N$. Put $Q = \lfloor N/L \rfloor$ and choose q using Lemma I.16.1. Then $qL \leq N$. Let $d \in [-1/2, 1/2]$ be a representative of $q\theta$ with $|d| \leq 1/(Q+1)$. Apply Lemma I.16.2. On a cell of length $\ell \leq 2L-1$, the phase has a real lift whose span is

$$(\ell-1)|d| \leq \frac{2L-2}{Q+1} \leq \frac{2L(L-1)}{N} \leq \varepsilon.$$

The middle inequality uses $Q+1 > N/L$; allowing a weak inequality also covers $L=1$. $\square$

The largest permitted integer in this proposition is

$$L_*(N, \varepsilon) = \left\lfloor \frac{1 + \sqrt{1 + 2\varepsilon N}}{2} \right\rfloor. \tag{I.16.2}$$

There is no exceptional small-scale regime: $L_* = 1$ automatically covers it. With $u = \varepsilon N/2$, maximality gives $u < L_*(L_*+1) \leq 2L_*^2$, and hence

$$L_* \geq \sqrt{\varepsilon N/4}, \quad L_* > \sqrt{\varepsilon N/2} - \frac{1}{2}. \tag{I.16.3}$$

The exact integer value, rather than either weakened estimate, is preferable in implementations.

I.16.2 The optimal constant with the original upper bound

Theorem I.16.4 (Sharp capped partition). *Suppose $0 < \varepsilon \leq 1$ and $\varepsilon N \geq 1$. Every real linear phase on I_N admits a partition into arithmetic progressions, with a common positive difference, such that each image lies in an arc of width at most ε and each cell satisfies*

$$\frac{1}{3}\sqrt{\varepsilon N} \leq |P| \leq \sqrt{\varepsilon N}. \quad (\text{I.16.4})$$

The lower constant $1/3$ is the largest possible universal constant if the upper constant remains one, even if different cells are allowed different common differences.

Consequently, under the hypotheses of the corrected Gowers Lemma 2.3, its lower bound can be improved from $\sqrt{rs/(16M)}$ to

$$\sqrt{rs/(9M)}, \quad (\text{I.16.5})$$

with the same image-diameter bound s and the same upper length bound $\sqrt{rs/M}$.

Proof. Write $t = \varepsilon N$, $U = \lfloor \sqrt{t} \rfloor$, and $K = \lceil U/2 \rceil$. Then $K \geq 1$ and

$$2K(K-1) \leq U^2 \leq t, \quad 2K-1 \leq U.$$

Proposition I.16.3, with $L = K$, gives cells of lengths between K and $2K-1$, and hence at most $\sqrt{t}$. Also

$$\sqrt{t} < U + 1 \leq 2K + 1 \leq 3K.$$

Thus every cell has length strictly greater than $\sqrt{t}/3$, which implies the stated weak bound.

For sharpness, take $N = 9$, a constant phase, and $\varepsilon = 1 - 1/M$ with $M \geq 10$. Then $t = 9(1 - 1/M) < 9$ and $t \rightarrow 9$ as $M \rightarrow \infty$. A putative lower constant $c > 1/3$ would eventually require every cell to have at least two points, while the upper bound $\sqrt{t} < 3$ forces every cell to have exactly two points. Nine points cannot be partitioned into pairs. This argument uses cardinalities alone and therefore applies to arbitrary allowed cell shapes.

For the modular specialization, use $N = r$, $\varepsilon = s/M$, and slope a/M for an affine map $x \mapsto ax + b \pmod{M}$. The constructed lift has integral span after multiplication by M , so its span at most s is exactly the required discrete diameter bound. The sharpness example becomes $r = 9$, $s = M - 1$, and constant affine map; it respects $r, s \leq M$ and $rs \geq M$. $\square$

Remark I.16.5. The improvement $16 \rightarrow 9$ in the denominator is an actual strengthening of the corrected statement, not a change of upper-bound convention. By contrast, using L_* in (I.16.2) gives a larger minimum but permits a larger maximum, approximately $\sqrt{2\varepsilon N}$. The two formulations serve different purposes. The sharpness of $1/3$ concerns the capped formulation only.

[write] **6 October 2026.** *Checked against the formal statement at HEAD. lemma_2_3 (Sections01_03.lean, line 141; exact companion lemma_2_3_holds, Proofs02Partition.lean, line 561) assumes $0 < N$, $0 < r$, $0 < s$, $r \leq N$, $s \leq N$, $N \leq rs$ and an affine map $x \mapsto ax + b$ from $\{0, \dots, r-1\}$ to $\mathbb{Z}_N$. It concludes a partition into proper progressions with image diameter at most s (`diameterAtMost`) and $\sqrt{rs/(16N)} \leq |P_j| \leq \sqrt{rs/N}$. Lean's N is the modulus M here. Under exactly these hypotheses $\varepsilon = s/M \leq 1$ and $\varepsilon r = rs/M \geq 1$, so Theorem I.16.4 gives the lower bound $\sqrt{rs/(9M)}$. The nine-point example ($r = 9$, $s = M - 1$, $M \geq 10$, constant map) satisfies them, since $9 \leq M$, $M - 1 \leq M$ and $M \leq 9(M - 1)$. So $1/3$ is optimal for the exact Lean formulation, as the theorem says. The improvement is not formalized.*

I.16.3 A strengthened Corollary 2.4 with the same upper length

Write $e(t) = \exp(2\pi it)$. If the values of $e(\phi(x))$ on a cell are contained in an arc of width ε , replacing the phase by its unit midpoint produces pointwise error at most $\pi\varepsilon$.

Corollary I.16.6. *Let $f : I_r \rightarrow \mathbb{C}$ satisfy $|f| \leq 1$, let $\phi(x) = \theta x + \beta$ be real linear, and suppose*

$$\left| \sum_{x \in I_r} f(x) e(-\phi(x)) \right| \geq \alpha r > 0, \quad \alpha r \geq 4\pi.$$

There is a partition into J arithmetic progressions for which

$$\sqrt{\alpha r / (36\pi)} \leq |P_j| \leq \sqrt{\alpha r / (4\pi)}, \quad (\text{I.16.6})$$

$$J \leq \sqrt{36\pi r / \alpha}, \quad (\text{I.16.7})$$

$$\sum_{j=1}^J \left| \sum_{x \in P_j} f(x) \right| \geq \frac{3}{4} \alpha r. \quad (\text{I.16.8})$$

Proof. We have $\alpha \leq 1$. Apply Theorem I.16.4 with $N = r$ and $\varepsilon = \alpha/(4\pi)$. The length and count bounds follow. On every cell choose a unit midpoint z_j for the conjugated phase arc. Then

$$\alpha r \leq \sum_j \left| \sum_{x \in P_j} f(x) \right| + \frac{\alpha}{4} \sum_x |f(x)| \leq \sum_j \left| \sum_{x \in P_j} f(x) \right| + \frac{\alpha}{4} r.$$

□

Thus the corrected Corollary 2.4 constants 128π and $\alpha/2$ can be replaced by 36π and $3\alpha/4$, respectively, without changing the upper length bound. Neither a prime modulus nor a rational slope is needed. This statement does not assert optimality of the new correlation constant or of 36π .

[write] **6 October 2026.** Compared with `corollary_2_4` (`Sections01_03.lean`, line 158; exact companion `corollary_2_4_holds`, `Proofs02PhasePartition.lean`, line 276), which assumes $r \leq N$, $4\pi \leq \alpha r$ and a linear phase on $\mathbb{Z}_N$, and gives lengths in $[\sqrt{\alpha r / (128\pi)}, \sqrt{\alpha r / (4\pi)}]$, at most $\sqrt{128\pi r / \alpha}$ cells and retained correlation $(\alpha/2)r$. The corollary above keeps the upper length, improves 128π to 36π and $\alpha/2$ to $3\alpha/4$, and drops $r \leq N$, allowing any real linear phase (checked at the write against the Lean statement; not formalized).

I.17 Density-sensitive localization and popular increments

Source 05, delivered Section 3.

I.17.1 Optimal disk geometry for an arc

Lemma I.17.1 (The center need not lie on the unit circle). *Suppose a subset of the unit circle is contained in an arc of width $\varepsilon < 1/2$, with midpoint $e(\gamma)$. Set $a = \pi\varepsilon$. The disk with center $c = \cos(a)e(\gamma)$ and radius $\sin(a)$ contains the arc. No smaller disk contains both endpoints of the full arc.*

Proof. Rotate to $\gamma = 0$. For $|u| \leq a$, with $a < \pi/2$,

$$|e^{iu} - \cos a|^2 = 1 + \cos^2 a - 2 \cos a \cos u \leq \sin^2 a.$$

The endpoints have distance $2 \sin a$, giving the matching radius lower bound. $\square$

All expectations below are normalized by the total number of points. For a partition $\mathcal{P}$, let $g = \mathbb{E}(f \mid \mathcal{P})$ denote the function constant on each cell and equal to its average there.

Proposition I.17.2 (Correlation transfer with an explicit error budget). *Suppose ψ has modulus one and its values on each cell lie in an arc of width $\varepsilon < 1/2$. Let*

$$\alpha \leq |\mathbb{E}f\psi|, \quad B = \mathbb{E}|f|, \quad W = \mathbb{E}|g|.$$

Then

$$W \geq \frac{\alpha - B \sin(\pi\varepsilon)}{\cos(\pi\varepsilon)}. \quad (\text{I.17.1})$$

In particular, $W \geq \alpha - \pi\varepsilon B$ whenever this latter expression is nonnegative.

Proof. Choose the center c_P from Lemma I.17.1 on each cell. With $w_P = |P|/N$,

$$|\mathbb{E}f\psi| \leq \sum_P w_P |c_P| |\mathbb{E}_P f| + \sin(\pi\varepsilon) \mathbb{E}|f| = \cos(\pi\varepsilon)W + \sin(\pi\varepsilon)B.$$

Rearrange. The simpler bound follows from $\sin u \leq u$ and $\cos u \leq 1$. $\square$

The dependence on B is essential for sparse sets. For $f = \mathbf{1}_A - \delta$, where $\delta = |A|/N$, one has exactly

$$\mathbb{E}f = 0, \quad -\delta \leq f \leq 1 - \delta, \quad \mathbb{E}|f| = 2\delta(1 - \delta) = 2v. \quad (\text{I.17.2})$$

Bounding $\mathbb{E}|f|$ by one discards information of order δ when δ is small.

I.17.2 A sharp positive-mass conversion

Lemma I.17.3 (Capacity-sensitive extraction). *Let g be real on a finite probability space, with $\mathbb{E}g = 0$ and $-\delta \leq g \leq b = 1 - \delta$, where $0 < \delta < 1$. Put $S = \mathbb{E}g_+ = \mathbb{E}(-g)_+$. If $0 < s \leq S$, then*

$$\max g \geq \eta(s) := \frac{\delta s}{\delta - s}. \quad (\text{I.17.3})$$

For $0 < t < \eta(s)$,

$$\Pr(g \geq t) \geq \frac{s(1 + t/\delta) - t}{b - t}. \quad (\text{I.17.4})$$

In particular,

$$\Pr(g \geq \eta(s)/2) \geq \frac{s}{2(b - \eta(s)/2)}. \quad (\text{I.17.5})$$

The bound (I.17.3) is sharp as a consequence of the stated data alone.

Proof. Let $M = \max g$ and $p = \Pr(g > 0)$. Then $S \leq Mp$ and $S \leq \delta(1 - p)$. Consequently $S/M + S/\delta \leq 1$, whence $M \geq \delta S/(\delta - S)$. The right-hand side is increasing in S . Here $S \leq \delta(1 - \delta) < \delta$, so all denominators are positive.

For popularity, write $w = \Pr(g \geq t)$. The positive set has mass at most $1 - S/\delta$, and hence

$$S \leq t(1 - S/\delta) + (b - t)w.$$

Rearranging and using $S \geq s$ gives (I.17.4). Setting $t = \eta(s)/2$ gives (I.17.5). Finally, for $s \leq \delta(1 - \delta)$, the two-point distribution taking values $-\delta$ and $\eta(s)$ with probabilities s/δ and $1 - s/\delta$ has mean zero and positive mass exactly s . It attains (I.17.3). $\square$

[write] **6 October 2026.** (I.17.3) and (I.17.4) are the case $a = \delta$, $b = 1 - \delta$ of the sharp bound printed first as source 02's Theorem I.9.1, here with the event $g \geq t$ in place of $g > t$; since $\Pr(g \geq t) \geq \Pr(g > t)$, this form is weaker. Credited to sources 01, 02, 03 and 05 (note after source 01's Theorem I.4.1). The popularity bound (I.17.5) is source 05's.

I.17.3 A strengthened Fourier density-increment theorem

We normalize the Fourier transform on $\mathbb{Z}/N\mathbb{Z}$ by

$$\widehat{h}(r) = \frac{1}{N} \sum_{x=0}^{N-1} h(x)e(-rx/N).$$

The repository uses an unnormalized transform; its hypothesis $|\widehat{A}(r)| \geq \alpha N$ is our hypothesis $|\widehat{\mathbf{1}}_A(r)| \geq \alpha$.

Theorem I.17.4 (Density-sensitive Fourier increment). *Let $A \subset \mathbb{Z}/N\mathbb{Z}$ have density $0 < \delta < 1$. Suppose that for a nonzero frequency r , $|\widehat{\mathbf{1}}_A(r)| \geq \alpha > 0$. Put*

$$v = \delta(1 - \delta), \quad \varepsilon_0 = \frac{\alpha}{4\pi v}, \quad L = \left\lceil \frac{1 + \sqrt{1 + \alpha N/(2\pi v)}}{2} \right\rceil, \quad \eta = \frac{\delta\alpha}{4\delta - \alpha}. \quad (\text{I.17.6})$$

There is a partition of the standard representatives I_N into progressions of a common positive difference, all of lengths in $[L, 2L - 1]$, with the following properties:

- (i) *Some cell P has density at least $\delta + \eta$.*
- (ii) *The union of cells having density at least $\delta + \eta/2$ has size at least*

$$\frac{\alpha}{8(1 - \delta - \eta/2)} N. \quad (\text{I.17.7})$$

In particular, a progression exists with

$$|P| \geq \sqrt{\frac{\alpha N}{16\pi\delta(1 - \delta)}}, \quad \frac{|A \cap P|}{|P|} \geq \delta + \frac{\delta\alpha}{4\delta - \alpha} \geq \delta + \frac{\alpha}{4}. \quad (\text{I.17.8})$$

No primality assumption or lower-scale hypothesis is needed.

Proof. The sum of a nontrivial character on a finite cyclic group is zero, even when N is composite. Therefore

$$\alpha \leq \min(\delta, 1 - \delta), \quad \widehat{\mathbf{1}_A - \delta}(r) = \widehat{\mathbf{1}_A}(r).$$

In particular $\alpha/v \leq 2$, so $0 < \varepsilon_0 \leq 1/(2\pi) < 1/2$. Proposition I.16.3 gives the partition with $L = L_*(N, \varepsilon_0)$. Apply Proposition I.17.2 to $f = \mathbf{1}_A - \delta$ and the conjugated character. By (I.17.2),

$$W \geq \alpha - \pi\varepsilon_0(2v) = \alpha/2.$$

The block averages g are real and mean zero, so $\mathbb{E}g_+ = W/2 \geq \alpha/4$. Lemma I.17.3, with $s = \alpha/4$, proves (i) and (ii). Finally (I.16.3) gives $L \geq \sqrt{\varepsilon_0 N}/4$, which is exactly the first bound in (I.17.8). Since $4\delta - \alpha \leq 4\delta$, the increment is at least $\alpha/4$. $\square$

Remark I.17.5 (There is no hidden singleton exception). When the input scale is small, the formula gives $L = 1$ and the construction uses singletons. The theorem remains valid: all estimates follow from the same proof. In particular, the upper bound $\alpha \leq 1 - \delta$ prevents an impossible required density above one.

I.17.4 The full oscillation–increment trade-off

The preceding choice of ε_0 is convenient rather than optimal. For any $0 < \varepsilon < 1/2$ with $2v \sin(\pi\varepsilon) < \alpha$, the same argument gives lengths at least $L_*(N, \varepsilon)$ and positive mass at least

$$s(\varepsilon) = \frac{\alpha - 2v \sin(\pi\varepsilon)}{2 \cos(\pi\varepsilon)}. \quad (\text{I.17.9})$$

The corresponding increment is $\delta s(\varepsilon)/(\delta - s(\varepsilon))$, and the popularity formula (I.17.4) remains available. Thus length and increment can be optimized against each other at the final finite step.

For example, using

$$\varepsilon = \frac{1}{\pi} \arcsin\left(\frac{\alpha}{4v}\right)$$

gives $s(\varepsilon) = \alpha/[4\sqrt{1 - (\alpha/(4v))^2}]$, while also giving an arc width at least ε_0 . This mildly improves both quantities in Theorem I.17.4. The gain is only a constant-factor refinement; it should not be confused with the density factor $v^{-1/2}$ in the progression length.

The proof also extends to any weight $a : I_N \rightarrow [0, 1]$ of average δ , since $\mathbb{E}|a - \delta| \leq 2\delta(1 - \delta)$. The latter follows pointwise from $|u - \delta| \leq u + \delta - 2u\delta$ for $0 \leq u \leq 1$. The interpretation is then a local increase of the average weight, not of the cardinality of a subset.

I.17.5 Comparison with the actual repository baseline

For large scales, the ratio of the exact new minimum length to the stronger length already present inside the repository proof is asymptotically

$$\frac{\sqrt{\alpha N/(8\pi v)}}{\sqrt{\alpha N/(128\pi)}} = \frac{4}{\sqrt{v}}. \quad (\text{I.17.10})$$

Even the uniform square-root bound in (I.17.8) improves that baseline by a factor $\sqrt{8/v}$. These are local length comparisons under the same Fourier hypothesis.

δ	Repository statement	Repository proof	New exact L	Length ratio
0.5	6233.47	24933.89	199471	8.00
0.1	49.87	4986.78	66490	13.33
0.01	0.0499	498.68	20048	40.20
0.001	0.0000499	49.87	6311	126.55

Table 4: The statement uses $\sqrt{\alpha^3 N/(128\pi)}$; the proof uses $\sqrt{\alpha N/(128\pi)}$. The last column compares the new exact integer bound with the proof-level expression.

Table 4 evaluates the formulas at $N = 10^{12}$ and $\alpha = \delta^2$. The numbers are conditional bounds, not a claim that a particular set with those parameters was constructed. Values below one in the older expression are simply vacuous real lower bounds on an integer length.

[write] 6 October 2026. The first row was recomputed at the write: 6233.47, 24933.89, $\lfloor (1 + \sqrt{1 + \alpha N/(2\pi v)})/2 \rfloor = 199471$ and ratio 8.00; the intake recomputed all four rows. Source 01's Theorem I.6.1 gives, under the same hypothesis, an increment at least $3/2$ times $\delta\alpha/(4\delta - \alpha)$ at a length that is not asymptotically shorter (note after Table 3); Theorem I.17.4 keeps its popularity bound (ii), its exact integer length and the trade-off (I.17.9).

I.18 Simultaneous phases and an optimal exponent

Source 05, delivered Section 4.

I.18.1 An anisotropic integer formulation

Sequentially applying a square-root localization m times can leave an exponent 2^{-m} . A simultaneous pigeonhole argument instead gives $1/(m+1)$. This is the elementary linear-phase analogue of the simultaneous localization problem that appears in more advanced density-increment arguments.

Lemma I.18.1 (Rectangular pigeonholing). *For real $\theta_1, \dots, \theta_m$ and positive integers $Q_1, \dots, Q_m$, there exists*

$$1 \leq q \leq \prod_{i=1}^m Q_i, \quad \|q\theta_i\|_{\mathbb{R}/\mathbb{Z}} < 1/Q_i \quad (1 \leq i \leq m).$$

Proof. Partition $[0, 1)^m$ into $\prod_i Q_i$ half-open rectangular boxes of side lengths $1/Q_i$. Two of the $\prod_i Q_i + 1$ points $j(\theta_1, \dots, \theta_m) \pmod{1}$ occupy the same box. Their index difference is the required q . $\square$

Theorem I.18.2 (Simultaneous linear-phase partition). *Let $N \geq 1$, $m \geq 1$, and $0 < \varepsilon_i \leq 1$. Define*

$$Q_i(L) = \max \left(1, \left\lceil \frac{2(L-1)}{\varepsilon_i} \right\rceil \right). \quad (\text{I.18.1})$$

For every positive integer L such that

$$L \prod_{i=1}^m Q_i(L) \leq N, \quad (\text{I.18.2})$$

there is a partition of I_N into progressions with one common positive difference, all of lengths in $[L, 2L - 1]$, such that every phase $\theta_i x + \beta_i$ lies in an arc of width ε_i on every cell. The number of cells is at most $\lfloor N/L \rfloor$.

In particular, one may always take

$$L = \max \left(1, \left\lfloor \left(\frac{N \prod_i \varepsilon_i}{2^m} \right)^{1/(m+1)} \right\rfloor \right). \quad (\text{I.18.3})$$

Proof. For $L = 1$ use singletons. Otherwise apply Lemma I.18.1 with $Q_i = Q_i(L)$. The resulting return q satisfies $qL \leq N$, so Lemma I.16.2 applies. Along a cell, a real lift of phase i has span at most $(2L - 2)/Q_i \leq \varepsilon_i$.

For the explicit bound, if the floor in (I.18.3) is at least two, then

$$Q_i(L) \leq \frac{2(L - 1)}{\varepsilon_i} + 1 \leq \frac{2L}{\varepsilon_i},$$

and hence $L \prod_i Q_i(L) \leq 2^m L^{m+1} / \prod_i \varepsilon_i \leq N$. The cases with $L = 1$ were handled separately. $\square$

The largest integer satisfying (I.18.2) is often appreciably better than (I.18.3). Its feasibility predicate is monotone, so it can be found by binary search. The theorem has no discarded exceptional set and gives a lower bound for *every* cell, rather than merely for the average length.

[write] **6 October 2026.** The same construction as source 01's Theorem I.5.1 and Corollary I.5.2 (note there); credited to both.

I.18.2 An explicit badly approximable vector

The optimality statement will be proved from scratch. This avoids importing a quantitative transference theorem with hidden constants.

Lemma I.18.3 (An algebraic obstruction). *Let $m \geq 1$, $\xi = 2^{1/(m+1)}$, and $\theta = (\xi, \xi^2, \dots, \xi^m)$. Then, for every integer $q \geq 1$,*

$$\max_{1 \leq i \leq m} \|q\xi^i\|_{\mathbb{R}/\mathbb{Z}} \geq c_m q^{-1/m}, \quad c_m = \frac{1}{2^{m+1} m (5m)^m}. \quad (\text{I.18.4})$$

Proof. Choose nearest integers p_i to $q\xi^i$ and put $h = \max_i |q\xi^i - p_i|$, $Q = \lceil q^{1/m} \rceil$. The $(Q + 1)^m > q$ vectors in $\{0, \dots, Q\}^m$ cannot give distinct residues of $\sum_i u_i p_i$ modulo q . Subtract two vectors with the same residue. This gives integers a_i , not all zero, with $|a_i| \leq Q$ and $q \mid \sum_i a_i p_i$. Define

$$a_0 = -q^{-1} \sum_i a_i p_i, \quad P(X) = a_0 + \sum_{i=1}^m a_i X^i.$$

Then P is a nonzero integer polynomial of degree at most m and

$$|P(\xi)| \leq \frac{mQh}{q}. \quad (\text{I.18.5})$$

Since $\xi^i < 2$ and $|p_i/q - \xi^i| \leq 1/(2q)$, we have $|a_0| \leq (5/2)mQ$.

The polynomial $X^{m+1} - 2$ is irreducible by Eisenstein's criterion at 2. Its roots are $\xi\zeta^j$, $0 \leq j \leq m$, where $\zeta = e(1/(m+1))$. The product

$$\prod_{j=0}^m P(\xi\zeta^j)$$

is a nonzero integer: it is the algebraic norm of the algebraic integer $P(\xi)$, equivalently an integer resultant. For $j \neq 0$,

$$|P(\xi\zeta^j)| \leq |a_0| + \sum_i |a_i| \xi^i \leq 5mQ.$$

Therefore $|P(\xi)| \geq (5mQ)^{-m}$. Combining with (I.18.5) gives $h \geq q/[m(5m)^m Q^{m+1}]$. Finally $Q \leq 2q^{1/m}$ yields (I.18.4). $\square$

Lemma I.18.4 (Unwrapping a flat progression). *Suppose $0 < \varepsilon < 1/2$ and the points $\beta, \beta + \theta, \dots, \beta + (\ell - 1)\theta$ lie in an arc of width ε , with $\ell \geq 2$. Then*

$$(\ell - 1) \|\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \varepsilon.$$

Proof. Choose the unique representatives of the points in a fixed real interval of length ε . Consecutive differences belong to $[-\varepsilon, \varepsilon]$ and are all congruent to θ modulo one. Since $2\varepsilon < 1$, they are the same real number. Thus the lifts form a real arithmetic progression of span $(\ell - 1) \|\theta\|_{\mathbb{R}/\mathbb{Z}}$, which is at most ε . $\square$

I.18.3 The sharp-order finite localization problem

For a vector $\theta \in \mathbb{R}^m$, let $F(\theta; N, \varepsilon)$ be the largest possible minimum cell length among all partitions of I_N into arithmetic progressions on which every $\theta_i x$ lies in an arc of width ε . Different cells may use different differences. Define

$$F_m(N, \varepsilon) = \inf_{\theta \in \mathbb{R}^m} F(\theta; N, \varepsilon).$$

This is a precise extremal formulation introduced for this note; we do not present it as a previously catalogued open problem.

Theorem I.18.5 (Optimal simultaneous localization exponent). *For $N \geq 1$ and $0 < \varepsilon < 1/2$,*

$$F_m(N, \varepsilon) \geq \max \left(1, \left\lfloor (2^{-m} \varepsilon^m N)^{1/(m+1)} \right\rfloor \right), \quad (\text{I.18.6})$$

$$F_m(N, \varepsilon) \leq 1 + (\varepsilon/c_m)^{m/(m+1)} (N - 1)^{1/(m+1)}. \quad (\text{I.18.7})$$

Consequently

$$F_m(N, \varepsilon) \asymp_m 1 + (\varepsilon^m N)^{1/(m+1)}. \quad (\text{I.18.8})$$

The exponent $1/(m+1)$ cannot be increased, even if one asks only for a single flat progression instead of a partition.

Proof. The lower bound is Theorem I.18.2 with equal widths. For the upper bound, take the explicit algebraic vector in Lemma I.18.3. If a flat progression has length $\ell \geq 2$ and difference q , then

$$q(\ell - 1) \leq N - 1, \quad c_m q^{-1/m} \leq \max_i \|q\theta_i\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{\varepsilon}{\ell - 1}.$$

Rearranging gives $\ell - 1 \leq (\varepsilon/c_m)^{m/(m+1)} (N - 1)^{1/(m+1)}$. This bounds every flat progression for that vector, hence every partition minimum. Singletons satisfy the same bound. The floor and additive one in the two estimates are absorbed into constants depending only on m , proving (I.18.8). $\square$

Remark I.18.6 (Scope of sharpness). This theorem proves sharpness for linear phases and uniform arc containment. It does not prove an optimal progression length for arbitrary density increments, which might be found without flattening a prescribed phase. It also gives a symmetric-width sharpness result, not a complete anisotropic lower-bound theory for independently varying ε_i .

[write] **6 October 2026.** At $m = 1$ source 01's Theorem I.7.2 gives the sharper bound $\ell \leq 1 + \sqrt{3\varepsilon(N-1)}$ with the golden-ratio frequency (note there).

I.19 Spectral energy, residual variance, and unequal tolerances

Source 05, delivered Section 5.

I.19.1 Why several correlations should be combined

A single Fourier coefficient gives an additive increment. A collection carrying energy comparable with δ^2 can instead give an increment comparable with δ . The useful issue here is not the existence of this classical principle, but how much phase precision it needs when the input is sparse.

Let $a : I_N \rightarrow [0, 1]$ have average $\delta \in (0, 1)$, set $f = a - \delta$ and $v = \delta(1 - \delta)$. Let $r_1, \dots, r_m$ be distinct nonzero frequencies, and write

$$\chi_i(x) = e(r_i x/N), \quad c_i = \widehat{f}(r_i), \quad E = \sum_i |c_i|^2 > 0, \quad \Sigma = \sum_i |c_i|. \quad (\text{I.19.1})$$

Zero coefficients may be removed. For a partition $\mathcal{P}$, put

$$a_P = \mathbb{E}(a \mid \mathcal{P}), \quad g = a_P - \delta, \quad V = \mathbb{E}g^2.$$

Orthogonality gives $E \leq \mathbb{E}|f|^2 \leq v$, and conditional expectation gives $0 \leq V \leq v$.

Lemma I.19.1 (Residual mass remembers the captured variance). *On every cell P , $\mathbb{E}_P|a - a_P| \leq 2a_P(1 - a_P)$. Consequently*

$$\mathbb{E}|a - a_P| \leq 2(v - V). \quad (\text{I.19.2})$$

For indicator functions, equality holds.

Proof. For $0 \leq u, b \leq 1$, $|u - b| \leq u + b - 2ub$. Average with $b = a_P = \mathbb{E}_P a$. Averaging over cells gives

$$\mathbb{E}a_P(1 - a_P) = \delta - \mathbb{E}a_P^2 = \delta - (\delta^2 + V) = v - V.$$

For $u \in \{0, 1\}$ the pointwise inequality is an equality. □

Theorem I.19.2 (Density-sensitive spectral extraction). *Suppose each χ_i is contained in an arc of width $\varepsilon_i < 1/2$ on every cell of $\mathcal{P}$, and set*

$$R = \sum_i |c_i| \sin(\pi \varepsilon_i). \quad (\text{I.19.3})$$

Then

$$E \leq \sqrt{EV} + 2R(v - V). \quad (\text{I.19.4})$$

If $R \leq E/(4v)$, then

$$V \geq V_0 := \frac{E}{(1 + \sqrt{1 - E/v})^2} \geq E/4, \quad (\text{I.19.5})$$

and some cell has average at least

$$\delta + V_0/\delta \geq \delta + E/(4\delta). \quad (\text{I.19.6})$$

Proof. Let $h = \sum_i c_i \chi_i$. With the normalized inner product $\langle u, v \rangle = \mathbb{E}(uv)$, orthogonality gives $\langle f, h \rangle = E$ and $\|h\|_2 = \sqrt{E}$. Decompose

$$E = \langle g, h \rangle + \langle a - a_P, h \rangle.$$

The first term has absolute value at most $\sqrt{VE}$. On each cell subtract from χ_i its disk center supplied by Lemma I.17.1. Since $a - a_P$ has zero mean on the cell, the subtraction changes no inner product. The residual term is therefore at most

$$\sum_i |c_i| \sin(\pi \varepsilon_i) \mathbb{E}|a - a_P| \leq 2R(v - V),$$

proving (I.19.4).

Assume $R \leq E/(4v)$. Since $v - V \geq 0$, divide the resulting inequality by E to obtain

$$1 \leq \sqrt{V/E} + \frac{1}{2} - \frac{V}{2v}.$$

With $x = \sqrt{V/E}$ and $\tau = E/v \in (0, 1]$, this is $\tau x^2 - 2x + 1 \leq 0$. Hence

$$x \geq \frac{1 - \sqrt{1 - \tau}}{\tau} = \frac{1}{1 + \sqrt{1 - \tau}},$$

which proves (I.19.5). Finally, if $D = \max_P a_P$, then $a_P^2 \leq D a_P$, and so $\delta^2 + V \leq D\delta$. This yields (I.19.6). $\square$

The conversion $D \geq \delta + V/\delta$ is sharp from the mean and variance alone: a random variable supported on 0 and D has variance $\delta(D - \delta)$. We do not claim that the complete phase-to-variance bound (I.19.5) is globally optimal.

I.19.2 A completely explicit progression corollary

Take equal widths

$$\varepsilon = \frac{E}{4\pi v \Sigma}. \quad (\text{I.19.7})$$

Since $E/\Sigma \leq \max_i |c_i| \leq \min(\delta, 1 - \delta)$, we have $\varepsilon \leq 1/(2\pi) < 1/2$. The inequality $\sin u \leq u$ gives $R \leq E/(4v)$. Apply Theorem I.18.2 with these widths and use the largest feasible L from (I.18.2). Then some ordinary arithmetic progression of length at least L has the increment (I.19.6). A simpler guaranteed choice is

$$L = \max \left(1, \left\lceil \left[\frac{N}{2^m} \left(\frac{E}{4\pi v \Sigma} \right)^m \right]^{1/(m+1)} \right\rceil \right). \quad (\text{I.19.8})$$

If $E \geq c\delta^2$, this gives an increment at least $c\delta/4$. Moreover $\Sigma \leq \sqrt{mE}$, so

$$\varepsilon \geq \frac{\sqrt{E}}{4\pi v\sqrt{m}} \geq \frac{\sqrt{c}}{4\pi(1-\delta)\sqrt{m}}.$$

Thus, under a fixed relative-energy hypothesis and fixed m , the required width need not shrink with δ . This is the density-sensitive feature lost by a generic unit-bound error estimate. It does not assert that every sparse progression-free set has such energy on a fixed-size spectrum.

I.19.3 Optimizing unequal phase widths

The sufficient linearized budget is

$$\sum_i b_i \varepsilon_i \leq B, \quad b_i = |c_i| > 0, \quad B = \frac{E}{4\pi v}. \quad (\text{I.19.9})$$

The guaranteed length in (I.18.3) increases with $\prod_i \varepsilon_i$. Fix a harmless cap $\varepsilon_i \leq 1/4$.

Proposition I.19.3 (Optimal allocation for the continuous bound). *If $B < \frac{1}{4} \sum_i b_i$, the product $\prod_i \varepsilon_i$ subject to (I.19.9) and $0 < \varepsilon_i \leq 1/4$ is maximized by*

$$\varepsilon_i = \min\left(\frac{1}{4}, \frac{\lambda}{b_i}\right), \quad \sum_i \min(b_i/4, \lambda) = B. \quad (\text{I.19.10})$$

If $B \geq \frac{1}{4} \sum_i b_i$, use all widths equal to $1/4$. When no cap is active,

$$\varepsilon_i = \frac{E}{4\pi m v |c_i|}, \quad \prod_i \varepsilon_i = \left(\frac{E}{4\pi m v}\right)^m \prod_i |c_i|^{-1}. \quad (\text{I.19.11})$$

Proof. Put $x_i = b_i \varepsilon_i$. Apart from a constant, the logarithm of the objective is $\sum_i \log x_i$. At a maximum with unused capacity excluded, $\sum_i x_i = B$. If two uncapped coordinates are unequal, moving a sufficiently small amount from the larger to the smaller strictly increases their product, preserves their sum, and respects both caps. Thus all uncapped x_i equal a common λ . A capped coordinate cannot exceed λ , since a small transfer from it to an uncapped coordinate would again improve the product. Consequently $x_i = \min(b_i/4, \lambda)$. Strict concavity of $\sum_i \log x_i$ makes these conditions sufficient as well; monotonicity of the displayed sum gives the unique relevant level. The uncapped formula follows immediately. $\square$

By the arithmetic–geometric mean inequality, the uncapped product improves the equal-width product by the factor

$$\frac{(\Sigma/m)^m}{\prod_i |c_i|} \geq 1.$$

This optimizes the continuous product bound, *not* the discontinuous exact ceiling problem (I.18.2). For finite implementations, evaluate the exact feasibility test after allocation; neighboring rational widths can sometimes do better. Also, adding frequencies increases energy but increases dimension. Choosing the best subset of frequencies remains a separate optimization problem.

I.19.4 An explicit extension to nonorthogonal phases

The orthogonality in Theorem I.19.2 is essential to its constants, but the argument admits a controlled extension. Let $\psi_1, \dots, \psi_m$ be arbitrary functions of modulus one on the finite probability space. Assume that $\kappa > 0$ satisfies

$$\left\| \sum_i z_i \psi_i \right\|_2^2 \leq \kappa \sum_i |z_i|^2 \quad \text{for every } (z_1, \dots, z_m) \in \mathbb{C}^m. \quad (\text{I.19.12})$$

Thus κ is a bound for the largest eigenvalue of the Gram matrix. One may always use $\kappa = m$, though a smaller certified bound is preferable. Set $c_i = \langle f, \psi_i \rangle$ and $E = \sum_i |c_i|^2 > 0$; retain v , V , and the arc-width budget R as before.

Corollary I.19.4 (Gram-controlled spectral extraction). *Under these hypotheses,*

$$E \leq \sqrt{\kappa E V} + 2R(v - V), \quad E \leq \kappa v. \quad (\text{I.19.13})$$

If $R \leq E/(4v)$, some cell has average at least

$$\delta + \frac{E}{\kappa \delta (1 + \sqrt{1 - E/(\kappa v)})^2} \geq \delta + \frac{E}{4\kappa \delta}. \quad (\text{I.19.14})$$

Proof. For $h = \sum_i c_i \psi_i$, the identity $\langle f, h \rangle = E$ still holds without orthogonality, while (I.19.12) gives $\|h\|_2 \leq \sqrt{\kappa E}$. Cauchy-Schwarz first yields $E^2 \leq \|f\|_2^2 \|h\|_2^2 \leq v \kappa E$, proving $E \leq \kappa v$. The same cellwise subtraction used in Theorem I.19.2 proves the first inequality in (I.19.13). Under the stated budget, put $x = \sqrt{\kappa V/E}$ and $\tau = E/(\kappa v)$. The resulting inequality is again $\tau x^2 - 2x + 1 \leq 0$. It gives $V \geq (E/\kappa)/(1 + \sqrt{1 - \tau})^2$. Apply $\max_P a_P \geq \delta + V/\delta$. $\square$

This includes polynomial phases and interval-restricted characters once their arc-flat partitions and a Gram bound are supplied. It does not supply either the required higher-degree partition or a useful small value of κ automatically.

I.20 Average-length partitions and exceptional-cell budgets

Source 05, delivered Section 6.

Higher-order localization statements sometimes control the number of cells rather than the minimum cell length. The following elementary interface keeps that distinction explicit and avoids unnecessarily replacing an average length by its square root.

Proposition I.20.1 (Discarding short cells quantitatively). *Let $a : I_N \rightarrow [0, 1]$ have average $\delta \in (0, 1)$, and let a partition with at most J cells satisfy*

$$\mathbb{E} |\mathbb{E}(a - \delta \mid \mathcal{P})| \geq 2s > 0.$$

For any real threshold $T > 0$ with $(1 - \delta)JT/N < s$, some cell of length at least T has average at least

$$\delta + s - (1 - \delta)JT/N. \quad (\text{I.20.1})$$

In particular, some cell has length at least

$$T = \frac{sN}{2(1 - \delta)J} \quad \text{and average at least } \delta + s/2. \quad (\text{I.20.2})$$

Proof. The total positive block mass is at least s . Cells of length less than T have total normalized size at most JT/N and therefore contribute at most $(1 - \delta)JT/N$ to that positive mass. On the remaining cells the positive mass is at least the difference in (I.20.1). Since their total normalized size is at most one, one remaining cell has positive average at least that difference. $\square$

If $N/J \geq N^\gamma$, this preserves length of order sN^γ , not merely $N^{\gamma/2}$, under the specified correlation hypothesis. This is an extraction statement only. It does not justify replacing short-cell steps inside a geometric or nilpotent factorization proof, where those steps may be needed for reasons other than density extraction.

More generally, Proposition I.17.2 and Lemma I.17.3 apply to any circle-valued phase once a valid arc-flat partition is supplied. The spectral extension to an arbitrary family requires the Gram correction in Corollary I.19.4, or an orthonormal family for the uncorrected theorem. The hard issue for higher degree is supplying the appropriate partition and controlling the Gram matrix; Theorem I.18.5 cannot simply be transferred to nonlinear phases.

I.21 Sharp density extraction from a partition

Source 06, delivered Section 5.

The last step of a partition argument converts a large sum of absolute cell integrals into one cell that is both large and positively biased. Gowers' Lemma 5.15 performs this conversion by discarding cells with a small average and then cells with a small size [1]. The following result determines the exact tradeoff when the total positive and negative integrals are retained. It applies to weighted partitions, asymmetric bounds on the function, and restrictions whose mean is no longer zero.

Let a finite measure space of total mass $W > 0$ be partitioned into M nonnull cells $P_1, \dots, P_M$. Write

$$w_j = \mu(P_j), \quad x_j = \frac{1}{w_j} \int_{P_j} f d\mu, \quad -a \leq x_j \leq b,$$

where $a, b > 0$. Define the signed integral, cell variation, and its positive and negative parts by

$$m = \sum_j w_j x_j, \quad v = \sum_j w_j |x_j|, \quad P_+ = \frac{v + m}{2}, \quad P_- = \frac{v - m}{2}. \quad (\text{I.21.1})$$

The quantity v is the variation of the *cell integrals*; it need not equal $\int |f|$. Feasibility of these data implies

$$\frac{P_+}{b} + \frac{P_-}{a} \leq W. \quad (\text{I.21.2})$$

Theorem I.21.1 (Exact extraction tradeoff). *Suppose $P_+, P_- > 0$. For $0 \leq \eta < b$, set*

$$S = \frac{P_+ - \eta(W - P_-/a)}{b - \eta}. \quad (\text{I.21.3})$$

If $S > 0$, then the cells with average strictly greater than η have total mass at least S :

$$\sum_{x_j > \eta} w_j \geq S. \quad (\text{I.21.4})$$

For $M \geq 3$, at least one such cell satisfies

$$w_j \geq \min \left\{ \frac{P_+}{b(M-1)}, \frac{S}{M-2} \right\}. \quad (\text{I.21.5})$$

For $M = 2$, the corresponding bound is $w_j \geq P_+/b$.

For fixed feasible data $W, a, b, P_+, P_-, \eta, M$, the right-hand side of (I.21.5) is the exact best universal lower bound on the largest mass of a cell with $x_j > \eta$.

Proof. Let $H = \{j : x_j > \eta\}$ and $s = \sum_{j \in H} w_j$. The negative cells have total mass at least P_-/a . The positive cells outside H have average at most η , so

$$P_+ \leq bs + \eta \left(W - \frac{P_-}{a} - s \right).$$

Rearrangement proves (I.21.4). Since $P_- > 0$, there is a negative cell, and consequently $|H| \leq M - 1$. If $|H| \leq M - 2$, the mass bound gives a high cell of mass at least $S/(M - 2)$. If $|H| = M - 1$, the single remaining cell is negative. In this case all positive integral lies in H , and

$$P_+ = \sum_{j \in H} w_j x_j \leq b \sum_{j \in H} w_j.$$

One of these $M - 1$ cells has mass at least $P_+/[b(M - 1)]$. The same argument proves the assertion for $M = 2$.

For sharpness, it suffices to construct weighted atomic spaces. For the first branch of (I.21.5), take $M - 1$ cells, each of mass $P_+/[b(M - 1)]$ and average b . Give the remaining cell mass $W - P_+/b$ and average

$$-\frac{P_-}{W - P_+/b}.$$

Condition (I.21.2) places this average in $[-a, 0)$. The data are exactly the prescribed ones, and the largest high cell has the first stated mass.

For the second branch, take $M - 2$ cells, each of mass $S/(M - 2)$ and average b . Add a cell of average exactly η and mass

$$\ell = W - P_-/a - S,$$

and a cell of average $-a$ and mass P_-/a . The definition of S gives

$$bS + \eta\ell = P_+,$$

and (I.21.2) gives $\ell \geq 0$. If the second branch is strictly smaller than the first, then $\ell > 0$: otherwise $S = P_+/b$, which makes the second branch larger. Thus the minimizing construction has M nonnull cells and attains exactly $S/(M - 2)$. At a tie the first construction suffices. This proves the asserted optimality, including the nonnull-cell requirement. $\square$

The strict inequality $x_j > \eta$ is useful here: the intermediate cell in the second extremizer has average exactly η . The theorem therefore has exact extremizers, rather than only limiting ones. If only a lower bound on v is available while m is known, it can be substituted in the formulas, because both terms in (I.21.5) increase with v .

[write] **6 October 2026; sources 06–10.** *Credits.* The mass bound (I.21.4) is the sharp one-sided bound of sources 01, 02, 03, 05 and 10 in mass form (note after Theorem I.4.1); the fixed- M optimum (I.21.5) with its extremizers is proved only here. The two extremizers were rechecked at the second write: in the first, $(M - 1)$ cells at b of mass $P_+/(b(M - 1))$ carry P_+ and the last cell carries $-P_-$ on mass $W - P_+/b$; in the second, $bS + \eta\ell = P_+$ with $\ell = W - P_-/a - S \geq 0$ by (I.21.2).

I.21.1 Balanced functions and indicator functions

For a probability space, a balanced function has $W = 1$ and $m = 0$. If its cell variation is at least α , then $P_+, P_- \geq \alpha/2$, and (I.21.3) gives

$$S \geq \frac{\alpha(a + \eta) - 2a\eta}{2a(b - \eta)}. \quad (\text{I.21.6})$$

The numerator is positive precisely when

$$\eta < \frac{a\alpha}{2a - \alpha}.$$

The denominator $2a - \alpha$ is positive, since feasibility implies $\alpha \leq 2ab/(a + b) < 2a$.

Corollary I.21.2 (An explicit strengthening of Lemma 5.15). *Let f take values in $[-1, 1]$ on an N -point set with $N \geq 1$, suppose $\sum_x f(x) = 0$, and let $P_1, \dots, P_M$ be a partition into nonempty cells with*

$$\sum_{j=1}^M \left| \sum_{x \in P_j} f(x) \right| \geq \alpha N, \quad 0 < \alpha \leq 1.$$

These assumptions force $M \geq 2$. If $M \geq 3$, there is a cell such that

$$\sum_{x \in P_j} f(x) > \frac{\alpha}{4} |P_j|, \quad |P_j| \geq N \min \left\{ \frac{\alpha}{2(M-1)}, \frac{\alpha(2+\alpha)}{2(4-\alpha)(M-2)} \right\}. \quad (\text{I.21.7})$$

For $M = 2$, the size bound is $|P_j| \geq \alpha N/2$.

Proof. Use normalized counting measure, $a = b = 1$, and $\eta = \alpha/4$. Formula (I.21.6) becomes

$$S \geq S_\alpha := \frac{\alpha(2+\alpha)}{2(4-\alpha)}.$$

Theorem I.21.1 now gives the result. □

Each entry in the minimum in (I.21.7) is strictly larger than $\alpha/(4M)$, the size guarantee in Gowers' lemma. There is also a simple way to improve both parameters together.

Corollary I.21.3 (Simultaneous improvement). *Under the hypotheses of Corollary I.21.2, define*

$$\tau(\alpha) = 1 - \frac{\alpha}{4} - \sqrt{1 - \alpha + \frac{\alpha^2}{16}}. \quad (\text{I.21.8})$$

Then

$$\frac{\alpha}{4} < \tau(\alpha) \leq \frac{\alpha}{2},$$

and some cell satisfies

$$\sum_{x \in P_j} f(x) > \tau(\alpha) |P_j|, \quad |P_j| \geq \frac{\tau(\alpha)}{M-1} N.$$

Moreover,

$$\tau(\alpha) = \frac{\alpha}{4} + \frac{3\alpha^2}{32} + O(\alpha^3) \quad (\alpha \rightarrow 0).$$

Proof. Put $P = \alpha/2$. The equation that makes the lower bound in (I.21.6) equal to its threshold is

$$\frac{P(1+\eta) - \eta}{1 - \eta} = \eta, \quad \text{or} \quad \eta^2 + (P - 2)\eta + P = 0.$$

Its smaller root is (I.21.8). The displayed interval for this root follows either by substitution at $\eta = \alpha/4$ and $\eta = \alpha/2$, or directly by squaring the nonnegative terms in (I.21.8). Thus the high cells have total mass at least $\tau(\alpha)$, and there are at most $M - 1$ of them. The expansion follows by the Taylor expansion of the square root. $\square$

[write] **6 October 2026; sources 06–10.** Corollary I.21.2 is compared with the $M - 1$ corollary of sources 02 and 03 in the note after Corollary I.9.3: it is at least as strong, and strictly stronger for $\alpha < 1$. Against the formal `lemma_5_15` (exact companion `lemma_5_15_holds`): that statement allows $\alpha \geq 0$ and empty cells and concludes $\alpha|P_j|/4 \leq \sum_{P_j} f$ with $|P_j| \geq \alpha N/(4M)$; the corollaries here need $\alpha > 0$ and nonempty cells. Not formalized.

For the balanced indicator $f = 1_A - \delta$ with $0 < \delta < 1$, the appropriate range is $[-\delta, 1 - \delta]$, rather than $[-1, 1]$. Keeping this asymmetry gives

$$S \geq \frac{\alpha(\delta + \eta) - 2\delta\eta}{2\delta(1 - \delta - \eta)}, \quad 0 \leq \eta < 1 - \delta. \quad (\text{I.21.9})$$

Here necessarily $\alpha \leq 2\delta(1 - \delta)$. At the maximal variation $\alpha = 2\delta(1 - \delta)$, the right-hand side of (I.21.9) equals δ for every $\eta < 1 - \delta$. This is exact: equality in the feasibility inequality forces every positive cell to have average $1 - \delta$ and every negative cell to have average $-\delta$.

The weighted sharpness examples also have an interpretation for actual subsets. For rational parameters with $a = \delta$ and $b = 1 - \delta$, replace each weighted cell by sufficiently many points and put a proportion $\delta + x_j$ of its points in A . The signed-mean condition makes the global density exactly δ . Thus the extremizers can be realized on finite uniform spaces whenever the parameters are rational; approximation gives the corresponding sharpness statement for real parameters.

I.21.2 Exceptional sets and signed drift

Restrictions of a balanced function usually cease to be balanced. The exact signed mean in (I.21.1) records this loss without inserting an additional unspecified error term.

Proposition I.21.4 (Extraction after discarding an exceptional set). *Suppose $\mu(\Omega) = 1$, $\int_{\Omega} f d\mu = 0$, and $-a \leq f \leq b$. Let the cells partition a subset of mass $1 - \rho$, and suppose their cell variation is at least α . For $0 \leq \eta \leq a$ and $\eta < b$, their high-cell mass satisfies*

$$\sum_{x_j > \eta} w_j \geq \frac{\alpha(a + \eta) - b\rho(a - \eta) - 2a\eta(1 - \rho)}{2a(b - \eta)}. \quad (\text{I.21.10})$$

More generally, a partition of mass W with $|m| \leq \varepsilon$ and $v \geq \alpha$ satisfies

$$\sum_{x_j > \eta} w_j \geq \frac{\alpha(a + \eta) - \varepsilon(a - \eta) - 2a\eta W}{2a(b - \eta)}. \quad (\text{I.21.11})$$

Proof. The argument proving (I.21.4) remains valid when one of P_+, P_- is zero. Rewriting its right-hand side in terms of v, m gives

$$S = \frac{v(a + \eta) + m(a - \eta) - 2a\eta W}{2a(b - \eta)}.$$

For the exceptional-set statement, the omitted integral belongs to $[-a\rho, b\rho]$, so its negative, the retained mean, belongs to $[-b\rho, a\rho]$. Since $a - \eta \geq 0$, substitution of $v \geq \alpha$, $m \geq -b\rho$, and $W = 1 - \rho$ proves (I.21.10). Substitution of $m \geq -\varepsilon$ proves (I.21.11). $\square$

Whenever either lower bound is positive, division by M gives a guaranteed cell size. If the observed positive and negative integrals are both nonzero, Theorem I.21.1 gives the stronger finite- M bound using their actual values. For example, $\alpha > \max(a, b)\rho$ guarantees both signs in the exceptional-set situation. These statements allow every discarded part of a structural argument to enter an explicit density budget.

[write] **6 October 2026; sources 06–10.** Exceptional sets are also treated by source 01’s Theorem I.2.3 (charging only the occupied mass) and source 10’s Theorem I.23.4; see the note after source 01’s theorem.

I.22 Conventions and normalization

Source 10, delivered Section 2.

All group averages in this report are normalized: $\mathbb{E}_{x \in G} f(x) = |G|^{-1} \sum_{x \in G} f(x)$. For a character χ of a finite abelian group G , write

$$\hat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}, \quad \|f\|_{L^p} = (\mathbb{E}|f|^p)^{1/p}.$$

The sum over the dual group in Parseval is an ordinary sum, so $\sum_{\chi} |\hat{f}(\chi)|^2 = \mathbb{E}|f|^2$. If $A \subset G$ has density δ , its balanced function is $b_A = \mathbf{1}_A - \delta$. We use both indicator sets and fractional sets $a : G \rightarrow [0, 1]$, with the latter explicitly identified.

For $s \geq 1$ define the normalized Gowers cube power by

$$\|f\|_{U^s}^{2^s} = \mathbb{E}_{x, h_1, \dots, h_s} \prod_{\omega \in \{0,1\}^s} \mathcal{C}^{|\omega|} f(x + \omega \cdot h),$$

where $\mathcal{C}$ is complex conjugation. This power is real and nonnegative: averaging one direction turns the remaining cube product into an absolute square. Thus the displayed expression defines its nonnegative 2^s -th root. In particular $\|f\|_{U^1} = |\mathbb{E}f|$ and $\|f\|_{U^2}^4 = \sum_{\chi} |\hat{f}(\chi)|^4$.

The source uses unnormalized Fourier transforms and a degree-indexed uniformity parameter. The conversion is as follows.

Source convention	Convention in this report
$\hat{f}_{\text{source}}(r)$	$N \hat{f}(r)$
Unnormalized k -AP count	$N^2 \Lambda_k$
u -uniform of degree d	$\ f\ _{U^{d+1}}^{2^{d+1}} \leq u$
Source gowersNorm at index $s \geq 1$	$N^{(s+1)/2^s} \ f\ _{U^s}$

Changing the sign of all difference variables changes neither cube averages nor norms. This reconciles the source’s $x - h$ convention with our $x + h$ convention.

A *proper* k -term modular progression here means k distinct points $x, x + d, \dots, x + (k - 1)d$. In a general finite group, merely requiring $d \neq 0$ is insufficient. An *ordinary* progression inside $[N]_0 = \{0, \dots, N - 1\}$ is an integer progression contained in that interval; the phase-partition results

produce such progressions directly. Finally, a Freiman homomorphism of order k is a map that preserves every equality between two sums of k domain elements, with repetitions allowed.

Two notational cautions prevent false gains. A claim about the average size of partition cells does not imply a lower bound for every cell, and a modular progression with distinct points is not automatically a nonwrapping integer progression. The proofs below specify the needed property each time.

I.23 Sharp density profiles on a partition

Source 10, delivered Section 3.

Let (X, μ) be a finite probability space, let $a : X \rightarrow [0, 1]$, and put $\delta = \mathbb{E}a \in (0, 1)$ and $v = \delta(1 - \delta)$. A partition $\mathcal{P}$ into cells of positive measure has cell densities $a_P = \mathbb{E}(a \mid P)$ and weights $w_P = \mu(P)$. Set

$$S_{\mathcal{P}} = \sum_{P \in \mathcal{P}} w_P |a_P - \delta|, \quad V_{\mathcal{P}} = \sum_{P \in \mathcal{P}} w_P (a_P - \delta)^2.$$

These are the first and second moments of the conditional density fluctuation. They retain information discarded by a single pigeonhole estimate.

Theorem I.23.1 (Optimal conversion of partition discrepancy). *If $M = \max_{P \in \mathcal{P}} a_P$, then*

$$S_{\mathcal{P}} \leq \frac{2\delta(M - \delta)}{M}, \tag{I.23.1}$$

$$M \geq \delta + \frac{\delta S_{\mathcal{P}}}{2\delta - S_{\mathcal{P}}}, \tag{I.23.2}$$

$$M \geq \delta + \frac{V_{\mathcal{P}}}{\delta}. \tag{I.23.3}$$

All three inequalities are sharp. Apart from zero-weight cells, equality in either moment upper bound occurs precisely when every a_P is 0 or M .

Proof. Since the mean of the a_P is δ , we have $M \geq \delta > 0$. Convexity, or a direct check on $[0, \delta]$ and $[\delta, M]$, gives

$$|u - \delta| \leq \delta + \left(1 - \frac{2\delta}{M}\right)u \quad (0 \leq u \leq M).$$

Averaging gives (I.23.1). Also $S_{\mathcal{P}} \leq 2\delta(1 - \delta) < 2\delta$, so rearrangement is legitimate and yields (I.23.2). The inequality $u^2 \leq Mu$ gives $V_{\mathcal{P}} \leq M\delta - \delta^2$, proving (I.23.3). Both pointwise inequalities are strict in the interior of $(0, M)$, except for the harmless degenerate case $M = \delta$. In that case equality forces $u = \delta$ almost surely. A distribution giving M weight δ/M and 0 weight $1 - \delta/M$ realizes equality. For rational data this is a literal finite weighted example, and rational approximations give uniform finite examples arbitrarily close to equality. $\square$

The second-moment bound is the elementary bounded-variable variance inequality. The contribution here is its use together with the exact first-moment conversion and the geometric bounds below; neither the variance inequality nor conditional expectation is claimed as a new method. The usual estimate $M \geq \delta + S_{\mathcal{P}}/2$ loses the factor $1/(1 - S_{\mathcal{P}}/(2\delta))$.

[write] **6 October 2026; sources 06–10.** (I.23.2) is the forced maximum of the sharp one-sided bound (source 01's (I.4.3) with $D = S_{\mathcal{P}}$; source 02's and source 03's $aT/(a - T)$), and Proposition I.23.2 below is that bound itself at $a = \delta$, $b = 1 - \delta$; six sources prove it (note after Theorem I.4.1). (I.23.3) is only here.

Proposition I.23.2 (Mass of cells exceeding a density threshold). *For $\delta < \tau < 1$, let $q_\tau = \sum_{P: a_P > \tau} w_P$. Then*

$$q_\tau \geq \max \left\{ 0, \frac{\tau S_{\mathcal{P}} - 2\delta(\tau - \delta)}{2\delta(1 - \tau)} \right\}.$$

Whenever the right-hand side is positive, its dependence on $\delta, \tau, S_{\mathcal{P}}$ is sharp.

Proof. For $u \in [0, 1]$,

$$|u - \delta| \leq \delta + \left(1 - \frac{2\delta}{\tau}\right)u + \frac{2\delta(1 - \tau)}{\tau} \mathbf{1}_{\{u > \tau\}}.$$

For $u \leq \tau$ this is the chord bound on $[0, \tau]$; for $u > \tau$ the difference between the left side and the first two terms is $2\delta(u/\tau - 1)$, at most $2\delta(1 - \tau)/\tau$. Averaging and rearranging proves the claim. Equality is attained by distributions supported on $\{0, \tau, 1\}$: give 1 weight q , τ weight $(\delta - q)/\tau$, and 0 the remaining weight. As $0 \leq q \leq \delta$ varies, these realize the entire positive-bound range. $\square$

I.23.1 Phase approximation with a density-sensitive error budget

Write $f = a - \delta$. Its mean is zero, and

$$\mathbb{E}|f| \leq 2v, \quad \mathbb{E}|f|^2 \leq v. \quad (\text{I.23.4})$$

Both are equalities for an indicator $a = \mathbf{1}_A$. The first follows from the chord inequality $|u - \delta| \leq \delta + (1 - 2\delta)u$ on $[0, 1]$; the second follows from $u^2 \leq u$.

Theorem I.23.3 (Correlation-to-density transfer). *Let $\phi : X \rightarrow \mathbb{C}$ be 1-bounded, and suppose $|\mathbb{E}f\phi| \geq \eta > 0$. Suppose that for each $P \in \mathcal{P}$ there is z_P in the closed unit disc such that $|\phi(x) - z_P| \leq \rho$ for $x \in P$. If $s = \eta - 2v\rho > 0$, then some cell satisfies*

$$a_P \geq \delta + \frac{\delta s}{2\delta - s}.$$

The tail bound of Proposition I.23.2 also holds with $S_{\mathcal{P}}$ replaced by s .

Proof. The triangle inequality and (I.23.4) give

$$\eta \leq \left| \sum_P z_P \int_P f d\mu \right| + \rho \mathbb{E}|f| \leq S_{\mathcal{P}} + 2v\rho.$$

Apply Theorem I.23.1 and Proposition I.23.2. $\square$

This is valid for arbitrary bounded phases, arbitrary finite weights, and fractional sets $a \in [0, 1]$. Its improvement over a crude 1-bounded argument is substantial at small density: a phase error of size ρ costs at most $2\delta(1 - \delta)\rho$, rather than ρ .

Theorem I.23.4 (Allowance for an exceptional set). *In Theorem I.23.3, suppose instead that the cells partition $X \setminus E$, where $\mu(E) = \varepsilon < \delta$, and that the phase approximation holds only there. Put $s = \eta - 2v\rho$ and assume $s > 2(1 - \delta)\varepsilon$. Then a retained cell satisfies*

$$a_P \geq \delta + \frac{\delta(s - 2(1 - \delta)\varepsilon)}{2\delta(1 - \varepsilon) - s}. \quad (\text{I.23.5})$$

Proof. Let $b = \int_E a \, d\mu \leq \varepsilon$. The chord inequality gives $\int_E |f| \, d\mu \leq \delta\varepsilon + (1 - 2\delta)b$. Writing S for the unnormalized discrepancy over retained cells, the correlation estimate gives $s \leq S + \int_E |f| \, d\mu$. If all retained densities were at most δ , the right side would be at most $2(1 - \delta)b$, contradicting the hypothesis. Thus their maximum M exceeds δ . The chord bound on $[0, M]$, now with retained mass $1 - \varepsilon$ and retained a -mass $\delta - b$, gives

$$s \leq 2\delta(1 - b) - \frac{2\delta(\delta - b)}{M}.$$

Consequently

$$M \geq \frac{2\delta(\delta - b)}{2\delta(1 - b) - s} \geq \frac{2\delta(\delta - \varepsilon)}{2\delta(1 - \varepsilon) - s}.$$

The first fraction is nonincreasing in b , because its derivative has the sign of $s - 2v \leq 0$. All denominators are positive: $s \leq 2v$ and $\varepsilon < \delta$. Subtracting δ proves the formula. $\square$

The exceptional-set formula makes the usable error budget explicit. An exceptional proportion ε can consume all guaranteed positive imbalance unless the residual correlation exceeds $2(1 - \delta)\varepsilon$. The formula concerns the transfer step; it does not assert optimality of a preceding geometric construction.

[write] **6 October 2026; sources 06–10.** Compare source 01's Theorem I.2.3, which charges only the occupied mass of the exceptional set, and source 06's Proposition I.21.4; the three are different models (note after source 01's theorem).

I.24 A constructive Fourier-to-progression tradeoff

Source 10, delivered Section 4.

Write $[N]_0 = \{0, \dots, N - 1\}$ and $e(t) = \exp(2\pi it)$. The next partition lemma avoids a large-scale side condition and keeps all integer rounding visible.

Lemma I.24.1 (An explicit phase partition). *Let $N \geq 1$, $\theta \in \mathbb{R}$, and $0 < \lambda \leq \pi/2$. Define*

$$L = \max \left\{ 1, \left\lfloor \sqrt{\frac{N\lambda}{2\pi}} \right\rfloor \right\}.$$

There is a partition of $[N]_0$ into ordinary integer arithmetic progressions, each of length between L and $2L - 1$, such that on every cell P the phase $e(\theta x)$ lies within distance $\sin \lambda$ of a complex number z_P with $|z_P| \leq 1$.

Proof. For $L = 1$, take singletons. Otherwise set $Q = \lfloor N/L \rfloor$. Among the $Q + 1$ points $0, \theta, \dots, Q\theta$ on $\mathbb{R}/\mathbb{Z}$, two have circular distance at most $1/(Q + 1)$: sort them around the circle and use the smallest circular gap, treating repeated points separately. Thus there is $1 \leq q \leq Q$ with $\|q\theta\|_{\mathbb{R}/\mathbb{Z}} \leq 1/(Q + 1)$. Each residue class modulo q inside $[N]_0$ has at least $\lfloor N/q \rfloor \geq L$ points. A string of $n \geq L$ points can be cut into blocks of sizes between L and $2L - 1$: write $n = aL + b$ with $a \geq 1$ and $0 \leq b < L$, use one block of length $L + b$, and $a - 1$ blocks of length L .

Choose a signed representative t of $q\theta$ with $|t| \leq 1/(Q + 1)$. On a block of $m \leq 2L - 1$ consecutive points of a residue class, the phase lies on an arc centered at its midpoint phase u_P , with half-angle

$$H = \pi(m - 1)|t| \leq \frac{2\pi(L - 1)}{Q + 1} < \frac{2\pi L^2}{N} \leq \lambda.$$

Set $z_P = (\cos H)u_P$. For a point on the arc making angle h with u_P , where $|h| \leq H \leq \pi/2$,

$$|e^{ih} - \cos H|^2 = 1 + \cos^2 H - 2 \cos H \cos h \leq \sin^2 H.$$

This proves the claimed disc approximation and all the length bounds. $\square$

Theorem I.24.2 (A one-parameter density–length frontier). *Let $a : [N]_0 \rightarrow [0, 1]$ have mean $\delta \in (0, 1)$, put $v = \delta(1 - \delta)$, and suppose*

$$\left| \frac{1}{N} \sum_{x=0}^{N-1} (a(x) - \delta)e(\theta x) \right| \geq \eta > 0.$$

For every $0 < \gamma < 1$, set

$$\begin{aligned} \lambda_\gamma &= \arcsin\left(\frac{(1 - \gamma)\eta}{2v}\right), \\ L_\gamma &= \max\left\{1, \left\lfloor \sqrt{\frac{N\lambda_\gamma}{2\pi}} \right\rfloor\right\}, \\ D_\gamma &= \frac{\delta\gamma\eta}{2\delta - \gamma\eta}. \end{aligned}$$

There exists an ordinary arithmetic progression $P \subset [N]_0$ with

$$L_\gamma \leq |P| \leq 2L_\gamma - 1, \quad \frac{1}{|P|} \sum_{x \in P} a(x) \geq \delta + D_\gamma.$$

Moreover, the cells of the phase partition whose density exceeds any $\tau \in (\delta, \delta + D_\gamma)$ cover at least

$$N \frac{\tau\gamma\eta - 2\delta(\tau - \delta)}{2\delta(1 - \tau)}$$

points. Their number is at least this quantity divided by $2L_\gamma - 1$.

Proof. By (I.23.4), $\eta \leq 2v$, so the arcsine is defined, strictly positive, and less than $\pi/2$. Apply Lemma I.24.1 with $\lambda = \lambda_\gamma$ and Theorem I.23.3 with $\rho = \sin \lambda_\gamma = (1 - \gamma)\eta/(2v)$. Then $s = \gamma\eta$. Proposition I.23.2 gives the final assertion. $\square$

Corollary I.24.3 (A convenient balanced choice). *If $A \subset \mathbb{Z}/N\mathbb{Z}$ has density $\delta \in (0, 1)$ and some nontrivial normalized Fourier coefficient has modulus at least $\eta > 0$, then A , represented inside $[N]_0$, has a progression with density at least*

$$\delta + \frac{\delta\eta}{4\delta - \eta}$$

and length at least

$$\max\left\{1, \left\lfloor \sqrt{\frac{N}{2\pi} \arcsin\left(\frac{\eta}{4\delta(1 - \delta)}\right)} \right\rfloor\right\}. \quad (\text{I.24.1})$$

No primality assumption on N is required.

The corresponding intermediate estimate already in the repository’s proof of Corollary 2.5 is length at least $(\eta N/(128\pi))^{1/2}$ and increment $\eta/4$, in its stated large-scale range. Thus merely replacing the printed η^3 by η would not be a new improvement relative to the supplied version. Our additional gain is the explicit density factor, the optimized cap conversion, improved geometric constants, and the unrestricted finite-scale statement. For $\eta/v \rightarrow 0$, the continuous expression underlying (I.24.1) is

$$\sqrt{\frac{\eta N}{8\pi v}} \left(1 + O((\eta/v)^2)\right).$$

Its ratio to the repository’s continuous lower bound is $(4/\sqrt{v})(1 + O((\eta/v)^2))$; floors may matter at small N . The density gain is $\eta/[4(1 - \eta/(4\delta))] \geq \eta/4$.

I.24.1 Numerical scale of the guarantees

Figure I.24.1 evaluates the explicit family in Theorem I.24.2. Increasing the retained correlation fraction γ increases the guaranteed density increment and reduces the minimum length supplied by this partition construction. The curve is a family of sufficient guarantees; optimality of its geometric constants is not asserted.

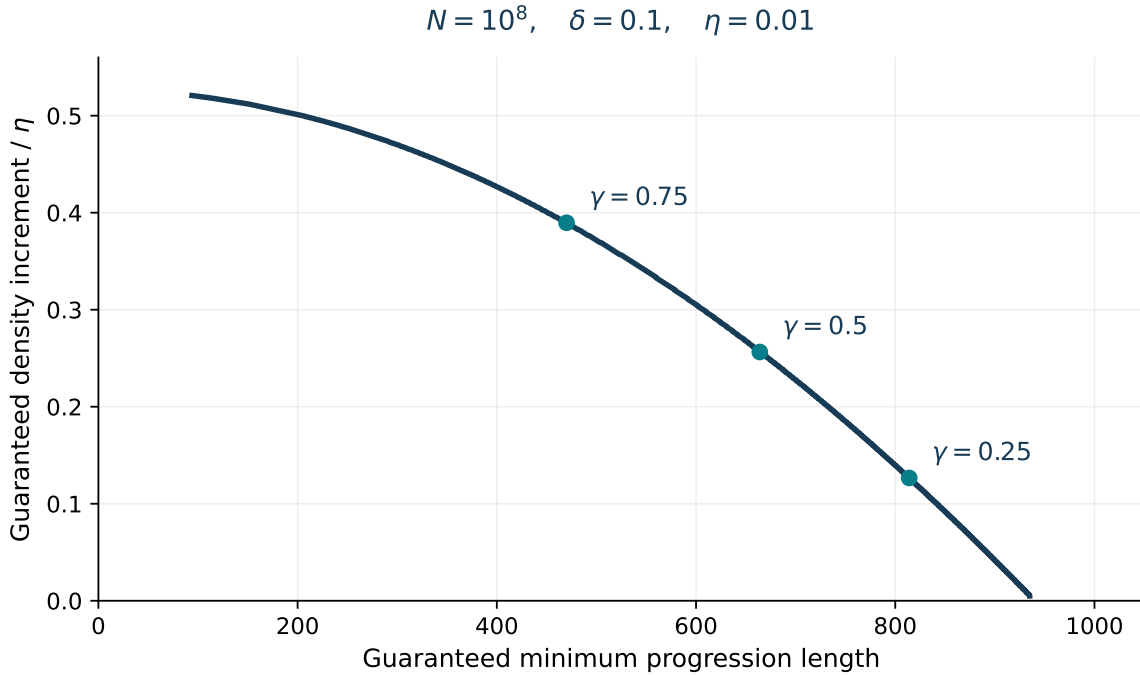


Figure I.24.1: The proved density–length tradeoff, evaluated at fixed $N = 10^8$, $\delta = 0.1$, and normalized correlation $\eta = 0.01$. The vertical axis divides the guaranteed increment by η . These points are evaluations of formulas, conditional on the correlation hypothesis, rather than measurements of extremal sets.

For a concrete comparison, the following table uses $\gamma = 1/2$ and $N = 10^8$ throughout. The older length column evaluates the continuous lower bound already present in the supplied proof; the new length column includes the floor in (I.24.1). All three parameter choices satisfy the older large-scale condition.

δ	η	Older length	New length	Older gain	New gain
0.5	0.05	111.51	892	0.0125	0.0128205
0.1	0.01	49.87	664	0.0025	0.00256410
0.01	0.0001	4.99	200	0.000025	0.0000250627

The improvement in the length guarantee is especially visible at low density because the phase approximation error is charged to $\mathbb{E}|\mathbf{1}_A - \delta| = 2\delta(1 - \delta)$. Each entry is conditional on the presence of the specified Fourier coefficient; the table does not claim that an arbitrary set of the given density has such a coefficient.

[write] **6 October 2026; sources 06–10.** Corollary I.24.3 has exactly source 01’s length (I.6.1) and source 05’s increment $\delta\eta/(4\delta - \eta)$ (Theorem I.17.4), so source 01’s Theorem I.6.1 gives an increment at least $3/2$ times as large at the same length; along the frontier of Theorem I.24.2 the comparison holds for $\gamma \leq 1/2$ and changes at $\gamma = 3/(4 - \kappa)$ (proof in the note after Table 3). The repository’s proof-level pair of Corollary 2.5 is `cor25_large_scale` (`Proofs02DensityIncrement.lean`); not formalized.

I.24.2 Using the exact variance in the U^2 inverse step

Proposition I.24.4. *Let A be a set of density $\delta \in (0, 1)$ in a finite abelian group, and $f = \mathbf{1}_A - \delta$. Then*

$$\max_{\chi \neq 1} |\hat{f}(\chi)| \geq \frac{\|f\|_{U^2}^2}{\sqrt{\delta(1 - \delta)}}.$$

More generally, a mean-zero function with $\|f\|_2^2 = V > 0$ has a nontrivial Fourier coefficient of modulus at least $\|f\|_{U^2}^2/\sqrt{V}$.

Proof. Fourier orthogonality gives $\|f\|_{U^2}^4 = \sum_{\chi} |\hat{f}(\chi)|^4$ and Parseval gives $\sum_{\chi} |\hat{f}(\chi)|^2 = V$. The trivial coefficient vanishes. Bound one of the two quadratic factors in each fourth power by the largest coefficient squared and divide by V . $\square$

In a cyclic group, this may be substituted directly into Corollary I.24.3. It is a variance-sensitive version of the standard U^2 inverse argument, not a replacement for the higher-order inverse theorem. The proof also explains why an apparent improvement obtained solely by changing normalized to unnormalized Fourier coefficients would be spurious.

Part II

The inverse step: energies, moments, restriction and dependent random choice

Contents. From many correlated derivatives to an approximate homomorphism. Sections II.1–II.4: source 02, delivered Sections 2–5 (a sharp weighted, mixed-function extension of Proposition 6.1 on every finite abelian group, its equality cases, deficits and rigidity). Sections II.5–II.6: source 03, delivered Sections 7–8 (a cubic dependent-random-choice certificate for Lemma 7.4, and an elementary replacement for Proposition 7.3). Sections II.7–II.10: source 04, delivered Sections 2–5 (its notation; exact second moments for Lemma 9.2; a polynomial restriction theorem for Lemma 9.3 and Corollary 9.4 over prime fields; arrangement moments for Lemmas 12.4, 14.7 and 14.8). The three sources treat different steps; no statement is printed twice. Source 04’s Section II.7 also fixes the cube notation used by its Part III sections.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
$\mathcal{E}$	02: $\mathcal{E}(w)$, additive energy of a weight on $G \times \widehat{G}$, normalized by N^{-3} ; 04: $\mathcal{E}_j(w) = M^{-1} \sum_{\chi} \widehat{w}(\chi) ^{2j}$ with the <i>unnormalized</i> transform, counting $2j$ -tuples; 03: $E(A)$ the additive energy of a set. <i>These are three different normalizations.</i>
$\mathcal{U}, T$	02: $\mathcal{U}(f) = \ f\ _{U^3}^8$ (normalized), $\mathcal{T}_w(f, g)$ a weighted derivative correlation, T its graph case; 04: $T_d(B)$ and $T_d^\phi(B)$ counts of additive and respected $2d$ -tuples.
κ, ρ, m	02: $\kappa = \mathcal{U}(f)/m(f)^4$, $m(f) = \mathbb{E} f ^2$; 03: $\kappa = 7/(3\sqrt{6})$, κ_* , $\rho = c/(2 - c)$, $m = V $ or $ A $; 04: $\kappa_d = 1/(2(400d)^d)$, $\rho = m/M$ with $m = A $.
R, L, Q	02: $R = \mathbb{E} K_u(s) ^2$; 04: R_j counts respected arrangements, L the kernel bandwidth, Q the kernel ratio sum (R14); 03: $L = D $. Source 04’s tags (M1)–(M7), (R1)–(R32), (A1)–(A14) label displays: <i>(R3) is a display, not a remainder R_3; (A1) is not Appendix A.</i>

Added in the second write (6 October 2026): sources 06–10. Contents. Sections II.11–II.19 are source 07’s chapter: a guide written for this report (Section II.11), source 07’s own notation table (its Appendix A, placed first), and its delivered Sections 2–8 (finite fibered domains and the directional error law; transport profiles; the coupled-path rigidity theorem; the sharp balanced threshold $1/(2k)$; the application to Gowers’s Section 10 with Freiman orders five and nine; the rational linear program for the starting law; collision-to-mode exactification, weighted laws and nonabelian groups). Sections II.20–II.21: source 10, delivered Sections 7–8 (a weighted energy extraction with $(c/4, 2^{10}c^{-5})$ in Proposition 7.3; the vertical-obstruction-fibre form of the Freiman extraction of Lemma 7.5, the order-eight retention $2^{-209}\gamma^{103}$ in Corollary 7.6, and a torsion budget for general finite codomains).

Overlaps (dated notes). Source 10’s Theorem II.20.1 and source 03’s Theorem II.6.3 are two elementary replacements of Proposition 7.3 under the same hypothesis; source 10’s is stronger in both conclusions (note after source 03’s theorem). Source 07 treats a step that no other source treats (Lemma 10.9); its relation to the Lean statement `lemma_10_9` is stated precisely in Section II.11 and in the note after Corollary II.17.6.

Reading conventions for the sections of sources 06–10 in this Part.

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
δ, δ_d, b_d	07: δ_d the directional defect of a direction d , b_d its boundary mass, δ a uniform bound on the δ_d . 07's δ is a failure probability, not a density.
$\mathcal{T}_s, \mathcal{C}, c(s)$	07: $\mathcal{T}_s(t)$ the transport profile, $\mathcal{C}(\mathbf{d})$ the cost of a word, $c(s)$, $A(s)$, $M(s)$ the coarse envelopes, ν an alternative starting law. 07's $\mathcal{C}$ is not the cube count $\mathcal{C}_d$ of 08 and 09 (Part III).
$\kappa, \tau, L, r_0, S, W$	07: κ the fibre-size ratio, L the one-step fibre-size variation, r_0 the internal lower bound, $\tau = L/r_0$; S the usable support, W the union of its fibres. 07's κ is none of the κ 's of 02, 03 and 04.
$\theta, \eta, \alpha, \rho, \sigma, M$	07: the parameters of Gowers's Lemmas 10.6–10.9 and of Section10.lean : $\theta = 10\eta^{1/5}$, $q(\theta) = 2\theta - \theta^2$, $\Delta = q(\theta) + \eta$; α, ρ, σ, M in (II.17.10) as there (M is the fibre-size cap, not a modulus). $\text{Fr}_k(D, H)$: Freiman homomorphisms of order k , the definition of the Lean FreimanHom .
$E(A), c, T(a, b)$	10: $E(A) = \sum_d r_A(d)^2$ and c as in 03's Theorem II.6.3; $w(a, b)$, $T(a, b)$ weighted codegrees, K a random subset, Z its bad pairs. 10's T is not 02's T .
$\Gamma, H_k, h_k, \gamma, F, V$	10: Γ a graph, H_k its vertical obstruction fibre and $h_k = H_k $, C a doubling constant, γ the graph-energy proportion, F the defect set and V the codomain of Theorem II.21.3. 10's h_k is not the $h_k = (k-2)2^{k-1} + 1$ of Part IV, and this γ is not 10's γ of Part I.

II.1 Normalization and finite Fourier identities

Source 02, delivered Section 2.

Let G be a finite abelian group of order $N \geq 1$, written additively, and let $\widehat{G}$ be its character group, written multiplicatively. Thus $(h, \chi) + (k, \psi)$ in $G \times \widehat{G}$ means $(h + k, \chi\psi)$. All averages over G are normalized; sums over $\widehat{G}$ are not:

$$\mathbb{E}_x a(x) = \frac{1}{N} \sum_{x \in G} a(x), \quad \widehat{a}(\chi) = \mathbb{E}_x a(x) \overline{\chi(x)}.$$

Fourier orthogonality gives

$$\mathbb{E}_x |a(x)|^2 = \sum_{\chi \in \widehat{G}} |\widehat{a}(\chi)|^2. \quad (\text{II.1.1})$$

For normalized convolution $(a * b)(s) = \mathbb{E}_h a(s - h)b(h)$, we have $\widehat{a * b} = \widehat{a} \widehat{b}$. These facts follow by expanding the sums and using $\mathbb{E}_x \chi(x) = 0$ for a nontrivial character and 1 for the trivial character.

Define

$$\Delta_h f(x) = f(x) \overline{f(x - h)}, \quad m(f) = \mathbb{E}_x |f(x)|^2,$$

and define the eighth power of the normalized U^3 norm by

$$\mathcal{U}(f) = \mathbb{E}_h \sum_{\chi \in \widehat{G}} |\widehat{\Delta_h f}(\chi)|^4. \quad (\text{II.1.2})$$

It also equals the usual cube average

$$\mathbb{E}_{x,a,b,c} \prod_{\omega \in \{0,1\}^3} \mathcal{C}^{|\omega|} f(x + \omega_1 a + \omega_2 b + \omega_3 c), \quad (\text{II.1.3})$$

where $\mathcal{C}$ denotes complex conjugation. One proof first expands the fourth Fourier moment in (II.1.2); character orthogonality leaves the additive quadruple constraint, which is parameterized by three variables. The remaining derivative variable gives (II.1.3), after changes of signs in the increment variables. In particular $\mathcal{U}(f)$ is nonnegative.

For $f, g : G \rightarrow \mathbb{C}$, write

$$A_{f,g}(h, \chi) = \mathbb{E}_x f(x) \overline{g(x-h)\chi(x)}.$$

This is just a Fourier transform of a mixed derivative, not an additional assumption about f or g .

Definition II.1.1 (Weighted correlation and energy). For any $w : G \times \widehat{G} \rightarrow [0, \infty)$, define

$$\mathcal{T}_w(f, g) = \mathbb{E}_h \sum_{\chi} w(h, \chi) |A_{f,g}(h, \chi)|^2, \quad (\text{II.1.4})$$

$$\mathcal{E}(w) = \frac{1}{N^3} \sum_{\substack{h_1+h_2=h_3+h_4 \\ \chi_1\chi_2=\chi_3\chi_4}} \prod_{j=1}^4 w(h_j, \chi_j). \quad (\text{II.1.5})$$

The factor N^{-3} in (II.1.5) is deliberate: it normalizes the energy of a full additive graph to 1. It is not the uniform-probability normalization on the group of order N^2 . If $w = \mathbf{1}_\Gamma$, then $N^3 \mathcal{E}(w)$ is the ordinary ordered additive-quadruple count of Γ .

II.2 A sharp weighted extension of Proposition 6.1

Source 02, delivered Section 3.

Theorem II.2.1 (Mixed energy-sensitive inverse step). *For every finite abelian group G , every $f, g : G \rightarrow \mathbb{C}$, and every nonnegative weight $w : G \times \widehat{G} \rightarrow [0, \infty)$,*

$$\boxed{\mathcal{T}_w(f, g)^4 \leq \min\{m(f)^4 \mathcal{U}(g), m(g)^4 \mathcal{U}(f)\} \mathcal{E}(w).} \quad (\text{II.2.1})$$

There is no pointwise boundedness assumption on f, g , and no graph or bounded-fiber assumption on w . The coefficient 1 cannot be decreased.

II.2.1 Three exact identities

For $u \in G$ define

$$F_u(s) = f(s) \overline{f(s-u)}, \quad D_u(s) = g(s) \overline{g(s-u)}, \quad V_u(h) = \sum_{\chi} w(h, \chi) \chi(u),$$

and put $K_u = D_u * V_u$. Use the normalized inner product

$$\langle F, K \rangle = \mathbb{E}_{u,s} F_u(s) \overline{K_u(s)}.$$

Then

$$\mathcal{T}_w(f, g) = \langle F, K \rangle, \quad (\text{II.2.2})$$

$$R := \mathbb{E}_{u,s} |K_u(s)|^2 = \mathbb{E}_u \sum_{\psi} |\widehat{D}_u(\psi)|^2 |\widehat{V}_u(\psi)|^2, \quad (\text{II.2.3})$$

$$\mathbb{E}_u \sum_{\psi} |\widehat{V}_u(\psi)|^4 = \mathcal{E}(w). \quad (\text{II.2.4})$$

For (II.2.2), expand the square in (II.1.4), set the second spatial variable equal to $s - u$, and obtain

$$\mathbb{E}_{u,s} f(s) \overline{f(s-u)} \mathbb{E}_h \overline{g(s-h)} g(s-u-h) \sum_{\chi} w(h, \chi) \overline{\chi(u)}.$$

This is exactly $\langle F, K \rangle$. Nonnegativity of w also makes the original expression visibly real and nonnegative. Identity (II.2.3) is Parseval and the convolution identity.

For (II.2.4), expand

$$\widehat{V}_u(\psi) = \frac{1}{N} \sum_{h, \chi} w(h, \chi) \chi(u) \overline{\psi(h)}.$$

The fourth power contributes N^{-4} . Summing over ψ contributes N and imposes $h_1 + h_2 = h_3 + h_4$. Averaging over u imposes $\chi_1 \chi_2 = \chi_3 \chi_4$ without another factor. The remaining normalization is N^{-3} , proving the identity.

II.2.2 Proof of the theorem

The first exact norm is

$$\|F\|_2^2 = \mathbb{E}_{u,s} |f(s)|^2 |f(s-u)|^2 = m(f)^2.$$

Cauchy–Schwarz applied to (II.2.2) therefore gives

$$\mathcal{T}_w(f, g)^2 \leq m(f)^2 R. \quad (\text{II.2.5})$$

Applying Cauchy–Schwarz to (II.2.3), this time on the measure space with average in u and counting measure in ψ , gives

$$R^2 \leq \left(\mathbb{E}_u \sum_{\psi} |\widehat{D}_u(\psi)|^4 \right) \left(\mathbb{E}_u \sum_{\psi} |\widehat{V}_u(\psi)|^4 \right) = \mathcal{U}(g) \mathcal{E}(w). \quad (\text{II.2.6})$$

Squaring (II.2.5) and using (II.2.6) proves the first bound in (II.2.1).

For the other bound, let $w^\vee(h, \chi) = w(-h, \chi^{-1})$. Directly changing variables in the mixed Fourier coefficient gives

$$|A_{f,g}(h, \chi)| = |A_{g,f}(-h, \chi^{-1})|.$$

Consequently $\mathcal{T}_w(f, g) = \mathcal{T}_{w^\vee}(g, f)$. Inversion is an automorphism of the abelian group $G \times \widehat{G}$, so $\mathcal{E}(w^\vee) = \mathcal{E}(w)$. Apply the first bound with f and g exchanged. Equality examples establishing the final assertion are given in Section II.3. $\square$

Corollary II.2.2 (Graph version). *Let $B \subseteq G$, $\phi : B \rightarrow \widehat{G}$, and $\Gamma = \{(h, \phi(h)) : h \in B\}$. If*

$$T = \mathbb{E}_h \mathbf{1}_B(h) |\widehat{\Delta_h f}(\phi(h))|^2 > 0,$$

then, writing $E(\Gamma)$ for the unnormalized additive energy,

$$\frac{E(\Gamma)}{N^3} \geq \frac{T^4}{m(f)^4 \mathcal{U}(f)}. \quad (\text{II.2.7})$$

In particular, the hypothesis (F.7.1) implies

$$E(\Gamma) \geq \frac{\alpha^4}{m(f)^4 \mathcal{U}(f)} N^3.$$

Proof. Take $g = f$ and $w = \mathbf{1}_\Gamma$. The unnormalized Fourier coefficient is N times our normalized coefficient, so the statistic in (F.7.1), divided by N^3 , is precisely T . Positive T makes the denominator nonzero by Theorem II.2.1. $\square$

[write] **6 October 2026.** *The formal statement is recovered.* The formal `proposition_6_1` (Section `ns06_07.lean`, line 37; exact companion `proposition_6_1_holds`, `Proofs05_10.lean`, line 505) is the case $|f| \leq 1$ on $\mathbb{Z}_N$ with a single-valued graph: there $m(f) \leq 1$ and $\mathcal{U}(f) \leq m(f)^4 \leq 1$ (Lemma II.3.1), so (II.2.7) gives $E(\Gamma) \geq \alpha^4 N^3$. The intake also tested Theorem II.2.1 on the non-cyclic groups $\mathbb{Z}/2 \times \mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$ and on $\mathbb{Z}/6$ (largest ratio of left to right side 0.084); the delivered suite uses cyclic groups only. *Notation:* this $\mathcal{E}(w)$ is not source 03's $\mathcal{E}(f)$ (Part III), source 04's $\mathcal{E}_j(w)$ (Section II.8) or source 05's captured energy E (Part I); see Section F.3.

II.3 The L^2 denominator and equality examples

Source 02, delivered Section 4.

Lemma II.3.1 (Two tetrahedra). *For every finite abelian group and every complex function on it,*

$$\mathcal{U}(f) \leq m(f)^4, \quad \text{equivalently} \quad \|f\|_{U^3} \leq \|f\|_2. \quad (\text{II.3.1})$$

Proof. Take the absolute value of the cube average (II.1.3). Divide its eight factors into the four vertices

$$x, \quad x+a, \quad x+b, \quad x+c$$

and the complementary four vertices

$$x+a+b, \quad x+a+c, \quad x+b+c, \quad x+a+b+c.$$

Cauchy–Schwarz bounds the cube average by the geometric mean of the averages of the squared products on these two lists. The first list is a bijective change of coordinates from G^4 to G^4 . The second is also bijective: if its coordinates are y_1, y_2, y_3, y_4 , then

$$c = y_4 - y_1, \quad b = y_4 - y_2, \quad a = y_4 - y_3, \quad x = y_1 + y_2 + y_3 - 2y_4.$$

Thus each squared-product average is $m(f)^4$, proving (II.3.1). No division by 2 occurs in the inverse map, so groups with 2-torsion are included. $\square$

Corollary II.3.2 (Variance-normalized energy). *Whenever the denominator is nonzero,*

$$\mathcal{E}(w) \geq \left(\frac{\mathcal{T}_w(f, g)}{m(f)m(g)} \right)^4.$$

For a graph and a balanced indicator $f = \mathbf{1}_A - \delta$, where $\delta = |A|/N \in (0, 1)$,

$$\boxed{\frac{E(\Gamma)}{N^3} \geq \frac{T^4}{[\delta(1-\delta)]^4 \mathcal{U}(f)} \geq \frac{T^4}{[\delta(1-\delta)]^8}.} \quad (\text{II.3.2})$$

Proof. Use Lemma II.3.1 in Theorem II.2.1. For a balanced indicator, direct averaging gives $m(f) = \delta(1-\delta)$. $\square$

The improvement in (II.3.2) is not a change in the fourth power of T . It is the retention of an exact variance denominator, together with a possibly smaller U^3 denominator. It is therefore misleading to describe it as a new exponent in Szemerédi's theorem. It can nevertheless be a large improvement in a local energy ledger.

Example II.3.3 (Quadratic-phase graph). Let $G = \mathbb{Z}_p$ for an odd prime p , put $q(x) = \exp(2\pi i x^2/p)$, and let $f = cq$, $c \neq 0$. Then

$$\Delta_h f(x) = |c|^2 \exp(2\pi i(2hx - h^2)/p).$$

Choose the graph $h \mapsto \chi_{2h}$. We have

$$m(f) = |c|^2, \quad \mathcal{U}(f) = |c|^8, \quad T = |c|^4, \quad E(\Gamma) = N^3.$$

Equality holds in both (II.2.1) and the variance-normalized bound. In particular the coefficient 1 in the main theorem is optimal even for a single-valued graph.

Example II.3.4 (A genuinely multivalued sharp example). Let $H \leq G$ have density $d = |H|/N$, let $f = \mathbf{1}_H$, and take $w = \mathbf{1}_{H \times H^\perp}$, where $H^\perp$ is the annihilator of H . The set $H \times H^\perp$ is a subgroup of $G \times \widehat{G}$ of size N , hence $\mathcal{E}(w) = 1$. Also

$$m(f) = d, \quad \mathcal{U}(f) = d^4, \quad \mathcal{T}_w(f, f) = d^2.$$

The cube formula gives $\mathcal{U}(f) = d^4$ because all eight vertices belong to H exactly when the base and three increments do. The Fourier coefficient of $\mathbf{1}_H$ is d on $H^\perp$ and 0 elsewhere, giving the value of $\mathcal{T}$. Again equality holds in the main theorem.

Example II.3.5 (Sharpness within balanced indicators). On an even cyclic group, take A to be the even residues. Then $f = \mathbf{1}_A - 1/2$ is one half of a character. For the graph with trivial frequency at every h ,

$$m(f) = \frac{1}{4}, \quad \mathcal{U}(f) = \frac{1}{256}, \quad T = \frac{1}{16}, \quad E(\Gamma)/N^3 = 1.$$

The original T^4 bound is $1/65536$, whereas (II.3.2) gives the exact value 1. This demonstrates a strict strengthening on an admissible balanced input, not merely on arbitrary unbounded functions.

II.4 Keeping the full derivative spectrum and tracking rigidity

Source 02, delivered Section 5.

II.4.1 A support-and-energy certificate without selecting one frequency

Proposition II.4.1. *Let $f \neq 0$, set $m = m(f)$ and $\kappa = \mathcal{U}(f)/m^4 \in (0, 1]$, and let $0 < \tau \leq 1$. Define*

$$\Gamma_\tau = \{(h, \chi) : |A_{f,f}(h, \chi)| \geq \tau m\}, \quad t_\tau = \frac{\mathcal{T}_{\mathbf{1}_{\Gamma_\tau}}(f, f)}{m^2}.$$

Then

$$|\Gamma_\tau| \leq N\tau^{-2}, \tag{II.4.1}$$

$$\frac{E(\Gamma_\tau)}{N^3} \geq \frac{t_\tau^4}{\kappa}, \tag{II.4.2}$$

$$\frac{E(\Gamma_\tau)}{|\Gamma_\tau|^3} \geq \frac{t_\tau^4 \tau^6}{\kappa} \quad \text{if } \Gamma_\tau \neq \emptyset. \tag{II.4.3}$$

Proof. Parseval, followed by averaging in h , gives the exact identity

$$\sum_{h,\chi} |A_{f,f}(h, \chi)|^2 = N \mathbb{E}_{h,x} |f(x)|^2 |f(x-h)|^2 = Nm^2.$$

This proves the support bound. Apply Theorem II.2.1 to the full set Γ_τ , not a chosen graph, and substitute $\mathcal{U}(f) = \kappa m^4$. Finally divide by the cube of the support bound. Nonzero f has $\mathcal{U}(f) > 0$: the $h = 0$ summand in (II.1.2) already has a nonzero Fourier coefficient. $\square$

The certificate (II.4.3) is suitable input for an additive-energy-to-structure theorem of Balog–Szemerédi–Gowers type. The subsequent structural step is not proved anew here. The point is that there is no preliminary loss from discarding all but one frequency in each fiber. Whether that benefit survives the later graph and progression constructions is a separate quantitative problem.

II.4.2 Exact deficits in the two Cauchy–Schwarz steps

With F, D, V, K, R as in Section II.2, set

$$D_1 = m(f)^2 R - \mathcal{T}_w(f, g)^2, \quad D_2 = \mathcal{U}(g) \mathcal{E}(w) - R^2.$$

Both are nonnegative, and direct algebra gives

$$m(f)^4 \mathcal{U}(g) \mathcal{E}(w) - \mathcal{T}_w(f, g)^4 = m(f)^4 D_2 + D_1 (m(f)^2 R + \mathcal{T}_w(f, g)^2). \quad (\text{II.4.4})$$

Thus the loss is decomposed exactly, rather than hidden in an unspecified constant.

Proposition II.4.2 (Quantitative alignment). *Suppose all quantities used to normalize below are nonzero and $0 \leq \varepsilon < 1$. If*

$$\mathcal{T}_w(f, g)^4 \geq (1 - \varepsilon) m(f)^4 \mathcal{U}(g) \mathcal{E}(w),$$

then

$$\begin{aligned} \left\| \frac{F}{\|F\|_2} - \frac{K}{\|K\|_2} \right\|_2^2 &\leq 2(1 - (1 - \varepsilon)^{1/4}), \\ \left\| \frac{a}{\|a\|_2} - \frac{b}{\|b\|_2} \right\|_2^2 &\leq 2(1 - \sqrt{1 - \varepsilon}), \end{aligned}$$

where $a(u, \psi) = |\widehat{D}_u(\psi)|^2$ and $b(u, \psi) = |\widehat{V}_u(\psi)|^2$. The first norm uses normalized measure on G^2 ; the second uses average in u and counting measure in ψ .

Proof. Let $A = \mathcal{T}^2/(m(f)^2 R)$ and $B = R^2/(\mathcal{U}(g) \mathcal{E}(w))$. The proof of Theorem II.2.1 gives $A, B \in [0, 1]$, and the hypothesis says $A^2 B \geq 1 - \varepsilon$. Hence $A \geq \sqrt{1 - \varepsilon}$ and $B \geq 1 - \varepsilon$. The normalized inner products of the first and second pairs are $\sqrt{A}$ and $\sqrt{B}$, respectively. They are real and nonnegative, by (II.2.2) and (II.2.3). Apply $\|v - w\|^2 = 2 - 2 \operatorname{Re} \langle v, w \rangle$ for unit vectors. $\square$

This is an alignment theorem for the proof’s intermediate functions, not a full classification of all near-extremal pairs (f, w) . The distinction matters: subgroup-localized examples and polynomial phases already occur, and broader extremizer theory is known [7].

II.4.3 An elementary exact rigidity endpoint

Proposition II.4.3. *Let $f \not\equiv 0$, and let $\phi : G \rightarrow \widehat{G}$ be defined on all of G . For its graph statistic, $T \leq m(f)^2$. Equality holds if and only if $f = cq$, with $|q| = 1$, and for each h*

$$\Delta_h q(x) = c_h \phi(h)(x), \quad |c_h| = 1.$$

In this situation ϕ is a homomorphism and all third multiplicative derivatives of q equal 1.

Proof. The chosen Fourier coefficient in each fiber has squared magnitude at most the sum of squared magnitudes in that fiber. The sum of all fibers is $m(f)^2$ after normalized averaging, by Parseval. Equality forces every $\Delta_h f$ to have Fourier support at the chosen single character. At $h = 0$, $\Delta_0 f = |f|^2$ is nonnegative and has positive mean. Its sole character must therefore be trivial, and $|f|^2 = m(f)$ everywhere. Write $f = cq$ with $|q| = 1$. Fourier inversion gives the stated derivative formula. Conversely that formula makes Parseval an equality in every fiber.

The identity

$$\Delta_{h+k} q(x) = \Delta_h q(x) \Delta_k q(x - h)$$

shows that $\phi(h+k) = \phi(h)\phi(k)$, since a nonzero scalar multiple of a character has a unique character frequency. A further derivative of a character is constant, and a derivative of a unit-modulus constant is 1; therefore third derivatives vanish multiplicatively. This definition of a quadratic phase does not impose an unjustified classical-polynomial representation in small characteristic. $\square$

II.5 Sharper common-neighborhood selection

Source 03, delivered Section 7.

Gowers' Lemma 7.4 is a convenient place to make a concrete quantitative improvement without changing the combinatorial objects in the argument. Its hypothesis controls the average intersection of a family of sets; its conclusion finds a subfamily in which almost every pair has a reasonably large intersection. The published proof samples five witnesses and estimates the useful and exceptional contributions separately [1]. Keeping those contributions together allows three witnesses to suffice.

The method remains dependent random choice, whose subsequent development is surveyed by Fox and Sudakov [11]. Our assertion here is a precise replacement for the numerical statement in the source paper. We do not assert that the resulting exponent is optimal among all possible selection arguments.

II.5.1 A cubic certificate

All pairs below are ordered, and diagonal pairs are included. For a finite family $A_1, \dots, A_n$ of subsets of a set V of cardinality $m > 0$, write

$$p_{xy} = \frac{|A_x \cap A_y|}{m}.$$

A pair will be called bad when $p_{xy} < a/2$, where $a > 0$ is the specified lower bound for the mean of the p_{xy} .

Lemma II.5.1 (Rational cubic certificate). *For every $a > 0$ and $p \geq 0$, define*

$$F_a(p) = \begin{cases} -9p^3, & 0 \leq p < a/2, \\ p^3, & p \geq a/2. \end{cases}$$

Then

$$F_a(p) \geq \frac{49}{12}a^2p - \frac{343}{108}a^3. \quad (\text{II.5.1})$$

Proof. On the good region $p \geq a/2$, the desired inequality is the tangent-line inequality for p^3 at $p = 7a/6$. More explicitly,

$$p^3 - \frac{49}{12}a^2p + \frac{343}{108}a^3 = \left(p - \frac{7a}{6}\right)^2 \left(p + \frac{7a}{3}\right) \geq 0.$$

On the interval $0 \leq p \leq a/2$, the function

$$-9p^3 - \frac{49}{12}a^2p + \frac{343}{108}a^3$$

is concave, so its minimum on that interval occurs at an endpoint. Its values at 0 and $a/2$ are, respectively,

$$\frac{343}{108}a^3 \quad \text{and} \quad \frac{-243 - 441 + 686}{216}a^3 = \frac{a^3}{108}.$$

Both are positive. At $p = a/2$, the actual function F_a takes the larger, good-region value, so the strict definition of a bad pair presents no boundary issue. $\square$

Theorem II.5.2 (Cubic replacement for Gowers' Lemma 7.4). *Let m, n be positive integers, let V have cardinality m , and let $0 < \delta \leq 1$. Suppose that $A_1, \dots, A_n \subseteq V$ satisfy*

$$\sum_{x,y=1}^n |A_x \cap A_y| \geq \delta^2 mn^2. \quad (\text{II.5.2})$$

There is a set $K \subseteq \{1, \dots, n\}$ such that

$$|K| \geq \frac{7}{3\sqrt{6}} \delta^3 n, \quad (\text{II.5.3})$$

and at least 90% of the pairs $(x, y) \in K^2$ satisfy

$$|A_x \cap A_y| \geq \frac{\delta^2 m}{2}. \quad (\text{II.5.4})$$

In particular, the conclusion holds when every A_x has cardinality at least δm .

Proof. Put $a = \delta^2$. Independently sample j_1, j_2, j_3 uniformly from V , allowing repetitions, and set

$$K = \{x : j_1, j_2, j_3 \in A_x\}.$$

Let $B(K)$ count the bad ordered pairs in K^2 . The probability that $(x, y) \in K^2$ is p_{xy}^3 . Consequently, linearity of expectation gives the exact identity

$$\begin{aligned}\mathbb{E}[|K|^2 - 10B(K)] &= \sum_{x,y=1}^n F_a(p_{xy}) \\ &\geq \frac{49}{12}a^2 \sum_{x,y=1}^n p_{xy} - \frac{343}{108}a^3n^2 \\ &\geq \left(\frac{49}{12} - \frac{343}{108}\right)a^3n^2 = \frac{49}{54}a^3n^2.\end{aligned}$$

The first inequality is Lemma II.5.1; the second uses (II.5.2). Some realization therefore satisfies

$$|K|^2 - 10B(K) \geq \frac{49}{54}\delta^6n^2 > 0.$$

It follows simultaneously that $|K| \geq \sqrt{49/54}\delta^3n$ and $B(K) \leq |K|^2/10$. These are the required conclusions.

For the final assertion, put $d(v) = |\{x : v \in A_x\}|$. Then

$$\sum_{x,y} |A_x \cap A_y| = \sum_{v \in V} d(v)^2 \geq \frac{1}{m} \left(\sum_{v \in V} d(v) \right)^2 \geq \delta^2 mn^2,$$

by Cauchy–Schwarz and the assumed lower bounds on $|A_x|$. □

The constant $7/(3\sqrt{6})$ is approximately 0.95258. Thus the original $2^{-1/2}\delta^5n$ is replaced by a larger bound with exponent 3, while both the half-mean intersection threshold and the 90% requirement are preserved. The improvement comes from a stronger estimate of the same penalized random variable, rather than from a change in the definition of success.

[write] **6 October 2026.** Compared with the formal statement (checked at HEAD): `lemma_7_4` (Section `ns06_07.lean`, line 167) has the exact companion `lemma_7_4_holds` (`Proofs07DRC.lean`, line 536), which samples five witnesses indexed by `Fin 5`, as Section V.13 describes. Theorem II.5.2 keeps its hypothesis, the threshold $\delta^2m/2$ and the 90% requirement, and replaces the size $2^{-1/2}\delta^5n$ by $7\delta^3n/(3\sqrt{6})$. Not formalized.

II.5.2 Weights and higher tuples

The certificate is independent of the counting measure on the index set. This gives a useful extension when vertices carry unequal importance or when intersections of more than two sets are under consideration.

Corollary II.5.3 (Weighted tuple selection). *Let X be a finite nonempty set, let $w_x \geq 0$ with $\sum_{x \in X} w_x = 1$, and let $(A_x)_{x \in X}$ be measurable subsets of a probability space (Ω, ν) . Fix an integer $r \geq 1$, and suppose*

$$\sum_{(x_1, \dots, x_r) \in X^r} w_{x_1} \cdots w_{x_r} \nu(A_{x_1} \cap \cdots \cap A_{x_r}) \geq a > 0.$$

There is a set $K \subseteq X$ with

$$w(K) := \sum_{x \in K} w_x \geq \left(\frac{49}{54}\right)^{1/r} a^{3/r}$$

such that the total product weight of tuples in K^r with $\nu(A_{x_1} \cap \cdots \cap A_{x_r}) < a/2$ is at most $w(K)^r/10$.

Proof. Sample three independent points of Ω and let K be the indices of the sets containing all three points. A fixed tuple lies in K^r with probability p^3 , where p is its intersection measure. Apply (II.5.1) to each tuple and sum with its product weight. The expected value of $w(K)^r$ minus ten times the bad-tuple weight is at least $49a^3/54$. Since X is finite, there are only finitely many possible sets K , and one attains at least this expectation. Taking an r th root gives the conclusion. $\square$

II.5.3 The exact scalar optimization

The simple rational point $7a/6$ was chosen to make the certificate easy to verify. One can determine the optimal supporting line for this argument. Let $z > 1$ be the unique solution of

$$2z^3 - 3z^2 = 9. \quad (\text{II.5.5})$$

The line tangent to p^3 at $p = za/2$ passes through the limiting bad point $(a/2, -9a^3/8)$. Applying that line instead of (II.5.1) gives

$$|K| \geq \kappa_* \delta^3 n, \quad \kappa_* = \sqrt{\frac{3z^2 - z^3}{4}} = 0.9537881585\dots \quad (\text{II.5.6})$$

Indeed, the tangent has slope $3(za/2)^2$ and intercept $-2(za/2)^3$. Equation (II.5.5) verifies its value at the bad endpoint. Concavity on the bad interval and convexity on the good interval again prove a global minorant. Its value at the mean a is $(3z^2 - z^3)a^3/4$. Existence and uniqueness of $z > 1$ follow from the fact that $2z^3 - 3z^2$ is strictly increasing on $(1, \infty)$, starts at -1 , and tends to infinity. Moreover $2 < z < 5/2$, so the expression under the square root is positive.

For clarity, the meaning of optimality here is limited and exact. When $za/2 \leq 1$, consider arbitrary scalar random variables $P \in [0, 1]$ with $\mathbb{E}P = a$. A two-point distribution, with one value tending to $(a/2)^-$ and the other equal to $za/2$, approaches the tangent-line lower bound for $\mathbb{E}F_a(P)$. Thus no larger bound follows from this scalar mean constraint alone. The infimum is approached because badness is defined by a strict inequality.

This observation also explains a restriction of a two-witness version of the same argument. Its penalized scalar function is p^2 on the good region and $-9p^2$ on the bad region. The analogous tangent point is ha , where $h = (1 + \sqrt{10})/2$. The tangent's value at a is

$$(2h - h^2)a^2 < 0.$$

For sufficiently small a , the corresponding two-point distribution lies in $[0, 1]$ and gives a negative limiting expectation. Consequently, two witnesses and this scalar first-moment certificate cannot produce a positive uniform lower bound at the original parameters. This does not exclude a better theorem exploiting properties of actual intersection matrices, nor does it prove optimality of the exponent 3.

II.5.4 A flexible one-witness alternative

For applications, keeping the threshold exactly at half the mean may be unnecessary. Allowing a smaller constant multiple of the mean gives a much larger selected set using just one witness. This is an explicit form of the established variant in Fox and Sudakov [11, Lemma 5.1].

Lemma II.5.4 (One-witness tradeoff). *Under (II.5.2), let $0 < \gamma < \varepsilon < 1$. There is a set $K \subseteq \{1, \dots, n\}$ satisfying*

$$|K| \geq \sqrt{1 - \gamma/\varepsilon} \delta n$$

such that at most an ε fraction of the ordered pairs in K^2 have intersection smaller than $\gamma\delta^2 m$.

Proof. Choose $j \in V$ uniformly and set $K = \{x : j \in A_x\}$. Then

$$\mathbb{E} |K|^2 = \sum_{x,y} p_{xy} \geq \delta^2 n^2.$$

If $B(K)$ counts the pairs in K^2 with $p_{xy} < \gamma\delta^2$, then

$$\mathbb{E} B(K) = \sum_{p_{xy} < \gamma\delta^2} p_{xy} \leq \gamma\delta^2 n^2.$$

Thus

$$\mathbb{E}[|K|^2 - \varepsilon^{-1} B(K)] \geq (1 - \gamma/\varepsilon)\delta^2 n^2 > 0.$$

A realization attaining at least this expectation gives both claims. $\square$

II.6 An elementary quantitative replacement for Proposition 7.3

Source 03, delivered Section 8.

The preceding selection statements have different uses. The cubic version strengthens the exact statement of Lemma 7.4. The one-witness version is more efficient for the later small-difference-set argument, because that argument can tolerate a smaller absolute constant in the common-neighbor threshold. We also remove a separate loss by counting the total mass of popular differences directly.

Throughout this section, A is a finite nonempty subset of an arbitrary abelian group and $m = |A|$. Put

$$r_A(d) = \left| \{(a, b) \in A^2 : a - b = d\} \right|, \quad E(A) = \sum_d r_A(d)^2.$$

We assume $E(A) \geq cm^3$, where $0 < c \leq 1$. The sums involve only finitely many nonzero terms, so no finiteness assumption on the ambient group is required.

II.6.1 Popular differences retain more edge mass

Lemma II.6.1 (Popular mass). *Let*

$$D = \{d : r_A(d) \geq cm/2\}, \quad M = \sum_{d \in D} r_A(d), \quad \rho = \frac{c}{2 - c}.$$

Then $D = -D$, $|D| \leq 2m/c$, and $M \geq \rho m^2$.

Proof. The symmetry follows from $r_A(d) = r_A(-d)$. Since $\sum_d r_A(d) = m^2$, the lower bound $r_A(d) \geq cm/2$ on D gives $|D| \leq 2m/c$. On D we have $r_A(d)^2 \leq m r_A(d)$, while off D we have $r_A(d)^2 \leq (cm/2) r_A(d)$. Therefore

$$cm^3 \leq E(A) \leq mM + \frac{cm}{2}(m^2 - M).$$

Rearranging proves $M \geq cm^2/(2 - c)$. $\square$

This estimate retains the edge density of the popular-difference relation. It avoids the loss from first counting popular values and then replacing each of their multiplicities by the smallest allowed multiplicity. In particular, the relation has density at least $\rho \geq c/2$.

II.6.2 A general walk-counting statement

Proposition II.6.2 (Compression by four edge differences). *Let $G \subseteq A \times A$ be a symmetric relation, with loops allowed, satisfying $|G| \geq \rho m^2$, where $0 < \rho \leq 1$. Suppose that every $(a, b) \in G$ has $a - b \in D$ for a finite set D , and put $L = |D|$. For $0 < \gamma < \varepsilon < \beta < 1/2$, set $s = \sqrt{1 - \gamma/\varepsilon}$. There is $A' \subseteq A$ such that*

$$|A'| \geq (1 - \varepsilon/\beta)spm, \quad (\text{II.6.1})$$

$$|A' - A'| \leq \frac{L^4}{(1 - 2\beta)\gamma^2 s \rho^5 m^3}. \quad (\text{II.6.2})$$

Proof. Write $N(a) = \{v : (a, v) \in G\}$. By symmetry, double-counting, and Cauchy–Schwarz,

$$\sum_{a, b \in A} |N(a) \cap N(b)| = \sum_{v \in A} |N(v)|^2 \geq \frac{|G|^2}{m} \geq \rho^2 m^3.$$

Lemma II.5.4 supplies $U \subseteq A$, of cardinality $u \geq spm$, with at most εu^2 bad ordered pairs. Here a pair is bad when its common G -neighborhood has cardinality smaller than $\gamma \rho^2 m$.

Let A' consist of the vertices of U that belong, in the first coordinate, to at most βu bad pairs in U^2 . Counting those pairs shows that at most $(\varepsilon/\beta)u$ vertices are excluded. This proves (II.6.1). Each $a \in A'$ has at least $(1 - \beta)u$ good partners in U . Consequently, every $a, b \in A'$ have at least $(1 - 2\beta)u$ common good partners $k \in U$.

Fix such a, b . For every common good partner k , choose

$$x \in N(a) \cap N(k), \quad y \in N(b) \cap N(k).$$

There are at least $(1 - 2\beta)u(\gamma \rho^2 m)^2$ triples (k, x, y) . Associate to each the four edge differences

$$d_1 = a - x, \quad d_2 = k - x, \quad d_3 = k - y, \quad d_4 = b - y.$$

They lie in D and satisfy

$$a - b = d_1 - d_2 + d_3 - d_4.$$

For the fixed endpoints a, b , the association is injective: one recovers successively $x = a - d_1$, $k = x + d_2$, and $y = k - d_3$.

For each element of $A' - A'$, fix one pair of endpoints representing it. The associated label quadruples for distinct differences are disjoint, because a label quadruple determines $d_1 - d_2 + d_3 - d_4$. Since there are only L^4 possible label quadruples,

$$|A' - A'| (1 - 2\beta)u\gamma^2 \rho^4 m^2 \leq L^4.$$

Using $u \geq spm$ proves (II.6.2). This counts walks with specified endpoints; repeated vertices and loops are allowed throughout, and cause no loss in the injection. $\square$

II.6.3 Explicit bounds and their scope

Theorem II.6.3 (Elementary historical improvement). *If $E(A) \geq c|A|^3$ with $0 < c \leq 1$, there is $A' \subseteq A$ satisfying*

$$|A'| \geq \frac{c}{8} |A|, \quad |A' - A'| \leq 2^{19} c^{-9} |A|. \quad (\text{II.6.3})$$

More precisely, with $\rho = c/(2 - c)$, one may take

$$|A'| \geq \frac{\rho m}{2\sqrt{2}}, \quad |A' - A'| \leq 6912\sqrt{2} \rho^{-5} c^{-4} m. \quad (\text{II.6.4})$$

Proof. Use D from Lemma II.6.1, and define $G = \{(a, b) \in A^2 : a - b \in D\}$. That lemma gives $|G| \geq \rho m^2$ and $L \leq 2m/c$. Apply Proposition II.6.2 with

$$\gamma = \frac{1}{12}, \quad \varepsilon = \frac{1}{6}, \quad \beta = \frac{1}{3}, \quad s = \frac{1}{\sqrt{2}}.$$

The size bound is $\rho m/(2\sqrt{2})$, and the difference bound becomes

$$|A' - A'| \leq \frac{432\sqrt{2} L^4}{\rho^5 m^3} \leq 6912\sqrt{2} \rho^{-5} c^{-4} m.$$

Finally $\rho \geq c/2$, so

$$|A'| \geq \frac{cm}{4\sqrt{2}} \geq \frac{cm}{8}, \quad |A' - A'| \leq 221184\sqrt{2} c^{-9} m < 2^{19} c^{-9} m.$$

This proves both versions. □

For comparison, Proposition 7.3 in the source paper gives a subset of size at least $2^{-20} c^{12} m$ with difference set of size at most $2^{38} c^{-24} m$ [1]. Theorem II.6.3 improves both exponents using only one sampled vertex, Cauchy–Schwarz, and the explicit counting injection above. Even if one changes only Lemma 7.4 to Theorem II.5.2 and leaves the remaining original proof intact, its formulas yield the weaker but direct improvements

$$|A'| \geq 2^{-14} c^8 m, \quad |A' - A'| \leq 2^{31} c^{-20} m.$$

Indeed, the original parameter $\delta = c^2/8$ then gives selected density $\alpha = \kappa \delta^4$, where $\kappa = 7/(3\sqrt{6})$; the original cleanup and walk count give $\alpha m/2$ and $120m/(\alpha \delta^4 c^4)$, respectively. The displayed relaxations use $\kappa > 1/2$ and $120/\kappa < 128$.

[write] 6 October 2026; sources 06–10. *A stronger elementary replacement (source 10).* Under the same hypothesis $E(A) \geq c|A|^3$, $0 < c \leq 1$, source 10’s Theorem II.20.1 gives $|A_*| \geq cm/4$ and $|A_* - A_*| \leq 2^{10} c^{-5} m$, on any abelian group, by weighted dependent random choice with the diagonal pairs kept. It is stronger than (II.6.3) in both conclusions: $c/4 > c/8$, and $2^{10} c^{-5} \leq 2^{19} c^{-9}$ because $c^4 \leq 1 \leq 2^9$. Both are far weaker than Reiher–Schoen [12], as both sources say. Both imply the formal `proposition_7_3` bounds $2^{-20} c^{12}$ and $2^{38} c^{-24}$ (for source 10: $c^{11} \leq 2^{18}$ and $c^{19} \leq 2^{28}$); neither is formalized. Lemma 7.4 is treated only here.

These are improvements to the historical proof architecture, rather than to the strongest available Balog–Szemerédi–Gowers theorem. For example, Reiher and Schoen prove that, for every $0 < \eta < 1/2$, the same energy hypothesis gives a subset satisfying

$$|A'| \geq (1 - \eta) c^{1/2} m, \quad |A' - A'| \leq 2^{33} \eta^{-9} c^{-4} |A'|$$

[12, Theorem 1.2]. Their result is substantially stronger. The purpose of the elementary version is to supply a short, fully explicit replacement that can be checked independently and integrated into the existing development with a small dependency set. A complete improvement to the final quantitative Szemerédi bound would also require propagating the new parameters through the later arguments; no such consequence follows from this section alone.

[write] **6 October 2026.** *Compared with the formal statement (checked at HEAD):* `proposition_7_3` (`Sections06_07.lean`, line 135; exact companion `proposition_7_3_holds`, `Proofs07BalogSzemeredi.lean`, line 996) is stated on $\mathbb{Z}^n$ with the bounds $2^{-20}c^{12}|A|$ and $2^{38}c^{-24}|A|$. For $0 < c \leq 1$, $c/8 \geq 2^{-20}c^{12}$ (as $c^{11} \leq 2^{17}$) and $2^{19}c^{-9} \leq 2^{38}c^{-24}$ (as $c^{15} \leq 2^{19}$), so Theorem II.6.3, which holds in every abelian group, implies the bounds of the formal statement. Not formalized.

II.7 Notation and normalization

Source 04, delivered Section 2.

Let G be a finite abelian group, written additively, and let $N = |G|$. We write $\mathbb{E}_{x \in G} g(x) = N^{-1} \sum_{x \in G} g(x)$. Characters are written multiplicatively. Except in the moment section, which explicitly introduces an unnormalized transform, Fourier coefficients are normalized:

$$\widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}.$$

Thus

$$\mathbb{E}_x |f(x)|^2 = \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2.$$

Sums over the dual group in this convention are ordinary sums, not averages.

For $d \geq 1$, put

$$\|f\|_{U^d(G)}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\epsilon \in \{0,1\}^d} \mathcal{C}^{|\epsilon|} f(x + \epsilon_1 h_1 + \dots + \epsilon_d h_d),$$

where $\mathcal{C}$ is complex conjugation. We omit G when it is clear. For $d = 1$ this means $\|f\|_{U^1} = |\mathbb{E} f|$. For $d \geq 2$ it is the Gowers norm. If f is real, conjugations can be omitted.

For a real function w , define its normalized cube count

$$Q_d(w) = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\epsilon \in \{0,1\}^d} w(x + \epsilon \cdot h).$$

For $w = 1_A$, $N^{d+1}Q_d(w)$ counts all ordered additive d -cubes, including degenerate cubes, exactly as in the corresponding count in the source. If $\delta = |A|/N$, its balanced function is $f = 1_A - \delta$.

The conversion to Gowers's parameter is important:

$$f \text{ is } \alpha\text{-uniform of degree } d-1 \iff \|f\|_{U^d}^{2^d} \leq \alpha.$$

In particular, the letter u below generally denotes a norm, while α denotes an upper bound on a power of that norm. Confusing these two quantities would conceal the improvement in the exponent.

We use the standard Gowers–Cauchy–Schwarz inequality in normalized form:

$$\left| \mathbb{E}_{x, h} \prod_{\epsilon} \mathcal{C}^{|\epsilon|} f_{\epsilon}(x + \epsilon \cdot h) \right| \leq \prod_{\epsilon} \|f_{\epsilon}\|_{U^d}.$$

It follows by applying Cauchy–Schwarz successively in the cube coordinates; this is Lemma 3.8 in the source. Its ordinary consequences include the triangle inequality and $\|f\|_{U^d} \leq \|f\|_{U^{d+1}}$. For clarity, the latter also follows by writing

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_{h_1, \dots, h_{d-1}} \left| \mathbb{E}_x \Delta_{h_1} \cdots \Delta_{h_{d-1}} f(x) \right|^2.$$

Square this identity and use Cauchy–Schwarz, then apply $|\mathbb{E}g|^4 \leq \|g\|_{U^2}^4$ to each derivative. The resulting right-hand side is $\|f\|_{U^{d+1}}^{2^{d+1}}$.

An ordered additive $2d$ -tuple of a set B is a tuple with $x_1 + \cdots + x_d = x_{d+1} + \cdots + x_{2d}$. A map $\phi : B \rightarrow H$ *respects* the tuple if the same equation holds after applying ϕ . All tuple counts in this article are ordered and allow repetitions unless expressly called nonexceptional.

II.8 Exact second moments and density-sensitive lifting

Source 04, delivered Section 3.

Several losses in the paper occur when a second moment is bounded using the size of the ambient space rather than the size of the set under consideration. Retaining the exact second moment is useful on its own, and becomes particularly effective when followed by the restriction argument in the next section. The underlying interpolation inequality is standard Hölder; the contribution here is the careful normalization, the resulting sharper statements, and their use in the subsequent selection theorem.

II.8.1 Weighted energies on finite abelian groups

Let H be a finite abelian group of order M , and let $\widehat{H}$ be its character group. For $w : H \rightarrow \mathbb{C}$, use the unnormalized transform

$$\widehat{w}(\chi) = \sum_{x \in H} w(x) \overline{\chi(x)}.$$

For an integer $j \geq 1$, define

$$\mathcal{E}_j(w) = \frac{1}{M} \sum_{\chi \in \widehat{H}} |\widehat{w}(\chi)|^{2j}.$$

Character orthogonality gives

$$\mathcal{E}_1(w) = \sum_{x \in H} |w(x)|^2.$$

For an indicator $w = 1_A$, the same orthogonality calculation shows that $\mathcal{E}_j(A)$ counts ordered tuples $(a_1, \dots, a_{2j}) \in A^{2j}$ satisfying

$$a_1 + \cdots + a_j = a_{j+1} + \cdots + a_{2j}.$$

Repeated entries are included throughout.

Theorem II.8.1 (Exact energy interpolation). *Let $j \geq 2$, and suppose that $\mathcal{E}_1(w) > 0$. Then*

$$\boxed{\mathcal{E}_j(w) \geq \frac{\mathcal{E}_2(w)^{j-1}}{\mathcal{E}_1(w)^{j-2}}.} \tag{M1}$$

For $j > 2$, equality holds if and only if all nonzero Fourier coefficients of w have the same modulus. In particular, if $A \neq \emptyset$, then

$$\mathcal{E}_j(A) \geq \frac{\mathcal{E}_2(A)^{j-1}}{|A|^{j-2}}. \quad (\text{M2})$$

For $j > 2$, equality in (M2) holds exactly when A is a coset of a subgroup.

Proof. Put $x_\chi = |\widehat{w}(\chi)|^2$. If $j = 2$, the assertion is an identity. For $j > 2$, Hölder gives

$$\sum_{\chi} x_{\chi}^2 = \sum_{\chi} x_{\chi}^{(j-2)/(j-1)} x_{\chi}^{j/(j-1)} \leq \left(\sum_{\chi} x_{\chi} \right)^{(j-2)/(j-1)} \left(\sum_{\chi} x_{\chi}^j \right)^{1/(j-1)}.$$

Raising to the power $j - 1$ and dividing by M^{j-1} proves (M1). Equality in Hölder says that the positive values of x_{χ} are equal.

Now let $w = 1_A$, and put $m = |A|$. Its trivial Fourier coefficient is m . Thus equality forces every nonzero Fourier coefficient to have modulus m . Let K be the subgroup generated by $A - A$. Equality in the triangle inequality shows that $|\widehat{1_A}(\chi)| = m$ precisely when χ is constant on A , equivalently when $\chi \in K^{\perp}$. Consequently Parseval gives

$$Mm = |K^{\perp}| m^2 = \frac{M}{|K|} m^2.$$

Hence $m = |K|$. Since A is contained in a coset of K , it is that entire coset. Conversely, the Fourier transform of a coset indicator has constant modulus on its support, so it gives equality. $\square$

The proof applies without change to any finite family of spectra: the index χ can be replaced by a pair (i, χ) . This observation will be used for arrangements, where the extra index records common sidelengths.

Corollary II.8.2 (Lifting respected graph energies). *Let B be a nonempty subset of a finite abelian group, and let $\phi : B \rightarrow K$ take values in another finite abelian group. Write $m = |B|$, let $0 \leq \gamma \leq 1$, and suppose that at least γm^3 ordered additive quadruples of B are respected by ϕ . Then, for every integer $j \geq 2$, at least*

$$\gamma^{j-1} m^{2j-1} \quad (\text{M3})$$

ordered additive $2j$ -tuples are respected by ϕ .

Proof. Apply Theorem II.8.1 to the indicator of the graph $\Gamma = \{(x, \phi(x)) : x \in B\}$. Its first energy is m , and its j -th energy is exactly the number in question. $\square$

For $B \subset \mathbb{F}_p$ of size βp , the source proof of Lemma 9.2 uses

$$\sum_{u,r} |\widehat{f_u}(r)|^2 \leq p^3.$$

The exact identity is

$$\sum_{u,r} |\widehat{f_u}(r)|^2 = p^2 |B| = \beta p^3. \quad (\text{M4})$$

Thus, under its unnormalized hypothesis of at least $\gamma_0 p^3$ respected quadruples, its conclusion strengthens to $\gamma_0^7 \beta^{-6} p^{15}$ respected 16-tuples. Under the hypothesis of Corollary 9.4, where $\gamma_0 = \gamma \beta^3$, this is

$$\gamma^7 \beta^{15} p^{15}, \quad (\text{M5})$$

rather than $\gamma^7 \beta^{21} p^{15}$. These comparisons refer to the displayed statements in the source and its repository transcription [1, 23].

[write] 6 October 2026. *Proof-level baseline* (checked at HEAD): in `Proofs09Moments.lean`, the private lemma `section9_second_moment_le` (line 480), used for `lemma_9_2_holds` (line 527), computes the exact value $N^2|B|$ of this sum (line 495) before weakening it to N^3 . So (M4) is already a proof-level fact in the repository, as the stronger Corollary 2.5 pair is (Part I); the exported `lemma_9_2` keeps the weaker bound, and (M5) and Corollary II.8.2 are not formalized.

II.8.2 A finite-size correction from the trivial character

There is a further refinement when the ambient group size matters. Its proof also identifies a useful limitation of using only the full first and second moments.

Proposition II.8.3 (Removing the trivial Fourier coefficient). *Let $A \subset H$, where $0 < m = |A| < M = |H|$, and put*

$$\rho = \frac{m}{M}, \quad \gamma = \frac{\mathcal{E}_2(A)}{m^3}.$$

Then $\gamma \geq \rho$, and for every integer $j \geq 2$,

$$\frac{\mathcal{E}_j(A)}{m^{2j-1}} \geq \rho + \frac{(\gamma - \rho)^{j-1}}{(1 - \rho)^{j-2}}. \quad (\text{M6})$$

The right side is at least γ^{j-1} . If only $\mathcal{E}_2(A) \geq \gamma_0 m^3$ is known, one may replace $\gamma - \rho$ by $\max\{\gamma_0 - \rho, 0\}$.

Proof. Remove the trivial character before applying the Hölder argument. The sums of the remaining squared and fourth Fourier moduli are

$$M \left(m - \frac{m^2}{M} \right), \quad M \left(\mathcal{E}_2(A) - \frac{m^4}{M} \right),$$

respectively. The remaining $2j$ -th moment is $M(\mathcal{E}_j(A) - m^{2j}/M)$. Substitution in the same interpolation inequality proves (M6). The second displayed quantity is nonnegative, giving $\gamma \geq \rho$. Finally, write

$$\gamma = \rho \cdot 1 + (1 - \rho) \frac{\gamma - \rho}{1 - \rho}$$

and apply convexity of $t \mapsto t^{j-1}$. □

This correction can be sharp away from subgroup examples. For the parabola $\Gamma = \{(x, x^2) : x \in \mathbb{F}_p\}$, with p odd, the trivial Fourier coefficient has modulus p , the coefficients at $(r, 0)$ with $r \neq 0$ vanish, and all coefficients whose second frequency is nonzero have modulus $\sqrt{p}$. Indeed, completing the square reduces the latter assertion to

$$\left| \sum_x e_p(ux^2) \right|^2 = \sum_{x,y} e_p(u(x-y)(x+y)) = p \quad (u \neq 0),$$

where $e_p(t) = \exp(2\pi it/p)$; the change of variables $(x, y) \mapsto (x - y, x + y)$ is invertible. It follows that

$$\mathcal{E}_2(\Gamma) = 2p^2 - p, \quad \mathcal{E}_j(\Gamma) = p^{2j-2} + (p-1)p^{j-1}. \quad (\text{M7})$$

All nonzero residual squared Fourier moduli agree, so Proposition II.8.3 is an equality in this example.

II.9 Polynomial restriction to approximate homomorphisms

Source 04, delivered Section 4.

We now replace the repeated degree-one filters in the proof of Lemma 9.3 by one filter of higher degree. The degree is chosen from the desired accuracy. Although more coefficient relations appear, their nonnegative Fourier weights have total mass one. A weighted error bound therefore avoids a loss proportional to the number of those relations. The retained number of configurations has a polynomial dependence with exponent linear in the number of entries, instead of the much larger exponent in the source.

For $B \subset \mathbb{F}_p$, let $T_d(B)$ denote its number of ordered additive $2d$ -tuples, and let $T_d^\phi(B)$ denote the number also satisfying

$$\phi(x_1) + \cdots + \phi(x_d) = \phi(x_{d+1}) + \cdots + \phi(x_{2d}). \quad (\text{R1})$$

Thus $T_d^\phi(B) = \mathcal{E}_d(\Gamma)$. An elementary upper bound that will be used repeatedly is

$$T_d(B) \leq |B|^{2d-1}, \quad (\text{R2})$$

because the first $2d - 1$ entries uniquely determine the last.

Theorem II.9.1 (Polynomial restriction theorem). *Let $d \geq 2$ be an integer, and let $0 < \alpha, \beta, \eta \leq 1$. Define*

$$\kappa_d = \frac{1}{2(400d)^d}, \quad \Lambda_d = 2 \left(1 + \binom{2d}{2} \right) (400d)^d. \quad (\text{R3})$$

Suppose that p is a prime with

$$p \geq \Lambda_d(\alpha\eta)^{-2d}\beta^{-1}. \quad (\text{R4})$$

Let $B \subset \mathbb{F}_p$, with $|B| = \beta p$, and let $\phi : B \rightarrow \mathbb{F}_p$ satisfy

$$T_d^\phi(B) \geq \alpha\beta^{2d-1}p^{2d-1}. \quad (\text{R5})$$

Then there is $B' \subset B$ for which

$$T_d(B') \geq \kappa_d \alpha^{2d} \eta^{2d-1} \beta^{2d-1} p^{2d-1}, \quad (\text{R6})$$

and

$$\frac{T_d^\phi(B')}{T_d(B')} \geq \frac{1}{1+\eta} \geq 1-\eta. \quad (\text{R7})$$

Equivalently, the size hypothesis is $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$. It depends on the number of available points rather than a separate density requirement. The theorem concerns preservation of additive configurations. It does not merely assert that B' has many points. The normalization by $|B|^{2d-1}$, rather than the ambient configuration count, is essential for its use after Corollary II.8.2.

II.9.1 The kernel and its coefficients

Lemma II.9.2 (A powered cosine with many useful coefficients). *Let d, L be positive integers, and put $q = 4dL^2$. Write*

$$K_q(t) = \cos^{2q}(\pi t) = \sum_{\ell=-q}^q b_\ell e^{2\pi i \ell t}, \quad b_\ell = 4^{-q} \binom{2q}{q+\ell}. \quad (\text{R8})$$

Then $0 \leq K_q \leq 1$, its coefficients are nonnegative and symmetric, and

$$b_0 \geq \frac{1}{2\sqrt{q}}, \quad \sum_{\ell=-q}^q \left(\frac{b_\ell}{b_0} \right)^{2d} \geq L + 1. \quad (\text{R9})$$

Proof. The expansion follows by applying the binomial theorem to $(e^{\pi i t} + e^{-\pi i t})^{2q}/2^{2q}$. For the central coefficient, write $u_q = 4^{-q} \binom{2q}{q}$. We have $u_1 = 1/2$ and

$$\frac{u_{q+1}}{u_q} = \frac{2q+1}{2q+2} \geq \sqrt{\frac{q}{q+1}}.$$

After squaring, the last inequality reduces to $(2q+1)^2 \geq 4q(q+1)$. Induction proves the stated lower bound for b_0 .

For $0 \leq \ell \leq L$,

$$\frac{b_\ell}{b_0} = \prod_{j=1}^{\ell} \frac{q-j+1}{q+j} = \prod_{j=1}^{\ell} \left(1 - \frac{2j-1}{q+j} \right) \geq 1 - \frac{\ell^2}{q}. \quad (\text{R10})$$

Here all factors lie in $[0, 1]$, and $\prod_j (1 - z_j) \geq 1 - \sum_j z_j$, while $\sum_{j=1}^{\ell} (2j-1)/(q+j) \leq \ell^2/q$. Bernoulli's inequality now gives

$$\left(\frac{b_\ell}{b_0} \right)^{2d} \geq 1 - \frac{2d\ell^2}{q} \geq \frac{1}{2}. \quad (\text{R11})$$

The central term contributes 1, and the $2L$ terms with $1 \leq |\ell| \leq L$ contribute at least L . $\square$

II.9.2 Repeated entries and weighted relation errors

A direct union bound over all the kernel's coefficient relations would be expensive, because its degree depends on the desired accuracy. Instead, we keep their Fourier weights. Positivity and the fact that the weights sum to one allow all the unwanted relations to be controlled together.

Lemma II.9.3 (A rank-two relation inside a finite set). *Let $d \geq 2$, let p be prime, let $B \subset \mathbb{F}_p$ have size $m > 0$, and put*

$$v = (\underbrace{1, \dots, 1}_d, \underbrace{-1, \dots, -1}_d) \in \mathbb{F}_p^{2d}.$$

For every $\epsilon \in \mathbb{F}_p^{2d} \setminus \mathbb{F}_p v$,

$$\#\{x \in B^{2d} : v \cdot x = \epsilon \cdot x = 0\} \leq m^{2d-2}. \quad (\text{R12})$$

The number of additive $2d$ -tuples in B with repeated entries is at most $\binom{2d}{2} m^{2d-2}$.

Proof. The matrix with rows v, ϵ has rank two, so it has an invertible 2-by-2 column minor. Fix the other $2d - 2$ coordinates arbitrarily in B . The two remaining coordinates are uniquely determined in $\mathbb{F}_p$; either both belong to B , or this choice gives no admissible completion. This proves (R12).

If $x_i = x_j$, the vector $e_i - e_j$ supplies a second relation. It is not a scalar multiple of v , since it has zero coordinates whereas every nonzero scalar multiple of v has none. Apply (R12) to each of the $\binom{2d}{2}$ pairs of positions and take a union bound. $\square$

Lemma II.9.4 (Weighted selection estimates). *Let $d \geq 2$, $q \geq 1$, and let $p > 2q$ be prime. Let $B \subset \mathbb{F}_p$ have size $m > 0$, and let $\phi : B \rightarrow \mathbb{F}_p$. Choose r, s independently and uniformly in $\mathbb{F}_p$. Conditional on this choice, retain each point $x \in B$ independently with probability*

$$K_q \left(\frac{rx + s\phi(x)}{p} \right). \quad (\text{R13})$$

Integer representatives do not matter because K_q is periodic. Put

$$P_0 = b_0^{2d}, \quad Q = \sum_{\ell=-q}^q \left(\frac{b_\ell}{b_0} \right)^{2d}, \quad C_d = 1 + \binom{2d}{2}. \quad (\text{R14})$$

Let X count all selected respected additive tuples, and let Y count all selected unrespected additive tuples. Then

$$\mathbb{E}X \geq P_0 Q T_d^\phi(B), \quad \mathbb{E}Y \leq P_0 (T_d(B) - T_d^\phi(B)) + C_d m^{2d-2}. \quad (\text{R15})$$

Proof. For a fixed tuple $x = (x_1, \dots, x_{2d})$, abbreviate its point probabilities by $P_{r,s}(x_j)$, and form the product with multiplicities

$$A_{r,s}(x) = \prod_{j=1}^{2d} P_{r,s}(x_j).$$

Its actual conditional survival probability is the product over the *distinct* entries of x . Since all point probabilities lie in $[0, 1]$, the actual probability is at least $A_{r,s}(x)$, equals it when the entries are distinct, and exceeds it by at most 1 otherwise. This observation accounts for repeated points without assuming that their inclusion events are independent.

Expand $A_{r,s}(x)$ using the nonnegative Fourier coefficients from (R8). For $\epsilon \in \{-q, \dots, q\}^{2d}$, put $w_\epsilon = \prod_j b_{\epsilon_j}$. Orthogonality gives

$$\mathbb{E}_{r,s} A_{r,s}(x) = \sum_{\epsilon} w_\epsilon 1_{\{\epsilon \cdot x = 0\}} 1_{\{\epsilon \cdot \phi(x) = 0\}}, \quad \sum_{\epsilon} w_\epsilon = \left(\sum_{\ell=-q}^q b_\ell \right)^{2d} = 1. \quad (\text{R16})$$

Here $\epsilon \cdot \phi(x)$ denotes $\sum_j \epsilon_j \phi(x_j)$. The last equality uses $K_q(0) = 1$. All weights in this identity are nonnegative.

If ϵ is a scalar multiple of v modulo p , the condition $p > 2q$ forces its integer entries to be $(\ell, \dots, \ell, -\ell, \dots, -\ell)$ for a single $\ell \in [-q, q]$. These terms have total weight $P_0 Q$. Every one contributes for a respected additive tuple. Thus its expected product with multiplicities is at least $P_0 Q$, and the same lower bound holds for its actual survival probability. Summing proves the assertion about $\mathbb{E}X$.

For an unrespected additive tuple, every nonzero such ℓ is invertible modulo p , so only $\ell = 0$ can contribute among these scalar multiples. Its weight is P_0 . For each remaining coefficient vector, Lemma II.9.3 bounds the number of domain tuples satisfying the additional relation by

m^{2d-2} . Requiring that they be unrespected, or imposing the additional image relation in (R16), can only reduce this number. Summing first over tuples and then over coefficient vectors, and using $\sum_{\epsilon} w_{\epsilon} = 1$, gives

$$\sum_{\substack{x \in B^{2d} \\ x \text{ additive and unrespected}}} \mathbb{E}_{r,s} A_{r,s}(x) \leq P_0(T_d(B) - T_d^{\phi}(B)) + m^{2d-2}.$$

Finally, the difference between actual survival and this product need only be charged to repeated tuples. Lemma II.9.3 bounds their number by $\binom{2d}{2} m^{2d-2}$. Adding that error proves the assertion about $\mathbb{E}Y$. $\square$

II.9.3 Proof of the polynomial restriction theorem

Proof of Theorem II.9.1. Set

$$a = \alpha\eta, \quad m = |B| = \beta p, \quad H = m^{2d-1}, \quad c = \kappa_d a^{2d}. \quad (\text{R17})$$

Choose

$$L = \left\lceil \frac{4}{a} \right\rceil, \quad q = 4dL^2. \quad (\text{R18})$$

Since $a \leq 1$,

$$L \leq \frac{5}{a}, \quad q \leq \frac{100d}{a^2}. \quad (\text{R19})$$

The size condition is $m \geq \Lambda_d a^{-2d}$. It implies $p > 2q$, since $p \geq m$, $\Lambda_d > 200d$, and $a^{-2d} \geq a^{-2}$. Furthermore, with $C_d = 1 + \binom{2d}{2}$, it gives

$$C_d m^{2d-2} \leq cH, \quad (\text{R20})$$

because $\Lambda_d = C_d / \kappa_d$.

Apply Lemma II.9.4 with this kernel. Lemma II.9.2 gives

$$Q \geq L + 1 \geq \frac{4}{a}, \quad P_0 \geq (4q)^{-d} \geq \frac{a^{2d}}{(400d)^d} = 2c. \quad (\text{R21})$$

The input hypothesis and the bound $T_d(B) \leq H$ therefore imply

$$\mathbb{E}X \geq P_0 Q \alpha H, \quad \mathbb{E}Y \leq P_0 H + C_d m^{2d-2}. \quad (\text{R22})$$

Consequently

$$\begin{aligned} \mathbb{E}[\eta X - Y] &\geq P_0(\eta Q \alpha - 1)H - C_d m^{2d-2} \\ &\geq 3P_0 H - C_d m^{2d-2} \geq 5cH \geq cH. \end{aligned} \quad (\text{R23})$$

Some realization B' thus satisfies

$$\eta X - Y \geq cH. \quad (\text{R24})$$

There is no omitted class of tuples here: X and Y include all additive tuples in B' , with repeated entries counted. Hence

$$T_d(B') = X + Y \geq X \geq \frac{cH}{\eta} = \kappa_d \alpha^{2d} \eta^{2d-1} H, \quad (\text{R25})$$

and

$$\frac{T_d^{\phi}(B')}{T_d(B')} = \frac{X}{X+Y} \geq \frac{1}{1+\eta} \geq 1-\eta. \quad (\text{R26})$$

This proves both conclusions. $\square$

The weighted estimate is what makes the size condition depend on $(\alpha\eta)^{-2d}$. Counting all coefficient vectors separately would introduce a power of the bandwidth q . Instead, each unwanted relation is bounded by the same m^{2d-2} , and its contribution is multiplied by its Fourier weight; the weights have total mass one. The separate error for repeated entries has the same lower power of m . No independence assumption is made for repeated coordinates.

The exponent η^{2d-1} in the conclusion also deserves attention. The survival calculation supplies a quantity c proportional to $(\alpha\eta)^{2d}$, but (R24) yields $X \geq cH/\eta$. Keeping this final factor improves the retained count by η^{-1} .

II.9.4 Order eight and the comparison with Corollary 9.4

Corollary II.9.5 (Explicit order-eight restriction). *Let $0 < \beta, \gamma, \eta \leq 1$, let p be prime, and let $B \subset \mathbb{F}_p$ have size βp . Suppose that $\phi : B \rightarrow \mathbb{F}_p$ respects at least $\gamma|B|^3$ additive quadruples. If*

$$\boxed{p \geq 2^{102} \gamma^{-112} \eta^{-16} \beta^{-1}}, \quad (\text{R27})$$

then there is $B' \subset B$ containing at least

$$\boxed{2^{-95} \gamma^{112} \eta^{15} \beta^{15} p^{15}} \quad (\text{R28})$$

ordered additive 16-tuples, at least a proportion $1 - \eta$ of which are respected by ϕ .

Proof. Corollary II.8.2 permits $\alpha = \gamma^7$ in Theorem II.9.1 with $d = 8$. Its exact constants are

$$\kappa_8 = \frac{1}{2 \cdot 3200^8}, \quad \Lambda_8 = 242 \cdot 3200^8. \quad (\text{R29})$$

The integer inequalities

$$2 \cdot 3200^8 < 2^{95}, \quad 242 \cdot 3200^8 < 2^{102} \quad (\text{R30})$$

give the convenient powers of two in the statement. $\square$

The displayed count in source Corollary 9.4 is

$$\left(\frac{\gamma^7 \beta^6 \eta}{4} \right)^{2^{19}} \beta^{15} p^{15}. \quad (\text{R31})$$

The improvement has two distinct causes. Exact Parseval removes the β^6 inside the large power. The powered cosine replaces the repeated weak enrichment by one stronger selection, giving the powers $\gamma^{112} \eta^{15}$. Both changes preserve the same configuration normalization. The source corollary also inherits the sufficiently-large- p condition from Lemma 9.3; the present statement gives an explicit sufficient condition instead of suppressing it [1, 23].

This is a local improvement in the transition from a positive proportion of respected relations to an almost-homomorphism. Its relevance is not confined to the original exposition: later quantitative work also uses transitions of this kind, followed by additional steps that organize and extend the approximate homomorphism [6]. The present calculations do not by themselves supply an improved global bound for Szemerédi's theorem. Such a claim would require propagating the parameters through all later restrictions and density increments.

[write] **6 October 2026.** Rechecked at the write: at $\gamma = \eta = \beta = 1/2$, $(\gamma^7 \beta^6 \eta / 4)^{2^{19}} = 2^{-16 \cdot 2^{19}} = 2^{-8\,388\,608}$, while $2^{-95} \gamma^{112} \eta^{15} = 2^{-222}$ relative to $|B|^{15}$, under $p \geq 2^{102} \cdot 2^{112} \cdot 2^{16} \cdot 2 = 2^{231}$ (Section V.16). The formal `lemma_9_3` and `corollary_9_4` (Sections08_09.lean, lines 77 and 91; exact companions in `Proofs09Restriction.lean`, lines 376 and 416) quantify over all moduli $N \geq N_0$ without primality and are proved by another route; Corollary II.9.5 is a prime-field interface and does not prove them, as Section V.19 says. Not formalized.

II.9.5 What the exponent says about this selection method

The use of a higher-degree kernel raises a natural optimization question. There is a simple bound that explains why the powers in Theorem II.9.1 are natural for the particular expectation argument just used.

Proposition II.9.6 (A limitation of single-kernel amplification). *Let K be a real trigonometric polynomial with $0 \leq K \leq 1$, mean $b_0 > 0$, and Fourier coefficients b_ℓ . For $d \geq 2$, put*

$$Q_K = \frac{\sum_\ell |b_\ell|^{2d}}{b_0^{2d}}.$$

Then

$$Q_K \leq \frac{1}{b_0}. \quad (\text{R32})$$

Consequently, if $Q_K \geq 4/(\alpha\eta)$, the base survival probability b_0^{2d} is at most $(\alpha\eta/4)^{2d}$, and the scalar-multiple contribution $\sum_\ell |b_\ell|^{2d}$ to respected-tuple survival is at most $(\alpha\eta/4)^{2d-1}$.

Proof. Since $K \geq 0$, $|b_\ell| \leq b_0$. Parseval and $K^2 \leq K$ imply

$$\sum_\ell |b_\ell|^{2d} \leq b_0^{2d-2} \sum_\ell |b_\ell|^2 = b_0^{2d-2} \int_0^1 K(t)^2 dt \leq b_0^{2d-1}.$$

Divide by b_0^{2d} , and then use the assumed lower bound for Q_K . □

For an input with exactly αH respected tuples, the contribution guaranteed by the scalar-multiple coefficient relations is bounded by a constant times $\alpha^{2d} \eta^{2d-1} H$, under the stipulated amplification requirement. Theorem II.9.1 attains these parameter powers up to constants depending on d . This is an observation about that part of the single-kernel expectation budget. Additional relations and repeated entries can increase the actual expectation. The calculation is not a lower bound for every possible restriction procedure, does not rule out unusually favorable realizations, and does not establish optimality of the theorem's size threshold.

II.10 Arrangement moments retain the cross-section profile

Source 04, delivered Section 5.

The same principle applies to Lemmas 12.3 and 14.7, but their second moments carry multiplicities. The correct quantity to retain is the number of respected pairs of congruent cubes within a cross-section, rather than simply $|B|$.

II.10.1 The weighted representation

Let $B \subset \mathbb{F}_p^{k+1}$, with $k \geq 1$, let $\phi : B \rightarrow \mathbb{F}_p$, and write

$$B_r = \{y \in \mathbb{F}_p^k : (y, r) \in B\}.$$

For $h, y \in \mathbb{F}_p^k$, the axis-parallel cube with base y and sidelengths h has vertices

$$y + \epsilon \cdot h = (y_1 + \epsilon_1 h_1, \dots, y_k + \epsilon_k h_k), \quad \epsilon \in \{0, 1\}^k.$$

If all these vertices lie in B_r , define its value by

$$\Phi_h(y, r) = \sum_{\epsilon \in \{0, 1\}^k} (-1)^{\epsilon_1 + \dots + \epsilon_k} \phi(y + \epsilon \cdot h, r). \quad (\text{A1})$$

For each h , define the nonnegative integer weight

$$a_h(r, v) = \#\{y : y + \{0, 1\}^k \cdot h \subset B_r, \Phi_h(y, r) = v\}. \quad (\text{A2})$$

Let R_j count the respected j -arrangements: ordered lists of $2j$ cubes with common h , satisfying equal sums of the first and second halves of their cross-section indices, and equal sums of their cube values. Zero sidelengths and repeated vertices are included in this count, as in the moment identities of the source. Expanding the Fourier transform of a_h on $\mathbb{F}_p^2$ gives

$$R_j = \frac{1}{p^2} \sum_{h \in \mathbb{F}_p^k} \sum_{\chi \in \widehat{\mathbb{F}_p^2}} |\widehat{a}_h(\chi)|^{2j}. \quad (\text{A3})$$

In particular,

$$R_1 = \sum_{h, r, v} a_h(r, v)^2 \quad (\text{A4})$$

counts ordered congruent cube pairs in the same cross-section with equal values.

Theorem II.10.1 (Arrangement interpolation with profile). *Suppose that $R_2 > 0$. For every integer $j \geq 2$,*

$$R_j \geq \frac{R_2^{j-1}}{R_1^{j-2}}. \quad (\text{A5})$$

Define

$$\beta_r = \frac{|B_r|}{p^k}, \quad \beta = \frac{|B|}{p^{k+1}}, \quad \sigma = \frac{1}{p} \sum_r \beta_r^3. \quad (\text{A6})$$

Then

$$R_1 \leq \sigma p^{3k+1}, \quad \sigma \leq \beta. \quad (\text{A7})$$

Consequently, if $\theta > 0$ and $R_2 \geq \theta p^{5k+3}$, then

$$\boxed{R_j \geq \theta^{j-1} \sigma^{-(j-2)} p^{(2j+1)k+2j-1}}. \quad (\text{A8})$$

Proof. Apply the Hölder calculation of Theorem II.8.1 to the spectrum indexed by (h, χ) , using the common normalization p^{-2} in (A3). This gives (A5). The hypothesis $R_2 > 0$ ensures $R_1 > 0$, so the denominator is legitimate.

To bound R_1 , temporarily ignore the equality of cube values. A pair of cubes in B_r , with bases y, z and common sidelengths h , determines a triple

$$(y, z, y + h) \in B_r^3. \quad (\text{A9})$$

The last point is the opposite vertex of the first cube. This map is injective: the triple determines both bases and $h = (y + h) - y$. Hence there are at most $|B_r|^3$ such pairs, even before the value condition is imposed. Summing over r gives

$$R_1 \leq \sum_r |B_r|^3 = \sigma p^{3k+1}.$$

Since $0 \leq \beta_r \leq 1$, and $p^{-1} \sum_r \beta_r = \beta$, we have $\sigma \leq \beta$. Finally substitute these estimates into (A5). The ambient exponent is

$$(5k + 3)(j - 1) - (3k + 1)(j - 2) = (2j + 1)k + 2j - 1,$$

which proves (A8). $\square$

For $j = 8$, the replacement for Lemma 14.7 is therefore

$$R_8 \geq \theta^7 \sigma^{-6} p^{17k+15} \geq \theta^7 \beta^{-6} p^{17k+15}. \quad (\text{A10})$$

The source uses the same fourth-moment hypothesis but discards the factor σ when bounding the second moment. If every cross-section has density β , then $\sigma = \beta^3$, so the gain is β^{-18} . More generally, $\beta_r \leq K\beta$ for every r implies

$$\sigma \leq K^2 \beta^3. \quad (\text{A11})$$

This records exactly which information about the cross-sections can be used by the interpolation step.

Using Corollary 14.6 as stated in the repository's corrected transcription, the density power in Lemma 14.8 improves from $7 \cdot 4^{k+1}$ to $7 \cdot 4^{k+1} - 6$:

$$R_8 \geq \beta^{7 \cdot 4^{k+1} - 6} \gamma^{21k \cdot 4^{k+1}} p^{17k+15}. \quad (\text{A12})$$

Keeping the profile gives the stronger factor $\beta^{7 \cdot 4^{k+1}} \sigma^{-6}$. The analogous use in Lemma 12.4 changes $\beta^{112} \gamma^{336} p^{32}$ to $\beta^{106} \gamma^{336} p^{32}$. In that section the relevant fibers are vertical; its two coordinate roles are exchanged relative to the notation above. These conclusions use only the improved moment step, with the preceding hypotheses and estimates retained [1, 23].

II.10.2 Equality examples and the scope of sharpness

The arrangement interpolation and the profile bound extend to axis-parallel cubes over any finite abelian group G ; the proof uses only its characters and additive operations. This extension supplies a sharp example for the factor β^{-6} . Let $|G| = N$, let $C \leq G$, and take

$$B = G^k \times C, \quad \phi = 0, \quad \beta = \frac{|C|}{N}.$$

The nonempty cross-sections are full. Direct counting gives

$$R_1 = \beta N^{3k+1}, \quad R_2 = \beta^3 N^{5k+3}, \quad R_j = \beta^{2j-1} N^{(2j+1)k+2j-1}. \quad (\text{A13})$$

Thus both the interpolation inequality and $\sigma \leq \beta$ are equalities. This proves sharpness for the finite-abelian formulation; it does not assert the existence of proper subgroups of a cyclic group of prime order.

The uniform-profile case also has equality examples over a prime field. Let V be a coordinate subspace of $\mathbb{F}_p^k$, and take $B = V \times \mathbb{F}_p$, with $\phi = 0$. Every cross-section has density $\beta = |V|/p^k$. A cube is contained in V precisely when its base and sidelength vector belong to V . Therefore

$$R_j = |V|^{2j+1} p^{2j-1}, \quad \sigma = \beta^3. \quad (\text{A14})$$

Again the moment and profile estimates are equalities. In particular, the improvement by β^{-18} in the uniform-profile order-eight bound is attained at the available densities of these examples.

These equality constructions identify which parts of the argument cannot be improved using only the displayed moments and profile. They leave open the use of additional geometric or distributional information in applications.

II.11 Source 07’s chapter: transport rigidity at Lemma 10.9 [write]

Added in the second write, 6 October 2026.

The next eight sections are source 07, a manuscript devoted to one step of Gowers’s Section 10: the passage, in Lemma 10.9, from a local model of differences to a Freiman homomorphism. Its spine, an approximate homomorphism made exact, is shared with source 02’s extension of Proposition 6.1, source 03’s Sections II.5–II.6, source 04’s restriction theorem and source 10’s fibre extraction (Section II.21), but no other source treats Lemma 10.9, and source 07 shares with them only the notion of Freiman order and a reference on carries. It is therefore printed as a chapter of its own. Its notation table (its Appendix A) comes first, then its delivered Sections 2–8 in order: the fibered model and the directional error law (Section II.13), transport profiles (Section II.14), the coupled-path theorem (Section II.15), sharpness of $\delta < 1/(2k)$ (Section II.16), the application to Section 10 (Section II.17), the linear program for the starting law (Section II.18) and further extensions (Section II.19). Its introduction is Section F.12; its verification, questions, conclusion and reproduction commands are in Part V (Sections V.28–V.31).

Reading conventions. In this chapter δ is a failure probability (a directional defect), not a density; κ is a ratio of fibre sizes; $\mathcal{T}_s$ is a transport profile and $\mathcal{C}$ the cost of a word (not the cube count of Part III); $\theta, \eta, \alpha, \rho, \sigma$ and M are the parameters of Gowers’s Lemmas 10.6–10.9 as recorded in `Section10.lean`. The table at the head of this Part lists the clashes with the rest of the report.

What is proved, in one paragraph. A violated relation between two words in the directions must consume total failure probability at least one along a pair of coupled paths with shared endpoints (Theorem II.15.1); with fibre-size ratio at most κ this forces a Freiman homomorphism of order k once $2[1 + (k-1)\kappa]\delta < 1$ (Corollary II.15.3), and for constant fibres the threshold $\delta < 1/(2k)$ cannot be increased (Theorem II.16.1). Under the hypotheses of Gowers’s Lemma 10.9 this gives order five (Corollary II.17.3), and order nine when the ambient fibre invariance of the construction is kept (Corollary II.17.6). Five and nine are guaranteed orders, not maximal ones, and the sharpness concerns δ , not η (the source’s own non-claims, Section F.12).

Relation to the formal statement (checked at HEAD 764740f07). The Lean proposition `lemma_10_9` (`Section10.lean`, line 335; exact companion `lemma_10_9_holds`, `Proofs10Shift.lean`, line 717) concludes `FreimanHom 2 B' psi`. Corollary II.17.3 has exactly its hypotheses and concludes order five; Corollary II.17.6 needs in addition the fibre lower bound, the ambient invariance and the parameter relations (II.17.10), which in `Section10.lean` are hypotheses of `lemma_10_6` and of the Section 10 set-up but not of `lemma_10_9`. The precise comparison is in the notes after Corollaries II.17.3 and II.17.6. Neither statement is formalized.

II.12 Notation and assumptions at a glance

Source 07, delivered Appendix A.

Symbol	Meaning
G, H	Finite abelian index group and arbitrary abelian value group; Section II.19 also treats nonabelian groups.
X_g, R_g	Ambient fiber at index g and its cardinality; ambient fibers may be empty.
S, W, Ω	Nonempty usable support, union of its full fibers, and $ W $. Every R_g with $g \in S$ is positive.
$\mu(g)$	R_g/Ω on S , zero elsewhere. It is the index law of a uniform source point.
D, ϕ, ψ	Allowed directions, point-value map, and proposed directional difference map.
$F_d(g), \delta_d, b_d$	Conditional failure, its μ -mean, and source mass leaving S . Exits always count as failures.
$\mathcal{T}_s(t)$	Largest translated mass of a bounded error function of original mean at most t .
$c(s), A(s), M(s)$	Likelihood-ratio bound, positive mass discrepancy, and translated internal mass.
κ	Uniform ratio bound for fiber sizes on S .
L, r_0, τ	Ambient one-step fiber-size variation, internal lower bound, and $\tau = L/r_0$.
θ, η	Nested agreement error and overlap error in the Section 10 application.
$q(\theta), \Delta$	$2\theta - \theta^2$ and the convenient budget $q(\theta) + \eta$.
ν	An alternative initial-index law, supported on S ; the local error information remains measured using μ .

II.13 Finite fibered domains and the correct error law

Source 07, delivered Section 2.

Let G be a finite abelian group and H an abelian group, not necessarily finite. For each $g \in G$, let X_g be a finite set. Assume these sets are pairwise disjoint, and put $R_g = |X_g|$. Empty fibers are allowed. Fix a nonempty set

$$S \subseteq \{g \in G : R_g > 0\}, \quad W = \bigsqcup_{g \in S} X_g, \quad \Omega = |W|.$$

Let $\phi : \bigsqcup_g X_g \rightarrow H$, let $D \subseteq G$, and let $\psi : D \rightarrow H$ be the candidate difference map. The source-index law is

$$\mu(g) = \begin{cases} R_g/\Omega, & g \in S, \\ 0, & g \notin S. \end{cases} \quad (\text{II.13.1})$$

Thus choosing a point uniformly from W is the same as choosing its index with law μ , then choosing uniformly within that fiber.

Definition II.13.1 (Directional defect). For $d \in D$ and $g \in S$, set

$$F_d(g) = \begin{cases} 1, & g + d \notin S, \\ \frac{\#\{(x, y) \in X_g \times X_{g+d} : \phi(y) - \phi(x) \neq \psi(d)\}}{R_g R_{g+d}}, & g + d \in S. \end{cases} \quad (\text{II.13.2})$$

The total defect and boundary mass are

$$\delta_d = \sum_{g \in S} \mu(g) F_d(g), \quad b_d = \sum_{\substack{g \in S \\ g+d \notin S}} \mu(g). \quad (\text{II.13.3})$$

Operationally, start with a uniform point $x \in W$. A step in direction d is an automatic failure if its destination index lies outside S . Otherwise independently choose a uniform point y in the destination fiber and test the difference. This is a genuine probability experiment, and $0 \leq b_d \leq \delta_d \leq 1$.

An equivalent counting formula is

$$\delta_d = b_d + \frac{1}{\Omega} \sum_{\substack{g \in S \\ g+d \in S}} \frac{\#\{(x, y) \in X_g \times X_{g+d} : \phi(y) - \phi(x) \neq \psi(d)\}}{R_{g+d}}. \quad (\text{II.13.4})$$

The cancellation of R_g is worth noting. Uniform source points, uniform indices, and uniform ordered pairs across all fibers give different statistics when fiber sizes vary. All subsequent bounds use (II.13.3), not an unnormalized pair-counting surrogate.

Declaring a departure from W to be a failure is conservative: some outside fibers may exist and still carry correct differences. The advantage is that no later step depends on an unproved assertion about those fibers. In particular, we never divide by the size of an empty fiber.

Definition II.13.2 (Freiman order). For $k \geq 1$, a map $\psi : D \rightarrow H$ is a Freiman homomorphism of order k if, for all $d_1, \dots, d_k, e_1, \dots, e_k \in D$,

$$\sum_{i=1}^k d_i = \sum_{i=1}^k e_i \implies \sum_{i=1}^k \psi(d_i) = \sum_{i=1}^k \psi(e_i).$$

This definition makes sense for an empty direction set, when the assertion is vacuous. If $D \neq \emptyset$, preservation of order k implies preservation of every smaller positive order: pad both sides with repetitions of one fixed element of D , then cancel their images.

II.14 Transport profiles instead of square-root losses

Source 07, delivered Section 3.

II.14.1 An exact finite optimization

For $s \in G$, the shifted law is $\mu_s(g) = \mu(g - s)$. We only need its restriction to S ; mass outside S corresponds to discarded locations.

Definition II.14.1 (Transport profile). For $0 \leq t \leq 1$, define

$$\mathcal{T}_s(t) = \max \left\{ \sum_{g \in S} \mu(g-s)a_g : 0 \leq a_g \leq 1, \quad \sum_{g \in S} \mu(g)a_g \leq t \right\}. \quad (\text{II.14.1})$$

Also define

$$c(s) = \max_{g \in S} \frac{\mu(g-s)}{\mu(g)}, \quad A(s) = \sum_{g \in S} (\mu(g-s) - \mu(g))_+, \quad M(s) = \sum_{g \in S} \mu(g-s). \quad (\text{II.14.2})$$

Here $x_+ = \max\{x, 0\}$. Every denominator in (II.14.2) is positive. The feasible set in (II.14.1) is nonempty and compact, so the maximum exists. The quantity $\mathcal{T}_s(t)$ is the largest translated mass that an error function of original mean at most t can acquire.

Proposition II.14.2 (Basic profile bounds). *The profile is nondecreasing, concave, and piecewise linear. Moreover,*

$$\mathcal{T}_0(t) = t, \quad \mathcal{T}_s(t) \leq \min\{c(s)t, t + A(s), M(s)\}. \quad (\text{II.14.3})$$

An exact evaluation is obtained by sorting the ratios $\lambda_g = \mu(g-s)/\mu(g)$ in decreasing order and filling original μ -mass up to t , using at most one fractional final coordinate.

Proof. Increasing the budget enlarges the feasible set. If a and a' are feasible for budgets t and t' , their convex combination is feasible for the corresponding convex combination of the budgets; the objective is linear. Taking maximizing vectors proves concavity.

For any feasible a ,

$$\sum_g \mu(g-s)a_g \leq c(s) \sum_g \mu(g)a_g \leq c(s)t.$$

Subtracting $\sum_g \mu(g)a_g$ and retaining only positive coefficient differences gives the bound $t + A(s)$. The condition $a_g \leq 1$ gives $M(s)$. For $s = 0$, the objective equals the constraint sum, and the constant vector $a_g = t$ attains t .

For the exact formula, use variables $x_g = \mu(g)a_g$. The problem becomes

$$\max \sum_g \lambda_g x_g, \quad 0 \leq x_g \leq \mu(g), \quad \sum_g x_g \leq t.$$

If $\lambda_g > \lambda_h$, $x_g < \mu(g)$, and $x_h > 0$, moving a small positive mass from h to g increases the objective. Repeating this exchange leaves all higher ratios filled before a lower one is used. Ties may be resolved arbitrarily. This proves the stated algorithm and its piecewise linearity, including zero-ratio coordinates. $\square$

Example II.14.3 (The profile can beat the coarse envelopes). On $G = \mathbb{Z}/2\mathbb{Z}$, take $S = G$ and $\mu = (1/3, 2/3)$. For the nonzero shift,

$$\mathcal{T}_1(t) = \begin{cases} 2t, & 0 \leq t \leq 1/3, \\ \frac{1}{2} + \frac{1}{2}t, & 1/3 \leq t \leq 1. \end{cases}$$

Here $c(1) = 2$ and $A(1) = 1/3$. At $t = 1/2$, the exact value is $3/4$, whereas $\min\{2t, t + 1/3, 1\} = 5/6$.

II.14.2 Comparison and additive invariance of fiber sizes

Lemma II.14.4 (Uniform and prefixwise transport). *Suppose $R_u/R_v \leq \kappa$ for all $u, v \in S$, where $\kappa \geq 1$. Then $c(s) \leq \kappa$ for every $s \in G$.*

Suppose in addition that, for some $r_0 > 0$, $L \geq 0$,

$$R_g \geq r_0 \quad (g \in S), \quad |R_{g+d} - R_g| \leq L \quad (g \in G, d \in D). \quad (\text{II.14.4})$$

Put $\tau = L/r_0$. If $s = d_1 + \dots + d_j$ with $d_i \in D$, then

$$c(s) \leq 1 + j\tau. \quad (\text{II.14.5})$$

When both hypotheses hold, the minimum of κ and $1 + j\tau$ is available.

Proof. If $g - s \notin S$, the ratio defining $c(s)$ is zero. Otherwise it equals R_{g-s}/R_g , which gives the first assertion. For the second, telescope (II.14.4) along the sequence from $g - s$ to g . All intermediate indices are allowed because invariance is assumed on all of G , not just on S . This gives $R_{g-s} \leq R_g + jL$, and division by $R_g \geq r_0$ proves the claim. $\square$

The uniform estimate does not become κ^j after j steps. We compare the translated law with the original law directly, rather than renormalizing the surviving sample at each step. This is one reason the path argument retains useful higher-order information.

II.14.3 A parameterized replacement for a nested shift estimate

Proposition II.14.5 (Average transport of a point property). *Let $Q \subseteq W$, with $|W \setminus Q|/\Omega \leq \varepsilon$. For an internal destination fiber put $a_g = |X_g \setminus Q|/R_g$. Start at an index with law μ and shift by d . The mean destination failure restricted to destinations in S is at most $\mathcal{T}_d(\varepsilon)$.*

Consequently, for every $u > 0$, a proportion at least

$$1 - b_d - \frac{\mathcal{T}_d(\varepsilon)}{u} \quad (\text{II.14.6})$$

of source points have a destination in S on whose fiber the fraction outside Q is at most u .

Proof. The original mean of a is $|W \setminus Q|/\Omega \leq \varepsilon$. The destination mean over internal shifts is

$$\sum_{\substack{g \in S \\ g+d \in S}} \mu(g) a_{g+d} = \sum_{h \in S} \mu(h-d) a_h \leq \mathcal{T}_d(\varepsilon).$$

Markov's inequality bounds the source mass of internal destinations with $a_{g+d} > u$ by this mean divided by u . Add the boundary mass b_d . $\square$

A square root appears if one insists on making the outer exceptional fraction and the inner failure threshold comparable. It is not intrinsic to the averaged statement. The next section uses the average itself, so no such balancing is necessary.

II.15 The coupled-path rigidity theorem

Source 07, delivered Section 4.

For a nonempty word $\mathbf{d} = (d_1, \dots, d_m)$ in D , define its prefixes $s_0 = 0$, $s_j = d_1 + \dots + d_j$, and its cost

$$\mathcal{C}(\mathbf{d}) = \sum_{j=1}^m \mathcal{T}_{s_{j-1}}(\delta_{d_j}). \quad (\text{II.15.1})$$

Theorem II.15.1 (Transport-sensitive polygon rigidity). *Let $\mathbf{d} = (d_1, \dots, d_m)$ and $\mathbf{e} = (e_1, \dots, e_n)$ be nonempty words in D , with $\sum_j d_j = \sum_j e_j$. If*

$$\mathcal{C}(\mathbf{d}) + \mathcal{C}(\mathbf{e}) < 1, \quad (\text{II.15.2})$$

then

$$\sum_{j=1}^m \psi(d_j) = \sum_{j=1}^n \psi(e_j). \quad (\text{II.15.3})$$

Equivalently, every relation violated by ψ has total cost at least one. Each profile in the hypothesis may be replaced by any valid upper bound from Proposition II.14.2.

Proof. We specify the entire finite experiment before using a union bound. Choose $r \in S$ with law μ . The first path has formal indices $r + s_0, \dots, r + s_m$, and the second has formal indices $r + t_0, \dots, r + t_n$, where $t_j = e_1 + \dots + e_j$. Their initial indices are the same, and their terminal indices are the same.

At the common initial position choose one uniform point $x \in X_r$. If the terminal index belongs to S , choose one uniform terminal point y in that fiber, independently of x . Use this same y for both paths. At all internal formal positions whose indices lie in S , choose uniform fiber points independently of each other, of x , and of y . Different formal positions remain independent even when their indices coincide. In particular, when the total shift is zero, the initial and terminal points are still two independent draws from the same fiber. Positions outside S receive dummy values and will be treated as failures.

For edge j on the first path, let E_j be the event that all the preceding indices $r + s_0, \dots, r + s_{j-1}$ lie in S , and either the next index leaves S , or its edge equation fails. “Preceding indices” means only index survival; it does *not* require preceding equations to hold. Thus this restriction depends only on r , not on the sampled fiber values.

Write $g = r + s_{j-1}$. For a surviving source index g , the conditional failure probability is exactly $F_{d_j}(g)$: its endpoint samples have the required independent uniform marginal laws. The mass of the possible starting index $r = g - s_{j-1}$ is $\mu(g - s_{j-1})$. Discarding the extra requirement that all earlier indices survive therefore gives

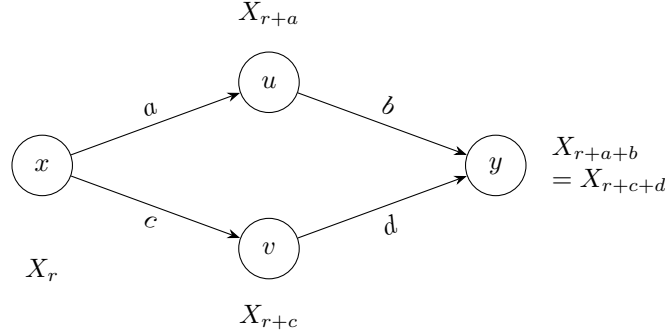
$$\begin{aligned} \mathbb{P}(E_j) &\leq \sum_{g \in S} \mu(g - s_{j-1}) F_{d_j}(g) \\ &\leq \mathcal{T}_{s_{j-1}}(\delta_{d_j}). \end{aligned} \quad (\text{II.15.4})$$

The last step uses $0 \leq F_{d_j} \leq 1$ and its original mean δ_{d_j} . The same construction for the second path gives events E'_j with the corresponding bounds.

If none of these events occurs, no path can leave S : its first exit would be one of the events. Every edge equation then holds. Telescoping along both paths yields

$$\sum_j \psi(d_j) = \phi(y) - \phi(x) = \sum_j \psi(e_j).$$

The union bound and (II.15.2) show that the probability of avoiding all events is positive, proving the asserted equality. Alternatively, if the equality fails, the union of the events is the whole sample space, so the sum of their upper bounds is at least one. $\square$



One shared start and one shared terminal; the two intermediate draws are independent, including when their fiber indices coincide.

For this four-edge diagram the criterion reads explicitly

$$\delta_a + \mathcal{T}_a(\delta_b) + \delta_c + \mathcal{T}_c(\delta_d) < 1. \quad (\text{II.15.5})$$

There is no independence assumption between edge failures. The union bound needs only their marginals. Conversely, conditioning each successive sample on all earlier equations being correct would change those marginals and would not justify (II.15.4).

Remark II.15.2 (Path-specific survival can be retained). For an edge with prefix s_{j-1} , the exact source-index weight in the proof is

$$\mu(g - s_{j-1}) \prod_{i=0}^{j-1} \mathbf{1}_S(g - s_{j-1} + s_i), \quad g \in S.$$

Replacing the translated weight in that edge's profile by this smaller weight gives another valid criterion. The simpler translation-only profile is useful because it depends on the prefix, not the full path history.

II.15.1 Higher-order consequences

Corollary II.15.3 (Explicit Freiman criteria). *Suppose $\delta_d \leq \delta$ for all $d \in D$, with $0 \leq \delta \leq 1$, and $k \geq 2$.*

If the fiber ratio on S is at most κ , then

$$2[1 + (k-1)\kappa]\delta < 1 \implies \psi \in \text{Fr}_k(D, H). \quad (\text{II.15.6})$$

Under (II.14.4),

$$[2k + \tau k(k-1)]\delta < 1 \implies \psi \in \text{Fr}_k(D, H). \quad (\text{II.15.7})$$

If both forms of control are available, it is enough that

$$2\delta \left[1 + \sum_{j=1}^{k-1} \min\{\kappa, 1 + j\tau\} \right] < 1. \quad (\text{II.15.8})$$

Proof. For a word of length k , the first cost is at most δ , because $\mathcal{T}_0(t) = t$. Every other cost is at most $\kappa\delta$, proving a one-path bound of $[1 + (k-1)\kappa]\delta$. Alternatively, Lemma II.14.4 bounds the $(j+1)$ -st cost by $(1 + j\tau)\delta$. Summing over $j = 0, \dots, k-1$ gives $[k + \tau k(k-1)/2]\delta$. Double the applicable one-path bound and use Theorem II.15.1 for an arbitrary order- k relation. Taking the smaller bound at each prefix proves (II.15.8). $\square$

In the important factor-two case, (II.15.6) is

$$(4k-2)\delta < 1. \quad (\text{II.15.9})$$

When the fibers have constant size on S , the sufficient condition is $2k\delta < 1$, even when S is a proper subset of G . Any boundary cost is already included in δ .

II.16 The balanced threshold is sharp

Source 07, delivered Section 5.

Theorem II.16.1 (Sharp uniform directional threshold). *For every $k \geq 2$, the universal strict threshold $\delta < 1/(2k)$ in the constant-fiber setting cannot be increased. More precisely, there are examples with full support, singleton fibers, identical finite cyclic source and target groups, and $\delta_d = 1/(2k)$ for every selected direction, whose candidate map is not a Freiman homomorphism of order k .*

Proof. Choose any integer $m \geq 2$ and put $N = 2km$. Take $G = H = \mathbb{Z}/N\mathbb{Z}$, $S = G$, and $X_r = \{r\}$. Using the representatives $0 \leq r < N$, define

$$\phi(r) = \left\lfloor \frac{r}{m} \right\rfloor \pmod{N}, \quad D = \{m, -m\}, \quad \psi(m) = 1, \quad \psi(-m) = -1. \quad (\text{II.16.1})$$

The two directions are distinct because $k \geq 2$.

For $0 \leq r < N - m$, shifting by m increases $\lfloor r/m \rfloor$ by one. On the final block of m residues, wraparound changes the value by $-(2k-1)$. This is not congruent to one modulo N , since their difference is $2k$ and $N = 2km > 2k$. Thus exactly m of the N source points fail, and $\delta_m = m/N = 1/(2k)$. Shifting by $-m$ has the analogous failure on the first block, so $\delta_{-m} = 1/(2k)$.

In the source group the k -term sums satisfy

$$\underbrace{m + \dots + m}_{k \text{ terms}} = \underbrace{(-m) + \dots + (-m)}_{k \text{ terms}},$$

because their difference is $2km = N$. Their proposed image sums are k and $-k$, whose difference $2k$ is nonzero modulo N . Therefore the order- k relation is violated. Its two-path budget is exactly $2k\delta = 1$, showing why strict inequality is necessary. $\square$

Example II.16.2 (An eight-point obstruction). For $k = m = 2$, the values of ϕ on $\mathbb{Z}/8\mathbb{Z}$ are

$$(0, 0, 1, 1, 2, 2, 3, 3).$$

With directions 2, 6 and labels 1, -1 , both defect fractions are $1/4$. Although $2 + 2 = 6 + 6$ modulo eight, the two image sums are 2 and 6. The domain has no missing fibers and no nonuniformity; the obstruction is a genuine accumulated wraparound error.

The example is naturally described as a carry or winding obstruction; carry phenomena and approximate homomorphisms also appear in [28]. What is established here is the exact extremal statement in Theorem II.16.1, with its specific directional sampling model.

The restriction to arbitrary selected direction sets matters. For example, if $D = G$, the constant-fiber criterion applied to the words (a, b) and $(a + b)$ shows that a uniform bound $3\delta < 1$ already forces $\psi(a + b) = \psi(a) + \psi(b)$. In particular, the sharp $1/4$ threshold for arbitrary-direction order-two relations is *not* a sharp threshold for full-group homomorphism recovery.

II.17 Application to Gowers's Section 10

Source 07, delivered Section 6.

II.17.1 The local input and its averaged consequence

The following formulation isolates the assumptions used from the repository interface [42]. Let $B \subseteq G$ be symmetric and $D = B' \subseteq B$. Suppose that W is a nonempty union of entire fibers, that their sizes on W vary by at most a factor of two, and that

$$|W \cap (W + d)| \geq (1 - \eta)|W| \quad (d \in B). \quad (\text{II.17.1})$$

Here $W + d$ denotes the union of the ambient fibers with indices in $S + d$; it is not a chosen bijection between individual fiber points. For each $d \in D$, suppose at least a fraction $1 - \theta$ of points $x \in W$ satisfy

$$\#\{y \in X_{r(x)+d} : \phi(y) - \phi(x) = \psi(d)\} \geq (1 - \theta)|X_{r(x)+d}|, \quad (\text{II.17.2})$$

where $r(x)$ denotes the index of x . For now assume $0 \leq \theta \leq 1$ and $\eta \geq 0$.

Because $-d \in B$, (II.17.1) implies

$$b_d = 1 - \frac{|W \cap (W - d)|}{|W|} \leq \eta. \quad (\text{II.17.3})$$

This is the precise point at which symmetry is used. The main path theorem itself does not require a symmetric direction set.

Lemma II.17.1 (Nested agreement to a single defect budget). *Under (II.17.2), put $q(\theta) = 2\theta - \theta^2$. Then*

$$\delta_d \leq 1 - (1 - \theta)(1 - \theta - b_d)_+ = \min\{1, q(\theta) + (1 - \theta)b_d\}. \quad (\text{II.17.4})$$

In particular, under (II.17.3),

$$\delta_d \leq q(\theta) + \eta. \quad (\text{II.17.5})$$

Proof. The good-source set in (II.17.2) has source-point mass at least $1 - \theta$. The set of sources with destination in S has mass $1 - b_d$. Their intersection has mass at least $(1 - \theta - b_d)_+$. On this intersection the destination fiber is nonempty, and the conditional success probability is at least $1 - \theta$. Therefore the overall success probability $1 - \delta_d$ is at least their product. Expanding gives the minimum formula, and $b_d \leq \eta$ gives (II.17.5). $\square$

This proof includes the case where (II.17.2) is vacuous on an empty ambient target fiber. Such a source lies outside the internal intersection and is charged to b_d . There is no hidden nonemptiness assumption.

For readable parameter comparisons we use the slightly larger budget

$$\Delta = q(\theta) + \eta. \quad (\text{II.17.6})$$

Every assertion below remains valid with the smaller budget from (II.17.4). Whenever a displayed criterion has the form $C\Delta < 1$ with $C \geq 1$, it automatically ensures $\Delta < 1$, as required when it is substituted for δ in the preceding results.

Theorem II.17.2 (A replacement exactification interface). *Under the local assumptions (II.17.1)–(II.17.2) and the factor-two fiber bound, for every $k \geq 2$,*

$$(4k - 2)(2\theta - \theta^2 + \eta) < 1 \implies \psi \in \text{Fr}_k(D, H). \quad (\text{II.17.7})$$

If ambient invariance (II.14.4) also holds, the alternative sufficient criterion is

$$[2k + \tau k(k - 1)](2\theta - \theta^2 + \eta) < 1. \quad (\text{II.17.8})$$

Proof. Use Lemma II.17.1, followed by Corollary II.15.3. No approximate-additivity or energy hypothesis beyond those already used to construct the local model is invoked here. $\square$

The density lower bound on B' in the repository's local-model definition is not needed by this exactification theorem. Its relevance lies in obtaining a large domain for later arguments, not in verifying a relation once its participating directions belong to B' .

II.17.2 Order five under the recorded Lemma 10.9 hypotheses

Corollary II.17.3 (Order five with no stronger smallness condition). *Under the local factor-two assumptions above, assume $\theta = 10\eta^{1/5}$ and $6\sqrt{\theta} < 1$. Then $\psi \in \text{Fr}_5(D, H)$.*

Proof. The assumptions give

$$\theta < \frac{1}{36}, \quad \eta < 360^{-5}.$$

Since q is increasing on $[0, 1]$,

$$q(\theta) < q(1/36) = \frac{71}{1296}.$$

The factor in (II.17.7) for $k = 5$ is eighteen. Hence

$$18\Delta < \frac{71}{72} + \frac{18}{360^5} < 1,$$

because $1296 < 360^5$. Apply Theorem II.17.2. $\square$

For the literal Lean statement there is a minor sign convention to address. If B' is empty, the order-five conclusion is vacuous. Otherwise choose $d \in B' \subseteq B$; nonemptiness of W and (II.17.1), together with $|W \cap (W + d)| \leq |W|$, force $\eta \geq 0$. Thus introducing a nonnegative real fifth root does not add an assumption in the nonvacuous case. The existing proof also derives this sign fact from overlap [43].

[write] 6 October 2026; sources 06–10. *Exactly the Lean hypotheses (checked at HEAD 764740f07).* lemma_10_9 (Section10.lean, line 335) assumes W nonempty, IsSection10ShiftRegular D W B eta (fibre saturation of W , symmetry of B , fibre sizes on W within a factor two, and $(1 - \eta)|W| \leq |W \cap D.\text{shift } W \ d|$ for $d \in B$), IsSection10LocalDifferenceModel D ϕ W B B' ψ θ with $\theta = 10\eta^{1/5}$ ($B' \subseteq B$, $(1 - \theta)|B| \leq |B'|$, and for $d \in B'$ the nested statement that a proportion $1 - \theta$ of $w \in W$ have a proportion $1 - \theta$ of z in the fibre over $\text{index}(w) + d$ with $\phi(z) - \phi(w) = \psi(d)$), and $6\sqrt{\theta} < 1$; it concludes FreimanHom 2 B' ψ , the Mathlib notion of the definition in Section II.13 with $G = H = \mathbb{Z}/N\mathbb{Z}$. These are the assumptions of Corollary II.17.3: the overlap is (II.17.1), the nested condition is (II.17.2), and the density condition on B' is not used (paragraph after Theorem II.17.2). So Corollary II.17.3 is the statement of lemma_10_9 with FreimanHom 5 in place of FreimanHom 2. It is not formalized; the companion lemma_10_9_holds (Proofs10Shift.lean, line 717) proves order two, and both files are unchanged since the pin ae0e89574.

Proposition II.17.4 (A larger allowance at the same factor-two interface). *Retain the local factor-two assumptions and $\theta = 10\eta^{1/5}$. The condition*

$$0 \leq \eta \leq 120^{-5} \tag{II.17.9}$$

is sufficient for $\psi \in \text{Fr}_2(D, H)$.

Proof. Now $\theta \leq 1/12$, so $q(\theta) \leq 23/144$. Therefore

$$6\Delta \leq \frac{23}{24} + \frac{6}{120^5} < 1.$$

The case $k = 2$ of (II.17.7) proves the assertion. □

The ratio of this explicit allowance to 360^{-5} is $(360/120)^5 = 243$. This compares sufficient conditions for the local step only; it does not assert a factor-243 improvement in any final progression-length or density estimate.

II.17.3 Retaining the full ambient invariance

The construction in Section 10 supplies stronger fiber control than the minimal shift-regular interface records. For its application here, the needed inequalities are the following [42]:

$$\begin{aligned} R_g &\geq \alpha^2 M / 16 \quad (g \in S), & |R_{g+d} - R_g| &\leq \sigma M \quad (g \in G, d \in B), \\ 0 < \alpha &\leq 1, \quad M > 0, \quad 0 \leq \rho &\leq \alpha^2 / 32, \quad 0 \leq \sigma &\leq \eta \rho \alpha^2 / 16. \end{aligned} \tag{II.17.10}$$

The relevant transport parameter is consequently

$$\tau = \frac{16\sigma}{\alpha^2} \leq \eta \rho \leq \eta. \tag{II.17.11}$$

Only these displayed consequences, not a new construction of W , are used in this subsection.

Theorem II.17.5 (An explicit higher-order range). *Assume the local model, the factor-two regularity, and (II.17.10). Set $\theta = 10\eta^{1/5}$. For every integer $k \geq 2$,*

$$0 \leq \eta \leq (40k)^{-5} \implies \psi \in \text{Fr}_k(D, H). \quad (\text{II.17.12})$$

More precisely, in the range $\eta \geq 0$ and $0 \leq \theta \leq 1$, it suffices that

$$[2k + \eta k(k-1)](2\theta - \theta^2 + \eta) < 1. \quad (\text{II.17.13})$$

Proof. The precise criterion follows from (II.17.11) and (II.17.8). To verify the convenient allowance (II.17.12), observe that

$$\theta \leq \frac{1}{4k}, \quad \Delta \leq \frac{1}{2k} - \frac{1}{16k^2} + \eta.$$

Using $k(k-1) \leq k^2$ and expanding gives

$$\begin{aligned} [2k + \eta k(k-1)]\Delta &\leq (2k + \eta k^2) \left(\frac{1}{2k} - \frac{1}{16k^2} + \eta \right) \\ &= 1 - \frac{1}{8k} + \frac{5}{2}k\eta - \frac{\eta}{16} + k^2\eta^2 \\ &\leq 1 - \frac{1}{8k} + \frac{5}{2}k\eta + k^2\eta^2. \end{aligned}$$

The hypothesis implies $\eta \leq 1/(64k^2)$, since $(40k)^5 \geq 64k^2$. Hence the last two positive terms are at most

$$\frac{5}{128k} + \frac{1}{4096k^2} \leq \frac{161}{4096k} < \frac{1}{8k}.$$

The strict inequality needed for (II.17.13) follows, including at the endpoint $\eta = (40k)^{-5}$. $\square$

Corollary II.17.6 (Order nine in the original parameter range). *Under the full assumptions of Theorem II.17.5, the original condition $6\sqrt{10\eta^{1/5}} < 1$ implies $\psi \in \text{Fr}_9(D, H)$.*

Proof. It gives $\eta < 360^{-5} = (40 \cdot 9)^{-5}$, so use Theorem II.17.5 with $k = 9$. $\square$

The distinction between Corollaries II.17.3 and II.17.6 is important. Order five follows from the assumptions exposed in the recorded local shift interface. Order nine additionally retains ambient invariance and a fiber lower bound from the construction. Omitting those extra hypotheses from an order-nine theorem would not be justified by this proof.

[write] **6 October 2026; sources 06–10.** *Order nine is not a corollary of the Lean statement.* Theorem II.17.5 and Corollary II.17.6 use, besides the hypotheses of `lemma_10_9` listed in the previous note, the package (II.17.10): fibre sizes at least $\alpha^2 M/16$ on S ; $|R_{g+d} - R_g| \leq \sigma M$ for all $g \in G$ and $d \in B$; $0 < \alpha \leq 1$, $M > 0$, $0 \leq \rho \leq \alpha^2/32$ and $0 \leq \sigma \leq \eta\rho\alpha^2/16$. None of these is a hypothesis of `lemma_10_9`. In `Section10.lean` they occur in other places: the fibre lower bound in `IsSection10RegularComponent`, the ambient invariance as `DomainInvariant D B (sigma * M)` and the bounds on α , M in `Section10Setup`, and $\rho \leq \alpha^2/32$, $\sigma \leq \eta\rho\alpha^2/16$ as hypotheses of `lemma_10_6` (there with $\sigma > 0$ and no explicit sign condition on ρ). A formal order-nine theorem therefore needs a new statement carrying these assumptions, for example one under the hypotheses of `lemma_10_6`, not a corollary of `lemma_10_9`. The order-five and order-nine statements are kept separate here, as the source keeps them.

II.17.4 A larger order-two allowance with ambient invariance

Proposition II.17.7 (The allowance 75^{-5}). *Under the full local and ambient assumptions of Theorem II.17.5, the condition*

$$0 \leq \eta \leq 75^{-5} \quad (\text{II.17.14})$$

is sufficient for $\psi \in \text{Fr}_2(D, H)$.

Proof. We have $\theta \leq 2/15$, so $q(\theta) \leq 56/225$. It follows that

$$\begin{aligned} (4 + 2\eta)\Delta &\leq (4 + 2\eta) \left(\frac{56}{225} + \eta \right) \\ &= \frac{224}{225} + \left(4 + \frac{112}{225} \right) \eta + 2\eta^2. \end{aligned}$$

Because $\eta \leq 75^{-5} < 1/4$, the last two terms are at most 5η . Indeed, $4 + 112/225 + 1/2 < 5$. Finally $5\eta \leq 5/75^5 < 1/225$, so the total is strictly below one. This is (II.17.13) for $k = 2$. $\square$

The two explicit endpoints and their ratio are

$$\begin{aligned} 360^{-5} &= \frac{1}{6046617600000} \approx 1.65382 \times 10^{-13}, \\ 75^{-5} &= \frac{1}{2373046875} \approx 4.21399 \times 10^{-10}, \\ \frac{75^{-5}}{360^{-5}} &= \left(\frac{24}{5} \right)^5 = 2548.03968. \end{aligned} \quad (\text{II.17.15})$$

These are transparent, rounded sufficient thresholds, not optimized roots. The exact inequalities (II.17.7) and (II.17.13), or the profiles themselves, should be used when constant-level optimization matters.

What does not follow. The earlier extraction of a large regular component, the size of B' , subsequent Bohr-set shrinkage, and the final density iteration all retain their own requirements. Enlarging the allowance at this one compatibility step does not remove them. In particular, neither the number 243 nor 2548.03968 is a claimed improvement factor in a global Szemerédi bound.

II.18 Choosing the initial law by a rational linear program

Source 07, delivered Section 7.

The original source law μ is useful because it is the law for which local errors are known. It need not be the best law from which to start a particular pair of paths. The distinction can be handled exactly.

Let ν be any probability law supported on S . Keep the same local functions F_d and their μ -means δ_d , but define

$$\mathcal{T}_s^\nu(t) = \max \left\{ \sum_{g \in S} \nu(g - s) a_g : 0 \leq a_g \leq 1, \quad \sum_{g \in S} \mu(g) a_g \leq t \right\}. \quad (\text{II.18.1})$$

Proposition II.18.1 (Alternative-root certificate). *Theorem II.15.1 remains valid after replacing every $\mathcal{T}_s$ by $\mathcal{T}_s^\nu$, provided that the same law ν is used for both paths. The choice of ν may depend on the relation being verified.*

Proof. Start the common initial index with law ν , still choosing uniformly within its fiber. The source-index weight for an edge of prefix s is bounded by $\nu(g - s)$, while F_d still has known μ -mean δ_d . Thus (II.15.4) becomes the corresponding bound with $\mathcal{T}_s^\nu$. All other steps are unchanged. $\square$

Lemma II.18.2 (An exact dual expression). *For every supported law ν , shift s , and $0 \leq t \leq 1$,*

$$\mathcal{T}_s^\nu(t) = \min_{\lambda \geq 0} \left[\lambda t + \sum_{g \in S} (\nu(g - s) - \lambda \mu(g))_+ \right]. \quad (\text{II.18.2})$$

Proof. For any feasible vector a and any $\lambda \geq 0$,

$$\begin{aligned} \sum_g \nu(g - s) a_g &= \lambda \sum_g \mu(g) a_g + \sum_g [\nu(g - s) - \lambda \mu(g)] a_g \\ &\leq \lambda t + \sum_g (\nu(g - s) - \lambda \mu(g))_+. \end{aligned}$$

This proves one inequality. For the reverse, order the ratios $\nu(g - s)/\mu(g)$ and use the filling construction in Proposition II.14.2. Choose λ to be the ratio of the possibly fractional final coordinate. Every positive coefficient in the second sum is then fully filled, every negative one is unfilled, and the remaining coordinate has zero coefficient. This makes the displayed bound an equality. At $t = 0$, choose λ at least the largest ratio; at $t = 1$, choose $\lambda = 0$. Budget boundaries and tied ratios give the same equality by the identical argument. $\square$

Theorem II.18.3 (Optimizing the root is a finite linear program). *Fix two equal-total words. Index all their edges by a finite set I , and let edge i have prefix s_i and direction d_i . The smallest alternative-root budget equals the optimum of*

$$\begin{aligned} &\text{minimize} \quad \sum_{i \in I} \left[\lambda_i \delta_{d_i} + \sum_{g \in S} z_{i,g} \right], \\ &\text{subject to} \quad \nu(g) \geq 0 \ (g \in S), \quad \sum_{g \in S} \nu(g) = 1, \quad \nu(g) = 0 \ (g \notin S), \\ &\quad \lambda_i \geq 0, \quad z_{i,g} \geq 0, \quad z_{i,g} \geq \nu(g - s_i) - \lambda_i \mu(g) \quad (i \in I, g \in S). \end{aligned} \quad (\text{II.18.3})$$

An optimum below one certifies the image relation. When the input laws and budgets are rational, an optimal rational certificate exists.

Proof. For fixed ν , independently minimizing over each λ_i and $z_{i,g}$ gives exactly the sum of the dual expressions (II.18.2). Minimizing also over ν therefore gives the desired smallest budget. Proposition II.18.1 proves the certificate claim.

For attainment and rationality, let $\mu_{\min} = \min_{g \in S} \mu(g) > 0$. One may restrict $0 \leq \lambda_i \leq 1/\mu_{\min}$ without worsening the objective: at the upper endpoint every coefficient $\nu(g - s_i) - \lambda_i \mu(g)$ is already nonpositive. Likewise one may restrict $0 \leq z_{i,g} \leq 1$, since the minimal feasible positive part never exceeds one. The resulting feasible set is a nonempty compact polytope. With rational data it is a rational polytope, and a linear objective attains its minimum at a rational vertex. $\square$

Taking $\nu = \mu$ recovers the earlier theorem, so this optimization never weakens that certificate. It is not claimed that every compatible relation admits a budget below one. The profiles optimize over arbitrary bounded error functions with the specified means; the errors of an actual map ϕ satisfy additional simultaneous constraints that this relaxation need not exploit.

For a computation one can use any candidate feasible rational solution of (II.18.3); an optimality proof is unnecessary when its objective is already below one. This separates discovering a certificate from checking one. The supplied script checks the exact primal–dual profile identity for many finite instances, but does not include a general LP solver.

II.19 Further proved extensions

Source 07, delivered Section 8.

II.19.1 Recovering labels from collision probabilities

So far the labels $\psi(d)$ were given. They can instead be selected from a concentrated directional difference distribution. For $h \in H$, define

$$p_d(h) = \sum_{\substack{g \in S \\ g+d \in S}} \mu(g) \frac{\#\{(x, y) \in X_g \times X_{g+d} : \phi(y) - \phi(x) = h\}}{R_g R_{g+d}}. \quad (\text{II.19.1})$$

This distribution has finite support even if H is infinite, and total mass $s_d = 1 - b_d$.

Proposition II.19.1 (Collision-to-mode exactification). *Assume $s_d > 0$ and put*

$$C_d = \sum_{h \in H} \left(\frac{p_d(h)}{s_d} \right)^2.$$

Choose $\psi(d)$ to maximize $p_d(h)$. Its directional defect satisfies

$$\delta_d = 1 - \max_h p_d(h) \leq 1 - s_d C_d. \quad (\text{II.19.2})$$

In particular, if $C_d \geq 1 - \varepsilon_d$, then

$$\delta_d \leq b_d + (1 - b_d)\varepsilon_d.$$

Substituting these budgets into Theorem II.15.1 or Corollary II.15.3 gives the corresponding exact relations among the selected modes.

Proof. For $q_h = p_d(h)/s_d$, the values q_h are a probability distribution. Hence

$$C_d = \sum_h q_h^2 \leq (\max_h q_h) \sum_h q_h = \max_h q_h.$$

Multiplying by s_d and using the definition of the defect proves the claim. If a selected label has defect below $1/2$, it is the unique label with that property, since two disjoint success events cannot both have probability greater than $1/2$. $\square$

The collision experiment is two independent samples from the normalized internal difference law. It is not automatically the same as a raw additive-energy statistic, which weights pairs of fibers differently. An application starting from energy must separately prove the necessary normalization or comparison. This proposition does not bypass that step.

II.19.2 Weighted conditional fiber laws

The finite uniform model is not essential to the main theorem. More generally, put any probability law π_g on each nonempty usable fiber, and any positive probability law μ on the finite support S . Define $F_d(g)$ by sampling independently from π_g and π_{g+d} on internal edges, and again set it equal to one on exits. Define δ_d by its μ -mean and keep the same transport profiles.

Proposition II.19.2 (Weighted version). *Theorems II.15.1 and II.18.3 remain valid for these arbitrary conditional laws. The fiber-size estimates of Lemma II.14.4 are replaced, where needed, by direct bounds on $\mu(g-s)/\mu(g)$.*

Proof. Choose the initial index with the stated law and every formal fiber point with its prescribed conditional law, still sharing the endpoints and keeping distinct formal internal positions independent. The conditional edge-failure probability is the newly defined $F_d(g)$. The proof of (II.15.4) and all transport optimizations depend only on its range and mean, not on uniformity of the fiber samples. $\square$

II.19.3 Nonabelian source and target groups

Theorem II.19.3 (Ordered word relations). *Let G be a finite group and H any group, now written multiplicatively. Use right steps $g \mapsto gd$, and test $\phi(x)^{-1}\phi(y) = \psi(d)$ on an internal edge. Define the transport profile with coefficient $\mu(gs^{-1})$ for prefix s . For two nonempty words with the same ordered product in G , a total profile budget below one forces equality of their ordered image products in H .*

Proof. For a prefix s , the equation $g = rs$ gives starting index $r = gs^{-1}$, so the edge marginal has precisely the claimed coefficient. Choose shared initial and terminal samples exactly as in Theorem II.15.1. On the event that all edge equations hold,

$$\prod_{j=1}^m \psi(d_j) = [\phi(x_0)^{-1}\phi(x_1)][\phi(x_1)^{-1}\phi(x_2)] \cdots [\phi(x_{m-1})^{-1}\phi(x_m)] = \phi(x_0)^{-1}\phi(x_m),$$

with products in their original order. Both paths have the same endpoints. The identical union bound now proves the result without any commutativity assumption. $\square$

This is a relation-wise, direction-wise statement. It should again be distinguished from a theorem deriving closeness to a global homomorphism from one averaged multiplication test, as in [27].

II.20 An explicit weighted replacement for the energy extraction step

Source 10, delivered Section 7.

The rather large powers in Proposition 7.3 of the source paper are not intrinsic to the energy extraction step. Two intermediate thresholding operations can be removed by retaining the representation multiplicities as weights. We give all constants, including the finite-size correction caused by diagonal pairs. This is a self-contained quantitative replacement for that proposition, not a claim to improve the modern Balog–Szemerédi–Gowers theorem.

For a nonempty finite subset A of an abelian group, write

$$m = |A|, \quad r_A(d) = |\{(a, b) \in A^2 : a - b = d\}|, \quad E(A) = \sum_d r_A(d)^2.$$

The ambient group need not be finite or torsion-free.

Theorem II.20.1 (An explicit weighted energy extraction). *Suppose $0 < c \leq 1$ and $E(A) \geq cm^3$. Then there is a nonempty subset $A_* \subseteq A$ such that*

$$|A_*| \geq \frac{c}{4}m, \quad |A_* - A_*| \leq 2^{10}c^{-5}m.$$

Proof. If $m \leq 32c^{-2}$, take $A_* = A$. Indeed, $|A - A| \leq m^2 \leq 32c^{-2}m \leq 2^{10}c^{-5}m$. We henceforth assume $m > 32c^{-2}$.

Define the symmetric matrix of weights

$$w(a, b) = \frac{r_A(a - b)}{m} \in [0, 1] \quad (a, b \in A)$$

and its weighted codegrees

$$T(a, b) = \sum_{v \in A} w(v, a)w(v, b).$$

Here diagonal pairs are retained throughout. The normalization was chosen so that

$$\sum_{a, b \in A} w(a, b) = \frac{E(A)}{m} \geq cm^2.$$

Call a pair (a, b) bad if $T(a, b) < c^2m/16$.

Choose v uniformly from A and, conditional on v , form a random subset $K \subseteq A$ by including each a independently with probability $w(v, a)$. If $d(v) = \sum_a w(v, a)$, then

$$\mathbb{E}(|K|^2 \mid v) = d(v)^2 + \sum_a w(v, a)(1 - w(v, a)) \geq d(v)^2.$$

Cauchy–Schwarz therefore gives

$$\mathbb{E}|K|^2 \geq \frac{1}{m} \sum_v d(v)^2 \geq c^2m^2.$$

Let Z be the number of ordered bad pairs in K^2 . For $a \neq b$, the probability that $a, b \in K$ is $T(a, b)/m$, hence is less than $c^2/16$ for a bad pair. There are at most m diagonal pairs, and each contributes at most one to the expectation. Consequently

$$\mathbb{E}Z \leq \frac{c^2m^2}{16} + m.$$

It follows that

$$\mathbb{E}(|K|^2 - 8Z) \geq \frac{c^2m^2}{2} - 8m > \frac{c^2m^2}{4}.$$

Fix one outcome with $|K|^2 - 8Z \geq c^2m^2/4$. This implies

$$|K| \geq cm/2, \quad Z \leq |K|^2/8.$$

Let A_* consist of those $a \in K$ for which at most $|K|/4$ elements $b \in K$ make (a, b) bad. Counting bad pairs by their first coordinate shows that $|A_*| \geq |K|/2 \geq cm/4$. Moreover, for every $a, b \in A_*$ there are at least $|K|/2$ elements $z \in K$ for which

$$T(a, z) \geq c^2 m/16, \quad T(b, z) \geq c^2 m/16.$$

Let $R(t)$ count the representations of t in the form $(p - q) - (r - s)$ with $p, q, r, s \in A$. For fixed $a, z \in A$,

$$R(a - z) \geq \sum_{u \in A} r_A(a - u) r_A(z - u) = m^2 T(a, z).$$

For completeness, the inequality involves no multiplicity error: each choice of u , a representation $p - q = a - u$, and a representation $r - s = z - u$ supplies a quadruple representing $a - z$; that quadruple recovers u from $u = a - (p - q)$.

For each of the above good choices of z , there are at least $c^4 m^6 / 2^8$ pairs of quadruples representing $a - z$ and $b - z$. Combining them gives representations of $a - b$ by ordered eight-tuples in A^8 . Representations coming from different z are disjoint: the first quadruple determines $a - z$ and hence determines z . Thus every $a - b$ with $a, b \in A_*$ has at least

$$\frac{|K|}{2} \frac{c^4 m^6}{2^8} \geq \frac{c^5 m^7}{2^{10}}$$

representations of this form. Choose one pair of endpoints (a, b) for every distinct member of $A_* - A_*$. An ordered eight-tuple has a unique value of the displayed alternating sum, so tuples counted for different differences are disjoint. There are only m^8 eight-tuples in total. Dividing proves $|A_* - A_*| \leq 2^{10} c^{-5} m$. $\square$

What is improved, and what is already known. The source paper gives $|A_*| \geq 2^{-20} c^{12} m$ and $|A_* - A_*| \leq 2^{38} c^{-24} m$. Theorem II.20.1 improves both quantitative guarantees and applies to arbitrary abelian groups. The mechanism is familiar: weighted dependent random choice, followed by counting many representations of each surviving difference. It should not be described as a new best theorem in additive combinatorics. For example, Reiher and Schoen prove that for every $\varepsilon \in (0, 1/2)$ one can obtain

$$|A'| \geq (1 - \varepsilon) c^{1/2} m, \quad |A' - A'| \leq 2^{33} \varepsilon^{-9} c^{-4} |A'|.$$

Their theorem is a substantially stronger modern benchmark [12, Theorem 1.2]. The advantage of the present replacement is a short proof with small, completely explicit constants that can be inserted directly into the source paper's chain of lemmas.

[write] **6 October 2026; sources 06–10.** Source 03's Theorem II.6.3 is the other elementary replacement, with $(c/8, 2^{19} c^{-9})$; this theorem is stronger in both conclusions for $0 < c \leq 1$ (note after source 03's theorem). Both imply the formal `proposition_7_3` bounds (`Sections06_07.lean`, line 135; exact companion `proposition_7_3_holds`); neither is formalized.

A parameterized version for later optimization. The same proof can retain its choices. Let $0 < \beta < \eta < 1/2$, $0 < \theta < \beta$, and $0 < \sigma < 1$ satisfy $\sigma^2 < 1 - \theta/\beta$. Put

$$M_0 = \frac{1}{\beta(1 - \theta/\beta - \sigma^2)c^2}.$$

If $m > M_0$, the proof with bad threshold $\theta c^2 m$ and potential $|K|^2 - \beta^{-1} Z$ gives

$$|A_*| \geq (1 - \beta/\eta)\sigma cm, \quad |A_* - A_*| \leq \frac{c^{-5}m}{(1 - 2\eta)\sigma\theta^2}.$$

For $m \leq M_0$ one can simply take $A_* = A$ and use $|A - A| \leq M_0 m$. The choice $(\beta, \eta, \theta, \sigma) = (1/8, 1/4, 1/16, 1/2)$ gives the theorem. The explicit m term in the random-pair calculation is essential: independence of the two membership events is available only off the diagonal. Omitting that term would leave a genuine gap for small sets.

II.21 A sharper obstruction-fibre form of the Freiman extraction

Source 10, delivered Section 8.

We next refine Lemma 7.5 of the source. This stage does not require the domain to be a cyclic group. What matters is that the *codomain* is the field of prime order.

Theorem II.21.1 (Extraction controlled by the vertical obstruction fibre). *Let G be an abelian group, let p be prime, let $B \subseteq G$ be finite and nonempty, and let $\phi : B \rightarrow \mathbb{F}_p$. Fix an integer $k \geq 1$, write $\Gamma = \{(x, \phi(x)) : x \in B\}$, and define*

$$H_k = \{y \in \mathbb{F}_p : (0, y) \in k\Gamma - k\Gamma\}, \quad h_k = |H_k|.$$

Then ϕ restricts to a Freiman homomorphism of order k on a subset $B' \subseteq B$ of size at least

$$|B'| \geq \frac{|B|}{k(h_k - 1) + 1}.$$

If, in addition, $|\Gamma - \Gamma| \leq C|\Gamma|$, then

$$h_k \leq C^{2k+1}, \quad |B'| \geq \frac{|B|}{kC^{2k+1}}.$$

Proof. The set H_k is symmetric and contains zero. If $h_k = 1$, then every relation between two sums of k elements of B already has zero vertical defect, so ϕ itself is a Freiman homomorphism of order k . Suppose henceforth that $h_k \geq 2$, and set

$$a = k(h_k - 1), \quad L = \left\lfloor \frac{p-2}{a} \right\rfloor.$$

For each integer j with $1 \leq j \leq kL$ and each $y \in H_k \setminus \{0\}$, the equation $jd = y$ excludes at most one $d \in \mathbb{F}_p^\times$. Indeed $kL \leq p-2$, so j is invertible. There are at most $kL(h_k - 1) = aL \leq p-2$ excluded nonzero choices of d , and there are $p-1$ nonzero choices available. Choose an allowed d . Symmetry of H_k ensures

$$jd \notin H_k \quad \text{whenever } 0 < |j| \leq kL.$$

For every $b \in \mathbb{F}_p$, consider

$$B_b = \{x \in B : \phi(x) \in b + d\{0, 1, \dots, L\}\}.$$

Since $L < p$ and $d \neq 0$, each $x \in B$ is counted in exactly $L+1$ of these sets. Some b thus satisfies $|B_b| \geq (L+1)|B|/p$. We claim

$$\frac{L+1}{p} \geq \frac{1}{a+1}.$$

If $a \geq p - 1$, then $L = 0$ and the assertion is immediate. If $a \leq p - 1$, Euclidean division of $p - 2$ by a gives $L + 1 \geq (p - 1)/a$, while $(p - 1)/a \geq p/(a + 1)$ because $p - 1 \geq a$.

Take $B' = B_b$. For each $x \in B'$ there is a unique integer $j(x) \in [0, L]$ such that $\phi(x) = b + dj(x)$. If $x_1 + \cdots + x_k = x'_1 + \cdots + x'_k$, then

$$d \left(\sum_{i=1}^k j(x_i) - \sum_{i=1}^k j(x'_i) \right) \in H_k.$$

The integer in parentheses lies in $[-kL, kL]$, and the chosen step d forces it to vanish. Hence the corresponding sums of ϕ values are equal, proving the Freiman property and the first bound.

For the small-doubling assertion, the map

$$H_k \times \Gamma \longrightarrow (k + 1)\Gamma - k\Gamma, \quad (y, \gamma) \longmapsto (0, y) + \gamma$$

is injective. Its first coordinate recovers γ because Γ is a graph, and then its second coordinate recovers y . The standard Plünnecke–Ruzsa inequality gives

$$|(k + 1)\Gamma - k\Gamma| \leq C^{2k+1} |\Gamma|.$$

For a small-difference hypothesis this follows by applying the sum-difference form of that inequality to the pair $-\Gamma, \Gamma$; see [36]. Consequently $h_k \leq C^{2k+1}$. Finally, $k(h_k - 1) + 1 \leq kC^{2k+1}$ for $k \geq 1$. $\square$

Why the fibre is the appropriate invariant. The set H_k records exactly the possible failures to respect length- k additive relations. The global size of a high sum-difference set is used only to bound h_k when no more specific information is available. In examples where the vertical defects occupy few values, the first bound in Theorem II.21.1 can be much stronger than its small-doubling corollary. If $H_k = \{0\}$ it correctly returns all of B , independently of the doubling constant. The dilation proof requires a prime codomain; the displayed argument does not extend unchanged to arbitrary composite cyclic groups.

Corollary II.21.2 (An explicit replacement for Corollary 7.6). *Let B_0 be a nonempty finite subset of an abelian group, let $\phi : B_0 \rightarrow \mathbb{F}_p$, and suppose the graph of ϕ has at least $\gamma|B_0|^3$ additive quadruples, where $0 < \gamma \leq 1$. For every integer $k \geq 1$ there is $B \subseteq B_0$ on which ϕ is a Freiman homomorphism of order k and*

$$|B| \geq \frac{2^{-24k-14}}{k} \gamma^{12k+7} |B_0|.$$

In particular, for order eight,

$$|B| \geq 2^{-209} \gamma^{103} |B_0|.$$

Proof. Apply Theorem II.20.1 to the graph, obtaining a subgraph Γ_1 with

$$|\Gamma_1| \geq \frac{\gamma}{4} |B_0|, \quad |\Gamma_1 - \Gamma_1| \leq 2^{10} \gamma^{-5} |B_0| \leq 2^{12} \gamma^{-6} |\Gamma_1|.$$

Set $C = 2^{12} \gamma^{-6}$ and use Theorem II.21.1. The retained proportion is at least

$$\frac{\gamma/4}{k(2^{12} \gamma^{-6})^{2k+1}} = \frac{2^{-24k-14}}{k} \gamma^{12k+7}.$$

When $k = 8$, $24k + 14 + \log_2 k = 209$ and $12k + 7 = 103$. $\square$

[write] **6 October 2026; sources 06–10.** *Checked at the second write.* The retained proportion is $(\gamma/4)/(k(2^{12}\gamma^{-6})^{2k+1}) = 2^{-2-12(2k+1)}k^{-1}\gamma^{12k+7}$, which at $k = 8$ is $2^{-209}\gamma^{103}$. *Against the formal statements* (HEAD 764740f07): `corollary_7_6` (Sections06_07.lean, line 197) assumes N prime, $|B_0| = \alpha N$ and $\gamma(\alpha N)^3 \leq$ the number of additive quadruples of the graph, and concludes a Freiman homomorphism of order eight on B with $|B| \geq 2^{-1882}\gamma^{1164}\alpha N$. The hypothesis forces $\gamma \leq 1$ (a graph on B_0 has at most $|B_0|^3$ additive quadruples), and then $2^{-209}\gamma^{103} \geq 2^{-1882}\gamma^{1164}$, so this corollary implies the formal conclusion with $\mathbb{F}_p = \mathbb{Z}/N\mathbb{Z}$. Likewise Theorem II.21.1 implies `lemma_7_5`'s $|B|/(8kC^{4k})$, since $C \geq 1$ when $|\Gamma - \Gamma| \leq C|\Gamma|$. Neither is formalized (exact companions `corollary_7_6_holds`, `lemma_7_5_holds` prove the printed constants).

A modern-input alternative. If the Reiher–Schoen theorem [12] is admitted as an external input instead of the self-contained energy extraction, the same fibre argument gives

$$|B| \geq \frac{(1-\varepsilon)\varepsilon^{9(2k+1)}}{k^{233(2k+1)}} \gamma^{8k+9/2} |B_0| \quad (0 < \varepsilon < 1/2).$$

For $k = 8$ the power of γ is $137/2$ rather than 103 . This is a corollary of a published result plus the preceding extraction calculation, and must be labelled as such. The self-contained and modern-input versions have different useful roles: the former has a short complete proof here, while the latter records a better presently available exponent. Neither by itself yields a new overall bound for Szemerédi's theorem; the later structural and iterative stages still have to be checked with their own quantified hypotheses.

Further questions arising specifically from these refinements. The obstruction fibres H_k may permit data-sensitive structure theorems stronger than replacing them immediately by C^{2k+1} . Natural precise projects are to bound $|H_k|$ from a graph's vertical energy profile, to study its growth with k , and to determine when an order- k homomorphism extends with small additional loss to order $k + 1$. The next subsection extends the prime-codomain dilation argument to a general finite abelian codomain by retaining the exact number of solutions of $jd = y$. For formalization, the two finite counting injections in the energy proof and the injection $H_k \times \Gamma \rightarrow (k + 1)\Gamma - k\Gamma$ are particularly suitable intermediate lemmas.

II.21.1 A torsion-sensitive extraction for general finite codomains

The prime-codomain dilation argument admits an explicit extension when the sizes of the multiplication fibres are retained. The resulting criterion records how the vertical defects meet the images jV .

Theorem II.21.3 (Extraction from a torsion and defect budget). *Let G be an abelian group, let $B \subseteq G$ be finite and nonempty, let V be a finite abelian group of order $n \geq 2$, and let $\phi : B \rightarrow V$. Fix an integer $k \geq 1$, and put*

$$\Gamma = \{(x, \phi(x)) : x \in B\}, \quad F = \{y \in V : (0, y) \in k\Gamma - k\Gamma\}.$$

For an integer $L \geq 0$ define

$$\mathcal{B}_{k,V,F}(L) = \sum_{j=1}^{kL} (|V[j]| |F \cap jV| - 1), \quad V[j] = \{v \in V : jv = 0\}.$$

If

$$\mathcal{B}_{k,V,F}(L) \leq n - 2, \quad (\text{II.21.1})$$

then ϕ restricts to a Freiman homomorphism of order k on a subset $B' \subseteq B$ of size at least

$$|B'| \geq \frac{L+1}{n} |B|. \quad (\text{II.21.2})$$

The subset can be chosen so that its image lies in $a + d\{0, \dots, L\}$ for some $a \in V$ and some $d \neq 0$. If $L > 0$, this step has order greater than kL . If $F = \{0\}$, the original map ϕ is already a Freiman homomorphism of order k on all of B , independently of the budget.

Proof. The defect set F is symmetric and contains zero. For $j \geq 1$ let

$$E_j = \{d \in V \setminus \{0\} : jd \in F\}.$$

Multiplication by j has image jV , and every nonempty fibre has $|V[j]|$ elements. Because $0 \in F$, the zero element is one of the preimages being counted. Removing it gives the exact identity

$$|E_j| = |V[j]| |F \cap jV| - 1. \quad (\text{II.21.3})$$

The union bound and (II.21.1) leave at least one nonzero d outside $\bigcup_{j=1}^{kL} E_j$. It satisfies

$$jd \notin F \quad \text{for } 1 \leq j \leq kL.$$

By symmetry of F , the same holds for $-kL \leq j \leq -1$. When $L > 0$, the inclusion $0 \in F$ also implies that the order of d is greater than kL . Hence the $L+1$ elements of $P = d\{0, \dots, L\}$ are distinct; this is immediate when $L = 0$.

For each $a \in V$ let

$$B_a = \{x \in B : \phi(x) \in a + P\}.$$

Every $x \in B$ occurs in exactly $L+1$ of these sets, since the possible translations are $a = \phi(x) - jd$ with $0 \leq j \leq L$. Thus

$$\sum_{a \in V} |B_a| = (L+1)|B|.$$

Choose a translate attaining at least this average, and put $B' = B_a$. For every $x \in B'$ there is a unique integer $t(x) \in [0, L]$ with $\phi(x) = a + dt(x)$.

Suppose $x_1 + \dots + x_k = x'_1 + \dots + x'_k$ with all the points in B' . Its vertical defect belongs to F and equals

$$dJ, \quad J = \sum_{i=1}^k t(x_i) - \sum_{i=1}^k t(x'_i), \quad |J| \leq kL.$$

The choice of d rules out every nonzero such integer J . Therefore $J = 0$, and the sums of the corresponding ϕ values agree. This proves the Freiman property and (II.21.2). Finally, if $F = \{0\}$, every length- k relation in the original domain has zero vertical defect by definition, proving the final assertion. $\square$

Corollary II.21.4 (Recovery of the prime-codomain estimate). *Under the hypotheses above, let $V = \mathbb{F}_p$ and $h = |F|$. There is a subset on which ϕ is a Freiman homomorphism of order k and whose cardinality is at least*

$$\frac{|B|}{k(h-1)+1}.$$

Proof. If $h = 1$, use all of B . Otherwise set

$$a = k(h - 1), \quad L = \left\lfloor \frac{p - 2}{a} \right\rfloor.$$

Then $kL \leq p - 2$, so multiplication by each $1 \leq j \leq kL$ is invertible. Every summand of the budget is $h - 1$, and

$$\mathcal{B}_{k, \mathbb{F}_p, F}(L) = kL(h - 1) \leq p - 2.$$

The theorem gives $(L + 1)|B|/p$. The integer inequality $p - 2 < a(L + 1)$ implies

$$p \leq a(L + 1) + 1 \leq (a + 1)(L + 1),$$

which proves the claimed bound, including the case $L = 0$. □

Scope of the criterion. The budget always allows $L = 0$, giving the constant-fibre bound $|B|/|V|$. Positive L requires a step of order greater than kL . For example, if $\exp(V) \leq k$, no positive L satisfies the budget: the summand with $j = \exp(V)$ is already $n - 1$. The separate case $F = \{0\}$ still returns the whole domain. Thus the theorem is a data-dependent extension of the dilation argument; it does not assert a uniform large-subset guarantee for every torsion profile. Sharper estimates for unions of the sets E_j , rather than their sum, are a natural way to improve this criterion.

Part III

Cube and progression counts

Contents. Counting consequences of uniformity. Sections III.1–III.2: source 03, delivered Sections 5–6 (progression counting by a density-weighted telescope; the exact fourth-order term of the centered cube count on every finite abelian group, its torsion correction, sharpness for sets, higher coefficients). Sections III.3–III.4: source 04, delivered Sections 6–7 (the sharp real centered gap $\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8$ on odd-order groups, the sharp leading constant $6\sqrt{2}$ of the three-cube bound, the fractional correction exponent $16/3$, and density-weighted progression counting with the lifting lemma).

Overlaps (dated notes). Cancellation through degree three and the quartic expansion (03 on every group, 04 on odd-order groups; $P_d = R_d$); the $\mathbb{Z}/3\mathbb{Z}$ cosine polynomial (identical in both); the generalized von Neumann bound and the telescoping theorem (independently written in both). **Cross-source answers.** Source 04’s Theorem III.3.4 answers source 03’s Research question V.14.1 for $d = 3$, with $c_3^{\text{odd}} = 2^{-1/2}$. Source 04’s Lemma III.4.5 is the only statement of the five sources that is formalized (`hasNatAP_of_short_modular_sequence`).

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
cube count	03: $Q_d(F)$, $m = 2^d$ vertices, $\eta = \ f\ _{U^d}$; 04: $Q_d(W)$, $q = 2^d$, $u = \ f\ _{U^d}$, $v = \ f\ _{U^2}$. <i>04’s q here is not its kernel degree q of Part II.</i>
R_d, P_d, T_d	03: R_d = number of ordinary parallelograms, T_d = binary tetrahedra, $K_d = R_d + T_d$, $\mathcal{R}_d$ the remainder; 04: P_d = number of parallelograms (= 03’s R_d), R_d, R_6, R_7 remainders. <i>04’s R_d is not 03’s R_d.</i>
$\mathcal{E}, B$	03: $\mathcal{E}(f) = \ f\ _{U^2}^4$, $\mathcal{E}_{(2)}(f)$ the order-two (quotient) energy, $B(S)$ the telescoping recursion; 04: $\mathcal{E}(\lambda)$ the additive energy of $\lambda_r = a_r ^2$, $B_5(f)$ the quintic correlation, $D(S)$ the recursion (= 03’s $B(S)$).
slopes, α	03: slopes a_i with invertible differences, Λ_a , $\eta = \ 1_A - \delta\ _{U^{k-1}}$; 04: admissible slopes c_i , Λ_c , u . Both: Gowers’s α -uniformity means $\ f\ _{U^d}^{2^d} \leq \alpha$; <i>α is a norm power, not a norm.</i>

Added in the second write (6 October 2026): sources 06–10. Contents. Section III.5: source 06, delivered Section 3 (a generalized von Neumann bound keeping two distinguished functions, first and second telescoping, a progression discrepancy quadratic in the uniformity norm, the sharp three-term variance factor, an explicit modular existence criterion). Sections III.6–III.15: source 08, delivered Sections 2–10 and Appendix A (cancellation through degree three, the exact quartic term with its order-two correction, the complete odd-order three-cube polynomial, the counterexample to the quartic lower bound with its exact certificate, sharpness for sets, Fourier stability, inhomogeneous weights, the character-kernel survival criterion). Sections III.16–III.27: source 09, delivered Sections 2–10, 13 and Appendices A–B, the fullest treatment of the cube count (complete four- and five-vertex classifications on every finite abelian group with Stirling multiplicities, the exact fourth and fifth orders, signs and sharpness through order six, the determinant-five configuration and the all-prime family, heterogeneous expansions, and the relation to progression bounds). Sections III.28, III.29 and III.30: source 10, delivered Sections 5, 6 and 9 (a torsion-sensitive mixed- L^p generalized von Neumann inequality and its sharpness; density-weighted counting and modular criteria; the exact dimension threshold for polynomial colourings).

Overlaps (dated notes at both ends). Cancellation through degree three: 03, 04, 08, 09. The exact quartic term on every group: 03, 08, 09 ($R_d = P_d$, T_d , $K_d = F_d = A_d$). The complete odd-order three-cube polynomial: 08’s Theorem III.9.2 contains 04’s support count and is the $d = 3$ case of 09’s Theorem III.21.1. The quartic lower bound with coefficient 12 is false: 03 (on $\mathbb{Z}/3\mathbb{Z}$), 08 (on $\mathbb{Z}/11\mathbb{Z}$, with the explicit ratio $15337/1296$) and 09 (on primes $q \geq 17$) show it; Remark W1 keeps it on record with the counterexample.

Sharpness for sets by rounding: 03, 04, 08, 09. Progression counting: the linear telescope of 03 and 04 is also 06's Theorem III.5.2 and 10's Theorem III.29.1; 06 adds a quadratic bound, 10 the torsion kernels.

Reading conventions for the sections of sources 06–10 in this Part.

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
cube count	08: $\mathcal{C}_d(F)$, $p = 2^d$, $t = \ f\ _{U^d}$, $\mathcal{D}_d(F) = \mathcal{C}_d(F) - \delta^p$; 09: $\mathcal{C}_d(F)$, $m = 2^d$, $\varepsilon = \ f\ _{U^d}$, $a_\chi = \widehat{f}(\chi)$. Both are 03's and 04's Q_d . 08's p is not a prime.
$R_d, P_d, T_d, F_d, A_d, K_d$	Parallelograms: 08's $R_d = 09$'s $P_d = 03$'s $R_d = 04$'s P_d . Non-parallelogram binary planes: 08's $T_d = 09$'s $T_d = 03$'s T_d . All binary planes: 08's $F_d = 09$'s $A_d = 03$'s K_d . 08's $K_d = \sum_{r \geq 4} \binom{p}{r}$ is not 03's K_d ; 08's F_d is not the floor count F_k of Part IV.
Q_d, S_d	09: Q_d the number of star supports, S_d of ternary supports. 08: $Q_d(f) = R_d E_4(f) + T_d E_{\langle 2 \rangle}(f)$, the quartic coefficient. <i>Neither is the cube count Q_d of 03 and 04.</i>
Fourier statistics	$E_4(f)$ (08) = $E_4(f)$ (09) = $\mathcal{E}(f)$ (03) = $\ f\ _{U^2}^4$; $E_{\langle 2 \rangle}(f)$ (08) = $\mathbb{T}_2(f)$ (09) = $\mathcal{E}_{\langle 2 \rangle}(f)$ (03); $S_5(f)$ (08) = $M_5(f)$ (09) = $B_5(f)$ (04); $\mathbb{T}_{3,5}(f)$ (09) = $M_{S_3}(f)$ (03); κ_d^{odd} (08) = c_d^{odd} (03); the subset moments $\Lambda_S(f)$ (08) = $T_S(f)$ (09) = $M_S(f)$ (03).
progressions	06: Λ_a , $\varepsilon = \ 1_A - \delta\ _{U^{k-1}}$ (a norm), $S_k(\delta)$, $B_k(\delta) = \sum_t (t+1)\delta^t$, E_k . 10: $\kappa_G(c) = G[c] $, the kernel product K , $\tau_j(G)$, B_j , D_j , $T_{k,G}(\delta)$, $S_k(\delta)$ (= 06's). 06's $B_k(\delta)$ is not the translation budget B_k of Part IV; 10's κ_G is none of the other κ 's.
colourings, $h(\ell)$	10, Section III.30: $k = 2m$, a prime $p > k$, a colouring Q with degrees D_ℓ , $s_\ell = \lceil D_\ell/2 \rceil$, $B = \sum_\ell s_\ell^2$, the field norm N_t (<i>this m is $k/2$</i>). 09: $h(\ell)$ the least support with isolated ℓ -torsion, $\mathcal{L}_S$ the row lattice, $K_G(S)$ the character constraint group.

III.1 Progression counting with fewer error terms

Source 03, delivered Section 5.

For integers $a_1, \dots, a_k$ and functions $g_i: G \rightarrow \mathbb{C}$, define

$$\Lambda_a(g_1, \dots, g_k) = \mathbb{E}_{x,t \in G} \prod_{i=1}^k g_i(x + a_i t).$$

We say that the slopes have invertible differences on G if each map $x \mapsto (a_i - a_j)x$, $i \neq j$, is an automorphism. For $G = \mathbb{Z}/N\mathbb{Z}$ this means $\gcd(a_i - a_j, N) = 1$. Consecutive slopes satisfy the condition whenever $\gcd(N, (k-1)!) = 1$, which is weaker than primality of N .

Lemma III.1.1 (Generalized von Neumann inequality). *Suppose the slopes have invertible differences and $k \geq 2$. For $|F_i| \leq 1$,*

$$|\Lambda_a(F_1, \dots, F_k)| \leq \|F_k\|_{U^{k-1}(G)}.$$

The same bound holds with any one of the functions distinguished.

Proof. For $k = 2$, the map $(x, t) \mapsto (x + a_1 t, x + a_2 t)$ is a bijection, so the average factors and is at most $|\mathbb{E} F_2| = \|F_2\|_{U^1}$. For the inductive step replace x by $y - a_1 t$, apply Cauchy–Schwarz in y , and remove the bounded factor $F_1(y)$. On expanding the square, write the two step variables as t and $t + u$. With $D_h F(z) = F(z) \overline{F(z+h)}$, the resulting expression is

$$\mathbb{E}_u \mathbb{E}_{y,t} \prod_{i=2}^k D_{(a_i - a_1)u} F_i(y + (a_i - a_1)t).$$

Its absolute value is at most $\mathbb{E}_u \|D_{(a_k - a_1)u} F_k\|_{U^{k-2}}$ by induction. Hölder and the invertibility of $a_k - a_1$ bound this by

$$\left(\mathbb{E}_h \|D_h F_k\|_{U^{k-2}}^{2^{k-2}} \right)^{1/2^{k-2}} = \|F_k\|_{U^{k-1}}^2.$$

This bounds the square of the original average. Permuting the indices allows any function to be distinguished. $\square$

Theorem III.1.2 (Density-weighted telescoping). *Assume the slopes have invertible differences. Let $0 \leq g_i \leq 1$, $\delta_i = \mathbb{E} g_i$, and $f_i = g_i - \delta_i$. For any permutation π of $\{1, \dots, k\}$,*

$$\left| \Lambda_{\mathbf{a}}(g_1, \dots, g_k) - \prod_i \delta_i \right| \leq \sum_{j=3}^k \left(\prod_{\ell=j+1}^k \delta_{\pi(\ell)} \right) \|f_{\pi(j)}\|_{U^{j-1}}. \quad (\text{III.1.1})$$

For $k = 2$ the error is zero.

Proof. Expand by replacing the functions with their means one at a time, starting with the last function in the chosen order. The exact difference is a sum of k terms. Term j has coefficient $\prod_{\ell > j} \delta_{\pi(\ell)}$ and an average containing $g_{\pi(1)}, \dots, g_{\pi(j-1)}, f_{\pi(j)}$. The first term vanishes because $f_{\pi(1)}$ has mean zero. The second vanishes by the two-variable bijection used in Lemma III.1.1. Apply that lemma to each remaining term, with its j slopes. $\square$

In the setting of Corollary 3.3, use the original index order. The normalized uniformity norms that occur in (III.1.1) are then at most its parameter α . Therefore its bound $2^k \alpha$ may be replaced by

$$\alpha \sum_{j=3}^k \prod_{i=j+1}^k \delta_i \leq (k-2)\alpha.$$

The operation is a telescoping expansion, which has one discrepancy per replacement. It avoids the unnecessary 2^k -term expansion of all balanced factors simultaneously. This is a standard multilinear technique; its role here is to record the stronger bound with the exact inhomogeneous parameters used by the source.

III.1.1 Optimizing the order of replacement

If all available norms in (III.1.1) are bounded by the same η , arrange the densities in nonincreasing order. This minimizes the displayed upper bound: every tail product then consists of the smallest densities of the required size. When distinct functions have different norms at different levels, define for a set of indices S

$$B(S) = 0 \quad (|S| \leq 2), \quad B(S) = \min_{i \in S} \{ \|f_i\|_{U^{|S|-1}} + \delta_i B(S \setminus \{i\}) \} \quad (|S| \geq 3).$$

This finite recursion computes the best bound among these telescoping orders in $O(k2^k)$ arithmetic operations once the norms are supplied. It is an optimization of the bound, not a claim that the bound is the exact counting error. For fixed small k , this separates analytic estimates from a finite, reproducible choice of order.

Corollary III.1.3 (An explicit modular progression criterion). *Let $k \geq 3$, let $|G| = N$, and suppose multiplication by each of $1, \dots, k-1$ is an automorphism of G . Let $A \subset G$ have density $0 < \delta \leq 1$, and set*

$$\eta = \|1_A - \delta\|_{U^{k-1}}, \quad S_{k-2}(\delta) = 1 + \delta + \dots + \delta^{k-3}.$$

The number of ordered pairs (x, t) with $t \neq 0$ and $x, x+t, \dots, x+(k-1)t \in A$ is at least

$$N^2(\delta^k - \eta S_{k-2}(\delta)) - \delta N. \quad (\text{III.1.2})$$

In particular, if

$$\eta \leq \frac{\delta^k}{2S_{k-2}(\delta)}, \quad N \geq 4\delta^{1-k}, \quad (\text{III.1.3})$$

there are at least $\delta^k N^2/4$ such pairs. Each has k distinct vertices. In the notation of the source, it suffices that

$$\alpha \leq \left(\frac{\delta^k}{2S_{k-2}(\delta)} \right)^{2^{k-1}}$$

when A is α -uniform of degree $k-2$.

Proof. Gowers norm monotonicity gives $\|1_A - \delta\|_{U^{j-1}} \leq \eta$ for $3 \leq j \leq k$. Apply Theorem III.1.2 to equal functions and consecutive slopes. Exactly $|A| = \delta N$ parameter pairs have $t = 0$. For $t \neq 0$, equality of vertices with indices i, j would imply $(i-j)t = 0$, contrary to the automorphism assumption. Under (III.1.3), the right side of (III.1.2) is at least $\delta^k N^2/4$. $\square$

Remark III.1.4 (Modular and integer progressions). Corollary III.1.3 concerns progressions in the group. They do not automatically become ordinary integer progressions under the representative map to $\{0, \dots, N-1\}$. One sufficient rectification condition is that every representative vertex lies in an interval of length H with $2(H-1) < N$. Its consecutive second differences are zero modulo N and have absolute value below N , so they are zero as integers. This proves rectification, but obtaining the necessary interval control is additional information. The general modular counting criterion by itself supplies no such localization.

[write] **6 October 2026.** This rectification argument is formalized, in the form of source 04's Lemma III.4.5 (note there): Lean's `hasNatAP_of_short_modular_sequence` takes representatives in $\{0, \dots, n\}$ with $2n < N$. The remark's interval of H consecutive integers reduces to that case by a translation, a step not stated in Lean. Sections III.1 and III.4 prove the same telescoping bound independently; see the note after Theorem III.4.3.

Remark III.1.5 (Why small torsion cannot simply be ignored). For $G = \mathbb{Z}/2\mathbb{Z}$, slopes 0, 1, 2 have a noninvertible difference. Taking the nontrivial real character f , the average $\mathbb{E}_{x,t} f(x)f(x+2t)$ equals one, although $\mathbb{E}f = 0$. The two centered factors are evaluated at the same point. Thus the pair cancellation used above really can fail when the slope hypothesis is removed.

III.2 The exact fourth-order term in the cube count

Source 03, delivered Section 6.

Gowers' Lemma 3.10 bounds a cube count by applying the triangle inequality to the Gowers norm. The resulting estimate is convenient, but it discards the mean-zero condition on the balanced function after that condition has been used to define uniformity. Retaining it has a substantial local

effect: the first three error terms vanish, and the fourth can be computed exactly. The computation also identifies precisely what changes when the ambient finite abelian group has a nontrivial quotient of exponent two.

All results in this section concern the same labelled cubes as [1, Lemmas 3.7–3.10]; coincident vertices are allowed. The refinements are elementary consequences of the cube expansion, character orthogonality, and Gowers–Cauchy–Schwarz. Their proofs are included in full. They are useful sharpenings of the displayed lemma; we do not claim a new inverse theorem or a change in the best known quantitative bounds for Szemerédi’s theorem.

III.2.1 Normalization and the centered expansion

Let G be a finite abelian group, with all averages normalized by the cardinality of their domain. Write $V_d = \{0, 1\}^d$, $m = 2^d$, and

$$L_v(x, h) = x + \sum_{i=1}^d v_i h_i \quad (v \in V_d).$$

For a real function $F : G \rightarrow \mathbb{R}$, define

$$Q_d(F) = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{v \in V_d} F(L_v(x, h)). \quad (\text{III.2.1})$$

Thus $Q_d(F) = \|F\|_{U^d}^m$; the usual conjugations in the definition of U^d have no effect for real functions. In particular, $Q_d(F) \geq 0$. If $F = \mathbf{1}_A$, the number of labelled cubes in A is $|G|^{d+1} Q_d(F)$.

Put $\delta = \mathbb{E}F$ and $f = F - \delta$. For $S \subseteq V_d$, put

$$M_S(f) = \mathbb{E}_{x, h} \prod_{v \in S} f(L_v(x, h)), \quad M_\emptyset(f) = 1. \quad (\text{III.2.2})$$

The exact finite expansion is

$$Q_d(\delta + f) = \sum_{S \subseteq V_d} \delta^{m-|S|} M_S(f). \quad (\text{III.2.3})$$

No limiting argument or regularity assumption is involved.

For real functions $g_v : G \rightarrow \mathbb{R}$, the normalized Gowers–Cauchy–Schwarz inequality gives

$$\left| \mathbb{E}_{x, h} \prod_{v \in V_d} g_v(L_v(x, h)) \right| \leq \prod_{v \in V_d} \|g_v\|_{U^d}. \quad (\text{III.2.4})$$

This is [1, Lemma 3.8] with normalized sums; its proof by successive Cauchy–Schwarz inequalities works on every finite abelian group. Taking $g_v = f$ on S and $g_v = 1$ elsewhere shows that, if $\eta = \|f\|_{U^d}$,

$$|M_S(f)| \leq \eta^{|S|}. \quad (\text{III.2.5})$$

We use also the standard monotonicity $\|f\|_{U^2} \leq \|f\|_{U^d}$ for $d \geq 2$.

Lemma III.2.1 (Cancellation through degree three). *If $\mathbb{E}f = 0$ and $1 \leq |S| \leq 3$, then $M_S(f) = 0$ on every finite abelian group.*

Proof. For $|S| = 1$, average over x . Otherwise choose a coordinate in which the distinct selected vertices are not all equal. For two or three vertices, one side of this coordinate contains exactly one selected vertex. Reflect that coordinate if necessary, so that this vertex is the only one with coordinate 1. A coordinate reflection preserves the average: it corresponds to the bijective change of variables $x' = x + h_i$, $h'_i = -h_i$, leaving the other h_j unchanged. After the reflection, h_i occurs in exactly one factor in the product defining M_S . Its average is $\mathbb{E}f = 0$. $\square$

Corollary III.2.2 (First refinement of Lemma 3.10). *For $d \geq 2$, $\delta \geq 0$, and real mean-zero f ,*

$$|Q_d(\delta + f) - \delta^m| \leq \sum_{j=4}^m \binom{m}{j} \delta^{m-j} \eta^j. \quad (\text{III.2.6})$$

Consequently the cube-count error is $O_{d,\delta}(\eta^4)$ as $\eta \rightarrow 0$ with d and positive δ fixed.

Proof. Expand using (III.2.3), discard the terms that vanish by Lemma III.2.1, and apply (III.2.5) to every remaining term. $\square$

The coefficient $\binom{m}{4}$ in (III.2.6) is itself far from necessary. Most four-vertex supports vanish as well.

III.2.2 Fourier energies and the quotient by doubling

Let $\widehat{G}$ be the character group, with characters written multiplicatively, and use normalized Fourier coefficients

$$\widehat{f}(\chi) = \mathbb{E}_{x \in G} f(x) \overline{\chi(x)}.$$

Write 1 for the trivial character. For real mean-zero f , set

$$\mathcal{E}(f) = \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^4 = \|f\|_{U^2}^4, \quad (\text{III.2.7})$$

$$\mathcal{E}_{(2)}(f) = \sum_{\substack{\chi \in \widehat{G} \setminus \{1\} \\ \chi^2 = 1}} |\widehat{f}(\chi)|^4. \quad (\text{III.2.8})$$

The omitted trivial-character term is zero. In particular,

$$0 \leq \mathcal{E}_{(2)}(f) \leq \mathcal{E}(f) \leq \eta^4. \quad (\text{III.2.9})$$

The second energy has a concrete quotient interpretation. Let $2G = \{2x : x \in G\}$ and

$$P_2 f(x) = \mathbb{E}_{z \in 2G} f(x + z).$$

This is the conditional average of f on the cosets of $2G$. A character survives this average exactly when it annihilates $2G$, equivalently when $\chi^2 = 1$. Fourier orthogonality therefore gives

$$\mathcal{E}_{(2)}(f) = \|P_2 f\|_{U^2(G)}^4. \quad (\text{III.2.10})$$

Thus the correction below measures the part of f visible in $G/2G$. If $|G|$ is odd, $2G = G$ and this correction is zero. If every element of G has order dividing two, then $P_2 f = f$.

III.2.3 A complete classification of four-vertex supports

Call a four-element subset $S \subseteq V_d$ an *ordinary parallelogram* if its vertices can be ordered a, b, c, e with $a + b = c + e$ as vectors in $\mathbb{Z}^d$. Call it a *binary tetrahedron* if

$$\bigoplus_{v \in S} v = 0$$

under coordinatewise addition modulo two, but S is not an ordinary parallelogram. This terminology describes the support in the Boolean cube, rather than a nondegeneracy condition on its image in G .

Lemma III.2.3 (The three possible fourth-order contributions). *For real mean-zero f and a four-element $S \subseteq V_d$,*

$$M_S(f) = \begin{cases} \mathcal{E}(f), & S \text{ is an ordinary parallelogram,} \\ \mathcal{E}_{(2)}(f), & S \text{ is a binary tetrahedron,} \\ 0, & \bigoplus_{v \in S} v \neq 0. \end{cases} \quad (\text{III.2.11})$$

These cases exhaust all four-element subsets.

Proof. If the XOR of the four vertices is nonzero, some coordinate has an odd number of ones, namely one or three. Reflect this coordinate when there are three. As in Lemma III.2.1, one difference variable then appears in exactly one centered factor, so the average vanishes.

Suppose that the XOR is zero. Reflect coordinates so that one chosen vertex becomes 0, and denote the other three vertices by u, v, w . The reflection preserves both the averages and the property of being an ordinary parallelogram. We have $u + v + w = 0$ over $\mathbf{F}_2$. Accordingly the only possible coordinate patterns across these three rows are

$$000, \quad 110, \quad 101, \quad 011.$$

At least two of the three nonzero patterns occur, because u, v, w are nonzero and pairwise distinct. Sum the independent variables h_i over coordinates of each of the three nonzero types, naming the sums r, s, t respectively. Every nonempty sum is uniform on G , and the nonempty sums are mutually independent.

If exactly two types occur, the four arguments, in an appropriate order, are

$$x, \quad x + r, \quad x + s, \quad x + r + s.$$

Their coefficient vectors form an ordinary parallelogram, and the average of the product of the four real f values is $\|f\|_{U^2}^4 = \mathcal{E}(f)$.

If all three types occur, the four arguments are

$$x, \quad x + r + s, \quad x + r + t, \quad x + s + t. \quad (\text{III.2.12})$$

There is no pairing with equal sums of coefficient vectors. For example, each possible pairing is ruled out by one of the three occupied coordinate types. Thus this is precisely the binary tetrahedron case.

It remains to evaluate (III.2.12). Substitute the Fourier expansion of f into all four factors, assigning characters $\chi_0, \chi_1, \chi_2, \chi_3$ in the displayed order. Averaging over r, s, t imposes

$$\chi_1 \chi_2 = 1, \quad \chi_1 \chi_3 = 1, \quad \chi_2 \chi_3 = 1.$$

Hence $\chi_1 = \chi_2 = \chi_3 = \chi$ with $\chi^2 = 1$. The average over x then forces $\chi_0 = \chi$. The correlation is therefore

$$\sum_{\chi^2=1} \widehat{f}(\chi)^4.$$

For such characters $\chi(x) \in \{-1, 1\}$, so $\widehat{f}(\chi)$ is real. The trivial character has coefficient zero, and the last display equals (III.2.8). $\square$

Remark III.2.4 (Why integer rank alone is insufficient). The four vectors in (III.2.12), including the coefficient of x , form a matrix of determinant ± 2 . They are independent over $\mathbb{Q}$ but need not give independent uniform values on G . Discarding all rationally independent supports would incorrectly omit $\mathcal{E}_{(2)}(f)$. The three Smith types for a four-vertex support are $(1, 1, 1, 0)$, $(1, 1, 1, 2)$, and $(1, 1, 1, 1)$, corresponding respectively to an ordinary parallelogram, a binary tetrahedron, and the final case of Lemma III.2.3. In the first type the final zero records rank three; for $d = 2$ the rectangular matrix has only three nonzero invariant factors.

Lemma III.2.5 (Exact multiplicities). *The numbers of ordinary parallelograms and binary tetrahedra in V_d are respectively*

$$R_d = \frac{6^d - 2 \cdot 4^d + 2^d}{8}, \quad (\text{III.2.13})$$

$$T_d = \frac{8^d - 3 \cdot 6^d + 3 \cdot 4^d - 2^d}{24}. \quad (\text{III.2.14})$$

Their sum is

$$K_d = R_d + T_d = \frac{m(m-1)(m-2)}{24}. \quad (\text{III.2.15})$$

Proof. Count ordered quadruples satisfying $a + b = c + e$ in $\mathbb{Z}^d$. There are six possible coordinate patterns, so the total is 6^d . The degeneracies $a = c$, $b = e$ and $a = e$, $b = c$ each contribute 4^d , and their intersection contributes 2^d . These are all degeneracies: for example, $a = b$ forces $c = e = a$ coordinatewise. An ordinary parallelogram with four distinct vertices has exactly one pairing into two equal-sum pairs. Ordering both pairs and ordering the pairs gives eight ordered solutions. This proves (III.2.13).

For the XOR count, choose distinct ordered vertices a, b, c arbitrarily and put $e = a \oplus b \oplus c$. The fourth vertex differs from each of the first three. Every unordered four-element XOR-zero subset is counted 24 times. The total is thus $m(m-1)(m-2)/24$. Every ordinary parallelogram has XOR zero, and the other XOR-zero subsets are exactly the binary tetrahedra. Subtracting R_d gives (III.2.14). $\square$

For orientation, the first coefficients are

d	m	R_d	T_d	K_d
2	4	1	0	1
3	8	12	2	14
4	16	100	40	140
5	32	720	520	1240

For comparison, $\binom{m}{4}$ equals 70, 1820, 35960 when $d = 3, 4, 5$. Even after the first three terms have been removed, counting every fourth-order support by its absolute upper bound wastes most of the available cancellation.

III.2.4 The sharpened cube-count theorem

Theorem III.2.6 (Exact fourth-order expansion with explicit remainder). *Let $d \geq 2$, $m = 2^d$, let G be a finite abelian group, and let $F : G \rightarrow \mathbb{R}$ have mean $\delta \geq 0$. Put $f = F - \delta$ and $\eta = \|f\|_{U^d}$. Then*

$$Q_d(F) = \delta^m + \delta^{m-4}(R_d\mathcal{E}(f) + T_d\mathcal{E}_{(2)}(f)) + \mathcal{R}_d(F), \quad (\text{III.2.16})$$

where

$$|\mathcal{R}_d(F)| \leq \sum_{j=5}^m \binom{m}{j} \delta^{m-j} \eta^j. \quad (\text{III.2.17})$$

An empty sum is zero. In particular, the formula for $d = 2$ is exact with no remainder.

Proof. Use (III.2.3). The constant term is δ^m , and the terms of degrees one, two, and three vanish by Lemma III.2.1. Lemmas III.2.3 and III.2.5 give the total fourth-order term. The terms of degrees at least five satisfy (III.2.17) by (III.2.5) and the triangle inequality. $\square$

The formula holds in particular for arbitrary weights $0 \leq F \leq 1$, as well as for characteristic functions. The energies in the fourth term are nonnegative, even though the higher-order remainder can have either sign.

Corollary III.2.7 (Uniformity-parameter form). *Let $A \subseteq G$ have density δ , and suppose $\|\mathbf{1}_A - \delta\|_{U^d}^m \leq \alpha$. Then*

$$Q_d(\mathbf{1}_A) \leq \delta^m + K_d \delta^{m-4} \alpha^{4/m} + \sum_{j=5}^m \binom{m}{j} \delta^{m-j} \alpha^{j/m}. \quad (\text{III.2.18})$$

If $|G|$ is odd, the coefficient K_d in the fourth-order term can be replaced by R_d . The latter replacement also holds on an arbitrary group whenever $P_2(\mathbf{1}_A - \delta) = 0$.

Proof. Apply (III.2.9) and Theorem III.2.6. The vanishing of the quotient energy follows from (III.2.10). $\square$

In the notation of Gowers' Lemma 3.10, the old upper bound is $(\delta + \alpha^{1/m})^m$. For fixed d and positive δ , its displayed excess is of order $\alpha^{1/m}$; the new bound has excess of order $\alpha^{4/m}$. This comparison concerns the strength of the cube-count estimate under exactly the same norm hypothesis. It does not replace the arithmetic-progression counting or inverse arguments elsewhere in the proof.

For example, on an odd-order group and with $d = 3$,

$$Q_3(F) = \delta^8 + 12\delta^4 \|f\|_{U^2}^4 + \mathcal{R}_3(F), \quad (\text{III.2.19})$$

$$|\mathcal{R}_3(F)| \leq 56\delta^3 \eta^5 + 28\delta^2 \eta^6 + 8\delta \eta^7 + \eta^8. \quad (\text{III.2.20})$$

On an arbitrary group the exact fourth-order term has the additional $2\delta^4 \mathcal{E}_{(2)}(f)$.

If one prefers a single remainder expression, Taylor's theorem applied to $(\delta + t)^m$ gives, for $m \geq 5$,

$$\sum_{j=5}^m \binom{m}{j} \delta^{m-j} \eta^j \leq \binom{m}{5} \eta^5 (\delta + \eta)^{m-5}. \quad (\text{III.2.21})$$

The polynomial bound (III.2.17) usually retains more useful information when the density is small.

III.2.5 Sharpness and a necessary torsion correction

Proposition III.2.8 (The fourth power cannot be increased). *Fix $d \geq 2$ and $0 < \delta < 1$. Even if one restricts to odd-order finite abelian groups and weights $F : G \rightarrow [0, 1]$ of mean δ , there is no bound*

$$Q_d(F) - \delta^m = O_{d,\delta}(\|F - \delta\|_{U^d}^q) \quad (q > 4)$$

valid as the centered norm tends to zero. Among all finite abelian groups, K_d is the optimal leading coefficient in (III.2.18), expressed in terms of $\eta = \alpha^{1/m}$.

Proof. On $G = \mathbb{Z}/3\mathbb{Z}$, let $g(x) = \cos(2\pi x/3)$ and $F_t(x) = \delta + tg(x)$. For positive sufficiently small t , the weights lie in $[0, 1]$ and have mean δ . The two nonzero Fourier coefficients of g are $1/2$, so $\mathcal{E}(g) = 1/8$. Theorem III.2.6 gives

$$Q_d(F_t) - \delta^m = \frac{R_d}{8} \delta^{m-4} t^4 + O_{d,\delta}(t^5), \quad (\text{III.2.22})$$

where the remainder is identically zero when $d = 2$. Since $\|F_t - \delta\|_{U^d} = t \|g\|_{U^d}$ and $\|g\|_{U^d} > 0$, the exponent $q > 4$ is impossible.

For the coefficient over unrestricted groups, take a nontrivial character $\chi : G \rightarrow \{-1, 1\}$ and $F_t = \delta + t\chi$. Then $\|t\chi\|_{U^d} = t$ and $\mathcal{E}(t\chi) = \mathcal{E}_{(2)}(t\chi) = t^4$. Therefore

$$Q_d(F_t) - \delta^m = K_d \delta^{m-4} t^4 + O_{d,\delta}(t^5). \quad (\text{III.2.23})$$

Any smaller universal leading coefficient would fail for small t . □

The same example proves that the torsion correction cannot be dropped. For a particularly transparent case, on $\mathbb{Z}/2\mathbb{Z}$ and with $d = 3$ one can compute the whole polynomial:

$$Q_3(\delta + t\chi) = \delta^8 + 14\delta^4 t^4 + t^8. \quad (\text{III.2.24})$$

Indeed, a subset S contributes precisely when both $|S|$ and every coordinate sum of its vertices are even. For the three-dimensional cube the only nonempty such supports have sizes four and eight: there are fourteen supports of size four and one of size eight. For size six, taking the complement would give two distinct vertices with equal XOR, which is impossible. Formula (III.2.24) directly distinguishes the correct coefficient 14 from the odd-order coefficient 12.

Remark III.2.9 (What sharpness has and has not shown). The identity determines the exact fourth-order coefficient in terms of the two Fourier energies. The constant K_d is also sharp when those energies are bounded only by η^4 over all groups. We have not asserted that R_d is the optimal leading constant in terms of η^4 when all groups are required to have odd order. That separate optimization involves the possible ratios $\|f\|_{U^2}^4 / \|f\|_{U^d}^4$ for real centered functions.

[write] **6 October 2026.** *Settled for $d = 3$ by source 04.* By source 04's Theorem III.3.4, $\|f\|_{U^2}^4 \leq 2^{-1/2} \|f\|_{U^3}^4$ for real centered f on odd-order groups, with equality for $f = 2\text{Re}(c\xi)$. Hence on odd-order groups the best leading constant in terms of $\eta^4 = \|f\|_{U^3}^4$ is $R_3 \cdot 2^{-1/2} = 6\sqrt{2} < 12 = R_3$ (source 04's Theorem III.3.6 and Proposition III.3.7); the proof is in the note after Theorem III.3.4. For $d \geq 4$ the question remains open (Research question V.14.1).

III.2.6 Sharpness can be transferred to actual sets

The use of small real weights in the previous proposition is not an essential restriction on the exponent. The following rounding argument records the quantitative reason.

Lemma III.2.10 (Rounding a bounded weight in a cube norm). *Let $p : G \rightarrow [0, 1]$ and $d \geq 2$, and choose a random subset A by including the elements of G independently with probabilities $p(x)$. For $e = \mathbf{1}_A - p$ and $m = 2^d$,*

$$\mathbb{E}_A \|e\|_{U^d}^m \leq \frac{\binom{m}{2}}{|G|}. \quad (\text{III.2.25})$$

Here $\mathbb{E}_A$ denotes expectation over the choice of the set, distinct from the averages over group elements.

Proof. Expand the Gowers norm as a cube average. When its m vertex values are pairwise distinct, independence and $\mathbb{E}_A e(x) = 0$ show that the expected product is zero. Every other product has absolute value at most one. For distinct vertices $u, v \in V_d$, the difference $L_u - L_v$ has a coefficient 1 or -1 on at least one of the independent variables h_i , and is therefore uniform on G . The probability of a collision between this pair is $|G|^{-1}$. The union bound over pairs proves (III.2.25). $\square$

Corollary III.2.11 (Optimal order and universal constant for sets). *For every fixed $d \geq 2$ and $0 < \delta < 1$, there are finite abelian groups G_n with $|G_n| \rightarrow \infty$ and subsets $A_n \subseteq G_n$ whose densities δ_n tend to δ , such that, writing $\eta_n = \|\mathbf{1}_{A_n} - \delta_n\|_{U^d}$, we have $\eta_n \rightarrow 0$ and*

$$\frac{Q_d(\mathbf{1}_{A_n}) - \delta_n^m}{\delta_n^{m-4} \eta_n^4} \rightarrow K_d. \quad (\text{III.2.26})$$

If the groups are required to have odd order, the power four is still optimal, although this argument does not identify the best leading constant in that restricted setting.

Proof. First take $G_n = (\mathbb{Z}/2\mathbb{Z})^n$, $N = |G_n|$, and a nontrivial real character χ_n . Put $t = N^{-1/(10m)}$ and $p = \delta + t\chi_n$, which lies in $[0, 1]$ for large n . Lemma III.2.10 and Markov's inequality give a realization of A_n with

$$\|\mathbf{1}_{A_n} - p\|_{U^d} \leq N^{-1/(2m)} = t^5$$

for all sufficiently large n , since the failure probability is at most $\binom{m}{2}N^{-1/2}$. The inequality $|\mathbb{E}e| \leq \|e\|_{U^d}$ implies $|\delta_n - \delta| \leq t^5$. After centering, the error in U^d is at most $2t^5$, so $\eta_n = t + O(t^5)$.

Both $\mathbf{1}_{A_n}$ and p have U^d norm at most one. The norm triangle inequality and $|a^m - b^m| \leq m|a - b|$ for $a, b \in [0, 1]$ show that

$$|Q_d(\mathbf{1}_{A_n}) - Q_d(p)| \leq mt^5.$$

The change from δ^m to δ_n^m is at most mt^5 as well. Combining these estimates with (III.2.23) proves (III.2.26).

For odd-order groups use $G_n = (\mathbb{Z}/3\mathbb{Z})^n$ and let g_n be the cosine of a nontrivial character. Its U^d norm is the same fixed positive number as the function g on $\mathbb{Z}/3\mathbb{Z}$ in Proposition III.2.8. Round $p = \delta + N^{-1/(10m)}g_n$ in exactly the same way. Equation (III.2.22) gives a strictly positive limit after division by η_n^4 . Consequently no exponent greater than four can hold uniformly even for characteristic functions on odd-order groups. $\square$

Corollary III.2.12 (Exactly prescribed compatible rational densities). *If $0 < \delta < 1$ is rational, the sharp universal limit in (III.2.26) can be achieved with $\delta_n = \delta$ for every n . If the denominator of δ is odd, the odd-order sharpness of the fourth power can likewise be achieved at exactly that density.*

Proof. Write $\delta = a/q$ and use $G_n = (\mathbb{Z}/q\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})^n$, so $N = |G_n|$ makes δN an integer. Let χ_n be a nontrivial real character, put $t = N^{-1/(10m^2)}$, and round $p = \delta + t\chi_n$. The same Markov argument as before gives a realization A with

$$\|\mathbf{1}_A - p\|_{U^d} \leq N^{-1/(2m)} = t^{5m}.$$

Its density differs from δ by at most t^{5m} . Add or remove at most Nt^{5m} memberships to obtain a set B of cardinality exactly δN . If $e = \mathbf{1}_B - \mathbf{1}_A$ is supported on a fraction θ of the group, then $|e| \leq 1$ and

$$\|e\|_{U^d}^m \leq \mathbb{E}_{x,h} \left| \prod_v e(L_v(x, h)) \right| \leq \mathbb{E}_x \mathbf{1}_{\text{supp } e}(x) = \theta.$$

Thus the alteration costs at most t^5 in U^d norm. We obtain $\|\mathbf{1}_B - p\|_{U^d} = O(t^5)$, $\|\mathbf{1}_B - \delta\|_{U^d} = t + O(t^5)$, and $Q_d(\mathbf{1}_B) = Q_d(p) + O(t^5)$. Equation (III.2.23) gives the required limit. If q is odd, replace the second factor of G_n by $(\mathbb{Z}/3\mathbb{Z})^n$ and use the cosine of an order-three character. The same alteration argument and (III.2.22) prove optimality of the fourth power on odd-order groups at exactly the chosen density. $\square$

III.2.7 An exact formula for every higher coefficient

The fourth-order theorem is the first nontrivial instance of a complete Fourier description of the centered expansion. For a support $S \subseteq V_d$, define

$$\mathcal{K}_S(G) = \left\{ (\chi_v)_{v \in S} \in (\widehat{G} \setminus \{1\})^S : \prod_{v \in S} \chi_v = 1, \quad \prod_{v \in S} \chi_v^{v_i} = 1 \ (1 \leq i \leq d) \right\}.$$

All assigned characters are required to be nontrivial because f is centered. Character orthogonality gives the exact identity

$$M_S(f) = \sum_{(\chi_v) \in \mathcal{K}_S(G)} \prod_{v \in S} \widehat{f}(\chi_v). \quad (\text{III.2.27})$$

To prove it, expand each factor in (III.2.2); averaging over x gives the first character equation, and averaging over h_i gives the equation indexed by i . These equations are also sufficient for a Fourier term to survive. This is a finite exact formula, and it is valid without assumptions on the exponent or rank of G .

For groups of prime exponent, the equations can be studied as linear relations among the vectors $(1, v)$ over the prime field. For general finite abelian groups, their integer relation matrices and Smith invariant factors retain the additional torsion information. At four vertices, the only nontrivial invariant factor that can arise is two; this is why the single quotient energy in (III.2.8) suffices. Higher coefficients offer a concrete continuation: classify their support matrices, remove the supports for which (III.2.27) is empty, and estimate the remaining explicit Fourier correlations. The formula supplies an algorithmic route without assuming that the resulting classification will have a simple closed form.

III.2.8 The fifth-order remainder is also necessary

Proposition III.2.13 (Sharp order of the remainder). *Fix $d \geq 3$ and $0 < \delta < 1$. For weights $F : G \rightarrow [0, 1]$ on odd-order finite abelian groups, the remainder after the exact fourth-order term in Theorem III.2.6 cannot in general be bounded by $O_{d,\delta}(\|F - \delta\|_{U^d}^q)$ for any $q > 5$.*

Proof. Let $G = \mathbb{Z}/3\mathbb{Z}$, $g(x) = \cos(2\pi x/3)$, and $F_t = \delta + tg$ for positive sufficiently small t . Both nontrivial Fourier coefficients of g equal $1/2$, and its trivial coefficient is zero. Fourier expansion and character orthogonality show that every support moment $M_S(g)$ is nonnegative: each surviving term is a product of nonnegative Fourier coefficients.

In dimension at least three, consider the five-vertex support

$$S = \{0, e_1, e_2, e_3, e_1 + e_2 + e_3\}.$$

Its moment is

$$\mathbb{E}_{x,a,b,c} g(x)g(x+a)g(x+b)g(x+c)g(x+a+b+c) = \frac{1}{16}.$$

Indeed, assigning a nontrivial character χ to the last factor forces the three middle characters to be χ^{-1} and the first character to be χ^2 . Each of the two choices for χ therefore contributes $(1/2)^5$. Consequently

$$c_d := \sum_{|S|=5} M_S(g) \geq \frac{1}{16} > 0.$$

The exact finite expansion gives

$$\mathcal{R}_d(F_t) = c_d \delta^{m-5} t^5 + O_{d,\delta}(t^6),$$

whereas $\|F_t - \delta\|_{U^d} = t \|g\|_{U^d}$. Since $\|g\|_{U^d} > 0$, no exponent $q > 5$ can hold. $\square$

In dimension three the entire polynomial is especially simple:

$$Q_3(\delta + tg) = \delta^8 + \frac{3}{2}\delta^4 t^4 + \frac{1}{2}\delta^3 t^5 + \frac{1}{2}\delta^2 t^6 + \frac{1}{32}t^8. \quad (\text{III.2.28})$$

For a direct verification, put $a = \delta + t$ and $b = \delta - t/2$, the two possible weight values. Group the triples (h_1, h_2, h_3) by their number r of nonzero entries. Coordinate reflections allow each nonzero difference to be replaced by 1 when summing over x . For $r = 0, 1, 2, 3$ there are respectively 1, 6, 12, 8 difference triples, and their sums over x are respectively

$$a^8 + 2b^8, \quad 2a^4b^4 + b^8, \quad a^4b^4 + 2a^2b^6, \quad a^2b^6 + 2a^3b^5.$$

It follows that

$$81Q_3(F_t) = a^8 + 8b^8 + 24a^4b^4 + 32a^2b^6 + 16a^3b^5.$$

Substitution and expansion give (III.2.28). Its t^5 coefficient supplies a concrete witness to Proposition III.2.13.

[write] **6 October 2026.** (III.2.28) is identical to source 04's (III.3.19), computed there by counting the 81 cube parameters by the number of vertices at 0; credited to both. The five-vertex support S above is the representative $\{000, 100, 010, 001, 111\}$ of source 04's eight surviving quintic supports (Lemma III.3.3), so $M_S(g) = B_5(g)$; the value $1/16$ is (III.3.8) with $a_{\pm 1} = 1/2$ and $2\xi = -\xi$ on $\mathbb{Z}/3\mathbb{Z}$.

Replacing t by $-t$ in (III.2.28) makes its fifth-order term negative. For sufficiently small positive t , this gives

$$Q_3(\delta - tg) < \delta^8 + 12\delta^4 \|tg\|_{U^2}^4.$$

Thus the exact quartic term is not, by itself, an unconditional lower bound for the whole excess; the sign of the remainder matters.

Remark W1 (The quartic lower bound with coefficient 12 is false; second write, 6 October 2026). For real $F = \delta + f$ with $\mathbb{E}f = 0$ on an odd-order group, the exact quartic term invites the lower bound

$$Q_3(F) \stackrel{?}{\geq} \delta^8 + 12\delta^4 \|f\|_{U^2}^4 \quad (0 \leq F \leq 1).$$

It is false. The display above shows this on $\mathbb{Z}/3\mathbb{Z}$ for sufficiently small t (source 03). Source 08 (Theorem III.10.1) and source 09 (Proposition III.22.2) prove it on $\mathbb{Z}/11\mathbb{Z}$ and on $\mathbb{Z}/q\mathbb{Z}$ for every prime $q \geq 17$, and source 08 gives an explicit instance with comfortable margins. *Counterexample.* On $G = \mathbb{Z}/11\mathbb{Z}$ let $h(x) = \cos(2\pi x/11) - \cos(4\pi x/11)$, $\delta = 1/2$, $\eta = 1/12$ and $F = \delta + \eta h$, so $1/3 \leq F \leq 2/3$. The nonzero Fourier coefficients of h are $\hat{h}(\pm 1) = 1/2$ and $\hat{h}(\pm 2) = -1/2$, so $\|\eta h\|_{U^2}^4 = \eta^4/4$, and the exact cube polynomial is

$$Q_3(\delta + \eta h) = \delta^8 + 3\delta^4 \eta^4 - \frac{1}{2}\delta^3 \eta^5 + \frac{3}{2}\delta^2 \eta^6 + \frac{1}{4}\eta^8.$$

Hence $(Q_3(F) - \delta^8)/(\delta^4 \|F - \delta\|_{U^2}^4) = 12 - 2s + 6s^2 + s^4$ with $s = \eta/\delta = 1/6$, which equals $15337/1296 \approx 11.834 < 12$. The polynomial is source 08's (III.10.3) and source 09's (III.22.4) at $\sigma = -1$. *Independent check at the write:* the polynomial was recomputed exactly by summing the Fourier products over all 11^4 free frequency choices of the eight cube vertices (exact rational arithmetic), and the value $Q_3(F) = 0.0039151672\dots$ was confirmed by direct floating-point summation over all 11^4 cubes. What survives is the lower bound $Q_3(F) - \delta^8 \geq 2\delta^4 \|f\|_{U^2}^4$ (source 08's (III.12.2)); the best constant lies in $[2, 15337/1296]$ (source 08's question, Section V.2, item III.10).

III.2.9 A five-vertex correlation that detects order three

The next evaluated support illustrates why a higher-order classification will require more than the order-two energy. Consider the five vertices

$$S_3 = \{0000, 1110, 1101, 1011, 0111\} \subseteq \{0, 1\}^4.$$

For every real centered function f on a finite abelian group,

$$M_{S_3}(f) = \sum_{\substack{\chi \neq 1 \\ \chi^3 = 1}} \hat{f}(\chi)^4 \overline{\hat{f}(\chi)}. \quad (\text{III.2.29})$$

Here the four nonzero vertices are the all-ones vector with one entry removed. To prove (III.2.29), assign characters $\chi_0, \chi_1, \dots, \chi_4$ to the five factors, with χ_j corresponding to the vertex omitting coordinate j . Averaging over h_j imposes $\prod_{i \neq j, i \geq 1} \chi_i = 1$. Dividing any two such equations shows $\chi_1 = \dots = \chi_4 = \chi$; each equation then says $\chi^3 = 1$. The x -equation gives $\chi_0 = \chi^{-1}$. Fourier inversion and real-valuedness of f yield the formula.

This correlation vanishes if 3 does not divide $|G|$. On $\mathbb{Z}/3\mathbb{Z}$ with $f(x) = \cos(2\pi x/3)$ it equals $1/16$, and replacing f by $-f$ changes it to $-1/16$. In particular it is not a nonnegative energy of the fourth-order kind. We have evaluated this one support; we do not assert that it exhausts all possible fifth-order types.

[write] 6 October 2026; sources 06–10. *The fifth order is now classified (source 09).* Source 09's Theorem III.20.1 shows, on every finite abelian group and in every dimension, that a five-vertex centered moment can be nonzero only for two support types up to cube symmetry: the star $\{0, u, v, w, u + v + w\}$, whose moment is source 04's $B_5(f)$ ((III.3.5); source 08's S_5), and the ternary type $\{0, u + v + w, u + v + z, u + w + z, v + w + z\}$, whose moment is (III.2.29) (source 09's $T_{3,5}$). The support S_3 above is the ternary type

with one coordinate per block; the support S in Proposition III.2.13 is a star. Source 09's Theorem III.21.1 gives the exact fifth-order coefficient $\delta^{m-5}(Q_d M_5 + S_d T_{3,5})$ with $Q_d = 2^d \binom{d+1}{4}$ and $S_d = 2^d \binom{d+1}{5}$. The four-vertex Smith types of Remark III.2.4 agree with source 09's Theorem III.19.1 (indices one and two only), and source 08's Lemma III.8.1 proves the same classification. At six vertices source 09's Theorem III.23.1 exhibits a support with row-lattice index five (checked at the write: $\det B = -5$, $M_{Sr} = (10, 5, 5, 5, 5, 5)$, Smith form $1, 1, 1, 1, 1, 5$), so order-five torsion enters at order six. The exact quartic term of Theorem III.2.6 is also proved independently by source 08 (Theorem III.8.3) and source 09 (Theorem III.21.1), with the same coefficients $(R_d, T_d; \text{source 09 writes } P_d)$ and the same order-two energy.

III.3 Sharp centered refinements of the cube-counting lemma

Source 04, delivered Section 6.

Gowers' Lemma 3.10 bounds the number of d -dimensional cubes by applying the triangle inequality to the constant and balanced parts of an indicator. Centering contains additional information that this application of the triangle inequality discards. We retain it, first by identifying correlations that vanish, and then by proving a sharp comparison between the second and third uniformity norms of real centered functions. These results strengthen the stated cube-counting lemma; no historical priority beyond that comparison is asserted. As Gowers explains, this lemma is supplementary to his main density-increment argument, so an improvement here does not by itself imply an improved global bound in Szemerédi's theorem.

Throughout this section, averages on a finite abelian group G are normalized. For $d \geq 2$, put $q = 2^d$, $V_d = \{0, 1\}^d$, and

$$L_v(x, h) = x + v \cdot h, \quad Q_d(W) = \mathbb{E}_{x, h_1, \dots, h_d \in G} \prod_{v \in V_d} W(L_v(x, h)).$$

For real W , conjugation has no effect, so $Q_d(W) = \|W\|_{U^d}^q$. For an indicator, $|G|^{d+1} Q_d(1_A)$ is exactly Gowers' ordered cube count, including degenerate cubes. Write $W = \delta + f$, where $\delta \geq 0$, f is real, and $\mathbb{E}f = 0$; these are standing assumptions throughout the section. Set $u = \|f\|_{U^d}$. No bound on f is needed for the inequalities below.

III.3.1 Cancellation, parallelograms, and support counts

Lemma III.3.1 (Cancellation through degree three). *Any at most three distinct forms L_v are jointly uniform on the corresponding power of G , for every finite abelian group G . Consequently, if $\delta \geq 0$,*

$$|Q_d(\delta + f) - \delta^q| \leq \sum_{r=4}^q \binom{q}{r} \delta^{q-r} u^r. \quad (\text{III.3.1})$$

Proof. Reflect coordinates of the Boolean cube to move the first selected vertex to 0; these reflections correspond to invertible changes of the averaging variables. For three vertices the others are distinct nonzero Boolean vectors a, b . Two coordinates give a 2×2 minor of the matrix with rows a, b equal to 1 or -1 : use two coordinates distinguishing their supports, or one in their intersection and one in their difference if the supports are nested. Thus the map to $(x, x + a \cdot h, x + b \cdot h)$ is surjective, with equally sized fibers, on every abelian group. The cases of one or two vertices are immediate.

Expand the cube product according to the support on which f is chosen. Every term with one, two, or three factors of f vanishes. For a support of size r , the cube Cauchy–Schwarz inequality, with constant 1 at all other vertices, bounds its correlation in absolute value by u^r . Summation proves the assertion. $\square$

[write] **6 October 2026.** Lemma III.3.1 has the same statement and bound as source 03’s Lemma III.2.1 with Corollary III.2.2 (this u is 03’s η , this $q = 2^d$ is 03’s m). Second route: source 03 uses coordinate reflections, this proof joint uniformity through 2×2 minors. Credited to both.

Odd group order permits a much more precise fourth-order term.

Proposition III.3.2 (Exact quartic coefficient). *Suppose $|G|$ is odd, and put $v = \|f\|_{U^2}$. Then*

$$Q_d(\delta + f) = \delta^q + P_d \delta^{q-4} v^4 + R_d, \quad (\text{III.3.2})$$

$$|R_d| \leq \sum_{r=5}^q \binom{q}{r} \delta^{q-r} u^r, \quad (\text{III.3.3})$$

$$P_d = 2^{d-3}(3^d - 2^{d+1} + 1). \quad (\text{III.3.4})$$

In particular, $P_2 = 1$, $P_3 = 12$, $P_4 = 100$, and $P_5 = 720$.

Proof. Four distinct Boolean vertices are either affinely independent or form a parallelogram. Indeed, any three are affinely independent. If four are dependent, their affine span is a plane, and some projection onto two coordinate axes is injective on that plane. The four images must be all four vertices of the unit square. The affine inverse of this projection preserves its parallelogram relation.

If the four vertices are independent, a 3×3 Boolean minor after this translation has determinant ± 1 or ± 2 . The determinant bound follows by adjoining a row and column of ones, changing binary entries b to $1 - 2b$, and applying Hadamard’s inequality to the resulting 4×4 matrix. Multiplication by either determinant is invertible on an odd-order group. Thus the four forms are jointly uniform and their centered correlation is zero. On a parallelogram, any three forms are jointly uniform by Lemma III.3.1, while the fourth is their alternating sum. Its correlation is therefore exactly v^4 .

To count parallelograms, choose the k coordinates in which a pair of opposite vertices differs and fix the other coordinates. Around that center there are 2^{k-1} opposite pairs; choose two of them. The diagonal midpoint is unique, so

$$P_d = \sum_{k=2}^d \binom{d}{k} 2^{d-k} \binom{2^{k-1}}{2} = 2^{d-3}(3^d - 2^{d+1} + 1).$$

Bound the terms of degree at least five by cube Cauchy–Schwarz as before. $\square$

[write] **6 October 2026.** Proposition III.3.2 is the odd-order case of source 03’s Theorem III.2.6, which holds on every finite abelian group with the extra term $T_d \delta^{q-4} \mathcal{E}_{(2)}(f)$; that term vanishes on odd-order groups by (III.2.10). This P_d is source 03’s R_d : $2^{d-3}(3^d - 2^{d+1} + 1) = (6^d - 2 \cdot 4^d + 2^d)/8$ (both give 1, 12, 100, 720). Second route: affine independence and Boolean minors $\pm 1, \pm 2$ in place of source 03’s XOR classification. Do not read this remainder R_d as source 03’s count R_d , or as the binomial sum $R_d(H)$ of Section IV.6 (Section F.3).

A general organizing principle is useful. Over $G = \mathbb{F}_p^n$, consider the vector matroid of the coefficient vectors $(1, v)$. If a selected support has a coloop, its centered correlation vanishes. Indeed, linear algebra gives a direction on which that form has coefficient 1 and all other selected forms have

coefficient 0. Averaging along that direction averages one factor of f to zero. Thus only supports with no coloop need to be counted.

For $d = 3$ this count can be made completely explicit, and the result also holds on every odd-order abelian group.

Lemma III.3.3 (The surviving supports of a 3-cube). *Suppose that G has odd order. The numbers of supports with no coloop, in sizes $0, 1, \dots, 8$, are*

$$1, 0, 0, 0, 12, 8, 28, 8, 1.$$

All twelve quartic correlations equal $\|f\|_{U^2}^4$. The eight quintic correlations are equal to

$$B_5(f) = \mathbb{E}_{x,a,b,c} f(x)f(x+a)f(x+b)f(x+c)f(x+a+b+c). \quad (\text{III.3.5})$$

Consequently, with $u = \|f\|_{U^3}$ and $v = \|f\|_{U^2}$,

$$Q_3(\delta + f) = \delta^8 + 12\delta^4 v^4 + 8\delta^3 B_5(f) + R_6 + R_7 + u^8, \quad (\text{III.3.6})$$

$$|R_6| \leq 28\delta^2 u^6, \quad |R_7| \leq 8\delta u^7. \quad (\text{III.3.7})$$

Proof. The twelve coplanar quadruples are the six faces and six diagonal rectangles. A plane contains at most four Boolean vertices: some projection onto two coordinate axes is injective on the plane, and its Boolean image has at most four points. A five-set therefore has rank four and has a coloop exactly when it contains a coplanar quadruple. Each of the twelve quadruples has four extensions to a five-set. No five-set contains two such quadruples, since distinct planes meet in a line and a line contains at most two Boolean vertices. Thus $\binom{8}{5} - 12 \cdot 4 = 8$ five-sets survive. Every set of size at least six remains rank four after a deletion.

The eight surviving five-sets consist of one parity tetrahedron and one vertex of the opposite parity. Cube symmetries act transitively on them; a representative is $\{000, 100, 010, 001, 111\}$. The corresponding changes of averaging variables prove (III.3.5). These arguments remain valid on odd-order groups: any four independent cube forms have a minor ± 1 or ± 2 , hence are jointly uniform; an extra vertex outside a parallelogram is therefore independent of its three determining forms. Finally apply cube Cauchy–Schwarz to the 28 sextic and 8 septic supports. $\square$

For comparison, in characteristic two the support counts are $1, 0, 0, 0, 14, 0, 28, 8, 1$: the parity tetrahedra become two additional affine planes. The odd-order hypothesis is therefore a substantive restriction.

[write] **6 October 2026; sources 06–10.** *The complete three-cube polynomial in two more sources.* Source 08’s Theorem III.9.2 proves the exact identity behind (III.3.6) on every odd-order group, $Q_3(\delta + f) = \delta^8 + 12\delta^4 v^4 + 8\delta^3 S_5(f) + \delta^2(12A_6 + 12B_6 + 4C_6) + 8\delta A_7 + Q_3(f)$, with the same support counts $12, 8, 28, 8, 1$; its S_5 is B_5 here (the summands of the two Fourier forms are complex conjugates, and the sums are real). Source 09’s Theorem III.21.1 at $d = 3$ gives $\delta^4(12E_4 + 2T_2) + 8\delta^3 M_5$ on every finite abelian group; its T_2 vanishes in odd order. The characteristic-two counts 14 quartic supports above agree with source 08’s $F_3 = 14$ and source 09’s $A_3 = 14$. Sources 08 and 09 bound the quartic term by $12\delta^4 u^4$ in odd order; Theorem III.3.6 below has the sharp $6\sqrt{2}$.

III.3.2 A sharp gap between the real centered U^2 and U^3 norms

Use normalized Fourier coefficients

$$a_\xi = \widehat{f}(\xi) = \mathbb{E}_{x \in G} f(x) \overline{\xi(x)}, \quad \xi \in \widehat{G},$$

and write the dual group additively. Fourier inversion in (III.3.5) gives

$$B_5(f) = \sum_{\xi \in \widehat{G}} a_{2\xi} a_{-\xi}^3 a_\xi. \quad (\text{III.3.8})$$

For real centered f on an odd-order group, $a_{-\xi} = \overline{a_\xi}$, $a_0 = 0$, and the nonzero coefficients occur in distinct pairs. Therefore, with $\rho = \max_\xi |a_\xi|$,

$$|B_5(f)| \leq \rho \|f\|_{U^2}^4 \leq 2^{-1/4} \|f\|_{U^2}^5. \quad (\text{III.3.9})$$

Theorem III.3.4 (Sharp real centered norm gap). *Let G have odd order and let $f : G \rightarrow \mathbb{R}$ have mean zero. Then*

$$\|f\|_{U^3}^8 \geq 2 \|f\|_{U^2}^8. \quad (\text{III.3.10})$$

For nonzero f , equality holds exactly when

$$f(x) = 2 \operatorname{Re}(c\xi(x))$$

for a nontrivial character ξ and a nonzero complex number c .

Proof. We first use the Fourier identity

$$\begin{aligned} \|f\|_{U^3}^8 &= \sum_{s,t \in \widehat{G}} |T(s,t)|^2, \\ T(s,t) &= \sum_{r \in \widehat{G}} a_r \overline{a_{r+s}} \overline{a_{r+t}} a_{r+s+t}. \end{aligned} \quad (\text{III.3.11})$$

The Fourier coefficients are normalized, but all sums in (III.3.11) are unnormalized. To verify the identity, expand the original cube in Fourier series. The four orthogonality equations say precisely that its frequency variables form an affine cube:

$$r_v = r_{000} + v_1(r_{100} - r_{000}) + v_2(r_{010} - r_{000}) + v_3(r_{001} - r_{000}).$$

These equations are solved without division. Grouping the two parallel faces gives (III.3.11).

Put

$$\lambda_r = |a_r|^2, \quad S = \sum_r \lambda_r^2 = \|f\|_{U^2}^4, \quad M = \sum_r \lambda_r^4.$$

Retain only the two coordinate axes in the nonnegative sum (III.3.11). Since $T(0,t) = \sum_r \lambda_r \lambda_{r+t}$, this gives

$$\|f\|_{U^3}^8 \geq 2\mathcal{E}(\lambda) - S^2, \quad \mathcal{E}(\lambda) = \sum_{r+s=t+w} \lambda_r \lambda_s \lambda_t \lambda_w. \quad (\text{III.3.12})$$

The subtraction corrects the double counting of the origin.

The energy summands are nonnegative. Retain the union of three families:

$$\begin{aligned} \mathcal{A}: & \quad r = t, \quad s = w, \\ \mathcal{B}: & \quad r = w, \quad s = t, \\ \mathcal{C}: & \quad s = -r, \quad w = -t. \end{aligned}$$

Each family contributes S^2 , because λ is even. Each pairwise intersection contributes M . Their triple intersection has $2r = 0$; odd group order and $\lambda_0 = 0$ make its contribution zero. Thus

$$\mathcal{E}(\lambda) \geq 3S^2 - 3M,$$

and hence

$$\|f\|_{U^3}^8 \geq 5S^2 - 6M. \quad (\text{III.3.13})$$

Group nonzero frequencies into pairs $\{r, -r\}$, and let b_j denote λ_r^2 for a representative of each pair. Then

$$S = 2 \sum_j b_j, \quad M = 2 \sum_j b_j^2 \leq \frac{S^2}{2}.$$

This proves (III.3.10).

Equality in (III.3.10) forces equality in the last inequality, so only one Fourier pair can be nonzero. Conversely, a parallelogram on a two-point subset of an odd-order group is degenerate in at least one direction. If the Fourier support is $\{\xi, -\xi\}$, all nonzero terms in (III.3.11) therefore lie on its coordinate axes. Directly, the two norm powers are $8|a_\xi|^8$ and $4|a_\xi|^8$, respectively. $\square$

[write] **6 October 2026.** *This answers source 03's Research question V.14.1 for $d = 3$ (cross-source finding of the intake; proof at the write).* Let G have odd order and $f : G \rightarrow \mathbb{R}$ be centered and nonzero; then $|G| \geq 3$, since on the trivial group a centered function vanishes. By (III.3.10),

$$\frac{\|f\|_{U^2}^4}{\|f\|_{U^3}^4} = \left(\frac{\|f\|_{U^2}^8}{\|f\|_{U^3}^8} \right)^{1/2} \leq 2^{-1/2},$$

with equality exactly for $f = 2 \operatorname{Re}(c\xi)$, ξ a nontrivial character and $c \neq 0$; such f exist on every odd-order group with $|G| \geq 3$. Hence $c_3^{\text{odd}} = 2^{-1/2}$, the supremum is attained, and the attaining functions are the real parts of characters. The best odd-order leading coefficient in source 03's reduction is $R_3 c_3^{\text{odd}} = 12 \cdot 2^{-1/2} = 6\sqrt{2}$, which is Theorem III.3.6 with Proposition III.3.7. This also settles source 03's Remark III.2.9 at $d = 3$: $R_3 = 12$ is not the optimal odd-order constant, $6\sqrt{2}$ is. For $d \geq 4$ the question is open; it is source 04's own Question 3 (Section V.2, item III.1).

Corollary III.3.5 (Stability of the norm gap). *Suppose that G has odd order. If $f \neq 0$, $0 \leq \varepsilon < 3$, and*

$$\|f\|_{U^3}^8 \leq (2 + \varepsilon) \|f\|_{U^2}^8,$$

then some nontrivial character ξ , with $c = \widehat{f}(\xi)$, satisfies

$$\|f - 2 \operatorname{Re}(c\xi)\|_{U^2} \leq (\varepsilon/3)^{1/4} \|f\|_{U^2}. \quad (\text{III.3.14})$$

Proof. Equation (III.3.13) implies $M/S^2 \geq 1/2 - \varepsilon/6$. For the pair masses $p_j = 2b_j/S$, we have $\sum_j p_j = 1$ and $\sum_j p_j^2 \geq 1 - \varepsilon/3$. Since $\max_j p_j \geq \sum_j p_j^2$, one pair carries at least $1 - \varepsilon/3$ of the Fourier fourth-power mass. Removing it and using the Fourier formula for U^2 proves the claim. $\square$

This is a statement of stability in U^2 , not an assertion of approximation in L^2 . A large amount of Fourier square mass can be spread over many small coefficients.

III.3.3 The sharp leading cube bound and its models

Theorem III.3.6 (Refined form of Lemma 3.10 in dimension three). *Let G have odd order, $f : G \rightarrow \mathbb{R}$ have mean zero, $\delta \geq 0$, and $u = \|f\|_{U^3}$. Then*

$$\begin{aligned} Q_3(\delta + f) &\leq \delta^8 + 6\sqrt{2} \delta^4 u^4 + 2^{17/8} \delta^3 u^5 \\ &\quad + 28\delta^2 u^6 + 8\delta u^7 + u^8. \end{aligned} \quad (\text{III.3.15})$$

Consequently, if A is α -uniform of degree two in Gowers' convention,

$$\begin{aligned} Q_3(1_A) &\leq \delta^8 + 6\sqrt{2}\delta^4\alpha^{1/2} + 2^{17/8}\delta^3\alpha^{5/8} \\ &\quad + 28\delta^2\alpha^{3/4} + 8\delta\alpha^{7/8} + \alpha. \end{aligned} \tag{III.3.16}$$

Proof. In (III.3.6), use $v^4 \leq u^4/\sqrt{2}$ from Theorem III.3.4 and

$$|B_5(f)| \leq 2^{-1/4}v^5 \leq 2^{-7/8}u^5.$$

The remainder bounds give (III.3.15). Gowers' parameter satisfies $u^8 \leq \alpha$, which yields (III.3.16). $\square$

The leading error at fixed density is now $\alpha^{1/2}$, with an optimal constant, in place of the $\alpha^{1/8}$ term in $(\delta + \alpha^{1/8})^8$. In higher dimensions, monotonicity and Theorem III.3.4 similarly improve the quartic term in Proposition III.3.2 to $P_d\delta^{q-4}u^4/\sqrt{2}$ for $d \geq 3$; sharpness of that higher-dimensional constant is not established here.

Proposition III.3.7 (An exact model for the leading constant). *Let $p > 8$ be prime and $f(x) = t \cos(2\pi x/p)$ on $\mathbb{Z}/p\mathbb{Z}$. Then*

$$\|f\|_{U^2}^4 = t^4/8, \quad \|f\|_{U^3}^8 = t^8/32, \tag{III.3.17}$$

$$Q_3(\delta + f) = \delta^8 + \tfrac{3}{2}\delta^4t^4 + t^8/32 = \delta^8 + 6\sqrt{2}\delta^4u^4 + u^8. \tag{III.3.18}$$

For $0 < t \leq \min(\delta, 1 - \delta)$ this is a weight in $[0, 1]$.

Proof. The Fourier support is $\{1, -1\}$, with coefficients $t/2$. Odd-degree terms vanish, since an odd number of signs cannot sum to zero and $p > 8$ prevents modular aliasing. A six-set omits two distinct vertices. In a coordinate where those vertices differ, the selected set has exactly three ones, so no signed zero sum is possible there either.

In the full eight-vertex term, the positive-sign four-set must have two ones in each coordinate. There are exactly eight choices: six unions of two opposite-vertex pairs, and the two parity tetrahedra. To see completeness, if one opposite pair occurs, the remaining two vertices must also be opposite; otherwise, reflecting a selected vertex to 000 forces the other three to be 110, 101, 011. Thus the full term is $8(t/2)^8 = t^8/32$. The quartic term follows from Proposition III.3.2. $\square$

This proves that both the fourth power of u and its coefficient $6\sqrt{2}\delta^4$ are optimal. Moreover, a family attaining this leading coefficient as $u \rightarrow 0$ at fixed positive density must approach equality in Theorem III.3.4. Corollary III.3.5 then concentrates its Fourier fourth-power mass on one conjugate pair.

The next coefficient in (III.3.15) is also optimal over odd-order groups, but this assertion uses 3-torsion. On $\mathbb{Z}/3\mathbb{Z}$,

$$Q_3(\delta + t \cos(2\pi x/3)) = \delta^8 + \tfrac{3}{2}\delta^4t^4 + \tfrac{1}{2}\delta^3t^5 + \tfrac{1}{2}\delta^2t^6 + t^8/32. \tag{III.3.19}$$

For a direct verification, put $A = \delta + t$, $B = \delta - t/2$. Among the 81 cube parameters, the numbers with k vertices equal to zero are

k	0	2	3	4	8
number	8	32	16	24	1.

These counts follow by separating the cases of zero, one, two, or three nonzero increments. The cube count is therefore $(8B^8 + 32A^2B^6 + 16A^3B^5 + 24A^4B^4 + A^8)/81$; expansion gives (III.3.19). Since $u^8 = t^8/32$, its quintic term is exactly $2^{17/8}\delta^3u^5$. This optimality is not asserted for large prime cyclic groups, where there is no 3-torsion.

[write] **6 October 2026.** (III.3.19) is identical to source 03's (III.2.28), computed there by grouping difference triples; credited to both.

III.3.4 Without 3-torsion: a sharp fractional correction exponent

The distinction just observed leads to a further refinement. The fifth-order correlation requires a frequency and its double. A function near equality in the sharp norm gap concentrates on a pair; without 3-torsion, doubling leaves that pair. Balancing the resulting loss and gain produces a fractional power.

Theorem III.3.8 (A correction of order $u^{16/3}$). *Suppose $|G|$ is coprime to 6, $\mathbb{E}f = 0$, f is real, and $\delta > 0$. With $u = \|f\|_{U^3}$,*

$$Q_3(\delta + f) \leq \delta^8 + 6\sqrt{2}\delta^4u^4 + 6\delta^{8/3}u^{16/3} + 28\delta^2u^6 + 8\delta u^7 + u^8. \quad (\text{III.3.20})$$

The exponent 16/3 cannot be increased after subtracting the sharp leading term. The coefficient 6 is an explicit upper bound, not a claimed optimal constant. In the parameter $\alpha = u^8$, the correction has order $\alpha^{2/3}$.

Proof. The zero function is immediate. Otherwise put $S = \|f\|_{U^2}^4$, choose a largest Fourier pair $\{\xi, -\xi\}$, and set

$$\rho = |a_\xi|, \quad z = 1 - \frac{2\rho^4}{S}, \quad D = zS = \sum_{r \notin \{\xi, -\xi\}} |a_r|^4.$$

Here $0 \leq z < 1$. Since $M = \sum_r |a_r|^8 \leq \rho^4 S = (1 - z)S^2/2$, (III.3.13) gives $u^8 \geq (2 + 3z)S^2$. The convex function $g(z) = (1 + 3z/2)^{-1/2}$ lies below its chord on $[0, 1]$, so

$$g(z) \leq 1 - (1 - \sqrt{2/5})z \leq 1 - z/3.$$

Consequently,

$$12\delta^4 S \leq 6\sqrt{2}\delta^4 u^4 - 2\sqrt{2}\delta^4 u^4 z. \quad (\text{III.3.21})$$

There is no 3-torsion, so 2ξ and -2ξ are outside the selected pair. The two selected terms of (III.3.8) contribute at most $2\rho^4(D/2)^{1/4}$ in absolute value; the other terms contribute at most ρD . Thus

$$\begin{aligned} |B_5(f)| &\leq 2^{-1/4} S^{5/4} ((1 - z)z^{1/4} + (1 - z)^{1/4} z) \\ &\leq 2^{3/4} S^{5/4} z^{1/4} \leq 2^{1/8} u^5 z^{1/4}. \end{aligned}$$

Combine this with (III.3.21). For $A, K > 0$,

$$\sup_{z \geq 0} (Kz^{1/4} - Az) = \frac{3K^{4/3}}{4^{4/3}A^{1/3}}.$$

Taking $A = 2\sqrt{2}\delta^4 u^4$ and $K = 2^{25/8}\delta^3 u^5$ makes the right side $6\delta^{8/3}u^{16/3}$. Inserting the remainder estimates from (III.3.7) proves (III.3.20). $\square$

Proposition III.3.9 (A two-frequency obstruction). *Fix $0 < \delta < 1$ and a prime $p > 16$. There are weights W_u on $\mathbb{Z}/p\mathbb{Z}$, with mean δ and $\|W_u - \delta\|_{U^3} = u$, such that, as $u \downarrow 0$,*

$$Q_3(W_u) - \delta^8 = 6\sqrt{2}\delta^4 u^4 + (9/4)^{1/3}\delta^{8/3}u^{16/3} + o_\delta(u^{16/3}). \quad (\text{III.3.22})$$

Proof. For $f(x) = a \cos(2\pi x/p) + b \cos(4\pi x/p)$, with $a, b > 0$, Fourier inversion gives

$$\|f\|_{U^3}^8 = \frac{a^8 + 6a^4b^4 + b^8}{32}, \quad \|f\|_{U^2}^4 = \frac{a^4 + b^4}{8}, \quad B_5(f) = \frac{a^4b}{16}. \quad (\text{III.3.23})$$

In fact there is the complete identity

$$\begin{aligned} Q_3(\delta + f) &= \delta^8 + \frac{3}{2}\delta^4(a^4 + b^4) + \frac{1}{2}\delta^3a^4b + \frac{3}{2}\delta^2a^4b^2 \\ &\quad + \frac{a^8 + 6a^4b^4 + b^8}{32}. \end{aligned} \quad (\text{III.3.24})$$

Here is a finite counting proof of both this identity and the first formula in (III.3.23). Apply the Fourier cube identity (III.3.11) to $\delta + f$. Its frequency support is contained in $\{-2, -1, 0, 1, 2\}$, with coefficients $\delta, a/2, b/2$ at frequencies of absolute value 0, 1, 2, respectively. Since $p > 16$, the orthogonality equations have no modular aliasing, so the frequency variables are integer affine cubes in this five-point interval.

Reflecting directions with negative increments makes all increments nonnegative; their sum is at most four. Enumerate their nonzero magnitudes, choose their coordinate positions and signs, and then choose the initial vertex between -2 and 2 minus their sum. The possibilities are the empty list, the singletons (1), (2), (3), (4), the pairs (1, 1), (1, 2), (1, 3), (2, 2), and the triples (1, 1, 1), (1, 1, 2). If n_j is the number of vertices with frequency of absolute value j , this gives

(n_0, n_1, n_2)	number of frequency cubes
(8, 0, 0)	1
(4, 4, 0)	24
(4, 0, 4)	24
(3, 4, 1)	16
(2, 4, 2)	96
(0, 8, 0)	8
(0, 4, 4)	48
(0, 0, 8)	8

Each row contributes its count times $\delta^{n_0}(a/2)^{n_1}(b/2)^{n_2}$, proving (III.3.24). Setting $\delta = 0$ proves the stated U^3 formula. The remaining two formulas in (III.3.23) follow from the Fourier fourth moment and (III.3.8).

Let $c = (\sqrt{2}/6)^{1/3}$ and choose

$$b = c\delta^{-1/3}u^{4/3}, \quad a^4 = -3b^4 + \sqrt{32u^8 + 8b^8}.$$

For small u this is positive, $\delta + f$ takes values in $[0, 1]$, and (III.3.23) gives $\|f\|_{U^3} = u$ exactly. Since

$$a^4 = 4\sqrt{2}u^4 - 3b^4 + O(b^8/u^4),$$

equation (III.3.24) gives

$$Q_3(\delta + f) - \delta^8 = 6\sqrt{2}\delta^4 u^4 - 3\delta^4 b^4 + 2\sqrt{2}\delta^3 u^4 b + O_\delta(u^6).$$

Substitution of b produces the coefficient $-3c^4 + 2\sqrt{2}c = (9/4)^{1/3}$, proving (III.3.22). $\square$

For the local extremal problem over groups of order coprime to 6, these results determine the first coefficient and the next exponent. More precisely, subtract $6\sqrt{2}\delta^4 u^4$ from the supremum of the cube excess over weights of mean δ with centered U^3 norm at most u , and divide by $\delta^{8/3}u^{16/3}$. The liminf of this normalized correction is at least $(9/4)^{1/3}$, and its limsup is at most 6. Proving convergence and determining the limit remain focused open questions; existence of an asymptotic second coefficient is not asserted here.

III.3.5 Sharpness for indicator sets

Proposition III.3.10 (Rounding the sharpness examples). *For every fixed $0 < \delta < 1$, there are subsets A_j of prime cyclic groups, with densities $\delta_j \rightarrow \delta$ and $\alpha_j = \|1_{A_j} - \delta_j\|_{U^3}^8 \rightarrow 0$, such that*

$$Q_3(1_{A_j}) - \delta_j^8 \sim 6\sqrt{2}\delta^4\alpha_j^{1/2}. \quad (\text{III.3.25})$$

The obstruction to improving the exponent $16/3$ also persists for indicator sets on growing prime cyclic groups.

Proof. Independently include each $x \in \mathbb{Z}/p\mathbb{Z}$ with probability $w(x)$, where w is a prescribed weight of mean δ . At most $\binom{2^d}{2}/p$ of the normalized cube parameters identify two distinct vertices: each equality is one nontrivial linear equation with probability $1/p$. Away from these parameters, independence makes the expected product of indicators equal the product of their probabilities. The same argument applies to the centered variables $1_A(x) - \delta$. Thus the expectation error in the centered cube polynomial is at most $2\binom{2^d}{2}/p$.

Changing one inclusion decision changes a normalized cube polynomial by at most $2^d/p$. Indeed, telescope the product over its vertex occurrences; each occurrence has p^d possible remaining parameters, and the other factors are bounded by one. The bounded-difference variance inequality therefore gives variance $O_d(1/p)$. The empirical density has variance at most $1/(4p)$. Replacing δ by the empirical density δ_A changes the centered cube polynomial by at most $2^d|\delta_A - \delta|$. Consequently, for sufficiently large p there is a realization simultaneously approximating the prescribed density, cube count, and centered cube power to any specified positive error.

Apply this to the cosine weights of Proposition III.3.7, with $t_j \rightarrow 0$, and choose primes p_j sufficiently large that all errors are $o(t_j^8)$. Then

$$\alpha_j = t_j^8/32 + o(t_j^8), \quad Q_3(1_{A_j}) - \delta_j^8 = \frac{3}{2}\delta^4 t_j^4 + o(t_j^4),$$

which proves (III.3.25).

For the two-frequency example, choose approximation errors $o(u^{28/3})$. The error in replacing its prescribed u^4 by the rounded value is then $o(u^{16/3})$, since u^4 is the square root of the centered eighth power. The density and cube-count errors are smaller as well. Thus (III.3.22) transfers with its nonzero $u^{16/3}$ term. The 3-torsion quintic example can instead be rounded on $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$; it does not give quintic sharpness on prime cyclic groups. $\square$

III.4 Density-weighted progression counting

Source 04, delivered Section 7.

The expansion in Gowers's Corollary 3.3 uses every subset of the set of positions, producing a factor bounded by 2^k . An ordered telescoping expansion gives a smaller bound and keeps the density factors. This section first states the precise group hypothesis needed for the Cauchy–Schwarz argument.

Definition III.4.1. Integers $c_1, \dots, c_m$ are *admissible on G* if the map $x \mapsto (c_i - c_j)x$ is an automorphism of G for every $i \neq j$. For functions $a_i : G \rightarrow \mathbb{C}$, write

$$\Lambda_{\mathbf{c}}(a_1, \dots, a_m) = \mathbb{E}_{x,y} \prod_{i=1}^m a_i(x + c_i y).$$

For $G = \mathbb{Z}/N\mathbb{Z}$, admissibility is equivalent to $\gcd(c_i - c_j, N) = 1$ for all pairs. Consecutive slopes $0, 1, \dots, k-1$ are therefore admissible whenever $\gcd(N, (k-1)!) = 1$. Primality is sufficient but unnecessary.

Lemma III.4.2 (Generalized von Neumann bound). *Suppose $m \geq 2$, the slopes are admissible, and $|a_i| \leq 1$. For every j ,*

$$|\Lambda_{\mathbf{c}}(a_1, \dots, a_m)| \leq \|a_j\|_{U^{m-1}}.$$

Proof. Relabel so that the target is a_m . For $m = 2$, the map $(x, y) \mapsto (x + c_1 y, x + c_2 y)$ is a bijection: its inverse uses the automorphism $c_2 - c_1$. The average factors as $(\mathbb{E}a_1)(\mathbb{E}a_2)$, giving the assertion.

Assume $m > 2$ and the result for $m-1$. Replace x by $x - c_1 y$, and apply Cauchy–Schwarz in x , using $|a_1| \leq 1$. Writing $d_i = c_i - c_1$, this gives

$$|\Lambda_{\mathbf{c}}|^2 \leq \mathbb{E}_x \left| \mathbb{E}_y \prod_{i=2}^m a_i(x + d_i y) \right|^2 = \mathbb{E}_h \mathbb{E}_{x,y} \prod_{i=2}^m a_i(x + d_i y) \overline{a_i(x + d_i y + d_i h)}.$$

For each h , apply the inductive assertion to the $m-1$ derivative functions. Differences of the new slopes are still automorphisms. The triangle inequality bounds the last expression by

$$\mathbb{E}_h \|\Delta_{d_m h} a_m\|_{U^{m-2}}, \quad \Delta_t a(x) = a(x) \overline{a(x+t)}.$$

The map $h \mapsto d_m h$ is a bijection. Hölder's inequality and the derivative identity for Gowers norms give

$$\mathbb{E}_h \|\Delta_{d_m h} a_m\|_{U^{m-2}} \leq \left(\mathbb{E}_t \|\Delta_t a_m\|_{U^{m-2}}^{2^{m-2}} \right)^{1/2^{m-2}} = \|a_m\|_{U^{m-1}}^2.$$

Taking square roots completes the induction. □

Theorem III.4.3 (Telescoping with the density factors retained). *Let $k \geq 3$, let the slopes be admissible on G , and let $a_i : G \rightarrow [0, 1]$ have means δ_i . Put $b_i = a_i - \delta_i$ and $u_i = \|b_i\|_{U^{i-1}}$ for $i \geq 3$. Then*

$$\left| \Lambda_{\mathbf{c}}(a_1, \dots, a_k) - \prod_{i=1}^k \delta_i \right| \leq \sum_{j=3}^k \left(\prod_{i=j+1}^k \delta_i \right) u_j. \quad (*)$$

Every permutation of the positions gives a valid version, with the appropriate Gowers norm of the function in the j -th position.

Proof. Successively write $a_j = \delta_j + b_j$, starting with $j = k$ and ending with $j = 3$. Multilinearity gives the exact identity

$$\Lambda_{\mathbf{c}}(a_1, \dots, a_k) - \prod_i \delta_i = \sum_{j=3}^k \left(\prod_{i=j+1}^k \delta_i \right) \Lambda_{c_1, \dots, c_j}(a_1, \dots, a_{j-1}, b_j).$$

The remaining two-function average is exactly $\delta_1 \delta_2$ by pairwise admissibility. Each term is bounded by u_j , by Lemma III.4.2; all other functions in that term are 1-bounded. Taking absolute values proves the result. Relabeling preserves admissibility, proving the last assertion. $\square$

Under the source's hypotheses $u_j \leq \varepsilon$, the new error is

$$\varepsilon \sum_{j=3}^k \prod_{i=j+1}^k \delta_i \leq (k-2)\varepsilon.$$

This is a direct replacement for $2^k \varepsilon$, in the same normalization. When all densities equal δ , the factor is

$$S_{k-2}(\delta) := \sum_{r=0}^{k-3} \delta^r = \begin{cases} (1 - \delta^{k-2})/(1 - \delta), & \delta \neq 1, \\ k-2, & \delta = 1. \end{cases}$$

In particular it is at most $1/(1 - \delta)$ for $0 \leq \delta < 1$, independently of k , and the inequality is strict when $0 < \delta < 1$.

[write] **6 October 2026.** Source 03's Section III.1 proves the same generalized von Neumann bound (Lemma III.1.1; "slopes with invertible differences" there are "admissible slopes" here) and the same telescoping bound (Theorem III.1.2), by the same induction and expansion. Its order-optimization recursion $B(S)$ is $D(S)$ below. The two texts were written independently (the intake measured under 0.8% shared text); both are kept and credited. The first display of Corollary III.4.4 is source 03's (III.1.2). Source 03 concludes at least $\delta^k N^2/4$ progressions under $N \geq 4\delta^{1-k}$ (Corollary III.1.3), while source 04 concludes positivity under $N > 2\delta^{1-k}$. Source 04's exchange rule and source 03's real-character torsion example are each one source's alone. Source 04's $\mathbb{Z}/2$ example below has a complex a_3 and shows that the generalized von Neumann inequality itself fails.

III.4.1 Choosing the order of elimination

The preceding result has a small useful optimization principle. Suppose one has upper bounds $u_i(s) \geq \|b_i\|_{U^s}$, for all $2 \leq s \leq k-1$. For a subset S of positions define

$$D(S) = 0 \quad (|S| \leq 2), \quad D(S) = \min_{i \in S} (u_i(|S| - 1) + \delta_i D(S \setminus \{i\})) \quad (|S| \geq 3).$$

Induction on $|S|$ shows that $D(S)$ bounds the counting error for the subsystem indexed by S . Moreover it is precisely the smallest bound obtained by this family of telescoping estimates. Computing the recurrence requires at most $k2^k$ elementary comparisons, once the norm bounds have been supplied.

If a single upper bound v_i is used for each function at every remaining order, there is also a simple exchange rule. For two successive eliminations i, j , their contribution before the common later tail is either $v_i + \delta_i v_j$ or $v_j + \delta_j v_i$. Thus i should precede j exactly when

$$v_i(1 - \delta_j) \leq v_j(1 - \delta_i).$$

For a fixed pair of positions to retain, sort the other positions by $v_i/(1 - \delta_i)$, with the degenerate cases interpreted through the displayed inequality. One can compare the $\binom{k}{2}$ possible retained pairs. This optimizes the stated bound, and is not a claim of an optimal counting inequality among all methods.

III.4.2 An explicit supersaturation consequence

Corollary III.4.4 (A sharper uniformity threshold). *Let $A \subset G$ have density $0 < \delta \leq 1$, and suppose that the consecutive slopes $0, \dots, k-1$ are admissible, with $k \geq 3$. Put $u = \|1_A - \delta\|_{U^{k-1}}$. The number $T_k^*(A)$ of ordered k -term progressions with nonzero difference satisfies*

$$T_k^*(A) \geq N^2(\delta^k - uS_{k-2}(\delta)) - \delta N. \quad (\text{III.4.1})$$

In particular, if

$$u \leq \frac{\delta^k}{2S_{k-2}(\delta)} \quad \text{and} \quad N > 2\delta^{1-k},$$

then $T_k^*(A) > 0$.

Proof. Norm monotonicity bounds every U^{j-1} -norm in Theorem III.4.3 by u . There are N^2 ordered choices of start and difference. Exactly $|A| = \delta N$ of the counted progressions have difference zero, so subtracting them gives the asserted bound. The two displayed assumptions make its right-hand side positive. $\square$

Admissibility also ensures that a progression with nonzero difference has k distinct entries: a repetition would give $(i - j)y = 0$ with $0 < |i - j| < k$, contrary to the automorphism hypothesis.

[write] 6 October 2026; sources 06–10. *Two further counting refinements (sources 06 and 10).* Source 10's Corollary III.29.2 has error $\delta S_k(\delta)u$, a factor δ below the error $uS_{k-2}(\delta)$ of Corollary III.4.4 (the polynomials are equal: $S_k(\delta)$ there is $1 + \delta + \dots + \delta^{k-3}$), because its mixed L^p norms of indicators equal δ ; its Corollary III.29.3 needs $u \leq \delta^{k-1}/(2S_k(\delta))$ in place of $u \leq \delta^k/(2S_{k-2}(\delta))$. Source 06's Corollary III.5.5 is quadratic in u : error $\sqrt{\delta} B_k(\delta)u^2$ with $B_k(\delta) = \sum_{t \leq k-3} (t+1)\delta^t$, and its criterion needs $u^2 \leq \delta^{k-1/2}/(2B_k(\delta))$. In the paper's α (with $u^{2^{k-1}} \leq \alpha$) the sufficient conditions become $\alpha \leq (\delta^k/2S_{k-2})^{2^{k-1}}$ here, $\alpha \leq (\delta^{k-1}/2S_k)^{2^{k-1}}$ (source 10) and $\alpha \leq (\delta^{k-1/2}/2B_k)^{2^{k-2}}$ (source 06), against the formal **corollary_3_6**'s $(\delta/2)^{k2^k}$. The powers of δ are $k2^{k-1}$, $(k-1)2^{k-1}$, $(k-\frac{1}{2})2^{k-2}$ and $k2^k$; source 06's is the smallest for every $k \geq 3$, since $(k-\frac{1}{2})2^{k-2} < (k-1)2^{k-1}$ is $k > \frac{3}{2}$ (the constants $2S_{k-2} \leq 2(k-2)$ and $2B_k \leq (k-1)(k-2)$ differ). All need invertibility of $1, \dots, k-1$; source 10's Theorem III.28.2 and its criterion (III.29.6) work without it, with explicit kernel factors.

In the paper's uniformity parameter, the sufficient condition becomes

$$\alpha \leq \left(\frac{\delta^k}{2S_{k-2}(\delta)} \right)^{2^{k-1}}.$$

For fixed k and small δ , its density exponent is $k2^{k-1}$, compared with $k2^k$ in the sufficient condition $(\delta/2)^{k2^k}$ displayed in the inspected version of Corollary 3.6. This comparison concerns the modular progression step.

Lemma III.4.5 (Lifting from a short interval). *Let $A \subset \{0, \dots, n-1\}$ be viewed in $\mathbb{Z}/N\mathbb{Z}$, with $2(n-1) < N$. If A contains a nonconstant modular progression of length $k \geq 3$, then it contains an ordinary integer progression of length k and positive difference.*

Proof. Let $x_0, \dots, x_{k-1} \in \{0, \dots, n-1\}$ be the representatives. Each integer $x_i + x_{i+2} - 2x_{i+1}$ is divisible by N and has absolute value at most $2(n-1) < N$; it is therefore zero. Successive integer differences are equal. The common difference is nonzero because the modular progression is nonconstant. Reverse the sequence if that integer difference is negative. $\square$

The lemma specifies the extra hypothesis needed to draw an integer conclusion. A modular progression in an unrestricted subset of $\mathbb{Z}/N\mathbb{Z}$ can wrap around, so that conclusion should not be silently inserted into Corollary III.4.4.

[write] **6 October 2026.** *Already formalized.* This lemma is proved in Lean as `hasNatAP_of_short_modular_sequence` (`Proofs18IntervalTransfer.lean`, line 22), with the set form `hasNatAP_of_hasModAP_image` (line 92); both were added in commit 327773149, source 04's own pin. The Lean statement takes representatives $f(i) \leq n$ with $2n < N$ and a nonzero modular difference, for every length k : after $n-1 \mapsto n$ it is this lemma, and it also covers $k < 3$. Source 04 does not mention the formal version. It is the only statement of sources 01–05 that is formalized; source 03's rectification remark (Remark III.1.4) is the same argument.

Remark III.4.6 (Why invertibility cannot simply be omitted). For $G = \mathbb{Z}/2\mathbb{Z}$, take slopes 0, 1, 2, $a_3(0) = 1$, $a_3(1) = i$, $a_1 = \overline{a_3}$, and $a_2 = 1$. The forms at positions 0 and 2 coincide, so $\Lambda = 1$. Both normalized Fourier coefficients of a_3 have modulus $2^{-1/2}$, whence $\|a_3\|_{U^2} = 2^{-1/4} < 1$. Thus the stated generalized von Neumann inequality itself fails without an algebraic hypothesis. The obstruction is an algebraic dependence, and the admissibility condition states exactly what the present proof uses.

III.5 Quadratic discrepancy bounds for arithmetic progressions

Source 06, delivered Section 3.

The proof of Gowers's Theorem 3.2 is a repeated Cauchy–Schwarz argument. Keeping two functions in that argument, rather than discarding all but one of their norms, produces a quadratic error estimate. A second telescoping identity then exploits the exact disappearance of one- and two-function balanced correlations. This refines the displayed counting estimates in Section 3 of the paper and its repository transcription [1, 41].

III.5.1 Normalization and the precise hypothesis on the modulus

Let G be a finite abelian group of order N , with normalized averages. For an integer a , let $[a]: G \rightarrow G$ denote multiplication by a . For slopes $a_1, \dots, a_k$, define

$$\Lambda_a(F_1, \dots, F_k) = \mathbb{E}_{x,d \in G} \prod_{i=1}^k F_i(x + a_i d).$$

Throughout this section, assume that

$$[a_i - a_j] \text{ is an automorphism of } G \text{ whenever } i \neq j. \quad (\text{III.5.1})$$

This assumption is inherited by every subset of the slopes. It makes the map

$$(x, d) \mapsto (x + a_i d, x + a_j d)$$

a bijection of G^2 . In particular,

$$\mathbb{E}_{x,d} F_i(x + a_i d) F_j(x + a_j d) = (\mathbb{E} F_i)(\mathbb{E} F_j). \quad (\text{III.5.2})$$

For consecutive slopes $0, 1, \dots, k-1$, condition (III.5.1) means that multiplication by each of $1, \dots, k-1$ is invertible. For $G = \mathbb{Z}/N\mathbb{Z}$ it is equivalent to

$$\gcd(N, (k-1)!) = 1. \quad (\text{III.5.3})$$

Thus the counting results below apply to many composite moduli. For example, $N = 25$ is admissible for every $k \leq 5$. If $N \geq 2$ and the slopes are consecutive, every progression with $d \neq 0$ has k distinct terms: an equality between two terms would put d in the kernel of one of the automorphisms in (III.5.1).

For $s \geq 1$, use the normalized uniformity quantity

$$\|f\|_{U^s(G)}^{2^s} = \mathbb{E}_{x, h_1, \dots, h_s \in G} \prod_{\omega \in \{0,1\}^s} \mathcal{C}^{|\omega|} f(x + \omega \cdot h),$$

where $\mathcal{C}$ is complex conjugation. Thus $\|f\|_{U^1} = |\mathbb{E}f|$. The standard derivative identity is

$$\|f\|_{U^s}^{2^s} = \mathbb{E}_h \|\Delta_h f\|_{U^{s-1}}^{2^{s-1}}, \quad \Delta_h f(x) = f(x) \overline{f(x+h)} \quad (s \geq 2). \quad (\text{III.5.4})$$

We use the usual inequalities $\|f\|_{U^s} \leq \|f\|_{U^{s+1}}$ and $\|f\|_{U^s} \leq \|f\|_\infty$; they follow from the same Cauchy–Schwarz argument used in Section 3 of [1]. In the paper’s convention, degree- d α -uniformity is precisely

$$\|f\|_{U^{d+1}}^{2^{d+1}} \leq \alpha. \quad (\text{III.5.5})$$

III.5.2 Two distinguished functions

Theorem III.5.1 (Two-function uniformity estimate). *Assume (III.5.1) and $k \geq 2$. For distinct indices p, q and arbitrary complex functions $F_1, \dots, F_k$ on G ,*

$$|\Lambda_{\mathbf{a}}(F_1, \dots, F_k)| \leq \|F_p\|_{U^{k-1}} \|F_q\|_{U^{k-1}} \prod_{i \notin \{p, q\}} \|F_i\|_\infty. \quad (\text{III.5.6})$$

If $k \geq 3$ and $r \notin \{p, q\}$, then the stronger estimate

$$|\Lambda_{\mathbf{a}}(F_1, \dots, F_k)| \leq \|F_r\|_2 \|F_p\|_{U^{k-1}} \|F_q\|_{U^{k-1}} \prod_{i \notin \{p, q, r\}} \|F_i\|_\infty \quad (\text{III.5.7})$$

also holds. The coefficient 1 in (III.5.6) is sharp.

Proof. By homogeneity, it is enough to suppose that the functions other than F_p, F_q have supremum norm at most one. If any normalizing norm is zero, the corresponding function vanishes and there is nothing to prove. For $k = 2$, equation (III.5.2) gives the identity

$$|\Lambda_{(a_p, a_q)}(F_p, F_q)| = |\mathbb{E}F_p| |\mathbb{E}F_q| = \|F_p\|_{U^1} \|F_q\|_{U^1}.$$

Suppose (III.5.6) has been proved for $k-1$ functions. Choose $r \notin \{p, q\}$ and change x to $x - a_r d$ so that the r th slope becomes zero. Put $b_i = a_i - a_r$. Cauchy–Schwarz in x gives

$$\begin{aligned} |\Lambda_{\mathbf{a}}(F_1, \dots, F_k)|^2 &\leq \|F_r\|_2^2 \mathbb{E}_x \left| \mathbb{E}_d \prod_{i \neq r} F_i(x + b_i d) \right|^2 \\ &= \|F_r\|_2^2 \mathbb{E}_h \Lambda_{(b_i)_{i \neq r}}((\Delta_{b_i h} F_i)_{i \neq r}). \end{aligned}$$

The last average is real and nonnegative, as is apparent from the preceding line. Applying the triangle inequality and the induction hypothesis bounds it by

$$\mathbb{E}_h B(h)C(h), \quad B(h) = \|\Delta_{b_p h} F_p\|_{U^{k-2}}, \quad C(h) = \|\Delta_{b_q h} F_q\|_{U^{k-2}}.$$

All remaining differentiated functions have supremum norm at most one. Set $Q = 2^{k-2} \geq 2$. Hölder's inequality on the probability space G gives

$$\mathbb{E}_h B(h)C(h) \leq (\mathbb{E}_h B(h)^Q)^{1/Q} (\mathbb{E}_h C(h)^Q)^{1/Q}.$$

For $Q > 2$, this is Hölder with a third factor 1 and exponent $Q/(Q-2)$; for $Q = 2$, it is Cauchy–Schwarz. Both $[b_p]$ and $[b_q]$ are automorphisms. Therefore (III.5.4) yields

$$\mathbb{E}_h B(h)^Q = \|F_p\|_{U^{k-1}}^{2^{k-1}}, \quad \mathbb{E}_h C(h)^Q = \|F_q\|_{U^{k-1}}^{2^{k-1}}.$$

Taking square roots proves (III.5.7). Using $\|F_r\|_2 \leq \|F_r\|_\infty$ proves (III.5.6) and completes the induction. Restoring the supremum norms gives the homogeneous statements. Constant functions show that the coefficient in (III.5.6) cannot be decreased. $\square$

Bounding one of the two distinguished uniformity quantities by the supremum norm recovers the ordinary generalized von Neumann estimate. The gain in Theorem III.5.1 is useful when both distinguished functions are balanced and have small uniformity norm.

[write] **6 October 2026; sources 06–10.** The one-function case is the generalized von Neumann bound of sources 03 and 04 (Lemmas III.1.1 and III.4.2), with the same slope hypothesis (III.5.1). Source 10's Theorem III.28.2 replaces the L^∞ norms by mixed L^p norms and drops the invertibility of all differences except one, at the price of an explicit kernel factor. Keeping two distinguished functions is specific to this source.

III.5.3 First and second telescoping

Let $0 \leq F_i \leq 1$, let $\delta_i = \mathbb{E}F_i$, and put $f_i = F_i - \delta_i$.

Theorem III.5.2 (Indexed first-order estimate). *Suppose $k \geq 2$, assume (III.5.1), and suppose $\|f_j\|_{U^{j-1}} \leq \varepsilon_j$ for $3 \leq j \leq k$. Then*

$$\left| \Lambda_{\mathbf{a}}(F_1, \dots, F_k) - \prod_{i=1}^k \delta_i \right| \leq \sum_{j=3}^k \varepsilon_j \prod_{h=j+1}^k \delta_h. \quad (\text{III.5.8})$$

For $k = 2$, the empty sum is zero and the count is exact.

Proof. The pointwise telescoping identity is

$$\prod_{i=1}^k F_i - \prod_{i=1}^k \delta_i = \sum_{j=1}^k f_j \prod_{i < j} F_i \prod_{h > j} \delta_h.$$

Evaluate the i th function at $x + a_i d$ and average. The $j = 1$ summand vanishes because $\mathbb{E}f_1 = 0$. The $j = 2$ summand vanishes by (III.5.2). For $j \geq 3$, apply the one-function consequence of Theorem III.5.1 to the j slopes $a_1, \dots, a_j$. This bounds that summand by $\varepsilon_j \prod_{h > j} \delta_h$. $\square$

Under the indexed hypotheses of Gowers's Corollary 3.3, the right side of (III.5.8) is at most $(k-2)\alpha$, in the notation of that corollary, replacing its displayed $2^k\alpha$ bound. The symbol α there is a norm-scale parameter; conversion to the paper's degree-uniformity energy must use (III.5.5).

Theorem III.5.3 (Second-order estimate). *Suppose $k \geq 3$, assume (III.5.1), and suppose $\|f_i\|_{U^{k-1}} \leq \varepsilon_i$ for all i . For $2 \leq i < j \leq k$, set*

$$w_{ij} = \prod_{i < h < j} \delta_h \prod_{h > j} \delta_h, \quad \mu_i = \min_{r < i} \|F_r\|_2.$$

Then

$$\left| \Lambda_{\mathbf{a}}(F_1, \dots, F_k) - \prod_{i=1}^k \delta_i \right| \leq \sum_{2 \leq i < j \leq k} w_{ij} \mu_i \varepsilon_i \varepsilon_j. \quad (\text{III.5.9})$$

One may replace every μ_i by 1. For indicator functions, $\mu_i = \min_{r < i} \sqrt{\delta_r}$.

Proof. In the proof of Theorem III.5.2, telescope each prefix a second time:

$$\prod_{h < j} F_h = \prod_{h < j} \delta_h + \sum_{i < j} f_i \prod_{r < i} F_r \prod_{i < h < j} \delta_h.$$

The constant-prefix contribution vanishes after multiplication by $f_j(x + a_j d)$ and averaging. The $i = 1$ contribution vanishes by (III.5.2). The exact surviving expression for the discrepancy is consequently

$$\sum_{2 \leq i < j \leq k} w_{ij} \Lambda_{(a_1, \dots, a_i, a_j)}(F_1, \dots, F_{i-1}, f_i, f_j).$$

There are $i + 1$ functions in each average. Apply (III.5.7) with f_i, f_j distinguished and any F_r , $r < i$, supplying the L^2 factor. Norm monotonicity gives $\|f_i\|_{U^i} \leq \|f_i\|_{U^{k-1}}$ and the corresponding inequality for f_j . Minimizing over $r < i$ and summing proves the claim. $\square$

Remark III.5.4 (Ordering the inputs). If all balanced functions satisfy the common bound $\|f_i\|_{U^{k-1}} \leq \varepsilon$, the slope-function pairs may be reordered before either telescoping argument. A nonincreasing order of the densities minimizes the coefficient in (III.5.8), and also minimizes the coefficient in (III.5.9) when the μ_i are replaced by 1. For the first assertion, the coefficient is a sum of products of final segments. In an adjacent swap, the only changed final-segment term is a nonnegative common product times the later density; placing the smaller density later cannot increase it. For the second assertion, the coefficient is

$$\sum_{i=2}^{k-1} e_{k-i-1}(\delta_{i+1}, \dots, \delta_k),$$

where e_j is the elementary symmetric polynomial. Again, an adjacent swap changes only one final-segment summand, whose dependence on the changed endpoint is affine with a nonnegative coefficient. This is an optimization of these bounds among orders; it does not assert that their coefficients are universally sharp discrepancy constants.

III.5.4 One set and explicit coefficients

For $k \geq 3$, define the polynomials

$$S_k(\delta) = \sum_{t=0}^{k-3} \delta^t, \quad B_k(\delta) = \sum_{t=0}^{k-3} (t+1) \delta^t. \quad (\text{III.5.10})$$

Corollary III.5.5 (Quadratic progression discrepancy). *Let $A \subseteq G$ have density δ , let $\varepsilon = \|1_A - \delta\|_{U^{k-1}}$, and suppose $k \geq 3$ and (III.5.1). Then*

$$\left| \Lambda_a(1_A, \dots, 1_A) - \delta^k \right| \leq \min \left\{ S_k(\delta)\varepsilon, \sqrt{\delta} B_k(\delta)\varepsilon^2 \right\}. \quad (\text{III.5.11})$$

Proof. Apply Theorems III.5.2 and III.5.3, using norm monotonicity for the first. In the second theorem, $\mu_i = \sqrt{\delta}$ and $w_{ij} = \delta^{k-i-1}$. For fixed i there are $k-i$ choices of j , so its coefficient is

$$\sum_{i=2}^{k-1} (k-i)\delta^{k-i-1} = \sum_{t=0}^{k-3} (t+1)\delta^t = B_k(\delta).$$

□

For $0 \leq \delta < 1$, the closed forms are

$$S_k(\delta) = \frac{1 - \delta^{k-2}}{1 - \delta}, \quad B_k(\delta) = \frac{1 - (k-1)\delta^{k-2} + (k-2)\delta^{k-1}}{(1 - \delta)^2}.$$

At $\delta = 1$, the polynomial values are $k-2$ and $(k-1)(k-2)/2$. In particular,

$$B_k(\delta) \leq \frac{(k-1)(k-2)}{2}, \quad B_k(\delta) \leq \frac{1}{(1-\delta)^2} \quad (\delta < 1).$$

For fixed $\delta < 1$, this latter bound remains bounded as k grows. The quadratic dependence on ε strengthens the usual linear discrepancy estimate independently of the improvement in the coefficient.

Proposition III.5.6 (A sharp three-term variance factor). *Under the three-slope case of (III.5.1),*

$$\left| \Lambda_a(1_A, 1_A, 1_A) - \delta^3 \right| \leq \sqrt{\delta(1-\delta)} \|1_A - \delta\|_{U^2}^2. \quad (\text{III.5.12})$$

The coefficient 1 is sharp, even for consecutive slopes in cyclic groups of odd order.

Proof. Set $f = 1_A - \delta$. Every expansion term with one or two factors f vanishes by mean zero and (III.5.2). Thus the discrepancy equals $\Lambda_a(f, f, f)$. Apply (III.5.7) and use $\|f\|_2^2 = \delta(1-\delta)$.

For sharpness, take A to be a singleton in a cyclic group of odd order $N \geq 3$. Then $\delta = 1/N$, and its normalized three-term count is $1/N^2$. With normalized Fourier coefficients, the balanced singleton has coefficient 0 at the trivial character and modulus $1/N$ at each of the other $N-1$ characters. Therefore

$$\|f\|_{U^2}^4 = \frac{N-1}{N^4}, \quad \sqrt{\delta(1-\delta)} \|f\|_{U^2}^2 = \frac{N-1}{N^3} = \frac{1}{N^2} - \frac{1}{N^3}.$$

Equality holds in (III.5.12). □

III.5.5 An explicit modular existence criterion

Corollary III.5.7 (Exact subtraction of constant progressions). *Let $N \geq 2$, $k \geq 3$, and let $A \subseteq G$ have density $0 < \delta \leq 1$. Assume that multiplication by $1, \dots, k-1$ is invertible on G . Put*

$$E_k = \min \left\{ S_k(\delta)\varepsilon, \sqrt{\delta} B_k(\delta)\varepsilon^2 \right\}, \quad \varepsilon = \|1_A - \delta\|_{U^{k-1}}.$$

The number of nonconstant ordered k -term progressions in A is at least

$$N^2(\delta^k - E_k) - \delta N. \quad (\text{III.5.13})$$

In particular, such a progression exists whenever

$$\delta^k - E_k > \frac{\delta}{N}. \quad (\text{III.5.14})$$

A convenient sufficient condition is

$$N > 2\delta^{1-k}, \quad \varepsilon^2 \leq \frac{\delta^{k-1/2}}{2B_k(\delta)}. \quad (\text{III.5.15})$$

Consequently, if A is α -uniform of degree $k-2$ in the paper's convention, it is sufficient that

$$N > 2\delta^{1-k}, \quad \alpha \leq \left(\frac{\delta^{k-1/2}}{2B_k(\delta)} \right)^{2^{k-2}}. \quad (\text{III.5.16})$$

Proof. Corollary III.5.5 gives at least $N^2(\delta^k - E_k)$ ordered progressions. Exactly $|A| = \delta N$ of them have $d = 0$. Subtracting these proves (III.5.13); every remaining progression has distinct terms by (III.5.1). Under (III.5.15), $E_k \leq \delta^k/2$ and $\delta/N < \delta^k/2$, proving (III.5.14). Finally, (III.5.5) implies $\varepsilon^2 \leq \alpha^{1/2^{k-2}}$, giving (III.5.16). $\square$

For $k = 2$, no uniformity assumption is necessary: the number of nonconstant progressions is exactly $|A|(|A| - 1)$, so existence is equivalent to $|A| \geq 2$. For $\delta = 1$ and $N \geq 2$, a nonzero difference gives a progression immediately. For $N = 1$, no nonconstant progression exists. Empty and full sets have zero balanced function, so their progression counts are exact.

The conclusions of Corollary III.5.7 concern modular progressions. Transferring them to integer intervals requires a separate embedding without wraparound or a rectification argument. The improved uniformity threshold is a local improvement in the counting part of the proof; a stronger final quantitative Szemerédi theorem would also require corresponding control of the subsequent inverse and density-increment steps.

[write] **6 October 2026; sources 06–10.** Theorem III.5.2 is the telescope of sources 03 and 04 (Theorems III.1.2 and III.4.3); Remark III.5.4 is their optimal-order observation. The criteria of Corollary III.5.7 are compared with those of sources 03, 04 and 10 in the note after Corollary III.4.4: in the paper's α , (III.5.16) has the smallest power of δ . Not formalized; `corollary_3_3_holds` and `corollary_3_6_holds` keep the printed constants.

III.6 Normalization and the basic analytic tools

Source 08, delivered Section 2.

Let G be a finite abelian group, written additively, and let all expectations be normalized counting averages. Put

$$V_d = \{0, 1\}^d, \quad p = 2^d, \quad L_v(x, h) = x + \sum_{i=1}^d v_i h_i.$$

The variables $(x, h_1, \dots, h_d)$ are independent and uniform in G . Define the real multilinear cube average by

$$\Lambda_d((F_v)_{v \in V_d}) = \mathbb{E}_{x,h} \prod_{v \in V_d} F_v(L_v(x, h)), \quad \mathcal{C}_d(F) = \Lambda_d((F)_{v \in V_d}). \quad (\text{III.6.1})$$

For a set $A \subseteq G$, the unnormalized number of parameterized cubes in A is $|G|^{d+1} \mathcal{C}_d(\mathbf{1}_A)$. Degenerate cubes are included. Thus no assumption that the sampled vertices are distinct is hidden in the notation.

For a complex function g , the normalized Gowers norm is

$$\|g\|_{U^d(G)}^p = \mathbb{E}_{x,h} \prod_{v \in V_d} \mathcal{K}^{|v|} g(L_v(x, h)), \quad (\text{III.6.2})$$

where $\mathcal{K}$ is complex conjugation. For real functions,

$$\mathcal{C}_d(g) = \|g\|_{U^d(G)}^p. \quad (\text{III.6.3})$$

For $d = 1$ this is a seminorm, $\|g\|_{U^1} = |\mathbb{E}g|$; for $d \geq 2$ it is a norm.

We use the normalized Gowers–Cauchy–Schwarz inequality

$$\left| \mathbb{E}_{x,h} \prod_{v \in V_d} \mathcal{K}^{|v|} g_v(L_v) \right| \leq \prod_{v \in V_d} \|g_v\|_{U^d(G)}. \quad (\text{III.6.4})$$

This is the normalized form of the cube inequality in [1, Lemma 3.8]. Its repeated Cauchy–Schwarz proof uses only translation invariance of the uniform measure, so it applies to every finite abelian group. Real inputs allow us to omit the conjugations. Complex unconjugated inputs can be treated by replacing g_v with $\mathcal{K}^{|v|} g_v$.

Here are the consequences needed later. Padding one face of a $(d+1)$ -cube with the constant function 1, and the other face with g , gives from (III.6.4)

$$\|g\|_{U^d}^{2^d} \leq \|g\|_{U^{d+1}}^{2^d}.$$

Together with the U^1 identity this proves

$$|\mathbb{E}g| \leq \|g\|_{U^2} \leq \dots \leq \|g\|_{U^d} \leq \|g\|_{\infty}. \quad (\text{III.6.5})$$

Expanding the cube of $g + h$ and applying (III.6.4) term by term proves Minkowski’s inequality. These facts require no condition on the torsion of G .

Fix now a real function F with nonnegative mean, and write

$$\delta = \mathbb{E}F \geq 0, \quad f = F - \delta, \quad t = \|f\|_{U^d(G)}, \quad \mathcal{D}_d(F) = \mathcal{C}_d(F) - \delta^p. \quad (\text{III.6.6})$$

Most applications use $0 \leq F \leq 1$, but the polynomial identities hold for arbitrary real f of mean zero and arbitrary $\delta \geq 0$. By (III.6.5), $\mathcal{D}_d(F) \geq 0$ even without pointwise positivity of F .

For $S \subseteq V_d$, set

$$\Lambda_S(f) = \mathbb{E}_{x,h} \prod_{v \in S} f(L_v(x, h)), \quad \Lambda_{\emptyset}(f) = 1. \quad (\text{III.6.7})$$

Padding the other vertices with 1 in (III.6.4) gives the useful homogeneous estimate

$$|\Lambda_S(f)| \leq t^{|S|}. \quad (\text{III.6.8})$$

Unlike the estimate of every error term by t , this remembers the number of balanced factors.

III.7 Three exact cancellations and the fourth-order bound

Source 08, delivered Section 3.

Lemma III.7.1 (Three-vertex independence). *Let $1 \leq r \leq 3$, and let $v_1, \dots, v_r$ be distinct vertices of V_d . Then $(L_{v_1}, \dots, L_{v_r})$ is uniformly distributed on G^r . Consequently, for functions $g_1, \dots, g_r : G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_{x,h} \prod_{j=1}^r g_j(L_{v_j}) = \prod_{j=1}^r \mathbb{E} g_j.$$

Proof. Reflecting coordinate i of a cube is implemented by $x' = x + h_i$, $h'_i = -h_i$, leaving the other increments unchanged. This is a bijection of G^{d+1} preserving its uniform distribution. Reflect enough coordinates to send v_1 to 0.

For two vertices the other vertex is a nonzero binary vector u . At least one coefficient of $u \cdot h$ is 1, so this linear form is uniform and independent of x .

For three vertices write them as 0, u, v , with u, v distinct and nonzero. There are two coordinates i, j for which

$$\det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} = \pm 1.$$

Indeed, distinct nonzero binary vectors are linearly independent over $\mathbb{F}_2$; an invertible 2×2 binary minor has integer determinant ± 1 . After fixing all other increments, the map $(h_i, h_j) \mapsto (u \cdot h, v \cdot h)$ is therefore a bijection of G^2 : its inverse has integer coefficients. Together with the independent variable x , this makes $(x, x + u \cdot h, x + v \cdot h)$ uniform on G^3 . No division by 2 or another potentially noninvertible integer occurs. $\square$

Corollary III.7.2. *If $\mathbb{E}f = 0$ and $1 \leq |S| \leq 3$, then $\Lambda_S(f) = 0$. The same conclusion holds when the centered function at each selected vertex is allowed to be different.*

Theorem III.7.3 (Fourth-order refinement). *Let G be a finite abelian group, $d \geq 2$, $p = 2^d$, and let $F = \delta + f$ be real, where $\delta \geq 0$ and $\mathbb{E}f = 0$. With $t = \|f\|_{U^d(G)}$,*

$$0 \leq \mathcal{D}_d(F) \leq \sum_{r=4}^p \binom{p}{r} \delta^{p-r} t^r, \quad (\text{III.7.1})$$

$$\mathcal{C}_d(F) \leq (\delta + t)^p - p\delta^{p-1}t - \binom{p}{2} \delta^{p-2} t^2 - \binom{p}{3} \delta^{p-3} t^3. \quad (\text{III.7.2})$$

In dimension 2, the first estimate is an identity: $\mathcal{C}_2(F) = \delta^4 + \|f\|_{U^2}^4$.

Proof. The finite product expansion gives the exact identity

$$\mathcal{C}_d(\delta + f) = \sum_{S \subseteq V_d} \delta^{p-|S|} \Lambda_S(f). \quad (\text{III.7.3})$$

The empty subset contributes δ^p . Corollary III.7.2 removes all subsets of sizes 1, 2, 3. For each remaining subset, (III.6.8) bounds its absolute contribution by $\delta^{p-|S|} t^{|S|}$. There are $\binom{p}{r}$ subsets of size r . This proves the upper bound; the lower bound was established in (III.6.5). Expanding $(\delta + t)^p$ gives (III.7.2). When $d = 2$, the only nonempty surviving subset is the entire four-vertex cube, whose average is t^4 . $\square$

[write] **6 October 2026; sources 06–10.** The cancellation through degree three and (III.7.1) are also proved by source 03 (Lemma III.2.1, Corollary III.2.2), source 04 (Lemma III.3.1) and source 09 (Lemma III.17.1, Theorem III.17.2); four independent proofs. This t is 03's η , 04's u and 09's ε ; this $\mathcal{C}_d$ is their Q_d and $\mathcal{C}_d$. Not formalized; lemma_3_10_holds keeps the Minkowski bound.

III.7.1 Conversion to the original uniformity parameter

Define

$$K_d = \sum_{r=4}^p \binom{p}{r} = 2^p - 1 - p - \binom{p}{2} - \binom{p}{3}. \quad (\text{III.7.4})$$

For $F = \mathbf{1}_A$, $0 \leq \delta, t \leq 1$. If $t^p \leq \alpha \leq 1$, then

$$0 \leq \mathcal{C}_d(\mathbf{1}_A) - \delta^p \leq K_d \alpha^{4/p}. \quad (\text{III.7.5})$$

The original bound (F.13.1), after normalization and subtraction of δ^p , has a leading term of order $\alpha^{1/p}$ for fixed positive δ . Equation (III.7.5) replaces that exponent by $4/p$. For example, the exponents are 1, $1/2$, $1/4$, and $1/8$ in dimensions 2, 3, 4, 5, respectively. Theorem III.11.4 will show that this power cannot be increased in a bound uniform over groups and sets.

III.7.2 Density-sensitive versions

When $0 < t \leq \delta$, equation (III.7.1) gives

$$\mathcal{D}_d(F) \leq K_d \delta^{p-4} t^4. \quad (\text{III.7.6})$$

More precisely, for $\delta > 0$ put $s = t/\delta$. Then

$$\frac{\mathcal{D}_d(F)}{\delta^p} \leq B_p(s), \quad B_p(s) = \sum_{r=4}^p \binom{p}{r} s^r. \quad (\text{III.7.7})$$

This makes the relative scale t/δ , rather than t alone, explicit. For $0 < \varepsilon \leq 1$, a sufficient condition for relative error at most ε is

$$t^p \leq \delta^p \left(\frac{\varepsilon}{K_d} \right)^{p/4}. \quad (\text{III.7.8})$$

The condition implies $t \leq \delta$ because $K_d \geq 1$.

A useful closed bound, valid without $t \leq \delta$, is

$$\mathcal{D}_d(F) \leq \binom{p}{4} t^4 (\delta + t)^{p-4}. \quad (\text{III.7.9})$$

To verify it, compare coefficients using $\binom{p}{4} \binom{p-4}{r-4} = \binom{p}{r} \binom{r}{4} \geq \binom{p}{r}$. These convenient constants are not claimed to be optimal. The next section substantially reduces the fourth-order coefficient itself.

III.8 The exact quartic term and the role of two-torsion

Source 08, delivered Section 4.

III.8.1 Fourier notation

Write $\widehat{G}$ additively for the character group, and use $e_\chi(x)$ for the character corresponding to $\chi \in \widehat{G}$. Set

$$\begin{aligned}\widehat{f}(\chi) &= \mathbb{E}_x f(x) \overline{e_\chi(x)}, & E_4(f) &= \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^4, \\ E_{\langle 2 \rangle}(f) &= \sum_{\substack{\chi \in \widehat{G} \setminus \{0\} \\ 2\chi=0}} |\widehat{f}(\chi)|^4.\end{aligned}\tag{III.8.1}$$

For real f , $\widehat{f}(-\chi) = \overline{\widehat{f}(\chi)}$. Character orthogonality on a two-cube gives

$$E_4(f) = \|f\|_{U^2(G)}^4, \quad 0 \leq E_{\langle 2 \rangle}(f) \leq E_4(f) \leq t^4.\tag{III.8.2}$$

If G has odd order, $E_{\langle 2 \rangle}(f) = 0$. If G has exponent 2 and f has mean zero, $E_{\langle 2 \rangle}(f) = E_4(f)$.

For any $S \subseteq V_d$, expanding every factor by Fourier inversion yields

$$\Lambda_S(f) = \sum_{\substack{(\chi_v)_{v \in S} \in \widehat{G}^S \\ \sum_v \chi_v = 0, \sum_v v_i \chi_v = 0 \ (1 \leq i \leq d)}} \prod_{v \in S} \widehat{f}(\chi_v).\tag{III.8.3}$$

This formula is exact over every finite abelian group. In particular, one must solve the equations in $\widehat{G}$, not merely compute a rank over $\mathbb{Q}$.

III.8.2 Classification of four-vertex subsets

A set of four distinct binary vertices is an *ordinary parallelogram* if it can be labeled a, b, c, e with $a + b = c + e$ as an equality in $\mathbb{Z}^d$. It is a *binary affine plane* if its four vertices sum to 0 in $\mathbb{F}_2^d$. Every ordinary parallelogram is a binary affine plane, but the converse fails in dimension at least 3.

Lemma III.8.1 (Quartic classification). *Suppose $f : G \rightarrow \mathbb{R}$ has mean zero and $|S| = 4$. Then*

$$\Lambda_S(f) = \begin{cases} E_4(f), & S \text{ is an ordinary parallelogram,} \\ E_{\langle 2 \rangle}(f), & S \text{ is a binary affine plane but not an ordinary parallelogram,} \\ 0, & S \text{ is not a binary affine plane.} \end{cases}$$

Proof. Use cube reflections to write $S = \{0, u, v, w\}$ with distinct nonzero binary vectors u, v, w . Let the corresponding Fourier frequencies be $\lambda_0, \lambda_1, \lambda_2, \lambda_3$. The constraints in (III.8.3) are

$$\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 = 0, \quad u_i \lambda_1 + v_i \lambda_2 + w_i \lambda_3 = 0 \quad (1 \leq i \leq d).\tag{III.8.4}$$

First suppose the matrix with columns u, v, w has rank 2 over $\mathbb{Q}$. The vectors u, v have a unit 2×2 minor, and $w = au + bv$ for some rational a, b . Among the coordinate patterns (u_i, v_i) , at least two of 10, 01, 11 occur. If all three occur, the constraints that $a, b, a + b$ lie in $\{0, 1\}$ would force w to be 0, u , or v , which is excluded. If only 10, 01 occur, then $w = u + v$; if the two patterns are 10, 11 or 01, 11, then one of u, v, w is the sum of the other two. In every case the four points form a parallelogram. The unit minor shows that all solutions of (III.8.4) are the primitive parallelogram relation multiplied by a single character χ . The product of the four Fourier coefficients is $|\widehat{f}(\chi)|^4$.

Next suppose the matrix has rank 3 over $\mathbb{F}_2$. It has an odd 3×3 binary minor. Every 3×3 binary determinant has absolute value at most 2, so this minor is ± 1 and is invertible over every abelian group. Thus all four frequencies vanish, and the contribution is zero because $\hat{f}(0) = 0$. For completeness, the determinant bound follows by adjoining an all-one first row and column and replacing the binary block A by $1 - 2A$. The resulting 4×4 sign matrix has determinant $-8 \det A$ and absolute determinant at most 16 by Hadamard's inequality.

The remaining case has rank 3 over $\mathbb{Q}$ and rank 2 over $\mathbb{F}_2$. Since u, v, w are distinct and nonzero, their binary relation is $u + v + w = 0$. The only nonzero coordinate patterns (u_i, v_i, w_i) are 110, 101, 011. All three patterns must occur, since otherwise the rational rank would be at most 2. Equations (III.8.4) now give

$$\lambda_1 + \lambda_2 = \lambda_1 + \lambda_3 = \lambda_2 + \lambda_3 = 0.$$

Hence $\lambda_1 = \lambda_2 = \lambda_3 = \chi$, $2\chi = 0$, and $\lambda_0 = \chi$. The Fourier coefficient of an order-two character is real, and the trivial character contributes zero. The sum is precisely $E_{\langle 2 \rangle}(f)$. These cases exhaust all possibilities. $\square$

III.8.3 Counting the two types

Define

$$R_d = \frac{6^d - 2 \cdot 4^d + 2^d}{8}, \quad (\text{III.8.5})$$

$$F_d = \frac{2^d(2^d - 1)(2^d - 2)}{24}, \quad (\text{III.8.6})$$

$$T_d = F_d - R_d = \frac{8^d - 3 \cdot 6^d + 3 \cdot 4^d - 2^d}{24}. \quad (\text{III.8.7})$$

Here F_d is a combinatorial count and should not be confused with the function F .

Lemma III.8.2. *The numbers of ordinary parallelograms and binary affine planes in V_d are R_d and F_d , respectively. The number of non-parallelogram binary planes is T_d .*

Proof. For one binary coordinate, there are six ordered solutions of $a + b = c + e$: one with sum 0, four with sum 1, and one with sum 2. Thus there are 6^d ordered solutions in V_d . The repeated-vertex solutions are exactly those with $(a, b) = (c, e)$ or $(a, b) = (e, c)$, numbering $2 \cdot 4^d - 2^d$. Every remaining four-element parallelogram is counted eight times, corresponding to the orderings of its two diagonals and their interchange. This proves (III.8.5).

A binary affine plane has four possible origins and six ordered bases for its two-dimensional direction space. Choosing an origin and two independent directions in $\mathbb{F}_2^d$ gives $2^d(2^d - 1)(2^d - 2)$ choices, each plane occurring 24 times. This proves (III.8.6); subtraction gives (III.8.7). $\square$

d	$\binom{2^d}{4}$	R_d	T_d	F_d
2	1	1	0	1
3	70	12	2	14
4	1,820	100	40	140
5	35,960	720	520	1,240
6	635,376	4,816	5,600	10,416

The middle columns count all parallelograms, not just coordinate square faces. In particular, a three-dimensional Boolean cube contains twelve ordinary parallelograms, not six.

Theorem III.8.3 (Exact torsion-sensitive quartic expansion). *Under the hypotheses of Theorem III.7.3, put*

$$Q_d(f) = R_d E_4(f) + T_d E_{\langle 2 \rangle}(f).$$

Then

$$\mathcal{C}_d(\delta + f) = \delta^p + \delta^{p-4} Q_d(f) + \mathcal{R}_{\geq 5}, \quad (\text{III.8.8})$$

where

$$|\mathcal{R}_{\geq 5}| \leq \sum_{r=5}^p \binom{p}{r} \delta^{p-r} t^r. \quad (\text{III.8.9})$$

In particular,

$$\mathcal{D}_d(F) \leq F_d \delta^{p-4} t^4 + \sum_{r=5}^p \binom{p}{r} \delta^{p-r} t^r. \quad (\text{III.8.10})$$

If $|G|$ is odd, F_d in this upper bound can be replaced by R_d .

Proof. In (III.7.3), group the four-element subsets according to Lemma III.8.1 and use Lemma III.8.2. The remaining subsets have size at least five, so (III.6.8) gives (III.8.9). Finally, $E_{\langle 2 \rangle}(f) \leq E_4(f) \leq t^4$, with $E_{\langle 2 \rangle}(f) = 0$ in odd order. $\square$

Remark III.8.4 (What the torsion correction measures). For a group of exponent 2, the quartic coefficient is $F_d E_4(f)$. For an odd-order group it is $R_d E_4(f)$. In a general group, the additional contribution sees only the part of the Fourier spectrum lying on characters of order two. It is not a correction proportional to the total size of the two-torsion subgroup.

Remark III.8.5 (A sharp coefficient and a potentially non-sharp one). The universal leading coefficient F_d in (III.8.10) is sharp; a real order-two character attains it to fourth order (Proposition III.11.1). The coefficient R_d in the odd-order upper bound is the exact multiplier of $E_4(f)$, but its optimality as a multiplier of t^4 is a different question. We do not identify those two optimization problems.

[write] **6 October 2026; sources 06–10.** Theorem III.8.3 is source 03's Theorem III.2.6 (same R_d , T_d , and $E_{\langle 2 \rangle} = \mathcal{E}_{\langle 2 \rangle}$) and the quartic part of source 09's Theorem III.21.1 ($P_d = R_d$); three independent proofs, by Fourier kernels with binary minors here, by an XOR classification in source 03 and by lattice indices in source 09. The odd-order question in the last remark is source 03's Research question V.14.1 and this source's Question V.34.1; it is settled for $d = 3$ by source 04 (note there).

III.9 The complete three-cube expansion in odd order

Source 08, delivered Section 5.

In this section $d = 3$, $p = 8$, and G has odd order. Define

$$S_5(f) = \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \widehat{f}(\chi)^2 \overline{\widehat{f}(2\chi)}. \quad (\text{III.9.1})$$

The sum is real: the summand at $-\chi$ is the complex conjugate of the summand at χ . Set

$$\begin{aligned} A_6(f) &= \Lambda_{V_3 \setminus \{000, 001\}}(f), \\ B_6(f) &= \Lambda_{V_3 \setminus \{000, 011\}}(f), \\ C_6(f) &= \Lambda_{V_3 \setminus \{000, 111\}}(f), \\ A_7(f) &= \Lambda_{V_3 \setminus \{000\}}(f). \end{aligned}$$

Binary strings denote vertices, and Λ_S is still averaged over all four cube parameters.

Lemma III.9.1 (Five-vertex terms). *For a real mean-zero f on an odd-order group,*

$$\sum_{\substack{S \subseteq V_3 \\ |S|=5}} \Lambda_S(f) = 8S_5(f).$$

Proof. There are $\binom{8}{5} = 56$ five-element subsets. Exactly 48 contain an ordinary parallelogram: there are 12 parallelograms, each with four choices of an additional vertex, and no five-element set contains two different parallelograms. To justify the last claim, two parallelograms sharing three vertices lie in the same affine plane; an affine two-plane contains at most four vertices of a Boolean cube. One way to see the latter fact is to choose two coordinate projections that are injective on its two-dimensional direction space; they map its binary vertices injectively into $\{0, 1\}^2$.

A parallelogram together with a fifth vertex has just its parallelogram relation as a rational dependence. More importantly, this remains the only dependence over $\hat{G}$. Choose three affinely independent vertices of the parallelogram and the fifth vertex. Their 4×4 affine matrix has determinant ± 1 or ± 2 : reflect one chosen vertex to zero and use the binary 3×3 determinant bound from the proof of Lemma III.8.1. Multiplication by either determinant is invertible on an odd-order group. Subtracting a suitable multiple of the primitive parallelogram relation from any Fourier solution kills one parallelogram frequency, after which this invertible matrix kills all remaining frequencies. In particular, the fifth frequency is zero. Its factor $\hat{f}(0)$ makes the average vanish.

The other eight five-element sets are the cube symmetries of

$$S_* = \{000, 110, 101, 011, 111\}.$$

Indeed, for each of the eight vertices w , delete the three neighbors of w . These give eight distinct sets with no parallelogram. To check the latter property for S_* , its unique rational dependence, in the displayed order, is

$$(-1, 1, 1, 1, -2), \tag{III.9.2}$$

which has full support. A parallelogram would give a relation with one zero coefficient. The count $56 - 48 = 8$ proves exhaustion.

For S_* , the three coordinate equations show directly, over any character group, that the frequencies at 110, 101, 011 coincide, say at χ . The frequencies at 111 and 000 are -2χ and $-\chi$, respectively. Thus

$$\Lambda_{S_*}(f) = \sum_{\chi} \hat{f}(-\chi) \hat{f}(\chi)^3 \hat{f}(-2\chi) = S_5(f).$$

Cube reflections and coordinate permutations preserve the average, proving the asserted factor of eight. $\square$

Theorem III.9.2 (Exact three-cube polynomial). *Let G have odd order and $f : G \rightarrow \mathbb{R}$ have mean zero. For $\delta \geq 0$,*

$$\begin{aligned} \mathcal{C}_3(\delta + f) &= \delta^8 + 12\delta^4 E_4(f) + 8\delta^3 S_5(f) \\ &\quad + \delta^2 (12A_6(f) + 12B_6(f) + 4C_6(f)) \\ &\quad + 8\delta A_7(f) + \mathcal{C}_3(f). \end{aligned} \tag{III.9.3}$$

Consequently, with $t = \|f\|_{U^3(G)}$,

$$0 \leq \mathcal{D}_3(\delta + f) \leq 12\delta^4 t^4 + 8\delta^3 t^5 + 28\delta^2 t^6 + 8\delta t^7 + t^8. \tag{III.9.4}$$

Proof. The constant and degree-four terms follow from Theorem III.8.3, since $R_3 = 12$ and there is no order-two Fourier contribution. Lemma III.9.1 gives the fifth-degree term. A six-element subset is determined by its missing pair of vertices. Pairs at Hamming distances 1, 2, 3 form three symmetry classes of sizes 12, 12, 4, respectively. These give the three degree-six averages displayed above. The eight seven-element subsets are all equivalent under cube reflections, and the eighth-degree term is the full cube of f . This proves the identity. Apply (III.6.8) to each average, and use $E_4(f) \leq t^4$. $\square$

The naive fourth-order estimate in dimension three has coefficients 70, 56, 28, 8, 1 in degrees 4, 5, 6, 7, 8. The all-group quartic analysis changes the first coefficient to 14. In odd order the exact support analysis improves these coefficients to 12, 8, 28, 8, 1. In particular, if $t \leq \delta$,

$$\mathcal{D}_3(\delta + f) \leq 57\delta^4 t^4. \tag{III.9.5}$$

This is a convenient uniform corollary, not an assertion that 57 is the best possible constant.

[write] **6 October 2026; sources 06–10.** Theorem III.9.2 contains source 04's Lemma III.3.3 (the same support counts 12, 8, 28, 8, 1; this S_5 is 04's B_5) and is the $d = 3$ case of source 09's Theorem III.21.1 in odd order. Source 04's Theorem III.3.6 replaces the leading $12\delta^4 t^4$ by the sharp $6\sqrt{2}\delta^4 t^4$.

III.9.1 A Fourier-sensitive remainder

Let $m(f) = \max_{\chi} |\widehat{f}(\chi)|$. Directly from (III.9.1),

$$|S_5(f)| \leq m(f) E_4(f). \tag{III.9.6}$$

Together with the cube inequality, we also have $|S_5(f)| \leq t^5$. Thus (III.9.3) gives the two-sided estimate

$$\left| \mathcal{D}_3(\delta + f) - 12\delta^4 E_4(f) \right| \leq 8\delta^3 m(f) E_4(f) + 28\delta^2 t^6 + 8\delta t^7 + t^8. \tag{III.9.7}$$

This is useful when the fourth Fourier moment is substantially smaller than t^4 . It does not turn U^2 uniformity alone into control of the full three-cube count: the remaining t^6, t^7, t^8 terms still matter.

The formula also identifies a concrete interaction invisible in a simple norm bound. The fifth-order contribution couples frequencies χ and 2χ . Its sign depends on their relative phases. Even for real weights, that sign need not be positive.

III.10 A counterexample to the quartic global lower bound

Source 08, delivered Section 6.

The positive fourth-order term might suggest the stronger lower estimate

$$\mathcal{D}_3(F) \stackrel{?}{\geq} 12\delta^4 E_4(F - \delta) \quad (|G| \text{ odd}, \quad 0 \leq F \leq 1). \quad (\text{III.10.1})$$

It is false. This is a natural strengthening posed and settled within this memorandum, not a claim to settle a previously identified published conjecture.

Theorem III.10.1 (Negative fifth-order correction). *On $G = \mathbb{Z}/11\mathbb{Z}$, put*

$$h(x) = \cos(2\pi x/11) - \cos(4\pi x/11).$$

For every fixed $0 < \delta < 1$, the weight $F = \delta + \eta h$ belongs to $[0, 1]$ and violates (III.10.1) for all sufficiently small positive η .

Proof. The nonzero Fourier coefficients of h are

$$\hat{h}(1) = \hat{h}(-1) = \frac{1}{2}, \quad \hat{h}(2) = \hat{h}(-2) = -\frac{1}{2}.$$

Thus $\mathbb{E}h = 0$, $E_4(h) = 1/4$, and (III.9.1) gives $S_5(h) = -1/16$: only $\chi = 1, -1$ contribute, each with value $-1/32$. Applying (III.9.3) to $f = \eta h$ yields

$$\mathcal{D}_3(\delta + \eta h) - 12\delta^4 E_4(\eta h) = -\frac{1}{2}\delta^3 \eta^5 + O_\delta(\eta^6), \quad (\text{III.10.2})$$

which is negative for sufficiently small positive η . Since $|h| \leq 2$, the condition $2\eta \leq \min(\delta, 1 - \delta)$ guarantees $0 \leq F \leq 1$.

There is also a completely explicit safe range. Use $\|h\|_{U^3} \leq 2$ in the terms of degree at least six in (III.9.3). Their absolute value, divided by $\frac{1}{2}\delta^3 \eta^5$, is at most

$$3584 \frac{\eta}{\delta} + 2048 \left(\frac{\eta}{\delta}\right)^2 + 512 \left(\frac{\eta}{\delta}\right)^3.$$

This is less than 1 when $0 < \eta/\delta \leq 10^{-4}$. Combining that condition with the pointwise range condition proves the claim without any numerical evaluation of a Gowers norm. $\square$

III.10.1 An exact finite certificate and a larger explicit example

For the same function h , a finite Fourier calculation gives

$$\mathcal{C}_3(\delta + \eta h) = \delta^8 + 3\delta^4 \eta^4 - \frac{1}{2}\delta^3 \eta^5 + \frac{3}{2}\delta^2 \eta^6 + \frac{1}{4}\eta^8. \quad (\text{III.10.3})$$

Here is a compact way to reproduce and check that calculation. The four Fourier constraints on the eight cube vertices have the following complete parametrization, with arithmetic modulo 11:

v	000	001	010	100	011	101	110	111
λ_v	a	b	c	e	$-a - b - c$	$-a - b - e$	$-a - c - e$	$2a + b + c + e$.

(III.10.4)

Every frequency must lie in $D = \{0, 1, -1, 2, -2\}$. A zero frequency contributes δ ; a frequency ± 1 contributes $\eta/2$; a frequency ± 2 contributes $-\eta/2$. It therefore suffices to check $5^4 = 625$ tuples $(a, b, c, e) \in D^4$. The resulting signed counts are

Degree r	0	1	2	3	4	5	6	7	8
Positive products	1	0	0	0	48	0	96	0	64
Negative products	0	0	0	0	0	16	0	0	0

Dividing the signed count in degree r by 2^r gives (III.10.3). Appendix III.15 supplies the short exhaustive certificate; the accompanying verification program also checks it against an independent eight-frequency enumeration. The general negative-sign conclusion in Theorem III.10.1 does not depend on this enumeration.

For an explicit example with comfortable margins from 0 and 1, take $\delta = 1/2$ and $\eta = 1/12$. Then $1/3 \leq F \leq 2/3$ and

$$\frac{\mathcal{D}_3(F)}{\delta^4 E_4(F - \delta)} = 12 - 2s + 6s^2 + s^4 \Big|_{s=1/6} = \frac{15337}{1296} < 12. \quad (\text{III.10.5})$$

The distinction between a local Taylor coefficient and a global lower-bound coefficient is therefore substantive, not just a possible technical caveat.

[write] **6 October 2026; sources 06–10.** Kept on record as Remark W1 in source 03’s section, where source 03 notes the same failure on $\mathbb{Z}/3\mathbb{Z}$; source 09’s Proposition III.22.2 proves it on primes $q \geq 17$. The polynomial (III.10.3) and the ratio (III.10.5) were recomputed independently at the second write (all 11^4 frequency choices, exact arithmetic).

III.11 Sharpness of the fourth-order exponent

Source 08, delivered Section 7.

III.11.1 An exact order-two model

Proposition III.11.1. *Suppose G has a nontrivial character $\chi : G \rightarrow \{1, -1\}$. For $p = 2^d$, $d \geq 2$,*

$$\mathcal{C}_d(\delta + \eta\chi) = \frac{(\delta + \eta)^p + (\delta - \eta)^p + 2(p-1)(\delta^2 - \eta^2)^{p/2}}{2p}. \quad (\text{III.11.1})$$

Moreover, $\|\eta\chi\|_{U^d} = |\eta|$, and for fixed $\delta > 0$,

$$\mathcal{D}_d(\delta + \eta\chi) = F_d \delta^{p-4} \eta^4 + O_{d,\delta}(\eta^6).$$

Thus the leading constant F_d in the all-group bound (III.8.10) is sharp.

Proof. The $d+1$ signs $\chi(x), \chi(h_1), \dots, \chi(h_d)$ are independent and uniform. If every increment sign is 1, all cube values have the same character sign; the two choices of $\chi(x)$ give the first two terms in the numerator. There are $p-1$ other increment-sign patterns. For each, exactly $p/2$ vertices have each sign, so either base sign gives the product $(\delta^2 - \eta^2)^{p/2}$. This proves the exact formula.

A character has Gowers norm 1 in every dimension at least two. The coefficient of η^4 in (III.11.1) is

$$\frac{2\binom{p}{4} + 2(p-1)\binom{p/2}{2}}{2p} = \frac{p(p-1)(p-2)}{24} = F_d.$$

The coefficient of η^2 cancels and all odd coefficients vanish. The remainder is of degree at least six, with the zero remainder understood when $p = 4$. $\square$

[write] **6 October 2026; sources 06–10.** (III.11.1) is source 09's (III.22.1) (the two closed forms were expanded and compared at the second write for $d = 2, \dots, 5$). Its consequence, that the universal coefficient F_d is attained, is also source 03's optimality of K_d over all groups and source 09's Proposition III.22.1.

III.11.2 Odd cyclic groups and real cosine perturbations

To show that the exponent does not rely on two-torsion, let $N > p$ be an odd prime and put

$$h_N(x) = \cos(2\pi x/N), \quad x \in \mathbb{Z}/N\mathbb{Z}.$$

Then $\mathbb{E}h_N = 0$ and $E_4(h_N) = 1/8$. There is a positive constant c_d , depending only on d , such that

$$\|h_N\|_{U^d} = c_d \quad (N > p). \quad (\text{III.11.2})$$

Indeed, in the Fourier expansion of its full cube, every frequency is 1 or -1 . Every constraint is an integer of absolute value at most p ; since $N > p$, vanishing modulo N is equivalent to integer vanishing. The number of admissible frequency assignments is therefore independent of N . There is at least one assignment, namely the alternating cube assignment $\lambda_v = (-1)^{|v|}$, and every summand is positive. This proves (III.11.2) and $c_d > 0$.

For fixed $0 < \delta < 1$ and sufficiently small $\eta > 0$, the weight $F_N = \delta + \eta h_N$ takes values in $[0, 1]$. By Theorem III.8.3, uniformly for $N > p$,

$$\mathcal{D}_d(F_N) = \frac{R_d}{8} \delta^{p-4} \eta^4 + O_{d,\delta}(\eta^5), \quad \|F_N - \delta\|_{U^d}^p = c_d^p \eta^p. \quad (\text{III.11.3})$$

Since $R_d > 0$, no exponent larger than $4/p$ can replace $4/p$ in a uniform bound for these bounded real weights. The remainder could be improved in this particular example, but degree five already suffices for sharpness.

III.11.3 A rounding lemma in the Gowers norm

Weighted sharpness does not automatically imply sharpness for sets, so a separate argument is needed.

Lemma III.11.2 (Independent rounding). *Let $F : G \rightarrow [0, 1]$, $|G| = N$, and $d \geq 2$, $p = 2^d$. There exists a set $A \subseteq G$ such that*

$$\|\mathbf{1}_A - F\|_{U^d(G)} \leq \left(\frac{2\binom{p}{2}}{N} \right)^{1/p}. \quad (\text{III.11.4})$$

Proof. Choose the random variables $\mathbf{1}_A(y)$ independently, with success probabilities $F(y)$, and set $X(y) = \mathbf{1}_A(y) - F(y)$. They have mean zero under this random choice and satisfy $|X(y)| \leq 1$. For a fixed cube whose p sampled vertices are distinct, independence implies

$$\mathbb{E}_{\text{rand}} \prod_{v \in V_d} X(L_v) = 0.$$

For any pair of distinct Boolean vertices v, w , the difference $L_v - L_w$ is a linear form in the increments with at least one coefficient 1 or -1 . It is uniform in G , so $\Pr_{x,h}(L_v = L_w) = 1/N$. By the union bound, the probability that the sampled cube has any collision is at most $\binom{p}{2}/N$.

The product is bounded in absolute value by 1 in all cases. Interchanging the two finite averages therefore gives

$$0 \leq \mathbb{E}_{\text{rand}} \|X\|_{U^d}^p \leq \frac{\binom{p}{2}}{N}.$$

Markov's inequality shows that some realization has the moment at most twice this bound, which is (III.11.4). The random centering used here is pointwise in the probability space; it need not give $\mathbb{E}_{y \in G} X(y) = 0$ for an individual realization. $\square$

[write] **6 October 2026; sources 06–10.** Source 09's Lemma III.22.4 is the same random-rounding lemma with the constant $(\binom{p}{2}/|G|)^{1/p}$ (without the factor two of the Markov step); source 03 rounds on $(\mathbb{Z}/2\mathbb{Z})^n$ and $(\mathbb{Z}/3\mathbb{Z})^n$ with an exact density and source 04 on prime cyclic groups by variance. Four routes to sharpness for sets.

Lemma III.11.3 (Continuity of cube excess). *If $A \subseteq G$, $F : G \rightarrow [0, 1]$, $\delta = \mathbb{E}F$, $\delta_A = |A|/|G|$, and $q = \|\mathbf{1}_A - F\|_{U^d}$, then*

$$|\delta_A - \delta| \leq q, \quad (\text{III.11.5})$$

$$|\mathcal{D}_d(\mathbf{1}_A) - \mathcal{D}_d(F)| \leq 2pq, \quad (\text{III.11.6})$$

$$\| \|\mathbf{1}_A - \delta_A\|_{U^d} - \|F - \delta\|_{U^d} \| \leq 2q. \quad (\text{III.11.7})$$

Proof. The first assertion is (III.6.5) applied to $X = \mathbf{1}_A - F$. Telescope the difference between the two p -factor cube products. Each of the p terms has one factor X , while all other factors are F or $\mathbf{1}_A$, whose U^d norms are at most 1. Equation (III.6.4) gives $|\mathcal{C}_d(\mathbf{1}_A) - \mathcal{C}_d(F)| \leq pq$. Since $\delta, \delta_A \in [0, 1]$, $|\delta_A^p - \delta^p| \leq p|\delta_A - \delta| \leq pq$. This proves the excess estimate. Finally the difference of the centered functions is $X - (\delta_A - \delta)$, whose norm is at most $2q$ by Minkowski and the constant-function norm. The reverse triangle inequality proves the last claim. $\square$

Theorem III.11.4 (Sharpness for genuine sets in odd prime order). *Fix $d \geq 2$, $p = 2^d$, and $0 < \delta < 1$. There are primes $N_j \rightarrow \infty$ and sets $A_j \subseteq \mathbb{Z}/N_j\mathbb{Z}$ such that, with $\delta_j = |A_j|/N_j$ and $\alpha_j = \|\mathbf{1}_{A_j} - \delta_j\|_{U^d}^p$,*

$$\delta_j \rightarrow \delta, \quad \alpha_j \rightarrow 0, \quad \frac{\mathcal{D}_d(\mathbf{1}_{A_j})}{\alpha_j^{4/p}} \rightarrow \frac{R_d \delta^{p-4}}{8c_d^4} > 0.$$

For every $\beta > 4/p$, $\mathcal{D}_d(\mathbf{1}_{A_j})/\alpha_j^\beta \rightarrow \infty$. Thus a bound uniform over such sets, even with density restricted to any fixed neighborhood of δ , cannot have exponent larger than $4/p$.

Proof. Choose $\eta_j = c/j$ with $0 < c < \min(\delta, 1 - \delta)$. Choose a prime $N_j > \max(p, j^{8p})$ and set $F_j = \delta + \eta_j h_{N_j}$. Apply Lemma III.11.2 to obtain A_j with rounding error $q_j = O_d(j^{-8}) = o(\eta_j^4)$. By Lemma III.11.3 and (III.11.3),

$$\begin{aligned} \delta_j &= \delta + o(1), \\ \|\mathbf{1}_{A_j} - \delta_j\|_{U^d} &= c_d \eta_j + o(\eta_j), \\ \mathcal{D}_d(\mathbf{1}_{A_j}) &= \frac{R_d}{8} \delta^{p-4} \eta_j^4 + o(\eta_j^4). \end{aligned}$$

Divide the last relation by the fourth power of the norm in the second. For $\beta > 4/p$, the additional factor $\alpha_j^{4/p-\beta}$ tends to infinity. $\square$

Remark III.11.5 (The density statement is intentionally asymptotic). The sets in this theorem have densities tending to the prescribed δ ; we do not assert exact density δ in every prime cyclic group. For instance, density $1/2$ is impossible in an odd-order group. This does not weaken sharpness for a uniform bound such as (III.7.5).

III.12 Lower bounds and Fourier stability

Source 08, delivered Section 8.

The preceding upper estimates can be paired with a strengthened form of the basic lower cube count. This section records consequences of norm monotonicity and Parseval, rather than attributing a new general norm theorem to the present calculation.

Theorem III.12.1 (Fourier-moment lower bound). *For a real function $F = \delta + f$ on a finite abelian group, where $\delta \geq 0$ and $\mathbb{E}f = 0$, and for $d \geq 2$, $p = 2^d$,*

$$\mathcal{C}_d(F) \geq (\delta^4 + E_4(f))^{p/4}. \quad (\text{III.12.1})$$

In particular,

$$\mathcal{D}_d(F) \geq \frac{p}{4} \delta^{p-4} E_4(f). \quad (\text{III.12.2})$$

For $\delta > 0$, if $D = \mathcal{D}_d(F)$, then

$$E_4(f) \leq (\delta^p + D)^{4/p} - \delta^4 \leq \frac{4}{p} \delta^{4-p} D. \quad (\text{III.12.3})$$

Equality in $\mathcal{C}_d(F) \geq \delta^p$ holds if and only if F is constant.

Proof. The constant frequency of F is δ , while all other Fourier coefficients are those of f . Thus $\|F\|_{U^2}^4 = \delta^4 + E_4(f)$. Now $\|F\|_{U^d} \geq \|F\|_{U^2}$, giving (III.12.1). The binomial theorem, since $p/4$ is a positive integer, implies (III.12.2). Rearrange (III.12.1) to obtain the first inequality in (III.12.3). The concavity of $u \mapsto u^{4/p}$ gives the second inequality by the tangent line at $u = \delta^p$. If the cube count equals δ^p , then (III.12.1) forces $E_4(f) = 0$. Every Fourier coefficient of f is zero, so $f = 0$. The converse is immediate. $\square$

As a direct stability statement, every nontrivial Fourier coefficient obeys

$$\max_{\chi \neq 0} |\widehat{F}(\chi)| \leq \left(\frac{4}{p} \delta^{4-p} \mathcal{D}_d(F) \right)^{1/4} \quad (\delta > 0).$$

This is stability in Fourier space. It is not closeness in L^2 to a constant: for an indicator, the squared L^2 distance is exactly $\delta(1 - \delta)$, even along the near-minimal sequences in Theorem III.11.4. The displayed inequality itself controls U^2 , not the balanced U^d norm; a stronger norm-geometric conclusion would require an additional argument.

Corollary III.12.2 (A finite-size correction). *Suppose $|G| = N > 1$ and write $\sigma^2 = \mathbb{E}|F - \delta|^2$. Then*

$$\mathcal{C}_d(F) \geq \left(\delta^4 + \frac{\sigma^4}{N-1} \right)^{p/4}. \quad (\text{III.12.4})$$

For $A \subseteq G$ of density δ ,

$$\mathcal{C}_d(\mathbf{1}_A) \geq \left(\delta^4 + \frac{\delta^2(1-\delta)^2}{N-1} \right)^{p/4}. \quad (\text{III.12.5})$$

Proof. Parseval and $\widehat{f}(0) = 0$ give $\sum_{\chi \neq 0} |\widehat{f}(\chi)|^2 = \sigma^2$. Cauchy–Schwarz over the $N - 1$ nontrivial characters yields $E_4(f) \geq \sigma^4/(N - 1)$. Insert this into (III.12.1). For an indicator, $\sigma^2 = \delta(1 - \delta)$. $\square$

The lower coefficient $p/4$ in (III.12.2) and the local coefficient R_d in odd order are different quantities. Theorem III.10.1 shows explicitly why one cannot simply replace the former with the latter. Determining the best global lower coefficient remains a separate optimization problem.

III.13 Different weights at different vertices

Source 08, delivered Section 9.

The centering argument is not limited to counting cubes in a single set. Suppose each vertex has a real weight

$$F_v = \delta_v + f_v, \quad \delta_v = \mathbb{E}F_v, \quad \mathbb{E}f_v = 0, \quad t_v = \|f_v\|_{U^d}.$$

The means may be different and may be negative.

Theorem III.13.1 (Inhomogeneous fourth-order estimate). *With the notation above,*

$$\left| \Lambda_d((F_v)) - \prod_{v \in V_d} \delta_v \right| \leq \sum_{\substack{S \subseteq V_d \\ |S| \geq 4}} \prod_{v \notin S} |\delta_v| \prod_{v \in S} t_v. \quad (\text{III.13.1})$$

All terms with one, two, or three centered factors vanish identically.

Proof. Expand the product by the subset of vertices at which the centered factor is chosen. Lemma III.7.1 makes every term of sizes one to three a product of zero means. Pad the remaining selected inputs with constant 1 in (III.6.4) and take absolute values. $\square$

The quartic classification is also valid with different real centered inputs. If $S = \{a, b, c, e\}$ is an ordinary parallelogram with $a + b = c + e$, the associated mixed fourth-order average is

$$\sum_{\chi} \widehat{f}_a(\chi) \widehat{f}_b(\chi) \overline{\widehat{f}_c(\chi)} \overline{\widehat{f}_e(\chi)}. \quad (\text{III.13.2})$$

For a non-parallelogram binary plane it is

$$\sum_{\substack{\chi \neq 0 \\ 2\chi=0}} \prod_{v \in S} \widehat{f}_v(\chi). \quad (\text{III.13.3})$$

For every other four-set it vanishes. These statements follow from the same character equations as Lemma III.8.1. Hölder’s inequality bounds the absolute value of either expression by

$$\prod_{v \in S} \|f_v\|_{U^2}.$$

Thus one can improve (III.13.1) further by keeping the quartic Fourier expressions and using U^d norms only from degree five onward. Unlike the common-function quartic term, a mixed quartic expression need not be nonnegative. Accordingly, no lower bound by $\prod_v \delta_v$ is asserted in this inhomogeneous setting.

This version applies, for example, when each cube vertex must belong to a different set A_v . It isolates the contribution of the densities and shows that the first possible deviation involves four of the balanced vertex conditions, rather than just one.

III.14 Which centered terms can survive?

Source 08, delivered Section 10.

The four- and five-vertex calculations are instances of a general finite linear-algebra principle. It is useful both for organizing higher coefficients and for designing checkable certificates.

III.14.1 An exact character-kernel criterion

Let $a_1, \dots, a_r \in \mathbb{Z}^m$ be coefficient columns and let $L_j(z) = a_j \cdot z$ for $z \in G^m$. Let M be the matrix with these columns. For independently chosen functions $g_j : G \rightarrow \mathbb{C}$,

$$\mathbb{E}_{z \in G^m} \prod_{j=1}^r g_j(L_j(z)) = \sum_{\lambda \in \ker_{\widehat{G}} M} \prod_{j=1}^r \widehat{g}_j(\lambda_j), \quad (\text{III.14.1})$$

where $\ker_{\widehat{G}} M$ means the solutions of $M\lambda = 0$ in $\widehat{G}^r$. A vector has *full support* if none of its coordinates is zero.

Proposition III.14.1 (Universal vanishing criterion). *The following conditions are equivalent:*

- (i) *The kernel $\ker_{\widehat{G}} M$ has no full-support vector.*
- (ii) *The average in (III.14.1) is zero for every choice of complex mean-zero functions $g_1, \dots, g_r$.*
- (iii) *The average is zero for every common real mean-zero function $g_1 = \dots = g_r = f$.*

Proof. If no kernel vector has full support, every summand in (III.14.1) contains a factor $\widehat{g}_j(0) = 0$. Thus (i) implies (ii), and (ii) implies (iii).

Conversely, suppose $\lambda \in \ker_{\widehat{G}} M$ has full support. For the independently chosen complex functions one may simply take $g_j = e_{\lambda_j}$; their product along the forms is identically 1. To obtain a common real witness as well, take

$$f = \sum_{j=1}^r (e_{\lambda_j} + e_{-\lambda_j}).$$

This function is real and has mean zero. Its Fourier coefficients are all nonnegative real numbers, and every frequency λ_j has a strictly positive coefficient. Therefore all summands in (III.14.1) are nonnegative and the summand indexed by λ is positive. This contradicts (iii), proving (iii) implies (i). Rescaling gives bounded witness functions if desired. $\square$

For a cube subset S , use the matrix with columns $(1, v)$ for $v \in S$. Let $A_{d,r}(G)$ be the number of r -element subsets whose character kernel has a full-support vector. We then have the systematic improvement

$$|\mathcal{C}_d(\delta + f) - \delta^p| \leq \sum_{r=1}^p A_{d,r}(G) \delta^{p-r} t^r \quad (\delta \geq 0, \mathbb{E}f = 0). \quad (\text{III.14.2})$$

In particular, $A_{d,1} = A_{d,2} = A_{d,3} = 0$. For a nontrivial odd-order group $A_{d,4} = R_d$, while a nontrivial elementary abelian two-group has $A_{d,4} = F_d$. The coefficients $A_{d,r}$ are exact counts of potentially nonzero supports, but the resulting norm bound is not automatically optimal: the same function need not simultaneously saturate every surviving Gowers–Cauchy–Schwarz inequality.

III.14.2 The large-prime criterion and exceptional characteristics

The column-dependence structure of an integer matrix is often called its *column matroid*. Here only a simple part of that language is needed: a column is a *coloop* if it does not lie in the span of the other columns.

Proposition III.14.2. *Fix an integer matrix M with r columns. There is a finite set of primes $\mathcal{B}_M$ such that, for every prime $q \notin \mathcal{B}_M$ with $q > r$,*

$$\ker_{\mathbb{F}_q} M \text{ has a full-support vector} \iff M \text{ has no coloop over } \mathbb{Q}.$$

Proof. For M and for each matrix obtained by deleting one column, choose a nonzero minor of size equal to its rational rank; for a rank-zero matrix, use the empty minor 1. Let $\mathcal{B}_M$ contain the prime divisors of these finitely many nonzero integers. Outside $\mathcal{B}_M$, all these ranks are preserved modulo q , since the chosen minors remain nonzero and larger minors vanish identically over the integers. Consequently, the coloop condition agrees over $\mathbb{Q}$ and $\mathbb{F}_q$.

If the j th column is a coloop, every relation has $\lambda_j = 0$, so no full-support vector exists. Conversely, suppose there is no coloop and let $K = \ker_{\mathbb{F}_q} M$ have dimension k . For every j , the coordinate map $\lambda \mapsto \lambda_j$ is a nonzero linear functional on K . Its zero set has size q^{k-1} . The union of the r coordinate zero sets has at most $rq^{k-1} < q^k$ elements. Some vector in K avoids all of them. $\square$

The restriction $q > r$ is deliberate. Even when every coordinate functional is nonzero, finitely many coordinate hyperplanes can cover a vector space over a small field. Likewise, a rank computation over $\mathbb{Q}$ alone cannot detect the order-two quartic terms found in Section III.8. For a general finite abelian group, one must solve the character-group constraints themselves; integer normal forms and prime-primary decomposition provide one way to organize that computation.

This framework is not a substitute for true-complexity theory. It decides *identical vanishing caused by centering*. True-complexity results ask which uniformity norms control the remaining nonzero configuration averages [29, 14]. The two analyses can be used consecutively: first remove terms that vanish exactly, then apply the strongest available norm estimate to the terms that remain.

III.15 A short exact certificate for the counterexample

Source 08, delivered Appendix A.

The code below implements (III.10.4) directly. Every loop variable and every comparison is an integer. Because that parametrization is a bijection between four free frequencies and the Fourier kernel, the enumeration has no omitted solutions and no multiplicities to correct. The returned coefficients are exact rational numbers.

```
from collections import Counter
from fractions import Fraction
from itertools import product

D = {0, 1, 10, 2, 9} # residues 0, +1, -1, +2, -2 mod 11
sign = {1: 1, 10: 1, 2: -1, 9: -1}
counts = Counter()
for a, b, c, e in product(D, repeat=4):
```

```

frequencies = [
    a, b, c, e,
    (-a-b-c) % 11, (-a-b-e) % 11,
    (-a-c-e) % 11, (2*a+b+c+e) % 11,
]
if not all(k in D for k in frequencies):
    continue
degree = sum(k != 0 for k in frequencies)
product_sign = 1
for k in frequencies:
    if k != 0:
        product_sign *= sign[k]
counts[degree, product_sign] += 1

coefficients = [
    Fraction(counts[r, 1] - counts[r, -1], 2**r)
    for r in range(9)
]
assert coefficients == [
    1, 0, 0, 0, 3, Fraction(-1, 2), Fraction(3, 2), 0,
    Fraction(1, 4),
]

```

The coefficient of degree r multiplies $\delta^{8-r}\eta^r$. This certificate is elementary executable arithmetic, not a Lean kernel certificate. The full verification program includes additional independent checks and records their output in `verification_results.json`.

III.16 Normalized cube forms and the analytic input

Source 09, delivered Section 2.

All groups in this paper are finite abelian groups written additively; the trivial group is allowed. Averages are normalized. The character group $\hat{G}$ is written multiplicatively, with identity character 1. The normalized Fourier transform is

$$a_\chi = \hat{f}(\chi) = \mathbb{E}_{x \in G} f(x) \overline{\chi(x)}, \quad f(x) = \sum_{\chi \in \hat{G}} a_\chi \chi(x). \quad (\text{III.16.1})$$

For real f , $a_{\chi^{-1}} = \overline{a_\chi}$, and $\mathbb{E}f = 0$ means $a_1 = 0$.

Let $V_d = \{0, 1\}^d$, and put

$$L_v(x, h) = x + \sum_{i=1}^d v_i h_i \quad (x \in G, h \in G^d, v \in V_d).$$

For real F , define

$$\mathcal{C}_d(F) = \mathbb{E}_{x, h} \prod_{v \in V_d} F(L_v(x, h)). \quad (\text{III.16.2})$$

For complex functions the Gowers form has complex conjugation at odd vertices:

$$\Lambda_d((g_v)) = \mathbb{E}_{x, h} \prod_{v \in V_d} \mathcal{C}^{|v|} g_v(L_v(x, h)), \quad \|g\|_{U^d}^{2^d} = \Lambda_d((g)_v),$$

where $\mathcal{C}$ denotes conjugation. For real functions the conjugations disappear, so (III.16.2) is $\|F\|_{U^d}^{2^d}$. For $d \geq 2$ this is a norm power. The case $d = 1$ is the seminorm $|\mathbb{E}F|$ and will not be used for the main results.

Lemma III.16.1 (Gowers–Cauchy–Schwarz). *For arbitrary complex functions $g_v : G \rightarrow \mathbb{C}$,*

$$|\Lambda_d((g_v))| \leq \prod_{v \in V_d} \|g_v\|_{U^d}.$$

Consequently, if f is real and $S \subseteq V_d$, then

$$\left| \mathbb{E}_{x,h} \prod_{v \in S} f(L_v(x, h)) \right| \leq \|f\|_{U^d}^{|S|}. \quad (\text{III.16.3})$$

The first assertion is the normalized form of Gowers’ Lemma 3.8 [1]. Its iterated Cauchy–Schwarz proof works over every finite abelian group: at each coordinate, separate the two opposite faces, apply Cauchy–Schwarz, and duplicate the averaging variable. After all coordinates have been duplicated, each vertex function has its own full cube average. If one starts with the paper’s bounded version, scaling each of the finitely many functions proves the unrestricted version. The second assertion follows by setting $g_v = f$ on S and $g_v = 1$ elsewhere.

For completeness, the auxiliary norm facts needed below follow directly from this input. Taking one vertex function to be g and all others one gives $|\mathbb{E}g| \leq \|g\|_{U^d}$. Taking g on one face and one on the opposite face gives $\|g\|_{U^{d-1}} \leq \|g\|_{U^d}$ for $d \geq 2$, interpreting U^1 as above. Expanding the cube form of $g + h$ and applying Lemma III.16.1 termwise gives the triangle inequality. Finally, Fourier orthogonality gives

$$\|f\|_{U^2}^4 = \sum_{\chi \in \widehat{G}} |a_\chi|^4. \quad (\text{III.16.4})$$

These observations also prove $\mathcal{C}_d(F) \geq (\mathbb{E}F)^m$ for real F with nonnegative mean.

Definition III.16.2 (Subset moments and binomial tails). For $S \subseteq V_d$, put

$$T_S(f) = \mathbb{E}_{x,h} \prod_{v \in S} f(L_v(x, h)), \quad T_\emptyset(f) = 1.$$

For $0 \leq r \leq m$ and $\delta, \varepsilon \geq 0$, put

$$B_{m,r}(\delta, \varepsilon) = \sum_{s=r}^m \binom{m}{s} \delta^{m-s} \varepsilon^s.$$

The elementary but useful estimate

$$B_{m,r}(\delta, \varepsilon) \leq \binom{m}{r} \varepsilon^r (\delta + \varepsilon)^{m-r} \quad (\text{III.16.5})$$

follows by comparing the coefficient of $\delta^{m-s} \varepsilon^s$ on both sides: $\binom{m}{r} \binom{m-r}{s-r} = \binom{m}{s} \binom{s}{r} \geq \binom{m}{s}$. We use the usual convention that a zeroth power is one, including at zero. No division by the density is needed in the main estimates.

III.17 Balancing removes the first three orders

Source 09, delivered Section 3.

Lemma III.17.1 (Three-vertex uniformity). *Let $S \subseteq V_d$ with $1 \leq |S| \leq 3$. If (x, h) is uniform in G^{d+1} , then $(L_v(x, h))_{v \in S}$ is uniform in G^S . In particular, $T_S(f) = 0$ whenever $\mathbb{E}f = 0$.*

Proof. Flipping an abstract cube coordinate is an invertible change of variables: replace h_i by $-h_i$ and x by $x + h_i$ for every flipped coordinate, with all replacements understood simultaneously. Thus we may move one selected vertex to 0.

For two selected vertices, the second is a nonzero binary vector. One of its coordinates is one, so the two values are controlled by x and an independent group variable with coefficient one.

For three selected vertices, write the other two as distinct nonzero binary vectors u, v . After interchanging them if necessary, there is a coordinate i with $(u_i, v_i) = (1, 0)$. There is a coordinate j with $v_j = 1$. The two rows $(1, 0)$ and $(u_j, 1)$ have determinant one. Together with the base variable, these coordinates give an integer minor of determinant ± 1 for the three value maps. Fixing all other variables, the remaining map is an automorphism of G^3 followed by a translation. Hence its output is uniform. The one-vertex case is immediate. Independence now gives $T_S(f) = (\mathbb{E}f)^{|S|} = 0$. $\square$

The determinant-one argument is important. Rational linear independence alone would not imply this statement on arbitrary finite groups.

Theorem III.17.2 (Fourth-order replacement for Lemma 3.10). *Let $d \geq 2$, $m = 2^d$, $f : G \rightarrow \mathbb{R}$ have mean zero, and $\delta \geq 0$. Put $\varepsilon = \|f\|_{U^d}$. Then*

$$0 \leq \mathcal{C}_d(\delta + f) - \delta^m \leq B_{m,4}(\delta, \varepsilon) \leq \binom{m}{4} \varepsilon^4 (\delta + \varepsilon)^{m-4}. \quad (\text{III.17.1})$$

In particular, for $A \subseteq G$ of density δ and $\|\mathbf{1}_A - \delta\|_{U^d}^m \leq \alpha$,

$$\#\{(x, h) : L_v(x, h) \in A \ \forall v\} \leq |G|^{d+1} \left[\delta^m + \sum_{s=4}^m \binom{m}{s} \delta^{m-s} \alpha^{s/m} \right]. \quad (\text{III.17.2})$$

Proof. Multilinearity gives the exact identity

$$\mathcal{C}_d(\delta + f) = \sum_{S \subseteq V_d} \delta^{m-|S|} T_S(f). \quad (\text{III.17.3})$$

The empty subset contributes δ^m . Lemma III.17.1 removes all subsets of sizes one through three. Bound the others by (III.16.3) and count subsets of each size. The final inequality is (III.16.5); the lower bound was proved after Lemma III.16.1. For an indicator, the normalized count becomes the integer count after multiplication by $|G|^{d+1}$, and the polynomial upper bound is increasing in $\varepsilon \geq 0$. $\square$

[write] **6 October 2026; sources 06–10.** Theorem III.17.2 is also proved by sources 03, 04 and 08 (note after source 08's Theorem III.7.3); (III.17.2) is source 08's proposed formal target (V.32.1).

For $d = 2$ there is only one surviving subset. Fourier orthogonality therefore gives the exact identity

$$\mathcal{C}_2(\delta + f) = \delta^4 + \|f\|_{U^2}^4. \quad (\text{III.17.4})$$

For larger d , the surviving terms are not arbitrary. Most four- and five-vertex subsets still contribute zero. Their complete classification is the main task of the next sections.

III.18 Integer value lattices and Fourier constraints

Source 09, delivered Section 4.

III.18.1 A general exact formula

Order a subset $S = \{v_1, \dots, v_s\} \subseteq V_d$ and form the $(d+1) \times s$ integer matrix

$$M_S = \begin{pmatrix} 1 & \cdots & 1 \\ v_1 & \cdots & v_s \end{pmatrix}. \quad (\text{III.18.1})$$

Let $\mathcal{L}_S$ be the integer lattice generated by its rows in $\mathbb{Z}^s$. Define the character constraint group

$$K_G(S) = \left\{ (\chi_j) \in \widehat{G}^s : \prod_j \chi_j = 1, \quad \prod_j \chi_j^{(v_j)_i} = 1 \ (1 \leq i \leq d) \right\}. \quad (\text{III.18.2})$$

Proposition III.18.1 (Fourier–lattice identity). *For functions $f_j : G \rightarrow \mathbb{C}$ with Fourier coefficients $a_{j,\chi}$,*

$$\mathbb{E}_{x,h} \prod_{j=1}^s f_j(L_{v_j}(x, h)) = \sum_{(\chi_j) \in K_G(S)} \prod_{j=1}^s a_{j,\chi_j}. \quad (\text{III.18.3})$$

If all f_j have mean zero, only tuples with every $\chi_j \neq 1$ can contribute.

Proof. Insert Fourier inversion for every factor. The average over x vanishes unless $\prod_j \chi_j = 1$. Independently, the average over h_i vanishes unless $\prod_j \chi_j^{(v_j)_i} = 1$. Every surviving character average is one. All sums are finite, so no interchange issue arises. $\square$

If $\mathcal{L}_S = \mathbb{Z}^s$, then the constraints force all $\chi_j = 1$. More generally, when M_S has full column rank, the quotient $\mathbb{Z}^s / \mathcal{L}_S$ is finite; its invariant factors specify the possible torsion characters. For example, an index-two quotient permits only one order-two parameter. An index-three quotient permits only one order-three parameter. The support of the corresponding relation still matters: a forced trivial character at even one vertex kills the moment.

When the rank is $s - 1$, there can instead be a free character parameter coming from a real affine relation. If the row lattice is saturated in its rational span, that relation describes the entire constraint group, without additional torsion components. These are the cases that give ordinary fourth moments and the quintic statistic below.

III.18.2 Small determinant bounds

Lemma III.18.2. *Every 2×2 binary matrix has determinant of absolute value at most one; every 3×3 binary matrix has determinant of absolute value at most two; every 4×4 binary matrix has determinant of absolute value at most three.*

Proof. The first claim follows from the determinant formula. For an $n \times n$ binary matrix B , introduce the $(n+1) \times (n+1)$ sign matrix

$$H = \begin{pmatrix} 1 & \mathbf{1}^T \\ \mathbf{1} & J - 2B \end{pmatrix}.$$

Subtracting its first row from each remaining row shows $\det H = (-2)^n \det B$. Hadamard's determinant inequality gives

$$|\det B| \leq \frac{(n+1)^{(n+1)/2}}{2^n}.$$

For $n = 3$ this is two; for $n = 4$ it is less than four, so the integral determinant is at most three in absolute value. $\square$

Any minor of M_S which includes the all-ones row can be reduced, after moving one vertex to zero by cube-coordinate flips, to a binary minor one size smaller. This permits Lemma III.18.2 to control both primitive affine relations and possible lattice indices.

Lemma III.18.3 (A plane contains at most four cube vertices). *An affine subspace of real dimension r contains at most 2^r vertices of a Boolean cube. In particular, five distinct cube vertices have augmented rank at least four.*

Proof. Choose r coordinate functionals whose restrictions are independent on the direction space. Projection onto these coordinates is injective on the affine subspace. A cube vertex projects to an element of $\{0, 1\}^r$, of which there are 2^r . $\square$

III.19 The complete four-vertex classification

Source 09, delivered Section 5.

Vectors denoted by u, v, w, z in a normal form below are binary vectors with pairwise disjoint, nonempty supports. Coordinates unused by any of them are permitted. Cube symmetries mean coordinate permutations and coordinate flips, not arbitrary affine transformations of the cube.

Theorem III.19.1 (Four selected vertices). *For real mean-zero f , a four-vertex moment can be nonzero only in the following two types, up to cube symmetries:*

Type	Normal form	Moment
Parallelogram	$\{0, u, v, u + v\}$	$\sum_{\chi} a_{\chi} ^4$
Binary torsion	$\{0, u + v, u + w, v + w\}$	$\sum_{\chi^2=1} a_{\chi} ^4$

All other four-vertex moments vanish over every finite abelian group.

Proof. Any three selected vertices have augmented rank three and a primitive minor, by Lemma III.17.1. Thus the four-vertex rank is three or four.

Suppose first that it is three. A primitive relation has no zero coefficient, since any three vertices are independent. Choose three independent rows including the all-ones row. The cofactors giving its relation have absolute value at most one: after a cube flip, they are 2×2 binary determinants. Hence the relation has four coefficients ± 1 . Their sum is zero, so two have each sign. Moving one vertex to zero reduces the relation to $u + v = w$, and binarity forces u and v to have disjoint supports. Both are nonempty. The value relation is an ordinary parallelogram relation, and the primitive three-vertex minor excludes additional torsion. Formula (III.18.3) gives $\sum_{\chi} a_{\chi}^2 a_{\chi^{-1}}^2 = \sum_{\chi} |a_{\chi}|^4$.

Suppose next that the rank is four. Some full minor including the all-ones row is nonzero, and it has absolute value at most two. Therefore the lattice index is one or two. Index one makes the moment zero. At index two the nonzero relation modulo two is unique. Its support has even size because of the all-ones row; it cannot have size two because the vertices are distinct. It therefore uses all four vertices.

Move one vertex to zero. In each coordinate, the remaining three bits have even sum. The only patterns are 000, 110, 101, 011. Full rank requires all three nonzero patterns to occur. Grouping equal patterns produces precisely $\{0, u + v, u + w, v + w\}$ with three disjoint nonempty blocks. Its quotient has order two, and its character constraints are (χ, χ, χ, χ) with $\chi^2 = 1$. Such a Fourier coefficient is real, so its fourth power is $|a_\chi|^4$. $\square$

Proposition III.19.2 (Counting the four-vertex types). *The numbers of parallelogram and binary-torsion subsets of V_d are respectively*

$$P_d = \frac{2^d}{8}(3^d - 2 \cdot 2^d + 1) = \frac{6^d - 2 \cdot 4^d + 2^d}{8}, \quad (\text{III.19.1})$$

$$T_d = \frac{2^d}{24}(4^d - 3 \cdot 3^d + 3 \cdot 2^d - 1) = \frac{8^d - 3 \cdot 6^d + 3 \cdot 4^d - 2^d}{24}. \quad (\text{III.19.2})$$

Proof. For a parallelogram, choose a base vertex in 2^d ways and assign each coordinate to the unused block, the support of u , or the support of v , requiring both active blocks to occur. Inclusion–exclusion gives $3^d - 2 \cdot 2^d + 1$ assignments. Each resulting subset is counted for four choices of base and two orders of the active blocks, so divide by eight.

For the second type, choose a base and assign coordinates to an unused block or one of three nonempty active blocks. There are $4^d - 3 \cdot 3^d + 3 \cdot 2^d - 1$ assignments. Each subset has four base choices and $3!$ orders of the blocks, so divide by twenty-four. $\square$

In particular, the fourth-order coefficient is never negative. Its arithmetic dependence is entirely through the order-two part of the character group. There are no other fourth-order corrections, even on composite groups with large exponents.

[write] **6 October 2026; sources 06–10.** Theorem III.19.1 and Proposition III.19.2 are source 03’s Lemma III.2.3 with its counts and source 08’s Lemmas III.8.1 and III.8.2: P_d is 03’s and 08’s R_d , T_d is theirs, and $P_d + T_d = A_d$ is 08’s F_d and 03’s K_d (checked for $d = 2, \dots, 7$ at the second write). Three independent proofs.

III.20 The complete five-vertex classification

Source 09, delivered Section 6.

Theorem III.20.1 (Five selected vertices). *For real mean-zero f , every five-vertex moment vanishes except possibly for the following two types, up to cube symmetries:*

$$\text{star:} \quad \{0, u, v, w, u + v + w\}, \quad (\text{III.20.1})$$

$$\text{ternary torsion:} \quad \{0, u + v + w, u + v + z, u + w + z, v + w + z\}. \quad (\text{III.20.2})$$

The star moment is

$$\sum_{\chi \in \widehat{G}} |a_\chi|^2 a_\chi^2 \overline{a_{\chi^2}}, \quad (\text{III.20.3})$$

and the ternary-torsion moment is

$$\sum_{\chi^3=1} a_\chi^4 \overline{a_\chi}. \quad (\text{III.20.4})$$

Proof. By Lemma III.18.3, the augmented rank is four or five.

Rank four, with a zero in the unique relation. The relation must involve exactly four vertices, and those four form a parallelogram. Put them in the form $0, u, v, u + v$, with u, v supported on disjoint nonempty coordinate blocks U, V . Let w be the fifth vertex. It lies outside their real affine plane. Therefore at least one of the following holds: w is nonzero outside $U \cup V$; its bits vary within U ; or its bits vary within V .

In the first case, a coordinate outside $U \cup V$, together with one coordinate in each of U, V , gives a determinant-one minor for $0, u, v, w$. In the second case, take two coordinates in U where the bits of w differ and one in V ; their determinant is ± 1 . The third case is symmetric. Thus the four values at $0, u, v, w$ are jointly uniform over every group. The fourth parallelogram value is determined by the first three, while the value at w is independent. Its mean-zero factor kills the entire moment. This argument rules out hidden torsion, rather than merely using rational independence.

Rank four, with all five relation coefficients nonzero. Choose four independent rows including the all-ones row. Every cofactor has absolute value at most two, by Lemma III.18.2. The primitive relation therefore has coefficients of absolute values one or two, with sum zero. Up to an overall sign and permutation, the only possibilities are

$$(1, 1, 1, -1, -2), \quad (2, 2, -2, -1, -1).$$

Indeed, the number of odd coefficients is even; a primitive vector cannot have all coefficients even. Enumerating one or three even coefficients gives these two possibilities.

The second is impossible for distinct cube vertices. In any coordinate, the zero-sum condition forces the two vertices with coefficient -1 to have equal bits, since all other contributions are even. Those two vertices would therefore coincide. For the first relation, move the coefficient -2 vertex to zero. If u, v, w are the coefficient-1 vertices and a is the coefficient -1 vertex, the relation reads $u + v + w = a$. Binariness forces the supports of u, v, w to be disjoint; distinctness makes them nonempty. This gives (III.20.1).

There is no extra torsion in this case. The cofactor vector is an integer multiple of the primitive relation. Since one coefficient has absolute value two and all cofactors have absolute value at most two, that multiple has absolute value one. Some cofactor is consequently ± 1 , so the row lattice is saturated. In the order $(0, u, v, w, u + v + w)$, a primitive relation is $(-2, 1, 1, 1, -1)$. The character tuple is therefore $(\chi^{-2}, \chi, \chi, \chi, \chi^{-1})$. Its Fourier product is $a_\chi^3 \overline{a_\chi} \overline{a_{\chi^2}}$, which is (III.20.3).

Rank five. A nonzero full minor including the all-ones row has absolute value at most three. The row-lattice index is therefore one, two, or three. Index one gives zero. At index two the unique nonzero relation modulo two has even support; distinctness excludes support two, so its support is four. The remaining vertex has a forced trivial character and the moment vanishes.

At index three, take a nonzero relation modulo three. Its coefficients are $0, 1, -1$. It cannot have support at most three, by the primitive three-vertex property. If it had support four, the all-ones row would force two plus signs and two minus signs. In each binary coordinate the corresponding integer sum lies in $[-2, 2]$; divisibility by three would force the sum to be zero as an integer. This would be a real parallelogram relation, contradicting rank five. Hence the relation has full support.

The all-ones row now forces four coefficients of one sign and one of the other. After rescaling the relation modulo three, write them as $(-1, 1, 1, 1, 1)$, and move the exceptional vertex to zero.

Each coordinate of the other four vertices has either zero or three ones. Thus its pattern is 0000 or the complement of a unit vector. Rank five requires all four nonzero pattern types. Grouping equal patterns gives (III.20.2), with four nonempty disjoint blocks. The constraint tuples are $(\chi^{-1}, \chi, \chi, \chi, \chi)$ with $\chi^3 = 1$. Their Fourier products give (III.20.4). $\square$

Proposition III.20.2 (Counting the five-vertex types). *The numbers of star and ternary-torsion subsets are respectively*

$$Q_d = \frac{2^d}{6}(4^d - 3 \cdot 3^d + 3 \cdot 2^d - 1), \quad (\text{III.20.5})$$

$$S_d = \frac{2^d}{24}(5^d - 4 \cdot 4^d + 6 \cdot 3^d - 4 \cdot 2^d + 1). \quad (\text{III.20.6})$$

Proof. For a star, the coefficient- -2 vertex is intrinsically distinguished. Choose this vertex and then choose three nonempty active coordinate blocks. Only their $3!$ orders produce multiple counting. This gives (III.20.5).

For a ternary configuration the exceptional coefficient in its unique modulo-three relation is distinguished. Choose that vertex and four nonempty active blocks, dividing by $4!$. Inclusion-exclusion gives (III.20.6). Unlike the binary-torsion case, one does not divide by the number of vertices: the exceptional vertex is unique. $\square$

The Stirling expressions (F.14.5) follow without any algebraic guessing. Assign d labeled coordinates to k nonempty active blocks and one possibly empty inactive block. Adjoin one extra labeled element to the inactive block. After forgetting the names of the active blocks, this is a partition of $d + 1$ elements into $k + 1$ nonempty blocks, with the block of the extra element distinguished. The number of assignments is therefore $k! \left\{ \begin{matrix} d+1 \\ k+1 \end{matrix} \right\}$.

[write] **6 October 2026; sources 06–10.** The star moment (III.20.3) is source 04's $B_5(f)$ ((III.3.8)) and source 08's $S_5(f)$; the ternary moment (III.20.4) is source 03's (III.2.29). This classification answers the fifth-order part of source 03's Research question on the fifth and sixth coefficients (note there).

III.21 Exact expansion and quantitative consequences

Source 09, delivered Section 7.

Define

$$E_4(f) = \sum_{\chi} |a_{\chi}|^4, \quad (\text{III.21.1})$$

$$T_2(f) = \sum_{\chi^2=1} |a_{\chi}|^4, \quad (\text{III.21.2})$$

$$M_5(f) = \sum_{\chi} |a_{\chi}|^2 a_{\chi}^2 \overline{a_{\chi^2}}, \quad (\text{III.21.3})$$

$$T_{3,5}(f) = \sum_{\chi^3=1} a_{\chi}^4 \overline{a_{\chi}}. \quad (\text{III.21.4})$$

The square in a_{χ}^2 means the square of a Fourier coefficient; the subscript in a_{χ^2} means evaluation at the squared character. These operations must not be confused. Both quintic statistics are real: conjugation permutes the summands by $\chi \mapsto \chi^{-1}$. They need not be nonnegative.

Theorem III.21.1 (Exact fourth and fifth orders). *Let G be finite abelian, $d \geq 3$, $m = 2^d$, $\delta \geq 0$, and $f : G \rightarrow \mathbb{R}$ have mean zero. With $\varepsilon = \|f\|_{U^d}$,*

$$\begin{aligned} \mathcal{C}_d(\delta + f) &= \delta^m + \delta^{m-4}(P_d \mathbf{E}_4(f) + T_d \mathbf{T}_2(f)) \\ &\quad + \delta^{m-5}(Q_d \mathbf{M}_5(f) + S_d \mathbf{T}_{3,5}(f)) + R_6, \end{aligned} \quad (\text{III.21.5})$$

where

$$R_6 = \sum_{\substack{S \subseteq V_d \\ |S| \geq 6}} \delta^{m-|S|} T_S(f), \quad |R_6| \leq B_{m,6}(\delta, \varepsilon).$$

The coefficients are given by (F.14.5) and satisfy $P_d + T_d = A_d$ and $Q_d = 4T_d$.

Proof. Start with (III.17.3). Apply Lemma III.17.1, Theorems III.19.1 and III.20.1, and the counting Propositions III.19.2 and III.20.2. The remaining subsets are exactly those defining R_6 . Apply Lemma III.16.1 to each one. The two coefficient identities follow from (III.19.1)–(III.20.6). $\square$

d	P_d	T_d	Q_d	S_d
2	1	0	0	0
3	12	2	8	0
4	100	40	160	16
5	720	520	2080	480
6	4816	5600	22400	8960
7	30912	54432	217728	134400

III.21.1 Useful special cases

For three-dimensional cubes,

$$\mathcal{C}_3(\delta + f) = \delta^8 + \delta^4(12\mathbf{E}_4 + 2\mathbf{T}_2) + 8\delta^3\mathbf{M}_5 + R_6, \quad (\text{III.21.6})$$

with

$$|R_6| \leq 28\delta^2\varepsilon^6 + 8\delta\varepsilon^7 + \varepsilon^8.$$

For four-dimensional cubes the fourth and fifth corrections are

$$\delta^{12}(100\mathbf{E}_4 + 40\mathbf{T}_2), \quad \delta^{11}(160\mathbf{M}_5 + 16\mathbf{T}_{3,5}).$$

If $|G|$ is odd, then $\mathbf{T}_2 = 0$. If $3 \nmid |G|$, then $\mathbf{T}_{3,5} = 0$. These vanishings include the trivial character because $a_1 = 0$. If G has exponent two, every $\chi^2 = 1$, so $\mathbf{M}_5 = 0$ and $\mathbf{T}_2 = \mathbf{E}_4$; there is also no ternary term. If G has exponent three, then $\chi^2 = \chi^{-1}$ and $\mathbf{M}_5 = \mathbf{T}_{3,5}$.

III.21.2 A purely Gowers-norm upper bound

Corollary III.21.2. *Under the hypotheses of Theorem III.21.1,*

$$\mathcal{C}_d(\delta + f) \leq \delta^m + A_d \delta^{m-4} \varepsilon^4 + (Q_d + S_d) \delta^{m-5} \varepsilon^5 + B_{m,6}(\delta, \varepsilon). \quad (\text{III.21.7})$$

If $|G|$ is coprime to six, A_d can be replaced by P_d and $Q_d + S_d$ by Q_d .

Proof. Each nonzero type is a subset moment and is bounded in absolute value by ε to the size of its support. Sum over the counted subsets. For a coprime-to-six group the torsion terms vanish. $\square$

For an indicator with $\varepsilon^m \leq \alpha$, this gives the explicit replacement for (F.14.1) obtained by multiplying

$$\delta^m + A_d \delta^{m-4} \alpha^{4/m} + (Q_d + S_d) \delta^{m-5} \alpha^{5/m} + \sum_{s=6}^m \binom{m}{s} \delta^{m-s} \alpha^{s/m}$$

by N^{d+1} . The separate exact formula (III.17.4) applies when $d = 2$.

III.21.3 Keeping Fourier information instead of discarding it

Let

$$\tau = \|f\|_{U^2}, \quad \rho = \max_{\chi} |a_{\chi}|, \quad \mathsf{T}_{3,4}(f) = \sum_{\chi^3=1} |a_{\chi}|^4.$$

Then $\rho \leq \tau \leq \varepsilon$, and directly from the definitions,

$$|\mathsf{M}_5| \leq \rho \tau^4, \quad |\mathsf{T}_{3,5}| \leq \rho \mathsf{T}_{3,4} \leq \rho \tau^4. \quad (\text{III.21.8})$$

Consequently,

$$\begin{aligned} \mathcal{C}_d(\delta + f) - \delta^m &\leq \delta^{m-4} (P_d \tau^4 + T_d \mathsf{T}_2) \\ &\quad + \delta^{m-5} \rho (Q_d \tau^4 + S_d \mathsf{T}_{3,4}) + B_{m,6}(\delta, \varepsilon). \end{aligned} \quad (\text{III.21.9})$$

This is useful when ordinary Fourier uniformity is substantially stronger than the available higher-order uniformity. The exact formula shows that all terms through degree five are ordinary Fourier statistics; one should not pay a higher-order uniformity bound for them unnecessarily.

III.21.4 Certificates of nonuniformity

For $F : G \rightarrow [0, 1]$, put $\Delta = \mathcal{C}_d(F) - \delta^m \geq 0$. Since $|f| \leq 1$, one has $\varepsilon \leq 1$. Therefore

$$\varepsilon \geq \left[\frac{\Delta}{\binom{m}{4} (1 + \delta)^{m-4}} \right]^{1/4}. \quad (\text{III.21.10})$$

This follows from Theorem III.17.2 and (III.16.5). A sharper, still explicit denominator follows from (III.21.7) by replacing every ε^s , $s \geq 4$, by ε^4 .

If the fourth and fifth Fourier corrections have been measured, define

$$D_6 = \mathcal{C}_d(F) - \delta^m - \delta^{m-4} (P_d \mathsf{E}_4 + T_d \mathsf{T}_2) - \delta^{m-5} (Q_d \mathsf{M}_5 + S_d \mathsf{T}_{3,5}).$$

Then

$$\varepsilon \geq \left[\frac{|D_6|}{\binom{m}{6} (1 + \delta)^{m-6}} \right]^{1/6} \quad (d \geq 3). \quad (\text{III.21.11})$$

These are lower bounds on a uniformity norm. They do not, by themselves, produce a polynomial phase, a nilsequence, or a density increment. Calling them inverse theorems would overstate their content.

III.21.5 Local derivatives at a constant

For a fixed mean-zero direction f , the map $t \mapsto \mathcal{C}_d(\delta + tf)$ is a polynomial. Its first three derivatives at zero vanish, and

$$\begin{aligned} \left. \frac{d^4}{dt^4} \mathcal{C}_d(\delta + tf) \right|_{t=0} &= 4! \delta^{m-4} (P_d \mathbf{E}_4(f) + T_d \mathbf{T}_2(f)), \\ \left. \frac{d^5}{dt^5} \mathcal{C}_d(\delta + tf) \right|_{t=0} &= 5! \delta^{m-5} (Q_d \mathbf{M}_5(f) + S_d \mathbf{T}_{3,5}(f)). \end{aligned}$$

Thus the cube-count functional is flat through third order along the mean-zero hyperplane at a positive constant. Its first nonzero variation is quartic, and its arithmetic anisotropy is already visible there. This is a statement about a specific functional at a specific base point, not a claim of vanishing Hessians at arbitrary functions.

III.22 Sharpness, signs, and transfer to genuine sets

Source 09, delivered Section 8.

III.22.1 The fourth-order exponent and its universal coefficient

Proposition III.22.1. *For every $d \geq 2$ and $0 < \delta < 1$, no uniform replacement of the fourth-order error by $O_{d,\delta}(\varepsilon^{4+\eta})$, $\eta > 0$, is possible. This remains true when groups are restricted to have odd order. If all finite abelian groups are allowed, the coefficient A_d of $\delta^{m-4}\varepsilon^4$ in (III.21.7) is optimal as a leading small- ε coefficient.*

Proof. Fix any nonzero real mean-zero function g on a finite group and put $F_t = \delta + tg$, with $t > 0$ small enough that $0 \leq F_t \leq 1$. Its balanced norm is $t\|g\|_{U^d}$. The fourth-order coefficient in Theorem III.21.1 is positive because $P_d > 0$ and $\mathbf{E}_4(g) > 0$. For $d = 2$, use (III.17.4) instead. Thus the excess is a positive constant times $t^4 + O(t^5)$ and cannot be of order $t^{4+\eta}$. A nonzero cosine on a cyclic group of odd prime order at least seventeen gives the same conclusion among odd-order groups.

For the coefficient assertion, take a nontrivial order-two character $\chi : G \rightarrow \{\pm 1\}$ and let $g = \chi$. Then $\|g\|_{U^d} = 1$, $\mathbf{E}_4(g) = \mathbf{T}_2(g) = 1$, and the leading coefficient is exactly $P_d + T_d = A_d$. This rules out a smaller universal leading coefficient. $\square$

There is also a closed formula for the last example:

$$\mathcal{C}_d(\delta + t\chi) = 2^{-d-1}((\delta + t)^m + (\delta - t)^m) + (1 - 2^{-d})(\delta^2 - t^2)^{m/2}. \quad (\text{III.22.1})$$

To prove it, condition on $(\chi(h_1), \dots, \chi(h_d))$. With probability 2^{-d} all signs are positive and every vertex has sign $\chi(x)$, whose two values are equiprobable. Otherwise exactly half the cube vertices have each sign. In dimension three, (III.22.1) reduces to

$$\mathcal{C}_3(\delta + t\chi) = \delta^8 + 14\delta^4 t^4 + t^8.$$

This also explains why deleting the binary-torsion term would give an incorrect fourth-order coefficient.

III.22.2 A negative quintic correction

Proposition III.22.2. *The fifth-order term in Theorem III.21.1 can have either sign, even on groups of prime order greater than three. In particular, the inequality*

$$\mathcal{C}_3(\delta + f) \geq \delta^8 + 12\delta^4 \|f\|_{U^2}^4 \quad (\text{III.22.2})$$

is false in general, even for $[0, 1]$ -valued functions on such groups.

Proof. Let $q \geq 17$ be prime and, for $\sigma \in \{1, -1\}$, put

$$g_\sigma(x) = \cos(2\pi x/q) + \sigma \cos(4\pi x/q).$$

Its only nonzero Fourier coefficients are $a_{\pm 1} = 1/2$ and $a_{\pm 2} = \sigma/2$, with frequencies written additively in this calculation. Therefore

$$\mathbf{E}_4(g_\sigma) = \frac{1}{4}, \quad \mathbf{M}_5(g_\sigma) = \frac{\sigma}{16}, \quad \mathbf{T}_2(g_\sigma) = \mathbf{T}_{3,5}(g_\sigma) = 0.$$

Apply (III.21.6) to $f = tg_\sigma$. At any fixed $0 < \delta < 1$,

$$\mathcal{C}_3(\delta + tg_\sigma) = \delta^8 + 3\delta^4 t^4 + \frac{\sigma}{2} \delta^3 t^5 + O_\delta(t^6). \quad (\text{III.22.3})$$

For all sufficiently small $t > 0$, one has $0 \leq \delta + tg_\sigma \leq 1$ because $|g_\sigma| \leq 2$. Taking $\sigma = -1$ makes the difference between the left side of (III.22.2) and its proposed right side negative. The $O(t^6)$ bound is explicitly supplied by (F.14.6); no finite numerical experiment is needed for this counterexample. $\square$

The same example proves that, after the exact fourth-order correction has been removed, a fifth-order remainder cannot in general be replaced by a higher power. More strongly, for every $d \geq 3$ the star count Q_d is positive, so the same two-frequency example gives a nonzero fifth-order term in that dimension.

An exact finite calculation, reproduced in Appendix III.26, yields

$$\mathcal{C}_3(\delta + tg_\sigma) = \delta^8 + 3\delta^4 t^4 + \frac{\sigma}{2} \delta^3 t^5 + \frac{3}{2} \delta^2 t^6 + \frac{1}{4} t^8. \quad (\text{III.22.4})$$

For example, when $0 < t \leq \delta/4$, the negative fifth-order term in the $\sigma = -1$ case dominates the displayed positive sixth- and eighth-order terms. The asymptotic proof above is logically independent of this stronger computational identity.

[write] **6 October 2026; sources 06–10.** (III.22.2) is the bound refuted also by source 03 (on $\mathbb{Z}/3\mathbb{Z}$) and source 08 (on $\mathbb{Z}/11\mathbb{Z}$, Theorem III.10.1); Remark W1 keeps it on record with an explicit counterexample. The proposition’s “groups of prime order greater than three” is witnessed by the proof for primes $q \geq 17$; on $\mathbb{Z}/11\mathbb{Z}$ source 08’s example is the case $\sigma = -1$ and its polynomial is (III.22.4). On $\mathbb{Z}/5\mathbb{Z}$ the fifth-order argument does not apply to g_{-1} : there $2 \cdot 2 = -1$, the coefficients at ± 2 also contribute, and $\mathbf{M}_5(g_{-1}) = 0$ (checked at the second write).

III.22.3 The sixth-order remainder is also necessary

Proposition III.22.3. *For every $d \geq 3$ and $0 < \delta < 1$, the sixth-order remainder after the exact fourth and fifth corrections cannot be uniformly replaced by $O_{d,\delta}(\varepsilon^{6+\eta})$ for any $\eta > 0$.*

Proof. Use g_+ from Proposition III.22.2. Every Fourier coefficient is nonnegative, so every subset moment is nonnegative by Proposition III.18.1. Consider the six vertices of a three-dimensional cube obtained by deleting the adjacent pair $(0, 0, 0), (1, 0, 0)$. On the three remaining vertices in the first slice assign frequencies $(1, 1, -2)$, and on the corresponding vertices in the second slice assign $(-1, -1, 2)$. Their total frequency is zero; the sum in the first cube coordinate is zero; and the sums in each of the other two coordinates cancel between the slices. This is an admissible Fourier tuple with positive coefficient product. Thus that six-vertex moment is strictly positive.

For $d > 3$, embed this subset in a three-dimensional face and give the remaining coordinates value zero. The same tuple still satisfies all constraints. Hence the coefficient of t^6 in R_6 for $f = tg_+$ is positive, and no other sixth-order contribution can cancel it. Since $\varepsilon = t\|g_+\|_{U^d}$, the claimed stronger remainder is impossible. $\square$

III.22.4 Why weighted examples are not a loophole

The following rounding lemma transfers these phenomena to indicators. It also quantifies the size of the auxiliary group needed for the transfer.

Lemma III.22.4 (Random rounding in a Gowers norm). *Let $F : G \rightarrow [0, 1]$, let $d \geq 2$, and let $m = 2^d$. There exists $A \subseteq G$ such that*

$$\|\mathbf{1}_A - F\|_{U^d} \leq \zeta := \left(\frac{\binom{m}{2}}{|G|} \right)^{1/m}. \quad (\text{III.22.5})$$

Writing $\delta = \mathbb{E}F$ and $\delta_A = |A|/|G|$, one also has

$$|\delta_A - \delta| \leq \zeta, \quad (\text{III.22.6})$$

$$|\mathcal{C}_d(\mathbf{1}_A) - \mathcal{C}_d(F)| \leq m\zeta, \quad (\text{III.22.7})$$

$$\|(\mathbf{1}_A - \delta_A) - (F - \delta)\|_{U^d} \leq 2\zeta. \quad (\text{III.22.8})$$

Proof. Choose independent Bernoulli variables $\mathbf{1}_A(x)$ of means $F(x)$, and set $R(x) = \mathbf{1}_A(x) - F(x)$. For a fixed parameterized cube whose m evaluated vertices are distinct, independence and $\mathbb{E}_A R(x) = 0$ give $\mathbb{E}_A \prod_v R(L_v) = 0$. If vertices collide, the absolute value of the expected product is at most one.

For two distinct abstract vertices v, w , the equation $L_v = L_w$ has probability $1/|G|$: one coefficient in $\sum_i (v_i - w_i)h_i$ is 1 or -1 . The union bound gives collision probability at most $\binom{m}{2}/|G|$. Therefore

$$0 \leq \mathbb{E}_A \|R\|_{U^d}^m \leq \frac{\binom{m}{2}}{|G|}.$$

Some realization satisfies (III.22.5). Now $|\mathbb{E}R| \leq \|R\|_{U^d}$ proves (III.22.6). Telescope the two cube products in (III.22.7); each of the m terms has one factor R , and all other factors are either F or $\mathbf{1}_A$, with Gowers norm at most one. Lemma III.16.1 bounds each term by ζ . Finally, subtract the mean of R and apply the triangle inequality to prove (III.22.8). $\square$

Corollary III.22.5 (Transfer of the sharpness examples). *The fourth-order exponent obstruction, the optimal universal leading coefficient A_d , the fifth-order sign obstruction, and the necessity of a sixth-order remainder persist for sequences of actual sets in growing finite abelian groups. Their densities tend to any prescribed $\delta \in (0, 1)$. The first exponent obstruction also persists in odd-order groups.*

Proof. Lift a fixed weighted example on H to $H \times K$ using the projection to H . All cube averages and Gowers norms of the lifted function are unchanged. Let $t \rightarrow 0$ and choose $|K|$ sufficiently large that the rounding error satisfies $\zeta = o(t^6)$; larger polynomial choices of $|K|$ suffice.

The balanced norm changes by at most 2ζ , the density by at most ζ , and the full cube count by at most $m\zeta$. Each of the fourth- and fifth-order statistics is itself a fixed subset moment. Telescoping it and using Gowers–Cauchy–Schwarz shows that it changes by $O_d(\zeta)$, since the balanced functions involved are bounded by one. Thus the nonzero terms at orders four, five, and six persist. For the odd-order assertion take both H and K of odd order. $\square$

Remark III.22.6 (Exact rational densities). For a prescribed rational $\delta = a/b \in (0, 1)$, the same obstructions can be realized with density exactly δ . Choose $|H \times K|$ divisible by b . After rounding, add or delete at most $\zeta|G|$ points to obtain the required cardinality. If E is the edit set, the difference D of the two indicators is supported on E and bounded by one. Hence

$$\|D\|_{U^d}^m \leq \mathbb{E}_{x,h} \left| \prod_v D(L_v) \right| \leq \frac{|E|}{|G|} \leq \zeta.$$

The total approximation error in U^d is at most $\zeta + \zeta^{1/m}$. Choosing $\zeta = o(t^{6m})$ makes this $o(t^6)$ and preserves all the sharpness conclusions at the exact fixed density. For an odd-order construction this exact-density variant applies when b is odd; arbitrary densities can still be approached by the preceding corollary. Irrational densities, of course, cannot be the exact density of a finite set.

[write] **6 October 2026; sources 06–10.** Rounding to sets is also done by sources 03, 04 and 08 (note after source 08’s Lemma III.11.2). Lemma III.22.4 has constant $((\binom{m}{2})/|G|)^{1/m}$ by taking a realization at most the mean; source 08 takes twice the mean.

III.23 A sixth-order order-five obstruction and an all-prime family

Source 09, delivered Section 9.

The small-support expansion is especially simple on groups whose order is coprime to six. That simplification does not continue indefinitely. There is already an order-five torsion configuration on six selected vertices.

Theorem III.23.1 (First possible isolated order-five contribution). *There exists a six-vertex subset of V_5 with full augmented rank and row-lattice index five, whose mean-zero moment can be nonzero on groups with order-five characters but is zero when multiplication by five is invertible on G . No such order-five-only contribution occurs on fewer than six selected cube vertices.*

Proof. Consider the binary matrix

$$B = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}. \quad (\text{III.23.1})$$

Its determinant is -5 . Take S to consist of the zero vector and the five columns of B . Its augmented matrix is

$$M_S = \begin{pmatrix} 1 & \mathbf{1}^T \\ 0 & B \end{pmatrix}, \quad |\det M_S| = 5.$$

For

$$r = (2, 1, 3, 2, 1, 1)^T$$

one checks exactly that

$$M_S r = (10, 5, 5, 5, 5, 5)^T. \quad (\text{III.23.2})$$

Every coordinate of r is nonzero modulo five. Since the row quotient has order five, the full character constraint group is

$$K_G(S) = \{(\chi^{r_0}, \dots, \chi^{r_5}) : \chi^5 = 1\}.$$

Consequently,

$$T_S(f) = \sum_{\chi^5=1} \prod_{j=0}^5 a_{\chi^{r_j}}. \quad (\text{III.23.3})$$

If multiplication by five is invertible on G , the only parameter is the trivial character, and the moment is zero because $a_1 = 0$. On $G = \mathbb{Z}/5\mathbb{Z}$, take $f = 5\mathbf{1}_{\{0\}} - 1$. Its trivial Fourier coefficient is zero and its four nontrivial Fourier coefficients are one. Thus (III.23.3) is exactly four.

Finally, Lemma III.17.1 and the complete four- and five-vertex classifications show that the only isolated torsion contributions before degree six have orders two and three. In particular, none can be supported purely on order-five characters. This proves the minimal support-size assertion. $\square$

The determinant in (III.23.1) can be checked by ordinary integer elimination; the accompanying exact script additionally records the Smith invariants $1, 1, 1, 1, 1, 5$ of M_S . A direct elimination certificate is given in Appendix III.27. Rescaling the displayed f by a sufficiently small t and adding a density gives a $[0, 1]$ -valued example, so this obstruction belongs to the same perturbative counting problem as the main theorem.

[write] **6 October 2026; sources 06–10.** *Checked at the second write* with SymPy 1.14.0: $\det B = -5$, $\det M_S = -5$, $M_S r = (10, 5, 5, 5, 5, 5)$, the Smith normal form of M_S is $\text{diag}(1, 1, 1, 1, 1, 5)$, the five columns of B are distinct nonzero binary vectors, and the moment (III.23.3) equals 4 for $f = 5\mathbf{1}_{\{0\}} - 1$ on $\mathbb{Z}/5\mathbb{Z}$. At four vertices only the invariant factors one and two occur (source 03's Remark III.2.4), at five also three (Theorem III.20.1); index five first occurs here, at six vertices.

Proposition III.23.2 (An isolated torsion configuration for every prime). *For every prime ℓ , there is a subset of $V_{\ell+1}$ with $\ell + 2$ vertices, full augmented rank, and row-lattice quotient $\mathbb{Z}/\ell\mathbb{Z}$, whose constraint relation has full support modulo ℓ .*

Proof. Put $n = \ell + 1$, and take the zero vector together with the n columns of $J - I_n$. The eigenvalues of $J - I_n$ are $n - 1 = \ell$ on the all-ones direction and -1 on its complementary hyperplane. Hence its determinant is $(-1)^{n-1}\ell$. Its augmented matrix has the same determinant and therefore a quotient of order ℓ . The relation modulo ℓ is

$$(-1, 1, \dots, 1),$$

because its sum is $n - 1 = \ell$ and each coordinate row has ℓ ones. Every coefficient is nonzero. The moment is

$$\sum_{\chi^\ell=1} a_\chi^{\ell+1} \overline{a_\chi}.$$

It is nonzero for $f = \ell\mathbf{1}_{\{0\}} - 1$ on $\mathbb{Z}/\ell\mathbb{Z}$. $\square$

For $\ell = 2, 3$ this reproduces the two small torsion normal forms. For $\ell = 5$ it uses seven vertices, whereas Theorem III.23.1 uses six. Thus the uniform construction is not generally support-minimal.

One can make the resulting extremal problem precise. Let $h(\ell)$ be the least cardinality of an affinely independent Boolean-vertex set whose integer row quotient has ℓ -torsion and whose character kernel over $\mathbb{Z}/\ell\mathbb{Z}$ has a vector nonzero in every coordinate. The results above give

$$h(2) = 4, \quad h(3) = 5, \quad h(5) = 6, \quad h(\ell) \leq \ell + 2.$$

If $s = h(\ell)$, some nonzero full minor including the all-ones row is divisible by ℓ . The sign-matrix argument in Lemma III.18.2 gives the necessary condition

$$\ell \leq \frac{s^{s/2}}{2^{s-1}}. \quad (\text{III.23.4})$$

In particular, as ℓ grows, $h(\ell) \geq (2 - o(1)) \log \ell / \log \log \ell$. There is a wide gap between this elementary obstruction and the explicit general construction. We leave its systematic improvement as a research problem rather than pretending the small examples solve it.

III.24 Heterogeneous functions and an all-orders organizing principle

Source 09, delivered Section 10.

The method is not restricted to a single set placed at every vertex. Let $F_v = \delta_v + f_v$ be real functions indexed by V_d , with $\mathbb{E}f_v = 0$, $\delta_v \geq 0$, and $\varepsilon_v = \|f_v\|_{U^d}$.

Theorem III.24.1 (Heterogeneous expansion). *One has*

$$\begin{aligned} \mathbb{E}_{x,h} \prod_{v \in V_d} F_v(L_v) &= \prod_{v \in V_d} \delta_v \\ &+ \sum_{\substack{S \subseteq V_d \\ |S| \geq 4}} \left(\prod_{v \notin S} \delta_v \right) \mathbb{E}_{x,h} \prod_{v \in S} f_v(L_v). \end{aligned} \quad (\text{III.24.1})$$

Every four- or five-vertex term vanishes unless its vertex subset is one of the normal forms in Theorems III.19.1 and III.20.1. For a surviving term its exact value is the mixed Fourier sum in (III.18.3). The remainder after size five is bounded by

$$\sum_{\substack{S \subseteq V_d \\ |S| \geq 6}} \left(\prod_{v \notin S} \delta_v \right) \left(\prod_{v \in S} \varepsilon_v \right). \quad (\text{III.24.2})$$

Proof. Expand the product. Joint uniformity for up to three selected vertices makes those mixed moments a product of zero means. In the classification proofs, every vanishing case either has fully uniform values or forces a trivial character at one selected vertex, so the same vanishing holds for different mean-zero functions. Finally, apply Lemma III.16.1 to bound each remaining mixed term. $\square$

For example, a parallelogram with ordered relation $v_1 + v_4 = v_2 + v_3$ contributes

$$\sum_{\chi} a_{1,\chi} a_{4,\chi} a_{2,\chi^{-1}} a_{3,\chi^{-1}}.$$

A binary-torsion subset contributes $\sum_{\chi^2=1} \prod_{j=1}^4 a_{j,\chi}$. Thus the homogeneous formulas are not isolated coincidences; they are the diagonal specializations of a colored cube-count expansion.

At every order, Proposition III.18.1 gives a finite algorithm: enumerate a support S , compute its integer row lattice, identify which character constraints admit full-support tuples, and sum the corresponding mixed Fourier products. The difficult part is not the existence of an algorithm but the compression of the support types into dimension-uniform, human-readable formulas. Through degree five, the four normal forms achieve precisely that compression.

III.25 Relation to progression bounds and current conjecture status

Source 09, delivered Section 13.

The local result should be placed against the right global benchmark. Gowers' 2001 theorem is not the current quantitative endpoint. For example, Leng, Sah, and Sawhney obtain, for every fixed $k \geq 5$, an upper bound of the form

$$r_k(N) \ll N \exp(-(\log \log N)^{c_k})$$

with $c_k > 0$ [2]. The present manuscript does not compete with that bound: it sharpens one local cube-count inequality, without supplying a new inverse theorem or a new iteration.

A recent primary preprint of Milićević establishes a quasipolynomial inverse theorem for $U^k(\mathbb{F}_p^n)$ in the high characteristic regime $p \geq k$ [32]. That is another reason not to describe an elementary Fourier expansion as a breakthrough inverse theorem. Its relation to this manuscript is potentially complementary: low-order terms can be evaluated before an inverse theorem is applied to a residual, but a useful quantitative implementation would need a separate argument.

The conjectural lower bounds in Section 4 of Gowers' paper also require an up-to-date reading. Shi and Dong's 2026 preprint states a disproof of Gowers' conjectured progression lower bound for every even length at least six, and records the earlier length-four disproof [33]. Their higher-length uniformity condition is U^{k-2} uniformity, not merely ordinary Fourier uniformity. We cite this as the stated result of that preprint, rather than treating the old conjecture as still untouched or claiming its resolution here.

The cube lower bound $\mathcal{C}_d(F) \geq \delta^m$ is different. It follows from a norm inequality and remains valid. Proposition III.22.2 does not refute that bound: it refutes the stronger attempt to add the entire positive quartic correction as a global lower bound while ignoring the signed quintic term. Arithmetic progression counts and full cube counts are different linear-configuration averages; one must not transfer extremal conclusions between them without proof.

III.26 Exact two-frequency polynomial certificate

Source 09, delivered Appendix A.

For $q \geq 17$, all character constraints for eight frequencies from $\{-2, -1, 0, 1, 2\}$ can be checked as integer equalities: the total absolute frequency sum is at most sixteen, and every coordinate sum has absolute value at most eight. Thus no nonzero multiple of q can occur.

Order the eight vertices $v \in \{0, 1\}^3$. A term in the expansion of $\mathcal{C}_3(\delta + tg_\sigma)$ is indexed by frequencies $b_v \in \{-2, -1, 0, 1, 2\}$ satisfying

$$\sum_v b_v = 0, \quad \sum_v v_i b_v = 0 \quad (i = 1, 2, 3). \quad (\text{III.26.1})$$

Let s be the number of nonzero frequencies and r the number with absolute value two. Its contribution is $\delta^{8-s} t^s 2^{-s} \sigma^r$. The exact signed counts are

	s	0	1	2	3	4	5	6	7	8
$\sum 1$	1	1	0	0	0	48	16	96	0	64
$\sum (-1)^r$	1	1	0	0	0	48	-16	96	0	64

Multiplying the s th column by $2^{-s} \delta^{8-s} t^s$ proves (III.22.4). The following finite loop specifies the certificate without a floating-point operation:

```
for b in product(range(-2, 3), repeat=8):
    if sum(b) != 0:
        continue
    if any(sum(b[j] for j in coordinate_slice[i]) != 0
           for i in range(3)):
        continue
    s = sum(x != 0 for x in b)
    r = sum(abs(x) == 2 for x in b)
    positive[s] += 1
    signed[s] += (-1)**r
```

The complete executable version is included in `verify_expansion.py`, along with assertions for every table entry. This is a transparent exhaustive finite certificate; the all-dimension expansion and its sharpness proofs do not depend on trusting a numerical approximation to this table.

III.27 An integer determinant certificate for the order-five example

Source 09, delivered Appendix B.

Here is a direct check of the determinant in (III.23.1). Subtract row one from rows four and five, and expand along column one. The remaining 4×4 matrix is

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & -1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{pmatrix}.$$

Subtract its first row from each remaining row and expand again along column one. This gives

$$D = \begin{pmatrix} -1 & 1 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{pmatrix}, \quad \det D = -1 - 2 - 2 = -5.$$

All these row operations preserve the determinant, and both expansions use the upper-left entry with positive cofactor sign. Thus $\det B = -5$. Together with (III.23.2), this proves the index and kernel claims without reliance on the Smith-normal-form implementation.

III.28 A torsion-sensitive generalized von Neumann inequality

Source 10, delivered Section 5.

The reindexing step in the proof of Theorem 3.2 of the source paper is where the ambient modulus enters. Retaining the sizes of the relevant fibres gives an explicit estimate on every finite abelian group. Retaining the norms of the other functions at the same time gives a useful density improvement in the subsequent counting argument.

Throughout this section G is a nonempty finite abelian group, $N = |G|$, and all averages and L^p norms are normalized. For an integer c put

$$G[c] = \{x \in G : cx = 0\}, \quad \kappa_G(c) = |G[c]|.$$

Thus $\kappa_G(0) = N$, and multiplication by c is invertible exactly when $\kappa_G(c) = 1$. On a cyclic group, $\kappa_{\mathbb{Z}/N\mathbb{Z}}(c) = \gcd(N, c)$, with the usual convention $\gcd(N, 0) = N$.

For $f : G \rightarrow \mathbb{C}$ set $\Delta_h f(x) = f(x) \overline{f(x+h)}$, and write

$$\|f\|_{U^s}^{2^s} = \mathbb{E}_{x, h_1, \dots, h_s \in G} \prod_{\omega \in \{0,1\}^s} \mathcal{C}^{|\omega|} f(x + \omega \cdot h), \quad s \geq 1,$$

where $\mathcal{C}$ denotes complex conjugation. These powers are nonnegative. In particular $\|f\|_{U^1} = |\mathbb{E}f|$, which is a seminorm, and for $s \geq 2$ the recursion is

$$\|f\|_{U^s}^{2^s} = \mathbb{E}_h \|\Delta_h f\|_{U^{s-1}}^{2^{s-1}}. \quad (\text{III.28.1})$$

Lemma III.28.1 (The exact fibre identity). *For every $V : G \rightarrow [0, \infty)$ and every integer c ,*

$$\mathbb{E}_u V(cu) = \kappa_G(c) \mathbb{E}_h \mathbf{1}_{cG}(h) V(h) \leq \kappa_G(c) \mathbb{E}_h V(h).$$

Proof. Every element of the image cG has exactly $\kappa_G(c)$ preimages. Counting those preimages proves the equality, and $V \geq 0$ proves the inequality. $\square$

For a coefficient vector $\mathbf{a} = (a_1, \dots, a_k)$ define

$$\Lambda_{\mathbf{a}}(f_1, \dots, f_k) = \mathbb{E}_{x,d} \prod_{i=1}^k f_i(x + a_i d).$$

Theorem III.28.2 (General coefficients and mixed L^p control). *Let $k \geq 2$, let $a_1, \dots, a_k$ be integers, and suppose that multiplication by $a_k - a_{k-1}$ is invertible on G . Set*

$$K = \prod_{i=1}^{k-1} \kappa_G(a_k - a_i), \quad M = \left(\prod_{i=1}^{k-2} \|f_i\|_{L^{2^i}} \right) \|f_{k-1}\|_{L^{2^{k-2}}}.$$

For arbitrary functions $f_i : G \rightarrow \mathbb{C}$,

$$|\Lambda_{\mathbf{a}}(f_1, \dots, f_k)| \leq MK^{1/2^{k-1}} \|f_k\|_{U^{k-1}}. \quad (\text{III.28.2})$$

No boundedness hypothesis on the functions is needed.

Proof. For $k = 2$, the map

$$(x, d) \mapsto (x + a_1 d, x + a_2 d)$$

is a bijection of G^2 . Hence the average equals $(\mathbb{E}f_1)(\mathbb{E}f_2)$, and its absolute value is at most $\|f_1\|_1\|f_2\|_{U^1}$, as required.

Suppose $k \geq 3$ and assume the result for $k - 1$. Put $c_i = a_i - a_1$. Translate the first variable by $a_1 d$ and apply Cauchy–Schwarz in that variable. Expanding the square and writing the second difference variable as $d + h$ gives

$$\begin{aligned} |\Lambda_{\mathbf{a}}(f_1, \dots, f_k)|^2 &\leq \|f_1\|_2^2 \mathbb{E}_x \left| \mathbb{E}_d \prod_{i=2}^k f_i(x + c_i d) \right|^2 \\ &= \|f_1\|_2^2 \mathbb{E}_h \Lambda_{(c_2, \dots, c_k)}(\Delta_{c_2 h} f_2, \dots, \Delta_{c_k h} f_k). \end{aligned} \quad (\text{III.28.3})$$

The final average is real and nonnegative because it arose by expanding an average of squares. It is therefore bounded above by the average of the absolute values of its inner multilinear forms.

The induction applies to each such form: the last coefficient difference is still $a_k - a_{k-1}$. Its kernel product is

$$K' = \prod_{i=2}^{k-1} \kappa_G(a_k - a_i).$$

For every $p \geq 1$, Hölder’s inequality and translation invariance give

$$\|\Delta_h f\|_{L^p} \leq \|f\|_{L^{2p}}^2. \quad (\text{III.28.4})$$

Consequently the product of the non-target norms in the induction is at most M_{rest}^2 , where

$$M_{\text{rest}} = \left(\prod_{i=2}^{k-2} \|f_i\|_{L^{2^i}} \right) \|f_{k-1}\|_{L^{2^{k-2}}}.$$

Let $q = 2^{k-2}$. A further application of Hölder, followed by Lemma III.28.1 and (III.28.1), gives

$$\begin{aligned} \mathbb{E}_h \|\Delta_{(a_k - a_1)h} f_k\|_{U^{k-2}} &\leq \left(\mathbb{E}_h \|\Delta_{(a_k - a_1)h} f_k\|_{U^{k-2}}^q \right)^{1/q} \\ &\leq \kappa_G(a_k - a_1)^{1/q} \|f_k\|_{U^{k-1}}^2. \end{aligned}$$

Substituting these bounds into (III.28.3) and taking square roots proves (III.28.2), since $M = \|f_1\|_2 M_{\text{rest}}$ and $(K' \kappa_G(a_k - a_1))^{1/(2q)} = K^{1/2^{k-1}}$. $\square$

[write] 6 October 2026; sources 06–10. *Text repaired.* In the delivered source the subscript M_{rest} appears three times in this proof as “M_{\text{rest}}”, a line break and “m rest}”: apparently the two characters `\rm` of `\rm` were read as a carriage-return escape when the source was generated, and the delivered PDF prints “ $M_{m \text{ rest}}$ ”. Restored here as M_{rest} (byte scan at intake: no other control-character damage in sources 01–10).

Credits. For k -term progressions with consecutive slopes and invertible differences this is the generalized von Neumann bound of sources 03, 04 and 06 (Lemmas III.1.1, III.4.2, Theorem III.5.1) with L^∞ norms replaced by mixed L^p norms; the kernel factor $K^{1/2^{k-1}}$, which removes the invertibility assumption on all but one difference, is only here. It advances source 06’s Question 7 for one distinguished function, with explicit kernel constants that are shown sharp only for the families of Proposition III.28.4; the two-function form remains open.

The L^p exponents assigned to the non-target functions are 1 for $k = 2$, then $(2, 2)$, $(2, 4, 4)$, $(2, 4, 8, 8)$, and so forth. Their reciprocals sum to one. The functions and coefficients can be reordered

before applying the theorem, provided a coefficient with invertible difference from the target is left immediately before it. This permits optimization of the norm product when the functions have different densities or sizes.

Corollary III.28.3 (Every target in an arithmetic progression). *Let $k \geq 2$, let $f_0, \dots, f_{k-1} : G \rightarrow \mathbb{C}$ satisfy $|f_i| \leq 1$, and let $0 \leq j \leq k-1$. Then*

$$\left| \mathbb{E}_{x,d} \prod_{i=0}^{k-1} f_i(x + id) \right| \leq \left(\prod_{i \neq j} \kappa_G(i - j) \right)^{1/2^{k-1}} \|f_j\|_{U^{k-1}}. \quad (\text{III.28.5})$$

In particular, for an endpoint target the factor is $(\prod_{r=1}^{k-1} \kappa_G(r))^{1/2^{k-1}}$. It is one whenever the exponent of G is coprime to $(k-1)!$.

Proof. Every target has a neighboring coefficient with difference 1 or -1 . Place that neighbor last among the non-target functions and apply Theorem III.28.2. Its L^p product is at most one. The endpoint formula and the coprimality assertion follow directly. $\square$

For the internal target j , the kernel product is

$$\left(\prod_{r=1}^j \kappa_G(r) \right) \left(\prod_{r=1}^{k-1-j} \kappa_G(r) \right).$$

Thus coprimality of the exponent of G with $j!(k-1-j)!$ suffices for that individual estimate. The endpoint result in particular extends the prime-modulus statement to composite cyclic groups with no prime factor below k , and to arbitrary finite abelian groups satisfying the same restriction on their prime divisors.

The invertible-neighbor hypothesis in the general-coefficient theorem is essential to the theorem as stated. For $k = 2$, coefficients $(0, 2)$ on $G = \mathbb{Z}/2\mathbb{Z}$, and opposite nontrivial characters, the multilinear average is one whereas the target U^1 seminorm is zero.

III.28.1 Sharpness and an explicit obstruction

Proposition III.28.4 (Sharp torsion growth). *For three-term progressions on $G = (\mathbb{Z}/2\mathbb{Z})^n$, the endpoint factor $|G|^{1/4}$ in (III.28.5) is exactly sharp. More generally, for $k = p+1$ on $G = (\mathbb{Z}/p\mathbb{Z})^n$, with p prime, the endpoint factor $|G|^{1/2^p}$ has the optimal order in $|G|$, up to constants depending only on p .*

Proof. First let $G = (\mathbb{Z}/2\mathbb{Z})^n$, $N = 2^n$, and set

$$g(x_1, \dots, x_n) = i^{x_1 + \dots + x_n},$$

using representatives $x_i \in \{0, 1\}$. Every normalized Fourier coefficient of g has absolute value $N^{-1/2}$: in each coordinate the Fourier factors are $(1+i)/2$ and $(1-i)/2$. Therefore

$$\|g\|_{U^2}^4 = \sum_{\chi} |\widehat{g}(\chi)|^4 = N^{-1}.$$

Take the two endpoint functions to be $\bar{g}, g$, and the middle function to be one. Since $2d = 0$, the progression average equals one. The upper bound is exactly $N^{1/4}N^{-1/4} = 1$.

For the general assertion write $s = p$ and $N = p^n$. Choose independent uniform unit-circle phases $g(z)$ for the points $z \in G$. For a fixed s -dimensional cube, the expectation of its alternating product is either zero or one, according as the exponents of these independent phases fail to cancel or all cancel. If its 2^s vertices are all distinct, the expectation is zero. Each pair of distinct cube vertices coincides with probability $1/N$, because their difference contains a coefficient 1 or -1 on at least one independent uniform increment. The union bound gives

$$\mathbb{E}_g \|g\|_{U^s}^{2^s} \leq \frac{\binom{2^s}{2}}{N}.$$

There is consequently a circle-valued g with at most this Gowers power. Put $\bar{g}, g$ at the two endpoints of the $(p+1)$ -term pattern and put one at every intermediate position. The average is again one because $pd = 0$. Hence any factor valid for all such functions is at least

$$\left(\frac{N}{\binom{2^p}{2}} \right)^{1/2^p}.$$

The kernel product in the upper bound is exactly N , since the differences $1, \dots, p-1$ are invertible. This proves the asserted optimal order. The random estimate is useful for large N ; for small N its right-hand side may exceed the trivial upper bound one. $\square$

A small explicit example also illustrates why the modulus restriction in the source theorem matters. On $\mathbb{Z}/3\mathbb{Z}$ take $g = (1, 1, -1)$ and the four functions $(\bar{g}, 1, 1, g)$. Their four-term average is one. However,

$$\|g\|_{U^3}^8 = \frac{49}{81} < 1.$$

Indeed, the zero derivative is one, and each nonzero derivative is a translate of $(1, -1, -1)$, whose U^2 fourth power is $(1/3)^4 + 2(2/3)^4 = 11/27$. The recursion gives $(1 + 2 \cdot 11/27)/3 = 49/81$.

[write] **6 October 2026; sources 06–10.** Rechecked at the second write: $(1/3)^4 + 2(2/3)^4 = 33/81 = 11/27$ and $(1 + 22/27)/3 = 49/81$. The $\mathbb{Z}/2\mathbb{Z}$ torsion examples of sources 03 and 04 (Part III above) are of the same kind; here the factor is quantified and shown sharp on $(\mathbb{Z}/2\mathbb{Z})^n$.

III.29 Density-sensitive counting and modular progression criteria

Source 10, delivered Section 6.

For convenience write

$$\Lambda_k(F_1, \dots, F_k) = \mathbb{E}_{x,d} \prod_{i=1}^k F_i(x + (i-1)d).$$

The following replacement for the counting step keeps the densities and uses only $k-2$ error terms.

Theorem III.29.1 (A density-weighted telescoping estimate). *Let $k \geq 3$ and let $A_1, \dots, A_k \subseteq G$. Put $F_i = \mathbf{1}_{A_i}$, $\delta_i = \mathbb{E}F_i$, and $b_i = F_i - \delta_i$. For $3 \leq j \leq k$ set*

$$\tau_j(G) = \left(\prod_{r=1}^{j-1} \kappa_G(r) \right)^{1/2^{j-1}}, \quad B_j = \left(\prod_{i=1}^{j-2} \delta_i^{1/2^i} \right) \delta_{j-1}^{1/2^{j-2}}, \quad D_j = \prod_{i=j+1}^k \delta_i.$$

Then

$$\left| \Lambda_k(F_1, \dots, F_k) - \prod_{i=1}^k \delta_i \right| \leq \sum_{j=3}^k D_j B_j \tau_j(G) \|b_j\|_{U^{j-1}}. \quad (\text{III.29.1})$$

Proof. For $L_i = x + (i-1)d$ use the pointwise identity

$$\prod_{i=1}^k F_i(L_i) - \prod_{i=1}^k \delta_i = \sum_{j=1}^k \left(\prod_{i < j} F_i(L_i) \right) b_j(L_j) \left(\prod_{i > j} \delta_i \right).$$

The averaged term with $j = 1$ vanishes because $\mathbb{E}b_1 = 0$. The averaged term with $j = 2$ also vanishes: the two adjacent forms are jointly uniform on G^2 , so its average factors and contains $\mathbb{E}b_2 = 0$. Apply Theorem III.28.2 in natural coefficient order to every remaining j -term average. The target is b_j , its kernel factor is $\tau_j(G)$, and $\|\mathbf{1}_{A_i}\|_{L^p} = \delta_i^{1/p}$ makes its mixed norm product exactly B_j . The triangle inequality proves the claim. $\square$

For one or two forms the count equals its product of densities exactly. If the exponent of G is coprime to $(k-1)!$, all the factors $\tau_j(G)$ are one. In particular, under the uniformity assumptions of Corollary 3.3 of the source paper, one has $\|b_j\|_{U^{j-1}} \leq \alpha$, and its error term can be replaced by

$$\alpha \sum_{j=3}^k D_j B_j \leq (k-2)\alpha.$$

The displayed density-dependent sum is often appreciably smaller.

Corollary III.29.2 (One-set counting). *Let $k \geq 3$, let $A \subseteq G$ have density δ , and put*

$$\eta = \|\mathbf{1}_A - \delta\|_{U^{k-1}}, \quad T_{k,G}(\delta) = \sum_{j=3}^k \delta^{k-j} \tau_j(G).$$

Then

$$|\Lambda_k(A) - \delta^k| \leq \delta \eta T_{k,G}(\delta). \quad (\text{III.29.2})$$

In the coprime case, put

$$S_k(\delta) = \sum_{m=0}^{k-3} \delta^m.$$

Then $T_{k,G}(\delta) = S_k(\delta)$ and

$$|\Lambda_k(A) - \delta^k| \leq \delta S_k(\delta) \eta. \quad (\text{III.29.3})$$

For $0 \leq \delta < 1$, $S_k(\delta) = (1 - \delta^{k-2})/(1 - \delta)$, while $S_k(1) = k - 2$.

Proof. The reciprocals in the definition of B_j sum to one, so $B_j = \delta$ when all the sets are A . Gowers norm monotonicity gives $\|\mathbf{1}_A - \delta\|_{U^{j-1}} \leq \eta$ for $j \leq k$. Theorem III.29.1 now gives both estimates.

For completeness, monotonicity follows from two applications of Cauchy–Schwarz. Writing $g = \Delta_{h_1} \cdots \Delta_{h_{s-1}} f$, one has

$$|\mathbb{E}g|^4 = \left| \mathbb{E}_t \mathbb{E}_x g(x) \overline{g(x+t)} \right|^2 \leq \mathbb{E}_t \left| \mathbb{E}_x g(x) \overline{g(x+t)} \right|^2 = \|g\|_{U^2}^4.$$

Average this inequality over the h_i , and use $(\mathbb{E}_h |\mathbb{E}g|^2)^2 \leq \mathbb{E}_h |\mathbb{E}g|^4$. The two sides identify with the appropriate cube forms and yield $\|f\|_{U^s} \leq \|f\|_{U^{s+1}}$. $\square$

Corollary III.29.3 (An explicit modular progression criterion). *Let $k \geq 3$, suppose that the exponent of G is coprime to $(k-1)!$, and let $A \subseteq G$ have positive density δ . With η as above, the number of ordered pairs (x, d) for which all k terms are distinct and lie in A is at least*

$$\delta N^2 \left(\delta^{k-1} - S_k(\delta)\eta - \frac{1}{N} \right). \quad (\text{III.29.4})$$

In particular, A contains such a progression if

$$N(\delta^{k-1} - S_k(\delta)\eta) > 1.$$

A simpler sufficient pair of conditions is

$$\eta \leq \frac{\delta^{k-1}}{2S_k(\delta)}, \quad N > 2\delta^{1-k}. \quad (\text{III.29.5})$$

Proof. There are exactly $|A| = \delta N$ counted pairs with $d = 0$. If $d \neq 0$ and two terms coincide, then $rd = 0$ for some $1 \leq r \leq k-1$, contradicting invertibility of multiplication by r . Subtracting these exactly δN degenerate pairs from (III.29.3), after multiplying by N^2 , proves (III.29.4). Either displayed condition makes this lower bound strictly positive. $\square$

In the notation of the source paper, being α -uniform of degree $k-2$ means $\eta^{2^{k-1}} \leq \alpha$. Thus the sufficient uniformity condition in (III.29.5) can be written

$$\alpha \leq \left(\frac{\delta^{k-1}}{2S_k(\delta)} \right)^{2^{k-1}}, \quad N > 2\delta^{1-k}.$$

These improve the local modular-counting requirements. Incorporating them into a global quantitative Szemerédi argument also requires the later inverse, density-increment, and interval-transfer steps to be recomputed with their own hypotheses.

[write] **6 October 2026; sources 06–10.** Compared with the criteria of sources 03, 04 and 06 in the note after source 04’s Corollary III.4.4: the error here is δ times theirs, and in α terms the power of δ is $(k-1)2^{k-1}$, between source 06’s $(k - \frac{1}{2})2^{k-2}$ and source 04’s $k2^{k-1}$. Not formalized.

III.29.1 Repeated terms and the appropriate main term

The torsion-sensitive estimate also gives a distinct-point criterion without a coprimality assumption. Define

$$\mathcal{D}_k(G) = \bigcup_{r=1}^{k-1} G[r].$$

A progression repeats a point exactly when $d \in \mathcal{D}_k(G)$. For each such difference, at most δN choices of x can contribute, since the first point must belong to A . Hence a sufficient criterion for a progression with k distinct points in an arbitrary finite abelian group is

$$\delta^{k-1} - \eta T_{k,G}(\delta) > \frac{|\mathcal{D}_k(G)|}{N}. \quad (\text{III.29.6})$$

In small characteristic it is sometimes preferable to reduce the pattern before estimating it. On $(\mathbb{Z}/p\mathbb{Z})^n$, if $k > p$, the homogeneous indicator product is exactly the p -term product, since

coefficients repeat modulo p and indicators are idempotent. For a Bernoulli random set with inclusion probability δ , the exact expected normalized count is then

$$\left(1 - \frac{1}{N}\right) \delta^p + \frac{\delta}{N}.$$

Its leading independent-membership term is δ^p . This coefficient collision explains why a dimension-independent error tending to zero around δ^k cannot describe that pattern.

Further questions specific to these estimates. The sharpness argument determines the correct group-size order for $k = p + 1$; exact constants for larger k and general torsion profiles remain open within this investigation. A natural next problem is to retain the restricted images cG in Lemma III.28.1, instead of replacing each by all of G , and to exploit the overlap of these images. For heterogeneous sets, the telescoping order and the assignment of the L^p exponents can be optimized jointly; when all coefficient differences are invertible, the rearrangement inequality assigns the largest reciprocal exponents to the smallest densities within a fixed prefix. The finite fibre identity, mixed-norm induction, telescoping identity, and exact degenerate-pair count are also natural separate formalization targets.

III.30 An exact dimension threshold for polynomial colorings

Source 10, delivered Section 9.

There is a useful restricted question behind the recent constructions of sets with fewer arithmetic progressions than their uniformity might suggest: how many coordinates can a polynomial coloring control while forbidding reflection symmetry along every nontrivial progression? The construction of Shi and Dong uses a sum of odd-degree field norms [33]. We give an exact answer when the number of field-valued color coordinates and their total degrees are prescribed. The obstruction is a direct Chevalley–Warning argument [39, 40]; an analogous argument involving all homogeneous degrees appears in Fühner and Taranchuk [38]. Our purpose is to state the precise degree budget, prove its optimality, and identify the limit it places on this construction method. We make no claim of first discovery of these polynomial-method consequences.

Throughout this section, let $k = 2m \geq 4$ and let $p > k$ be prime. A coloring $Q : \mathbb{F}_p^d \rightarrow \mathbb{F}_p^r$ is *polynomial with coordinate degree bounds* $D_1, \dots, D_r$ if its ℓ th coordinate is represented by a polynomial Q_ℓ of total degree at most D_ℓ . A progression $(a, a + h, \dots, a + (k - 1)h)$ is *symmetrically colored* if

$$Q(a + ih) = Q(a + (k - 1 - i)h) \quad (0 \leq i < m). \quad (\text{III.30.1})$$

It is nontrivial when $h \neq 0$. The assumption $p > k$ guarantees that all k of its points are then distinct.

III.30.1 Odd coefficients encode reflection symmetry

Fix a center $z \in \mathbb{F}_p^d$ and introduce formal variables $H = (H_1, \dots, H_d)$ and T . For a polynomial P of total degree at most D , write

$$P(z + TH) = \sum_{j=0}^D C_{j,z}(H) T^j. \quad (\text{III.30.2})$$

The polynomial $C_{j,z}$ is homogeneous of degree j , or is zero. Indeed, in the expansion of a monomial, selecting the coefficient of T^j selects exactly j factors from the coordinates of H . This coefficient definition needs no differentiation or division by factorials.

Lemma III.30.1 (Reflection and the odd coefficient equations). *Suppose $1 \leq D \leq k$ and put $s = \lceil D/2 \rceil$. For $z, h \in \mathbb{F}_p^d$, the following are equivalent:*

1. *The progression with center z and direction h is symmetrically colored by P .*
2. $C_{1,z}(h) = C_{3,z}(h) = \cdots = C_{2s-1,z}(h) = 0$.
3. *The polynomial $P(z + Th)$ is even in T .*

Proof. Put $\alpha_i = i - \frac{1}{2} \in \mathbb{F}_p$ for $1 \leq i \leq m$. The progression with center z has points $z \pm \alpha_1 h, \dots, z \pm \alpha_m h$. Its reflection equalities are $P(z + \alpha_i h) = P(z - \alpha_i h)$.

The α_i are nonzero, and their squares are pairwise distinct: for $i \neq j$, the quantities $\alpha_i - \alpha_j = i - j$ and $\alpha_i + \alpha_j = i + j - 1$ are nonzero modulo p . In particular, $1 \leq i + j - 1 \leq k - 1 < p$.

Subtracting the two evaluations in (III.30.2) cancels all even powers, leaving

$$P(z + TH) - P(z - TH) = 2 \sum_{j=1}^s C_{2j-1,z}(H) T^{2j-1}.$$

Its degree is at most $2s - 1 \leq k - 1$, even when $D = k$. After division by $2\alpha_i$, the reflection equalities become

$$\sum_{j=1}^s C_{2j-1,z}(h) (\alpha_i^2)^{j-1} = 0 \quad (1 \leq i \leq m).$$

Since $s \leq m$, the first s equations form a square Vandermonde system with determinant

$$\prod_{1 \leq i < j \leq s} (\alpha_j^2 - \alpha_i^2) \neq 0.$$

Thus all its unknown odd coefficients vanish. Conversely, vanishing of these coefficients makes the polynomial even and implies every required reflection equality. $\square$

The important distinction is that the degree budget contains only $1, 3, \dots, 2s - 1$. An even top degree contributes no new reflection equation. This is why degree bounds $2s - 1$ and $2s$ lead to the same dimension threshold.

III.30.2 The zero-count obstruction

We recall the short finite-field proof of the form of Chevalley–Warning that we need.

Lemma III.30.2 (Chevalley–Warning in the needed form). *Let $F_1, \dots, F_t \in \mathbb{F}_p[H_1, \dots, H_d]$ vanish at the origin. Omit zero polynomials from the degree sum. If $\sum_{j=1}^t \deg F_j < d$, the number of their common zeros in $\mathbb{F}_p^d$ is divisible by p . Consequently there are at least p common zeros, including at least $p - 1$ nonzero common zeros.*

Proof. By Fermat's theorem, the polynomial function

$$I(H) = \prod_{j=1}^t (1 - F_j(H)^{p-1})$$

equals the indicator of the common zero set. Its total degree is less than $d(p-1)$. Every monomial of I therefore has at least one variable whose exponent e satisfies $0 \leq e < p-1$. For that exponent,

$$\sum_{u \in \mathbb{F}_p} u^e = 0 \quad \text{in } \mathbb{F}_p.$$

For $e = 0$ this is $p = 0$; for $1 \leq e < p-1$ it is the usual finite-field power sum. Summation of each monomial over $\mathbb{F}_p^d$ factors into the corresponding one-variable sums, and hence gives zero. It follows that $\sum_h I(h) = 0$ in $\mathbb{F}_p$, which is the required divisibility. The zero count is positive because the origin is a common zero, so it is at least p . $\square$

Theorem III.30.3 (Exact dimension threshold with heterogeneous degrees). *Let $r \geq 1$, let $1 \leq D_\ell \leq k$ for $1 \leq \ell \leq r$, and put*

$$s_\ell = \left\lceil \frac{D_\ell}{2} \right\rceil, \quad B = \sum_{\ell=1}^r s_\ell^2.$$

There exists a polynomial coloring $Q : \mathbb{F}_p^d \rightarrow \mathbb{F}_p^r$, with $\deg Q_\ell \leq D_\ell$, having no nontrivial symmetrically colored k -term arithmetic progression if and only if $d \leq B$.

If $d > B$, then for every such Q and every center z , the number of directions giving a symmetrically colored progression centered at z , including the zero direction, is divisible by p . In particular, there are at least $p-1$ nonzero such directions at every center.

Proof. For necessity, fix z and apply Lemma III.30.1 to each color coordinate. Reflection symmetry is equivalent to the common vanishing of

$$C_{\ell,1,z}(h), C_{\ell,3,z}(h), \dots, C_{\ell,2s_\ell-1,z}(h) \quad (1 \leq \ell \leq r).$$

These homogeneous polynomials vanish at the origin and have total degree sum at most

$$\sum_{\ell=1}^r \sum_{j=1}^{s_\ell} (2j-1) = \sum_{\ell=1}^r s_\ell^2 = B.$$

If $d > B$, Lemma III.30.2 gives the asserted divisibility and nonzero directions. Thus avoidance requires $d \leq B$.

For sufficiency we use the odd-degree norm construction of Shi and Dong [33, Sections 3–4], allowing a different top degree in each coordinate. For a positive odd integer t , identify $\mathbb{F}_p^t$ linearly with $\mathbb{F}_{p^t}$ and let $N_t : \mathbb{F}_p^t \rightarrow \mathbb{F}_p$ be the field norm. In degree one, take $N_1(u) = u$. We need that N_t is homogeneous of degree t and that $N_t(u) = 0$ holds exactly when $u = 0$.

Here is a polynomial description of those properties. If $\beta_1, \dots, \beta_t$ is a basis of $\mathbb{F}_{p^t}$ over $\mathbb{F}_p$, then

$$N_t(u_1, \dots, u_t) = \prod_{a=0}^{t-1} \left(\sum_{j=1}^t u_j \beta_j^{p^a} \right).$$

This is a product of t linear forms, and Frobenius permutes its factors, so its coefficients lie in $\mathbb{F}_p$. Its value is the norm of $\sum_j u_j \beta_j$, which is nonzero precisely when that field element is nonzero. In particular, its formal total degree is t , despite the alternative large-exponent expression for a field norm.

For color coordinate ℓ , introduce disjoint variable blocks

$$u_\ell^{(1)} \in \mathbb{F}_p, \quad u_\ell^{(3)} \in \mathbb{F}_p^3, \quad \dots, \quad u_\ell^{(2s_\ell-1)} \in \mathbb{F}_p^{2s_\ell-1}.$$

There are s_ℓ^2 variables in that coordinate's collection of blocks. On their full B -dimensional product define

$$Q_\ell(u) = \sum_{j=1}^{s_\ell} N_{2j-1}(u_\ell^{(2j-1)}). \quad (\text{III.30.3})$$

Its total degree is $2s_\ell - 1 \leq D_\ell$.

Suppose a progression with center z and direction h is symmetrically colored. Lemma III.30.1 says that every $Q_\ell(z + Th)$ is even. If h has a nonzero block in coordinate ℓ , let t be the largest odd index with $h_\ell^{(t)} \neq 0$. All higher-index blocks contribute constants, because their directions vanish; all lower-index blocks have degree less than t . The coefficient of T^t in $Q_\ell(z + Th)$ is therefore exactly

$$N_t(h_\ell^{(t)}) \neq 0.$$

Since t is odd, this contradicts evenness. Thus every direction block vanishes, and $h = 0$. This proves avoidance at $d = B$.

For $d < B$, restrict the construction to any linear embedding of $\mathbb{F}_p^d$ into $\mathbb{F}_p^B$. A nonzero direction remains nonzero under the embedding, and composition with a linear map does not increase total degree, so all the required properties are preserved. $\square$

Corollary III.30.4 (Equal degree bounds and optimal coordinate budget). *For $1 \leq D \leq k$, a coloring by r polynomial coordinates of total degree at most D can avoid all nontrivial symmetrically colored k -APs on $\mathbb{F}_p^d$ exactly when*

$$d \leq r \lceil D/2 \rceil^2.$$

In particular, for degree at most $k = 2m$ the exact condition is $d \leq rm^2$, and for $d \geq 1$ the least possible number of polynomial coordinates is $\lceil d/m^2 \rceil$.

At the equality dimension $d = B$, the coloring (III.30.3) is surjective onto $\mathbb{F}_p^r$. Indeed, each coordinate contains its own independent linear variable $u_\ell^{(1)}$. After choosing all other variables, the r linear variables are uniquely determined by the desired color. Thus every color class has exactly p^{B-r} points.

III.30.3 An upper bound for arbitrary degree budgets

When the degree exceeds the progression length, the reflection equations can vanish without all odd coefficients vanishing. Nevertheless, they give a useful explicit dimension bound.

Proposition III.30.5 (A dimension bound in every degree). *For a positive integer s , define*

$$B_m(s) = \begin{cases} s^2, & s \leq m, \\ m(2s - m), & s \geq m. \end{cases}$$

Let $D_1, \dots, D_r \geq 1$ be arbitrary, put $s_\ell = \lceil D_\ell/2 \rceil$, and suppose $Q : \mathbb{F}_p^d \rightarrow \mathbb{F}_p^r$ is polynomial with $\deg Q_\ell \leq D_\ell$. If

$$d > \sum_{\ell=1}^r B_m(s_\ell),$$

then at every prescribed center there are at least $p - 1$ nonzero directions producing symmetrically colored k -term progressions. Thus avoidance requires $d \leq \sum_{\ell=1}^r B_m(s_\ell)$.

Proof. Fix a center z and write the odd coefficients of one coordinate as $C_{2j+1,z}(H)$, for $0 \leq j < s$, with $s = \lceil D/2 \rceil$. Let $y_i = \alpha_i^2$ for the nonzero half-integer parameters used in Lemma III.30.1, and put $v_j = (y_i^j)_{i=1}^m \in \mathbb{F}_p^m$. After dividing each reflection equality by $2\alpha_i$, the vector of reflection equations is

$$E(H) = \sum_{j=0}^{s-1} v_j C_{2j+1,z}(H) = 0.$$

If $s \leq m$, these are equivalent to the individual odd coefficient equations, whose total degree sum is s^2 .

If $s > m$, form the square matrix

$$V = (v_{s-m}, v_{s-m+1}, \dots, v_{s-1}).$$

It is the usual Vandermonde matrix multiplied on the left by the invertible diagonal matrix with entries y_i^{s-m} , so it is invertible. The equivalent system $V^{-1}E(H) = 0$ has equations

$$F_j(H) = C_{2(s-m+j)+1,z}(H) + \sum_{q=0}^{s-m-1} b_{j,q} C_{2q+1,z}(H) = 0, \quad 0 \leq j < m,$$

for scalars $b_{j,q} \in \mathbb{F}_p$. Every coefficient in the second sum has strictly smaller degree than the displayed first coefficient. Consequently

$$\deg F_j \leq 2(s-m+j) + 1, \quad \sum_{j=0}^{m-1} \deg F_j \leq m(2s-m).$$

Zero polynomials may again be omitted. Although the transformed equations need not be homogeneous, they all vanish at the origin. Applying this procedure independently to each color coordinate produces a system equivalent to reflection symmetry with total degree sum at most $\sum_{\ell} B_m(s_\ell)$. The conclusion follows from Lemma III.30.2. $\square$

For four-term progressions and a scalar degree bound $D = 5$ or 6 , Proposition III.30.5 gives $d \leq 8$. The cubic construction already attains dimension 4, so the corresponding maximum dimension lies between 4 and 8. Determining that maximum is a concrete question beyond the exact low-degree theorem. No matching construction is asserted for the arbitrary-degree bound.

III.30.4 An explicit four-dimensional example

For $k = 4$, both scalar degree bounds $D = 3$ and $D = 4$ have exact dimension threshold 4. Over $\mathbb{F}_5$, the polynomial $T^3 + T + 1$ has no root and is therefore irreducible. If $\theta^3 + \theta + 1 = 0$, the norm of

$a + b\theta + c\theta^2$ is

$$\begin{aligned} N_3(a, b, c) &= \det \begin{pmatrix} a & -c & -b \\ b & a-c & -b-c \\ c & b & a-c \end{pmatrix} \\ &= a^3 - 2a^2c + ac^2 + ab^2 + 3abc - b^3 - bc^2 + c^3 \pmod{5}. \end{aligned} \quad (\text{III.30.4})$$

The determinant is that of multiplication by $a + b\theta + c\theta^2$ in the basis $1, \theta, \theta^2$. Consequently

$$Q(x_0, x_1, x_2, x_3) = x_0 + N_3(x_1, x_2, x_3)$$

is a five-coloring of $\mathbb{F}_5^4$ with five color classes of 125 points and no nontrivial symmetrically colored 4-AP.

The accompanying exact-arithmetic verification checks the determinant formula, the absence of nonzero norm zeros, and all $625 \cdot 624 = 390,000$ ordered nontrivial progressions. It finds no symmetrically colored progression. This finite check verifies the displayed example; the theorem in arbitrary dimension and characteristic follows from the preceding proof.

III.30.5 What the threshold does and does not settle

The same result holds for any prescribed centrally symmetric pattern $\{\pm\alpha_1, \dots, \pm\alpha_m\}$ in $\mathbb{F}_p$ with $\alpha_i \neq 0$ and pairwise distinct α_i^2 . The conditions $Q(z + \alpha_i h) = Q(z - \alpha_i h)$ are again equivalent to the same odd coefficient equations, by the same Vandermonde argument. Hence the optimal dimension depends on the degree budgets and the number of reflection pairs, rather than on equal spacing itself.

For degree at most k , the theorem shows that the formal palette $\mathbb{F}_p^r$ must satisfy $r/d \geq 4/k^2$, and the equality construction uses all p^r colors. This is a sharp statement about a prescribed number of polynomial color coordinates. It is not a lower bound on the actual image size of arbitrary maps with redundant color coordinates, nor an optimality theorem for unrestricted colorings of intervals or cyclic groups. In particular, the auxiliary digit colors in Shi and Dong's cyclic construction are outside this polynomial coordinate model. The possibility of $N^{o(1)}$ colors in the Deng–Tidor–Zhao symmetric-progression problem remains outside the conclusion [37].

The restriction $D_\ell \leq k$ also matters. With higher odd degrees, agreement at the k progression parameters need not make the entire odd polynomial vanish. Thus higher-degree constructions and nonpolynomial color data provide specific directions beyond the class solved here. Within the stated class, however, both sides of the optimization problem are complete: the odd-degree sum gives a universal obstruction, and the known norm construction attains it.

Part IV

Floor patterns and the Proposition 17.7 phase extraction

Contents. The number of floor patterns of subset sums, which controls how a cube meets a partition into progressions (Corollaries 17.5–17.6 and Proposition 17.7). Sections IV.1–IV.5: source 01, delivered Sections 9–13 (exact range factorization, quadratic-exponent upper and lower bounds, $\log_2 C_k \sim k^2$ via the classical threshold-function count, affine patterns and translation classes, polynomial features, certified $C_1, \dots, C_4 = 1, 2, 10, 154$). Section IV.6: source 04, delivered Section 8 (the same counts as F_k , the stronger elementary lower bound $2^{k(k-1)/2}$, exact modular range dependence, few profiles at a fixed slope, and a local phase-extraction theorem with coefficient $2^{-k^2-k \log_2 k - O(k)}$ at the stage of Proposition 17.7).

Overlaps. Sources 01 and 04 are independent manuscripts; they share only this count ($C_k = F_k$), which they prove independently with different certificate formats. Dated notes credit the shared lemmas and record that source 01’s asymptotic answers source 04’s Question 8.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
C_k, F_k, A_k	01: C_k fractional carry patterns, A_k affine patterns; 04: F_k (= 01’s C_k) and A_k (= 01’s A_k).
H_k, J_k, Q_k	Both: $H_k = (k-2)2^{k-1} + 1$; 01: $J_k = k2^{k-1}$; 04: $Q_k = k2^{k-1}$ (= 01’s J_k).
region bound	01: $\mathcal{R}(h, d) = \sum_{j \leq d} \binom{h}{j}$; 04: $R_d(H) = \sum_{j \leq d} \binom{H}{j}$. <i>The arguments are in the other order: 01’s $\mathcal{R}(H, d)$ is 04’s $R_d(H)$.</i>
B_k	01: translation budget $(2(k+1))^k \mathcal{R}(k2^{k-1}, k+1)$; 04: $(2k)^k R_{k+1}(k2^{k-1})$. <i>Different numbers</i> (note after Corollary IV.6.4).
ranges, M	01: coefficients in $(-r_i, r_i)$, M a modulus, T_n threshold counts, M_n nonnegative ones; 04: (a_i, b_i) with $m_i = b_i - a_i$, M a modulus, $\Omega_k = \{0, 1\}^k$, L labels and K local multiplicity in Lemma IV.6.6.

Added in the second write (6 October 2026): sources 06–10. Contents. Section IV.7: source 06, delivered Section 2 (exact separation of range and carries for every finite $V \subset \mathbb{Z}_{\geq 0}^k$ containing 0 and the unit vectors, the modular count $\min(2r, M)^k C(V; M)$, affine carries and translation classes, and $\log_2 C_k \sim k^2$ uniformly modulo every $M \geq 2$). Section IV.8: delivered Section 4 (localized phase removal at Proposition 17.7 with coefficient $2^{-2k^2 - O(k \log k)}$). Section IV.9: delivered Section 7 (the certified carry counts $C_0, \dots, C_4 = 1, 1, 2, 10, 154$ and $D_0, \dots, D_3 = 1, 2, 6, 38$, the uniformity and density checks, and the size of the improvements).

Overlaps (dated notes). The counts C_k are certified independently by sources 01 and 06 up to $k = 4$ ($C_4 = 154$ twice, with different certificate formats); D_k agrees with 04’s A_k for $k \leq 3$. Exact range scaling (01, 04, 06), the modular count (04, 06), general coefficient families (01’s polynomial features, 06’s V), $\log_2 C_k \sim k^2$ through the threshold-function count (01, 06). At Proposition 17.7 three coefficients are proved: 04’s $2^{-k^2-k \log_2 k - O(k)}$, 06’s $2^{-2k^2 - O(k \log k)}$, and 01’s weaker selection interface (note after Corollary IV.6.8).

Reading conventions for the sections of sources 06–10 in this Part.

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
C_k, D_k, L_k	06: C_k ($= 01$'s $C_k = 04$'s F_k), D_k affine carry patterns ($D_1, D_2, D_3 = 2, 6, 38 = 04$'s A_1, A_2, A_3), $C_{k,M}, D_{k,M}$ their reductions modulo M , $L_k(r, M)$ translation classes, $C(V), D(V)$ for a general family V .
$h_k, \tilde{h}_k, \mathcal{R}_d(m)$	06: $h_k = (k-2)2^{k-1} + 1$ ($= H_k$ of 01, 04), $\tilde{h}_k = k2^{k-1}$ ($= 01$'s J_k , 04's Q_k), $\mathcal{R}_d(m) = \sum_{j \leq d} \binom{m}{j}$ ($= 01$'s $\mathcal{R}(m, d) = 04$'s $R_d(m)$). <i>In $\mathcal{R}_d(m)$ the subscript is the dimension.</i>
$K_k, c_k, \mathcal{E}_d$	06: $K_k = (2(k+1))^k \mathcal{R}_{k+1}(k2^{k-1})$ ($= 01$'s B_k) and $c_k = 1/(8K_k^2)$; $\mathcal{E}_d(g) = p^{d+1} \ g\ _{U^d}^{2^d}$ an unnormalized cube energy; q, M, w, ℓ the partition scales. <i>06's c_k is not 04's c_k of Corollary IV.6.8; this $\mathcal{E}_d$ is none of the $\mathcal{E}$'s of Parts II–III.</i>
$T(n), T_+(n)$	06: the number of Boolean threshold functions and of those with nonnegative weights (01: T_n, M_n).

IV.1 Fractional carry patterns and exact range dependence

Source 01, delivered Section 9.

We turn to the floor-pattern interface. For $k \geq 1$ and $x = (x_1, \dots, x_k) \in [0, 1)^k$, define

$$c_x(S) = \left\lfloor \sum_{i \in S} x_i \right\rfloor, \quad S \subseteq [k], \quad C_k = \#\{c_x : x \in [0, 1)^k\}. \quad (\text{IV.1.1})$$

The values at the empty set and at singletons are always zero. We call c_x a *fractional carry pattern*. The terminology refers only to this finite function; no probabilistic assumption on x is made. For completeness set $C_0 = 1$.

IV.1.1 Why boundary points do not introduce extra patterns

A hyperplane-region count normally concerns strict inequalities. Floor functions also take values on the hyperplanes, so that issue must be settled before counting chambers.

Lemma IV.1.1 (Positive perturbation). *Every fractional carry pattern has a representative $y \in (0, 1)^k$ such that $\sum_{i \in S} y_i$ is nonintegral for every nonempty $S \subseteq [k]$.*

Proof. Given $x \in [0, 1)^k$, put $y_i = x_i + \varepsilon$. There are finitely many coordinates and subset sums. Choose $\varepsilon > 0$ so small that $x_i + \varepsilon < 1$ for every i and

$$|S|\varepsilon < \left\lfloor \sum_{i \in S} x_i \right\rfloor + 1 - \sum_{i \in S} x_i \quad (\emptyset \neq S \subseteq [k]).$$

Every right-hand side is strictly positive, including when the subset sum is an integer. The perturbation moves the sum positively but not to the next integer. Its floor is unchanged, and it is no longer an integer. All coordinates become strictly positive. $\square$

Theorem IV.1.2 (Exact separation of the coefficient range). *Let $r_1, \dots, r_k$ be positive integers and let $\mathcal{F}(\mathbf{r})$ be the family of functions*

$$S \mapsto \left\lfloor \sum_{i \in S} \alpha_i \right\rfloor, \quad -r_i < \alpha_i < r_i.$$

Then

$$|\mathcal{F}(\mathbf{r})| = \left(\prod_{i=1}^k 2r_i \right) C_k. \quad (\text{IV.1.2})$$

Proof. Write $\alpha_i = m_i + x_i$, where $m_i = \lfloor \alpha_i \rfloor$ and $x_i \in [0, 1)$. Necessarily

$$m_i \in \{-r_i, -r_i + 1, \dots, r_i - 1\}.$$

For every subset,

$$F_\alpha(S) = \sum_{i \in S} m_i + c_x(S). \quad (\text{IV.1.3})$$

The singleton values of F_α recover all m_i ; after subtracting their subset sums, they also recover c_x . Thus this decomposition is injective as a coding of the resulting functions.

Conversely, take any permitted integer vector m and any carry pattern. By Lemma IV.1.1, choose a representative $x \in (0, 1)^k$. Then $-r_i < m_i + x_i < r_i$ for every i , even when $m_i = -r_i$. Equation (IV.1.3) realizes the chosen pair. Hence there is a bijection between the pattern family and the Cartesian product of the integer vectors with the C_k carry patterns. $\square$

The endpoint convention is essential for the exact statement. The proof handles the open interval $(-r_i, r_i)$ precisely because each fractional pattern admits an interior representative. For nonintegral range bounds $R_i > 0$, replacing r_i by $\lceil R_i \rceil$ gives an immediate upper bound, but generally not the same exact factorization.

IV.1.2 A hyperplane-arrangement upper bound

Inside the open unit cube, a nonempty subset S produces integer threshold hyperplanes

$$\sum_{i \in S} x_i = j, \quad 1 \leq j \leq |S| - 1.$$

There are exactly

$$H_k = \sum_{\emptyset \neq S \subseteq [k]} (|S| - 1) = k2^{k-1} - (2^k - 1) = (k - 2)2^{k-1} + 1 \quad (\text{IV.1.4})$$

such hyperplanes. Different subsets have different normal vectors, and none of these hyperplanes is a duplicate of another.

Lemma IV.1.3 (Region bound). *The number of regions cut out by h affine hyperplanes in $\mathbb{R}^d$ is at most*

$$\mathcal{R}(h, d) = \sum_{j=0}^d \binom{h}{j}.$$

The same bound holds for their regions restricted to an open convex set.

Proof. Let $R(h, d)$ be the maximal possible number of regions. When one adds a hyperplane, each newly created region comes from splitting an old region that meets it. The old hyperplanes induce on the new hyperplane an arrangement of at most $h - 1$ hyperplanes in dimension $d - 1$; coincident or empty intersections create no additional regions. Therefore

$$R(h, d) \leq R(h - 1, d) + R(h - 1, d - 1).$$

The initial values $R(0, d) = 1$ and $R(h, 0) = 1$, together with Pascal's identity, prove the bound by induction. A region is an intersection of open half-spaces and is convex. Intersecting it with an open convex set cannot split it into several nonempty connected pieces, so restriction cannot increase the number. $\square$

Theorem IV.1.4 (Quadratic-exponent carry bound). *For every $k \geq 1$,*

$$C_k \leq \mathcal{R}(H_k, k). \quad (\text{IV.1.5})$$

For $k \geq 3$ this implies

$$\log_2 C_k \leq k^2 + (\log_2 e - 1)k. \quad (\text{IV.1.6})$$

Proof. Lemma IV.1.1 realizes every carry pattern off the threshold hyperplanes. Off those hyperplanes, the floor values determine, and are determined by, the sides of all the listed integer thresholds. A given sign pattern is an intersection of open half-spaces with the cube and is convex. Thus distinct carry patterns correspond exactly to nonempty regions of this arrangement inside the open cube. Apply Lemma IV.1.3.

For $h \geq k \geq 1$, set $t = k/h \leq 1$. Then

$$t^k \sum_{j=0}^k \binom{h}{j} \leq \sum_{j=0}^k \binom{h}{j} t^j \leq (1+t)^h \leq e^k.$$

Hence $\mathcal{R}(h, k) \leq (eh/k)^k$. For $k \geq 3$, $H_k \geq k$ and $H_k \leq k2^{k-1}$. Substitute these inequalities and take binary logarithms. $\square$

Corollary IV.1.5 (A strengthened floor-pattern bound). *For positive integers r_i ,*

$$|\mathcal{F}(\mathbf{r})| \leq \left(\prod_i 2r_i \right) \mathcal{R}(H_k, k).$$

In particular, for $r_i = r$ and $k \geq 3$,

$$\log_2 |\mathcal{F}(r, \dots, r)| \leq k \log_2(2r) + k^2 + (\log_2 e - 1)k. \quad (\text{IV.1.7})$$

Proof. Combine Theorems IV.1.2 and IV.1.4. $\square$

k	H_k	old log bound, $r = 1$	new log bound, $r = 1$
2	1	16	3.000
3	5	54	7.700
4	17	128	15.650
8	769	1024	69.358
12	20481	3456	155.025
16	458753	8192	272.667

Table 5: Binary logarithms of the bounds 2^{2k^3} and $2^k \mathcal{R}(H_k, k)$ for the coefficient range $(-1, 1)^k$. These upper bounds are not asserted to be exact counts.

IV.2 Lower bounds and the optimal entropy scale

Source 01, delivered Section 10.

IV.2.1 An elementary independent-choice construction

Theorem IV.2.1 (Elementary quadratic-exponent lower bound). *For $k \geq 1$,*

$$C_k \geq 2^{\lfloor k^2/4 \rfloor}. \quad (\text{IV.2.1})$$

Proof. The assertion for $k = 1$ is immediate. Write $k = m + n$ with $m, n \geq 1$. Use m anchor coordinates

$$x_j = \frac{2^{j-1}}{2^m} \quad (1 \leq j \leq m).$$

Their subset sums are precisely $t/2^m$, for $0 \leq t \leq 2^m - 1$. For each of the remaining n coordinates independently choose

$$y_i = \frac{h_i + 1/2}{2^m}, \quad 0 \leq h_i \leq 2^m - 1.$$

For an anchor subset with sum $t/2^m$,

$$\left\lfloor y_i + \frac{t}{2^m} \right\rfloor = \begin{cases} 0, & h_i + t < 2^m, \\ 1, & h_i + t \geq 2^m. \end{cases}$$

This threshold, as t ranges over all anchor subset sums, recovers h_i : for $h_i = 0$ there is no value 1, and otherwise the first value 1 occurs at $t = 2^m - h_i$. Therefore different choices of the vector $(h_1, \dots, h_n)$ give different carry patterns. There are 2^{mn} choices. Taking $m = \lfloor k/2 \rfloor$ and $n = \lceil k/2 \rceil$ proves the theorem. $\square$

This construction is independent of any threshold-function enumeration. Together with (IV.1.6), it already determines the order $\log C_k = \Theta(k^2)$ by elementary arguments.

IV.2.2 Embedding Boolean threshold functions

Let T_n denote the number of Boolean linear threshold functions on $\{0, 1\}^n$ admitting a strict representation

$$g(\varepsilon) = \mathbf{1}_{\sum_i w_i \varepsilon_i > t}, \quad \sum_i w_i \varepsilon_i \neq t \text{ for all } \varepsilon.$$

Let M_n count the nonconstant such functions with a representation using nonnegative weights. We need only a simple relation between these two counts and carry patterns.

Lemma IV.2.2 (Thresholds inject into one higher carry dimension). *For every $n \geq 1$,*

$$C_{n+1} \geq M_n \geq 2^{-n} T_n - 2. \quad (\text{IV.2.2})$$

Proof. If a coefficient w_i is negative, replace the Boolean variable ε_i by $1 - \varepsilon_i$ and adjust the constant threshold. This makes that coefficient nonnegative. Thus every threshold function is a coordinate-flipped version of a nonnegative-weight threshold function, with at most 2^n choices of coordinate flips. Including the two constant functions gives $T_n \leq 2^n(M_n + 2)$.

For the first inequality, fix a nonconstant nonnegative-weight threshold function. Strictness and nonconstancy imply $0 < t < \sum_i w_i$. Choose $c > 0$ such that $c \sum_i w_i < 1$, and set

$$x_0 = 1 - ct, \quad x_i = cw_i \quad (1 \leq i \leq n).$$

All coordinates lie in $[0, 1)$, and $x_0 \in (0, 1)$. For $S \subseteq [n]$, the sum $x_0 + \sum_{i \in S} x_i$ lies strictly between 0 and 2, is not 1, and satisfies

$$\left\lfloor x_0 + \sum_{i \in S} x_i \right\rfloor = \mathbf{1}_{\sum_{i \in S} w_i > t}.$$

Therefore the carry pattern restricted to subsets containing coordinate 0 recovers the original Boolean function. Distinct functions have distinct resulting carry patterns, regardless of the chosen strict representations and sufficiently small scaling constants. $\square$

The classical external input is

$$\log_2 T_n = n^2 - n \log_2 n + O(n). \quad (\text{IV.2.3})$$

The leading n^2 asymptotic is due to Zuev; the refinement in (IV.2.3) is attributed to Kahn–Komlós–Szemerédi. Baldi and Vershynin explicitly record both results and also recover the leading asymptotic as the degree-one case of their theorem [8, Section 1.4 and Theorem 1.1]. We use this input, rather than claiming to reprove it; the original references are [9, 10].

Theorem IV.2.3 (Optimal leading carry entropy). *Using the classical input (IV.2.3),*

$$k^2 - k \log_2 k - O(k) \leq \log_2 C_k \leq k^2 + (\log_2 e - 1)k. \quad (\text{IV.2.4})$$

In particular $\log_2 C_k \sim k^2$. Consequently, for arbitrary positive integer ranges $r_1, \dots, r_k$,

$$\log_2 |\mathcal{F}(\mathbf{r})| = \sum_{i=1}^k \log_2(2r_i) + k^2 + o(k^2), \quad (\text{IV.2.5})$$

where the error depends only on k , not on the ranges.

Proof. For $n = k - 1$, Lemma IV.2.2 and (IV.2.3) give

$$\log_2 C_k \geq (k - 1)^2 - (k - 1) \log_2(k - 1) - O(k).$$

The subtraction of two constant functions is negligible at this scale, and the factor 2^{-n} costs only $O(k)$ in the logarithm. The right side is $k^2 - k \log_2 k - O(k)$. Combine this with Theorem IV.1.4. Finally take logarithms in the exact identity (IV.1.2); its range term has no approximation error. $\square$

The leading coefficient 1 in (IV.2.4) is therefore settled by the stated classical input. The present estimates leave a possible $k \log k$ correction unresolved. For a completely elementary version of the article, omit (IV.2.3) and replace the asymptotic claim by the bounds in (F.6.5); every local density theorem is unaffected.

[write] 6 October 2026. This theorem answers source 04’s Question 8 (dated note there, Section V.17): source 04’s F_k is C_k and its A_k is the A_k of Section IV.3. Source 04’s elementary lower bound $F_k \geq 2^{k(k-1)/2}$ (Theorem IV.6.2) is stronger than Theorem IV.2.1 for every $k \geq 3$ (note there).

[write] 6 October 2026; sources 06–10. Source 06’s Theorem IV.7.6 proves the same leading entropy through the same external threshold-function count (Zuev), and extends it: $\log_2 C_{k,M}$ and $\log_2 D_{k,M}$ are $(1 + o(1))k^2$ uniformly in the modulus $M \geq 2$, and so is the number $L_k(r, M)$ of translation classes. It does not state the secondary term $-k \log_2 k - O(k)$ given here. Credited to both.

IV.3 Affine patterns, translation classes, and general features

Source 01, delivered Section 11.

IV.3.1 Fractional affine offsets

For $t = (t_0, t_1, \dots, t_k) \in [0, 1)^{k+1}$ define

$$a_t(S) = \left\lfloor t_0 + \sum_{i \in S} t_i \right\rfloor, \quad A_k = \#\{a_t : t \in [0, 1)^{k+1}\}.$$

Here $a_t(\emptyset) = 0$. Let

$$J_k = \sum_{S \subseteq [k]} |S| = k2^{k-1} \quad (k \geq 1). \quad (\text{IV.3.1})$$

Theorem IV.3.1 (Affine fractional-pattern bound). *For $k \geq 1$,*

$$C_k \leq A_k \leq \mathcal{R}(J_k, k+1). \quad (\text{IV.3.2})$$

In particular, with the classical input of Theorem IV.2.3, $\log_2 A_k \sim k^2$.

Proof. Taking $t_0 = 0$ shows $A_k \geq C_k$. For the upper bound, perturb all $k+1$ coordinates by a common sufficiently small positive number. Each nonempty affine sum increases by $(|S|+1)\varepsilon$ but does not reach the next integer, and t_0 remains less than one. Thus every pattern has an interior representative off all integer threshold hyperplanes. For each subset S the internal thresholds are

$$t_0 + \sum_{i \in S} t_i = j, \quad 1 \leq j \leq |S|.$$

Their number is J_k . Apply Lemma IV.1.3 in dimension $k+1$. For $k \geq 2$, $J_k \geq k+1$, and the binomial-sum estimate gives

$$\log_2 A_k \leq (k+1) \log_2 \frac{ek2^{k-1}}{k+1} = k^2 + O(k).$$

Use the lower bound from C_k for the leading asymptotic. □

IV.3.2 Reduction modulo a modulus and quotienting by translations

For a positive integer M , let $\mathcal{A}_M(\mathbf{r})$ consist of

$$S \mapsto \left\lfloor \alpha_0 + \sum_{i \in S} \alpha_i \right\rfloor \pmod{M}, \quad \alpha_0 \in \mathbb{R}, \quad -r_i < \alpha_i < r_i.$$

The group $\mathbb{Z}/M\mathbb{Z}$ acts on these functions by adding a constant to all values. This action preserves the family, since α_0 is unrestricted.

Theorem IV.3.2 (Modulus-free translation-class budget). *For positive integers r_i and M ,*

$$|\mathcal{A}_M(\mathbf{r})| \leq M \left(\prod_i 2r_i \right) \mathcal{R}(J_k, k+1), \quad (\text{IV.3.3})$$

$$|\mathcal{A}_M(\mathbf{r})/(\mathbb{Z}/M\mathbb{Z})| \leq \left(\prod_i 2r_i \right) \mathcal{R}(J_k, k+1). \quad (\text{IV.3.4})$$

Proof. Write $\alpha_0 = m_0 + t_0$ and $\alpha_i = m_i + t_i$ with fractional parts in $[0, 1)$. Every pattern equals

$$m_0 + \sum_{i \in S} m_i + a_t(S) \pmod{M}.$$

There are M choices of m_0 modulo M , at most $\prod_i 2r_i$ integer coefficient vectors, and A_k fractional affine patterns. This proves (IV.3.3).

Every translation orbit has a unique representative with value zero at the empty set: subtract the original value there. This cancels m_0 while retaining the information in t_0 . The remaining possibilities are counted by at most $(\prod_i 2r_i)A_k$. Apply Theorem IV.3.1. Equivalently, the constant-translation action is free, as can be checked at the empty set, so every orbit has exactly M members. $\square$

For $M = 1$, the actual orbit count is one; the inequality remains valid. One must not model translation classes by simply discarding the fractional part of the affine offset. Canonical normalization at the empty set removes the *integer* constant modulo M , not its fractional part. Modular collisions may make either inequality strict, which is why no exact product formula is asserted in this affine modular setting.

IV.3.3 The combinatorial input for phase removal

At the relevant floor-pattern step of the proof of Proposition 17.7, the cell-index maps have affine subset-sum form, with coefficient range bounded by $k+1$ [1, proof of Proposition 17.7]. Consequently a valid translation-class budget is

$$B_k = (2(k+1))^k \mathcal{R}(k2^{k-1}, k+1). \quad (\text{IV.3.5})$$

Its logarithm is at most $k^2 + k \log_2 k + O(k)$, rather than a cubic polynomial. The following elementary statement makes precise what substituting this budget does, and what it does not do.

Proposition IV.3.3 (A transparent selection interface). *Suppose $z_1, \dots, z_L \in \mathbb{C}$ and $|\sum_{\ell=1}^L z_\ell| \geq U$. Then some ℓ satisfies $|z_\ell|^2 \geq U^2/L^2$. If the terms are indexed by translation classes from Theorem IV.3.2 with $r_i = k+1$, the denominator may be replaced by B_k^2 . More generally, any correctly established estimate $E \geq K/L^2$, with $K \geq 0$ and L the actual number of such classes, implies $E \geq K/B_k^2$.*

Proof. The triangle inequality gives $U \leq \sum_\ell |z_\ell| \leq L \max_\ell |z_\ell|$. Square, and then use $L \leq B_k$ from Theorem IV.3.2. The last assertion follows because $x \mapsto x^{-2}$ is decreasing for positive x . Zero classes cannot carry positive U ; the vacuous zero-signal case is handled separately. $\square$

Thus the combinatorial L^{-2} loss at this interface has a budget $2^{-2k^2-2k \log_2 k - O(k)}$. This is a local substitution conditional on the other estimates in the application being valid. We do not promote it to an improved version of every quantitative statement in Sections 17–18: localization, phase removal, uniformity normalization, and the earlier structural parameters must all be propagated separately.

IV.3.4 Nonnegative integer features and effective rank

The Boolean cube is not essential to the upper-bound mechanism. Let $V \subseteq \mathbb{N}^k \setminus \{0\}$ be a finite set of row vectors and consider

$$x \mapsto (\lfloor v \cdot x \rfloor)_{v \in V}, \quad x \in [0, 1]^k.$$

Put $H(V) = \sum_{v \in V} (\|v\|_1 - 1)$ and $d = \dim_{\mathbb{R}} \text{span } V$.

Proposition IV.3.4 (Rank-sensitive feature bound). *The number of these patterns is at most $\mathcal{R}(H(V), d)$. If V contains every coordinate unit vector, allowing $\alpha_i \in (-r_i, r_i)$ multiplies the number of fractional patterns by exactly $\prod_i 2r_i$.*

Proof. Every nonzero row has positive dot product with the all-ones vector. The common positive perturbation of Lemma IV.1.1 therefore preserves all floors while removing every integral equality and every boundary coordinate. Row v contributes at most $\|v\|_1 - 1$ internal integer hyperplanes; duplicates can only reduce the count. The linear map $x \mapsto (v \cdot x)_{v \in V}$ has rank d . Its image of the open cube is an open convex set relative to its d -dimensional image, and the relevant threshold hyperplanes restrict to affine hyperplanes in that image. Lemma IV.1.3 proves the bound. For the last assertion, $v \cdot m$ is an integer and the unit-vector coordinates recover m_i . Interior representatives prove surjectivity exactly as in Theorem IV.1.2. $\square$

Example IV.3.5 (Why arbitrary signed features need a boundary argument). Near $x = 0$, the map $x \mapsto (\lfloor x \rfloor, \lfloor -x \rfloor)$ has three patterns: $(0, -1)$ for $x > 0$, $(-1, 0)$ for $x < 0$, and $(0, 0)$ at zero. The one hyperplane has only two open regions. Thus a chamber count alone misses a floor pattern when opposite orientations occur. A sufficient replacement for nonnegative rows is a common direction u with $v \cdot u > 0$ for every row, together with room to perturb the parameter domain in that direction. Without such a condition one must count the relevant boundary strata as well.

[write] **6 October 2026; sources 06–10.** Source 06's Theorem IV.7.3 is the same separation of range and carries for any finite $V \subset \mathbb{Z}_{\geq 0}^k$ that contains 0 and the unit vectors, with the modular count $\min(2r, M)^k C(V; M)$ and the region bound $\mathcal{R}_k(h(V))$ in the ambient dimension k ; Proposition IV.3.4 is rank-sensitive (the dimension d of the feature span). Both sources make the same point about signed families (the example above; the paragraph after source 06's theorem).

IV.4 A polynomial generalization on the Boolean cube

Source 01, delivered Section 12.

The feature bound also gives an extension in which the input dimension and coefficient dimension are different. It is useful when a localization step produces low-degree polynomial floor functions rather than affine ones.

Fix $1 \leq d \leq n$ and a positive integer r . Let $\mathcal{P}_{n,d}(r)$ be the family of integer-valued functions on $\{0, 1\}^n$ of the form

$$\varepsilon \mapsto \left| \sum_{\substack{S \subseteq [n] \\ |S| \leq d}} \alpha_S \prod_{i \in S} \varepsilon_i \right|, \quad -r < \alpha_S < r. \quad (\text{IV.4.1})$$

The constant monomial corresponds to $S = \emptyset$. Put

$$m = \sum_{j=0}^d \binom{n}{j}, \quad H_{n,d} = \sum_{j=1}^d \binom{n}{j} 2^{n-j}.$$

Let $T(n, d)$ count strict Boolean polynomial threshold functions of degree at most d ; on a Boolean cube a polynomial can be multilinearized without changing its values or increasing its degree.

Theorem IV.4.1 (Polynomial floor-pattern entropy). *For $1 \leq d \leq n$ and positive integer r ,*

$$T(n, d) \leq |\mathcal{P}_{n,d}(r)| \leq (2r)^m \mathcal{R}(H_{n,d}, m). \quad (\text{IV.4.2})$$

For fixed $d \geq 1$, using the polynomial threshold-function theorem of Baldi–Vershynin, if $r = r(n)$ is a positive integer with $\log r = o(n)$, then

$$\log_2 |\mathcal{P}_{n,d}(r)| = (1 + o(1)) n \sum_{j=0}^d \binom{n}{j} = (1 + o(1)) \frac{n^{d+1}}{d!}. \quad (\text{IV.4.3})$$

Proof. Index coefficient space by the m monomials of degree at most d . For an input ε , the evaluation vector has entries $\prod_{i \in S} \varepsilon_i \in \{0, 1\}$. It is nonzero because of the constant monomial. The sum of its coordinates counts the active monomials at ε . Summing the number of internal thresholds over all inputs gives

$$\sum_{\varepsilon \in \{0,1\}^n} (\#\{S : |S| \leq d, S \subseteq \text{supp } \varepsilon\} - 1) = \sum_{j=1}^d \binom{n}{j} 2^{n-j} = H_{n,d}.$$

Separate each coefficient into an integer part and a fractional part. There are at most $(2r)^m$ integer vectors. For each vector, fractional patterns are bounded by the hyperplane count in dimension m , using positive perturbation and Lemma IV.1.3. This proves the upper bound. It does not assert that different integer vectors always give disjoint pattern families.

For the lower bound, take a strict multilinear polynomial threshold function, represented by a polynomial p with no zero values on the Boolean cube. Multiply p by a sufficiently small positive c so that every coefficient of cp lies in $(-r, r)$ and $0 < |cp(\varepsilon)| < 1$ at every input. Then

$$\lfloor cp(\varepsilon) \rfloor = \begin{cases} -1, & p(\varepsilon) < 0, \\ 0, & p(\varepsilon) > 0. \end{cases}$$

Thus the integer-valued floor function encodes the threshold function. Distinct threshold functions give distinct floors.

For fixed d , the external theorem gives $\log_2 T(n, d) = (1 + o(1))nm$ [8, Theorem 1.1]. For sufficiently large n , $H_{n,d} \geq m$ and $H_{n,d} \leq 2^m$. The binomial estimate from Theorem IV.1.4 therefore yields

$$\log_2 |\mathcal{P}_{n,d}(r)| \leq m \log_2(2r) + m \log_2(eH_{n,d}/m) \leq nm + m \log_2(2r) + m \log_2 e.$$

Since $\log r = o(n)$, this matches the lower bound to leading order. Finally $m = (1 + o(1))n^d/d!$ for fixed d . $\square$

The finite upper bound in (IV.4.2) is elementary; only the matching leading lower asymptotic uses the external threshold theorem. This generalization also shows why using the square of the coefficient dimension as a universal exponent can be wasteful: for $d > 1$, the sharp leading entropy is on the scale n^{d+1} rather than $m^2 \asymp n^{2d}$.

IV.5 Exact finite enumeration and reproducible checks

Source 01, delivered Section 13.

IV.5.1 Small carry counts

Proposition IV.5.1 (Certified small dimensions). *The first four carry counts are*

$$C_1 = 1, \quad C_2 = 2, \quad C_3 = 10, \quad C_4 = 154. \quad (\text{IV.5.1})$$

The first three values have the elementary proof below. The fourth is proved by an exhaustive finite rational-certificate enumeration supplied with this article; its certificate format and exact verification argument are given in the following subsection.

Proof of the first three values. For $k = 1$ every floor is zero. For $k = 2$, only $\lfloor x_1 + x_2 \rfloor \in \{0, 1\}$ varies, and both values occur.

For $k = 3$, Lemma IV.1.1 allows us to avoid integral sums. If $x_1 + x_2 + x_3 < 1$, every pair floor is zero. If the total is greater than 2, every pair sum is greater than 1, since the remaining coordinate is less than 1. These give two patterns. When the total lies in $(1, 2)$, all eight assignments of pair floors occur. Up to coordinate permutations, representatives with respectively zero, one, two, and three pair floors equal to one are

$$(0.4, 0.4, 0.4), \quad (0.7, 0.5, 0.1), \quad (0.8, 0.4, 0.4), \quad (0.6, 0.6, 0.6).$$

There are $1 + 3 + 3 + 1 = 8$ such pair patterns, all with total floor one. The two outer patterns are realized by $(0.1, 0.1, 0.1)$ and $(0.9, 0.9, 0.9)$. Hence $C_3 = 10$. $\square$

For example, Theorem IV.1.2 now gives exactly $C_4(2r)^4 = 154(2r)^4$ patterns with four coefficients in $(-r, r)$. At $r = 1$ the exact count is 2464, while the simple new arrangement upper bound is 51424. No assertion is made that these finite values or their sequence have not appeared elsewhere.

[write] 6 October 2026. $C_1, C_2, C_3 = 1, 2, 10$ are certified independently in source 04 (as F_1, F_2, F_3 , Section IV.6), together with $A_1, A_2, A_3 = 2, 6, 38$. $C_4 = 154$ is certified only here. At intake the sequence 1, 2, 10, 154 matched only OEIS A134088 (1, 1, 2, 10, 154, 7994, ..., a power-series coefficient sequence); its next term 7994 differs from the uncertified recount $C_5 = 8410$ (note in Section IV.6), so the match looks like a coincidence. 2, 6, 38 gave no relevant hit. This is the intake's search of 6 October 2026, not repeated at the write; nothing was submitted to the OEIS.

[write] 6 October 2026; sources 06–10. *Stale since the second write: $C_4 = 154$ is also certified by source 06.* The sentence “ $C_4 = 154$ is certified only here” was true of sources 01–05. Source 06 certifies $C_0, \dots, C_4 = 1, 1, 2, 10, 154$ and $D_0, \dots, D_3 = 1, 2, 6, 38$ with its own exact format (215 rational chamber witnesses and 1,165 nonnegative dual certificates, checked by a standard-library verifier; Section IV.9), so $C_4 = 154$ is now certified twice, independently. Source 06's D_k (affine carries $\lfloor t_0 + v \cdot t \rfloor$) is the A_k here and in source 04 for $k \leq 3$ by the values; source 06 does not compute $D_4 = A_4$ or C_5 , which remain uncertified (Section V.2, item IV.2).

IV.5.2 A solver-independent rational certificate

Order the H_k hyperplanes by increasing binary mask of S , and then by increasing integer threshold. Build a binary tree of strict signs. For each partial sign word, use variables $z = (x_1, \dots, x_k, \epsilon)$ and inequalities

$$\begin{array}{ll} -x_i + \epsilon \leq 0, & x_i + \epsilon \leq 1, \\ v \cdot x + \epsilon \leq j & \text{for the sign } v \cdot x < j, \\ -v \cdot x + \epsilon \leq -j & \text{for the sign } v \cdot x > j. \end{array}$$

Write this finite system as $Az \leq b$. A sign word is strictly feasible exactly when this system has a solution with $\epsilon > 0$.

There are two certificate types. A *primal certificate* is a rational vector z satisfying $Az \leq b$ and $z_{k+1} > 0$. A *dual certificate* is a rational vector $\lambda \geq 0$ satisfying

$$\lambda^T A = (0, \dots, 0, 1), \quad \lambda^T b \leq 0. \quad (\text{IV.5.2})$$

The latter proves that every solution of $Az \leq b$ has $\epsilon = \lambda^T Az \leq \lambda^T b \leq 0$, so the sign word is strictly infeasible. This implication needs no floating-point tolerance and no appeal to an optimizer’s status message.

The generator uses a numerical linear program only to propose rational primal or dual vectors. Every proposal is checked with exact rational arithmetic before being accepted. It branches both ways at every feasible partial word and never branches below an infeasible word. The verifier reconstructs all inequalities from the sign words, checks every rational identity, checks that both children of every feasible nonterminal node are present, and rejects extra or unreachable nodes. It then counts feasible leaves. By induction on tree depth, this is an exhaustive count of all regions in the open cube. Theorem IV.1.4’s region-pattern correspondence turns that count into C_k .

Dimension k	Hyperplanes H_k	Certificate records	Feasible leaves C_k
1	0	1	1
2	1	3	2
3	5	49	10
4	17	2061	154

Table 6: Exact certificate results in `certificates/carry.json`. The count of records includes both primal and dual certificates.

IV.5.3 Commands and the limits of computational evidence

From the package directory, verification uses only the Python standard library:

```
python code/carry_certificates.py verify certificates/carry.json
```

Regeneration additionally needs NumPy and SciPy:

```
python code/carry_certificates.py generate --max-k 4 \
--out certificates/carry.json
```

The supplementary checks use NumPy and SymPy:

```
python code/check_refinements.py
```

They verify six symbolic identities, 210 broken-chord parameter cases, 210 equality constructions, 175 maximum-density root cases, 2000 sampled weighted-cell inequalities, and 24 concrete progression-partition cases. The parameter grids and random seed are fixed in the script. These are useful error detectors, not substitutes for the general proofs. In contrast, the finite carry enumeration has exact certificates for every branch and a mathematical completeness argument, though its implementation is still ordinary Python rather than kernel-checked proof code.

The verifier is independent of the numerical optimizer, not an independent second implementation of every aspect of the encoding. Its small trusted computational core includes Python integer and rational arithmetic, the constraint generator, and the tree checks. This makes the remaining trust boundary explicit.

IV.6 Floor patterns and efficient extraction of a progression profile

Source 04, delivered Section 8.

Corollaries 17.5 and 17.6 of Gowers’s paper bound the number of functions obtained by taking integer parts of linear expressions on a Boolean cube [1]. These estimates control the number of ways that a cube can meet a partition into arithmetic progressions. We isolate the fractional parts of the coefficients, determine the exact dependence on their integer ranges, and prove matching quadratic orders of magnitude for the logarithm of the remaining complexity. A second observation reduces the loss incurred when selecting one progression profile.

The edited repository already uses a joint hyperplane count in its proof of Proposition 17.7 [23]. The refinements below go further by separating integer and fractional parameters and distinguishing the number of globally possible profiles from the number possible at a fixed slope. The arguments establish the stated bounds; they do not establish historical priority for these elementary refinements.

IV.6.1 Fractional patterns and their boundary values

Write $\Omega_k = \{0, 1\}^k$ and put

$$F_k = \# \left\{ (\lfloor \epsilon \cdot t \rfloor)_{\epsilon \in \Omega_k} : t \in [0, 1)^k \right\},$$

$$A_k = \# \left\{ (\lfloor t_0 + \epsilon \cdot t \rfloor)_{\epsilon \in \Omega_k} : (t_0, t) \in [0, 1)^{k+1} \right\}.$$

Set $F_0 = A_0 = 1$, and define

$$R_d(H) = \sum_{j=0}^d \binom{H}{j}, \quad H_k = (k-2)2^{k-1} + 1, \quad Q_k = k2^{k-1},$$

where $\binom{H}{j} = 0$ for $j > H$.

[write] **6 October 2026.** *Source defect corrected in the write.* In both displays above the delivered source writes the subscript `_epsilon\in\Omega_k`, without the backslash before `epsilon`, so the delivered PDF prints the word “epsilon”; the intended $\epsilon \in \Omega_k$ is restored here. Notation: F_k is source 01’s C_k , A_k is source 01’s A_k , Q_k is source 01’s J_k , and $R_d(H)$ is source 01’s $\mathcal{R}(H, d)$, with the arguments in the other order.

Lemma IV.6.1 (Boundary removal). *Every pattern counted by F_k or A_k has a representative in the open parameter cube which lies on none of its nonconstant integer-threshold hyperplanes.*

Proof. Add a sufficiently small positive multiple of the all-ones vector to the parameters. Every nonconstant form under consideration increases strictly. An integer value moves into the interior of the same floor interval; a noninteger value remains in its floor interval if the step is sufficiently small. There are finitely many forms and every coordinate is initially less than 1, so one positive step works for all forms and coordinates. The identically zero form in the definition of F_k requires no adjustment. $\square$

Theorem IV.6.2 (Quadratic order of fractional complexity). *For every $k \geq 1$,*

$$2^{k(k-1)/2} \leq F_k \leq R_k(H_k), \quad \prod_{j=0}^{k-1} (2^j + 1) \leq A_k \leq R_{k+1}(Q_k). \quad (\text{IV.6.1})$$

Consequently $\log_2 F_k = \Theta(k^2)$ and $\log_2 A_k = \Theta(k^2)$. In particular,

$$F_k \leq e^k 2^{k(k-1)} \quad (k \geq 3), \quad A_k \leq e^{k+1} 2^{k^2-1} \quad (k \geq 2). \quad (\text{IV.6.2})$$

Proof. For a nonempty subset $S \subseteq \{1, \dots, k\}$ of size s , the form $\sum_{i \in S} t_i$ crosses precisely the possible thresholds $1, \dots, s-1$ inside the open cube. The resulting arrangement has

$$\sum_{s=1}^k \binom{k}{s} (s-1) = k2^{k-1} - (2^k - 1) = H_k$$

hyperplanes. A pattern is constant in each chamber. Its strict feasible region is an intersection of open halfspaces and is therefore convex, so a single pattern cannot occupy two chambers. Lemma IV.6.1 also accounts for all patterns originally attained on boundaries. Thus F_k is the number of chambers meeting the open cube. The standard arrangement bound, also stated as Lemma 17.4 in [1], gives $F_k \leq R_k(H_k)$.

With an intercept, the form $t_0 + \sum_{i \in S} t_i$ has thresholds $1, \dots, s$, and there are $Q_k = \sum_s s \binom{k}{s}$ relevant hyperplanes in dimension $k+1$. This gives the bound for A_k . For $H \geq d \geq 1$, $R_d(H) \leq (eH/d)^d$: apply $(1+u)^H \leq e^{Hu}$ with $u = d/H$, observing that $u^j \geq u^d$ for $j \leq d$. Applying this with $H_k/k \leq 2^{k-1}$ and $Q_k/(k+1) \leq 2^{k-1}$ proves (IV.6.2).

For the lower bound, fix one pattern counted by F_{k-1} . Its region contains a nonempty open set. Choose a representative in that set whose 2^{k-1} subset sums have pairwise distinct fractional parts. Such a choice is possible because the prohibited equalities are finitely many proper hyperplanes

$$\sum_i (\epsilon_i - \epsilon'_i) t_i = j, \quad \epsilon \neq \epsilon', \quad j \in \mathbb{Z}.$$

Only finitely many j can occur in the bounded parameter cube. In particular, no nonempty subset sum has fractional part zero. As t_k varies through $(0, 1)$, each nonempty old subset sum produces one jump, at $1 - \{\sum_{i \in S} t_i\}$. The $2^{k-1} - 1$ jump locations are distinct, and hence there are 2^{k-1} different extension patterns. Extensions of different old patterns remain different on the face $\epsilon_k = 0$. Therefore $F_k \geq 2^{k-1} F_{k-1}$, which proves the first lower bound.

For A_k , choose an old representative for which the 2^{k-1} affine sums $t_0 + \sum_{i \in S} t_i$ have distinct nonzero fractional parts. Varying t_k now gives 2^{k-1} distinct jumps and $2^{k-1} + 1$ extensions. Thus $A_k \geq (2^{k-1} + 1) A_{k-1}$, proving the second lower bound. $\square$

[write] 6 October 2026. *Credits and a comparison.* The upper bound $F_k \leq R_k(H_k)$ is source 01's Theorem IV.1.4, and $F_k \leq e^k 2^{k(k-1)}$ is the same as 01's $\log_2 C_k \leq k^2 + (\log_2 e - 1)k$; the affine upper bound is 01's Theorem IV.3.1. The lower bound $2^{k(k-1)/2}$ is stronger than source 01's elementary $2^{\lfloor k^2/4 \rfloor}$ (Theorem IV.2.1) for every $k \geq 3$, and equal to it for $k = 1, 2$. *Proof:* $k(k-1)/2 - k^2/4 = k(k-2)/4$, which is positive for $k \geq 3$, and $\lfloor k^2/4 \rfloor \leq k^2/4$; for $k = 1, 2$ both exponents are 0 and 1. The product lower bound for A_k is source 04's alone. Using the classical threshold-function count, source 01 proves $\log_2 F_k \sim k^2$ and $\log_2 A_k \sim k^2$ (Theorems IV.2.3 and IV.3.1). This answers Question 8 below (dated note there).

IV.6.2 Exact dependence on integer ranges

Theorem IV.6.3 (Integer-part scaling). *Let $a_i < b_i$ be integers and $m_i = b_i - a_i$. The number of functions*

$$\epsilon \longmapsto \left\lfloor \sum_{i=1}^k \epsilon_i \alpha_i \right\rfloor, \quad \alpha_i \in (a_i, b_i),$$

is exactly $F_k \prod_i m_i$. In particular, the count in Gowers's Corollary 17.5 is exactly $(2r)^k F_k$ for a positive integer r .

Let $F_{k,M}$ count the fractional patterns in the definition of F_k after reduction modulo a positive integer M . The corresponding number of functions with values in $\mathbb{Z}/M\mathbb{Z}$ is exactly

$$F_{k,M} \prod_{i=1}^k \min(m_i, M). \quad (\text{IV.6.3})$$

Moreover, $F_{k,M} = F_k$ whenever $M \geq k$.

Proof. Write $\alpha_i = n_i + t_i$, where $n_i = \lfloor \alpha_i \rfloor$ and $0 \leq t_i < 1$. Then

$$\lfloor \epsilon \cdot \alpha \rfloor = \epsilon \cdot n + \lfloor \epsilon \cdot t \rfloor.$$

The singleton values recover n_i exactly, since $\lfloor t_i \rfloor = 0$. Subtracting their linear contribution recovers the fractional pattern. Conversely, every fractional pattern has an interior representative by Lemma IV.6.1; this representative works in every integer cell, including those with an excluded left endpoint. There are precisely $\prod_i m_i$ integer choices.

Modulo M , the same singleton argument recovers the residues of the n_i , and each interval of m_i consecutive integers has $\min(m_i, M)$ residues. Subtracting these residues recovers the fractional residue pattern, proving (IV.6.3). Finally, every fractional floor belongs to $\{0, \dots, k-1\}$; reduction modulo $M \geq k$ is injective on this range. $\square$

For arbitrary real endpoints, the same argument gives the upper bound $F_k \prod_i (\lceil b_i \rceil - \lfloor a_i \rfloor)$; equality need not hold because the extreme integer cells may be truncated.

[write] **6 October 2026.** With $a_i = -r_i$, $b_i = r_i$ the first assertion is source 01's Theorem IV.1.2, by the same decomposition and interior-representative argument (source 01's Lemma IV.1.1 is Lemma IV.6.1); the arbitrary integer intervals and the modular count (IV.6.3) are source 04's.

Corollary IV.6.4 (Affine modular bound). *With the integer endpoints of Theorem IV.6.3, let α_0 vary over all real numbers. The number of functions*

$$\epsilon \longmapsto \lfloor \alpha_0 + \epsilon \cdot \alpha \rfloor \pmod{M}$$

is at most $MA_k \prod_i \min(m_i, M)$. The number of equivalence classes under addition of a common constant is at most $A_k \prod_i \min(m_i, M)$. Both assertions remain valid if every slope interval (a_i, b_i) is replaced by $[a_i, b_i]$.

Proof. Writing $\alpha_0 = n_0 + t_0$ gives

$$\lfloor \alpha_0 + \epsilon \cdot \alpha \rfloor = n_0 + \epsilon \cdot n + \lfloor t_0 + \epsilon \cdot t \rfloor.$$

Choose $n_0 \pmod{M}$, the allowed slope residues, and one of the A_k fractional patterns. This proves the upper bound. Translation by a constant acts freely, as seen at $\epsilon = 0$, and all M translates occur by changing n_0 .

To handle closed slope intervals, move each endpoint slope inward with velocity $d_i = 1$ at a_i and $d_i = -1$ at b_i , leaving other slopes fixed. Move t_0 upward with velocity $k + 1$. Every affine form has strictly positive velocity $k + 1 + \epsilon \cdot d \geq 1$. A sufficiently small movement preserves every floor and keeps $t_0 < 1$, while moving all slope endpoints into the open intervals. $\square$

In the symmetric case, Corollary IV.6.4 replaces the bound $M2^{2r(k+1)^3}$ in Gowers's Corollary 17.6 by

$$M \min(2r, M)^k R_{k+1}(k2^{k-1}).$$

The dependence on r is polynomial, and the exponent governing the fractional complexity is quadratic in k . Theorem IV.6.2 shows that this quadratic order cannot be reduced to $o(k^2)$.

[write] 6 October 2026. Source 01's Theorem IV.3.2 gives the two bounds of Corollary IV.6.4 with $\prod_i 2r_i$ and $\mathcal{R}(k2^{k-1}, k + 1)$ in place of $\prod_i \min(m_i, M)$ and A_k ; source 04's are never larger. *Notation clash* B_k . Source 01's translation budget $B_k = (2(k + 1))^k \mathcal{R}(k2^{k-1}, k + 1)$ (IV.3.5) and source 04's $B_k = (2k)^k R_{k+1}(k2^{k-1})$ (Theorem IV.6.7) differ. Source 01 uses Gowers's coefficient range $k + 1$ with open intervals; source 04 uses the closed slope interval $[-k, k]$ of its phase-extraction proof, through the closed-interval clause of this corollary. Both are valid under their hypotheses. Source 04's is smaller by the factor $((k + 1)/k)^k < e$.

IV.6.3 Small cases with exact certificates

The first values are

k	1	2	3
F_k	1	2	10
A_k	2	6	38

For example, in the definition of F_3 the three pair-sum threshold indicators attain all eight Boolean patterns. If all three are zero, the total floor can be 0 or 1; if all are one, it can be 1 or 2. For each of the other six patterns, the total floor is 1. This gives $F_3 = 2 + 2 + 6 = 10$. Similarly, the two singleton values for A_2 give two possibilities for the pair value when they agree and one when they disagree, giving $A_2 = 6$.

The accompanying files `pattern_census.py` and `pattern_census.json` certify all six counts. The arrangement is constructed one hyperplane at a time, and both possible children of every surviving chamber are checked. For a system of strict inequalities $Cx < b$, a feasible branch has a rational point satisfying all inequalities strictly. An infeasible branch has rational weights $y \geq 0$ satisfying

$$\sum_i y_i = 1, \quad C^\top y = 0, \quad b^\top y \leq 0.$$

Multiplying the strict inequalities by these weights would yield $0 < b^\top y \leq 0$, a contradiction. All 450 branch certificates are checked with exact fractions. Numerical linear programming proposes the certificates during generation; the verifier itself uses only Python's standard library and runs with

```
python -S pattern_census.py --verify pattern_census.json
```


[write] **6 October 2026.** Source 01 certifies the same F_1, F_2, F_3 (its C_k) and also $C_4 = F_4 = 154$ (Proposition IV.5.1), with a different certificate format. The census is shipped as `code/04-restriction-cubes-pattern_census.py` with `data/04-restriction-cubes-pattern_census.json`; the command above runs on a copy with the delivered names (README). At intake an independent recount reproduced $C_1, \dots, C_4$ and A_1, A_2, A_3 , and found $C_5 = 8410$ and $A_4 = 770$. It used value-branching floating-point linear programs and re-evaluated every feasibility witness exactly, but the infeasibility side rests on floating-point solver status, so these two values are *not certified*; they are recorded in Section V.2, item IV.2. They are consistent with the bounds: $2^{10} = 1024 \leq C_5 \leq R_5(H_5) = R_5(49) = 2\,138\,410$ and $2 \cdot 3 \cdot 5 \cdot 9 = 270 \leq A_4 \leq R_5(Q_4) = R_5(32) = 242\,825$.

[write] **6 October 2026; sources 06–10.** Source 06 (Section IV.9) certifies $C_4 = 154$ independently of source 01, and $D_1, D_2, D_3 = 2, 6, 38$ (its affine carry counts, the A_k here); it does not compute C_5 or $D_4 = A_4$, so the two intake values stay uncertified.

IV.6.4 Few profiles at a fixed slope

Lemma IV.6.5 (Intercept profiles). *For fixed $c \in \mathbb{R}^k$, the functions*

$$\epsilon \mapsto \lfloor \beta + \epsilon \cdot c \rfloor, \quad 0 \leq \beta < 1,$$

take at most 2^k different values. The same bound holds after reduction modulo M . Over the integers the count is exactly one plus the number of distinct nonzero fractional parts among the nonempty subset sums $\epsilon \cdot c$.

Proof. Each noninteger subset sum gives one jump in $(0, 1)$, at $1 - \{\epsilon \cdot c\}$. Integer subset sums give no interior jump. There are at most $2^k - 1$ distinct jump locations. The floor function is right-continuous, so a jump does not create an extra isolated pattern. Reduction modulo M can only identify patterns. Generic slopes make all subset sums distinct modulo 1, showing that the integer bound is sharp. $\square$

Lemma IV.6.6 (Extraction from locally few classes). *Let $z_{s,a}$ be complex numbers on finite index sets, with each pair (s, a) assigned one of L labels. Suppose that at most K labels occur for each fixed a . Set*

$$E = \sum_a \left| \sum_s z_{s,a} \right|^2, \quad E_j = \sum_a \left| \sum_{s: j(s,a)=j} z_{s,a} \right|^2.$$

Then $E \leq K \sum_j E_j$, and some label satisfies $E_j \geq E/(KL)$.

Proof. For each a , apply Cauchy–Schwarz to its at most K class sums. Sum over a , then average over the L labels. $\square$

IV.6.5 A quantitative local phase-extraction statement

Here is a precise form of the improvement at the stage of Proposition 17.7. For a function u on $G = \mathbb{F}_p$, write

$$\mathcal{U}_s(u) = \sum_{x, h_1, \dots, h_s \in G} \prod_{\omega \in \Omega_s} \mathcal{C}^{|\omega|} u(x - \omega \cdot h),$$

where $\mathcal{C}$ denotes complex conjugation. This quantity is nonnegative; it is $p^{s+1} \|u\|_{U_s}^{2s}$. We use the usual Gowers–Cauchy–Schwarz inequality for its mixed version [1, Lemma 3.8].

Theorem IV.6.7 (Local extraction with a quadratic exponential loss). *Let $k \geq 1$, let $p > k + 1$ be prime, and let $m = 2h + 1 > 3k$ satisfy $m \leq \sqrt{p}$. Put $I = \{-h, \dots, h\} \subseteq G$. Let $r \in G^k$, let $\tau : G^k \rightarrow G$ be a multi-affine polynomial, and let $f : G \rightarrow \mathbb{C}$. For $\epsilon \in \Omega_k$, put $f_\epsilon(s) = f(s - \epsilon \cdot r)$. Suppose $\alpha > 0$ and*

$$\sum_{a \in I^k} \left| \sum_{s \in G} e_p(-s\tau(a)) \prod_{\epsilon \in \Omega_k} \mathcal{C}^{|\epsilon|} f_\epsilon(s - \epsilon \cdot a) \right|^2 \geq \alpha p^2 m^k, \quad e_p(t) = \exp(2\pi i t/p). \quad (\text{IV.6.4})$$

Set

$$q = \lceil h/k \rceil, \quad M = \lfloor p/q \rfloor, \quad B_k = (2k)^k R_{k+1} (k2^{k-1}).$$

There is a polynomial P of degree at most $k + 1$ and a partition $(R_i)_{i \in \mathbb{Z}/M\mathbb{Z}}$ into modular arithmetic progressions of lengths q or $q + 1$, each at least $m/(3k)$, such that

$$\sum_i \mathcal{U}_{k+1}(1_{R_i} f e_p(-P)) \geq \frac{\alpha p^2 m^k}{2^k B_k M} \geq \frac{\alpha}{9k^2 2^k B_k} m^{k+2} M. \quad (\text{IV.6.5})$$

Proof. Put $w = p/M$. Since $q \leq \sqrt{p}$, division $p = qM + r_0$ gives $0 \leq r_0 < q \leq M$. The intervals

$$Q_i = \{x \in \{0, \dots, p-1\} : iw \leq x < (i+1)w\}, \quad 0 \leq i < M,$$

therefore form a partition with lengths q or $q + 1$. Also $w \geq q \geq h/k \geq m/(3k)$, where the last inequality follows from $h = (m-1)/2$ and $m \geq 3$.

For each (s, a) , the unique progression profile of its cube is

$$j_{s,a}(\epsilon) = \lfloor s/w - \epsilon \cdot a/w \rfloor \pmod{M}.$$

This formula is valid across modular wraparound because $p/w = M$. Normalize by subtracting $j_{s,a}(0)$. The resulting profile has intercept $\{s/w\}$ and slopes $-a_i/w \in [-k, k]$. Corollary IV.6.4 gives at most $L \leq B_k$ normalized profiles globally. Lemma IV.6.5 gives at most 2^k for each fixed a .

Apply Lemma IV.6.6 to the integrand in (IV.6.4). There is a normalized profile $j : \Omega_k \rightarrow \mathbb{Z}/M\mathbb{Z}$ whose class sum has squared energy at least $\alpha p^2 m^k / (2^k L)$. Write this class sum as a sum over its M translates. A further Cauchy–Schwarz inequality gives

$$\sum_i \sum_{a \in I^k} |T_i(a)|^2 \geq \frac{\alpha p^2 m^k}{2^k L M}, \quad (\text{IV.6.6})$$

where

$$T_i(a) = \sum_s e_p(-s\tau(a)) \prod_{\epsilon \in \Omega_k} \mathcal{C}^{|\epsilon|} (1_{Q_{j(\epsilon)+i}} f_\epsilon)(s - \epsilon \cdot a).$$

The identity of this expression with the class sum uses the disjointness of the Q_i : exactly one product of their indicators is nonzero for any given (s, a) .

Extend the sum in (IV.6.6) to all $a \in G^k$, which increases it. We record the phase decomposition explicitly. For suitable polynomials P_ω of degree at most $k + 1$,

$$u\tau(a) = \sum_{\omega \in \Omega_{k+1}} (-1)^{|\omega|} P_\omega(s - \omega \cdot (a, u)). \quad (\text{IV.6.7})$$

Indeed, expand τ into squarefree monomials. For a monomial involving the coordinates indexed by $J \subseteq \{1, \dots, k+1\}$, use

$$\prod_{j \in J} b_j = \frac{1}{|J|!} \sum_{\nu \in \{0,1\}^J} (-1)^{|\nu|} \left(s - \sum_{j \in J} \nu_j b_j \right)^{|J|}.$$

Here J includes the coordinate of u ; all required factorials are invertible because $p > k+1$. Adding these identities proves (IV.6.7).

Expand $|T_i(a)|^2$ and write its two base points as s and $s - u$. Using (IV.6.7) makes the resulting expression the mixed $(k+1)$ -cube form of

$$h_{i,\omega} = 1_{Q_{j(\epsilon)+i}} f_{\epsilon} e_p(-P_{\omega}), \quad \omega = (\epsilon, b), \quad b \in \{0, 1\}.$$

Gowers–Cauchy–Schwarz and the arithmetic–geometric mean inequality give

$$\sum_{a \in G^k} |T_i(a)|^2 \leq \prod_{\omega} \mathcal{U}_{k+1}(h_{i,\omega})^{1/2^{k+1}} \leq 2^{-(k+1)} \sum_{\omega} \mathcal{U}_{k+1}(h_{i,\omega}).$$

For a fixed ω , summing over i permutes all blocks, so that $\sum_i \mathcal{U}_{k+1}(h_{i,\omega})$ is independent of $j(\epsilon)$. Averaging over ω and using (IV.6.6) produces one $\omega = (\epsilon, b)$ with

$$\sum_i \mathcal{U}_{k+1}(1_{Q_i} f_{\epsilon} e_p(-P_{\omega})) \geq \frac{\alpha p^2 m^k}{2^k L M}.$$

Finally put $t = \epsilon \cdot r$, replace $P_{\omega}(z)$ by $P_{\omega}(z + t)$, and translate every block by $-t$. Translation invariance of the cube form gives the first inequality in (IV.6.5). The second follows from $L \leq B_k$ and $p/M = w \geq m/(3k)$. $\square$

Corollary IV.6.8 (Explicit strict nonuniformity coefficient). *Under Theorem IV.6.7, the coefficient*

$$c_k = \frac{1}{18k^2 2^k (2k)^k R_{k+1} (k 2^{k-1})}$$

gives strict failure of the inequality $\sum_i \mathcal{U}_{k+1}(1_{R_i} f_{\epsilon} e_p(-P)) \leq c_k \alpha m^{k+2} M$. For $k \geq 2$,

$$\log_2(1/c_k) \leq k^2 + k \log_2 k + (2 + \log_2 e)k + 2 \log_2 k + \log_2(9e).$$

Proof. The lower bound in (IV.6.5) is twice the proposed threshold and is positive. The logarithmic estimate follows from $R_{k+1}(Q_k) \leq (e Q_k / (k+1))^{k+1} \leq e^{k+1} 2^{k^2-1}$. $\square$

This gives a local coefficient of size $2^{-k^2 - k \log_2 k - O(k)}$ at the stage where Proposition 17.7 uses $2^{-2(k+1)^3}$. The statement above specifies the field characteristic and the multi-affine extension used in the phase argument. It does not recompute the later inverse-theorem induction or the final bound for $r_k(N)$. Those are separate problems, and the local improvement alone does not supply a new global quantitative Szemerédi theorem.

[write] **6 October 2026.** *Compared with the formal statement* (checked at HEAD): `proposition_17_7` (Sections17_18.lean, line 140; exact companion `proposition_17_7_holds`, `Proofs17LocalizedPhaseRemoval.lean`, line 2900) has the coefficient $2^{-2(k+1)^3}$. Its proof already bounds the profiles by a joint region count (`hyperplaneRegionCount`), as said at the start of this section. Source 01’s Proposition IV.3.3

is a weaker selection interface at the same step: its loss is B_k^2 with source 01's B_k , against $2^k B_k$ here. Not formalized; nothing is propagated to Theorem 18.1, which is open in the ledger.

[write] **6 October 2026; sources 06–10.** *A third coefficient (source 06).* Source 06's Theorem IV.8.1 proves the localized phase removal with $c_k = 1/(8K_k^2)$, $K_k = (2(k+1))^k \mathcal{R}_{k+1}(k2^{k-1})$ (source 01's B_k), so $\log_2(1/c_k) = 2k^2 + O(k \log k)$ (Corollary IV.8.2). It keeps the full L^2 loss over pattern classes, as source 01's selection interface does, and is weaker than Corollary IV.6.8. At $k = 8$ (recomputed at the second write): $\log_2(1/c_8) \approx 121.66$ here, ≈ 212.70 in source 06, and $2(k+1)^3 = 1458$ in `proposition_17_7`. Not formalized.

IV.7 Exact floor-pattern counts and asymptotically sharp carry entropy

Source 06, delivered Section 2.

We first prove a slightly more general result than the Boolean-cube statement. Let $V \subset \mathbb{Z}_{\geq 0}^k$ be finite and contain 0 and the coordinate vectors $e_1, \dots, e_k$. Put

$$h(V) = \sum_{v \in V \setminus \{0\}} (\|v\|_1 - 1), \quad \tilde{h}(V) = \sum_{v \in V} \|v\|_1.$$

These are counts with multiplicity of the possible integer-level hyperplanes in an open unit cube; coincident hyperplanes can only improve the resulting upper bounds.

Definition IV.7.1 (Carry families). For $t \in [0, 1)^k$ define $c_t : V \rightarrow \mathbb{Z}$ by $c_t(v) = \lfloor v \cdot t \rfloor$. Let

$$C(V) = \#\{c_t : t \in [0, 1)^k\}, \quad C(V; M) = \#\{c_t \bmod M : t \in [0, 1)^k\}.$$

For $t_0 \in [0, 1)$ also define $d_{t_0, t}(v) = \lfloor t_0 + v \cdot t \rfloor$, and let $D(V)$ and $D(V; M)$ be the corresponding cardinalities. For $V = \{0, 1\}^k$ abbreviate these quantities as $C_k, C_{k, M}, D_k, D_{k, M}$.

The word “carry” describes precisely the integer part remaining after the integer parts of the coefficients have been removed. All these families are finite, since every coordinate takes values in a finite integer interval.

Lemma IV.7.2 (One-sided removal of boundary points). *Every carry pattern c_t is realized by a point of $(0, 1)^k$ for which $v \cdot t$ is nonintegral for all $v \in V \setminus \{0\}$. Every affine carry pattern is likewise realized in $(0, 1)^{k+1}$ away from all its integer-level hyperplanes.*

Proof. Replace t by $t + \eta(1, \dots, 1)$ for a sufficiently small $\eta > 0$. The new point remains below 1 in every coordinate. For every nonzero $v \in V$, the sum increases by $\eta\|v\|_1 > 0$. Choose η small enough that it is less than the distance to the next integer, divided by $\|v\|_1$, for every such v . A sum initially at an integer moves to the side having that same floor; every other sum remains between its original neighboring integers. The same argument for the coefficient vectors $(1, v)$ proves the affine assertion. Finiteness makes a common positive choice of η possible. $\square$

Theorem IV.7.3 (Exact separation of range and carries). *Let k, r, M be positive integers. The number of functions*

$$v \mapsto \lfloor v \cdot \alpha \rfloor, \quad \alpha \in (-r, r)^k,$$

is exactly $(2r)^k C(V)$. The number after reduction modulo M is exactly

$$\min(2r, M)^k C(V; M). \tag{IV.7.1}$$

Moreover,

$$C(V; M) \leq C(V) \leq \mathcal{R}_k(h(V)), \quad D(V; M) \leq D(V) \leq \mathcal{R}_{k+1}(\tilde{h}(V)). \quad (\text{IV.7.2})$$

If $M \geq \max_{v \in V} \|v\|_1$, then $C(V; M) = C(V)$.

Proof. Write $\alpha = q + t$ with $q_i = \lfloor \alpha_i \rfloor$ and $0 \leq t_i < 1$. Then $q_i \in \{-r, \dots, r-1\}$ and

$$\lfloor v \cdot \alpha \rfloor = v \cdot q + c_t(v).$$

At $v = e_i$ the carry is zero, so the resulting pattern recovers q_i ; subtracting $v \cdot q$ then recovers c_t . Conversely every q in the stated integer box and every carry pattern are simultaneously realizable: use Lemma IV.7.2 to choose all $0 < t_i < 1$. In particular $q_i = -r$ causes no endpoint difficulty. This proves exact factorization.

Modulo M , the coordinate evaluations recover the residues $q_i \bmod M$. There are exactly $\min(2r, M)$ available residues in each coordinate. After these are recovered, the residual carry modulo M is recovered as well. Conversely all pairs are realizable, proving (IV.7.1). The possible nonzero carry values lie between 0 and $\|v\|_1 - 1$; hence reduction is injective on these coordinate ranges under the final hypothesis on M .

Inside the open unit cube, the only relevant hyperplanes for c_t are $v \cdot t = j$ with $1 \leq j \leq \|v\|_1 - 1$. Their number is at most $h(V)$. Each pattern is realized off these hyperplanes by Lemma IV.7.2, and each open region has a constant pattern. In fact a fixed pattern defines an intersection of strict linear inequalities, so it cannot split into disconnected pieces in the cube. The arrangement bound proves the first estimate. For affine carries the thresholds are $t_0 + v \cdot t = j$, $1 \leq j \leq \|v\|_1$, giving $\tilde{h}(V)$ hyperplanes in dimension $k + 1$. $\square$

The nonnegativity assumption on V has a specific purpose. If both v and $-v$ are allowed, a parameter with $v \cdot t = 0$ gives floors $(0, 0)$ that a generic perturbation need not preserve. Thus an extension to arbitrary signed coefficient families requires explicit treatment of lower-dimensional faces; the preceding proof does not silently cover that setting.

[write] **6 October 2026; sources 06–10.** Theorem IV.7.3 for $V = \{0, 1\}^k$ is source 01's Theorem IV.1.2 and source 04's Theorem IV.6.3; the modular count (IV.7.1) is source 04's modular count; the general family V corresponds to source 01's Proposition IV.3.4, which bounds by the rank of the feature span instead of k (note there). Source 01's example before its Section IV.4 is the signed-family obstruction described in the last paragraph. The bound $C_k \leq \mathcal{R}_k(h_k)$ is $C_k \leq \mathcal{R}(H_k, k) = R_k(H_k)$ of sources 01 and 04.

IV.7.1 The Boolean cube and the improvement of Corollary 17.5

For the Boolean cube, counting by support size gives

$$h_k := h(\{0, 1\}^k) = \sum_{s=2}^k \binom{k}{s} (s-1) = (k-2)2^{k-1} + 1, \quad (\text{IV.7.3})$$

$$\tilde{h}_k := \tilde{h}(\{0, 1\}^k) = \sum_{s=1}^k \binom{k}{s} s = k2^{k-1}. \quad (\text{IV.7.4})$$

The formula for h_k also gives $h_1 = 0$. Consequently,

$$\# \left\{ \varepsilon \mapsto \lfloor \varepsilon \cdot \alpha \rfloor : \alpha \in (-r, r)^k \right\} = (2r)^k C_k \leq (2r)^k \mathcal{R}_k(h_k). \quad (\text{IV.7.5})$$

This replaces the source's 2^{2rk^3} upper bound by a polynomial in r with the exact exponent and exact coefficient C_k . For example, $C_1 = 1$, $C_2 = 2$, and $C_3 = 10$. For $k = 3$, if the total fractional sum is below 1 all pair carries vanish, and if it is at least 2 all pair carries are 1. When the total is between 1 and 2, all eight patterns of the three pair carries occur. To realize respectively zero, one, two, or three positive pair carries in this middle slab, use $(.4, .4, .4)$, $(.6, .6, .1)$, $(.8, .3, .3)$, or $(.6, .6, .6)$ and permute coordinates. This proves the count $1 + 8 + 1 = 10$ directly.

The modular formula is exact even when the modulus is small:

$$\#\{\lfloor \varepsilon \cdot \alpha \rfloor \bmod M\} = \min(2r, M)^k C_{k,M}. \quad (\text{IV.7.6})$$

In particular, once $2r \geq M$ no further range loss is necessary. For $M \geq k$ the coefficient is the unreduced C_k .

IV.7.2 Affine patterns and translation classes

For $\alpha_0 \in \mathbb{R}$ and $\alpha \in (-r, r)^k$, consider

$$F_{\alpha_0, \alpha}(\varepsilon) = \lfloor \alpha_0 + \varepsilon \cdot \alpha \rfloor \pmod{M}.$$

Two such maps are equivalent if they differ by a constant in $\mathbb{Z}/M\mathbb{Z}$. Let $L_k(r, M)$ be the number of equivalence classes that occur.

Theorem IV.7.4 (Affine and modular carry bound). *For positive integers k, r, M ,*

$$L_k(r, M) \leq \min(2r, M)^k D_{k,M} \leq \min(2r, M)^k \mathcal{R}_{k+1}(k2^{k-1}). \quad (\text{IV.7.7})$$

The number of distinct maps is exactly $ML_k(r, M)$, and hence is at most $M \min(2r, M)^k \mathcal{R}_{k+1}(k2^{k-1})$.

Proof. Write $\alpha_0 = q_0 + t_0$ and $\alpha_i = q_i + t_i$ with fractional parts in $[0, 1)$. Then

$$F_{\alpha_0, \alpha}(\varepsilon) = q_0 + \varepsilon \cdot q + d_{t_0, t}(\varepsilon) \pmod{M}.$$

The constant q_0 disappears on passing to translation classes; the remaining choices give the upper bound. Unlike the nonaffine case, this decomposition need not be injective, because affine carries at singletons can be 1. Finally the family is invariant under addition of every constant modulo M , and that action is free: evaluation at $\varepsilon = 0$ detects the constant. Each orbit therefore has exactly M maps. $\square$

IV.7.3 Matching leading-order entropy

The next result determines the scale of the carry coefficient rather than merely improving an upper bound. The one external asymptotic input is the theorem of Zuev that the number $T(n)$ of Boolean linear threshold functions on $\{0, 1\}^n$ satisfies

$$\log_2 T(n) = (1 + o(1))n^2 \quad (n \rightarrow \infty) \quad [9, 8]. \quad (\text{IV.7.8})$$

A strict threshold representation can always be used: a separator on a finite cube can be moved slightly so that no value is zero.

Lemma IV.7.5 (Embedding thresholds into carries). *Let $T_+(n)$ denote the number of threshold functions admitting a representation with nonnegative coordinate weights. Then*

$$T(n) \leq 2^n T_+(n), \quad D_{n,M} \geq T_+(n) - 1, \quad C_{n+1,M} \geq D_{n,M} \quad (M \geq 2).$$

The last two inequalities also hold without modular reduction.

Proof. Flip each input coordinate whose coefficient in a chosen threshold representation is negative. The resulting function has nonnegative weights. Recording the subset of flipped coordinates and the resulting function recovers the original function. There are at most 2^n choices of the subset, proving the first inequality.

A nonconstant such function can be written $g(\varepsilon) = \mathbf{1}_{\{w \cdot \varepsilon > \theta\}}$ with all $w_i > 0$ and $0 < \theta < \sum_i w_i$. Zero weights can be perturbed to small positive ones because the finite strict separator has a positive margin. Choose $\lambda > 0$ sufficiently small that

$$t_0 = 1 - \lambda\theta \in (0, 1), \quad t_i = \lambda w_i \in (0, 1), \quad \sum_i t_i < 1.$$

All affine sums lie in $(0, 2)$, and

$$\lfloor t_0 + \varepsilon \cdot t \rfloor = g(\varepsilon).$$

The constant-zero function is also realizable by small positive parameters. Only the constant-one function must be omitted, because an affine carry at the zero vertex is 0. Distinct Boolean outputs remain distinct modulo every $M \geq 2$. This proves $D_{n,M} \geq T_+(n) - 1$. Finally restricting a nonaffine carry in $n + 1$ variables to the face whose first coordinate is 1 produces an arbitrary affine carry in n variables. This restriction is surjective and gives the final bound. $\square$

Theorem IV.7.6 (Asymptotically sharp carry entropy). *As $k \rightarrow \infty$,*

$$\log_2 C_k = (1 + o(1))k^2, \quad \log_2 D_k = (1 + o(1))k^2. \quad (\text{IV.7.9})$$

Both formulas remain valid for $C_{k,M}$ and $D_{k,M}$, uniformly for integer choices $M = M(k) \geq 2$. In particular, for fixed positive integer r the exact nonaffine count is

$$(2r)^k 2^{(1+o(1))k^2}.$$

For fixed $r \geq 1$ and $M \geq 2$, $\log_2 L_k(r, M) = (1 + o(1))k^2$ as well.

Proof. Equations (IV.7.2)–(IV.7.4) and (F.11.2) give $\log_2 C_k \leq k^2 + O(k)$ and $\log_2 D_k \leq k^2 + O(k)$; the finitely many small dimensions where $m < d$ can be treated directly. Lemma IV.7.5 and (IV.7.8) give

$$C_{k,M} \geq 2^{-(k-1)} T(k-1) - 1, \quad D_{k,M} \geq 2^{-k} T(k) - 1.$$

These lower bounds are independent of $M \geq 2$, while the upper bounds are the unreduced ones, proving the uniform assertion. For translation classes, the threshold patterns from the lemma have value 0 at the zero vertex. Two distinct such patterns cannot differ by a constant modulo $M \geq 2$. They are realized with all slopes in $(0, 1) \subset (-r, r)$, so $L_k(r, M) \geq T_+(k) - 1$. Combine this with Theorem IV.7.4. $\square$

Thus the cubic dimension exponent of the coarse source count can be replaced by a quadratic one, but a uniform subquadratic exponent for these pattern families is impossible. This lower bound concerns the number of patterns, not an optimality assertion for every proof of localized phase removal. A proof that avoids enumerating every pattern could conceivably have smaller analytical losses.

[write] **6 October 2026; sources 06–10.** Theorem IV.7.6 is source 01’s Theorem IV.2.3 (same external input, Zuev’s count), extended uniformly in the modulus and to translation classes; source 01 also bounds the secondary term from below by $-k \log_2 k - O(k)$. Together they answer source 04’s Question 8 (note there). Source 04’s elementary lower bound $2^{k(k-1)/2}$ (Theorem IV.6.2) is a third, non-asymptotic lower bound.

IV.8 A quadratic-exponent loss in localized phase removal

Source 06, delivered Section 4.

The carry bound has a direct analytical consequence. We give the argument because a better combinatorial count alone does not justify changing a downstream constant. The following formulation makes the partition scale explicit and supplies a replacement for the local loss in the source’s Proposition 17.7.

Let p be prime and write $e_p(t) = \exp(2\pi it/p)$. For a function on $\mathbb{F}_p$ define its unnormalized cube energy by

$$\mathcal{E}_d(g) = \sum_{s \in \mathbb{F}_p} \sum_{a \in \mathbb{F}_p^d} \prod_{\varepsilon \in \{0,1\}^d} \mathcal{C}^{|\varepsilon|} g(s - \varepsilon \cdot a) = p^{d+1} \|g\|_{U^d}^2.$$

Here $\mathcal{C}$ denotes complex conjugation. This is a nonnegative real number. For a partition $\mathcal{Q} = (Q_i)_{i=0}^{M-1}$ put

$$\mathcal{E}_d(g; \mathcal{Q}) = \sum_i \mathcal{E}_d(\mathbf{1}_{Q_i} g).$$

If every cell has length at most ℓ , the partition-uniformity normalization at degree $d-1$ is $M\ell^{d+1}$. Define the explicit integer

$$K_k = [2(k+1)]^k \mathcal{R}_{k+1}(k2^{k-1}), \quad c_k = \frac{1}{8K_k^2}. \quad (\text{IV.8.1})$$

The letter K_k here denotes a pattern bound; it is unrelated to the large exponents in the polynomial-partition section of the source.

Theorem IV.8.1 (Localized phase removal with the carry bound). *Let $k \geq 1$, let $p > k+1$ be prime, and let m be odd with $1 \leq m \leq \sqrt{p}$. Let*

$$P = \prod_{i=1}^k (r_i + I), \quad I = \{-h, \dots, h\}, \quad 2h+1 = m.$$

All sets are viewed in $\mathbb{F}_p$. Suppose $|f| \leq 1$, $\alpha > 0$, and σ is the restriction to P of a polynomial separately of degree at most one in each of its k variables. Assume

$$\sum_{x \in P} \left| \sum_{s \in \mathbb{F}_p} \prod_{\varepsilon \in \{0,1\}^k} \mathcal{C}^{|\varepsilon|} f(s - \varepsilon \cdot x) e_p(-s\sigma(x)) \right|^2 \geq \alpha p^2 m^k. \quad (\text{IV.8.2})$$

Then there is a polynomial ϕ of degree at most $k + 1$ and a partition $\mathcal{Q}$ into modular intervals, each of length at least $m/(3k)$ and at most some integer ℓ , such that

$$\mathcal{E}_{k+1}(fe_p(-\phi); \mathcal{Q}) > c_k \alpha M \ell^{k+2}. \quad (\text{IV.8.3})$$

The same assertion holds for input progressions with any common nonzero step, with modular progressions in the conclusion.

Proof. Small boxes. If $m \leq 3k$, use the singleton partition and $\phi = 0$. For each x , Cauchy–Schwarz in s bounds the Fourier square in (IV.8.2) by

$$p \sum_s \prod_{\varepsilon \in \{0,1\}^k} |f(s - \varepsilon \cdot x)|^2.$$

Arithmetic–geometric mean, followed by translation invariance of the s -sum, bounds this by $p \sum_s |f(s)|^{2^{k+1}}$. Thus the hypothesis implies $\sum_s |f(s)|^{2^{k+1}} \geq \alpha p$. This is exactly the singleton-partition cube energy, while $M = p$, $\ell = 1$, and $c_k < 1$. The singleton length is at least $m/(3k)$, as required.

The partition and its pattern classes. Suppose $m > 3k$. Put

$$q = \lceil h/k \rceil, \quad M = \lfloor p/q \rfloor, \quad w = p/M, \quad \ell = q + 1,$$

and, using integer representatives, set

$$Q_j = \{s \in \mathbb{F}_p : jw \leq s < (j+1)w\}, \quad 0 \leq j < M.$$

Since $q \leq \sqrt{p}$, one has $q \leq M$. Writing $p = qM + b$ gives $0 \leq b < q \leq M$, so the cell lengths are q or $q + 1$. Also

$$q \geq h/k \geq m/(3k), \quad w \geq q \geq \ell/2, \quad m \geq \ell. \quad (\text{IV.8.4})$$

Set $f_\varepsilon(s) = f(s - \varepsilon \cdot r)$ and $\tau(a) = \sigma(a + r)$, taking the indicated global multiaffine extension. In the expansion obtained by partitioning each f_ε , a nonzero contribution at (s, a) has cell labels

$$j(\varepsilon) = \lfloor s/w - \varepsilon \cdot (a/w) \rfloor \pmod{M}.$$

This identity is well defined for representatives because $p = Mw$. Moreover $|a_i/w| \leq h/q \leq k < k+1$. By Theorem IV.7.4, the number L of possible label functions up to simultaneous translation is at most K_k .

Keeping the pattern loss explicit. Choose one representative j_ν of each translation class and define

$$T_{\nu,i}(a) = \sum_s e_p(-s\tau(a)) \prod_{\varepsilon \in \{0,1\}^k} \mathcal{C}^{|\varepsilon|}(\mathbf{1}_{Q_{j_\nu(\varepsilon)+i}} f_\varepsilon)(s - \varepsilon \cdot a).$$

Indices on cells are modulo M . The original inner sum is $\sum_\nu \sum_i T_{\nu,i}(a)$. Cauchy–Schwarz over the L classes, selection of one class, and then Cauchy–Schwarz over its M translates give a ν such that

$$\sum_{i=0}^{M-1} \sum_{a \in I^k} |T_{\nu,i}(a)|^2 \geq \frac{\alpha p^2 m^k}{L^2 M}. \quad (\text{IV.8.5})$$

Extend the a -sum to $\mathbb{F}_p^k$; every added term is nonnegative.

Removing the multiaffine phase. Expanding each square and putting $y = s - t$ makes its phase $e_p(-y\tau(a))$. Every monomial in $y\tau(a)$ is square-free and has degree at most $k + 1$. For a square-free monomial on a nonempty coordinate set J , finite differences give

$$|J|! \prod_{j \in J} z_j = \sum_{\varepsilon \in \{0,1\}^J} (-1)^{|\varepsilon|} (s - \varepsilon \cdot z)^{|J|}.$$

All the factorials are invertible because $p > k + 1$. Regard this sum as a sum on the full cube by assigning zero to unused vertices, and add the identities for the monomials. We obtain polynomials ϕ_η of degree at most $k + 1$ such that

$$y\tau(a) = \sum_{\eta \in \{0,1\}^{k+1}} (-1)^{|\eta|} \phi_\eta(s - \eta \cdot (a, y)).$$

Consequently each expanded square in (IV.8.5) is a mixed cube form in the functions

$$\mathbf{1}_{Q_{j_\nu(\eta') + i}} f_{\eta'} e_p(-\phi_\eta), \quad \eta' = (\eta_1, \dots, \eta_k).$$

Apply the Gowers–Cauchy–Schwarz inequality to this form and then arithmetic–geometric mean to the resulting 2^{k+1} energy factors. Summing in i and averaging in η shows that for some η , with $g = f_{\eta'} e_p(-\phi_\eta)$,

$$\sum_i \mathcal{E}_{k+1}(\mathbf{1}_{Q_i} g) \geq \frac{\alpha p^2 m^k}{L^2 M}. \quad (\text{IV.8.6})$$

The shift of the cell index by $j_\nu(\eta')$ disappears because it permutes the M cells. The Gowers–Cauchy–Schwarz inequality used here is the standard cube inequality proved in Section 3 of the source; it applies to all complex functions, including these restrictions.

Divide (IV.8.6) by $M\ell^{k+2}$. Since $p/M = w$, (IV.8.4) yields

$$\frac{\mathcal{E}_{k+1}(g; \mathcal{Q})}{M\ell^{k+2}} \geq \frac{\alpha}{L^2} \left(\frac{w}{\ell}\right)^2 \left(\frac{m}{\ell}\right)^k \geq \frac{\alpha}{4K_k^2} > c_k \alpha.$$

Finally translate the partition by $-\eta' \cdot r$ and replace $\phi_\eta(s)$ by $\phi_\eta(s + \eta' \cdot r)$ to express g as a polynomial twist of the original f . A translation does not change cell lengths or cube energies. For a common nonzero input step, first dilate all variables to make that step 1; multiplication by the step is a bijection on $\mathbb{F}_p$, and the Fourier phase and the polynomial change by the same invertible substitution. $\square$

Corollary IV.8.2 (Size of the new local loss). *The constant in (IV.8.3) satisfies*

$$\log_2(c_k^{-1}) = 3 + 2k \log_2(2(k+1)) + 2 \log_2 \mathcal{R}_{k+1}(k2^{k-1}) = 2k^2 + O(k \log k).$$

Thus the explicit local loss has a quadratic leading exponent, compared with the cubic exponent $2(k+1)^3$ in the displayed source formulation.

The last equality follows directly from the binomial-sum upper bound and its largest term; alternatively, upper and lower bounds of order $k^2 + O(k \log k)$ for its logarithm suffice. The sharper raw estimate (IV.8.6) can be retained when the actual number of classes or the observed cell-size ratios are known. No asymptotic approximation is needed to use the explicit constant (IV.8.1).

This proof accounts for the scale of every partition cell and for the factor L^2 . Later structural extraction and stopping thresholds are separate inputs. Improving this local loss does not, on its own, justify substituting a new exponent into the final Szemerédi theorem.

[write] 6 October 2026; sources 06–10. Compared with the other two coefficients at this step. Source 04’s Theorem IV.6.7 with Corollary IV.6.8 has the stronger coefficient $2^{-k^2 - k \log_2 k - O(k)}$ (fixed-slope profiles, loss $2^k B_k$ in place of K_k^2); source 01’s Proposition IV.3.3 is a selection interface with the same L^2 loss as here. At $k = 8$: about $2^{-121.66}$ (04), $2^{-212.70}$ (here; recomputed at the second write), and 2^{-1458} in the formal `proposition_17_7` (exact companion `proposition_17_7_holds`). The numerical lemma “ $L^2 c < 1/4$ ” cited in Section V.26 is in `Proofs17LocalizedPhaseRemoval.lean`. Not formalized.

IV.9 Exact computations and reproducible verification

Source 06, delivered Section 7.

The general theorems do not depend on experimental extrapolation. Computations serve two narrower purposes: certifying small carry counts that are useful examples, and checking the quantitative inequalities on finite configurations that exercise their normalizations and endpoints.

IV.9.1 Certified carry counts

The included certificate package gives the following values. The zero-dimensional entries use the natural empty-cube convention.

k	C_k	$\mathcal{R}_k(h_k)$	D_k	$\mathcal{R}_{k+1}(\tilde{h}_k)$
0	1	1	1	1
1	1	1	2	2
2	2	2	6	15
3	10	26	38	794
4	154	3214	not computed	not tabulated

The entry “not computed” describes this package’s scope; it is not a claim that the value is unknown in the literature. A short initial sequence match is not sufficient to identify these objects with an existing integer sequence.

[write] 6 October 2026; sources 06–10. $C_1, \dots, C_4$ agree with source 01’s certified values (Proposition IV.5.1), so $C_4 = 154$ is certified twice with different formats; D_1, D_2, D_3 agree with source 04’s certified A_1, A_2, A_3 . The intake reran the verifier (all 215 witnesses and 1,165 dual certificates exact). C_5 and D_4 are not certified by any source; the intake’s floating-point values $C_5 = 8410$ and $A_4 = 770$ are item IV.2 of Section V.2.

The enumeration adds the relevant hyperplanes one at a time and branches on their two strict sides. For a sign prefix in dimension d , encode the unit-cube and selected side constraints as

$$A_i \cdot x + t \leq b_i.$$

A rational point x with a rational $t > 0$ proves the existence of a strictly feasible chamber. To certify that a branch has no strict solution, the package supplies rational multipliers $y_i \geq 0$ satisfying

$$\sum_i y_i = 1, \quad \sum_i y_i A_i = 0, \quad \sum_i y_i b_i \leq 0.$$

Multiplying and summing the inequalities would force $t \leq \sum_i y_i b_i \leq 0$, a contradiction. This is an exact certificate, not a numerical tolerance test. The verifier checks the entire binary prefix tree:

every branch either ends in such a certificate or reaches a strictly witnessed full pattern. It also checks that the resulting carry maps are distinct.

The final data contain 215 rational chamber witnesses and 1,165 exact nonnegative dual certificates across all tabulated cases. A numerical linear-programming solver is used only to propose rational certificates. The separate verifier uses Python’s standard library and `fractions.Fraction`; it verifies every relation exactly and rejects execution with Python’s assertion-disabling optimization flag. Lemma IV.7.2 is what makes this open-chamber enumeration complete for the original floor patterns, including their boundary values.

IV.9.2 Uniformity and density checks

The uniformity checker evaluates the finite-group progression counts and cube energies with integers and rational arithmetic. It raises the inequalities to integer powers instead of numerically approximating Gowers-norm roots. It checks all subsets of $\mathbb{Z}/5\mathbb{Z}$ for lengths 3, 4, and 5, all subsets of $\mathbb{Z}/7\mathbb{Z}$ for lengths 3 and 4, and eight specified subsets of $\mathbb{Z}/25\mathbb{Z}$ for lengths 3 and 4. All 368 set-length cases satisfy both the linear and quadratic bounds. It also checks 160 two-term cases and 168 three-term variance bounds, including 32 equality cases. The convenient sufficient existence criterion fires in 62 checked cases; every one contains a nonconstant progression.

The weighted extraction checker exercises 22,752 rational indicator/partition configurations under each of the ranges $[-1, 1]$ and $[-\delta, 1 - \delta]$, for 45,504 exact range cases, and verifies the two extremal constructions. These finite checks are supplementary to the proofs: unlike a carry-count certificate, a list of finite examples does not prove a universal inequality. The accompanying JSON files state the exact tested families and counts.

IV.9.3 Magnitude of the explicit improvements

Figure IV.9.1 evaluates the formulas proved above. In its left panel the new count bound is $2^k \mathcal{R}_k(h_k)$, so it uses the general arrangement estimate rather than small exact counts. The right panel uses the fully explicit c_k from (IV.8.1).

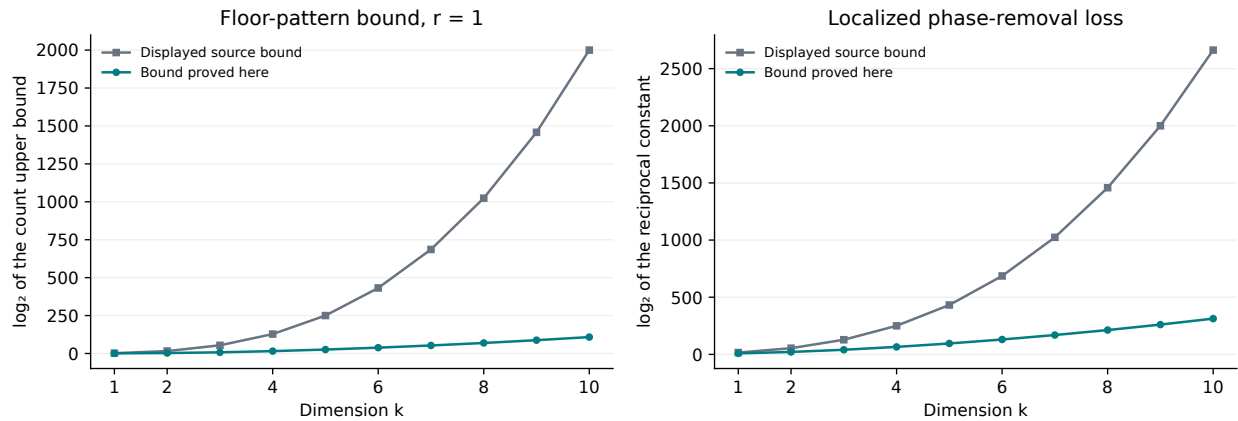


Figure IV.9.1: Comparison of logarithmic bounds. Lower values mean a smaller count upper bound (left) or a larger guaranteed local correlation constant (right). These are comparisons of local statements, not plots of final Szemerédi thresholds.

For example, at $k = 4, r = 1$, the displayed floor-count exponent 128 is replaced by the arrangement-based exponent approximately 15.650, and the exact count is still smaller, namely $2^4 C_4 = 2464$. At $k = 8$, the displayed local phase loss is 2^{-1458} , whereas (IV.8.1) gives a constant approximately $2^{-212.704}$. These comparisons quantify genuine local savings while leaving the later structural dependencies explicit.

Part V

Lean interface, ledger crosswalk and research questions

Contents. Sections V.1 and V.2 are the write’s: the crosswalk of every comparison with the formal project, checked against the current ledger, and the deduplicated list of the sources’ 46 research questions with credits (Vladimir’s standing rule of 4 October 2026). Then each source’s remaining sections, in source order: 01’s formalization interface, questions, conclusions and appendices (delivered Sections 14–16, Appendices A–B); 02’s (Sections 10–12, Appendix A); 03’s iteration bookkeeping, Lean route, questions and reproducibility (Sections 9–11, Appendix A); 04’s quantitative comparisons, questions, closing perspective and appendices (Sections 9–11, Appendices A–B); 05’s Lean relation, computation, questions and conclusion (Sections 7–10). The sources’ question sections are printed as delivered, with dated notes where another source answers a question.

Reading conventions of this Part (the full dictionary is Section F.3).

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
pins, counts	01, 03: pin 3a25020a6; 04, 05: pin 327773149; 02: observed branch reference 66c02f8a1. The ledger counts quoted by the sources (88 exact, 32 open) are those of the pins; at HEAD they are 98 and 22 (Section V.1).
Qk , (Q1), (Q2)	04: its questions are numbered Q1–Q12 and two displays are tagged (Q1), (Q2); 01: numbered Questions (Section V.4); 03: numbered Research questions (Section V.14); 02, 05: subsections and paragraphs. Item names in Section V.2 are the write’s.

Added in the second write (6 October 2026): sources 06–10. Contents. The write’s Sections V.1 and V.2 are extended by dated additions: the crosswalk to the ledger at HEAD of this write, and the 50 research questions of sources 06–10 (plus two inline question paragraphs of source 10) merged into the theme lists. Then the remaining sections of the new sources, in source order: 06’s iteration potential, Lean map and questions (delivered Sections 6, 8, 9); 07’s verification, questions, conclusion and reproduction commands (Sections 9–11, Appendix B); 08’s Lean integration, checks, questions, conclusion and build provenance (Sections 11–14, Appendix B); 09’s checks, Lean integration, questions, conclusion and dependency map (Sections 11, 12, 14, 15, Appendix C); 10’s iteration bookkeeping, Lean map, questions and provenance (Sections 10–12, Appendix A).

Reading conventions for the sections of sources 06–10 in this Part.

Symbol	Meanings in this Part, by source (<i>tempting false reading</i> in italics)
pins, counts (06–10)	06, 09, 10: pin 7d99e5dee; 07: pin ae0e89574; 08: no commit, blob identifiers of 7d99e5dee. The ledger counts quoted by 06 and 10 (95 exact, 25 open) and 06’s “140 modules, 1,365 theorems” are those of 7d99e5dee; at HEAD 764740f07 they are 99 and 21, and FORMALIZATION_STATUS.txt reports 196 modules and 1,805 audited theorems (Section V.1).
iteration	06: F , G the increment and the charge of Lemma V.25.1, c, p, q, a_0 the local rule, $A = \log(1/a_0)$, $L = \log(1/\delta_0)$, $B(\delta_0)$ the double-logarithmic budget. 10: c , a the increment rule, ϑ_i , b_i the length rule, P_t , M_t . 06’s L is a <i>logarithm</i> , not a <i>length</i> ; p is an <i>exponent</i> , not a <i>prime</i> .

V.1 The formal project: ledger crosswalk [write]

Added in the write, 6 October 2026.

Every comparison that a source makes with a corrected statement of the formal project was checked against the ledger and the Lean declarations at HEAD 25df8755d (6 October 2026). File and line numbers below are at that commit. Statements are `Sections*.lean` propositions; “exact” means the ledger lists an exact companion theorem. The ledger then records 98 exact companions and 22 open statements; at all three pins it recorded 88 and 32. The last column gives the formal status of the *improvement*. All of the comparisons are accurate at HEAD, except the stale ones the table names.

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Lemma 2.3	<code>lemma_2_3</code> (Sections01_03:141); <code>lemma_2_3_holds</code> (Proofs02Partition:561) (exact)	constant $1/4 \rightarrow 1/3$, optimal under the Lean hypotheses (05, Thm. I.16.4)	not formalized
Cor. 2.4	<code>corollary_2_4</code> (:158); <code>corollary_2_4_holds</code> (Proofs02PhasePartition:276); also <code>corollary_2_4_before_rou</code> <code>nding_repair_false</code> (exact)	$128\pi \rightarrow 36\pi$, $\alpha/2 \rightarrow 3\alpha/4$, same upper length, $r \leq N$ dropped (05, Cor. I.16.6)	not formalized
Cor. 2.5	<code>corollary_2_5</code> (:180); <code>corollary_2_5_holds</code> (Proofs02 DensityIncrement:517); proof-level pair $\sqrt{\alpha N/128\pi}$, $\alpha/4$ in private <code>cor25_large_scale</code> (:375) (exact)	density-sensitive increments (01, 03, 05); 03 and 05 compare with the proof-level pair, 01 with the statement (note after Table 3)	not formalized
Thm. 3.2, Cor. 3.3	<code>theorem_3_2_holds</code> (Proofs03Pr ogressionAverage:381); <code>corollary_3_3</code> (:276), <code>corollary_3_3_holds</code> (Proofs03ProgressionCount:349) (exact; N prime, $k \leq N$)	$2^k \alpha \rightarrow (k-2)\alpha$ in normalized terms; admissible slopes (03, 04, Part III)	not formalized
Cor. 3.6	<code>corollary_3_6</code> (:310); <code>corollary_3_6_holds</code> (Proofs03 ProgressionExistence:157) (exact)	density exponent $k2^{k-1}$ against $k2^k$ (04, Cor. III.4.4)	not formalized
Lemma 3.8	<code>lemma_3_8_holds</code> (Proofs03GowersCS:513); <code>gowers_cauchy_schwarz</code> (:485); <code>gowersNorm_add_le</code> (Proofs03Minkowski:131) (exact)	used as tools (03, 04)	—
Lemma 3.10	<code>lemma_3_10</code> (:340); <code>lemma_3_10_holds</code> (Proofs03CubeUpper:82) (exact)	excess $\alpha^{1/2^d} \rightarrow \alpha^{4/2^d}$ (03); leading $6\sqrt{2}\delta^4 u^4$ for $d=3$ (04)	not formalized
Cor. 5.6	<code>corollary_5_6</code> (Section05:205); <code>corollary_5_6_holds</code> (Proofs05FullPartition:85) (exact at HEAD, open at every pin)	02 imports it as $PP(K, R_0)$ with the same constants; 02’s “not Lean-verified” is stale (note in Section I.11)	02’s Thm. I.11.1 not formalized

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Cor. 5.8	<code>corollary_5_8</code> (Section05:242) (open); <code>corollary_5_8_with_scale_holds</code> (Proofs05FullPartition:94) proves a scale-qualified repair	02 declines to restate it (Section I.10)	—
Lemma 5.15	<code>lemma_5_15</code> (Section05:352); <code>lemma_5_15_holds</code> (Proofs05_10:735) (exact; $\alpha \geq 0$, empty cells allowed)	size $\alpha N/(4M) \rightarrow$ $\alpha(2+\alpha)N/(2(4-\alpha)(M-1))$ for $\alpha > 0$, nonempty cells (02, 03)	not formalized
Prop. 6.1	<code>proposition_6_1</code> (Sections06_07:37); <code>proposition_6_1_holds</code> (Proofs05_10:505) (exact)	recovered by 02's Thm. II.2.1 with $m(f) \leq 1$, $\mathcal{U} \leq m^4$	not formalized
Prop. 7.3	<code>proposition_7_3</code> (:135); <code>proposition_7_3_holds</code> (Proofs07BalogSzemerédi:996); also <code>proposition_7_3_group_holds</code> (exact)	$(2^{-20}c^{12}, 2^{38}c^{-24}) \rightarrow$ $(c/8, 2^{19}c^{-9})$ (03)	not formalized
Lemma 7.4	<code>lemma_7_4</code> (:167); <code>lemma_7_4_holds</code> (Proofs07DRC:536), five witnesses (exact)	$2^{-1/2}\delta^5 n \rightarrow 7\delta^3 n/(3\sqrt{6})$ (03)	not formalized
Lemma 9.2	<code>lemma_9_2</code> (Sections08_09:69); <code>lemma_9_2_holds</code> (Proofs09Moments:527); exact $N^2 B $ at line 495 (exact)	(M4) exact Parseval is proof-level; (M5) $\gamma^7\beta^{15}p^{15}$ (04)	not formalized
Lemma 9.3, Cor. 9.4	<code>lemma_9_3</code> (:77), <code>corollary_9_4</code> (:91); companions <code>Proofs09Restriction:376, :416</code> (exact; <i>no primality</i>)	prime-field restriction, $2^{-95}\gamma^{112}\eta^{15} B ^{15}$ (04); does not prove the Lean statements, as 04 says	not formalized
Lemma 12.4	<code>lemma_12_4</code> (Sections12_13:158); <code>lemma_12_4_holds</code> (Proofs12Amplification:21) (exact)	$\beta^{112} \rightarrow \beta^{106}$ (04, (A12))	not formalized
Cor. 14.6, Lemmas 14.7, 14.8	<code>corollary_14_6</code> (Sections14_15:266), <code>lemma_14_7</code> (:279), <code>lemma_14_8</code> (:292); companions <code>Proofs14Arrangements:838, Proofs14HigherArrangements:712,</code> <code>Proofs14ProductToArrangement</code> <code>s:18</code> (exact)	profile σ ; $7 \cdot 4^{k+1} \rightarrow 7 \cdot 4^{k+1} - 6$ (04, Thm. II.10.1)	not formalized
Lemma 17.4, Cors. 17.5, 17.6	<code>lemma_17_4_holds</code> (Proofs17Regions:691); <code>corollary_17_5</code> (Sections17_18:103), <code>corollary_17_6</code> (:117); companions <code>Proofs17FinitePatterns:396,</code> <code>:501</code> (exact)	$2^{2rk^3} \rightarrow (2r)^k C_k$; $M2^{2r(k+1)^3} \rightarrow$ $M \min(2r, M)^k R_{k+1}(k2^{k-1})$ (01, 04, Part IV)	not formalized

Gowers	Lean statement; companion (status)	Source claim here	Improvement
Prop. 17.7	<code>proposition_17_7 (:140);</code> <code>proposition_17_7_holds</code> (<code>Proofs17LocalizedPhaseRemoval:2900</code>), joint	$2^{-2(k+1)^3} \rightarrow 2^{-k^2 - k \log_2 k - O(k)}$ (04); weaker interface (01)	not formalized
Cor. 16.11, Thms. 1.3, 18.1, 18.2	<code>hyperplaneRegionCount</code> (exact) all open ; <code>theorem_18_1</code> 's docstring warns about the tower estimate of Cor. 16.11	01's dependency warning (Section V.3)	—
Section 18 (interval transfer)	<code>hasNatAP_of_short_modular_sequence</code> (<code>Proofs18IntervalTransfer:22</code>), <code>hasNatAP_of_hasModAP_image</code> (:92), added in 327773149	04's Lemma III.4.5; 03's Remark III.1.4	formalized (same statement, $n - 1 \mapsto n$)

Normalizations. The Lean development uses the unnormalized Fourier transform and the unnormalized Gowers norm; the intake found the sources' conversion tables accurate (source 02's Section V.8 and Appendix, source 03's Section V.13); the write did not re-derive them. `Definitions.lean` has changed since the pins only by an added `Box.IsProper`, which no source uses; every other file that a source cites is unchanged.

Formal status of this report. As stated in Section F.5: the lifting lemma is formalized and was formalized before the sources arrived; nothing else here is. The reciprocal pointer for `Combinatorics/Ramsey/FORMALIZATION_STATUS.txt` is left to the session that maintains the Lean development.

[write] **6 October 2026; sources 06–10.** *The ledger at the second write.* At HEAD 764740f07 (6 October 2026) the ledger records 99 exact companions and 21 open statements, and `FORMALIZATION_STATUS.txt` reports 196 modules reachable from the audit and 1,805 audited public theorems. Since the first write's HEAD 25df8755d the statements closed are: `lemma_16_9`. Since 7d99e5dee, the pin of sources 06, 09 and 10 (95 exact, 25 open, 140 modules, 1,365 theorems), 4 statements have closed: `lemma_16_1`, `lemma_16_5`, `lemma_16_6` and `lemma_16_9`; since ae0e89574, the pin of source 07, 4: `lemma_16_1`, `lemma_16_5`, `lemma_16_6` and `lemma_16_9`. The table above keeps its file and line numbers at 25df8755d; the rows below give them at 764740f07. The Lean development is being extended by another session as this is written, so the counts may already have moved. Every comparison made by sources 06–10 with a formal statement was checked against these declarations and is accurate, except the stale counts corrected in the notes after source 06's Section F.11 and source 10's Section F.15.

Gowers	Lean statement; companion (status)	Claim of sources 06–10	Improvement
Cor. 2.5	<code>corollary_2_5</code> (<code>Sections01_03:180</code>); <code>corollary_2_5_holds</code> (<code>Proofs02DensityIncrement:517</code>) (exact)	increment $\delta\eta/(4\delta - \eta)$ at source 01's length; the frontier D_γ, L_γ (10, Thm. I.24.2)	not formalized
Thm. 2.6	<code>theorem_2_6 (:189);</code> <code>theorem_2_6_holds</code> (<code>Proofs02Roth:1238</code>) (exact)	step count $321(\delta_0^{-1} - 1)$ against the transcription's $640\delta_0^{-1}$ (10, Prop. V.42.1)	not formalized

Gowers	Lean statement; companion (status)	Claim of sources 06–10	Improvement
Thm. 3.2, Cors. 3.3, 3.6	<code>theorem_3_2_holds</code> (Proofs03ProgressionAverage:381); <code>corollary_3_3</code> (:276), <code>corollary_3_3_holds</code> (Proofs03ProgressionCount:349); <code>corollary_3_6</code> (:310), <code>corollary_3_6_holds</code> (Proofs03ProgressionExistence:157) (exact)	two-target bound and quadratic discrepancy, $\alpha \leq (\delta^{k-1/2}/2B_k)^{2^{k-2}}$ (06); torsion kernels, error $\delta S_k \eta$, $\alpha \leq (\delta^{k-1}/2S_k)^{2^{k-1}}$ (10)	not formalized
Lemmas 3.7, 3.10	<code>lemma_3_7</code> (:320), <code>lemma_3_7_holds</code> (Proofs03Cubes:184); <code>lemma_3_10</code> (:340), <code>lemma_3_10_holds</code> (Proofs03CubeUpper:82) (exact)	excess of fourth order with exact quartic term (08, 09) and quintic term (09); exponent $4/2^d$ sharp for sets; lower bound $(\delta^4 + E_4)^{2^d/4}$ (08)	not formalized
Lemma 5.15	<code>lemma_5_15</code> (Section05:352); <code>lemma_5_15_holds</code> (Proofs05_10:735) (exact)	fixed- M optimum, $\tau(\alpha)$ (06); cap and tail (10)	not formalized
Prop. 7.3	<code>proposition_7_3</code> (Sections06_07:135); <code>proposition_7_3_holds</code> (Proofs07BalogSzemerédi:996) (exact)	$(c/4, 2^{10}c^{-5})$ on any abelian group (10)	not formalized
Lemma 7.5, Cor. 7.6	<code>lemma_7_5</code> (:183), <code>lemma_7_5_holds</code> (Proofs07FreimanRestriction:267); <code>corollary_7_6</code> (:197), <code>corollary_7_6_holds</code> (Proofs07AdditiveRestriction:100) (exact)	$ B /(k(h_k - 1) + 1)$; $2^{-209}\gamma^{103}$ against $2^{-1882}\gamma^{1164}$ (10); both imply the formal conclusions (note after Cor. II.21.2)	not formalized
Lemma 10.9	<code>lemma_10_9</code> (Section10:335); <code>lemma_10_9_holds</code> (Proofs10Shift:717) (exact; FreimanHom 2)	order five under exactly these hypotheses; order nine with the hypotheses of <code>lemma_10_6</code> and the set-up (07)	not formalized; order nine needs a new statement
Cors. 17.5, 17.6	as in the table above (Sections17_18:103, :117; Proofs17FinitePatterns:396, :501) (exact)	$(2r)^k C_k$, $\min(2r, M)^k C_{k,M}$, entropy uniform in M ; $C_4 = 154$ certified (06)	not formalized
Prop. 17.7	<code>proposition_17_7</code> (:140); <code>proposition_17_7_holds</code> (Proofs17LocalizedPhaseRemoval:2900) (exact)	$1/(8K_k^2) = 2^{-2k^2 - O(k \log k)}$ (06)	not formalized

Gowers	Lean statement; companion (status)	Claim of sources 06–10	Improvement
Cor. 16.11, Thms. 18.1, 18.2	<code>corollary_16_11</code> (Section16:893), <code>theorem_18_1</code> (Sections17_18:171), <code>theorem_18_2 (:190): open;</code> <code>section18_cyclic_density_incre- ment_holds_of_theorem_18_1</code> (Proofs18QuantitativeSzemeredy: i:169)	dependency warnings (06, 10); iteration potential (06)	—

[write] **6 October 2026; sources 06–10.** *Formal status after the second write.* No statement of sources 06–10 is formalized, and none gains formal status by its placement. The only formalized statement of the report is still the lifting lemma of source 04. Sources 06–10 ship no Lean files; their formalization plans are in Part V (source 07’s also as `07-transport-rigidity-FORMALIZATION.md`).

V.2 Further questions and research [write]

Added in the write, 6 October 2026.

This section applies Vladimir’s standing rule of 4 October 2026: no unproved or wrong claim is dropped; open claims are stated as questions with their source, the argument they came with and what is missing; wrong claims stay on record with a proof of their failure. The five sources ask 46 questions: source 01 ten (Section V.4, Questions 1–10 there in order), source 02 seven (Section V.9, delivered Subsections 11.1–11.7), source 03 nine (Section V.14, Research questions 1–9 there in order), source 04 twelve (Section V.17, Q1–Q12) and source 05 eight (Section V.23, paragraphs 1–8). They are printed there as delivered. Below they are grouped by theme, with every asking source credited; the counts per theme are 14, 9, 8, 7 and 8. Items are numbered within their theme, so later additions extend a list without renumbering it.

[write] **6 October 2026; sources 06–10.** *Sources 06–10.* They ask 50 further numbered questions: source 06 eleven (Section V.27, Questions 1–11 there), source 07 eight (Section V.29, Questions 1–8 there in order), source 08 ten (Section V.34, Questions 1–10 there in order), source 09 ten (Section V.39, Research questions 1–10, its Subsections 14.1–14.10) and source 10 eleven (Section V.44, items 1–11 there). Source 10 also ends its Sections III.29 and II.21 with a paragraph of questions; both are credited below. The intake dossier counted ten questions for source 10 and 95 in all; the count is 96. They are added to the themes below as new items or, where they ask the same thing, merged into an existing item by a dated sentence. The counts per theme become 17, 19, 33, 13 and 14 (the headings below keep the counts of sources 01–05).

Unproved or wrong claims in sources 06–10. The intake and the second write found no claim of sources 06–10 that is wrong, and no claim asserted as proved without a proof. One wrong statement is on record: the quartic lower bound with coefficient 12, which sources 08 and 09 pose and refute themselves and source 03 also refutes; Remark W1 keeps it with an explicit counterexample. Inputs used but not proved in the sources: Zuev’s threshold-function count (source 06, as for source 01), the Plünnecke–Ruzsa inequality (source 10, via Petridis) and, for source 10’s labelled modern-input variant only, the Reiher–Schoen theorem; these are published theorems. Source 10’s “independent review” is not shipped, and nothing rests on it. Statements of the intake corrected at the write: source 06’s improvement over the $M - 1$ corollary is strict only for $\alpha < 1$ (note after Corollary I.9.3); source 01’s increment dominates source 10’s by $3/2$ only for $\gamma \leq 1/2$ (note after Table 3); source 09’s date for version 2 of Shi–Dong is wrong (bibliography).

Answered or advanced inside the report. Source 08’s question whether $\kappa_3^{\text{odd}} = 1/\sqrt{2}$ is answered yes by source 04 (item III.1). Source 03’s question on the fifth coefficient is answered by source 09 (item III.2).

Source 02’s finite-cell problem is solved in the weighted model by source 06 (item I.6). Source 04’s Question 8 is answered a second time by source 06 (item IV.1). Source 06’s Question 7 is advanced by source 10 for one distinguished function (item III.9). Source 04’s Question 10 (non-admissible slopes) is advanced by source 10’s torsion factors (item III.7).

Unproved or wrong claims. The intake and the write found no claim of sources 01–05 that is wrong, and no claim asserted as proved without a proof. Three inputs are used but not proved in the sources. The classical threshold-function counts (Zuev; Kahn–Komlós–Szemerédi; Baldi–Vershynin) enter source 01’s asymptotics and are published theorems. The partition input $PP(K, R_0)$ of source 02 is now a formal theorem for polynomial phases on intervals (Section V.1). The “independent mathematical reviews” of source 03 are not shipped, and nothing here rests on them. Two values found at intake are not certified, $C_5 = 8410$ and $A_4 = 770$; they are item IV.2.

Answered inside the report (dated notes at the questions). Source 04’s Question 8, the leading constant of $\log_2 F_k/k^2$ and $\log_2 A_k/k^2$, is answered by source 01: it is 1, using the classical threshold count, and source 01 certifies $F_4 = C_4 = 154$. Source 03’s Research question on c_d^{odd} is answered for $d = 3$ by source 04’s Theorem III.3.4: $c_3^{\text{odd}} = 2^{-1/2}$, attained by real parts of characters. With it, source 03’s Remark III.2.9 is settled at $d = 3$. Source 04’s Question 3 is the same question as source 03’s for $d \geq 4$, and is merged with it (item III.1).

Theme I: density transfer and phase-flat partitions (14 questions)

- I.1** *Higher-degree phase partitions* (01 Question 8; 05 paragraph 6). The density transfer is proved for any short-arc phase (source 01’s Theorem I.2.1; source 05’s Proposition I.17.2 and Corollary I.19.4). Missing: polynomial or nilsequence partitions with a quantitative minimum cell length and small occupied exceptional mass (01), and density-sensitive oscillation budgets tracked through geometric complexity with all losses parameterized (05). Simultaneous linear approximation does not give them, and Theorem I.18.5 does not transfer to nonlinear phases.
- I.2** *The best transfer under arithmetic constraints* (01 Question 5; 05 paragraph 4). The extremizers of Theorem I.2.1 rotate cell arcs freely, and the disk radius and the capacity inequality are separately sharp (05). Asked: how much stronger the tail becomes for one fixed cyclic character on a common-step partition with nearly equal cells (01), and the exact feasible region of block positive mass, correlation, density and phase width (05).
- I.3** *Finite partition profiles and the size–increment trade-off* (05 paragraph 1; 01 Question 6). The universal constant $1/3$ is settled (Theorem I.16.4). Asked: the exact best minimum length for each (r, s, M) and slope, and the feasible pairs of minimum and maximum lengths (05); the best uniform pair of length and increment, combining (I.3.4) with an optimal finite partition, including the final incomplete blocks (01). [6 October 2026; sources 06–10] Source 10 (its Question 3) asks the same for its phase partition: the best universal minimum block length for a prescribed phase-disc radius with all integer rounding retained, first at fixed η/δ as $\delta \rightarrow 0$; its density cap is sharp, its passage from one approximate period to blocks of lengths $L, \dots, 2L - 1$ may not be.
- I.4** *Retaining more than one density statistic* (01 Question 7). Theorem I.4.2 solves the optimization once stop-loss statistics are chosen. Missing: structural lemmas that propagate those statistics through repeated localization at acceptable cost.
- I.5** *Heterogeneous phase budgets* (02 Subsection 11.5; 03 Research question 5). The analytic constraint is exact (source 02’s heterogeneous inequality in Section I.8; source 03’s Lemma I.14.3).

Missing: a refinement theorem whose cost depends on the oscillation allowed in each cell, so that budgets can be allocated by local mass. Any better $\text{PP}(K, R_0)$ transfers automatically to Theorem I.11.1.

- I.6** *The finite-cell extremal problem* (02 Subsection 11.6). Theorem I.9.1 solves the weighted total-mass problem. Open: the largest good cell when M and the allowed cell sizes are fixed, with integer cardinalities, which would sharpen the $M - 1$ corollary. [6 October 2026; sources 06–10] Advanced by source 06: with M fixed and the cell masses free, its Theorem I.21.1 gives the exact optimum $\min\{P_+/(b(M - 1)), S/(M - 2)\}$ with explicit extremizers (note after Corollary I.9.3). Still open: prescribed allowed cell sizes and integer cardinalities at a fixed N .
- I.7** *Constants in the simultaneous minimax theorem* (05 paragraph 2). The exponent $1/(m + 1)$ and the dependence on ε are sharp; the upper constant c_m is poor. Asked: the optimal leading constants, first for one or two phases (source 01's Theorem I.7.2 improves the one-phase constant), the effect of a common difference, and the order of limits.
- I.8** *Anisotropic lower bounds* (05 paragraph 3). Is there a matching minimax upper bound in all anisotropic regimes, with the regime decomposition that loose tolerances require?
- I.9** *Adaptive selection of frequencies and tolerances* (05 paragraph 5). Proposition I.19.3 solves the continuous allocation. Missing: subset selection and the exact ceiling problem, relative to certified Fourier inputs.
- I.10** *Interval-localized spectral extraction* (05 paragraph 8). Corollary I.19.4 corrects for the Gram matrix by its top eigenvalue only. Missing: certified conditioning bounds for specific frequency configurations, using more of the Gram matrix. [6 October 2026; sources 06–10] Source 10 (its Question 4) asks the companion problem: when several approximately constant characters or polynomial phases live on one partition, the best Gram-matrix lower bound for the conditional second moment, to be combined with its variance cap (I.23.3); the common partition has to be supplied by the geometry, not assumed.

Added in the second write, 6 October 2026 (sources 06–10):

- I.11** *Using the mass of all dense cells* (06 Question 8). Source 06's Theorem I.21.1 controls the total mass above every threshold, not only one cell, and its extremizers show that the scalar data cannot give more. Asked: whether a branching density-increment argument, keeping geometric information about the collection of high cells, improves a structural or counting bound. Related to item I.4.

The ten items carry the fourteen questions: 01 Questions 5–8, 02 Subsections 11.5–11.6, 03 Research question 5, and 05 paragraphs 1–6 and 8. [6 October 2026; sources 06–10] With sources 06–10 the eleven items carry seventeen questions: also 06 Question 8 and 10 Questions 3–4.

Theme II: the inverse step (9 questions)

- II.1** *A correlation-preserving structural theorem for multivalued spectra* (02 Subsection 11.1). Proposition II.4.1 keeps all large derivative frequencies. Missing: an energy-to-structure step that keeps a fixed fraction of t_τ and exploits κ , controlling both losses.

- II.2** *Near equality for the coupled pair (f, w)* (02 Subsection 11.2). Proposition II.4.2 gives two alignments, and the equality examples show distinct geometries. Asked: a dimension-free description of f and w jointly when Theorem II.2.1 is nearly an equality; results on near-maximal Gowers norms (Eisner–Tao) are relevant but do not classify the weight.
- II.3** *Carrying profiles through the proof* (02 Subsection 11.3; 04 Q7). Keep the distribution of $m(\Delta_{\mathbf{h}}f)$ and $\mathcal{U}(\Delta_{\mathbf{h}}f)$ across slices instead of their supremum (02), and the cross-section profile $\sigma = \mathbb{E}_r \beta_r^3$ through later restrictions (04). The difficulty in both: compatibility of structure between slices, and which operations preserve the profile.
- II.4** *The sharp fixed-threshold DRC profile* (03 Research question 3). The order of the best guaranteed $|K|/n$ under Lemma 7.4’s exact criterion as $\delta \rightarrow 0$, and whether the exponent 3 of Theorem II.5.2 can be improved using constraints of actual intersection matrices; the scalar argument only rules out a two-witness first-moment certificate.
- II.5** *Optimizing the elementary extraction* (03 Research question 4). Optimize $\gamma, \varepsilon, \beta$, walk lengths and popularity thresholds in Proposition II.6.2, judging the pair of conclusions together and keeping the comparison with Reiher–Schoen explicit.
- II.6** *Lowering the size threshold of the restriction theorem* (04 Q4), from $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$, by reflecting the additive structure of B in the exceptional relations or by a better kernel, with repeated entries handled.
- II.7** *The optimal retention law* (04 Q5). Proposition II.9.6 limits one expectation budget only. Asked: upper-bound examples or a better selection method, and simultaneous guarantees on $|B'|$, its energies and the respected density.
- II.8** *Restriction beyond prime cyclic groups* (04 Q6): versions for $\mathbb{Z}/N\mathbb{Z}$, $\mathbb{F}_q^n$ and other codomains, with hypotheses that quantify kernels of the coefficient maps.

Added in the second write, 6 October 2026 (sources 06–10):

- II.9** *The optimal nonuniform transport threshold* (07 Question 1). For fixed $k \geq 2$ and fibre ratio $\kappa > 1$, the largest universal δ forcing order- k relations, and the gain from a separate boundary bound. Proved: $1/(2[1 + (k-1)\kappa])$ (Corollary II.15.3); sharp only at $\kappa = 1$ (Theorem II.16.1). Suggested: a finite search over fibre sizes and actual value assignments, not independent error profiles.
- II.10** *The largest guaranteed Freiman order at Lemma 10.9* (07 Question 2), under the exact Lean hypotheses (order five proved) and under the ambient package (order nine proved). Five and nine are lower bounds; counterexamples must satisfy the nested local model, not only a scalar defect bound.
- II.11** *Stability of winding obstructions* (07 Question 3): must a violated relation with total budget close to one come from a few cuts or carry boundaries, after a structural reduction?
- II.12** *Efficient certificates for many relations* (07 Question 4): compressing the root-law linear programs of Theorem II.18.3 over a large family of relations, for instance by translation symmetry or a presentation of the generated group; one relation is a finite linear program, exponentially many are not efficient.

- II.13** *Joint optimization of extraction and transport* (07 Question 5): building the regular component and its source law together, so that the domain is large and the relevant profiles small, accounting for lost cardinality and the relation between energy weights and the chosen law.
- II.14** *Higher-dimensional coherence* (07 Question 6): shared-vertex sampling on cubes or other diagrams in place of nested exceptional sets in bilinear and multilinear exactification; the joint law must be specified first and every marginal proved.
- II.15** *The earlier exponent $\eta^{1/5}$* (07 Question 7): which losses before the local model are necessary, and whether the fifth root can be improved while keeping a large regular component and direction set. Source 07 removes the later square root only.
- II.16** *A sharp vertical-fibre restriction theorem* (10 Question 5 and the question paragraph at the end of Section II.21). How close is $|B|/(k(h_k - 1) + 1)$ to the best bound determined by the defect fibre alone; can the factor k be reduced under additive structure of H_k ; can $|H_k|$ be bounded from a vertical energy profile, how does it grow with k , and when does an order- k homomorphism extend to order $k + 1$ with small loss? Settled only for $H_k = \{0\}$.
- II.17** *Overlaps in the bad-step union* (10 Question 6): Theorem II.21.3 uses a union bound over $E_j = \{d \neq 0 : jd \in F\}$; an exact treatment of their intersections, for example for cyclic prime-power codomains via valuations, would improve the general-codomain retention.
- II.18** *Propagating a modern energy input* (10 Question 7): propagate the Reiher–Schoen retention (almost $\sqrt{\gamma}$, against $\gamma/4$ here) with the exact vertical fibre through the later Bohr-neighbourhood and local-linearity stages, producing a complete chain of inequalities rather than an isolated replacement of the exponent 1164. Related to item II.5.

The eight items carry the nine questions: 02 Subsections 11.1–11.3, 03 Research questions 3–4, and 04 Q4–Q7. [6 October 2026; sources 06–10] With sources 06–10 the eighteen items carry nineteen questions and one question paragraph: also 07 Questions 1–7, 10 Questions 5–7 and the paragraph at the end of 10’s Section II.21.

Theme III: cube and progression counts (8 questions)

- III.1** *The odd-order constant and the higher-dimensional norm gap* (03 Research question 1; 04 Q3). $c_d^{\text{odd}} = \sup \|f\|_{U^2}^4 / \|f\|_{U^d}^4$ over real centered f on odd-order groups (03), equivalently the best comparison of $\|f\|_{U^d}^{2d}$ with $\|f\|_{U^2}^{2d}$ for given group orders (04). *Settled for $d = 3$* (dated note after Theorem III.3.4): $c_3^{\text{odd}} = 2^{-1/2}$. Open for $d \geq 4$: monotonicity gives $\|f\|_{U^2}^4 \leq 2^{-1/2} \|f\|_{U^d}^4$ for $d \geq 3$ (04, after Theorem III.3.6), but its sharpness for $d \geq 4$ is not established. The best odd-order leading coefficient is $R_d c_d^{\text{odd}}$. [6 October 2026; sources 06–10] Source 08’s Question V.34.1 asks for the same constant (κ_d^{odd}) and whether $\kappa_3^{\text{odd}} = 1/\sqrt{2}$: yes, by source 04. Source 09’s Research question on the constrained envelope (item III.17) contains the odd-order leading coefficient relative to $\|f\|_{U^d}^4$ as a special case.
- III.2** *The fifth and sixth centered coefficients* (03 Research question 2). Classify the supports of sizes 5 and 6 for which (III.2.27) has solutions with no trivial character, with multiplicities and the quotient correlations they need; Lemma III.3.3 does this for $d = 3$ on odd-order groups, and (III.2.29) shows that order three enters. [6 October 2026; sources 06–10] *The fifth coefficient is answered by source 09* (Theorems III.20.1, III.21.1, Proposition III.20.2):

on every finite abelian group and for every $d \geq 3$, only the star and ternary supports survive, with multiplicities Q_d , S_d and correlations M_5 , $T_{3,5}$. Open: the sixth coefficient, asked also by source 08 (its third question, fifth and sixth orders for general d with their torsion dependence) and source 09 (its Research question 1, a dimension-uniform sextic expansion with a seventh-order remainder); source 09's Theorem III.23.1 shows that an isolated order-five term must appear there.

III.3 *The secondary extremal coefficient* (04 Q1). Between $(9/4)^{1/3}$ and 6 (display (Q1)); does the limit exist, and is the two-frequency value optimal?

III.4 *Stability for cube extremizers* (04 Q2), beyond the U^2 stability of Corollary III.3.5, and whether near-optimal secondary behaviour concentrates on $\{\pm\xi, \pm2\xi\}$.

III.5 *Quantitative rounding and derandomization* (03 Research question 8): the group size needed by the sharpness constructions, and an efficient deterministic rounding. [6 October 2026; sources 06–10] Also 08's sixth question: a deterministic construction with rounding error $o(\eta^4)$ in U^d , not only against single characters.

III.6 *Counting that rectifies automatically* (03 Research question 6): a localized version of Theorem III.1.2 that counts ordinary integer progressions with all boundary errors. The lifting step itself is formal (Lemma III.4.5); its interval hypothesis is what a modular norm does not give.

III.7 *Progression counting with more of the mixed information* (04 Q10): mixed norms, further cancellation, pairwise information, and non-admissible slopes with explicit subgroup averages. [6 October 2026; sources 06–10] Advanced by source 10: its Theorem III.28.2 needs only one invertible coefficient difference (next to the target), pays an explicit kernel factor $K^{1/2^{k-1}}$ for the others, and uses mixed L^p norms; the factor is sharp on $(\mathbb{Z}/2\mathbb{Z})^n$ for three terms (Proposition III.28.4). Explicit subgroup averages, and the pairwise information, remain open.

Added in the second write, 6 October 2026 (sources 06–10):

III.8 *The optimal density factor in progression counts* (06 Question 6). For $k = 3$ the variance factor of Proposition III.5.6 is sharp (singletons); for $k \geq 4$, the best coefficient in place of $\sqrt{\delta} B_k(\delta)$, first within repeated telescoping and mixed-norm Cauchy–Schwarz proofs. Source 10's mixed-norm bound $\delta S_k(\delta)\eta$ (Corollary III.29.2) is the corresponding linear statement.

III.9 *Torsion constants in the generalized von Neumann bound* (06 Question 7; 10 Questions 1–2 and the question paragraph at the end of Section III.29). The best constant C in $|\Lambda_{\mathbf{a}}(f)| \leq C \|f_j\|_{U^{k-1}}$ for a fixed group, pattern and target, first for four-term patterns on elementary abelian 3-groups (10); the two-target inequality of source 06's Theorem III.5.1 when slope differences have kernels (06); keeping the images cG of Lemma III.28.1 instead of G and exploiting their overlaps, exact constants for larger k and general torsion profiles, and the joint choice of telescoping order and L^p exponents (10). Source 10's Theorem III.28.2 answers the one-target form with explicit, not in general optimal, constants.

III.10 *A correct global lower bound for the cube excess* (08 Question 2; 09 Research question 5). The largest c with $Q_3(F) - \delta^8 \geq c\delta^4 \|F - \delta\|_{U^2}^4$ for all $0 \leq F \leq 1$ on odd-order groups; $2 \leq c \leq 15337/1296 < 12$ (08; the upper end is Remark W1). 09 asks for natural hypotheses under which $M_5 \geq 0$, or a sharper bound on its negative part, giving a corrected lower bound. Minimizing $12 - 2s + 6s^2 + s^4$ over the two-frequency family lowers the upper end slightly; that family is not known to be extremal.

- III.11** *Computing the survivor polynomial* (08 Question 4): $\sum_r A_{d,r}(G)z^r$ faster than by enumerating all subsets, keeping small-characteristic exceptions, with certificates a proof assistant can check.
- III.12** *Stability of quartic-coefficient extremizers* (08 Question 5; 09 Research question 4). Structure of f when $R_d E_4(f) + T_d E_{\langle 2 \rangle}(f)$ (09: $P_d E_4 + T_d T_2$) is nearly $A_d \|f\|_{U^d}^4$: concentration on one or a few characters; on odd-order groups the extremal model is not known. A different problem from item III.4.
- III.13** *Cube expansions localized to intervals, progressions or Bohr sets* (08 Question 7; 09 Research question 6). Which of the three exact cancellations survive when the cube is restricted, with explicit boundary terms; 09 suggests expanding around a weighted reference measure adapted to the interval.
- III.14** *Sparse majorized weights* (08 Question 8): a transfer of the inhomogeneous expansion (Theorem III.13.1) to sparse weights with an explicit error budget, expanding around the majorant model.
- III.15** *Other linear systems* (08 Question 10; 09 Research question 9). For a fixed system of integer linear forms, the least support size at which a centered term survives and the first torsion-sensitive terms, combined with true-complexity estimates to get the correct first nonzero power; progressions need not inherit the cube's three cancellations.
- III.16** *The prime-onset function* (09 Research question 2): $h(7)$, $h(11)$, and the gap between (III.23.4) and $h(\ell) \leq \ell + 2$; known $h(2) = 4$, $h(3) = 5$, $h(5) = 6$.
- III.17** *The constrained extremal envelope* (09 Research question 3): the largest $\mathcal{C}_d(F)$ at fixed d , δ , $\|F - \delta\|_{U^d}$, $\|F - \delta\|_{U^2}$ and $\max |\widehat{F - \delta}|$.
- III.18** *A residual-sensitive inverse step* (09 Research question 7): whether the sixth-order certificate (III.21.11) and estimates of the four Fourier statistics improve a concrete inverse or counting algorithm, with a full cost analysis.
- III.19** *Polynomial colourings of higher degree* (10 Question 8). For $D_\ell > k$ only the bound of Proposition III.30.5 is known; for $k = 4$ and a scalar degree 5 or 6 the largest dimension lies between 4 and 8; start with dimension five. Proposed by source 10, not an established conjecture.
- III.20** *Stability at the dimension threshold* (10 Question 9): must an avoiding colouring at $d = \sum_\ell \lceil D_\ell/2 \rceil^2$ resemble the odd-degree norm construction, first in the cubic four-dimensional case?
- III.21** *Actual colours against declared coordinates* (10 Question 10): can a map with more coordinates but a small image beat the full-palette exponent of Theorem III.30.3?
- III.22** *Unrestricted symmetric-progression colourings* (10 Question 11). Deng, Tidor and Zhao ask whether $N^{o(1)}$ colours suffice to avoid nontrivial symmetrically coloured four-term progressions [37]; Shi and Dong give $O(N^{1/4})$ [33]. An open question of the literature, recorded by source 10; its theorem only limits one bounded-degree polynomial model.

The seven items carry the eight questions: 03 Research questions 1, 2, 6 and 8, and 04 Q1–Q3 and Q10. [6 October 2026; sources 06–10] With sources 06–10 the twenty-two items carry thirty-three questions and one question paragraph: also 06 Questions 6–7; 08 Questions 1–8 and 10; 09 Research questions 1–7 and 9; 10 Questions 1–2 and 8–11, and the paragraph at the end of 10’s Section III.29.

Theme IV: floor patterns (7 questions)

- IV.1** *The second-order carry entropy* (01 Question 1; the remainder of 04 Q8). Is there a $-k \log_2 k$ term in $\log_2 C_k$, or is $\log_2 C_k = k^2 + O(k)$? Theorem IV.2.3 leaves exactly this scale; counting the hyperplane intersections that meet the open cube is the suggested route. The leading constant 1 is settled (dated note at 04’s Q8). [6 October 2026; sources 06–10] Source 06’s Question 2 asks the same; its Theorem IV.7.6 settles the leading constant independently and uniformly modulo M .
- IV.2** *Exact counts, arrangement structure, certified C_5 and A_4* (01 Question 2; 04 Q8, its computational part). A recurrence, an intersection-poset formula or a symmetry-reduced enumeration for C_k and A_k (01); certified counts for $k = 4$ (04): $C_4 = F_4 = 154$ is certified by source 01, A_4 is not. *Intake values, not certified:* $C_5 = 8410$ and $A_4 = 770$, from value-branching floating-point linear programs with every feasibility witness re-evaluated exactly. The infeasibility branches rest on solver status, not on rational certificates. Missing: rational certificates, for instance from source 01’s or source 04’s certificate generator, whose formats both handle these dimensions in principle. The values satisfy the proved bounds (note in Section IV.6). [6 October 2026; sources 06–10] Source 06’s Question 1 asks for a structural formula or recurrence for C_k and D_k (its D_k is the A_k here), from the certified $C_4 = 154$ and $D_3 = 38$, and warns that an OEIS match must identify the objects, not a prefix. Source 06 certifies $C_4 = 154$ a second time; it does not compute C_5 or $D_4 = A_4$.
- IV.3** *The modulus dependence and the realizable profiles* (01 Question 3; 04 Q9). When modular collisions stabilize, and the exact number of translation classes as a function of M (01); which global and fixed-slope profiles actually occur for slopes from a short discrete interval, whether an average multiplicity can replace the worst case 2^k , and small moduli (04). [6 October 2026; sources 06–10] Also source 06’s Questions 3 and 4: the ratio $C_{k,M}/C_k$ (or its logarithmic scale) for $M = 2, 3$, and an exact formula for the number $L_k(r, M)$ of affine translation classes in the saturated range $2r \geq M$.
- IV.4** *Reducing the selection loss* (01 Question 4). Which orthogonality or positivity hypotheses allow a phase-removal interface to lose L^{-1} instead of L^{-2} ; for arbitrary complex summands the L^{-2} loss is sharp. Source 04’s Lemma IV.6.6 already replaces B_k^2 by $2^k B_k$ through local multiplicity. [6 October 2026; sources 06–10] Source 06’s Question 5 asks the same for its Theorem IV.8.1: an orthogonality or energy-weighted decomposition that pays only for the classes carrying energy.
- IV.5** *Polynomial floor entropy beyond the leading term* (01 Question 9): secondary terms of $\log_2 |\mathcal{P}_{n,d}(r)|$ and the transition when $\log r \asymp n$.
- Added in the second write, 6 October 2026 (sources 06–10):*
- IV.6** *Carry entropy of other coefficient families* (06 Question 11): which finite families $V \subset \mathbb{Z}_{\geq 0}^k$ have carry entropy comparable to rank times the logarithm of their size, and, for signed

families, the lower-dimensional floor-boundary obstruction first (source 06’s remark after Theorem IV.7.3; source 01’s example before Section IV.4).

The five items carry the seven questions: 01 Questions 1–4 and 9, and 04 Q8–Q9. [6 October 2026; sources 06–10] With sources 06–10 the six items carry thirteen questions: also 06 Questions 1–5 and 11.

Theme V: integration and formalization (8 questions)

V.1 *A complete, audited propagation through one argument* (01 Question 10; 02 Subsection 11.4; 03 Research question 7; 04 Q11). All four sources ask for a dependency graph of the quantitative parameters, into which the local improvements are substituted. Every later threshold would then be recomputed under the same hypotheses, ending in an explicit final bound, compared with the original argument and with Leng–Sah–Sawhney. Source 02 proposes the length-four proof as a first target; source 03’s Proposition V.12.1 supplies the final bookkeeping. Source 01 adds that the issue flagged at Corollary 16.11 must be resolved first. That issue is still open at HEAD (`corollary_16_11`, `theorem_18_1`). [6 October 2026; sources 06–10] Also asked by source 06 (Question 10: close the structural estimates around Sections 13 and 16 and Theorem 18.1 first, keeping the local gains as parameters), source 07 (Question 8: which bottleneck dominates after its exactification, in a fully tracked bound) and source 09 (Research question 10: locate a step whose final bound actually depends on the cube excess). At HEAD of the second write `corollary_16_11`, `theorem_18_1` and `theorem_18_2` are still open.

V.2 *Formal libraries, and formalizing sharpness* (02 Subsection 11.7; 03 Research question 9; 04 Q12; 05 paragraph 7). A characteristic-independent library: weighted Fourier energy on $G \times \widehat{G}$ and real phase freezing (02); integer linear forms, centered multilinear averages and character constraints with torsion data (03); the low-risk interfaces first, namely moment interpolation, the telescope and the support enumeration, then the selection theorem (04); and sharpness as well as existence, starting with the parity obstruction for $1/3$ and the residual inequality (I.19.4) (05). The write adds one concrete gap. Source 02’s Theorem I.11.1 could become formal now that `corollary_5_6_holds` and `polynomial0n_affine_pullback` exist; what is missing is the transfer of cells back to the input progressions and the selection step. The lifting lemma (Lemma III.4.5) is already formal. [6 October 2026; sources 06–10] Also source 08 (Question 9: the elementary fourth-order bound and the quartic torsion formula as separate verified modules, with the unnormalized target (V.32.1)) and source 09 (Research question 8: a library of reindexing equivalences and integer-lattice certificates for Boolean affine forms, keeping non-prime groups).

Added in the second write, 6 October 2026 (sources 06–10):

V.3 *Finite corrections to the iteration potential* (06 Question 9). The leading cost of Theorem V.25.2 is optimal for its abstract rule; asked are a minimal discrete correction or an exact dynamic-programming potential with a usable closed form, and the treatment of density-dependent length prefactors and nonmonotone thresholds without worst-case replacement.

The two items carry the eight questions: 01 Question 10, 02 Subsections 11.4 and 11.7, 03 Research questions 7 and 9, 04 Q11–Q12, and 05 paragraph 7. [6 October 2026; sources 06–10] With sources 06–10 the three items carry fourteen questions: also 06 Questions 9–10, 07 Question 8, 08 Question 9 and 09 Research questions 8 and 10.

Further limitations stated by the sources (kept in place; listed here for the standing rule). Sharpness of the density transfer holds in the weighted model, not for arithmetic characters (01, Remark I.2.2). The constants 36π and $3/4$ are not claimed optimal (05). Sharpness of the simultaneous exponent is for linear phases, uniform arcs and symmetric widths (05). The coefficient 6 of the $u^{16/3}$ term is not claimed optimal, and the higher-dimensional constant $P_d/\sqrt{2}$ is not claimed sharp (04). The restriction theorem is prime-only and does not prove the Lean Lemma 9.3 or Corollary 9.4 (04). $\text{PP}(K, R_0)$ is an imported input (02). The elementary Balog–Szemerédi–Gowers bound is far weaker than Reiher–Schoen (03). R_d is not claimed to be the optimal odd-order constant (03; it is not, for $d = 3$). Finite and numerical checks prove nothing general (all five). No source has a new $r_k(N)$ bound, a priority claim, or Lean verification. [6 October 2026; sources 06–10] *Further limitations of sources 06–10* (kept in place). Source 06: the leading carry entropy rests on Zuev’s count; the ordering remark optimizes bounds, not discrepancy constants; the local phase loss is not substituted into a final theorem. Source 07: five and nine are guaranteed orders, not maximal ones; the sharpness is for the directional defect δ , not for η ; the threshold is not a replacement for an average homomorphism test; the factors 243 and 2548.03968 are not improvements of a global bound. Source 08: R_d is not claimed optimal as a multiplier of t^4 in odd order; 57 is a convenient constant; the sets in its sharpness theorem have densities tending to δ . Source 09: no originality claim; its sixth-order certificates are not inverse theorems. Source 10: no novelty claim for the polynomial method (Führer–Taranchuk credited), the colouring theorem concerns declared polynomial coordinates only, and the Reiher–Schoen variant is labelled as using a published input. All five: finite checks prove nothing general, no Lean compilation, no new bound for $r_k(N)$.

V.3 Relationship to the ProveIt formalization

Source 01, delivered Section 14.

V.3.1 The inspected snapshot and normalization

The source inspection is pinned to commit

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of VladimirReshetnikov/ProveIt, rather than to a moving branch [15]. The original paper was read through the public PDF mirror listed in the bibliography; selected repository files, not the entire repository build, were inspected.

The inspected `Definitions.lean` uses the unnormalized Fourier transform, a balanced indicator, finite partitions, natural-number progressions, and average cell size [16]. Consequently a hypothesis $|\widehat{1_A}(r)| \geq \alpha N$ corresponds to $\eta = \alpha$ in this article, not to $\eta = \alpha/N$. A sum of absolute balanced cell sums becomes our D after division by N .

The companion `Proofs02DensityIncrement.lean` develops the positive/negative mass split for the Fourier density increment [17]. The companion `Proofs17FinitePatterns.lean` uses a finite sign-pattern/Sauer–Shelah approach for the inherited floor-pattern bounds [18]. These are natural locations for stronger reusable interfaces; no assertion that they have been replaced or recompiled is made.

[write] **6 October 2026.** At HEAD (25df8755d, 6 October 2026) the inspected files are unchanged except `Definitions.lean`, which gained a `Box.IsProper` that no source uses. The ledger now records 98 exact

companions and 22 open statements, against 88 and 32 at this source’s pin. Of the statements compared here, `corollary_2_5`, `corollary_17_5`, `corollary_17_6` and `proposition_17_7` have exact companions, and `theorem_18_1` is open. Crosswalk: Section V.1.

V.3.2 Proposed formal interfaces

The dependencies can be organized in the following order.

Interface	Mathematical content	Reusable prerequisite
Real cell mass	$\sum g_j = 2 \sum (g_j)_+$ under $\sum g_j = 0$; weighted mean and occupied mass	Finite sums and order arithmetic
Cell phase envelope	Lemma I.1.2, first for an indicator, then for bounded weights	Complex norm, sine, finite convex combinations
Broken-chord transfer	Theorems I.2.1 and I.2.3 with explicit denominators	Convexity and mean identities
Integer partition	Theorem I.5.1, including the case of one block in a residue class	Finite pigeonhole and quotient/remainder
Fractional patterns	Positive perturbation and exact range factorization	Integer floor, finite minima of positive gaps
Pattern count	Hyperplane regions or an equivalent finite sign-family bound	Dimension and finite set cardinality
Translation quotient	Normalize a pattern by its empty-set value	Finite group action or an explicit canonical representative
Certificate checker	Rational primal/dual checks and exhaustive tree validation	Exact arithmetic and structural recursion

A practical first target is the $s = 0$ discrepancy theorem: its proof needs no trigonometry, no Fourier transform, and no structural additive combinatorics. The next target is the cancellation inequality, followed by the full envelope. Proving the statements over arbitrary finite weighted families prevents repeated normalization and counting arguments later.

For the carry side, one should formalize interior realization before using a strict sign-pattern theorem. The signed-feature counterexample in Section IV.3 is a small regression test against a false boundary shortcut. The exact coefficient-range factorization also needs the open-endpoint surjectivity argument, not merely an upper-bound map.

For progression lengths, retain natural-number inequalities $N \geq L \prod_i Q_i$ and $L \leq \ell \leq 2L - 1$ as long as possible. Introduce square roots and floors only in the corollary. This isolates small-scale cases rather than trying to recover them from real asymptotic estimates. If actual code is added, it should prove a new strengthened statement and derive the old statement as a corollary where appropriate; changing a catalogue definition alone is not a proof.

V.3.3 A dependency warning, not a claimed global repair

The inspected `Sections17_18.lean` records the numbered statements as proposition-valued definitions. Its note attached to Theorem 18.1 flags an unsupported tower estimate in the published Corollary 16.11 as a quantitative dependency [19]. This is a report of the repository’s warning, not an independent conclusion here that a published headline theorem is false. Our local theorems do not use that estimate. In particular, applying Theorem I.4.1 to a supplied partition is unconditional, but asserting the existence of that partition with a given tower exponent would require a separate validated source.

No Lean or Lake executable was available in the working environment, and no new Lean theorem was compiled in producing this package. The document contains a formalization plan, conventional mathematical proofs, and separately described computational checks. It does not present a proposition-valued specification as a kernel-verified theorem.

[write] **6 October 2026.** Checked at HEAD: `theorem_18_1` is still open in the ledger, and its docstring in `Sections17_18.lean` still warns about the unsupported tower estimate in the published Corollary 16.11 (line 168). `FORMALIZATION_STATUS.txt` still lists resolving “the documented Corollary 16.11 bound” among its open obligations, and `corollary_16_11` is open. The dependency warning above is therefore accurate.

V.4 Further research questions and concrete next steps

Source 01, delivered Section 15.

The questions below are posed by this report. They are unresolved within this development; this is not a certification that every formulation is new or currently open in the entire literature.

Question V.4.1 (The second-order carry entropy). *Does $\log_2 C_k$ have a $-k \log_2 k$ correction, or is $\log_2 C_k = k^2 + O(k)$? More generally, determine the coefficient of $k \log k$ if it exists.*

Theorem IV.2.3 leaves exactly this scale between its upper and lower estimates. A promising route is to count intersections of the internal subset-sum hyperplanes that actually meet the open cube, rather than use the unrestricted region bound. The elementary anchor construction is too sparse to decide the leading coefficient, let alone this correction. Threshold-function embeddings give a strong baseline but need not capture all simultaneous carry levels.

Question V.4.2 (Exact counts and arrangement structure). *Can one obtain a recurrence, a useful intersection-poset formula, or a symmetry-reduced exact enumeration for C_k and A_k ?*

The supplied certificate format scales by adding more branches, but its size grows rapidly. Coordinate permutations and complement transformations suggest reductions; any quotient must retain enough information to certify that all orbit representatives are covered. An exact-arithmetic checker in a proof assistant would make such computations useful as formal regression tests, not merely numerical evidence.

Question V.4.3 (The modulus dependence of affine patterns). *For fixed k and coefficient range, when do modular collisions stabilize, and what is the exact number of translation classes as a function of M ?*

Theorem IV.3.2 is uniform in M but deliberately discards these collisions. A finite list of integer-valued normalized patterns exists; modular coincidence is then controlled by divisibility of finitely

many coordinate differences. Determining those difference vectors could give an exact piecewise description, including a threshold beyond which no new collisions occur. The offset's fractional part must remain visible.

Question V.4.4 (Can the selection loss be reduced?). *What additional orthogonality or positivity assumptions allow a phase-removal interface to lose only L^{-1} rather than L^{-2} in energy, without changing the desired structural conclusion?*

For arbitrary complex summands the squared pigeonhole loss in Proposition IV.3.3 is sharp: equal aligned summands attain it. A stronger theorem therefore needs additional hypotheses, not just more careful algebra. One possible approach is to retain the sum of squared contributions across translation classes until after a structural selection step. Such a mechanism has not been established here.

Question V.4.5 (Sharpness under arithmetic phase constraints). *How much stronger can the density-tail bound become when ψ is one fixed cyclic character and the cells are proper progressions with a common step and nearly equal lengths?*

The extremizers of Theorem I.2.1 freely rotate cell arcs and assign endpoint phases to occupied points. A genuine character imposes spacing and compatibility restrictions. A quantitative analysis of those restrictions could improve constants or density dependence beyond the abstract sharp bound; it would not contradict that bound's optimality in the larger weighted model.

Question V.4.6 (The optimal finite size-increment tradeoff). *Can one combine the exact density tradeoff (I.3.4) with an optimal finite partition algorithm, including the final incomplete blocks, to characterize the best uniform pair of progression length and increment?*

Our integer partition is transparent but not claimed to have optimal constants. Theorem I.7.2 fixes the square-root exponent for one-frequency phase-flat methods. Remaining improvements may come from choosing unequal block lengths, exploiting rational frequencies, or paying for selected short cells through their occupied mass rather than their ambient size.

Question V.4.7 (Retaining more than one density statistic). *Which finite collection of stop-loss moments most efficiently preserves useful density-tail information through repeated localization steps?*

Theorem I.4.2 supplies an exact optimization method once the statistics are chosen. The research issue is to prove structural lemmas that propagate those statistics at acceptable cost. A two- or three-knot state could be substantially more informative than a single discrepancy number, but a global gain must include the cost of obtaining that state.

Question V.4.8 (Higher-degree phase partitions). *Can polynomial or nilsequence localization furnish partitions with both quantitative minimum cell sizes and small occupied exceptional mass, so that the sharp tail theorem applies without a large selection loss?*

The density transfer itself is already proved for any short-arc phase; what is missing is the required structural partition. Simultaneous linear approximation alone does not solve the higher-degree problem. The polynomial floor-pattern bound in Theorem IV.4.1 controls one finite complexity term but does not construct those partitions.

Question V.4.9 (Polynomial floor entropy beyond the leading term). *For fixed degree d , determine the secondary terms of $\log_2 |\mathcal{P}_{n,d}(r)|$, and identify the transition when $\log r$ is comparable to n .*

The finite upper bound separates a coefficient-range term from a region count, but unlike Theorem IV.1.2 it need not be an exact product. Understanding overlaps between different integer coefficient vectors is essential. The threshold-function lower bound alone does not settle the large-range contribution.

Question V.4.10 (A fully audited quantitative integration). *After independently checking the earlier structural parameter estimates, what explicit global improvement survives if the new density-transfer and translation-class bounds are propagated through one complete argument?*

The first deliverable should be a dependency graph with normalized quantities and verified recurrences, including the issue flagged at Corollary 16.11 in the inspected repository. Only then is it meaningful to compare a resulting global bound with modern benchmarks. This prevents a smaller local factor from being mistaken for a new global theorem.

V.5 Conclusions and result-status ledger

Source 01, delivered Section 16.

The work yields a reusable sharp transfer principle and a more economical floor-pattern count. The density calculation is sensitive to the actual balanced mass $2\delta(1 - \delta)$, to the full distribution of cell densities, and to occupied exceptional mass. The counting calculation separates integral coefficient freedom from fractional carries, treats boundary points correctly, and exhibits a threshold-complexity obstruction to a subquadratic exponent. Both refinements have explicit hypotheses and can be inserted at well-defined proof interfaces.

Claim	Status in this package
Sharp finite weighted density-tail theorem, including equality examples	Proved in Theorem I.2.1; arithmetic extremizers are not claimed.
Occupied-mass exceptional-set bound and optimal forced maximum	Proved in Theorems I.2.3 and I.3.1.
Explicit Fourier density increment with integer-exact length	Proved in Theorem I.6.1, for arbitrary real frequency and bounded weights; no prime-modulus restriction.
Exact range factorization and quadratic-exponent carry upper/lower bounds	Proved by elementary arguments in Theorems IV.1.2, IV.1.4, and IV.2.1.
$\log_2 C_k \sim k^2$ and polynomial-floor leading entropy	Deduced from explicitly identified classical threshold-function inputs; those inputs are not reproved here.
$C_4 = 154$	Exhaustive exact rational certificates with a Python verifier and a completeness argument; not Lean-verified.
Smaller translation-class budget	Proved in Theorem IV.3.2; subsequent global parameter propagation is not supplied.
Improved record bound for Szemerédi's theorem	Not claimed.

Claim	Status in this package
Priority of every formulation	Not certified by an exhaustive literature search.

V.6 Boundary and normalization checklist

Source 01, delivered Appendix A.

The formulas in this article are intended to be used with the following conditions kept explicit.

1. The phase results assume $0 < \delta < 1$. At $\delta = 0$ or 1 , a bounded function with that mean is constant and the balanced correlation is zero; formulas dividing by $\delta(1 - \delta)$ should not be used.
2. The short-arc theorem uses half-angle $\theta < \pi/2$. The progression partition uses full arc length r in turns, so $\theta = \pi r$, not $2\pi r$.
3. A normalized Fourier coefficient is the unnormalized coefficient divided by N . The normalized discrepancy is likewise the sum of cell balanced masses divided by N .
4. The density-tail event is $q > a$, strictly. Its stop-loss lower bound may vanish at the optimal forced maximum even though a cell with $q \geq a$ is guaranteed.
5. Exceptional occupied mass is $\mathbb{E}(h\mathbf{1}_E)$ in the original probability measure. It is not a conditional density on E .
6. The coefficient-range factorization assumes integral $r_i \geq 1$ and open intervals $(-r_i, r_i)$. Fractional coordinates can lie on a boundary only because Lemma IV.1.1 removes that boundary without changing the pattern.
7. Natural progressions have positive step and no repeated terms. Singletons handle the small-scale case; no nonintegral number is treated as a possible cell length.
8. The affine translation quotient retains the fractional offset. Its modulus-free upper bound does not assert absence of modular collisions.
9. The threshold-function asymptotic is an external theorem. The elementary finite pattern bounds and all density-transfer statements remain valid without it.

V.7 Source and artifact provenance

Source 01, delivered Appendix B.

The repository snapshot was read on 6 October 2026. The paper directory contains both a PDF and a TeX representation. The research arguments use the original paper’s relevant PDF pages and selected Lean source files, not an assumption that every statement in the repository has already been proved. The pinned commit identifies precisely which source comments and normalizations were inspected. No repository files were modified.

The package includes the article source, compiled PDF, certificate data, verification and supplementary-test scripts, result logs, and a README. The article’s source uses an internal

bibliography and does not require BibTeX or downloaded repository files to compile. No copies of the copyrighted source papers or font files are redistributed.

V.8 Formalization interface and reproducibility

Source 02, delivered Section 10.

V.8.1 What was inspected

The repository's `Definitions.lean` defines `fourier` using the unnormalized `ZMod.dft`, defines `difference` with $x - h$, and defines `balanced`, `IsPartition`, `UniformOfDegree`, and the unnormalized `gowersNorm`. The inspected `ProofInfrastructure.lean` provides, among other helpers, `IsPartition.sum_card` and `sum_balanced`. The inspected part of `Proofs02DensityIncrement.lean` contains private helpers for partition sums and the exact equality between total absolute mass and twice positive mass [20].

The normalized U^3 quantity in this article translates to the repository's degree-two uniformity numerator by

$$\mathcal{U}(f) = N^{-4} \sum_{a,b \in G} \left| \sum_{x \in G} \Delta_a \Delta_b f(x) \right|^2.$$

Thus the degree parameter of `UniformOfDegree` is 2 here, not 3. For the repository's unnormalized order-three Gowers norm, the eighth power is $N^4 \mathcal{U}(f)$. These normalizations should be established as explicit bridge lemmas before the energy theorem is imported.

V.8.2 Suggested dependency order

The most economical formalization starts with finite weighted real sums: positive/negative mass, the negative-support bound, and Theorem I.9.1. It then proves the real-phase lemma using positive and negative weighted averages. These results have no Fourier dependencies and can be added independently of a full higher-order inverse theorem.

For Theorem II.2.1, first formalize the three identities (II.2.2)–(II.2.4) over `ZMod N`. Keep all statements polynomial and multiplication-based, rather than dividing by $m(f)$ or $\mathcal{U}(f)$; this handles zero cases without extra hypotheses. A later generalization can replace `ZMod N` by an abstract finite abelian group and its character group.

Next prove the two-tetrahedron bijections in Lemma II.3.1, then add graph indicators and balanced indicators as specializations. The final arithmetic-progression wrapper should accept a proved $\text{PP}(K, R_0)$ hypothesis rather than hiding it behind an unverified claim about exact cell counts.

Care is needed with empty cells: the repository's general partition predicate does not itself require every cell to be nonempty. Empty cells must be removed before using cell means, $M - 1$, or a minimum cell length. Likewise, injectivity of a modular progression parameterization is not the same as nonwrapping in its standard integer representatives.

V.8.3 Verification status

No Lean compiler or Lake executable was available in the working environment. Accordingly, none of the new results is represented as kernel-checked, and no untested Lean proof is included. The source reading was targeted rather than a certification of every theorem in the supplied development. The mathematical proofs in this article are the primary deliverable.

The accompanying `verify.py` independently checks the weighted Fourier identities and both asymmetric forms of the energy inequality on finite cyclic groups, compares spectral and direct convolution energy for small groups, tests phase and tail inequalities, and evaluates exact sharpness examples. It uses the fixed seed 20261006. The run recorded in `verification_results.json` includes 490 random energy cases, 2,500 random phase cases, 3,000 zero-mean moment cases, 1,500 nonzero-mean moment cases, and 1,500 phase-budget optimization cases, together with three exact-form equality families evaluated numerically and a rational moment equality example. These experiments are regression checks, not proofs or substitutes for formal verification.

[write] **6 October 2026.** The suite is shipped as `code/02-energy-freezing-verify.py`; it overwrites `verification_results.json` beside itself, so the README's rerun route runs it on a copy. At intake it reproduced the recorded run identically (8,990 random cases and the exact families). The corrected Corollary 5.6 that Section I.11 imports is now formalized (note there).

V.9 Further research questions and proposed work

Source 02, delivered Section 11.

The following are proposed research targets suggested by the proven interfaces. Their inclusion is not a claim that each is a previously published open problem.

V.9.1 A correlation-preserving structural theorem for multivalued spectra

Proposition II.4.1 retains all significant derivative frequencies. Can the subsequent energy-to-structure and graph-selection steps preserve a fixed fraction of t_τ while exploiting the denominator κ ? The decisive issue is not merely obtaining a small-doubling subset: it is ensuring that the subset continues to carry the correlation needed for integration. A useful first goal is a theorem whose hypotheses include both additive energy and a specified nonnegative correlation weight, and whose conclusion controls both losses explicitly.

V.9.2 Classify near equality for the coupled pair (f, w)

Proposition II.4.2 gives two quantitative alignments, while Examples II.3.3–II.3.4 show distinct sharp geometries. Existing results about near-maximal Gowers norms [7] are relevant but do not by themselves state a classification of the additional spectral weight in our mixed inequality. One can ask for a dimension-free theorem describing w and f jointly when the main inequality is nearly saturated, with appropriate treatment of subgroups and quadratic phases.

V.9.3 Retain norm profiles through higher-order integration

The graph-energy step is repeatedly applied to derivatives in higher-degree arguments. A natural target is a weighted family theorem that keeps the distribution of $m(\Delta_{\mathbf{h}}f)$ and $\mathcal{U}(\Delta_{\mathbf{h}}f)$ instead of

replacing every slice by a supremum. The immediate difficulty is compatibility: good structural pieces from different slices must agree on overlapping configurations. A local improvement cannot be counted as a global gain until this compatibility cost is controlled.

V.9.4 A complete quantitative ledger for one progression length

An attainable next project is to propagate the present denominators, the sharp real-phase law, and the optimized budget through every step of the length-four proof, with one explicit choice of parameters. This would determine whether a real improvement survives the remaining inverse and recurrence estimates. Only after that full ledger is complete would it be justified to state an improved numerical threshold for that proof. Comparisons should distinguish improved constants within Gowers’s route from stronger bounds obtained by later methods [4, 2].

V.9.5 Sharper recurrence inputs and heterogeneous budgets

Theorem I.11.1 treats PP as a replaceable component. Any better value of K or R_0 transfers automatically to the progression certificate. A separate optimization problem assigns different diameter budgets to cells of different sizes and correlations, using the heterogeneous inequality in Section I.8. The local quadratic bound contains still more information than its linear envelope; exploiting it may improve nonuniform cases without altering the recurrence theorem.

V.9.6 The finite-cell extremal problem

Theorem I.9.1 solves the weighted total-good-mass problem sharply. It does not solve the problem of the largest good cell when M and all allowed cell sizes are fixed. The three-type extremizer explains why a negative cell and a low-positive cell can consume separate parts of the cell budget. A precise finite-dimensional optimization, including integer cardinalities, could strengthen the $M - 1$ corollary and produce exact finite- N certificates.

[write] 6 October 2026; sources 06–10. *Advanced by source 06.* With M fixed and the cell masses free (the weighted model), the problem is solved by source 06’s Theorem I.21.1: the exact best lower bound for the largest high cell is $\min\{P_+/(b(M - 1)), S/(M - 2)\}$, attained by explicit atomic extremizers; its second extremizer is a three-type configuration of exactly the kind described above (cells at b , one cell at the threshold, one negative cell). Source 06 realizes its extremizers on finite uniform spaces when the parameters are rational. Still open: prescribed allowed cell sizes and integer cardinalities at a fixed N (Section V.2, item I.6).

V.9.7 A characteristic-independent Lean library

The main analytic results work on every finite abelian group and the two-tetrahedron argument uses no field division. A useful library contribution would formalize weighted Fourier energy, its normalization on $G \times \widehat{G}$, and real-valued phase freezing independently of the prime-cyclic presentation. Regression examples should include even cyclic groups, subgroup spectral rectangles, empty weights, zero functions, and wrapping versus nonwrapping progression charts.

V.10 Conclusion

Source 02, delivered Section 12.

The strongest unconditional analytic conclusion of this article is the mixed weighted inequality (II.2.1), with its exact norm denominators and coefficient-one equality examples. The strongest elementary density conclusion is the sharp tail bound (I.9.1), coupled with the sharp aggregate real-phase law (I.8.3). Together they strengthen identifiable interfaces in Gowers’s argument, including Proposition 6.1 and Lemma 5.15, and yield an explicitly optimized progression certificate under a clearly stated partition input.

The remaining challenge is structural, not merely algebraic: local gains must survive extraction, integration, recurrence, and iteration. The article makes those gains available without presenting that uncompleted propagation as a new global theorem.

V.11 Source audit and normalization checklist

Source 02, delivered Appendix A.

The repository branch reference observed during this session was

66c02f8a1d26f553deac5a91dada1739e0d15064.

The specific inspected blobs are recorded below, so the source comparison is not dependent only on a moving branch name.

File below the supplied repository paths	Git blob SHA
<code>sz-thm-gowers-proof.tex</code>	41085a0efde8c1932e86e80c791984f52841c22e
<code>Definitions.lean</code>	202260e0d7ec6f033c1758347381c34a6f0d9212
<code>ProofInfrastructure.lean</code>	ebd9e7ed02d81de2e17b97b4223f861b785ed5d8
<code>Proofs02DensityIncrement.lean</code>	85b5c3ad8769b9769e1939ae5c05f0757f0678b5

[write] **6 October 2026.** Checked at HEAD (25df8755d): the four blob identifiers are those of commit 66c02f8a1. The transcription, `ProofInfrastructure.lean` and `Proofs02DensityIncrement.lean` are unchanged at HEAD; `Definitions.lean` has changed (blob 97113b7), by an added `Box.IsProper` that source 02 does not use. Source 02’s delivery README records no pin; this branch reference is its provenance (Section F.4).

The targeted paper passages were the corrected Corollary 5.6 and nearby phase-partition statements, Lemma 5.15, and Proposition 6.1. The transcription includes editorial changes; a theorem imported from that edited source is labeled as such above rather than silently attributed to the exact original printed formulation.

Quantity	This article	Paper/Lean convention
Fourier coefficient	$\mathbb{E}_x f(x) \overline{\chi(x)}$	$N \widehat{f}(\chi)$
Derivative-graph statistic	T	$N^3 T$
Order-three cube quantity	$\mathcal{U}(f)$	$N^4 \mathcal{U}(f)$
Graph additive energy	$\mathcal{E}(\mathbf{1}_\Gamma)$	$N^3 \mathcal{E}(\mathbf{1}_\Gamma)$
Balanced L^2 mass	$m(f) = \delta(1 - \delta)$	$N m(f)$ before averaging

V.12 How local improvements affect an iteration

Source 03, delivered Section 9.

The estimates above remove real losses, but they occur at different places in the proof. The cube theorem improves a counting consequence of uniformity. The DRC theorem improves extraction of a subfamily from intersection data. The Fourier theorem converts an already established correlation into a density increment on an ordinary progression. Their parameters are not interchangeable. In particular, a larger common-neighborhood core does not by itself provide an improved correlation theorem for arbitrary high Gowers norms.

Figure 1 displays two of the local comparisons using the literal formulas proved in the text. For cube counts, retaining the centering changes the first possible error power, while classifying its supports further reduces its coefficient. For DRC, retaining the original success criterion and changing the exponent from five to three gives a growing advantage as the density parameter decreases.

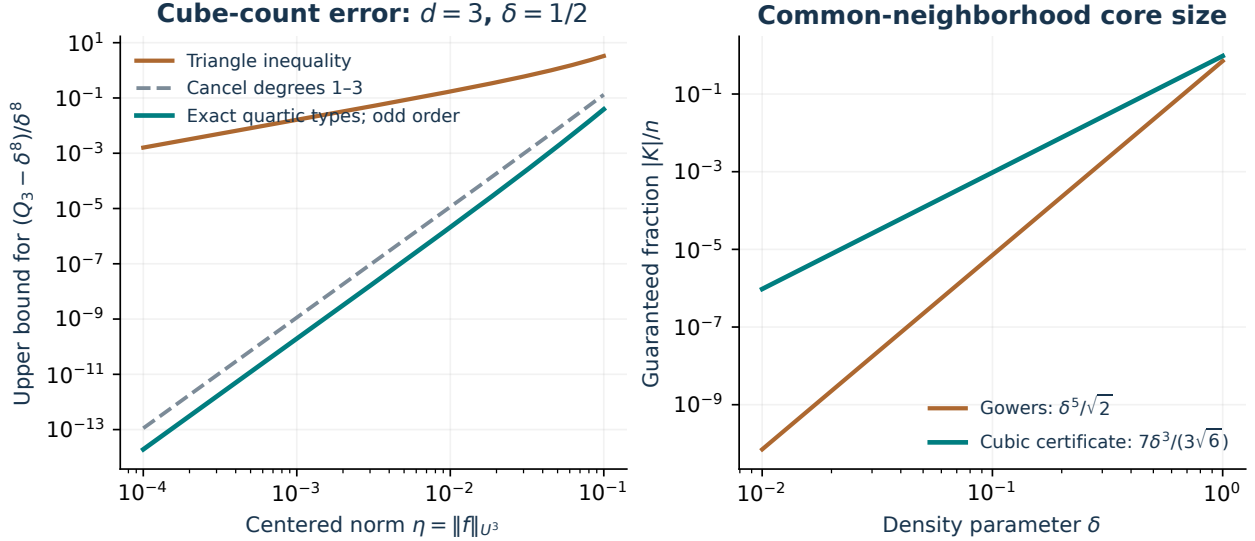


Figure 1: Evaluations of proved local bounds. Left: relative cube-count error bounds for $d = 3$, $\delta = 1/2$, comparing the triangle inequality, the cancellation of degrees one through three, and the exact quartic types on an odd-order group. Right: guaranteed core fractions at the same threshold $\delta^2 m/2$ and the same 90% good-pair requirement. These are upper or lower bounds as labelled, not measurements of extremal examples or claims that each curve is attainable.

V.12.1 An explicit abstract recurrence

The following elementary calculation explains which parameters must be tracked before any global conclusion can be stated. It is a conditional recurrence theorem, not a substitute for the analytic density increment theorem needed to supply its hypotheses.

Proposition V.12.1 (Density and length bookkeeping). *Let $c > 0$, $p > 1$, $C > 0$, $q \geq 0$, and $0 < \beta < 1$. Suppose positive numbers $\delta_i \leq 1$, $N_i > 0$ satisfy, for $0 \leq i < R$,*

$$\delta_{i+1} \geq \delta_i + c\delta_i^p, \quad N_{i+1} \geq C\delta_i^q N_i^\beta.$$

Then

$$R \leq \frac{(1+c)^p}{c(p-1)} (\delta_0^{1-p} - 1). \quad (\text{V.12.1})$$

Writing $B = \max\{0, -\log(C\delta_0^q)\}$, one also has

$$\log N_r \geq \beta^r \log N_0 - B \frac{1 - \beta^r}{1 - \beta} \quad (0 \leq r \leq R). \quad (\text{V.12.2})$$

Consequently, for an integer $T \geq R$ and $H \geq 1$, the condition

$$\log N_0 \geq \beta^{-T} \left(\log H + \frac{B}{1 - \beta} \right) \quad (\text{V.12.3})$$

ensures $N_r \geq H$ at every $r \leq R$.

Proof. The potential u^{1-p} decreases at a controlled rate. Since $\delta_i \leq 1$,

$$\begin{aligned} \delta_i^{1-p} - \delta_{i+1}^{1-p} &\geq (p-1) \int_{\delta_i}^{\delta_i + c\delta_i^p} t^{-p} dt \\ &\geq \frac{(p-1)c\delta_i^p}{(\delta_i + c\delta_i^p)^p} \geq \frac{(p-1)c}{(1+c)^p}. \end{aligned}$$

Summing and using $\delta_R^{1-p} \geq 1$ gives (V.12.1). The densities are increasing, so $\log N_{i+1} \geq \beta \log N_i - B$. Iterating this affine inequality gives (V.12.2). Under (V.12.3), $\beta^r \log N_0$ is at least $\log H + B/(1 - \beta)$ for $r \leq T$, which proves the last assertion. $\square$

For fixed constants, the familiar double-exponential length budget is visible in (V.12.3): the number of iterations occurs in the exponent of β^{-T} , which already bounds a logarithm of the starting size. Improving C or q changes the additive length cost B . Improving the density growth exponent p can change the scale of the number of iterations. These are different effects. To extract a better global Szemerédi bound from this report one would have to identify, and prove throughout the later sections, the corresponding updated values or functions of all these parameters. We have not carried out that full replacement of the higher-order inverse and localization arguments.

[write] **6 October 2026; sources 06–10.** *Two further iteration statements.* Source 10's Proposition V.42.1 bounds the number of steps of the same recurrence by $(\delta_0^{1-p} - 1)/(1 - (1+c)^{1-p})$, and source 06's Lemma V.25.1 with $G = 1$ by $1 + (\delta_0^{1-p} - 1)/(c(p-1)) + p \log(1/\delta_0)$ (its (V.25.3)). *Comparison (second write).* With $q = p - 1$, $1 - (1+c)^{-q} = q \int_0^c (1+u)^{-q-1} du$ lies between $qc(1+c)^{-q-1}$ and qc , so the coefficients of δ_0^{1-p} satisfy $1/(cq) \leq 1/(1 - (1+c)^{-q}) \leq (1+c)^p/(cq)$: source 06's leading coefficient is the smallest and source 10's lies between it and the one in (V.12.1); source 06's bound has the additional term $1 + p \log(1/\delta_0)$ and counts the densities $\delta_0, \dots, \delta_{t-1}$, whereas the bound here counts steps. For $p = 2$, $c = 1/320$ the coefficients are 320,

321 and $321^2/320 \approx 322.003$. Source 06 also proves that the leading term of its double-logarithmic length budget is optimal for its abstract local rule (its Section V.25), and source 10’s Proposition V.42.2 is the length recursion (V.12.2) with variable parameters.

The same distinction applies to the exact quartic identity. It is stronger information about cubes with a prescribed balanced Gowers norm. It does not imply that every set lacking arithmetic progressions has a large ordinary Fourier coefficient. Gowers’s motivating examples address precisely why higher-order structure is needed. The role of the improved count is to make the uniformity branch and its constants more precise.

V.13 The Lean development and a concrete formalization route

Source 03, delivered Section 10.

V.13.1 Pinned source and status of the inspection

All repository comparisons refer to revision

3a25020a654d196f7f1aa2e37649752c000d0e65

of ProveIt, inspected on 6 October 2026. The requested paper is at `Combinatorics/Ramsey/Papers/sz-thm-gowers-proof`; the Lean files are at `Combinatorics/Ramsey/Lean/GowersSzemerédi`. The adjacent source inventory, `gowers-proof-status.json`, reports 88 numbered statements with exact companion declarations and 32 open statements at that revision [22].

[write] 6 October 2026. *Stale since the pin.* At HEAD (25df8755d, 6 October 2026) the inventory records 98 exact companions and 22 open statements. The ten statements closed since 3a25020a6 are `corollary_5_6`, `corollary_5_7`, `lemma_5_9`, `lemma_5_10`, `corollary_5_11`, `lemma_5_14`, `lemma_16_1`, `lemma_16_3`, `lemma_16_5` and `lemma_16_6` (compared at the write). `theorem_1_3`, `theorem_18_1` and `theorem_18_2` are still open, so the conclusion of the next paragraph stands. Crosswalk: Section V.1.

Those counts are a source inventory, not a certification of the whole development. The facade exposes numbered results as propositions, usually with companion theorems whose names end in `_holds`. Defining a proposition neither proves it nor adds it as an axiom. The inventory explicitly separates exact companions from conditional, restricted, or otherwise related results. In particular, the quantitative headline `theorem_1_3` and final statements `theorem_18_1` and `theorem_18_2` are marked open there. This inspection therefore does not support describing the repository as a completed formal proof of the full quantitative theorem.

The files directly relevant to our comparisons contain substantive companion proofs. We inspected their definitions and proof scripts; we did not compile the Lean project. The new mathematical statements in this report have written proofs and independent mathematical reviews, together with the finite checks described in the appendix. No new machine-checked Lean proof is claimed or supplied.

[write] 6 October 2026. The “independent mathematical reviews” mentioned here and in Section V.15 are not part of the delivery and are not recorded anywhere in the repository; this report relies on the printed proofs, the intake’s checks and the write’s notes, not on them.

V.13.2 Normalization map

The following conversions prevent a common source of false improvements. Let $N = |G|$, and take $d \geq 1$.

Quantity	This report	Inspected repository convention
Fourier transform	$\mathbb{E}_x f(x) \overline{\chi(x)}$	Unnormalized sum, equal to N times this transform
Gowers norm	$\ f\ _{U^d}$	<code>gowersNorm</code> d $f = N^{(d+1)/2^d} \ f\ _{U^d}$
Uniformity parameter	$\ f_A\ _{U^d}^{2^d} \leq \alpha$	<code>UniformSetOfDegree</code> A $\alpha (d-1)$
Cube count	$Q_d(\mathbf{1}_A)$	<code>cubeCount</code> $= N^{d+1} Q_d(\mathbf{1}_A)$
Cube arguments	$x + \sum_i v_i h_i$	A minus convention in <code>cubeArgument</code> ; negating all h_i gives the same average

Both cube conventions count all parameter tuples, including degenerate cubes. This is why the rounding lemma can bound the exceptional tuples by a collision probability instead of changing the definition of the norm. Changing to unlabelled, injective, or symmetry-quotiented cubes would require an additional counting theorem.

V.13.3 Useful existing declarations

The source provides the following starting points in namespace `LeanProofs.GowersSzemerédi`. These are concrete interfaces to reuse or generalize, rather than assertions that the proposed additions have already been formalized.

Partition discrepancy and Fourier increment.

`Proofs02DensityIncrement.lean` proves `corollary_2_5_holds`. Its private `cor25_large_scale` already obtains the stronger $\rho/4$ increment and $\sqrt{\rho N/(128\pi)}$ length before weakening its conclusion. General finite weighted sums, followed by Theorem I.14.1, would expose the asymmetric information without repeating the Fourier argument.

Multilinear norms and cube expansion.

`Proofs03GowersCS.lean` contains the public homogeneous theorem `gowers_cauchy_schwarz`, which already allows functions beyond the unit disc. `Proofs03Minkowski.lean` contains the triangle inequality and private distributive expansion helpers. `Proofs03CubeUpper.lean` proves `lemma_3_10_holds` from that triangle inequality. Exporting a general expansion indexed by vertex subsets is a natural first step toward the exact quartic theorem.

Selection from many cells.

`Proofs05_10.lean` contains `lemma_5_15_holds`. Its assumptions and conclusion match the symmetric discrepancy extraction discussed in Section I.14. The general theorem should be stated with real nonnegative cell weights and asymmetric cell-sum bounds, then specialized to cardinalities.

Dependent random choice.

`Proofs07DRC.lean` proves `lemma_7_4_holds` using ordered samples with replacement, indexed by `Fin5`. Its internal score bound already has constant $11/16$ multiplying $\delta^{10} n^2$. The main new ingredient for the cubic theorem is a pointwise polynomial minorant, followed by the exact three-sample counting identity. Merely exposing the internal five-sample constant would give a smaller change.

V.13.4 Recommended order of implementation

First formalize the finite weighted selection theorem over an arbitrary finite index type. Prove positivity of its denominators once, and keep the pointwise inequalities separate from the division steps. This produces immediately reusable specializations to Lemma 5.15 and balanced indicators.

Second formalize the cubic certificate. Its good-region proof is a polynomial identity followed by nonnegativity; its bad-region proof may use concavity or a direct monotonicity argument. Keep the sampling identity generic in the number of witnesses. Specialization to three witnesses then replaces only the numerical part of the current DRC proof.

Third export the general cube expansion and prove the single-coordinate cancellation lemma. Nonempty supports of at most three selected vertices, or four vertices with nonzero XOR, leave one variable occurring in exactly one centered factor after a coordinate reflection. The corresponding reindexing is a bijection and uses no division by two. The first quartic upper bound then follows directly from Gowers–Cauchy–Schwarz, before any Fourier classification is needed.

Fourth formalize the two nonzero four-vertex types. One may begin over $\mathbb{Z}/N\mathbb{Z}$ to match existing files, then generalize to finite abelian groups. The extra quotient energy requires explicit handling of order-two characters; assuming an invertible doubling map is appropriate only for the odd-order specialization. The combinatorial formulas for R_d, T_d are separate finite enumeration theorems.

Finally integrate the progression-count telescope, the proper progression partition, and the four-edge compression statement. Each has a clear finite interface. The generalized von Neumann theorem should state invertibility of slope differences explicitly; the progression partition should carry its lower and upper integer lengths in its output. This order establishes useful lemmas early while avoiding dependence on the unfinished quantitative headline.

V.14 Further research questions and topics

Source 03, delivered Section 11.

The following questions are proposed continuations of the proved results. Some are deliberately scoped finite problems, while others concern the larger proof architecture. A question proposed here is not being presented as a previously published open problem unless a source is identified.

Research question V.14.1 (The odd-order leading constant). For fixed $d \geq 3$, determine

$$c_d^{\text{odd}} = \sup_{\substack{G \text{ finite abelian, } |G| \text{ odd} \\ 0 \neq f: G \rightarrow \mathbb{R}, \mathbb{E}f=0}} \frac{\|f\|_{U^2}^4}{\|f\|_{U^d}^4}.$$

The exact quartic theorem reduces the best leading coefficient on odd-order groups to $R_d c_d^{\text{odd}}$. Is the supremum attained or approached by a real part of a character, by several Fourier modes, or by a different construction?

The universal bound $c_d^{\text{odd}} \leq 1$ follows from monotonicity. Our cosine example gives a positive explicit lower bound. The order-two character that attains the unrestricted coefficient is unavailable here, so the sharp universal result does not settle this restricted problem. A practical first project is a systematic search over small odd cyclic groups, followed by an inequality explaining any apparent extremizer. Numerical optimization would be evidence, not a proof of the supremum.

[write] **6 October 2026.** *Settled for $d = 3$ (cross-source finding).* Source 04's Theorem III.3.4 gives $c_3^{\text{odd}} = 2^{-1/2}$, attained exactly by the real parts $2\operatorname{Re}(c\xi)$ of nontrivial characters (the question's first alternative); proof in the note after that theorem. The best odd-order leading coefficient for $d = 3$ is $R_3 c_3^{\text{odd}} = 6\sqrt{2}$, as source 04 proves directly. For $d \geq 4$ the question is open; it coincides with source 04's Question 3 (Section V.2, item III.1).

Research question V.14.2 (Classify the fifth and sixth centered coefficients). For each $r = 5, 6$, classify the supports $S \subseteq \{0, 1\}^d$ of size r for which the character system (III.2.27) can have solutions with no trivial character. Derive multiplicities as functions of d and identify the necessary quotient correlations and Fourier phase invariants.

The coordinate-reflection test removes every support with a coordinate occupied by exactly one selected vertex or by all but one. For surviving supports, Smith invariant factors identify possible small-torsion obstructions. A successful classification should distinguish actual vanishing of a centered correlation from mere rational independence of its defining forms. The fifth-order examples discussed at the end of Section III.2 give concrete starting cases.

[write] **6 October 2026; sources 06–10.** *Fifth order answered, sixth order open.* For $r = 5$ the question is answered by source 09's Theorems III.20.1 and III.21.1 with Proposition III.20.2: two support types, multiplicities Q_d and S_d , and the correlations $\mathbf{M}_5$ (order-two coupling χ, χ^2) and $\mathbf{T}_{3,5}$ (order-three characters). For $r = 6$ it remains open; source 09 shows that an isolated order-five term occurs there (Theorem III.23.1) and asks the same question (Section V.2, item III.2).

Research question V.14.3 (The sharp fixed-threshold DRC profile). Let $F(\delta)$ be the largest guaranteed fraction $|K|/n$ under the exact hypotheses and success criterion of Lemma 7.4. Determine the correct order of $F(\delta)$ as $\delta \downarrow 0$. Can the exponent 3 in Theorem II.5.2 be improved using constraints specific to intersection matrices?

The scalar optimization proves only that a particular two-witness first-moment certificate cannot give a positive uniform estimate at the original parameters. Actual overlap matrices are symmetric positive semidefinite Gram matrices of indicators and obey further incidence constraints. It is a real additional problem to determine whether those constraints rule out the scalar bad distributions. Conversely, a lower bound on the necessary exponent requires realizable families of sets, not just a distribution of formal overlap values.

Research question V.14.4 (Optimize the entire elementary extraction). Optimize the three parameters $\gamma, \varepsilon, \beta$ in Proposition II.6.2, and compare alternative walk lengths and popularity thresholds. Which size–difference-set tradeoffs can be achieved while retaining only elementary finite inequalities?

The meaningful objective is the pair of conclusions, including the normalization of the difference-set bound by $|A|$ or by $|A'|$. Optimizing one displayed exponent while allowing the selected set to shrink excessively can conceal a loss. Comparison with [12] should remain explicit; the purpose of this project would be a simple formalizable proof or a better constrained tradeoff, not to mistake an elementary refinement for an improvement of the strongest known theorem.

Research question V.14.5 (Allocate phase oscillation by local mass). Suppose a phase partition can be refined at a cost depending on the desired oscillation in each cell. Can one minimize the total refinement cost subject to

$$\sum_j \int_{P_j} |f| |\psi_j - z_j| d\mu \leq \lambda \rho$$

by using local information on $|f|$, rather than a single global oscillation tolerance?

Lemma I.14.3 already provides the exact analytic constraint. The unresolved work is geometric: prove a refinement theorem with a cost function sharp enough for this optimization to matter. One can start with linear phases and known cell masses, then study polynomial phases. The global density factor in Theorem I.15.2 is the simplest instance of retaining this information.

Research question V.14.6 (Local counting that rectifies automatically). Develop a version of Theorem III.1.2 for interval-supported functions or localized norms in which the conclusion directly counts ordinary integer progressions and includes all boundary errors.

The elementary second-difference test gives a clean sufficient condition for rectification, but the global modular norm does not enforce that condition. The target is an explicit statement whose hypotheses reflect both uniformity and support. Any proposed bound should be tested on sets concentrated near the ends of the interval and on small composite moduli before it is fed into the density increment iteration.

Research question V.14.7 (Propagate improvements through a verified dependency graph). Replace the improved local lemmas in the later sections of the proof, record every dependence of density, progression length, dimension, and correlation on earlier parameters, and obtain a complete explicit iteration theorem.

Proposition V.12.1 specifies the final bookkeeping for one common type of increment. The hard work is to prove the hypotheses with the newly propagated functions. An initial useful deliverable would be a symbolic parameter dependency graph accompanied by Lean theorems for each monotone substitution. This would reveal which improvements survive and which are absorbed by later losses, even if the final asymptotic class stays unchanged.

Research question V.14.8 (Quantitative rounding and efficient derandomization). Optimize the group-size requirement for the sharpness constructions at a given error tolerance, including the exact-density version of Corollary III.2.12. Can one derandomize the rounding with an efficient finite algorithm?

The existence problem at a compatible prescribed rational density is already solved by the slow perturbation and small alteration in Corollary III.2.12. Its size requirement is deliberately generous. Sampling directly with a fixed total, a sharper norm estimate for the alteration, or a deterministic conditional-expectation argument might improve it. Independence no longer applies unchanged after conditioning on cardinality, so that approach needs its own estimate.

Research question V.14.9 (A reusable formal library of centered configuration counts). Build a finite-abelian-group library whose basic objects are systems of integer linear forms, centered multilinear averages, character constraint sets, and explicit reindexing bijections. Recover the cube and progression results as specializations.

The key interface should retain integer torsion data. A proof over $\mathbb{Q}$ alone does not control evaluation on all finite groups. At the same time, the library should make cancellation by a single free variable easy, so simple cases do not require a full Smith normal form calculation. This is an achievable software and formalization project with reusable mathematical outputs, independent of completing the entire Gowers proof.

V.15 Reproducibility and verification

Source 03, delivered Appendix A.

The accompanying archive contains the complete LaTeX source and compiled PDF, the figure in vector PDF form, scripts for the figure and finite checks, and the recorded check output. No original third-party paper is repackaged. The repository revision and source paths are recorded so that the comparisons can be repeated against the same input.

V.15.1 What the scripts check

`verify_cubes.py` uses only the Python standard library and exact rational arithmetic. It enumerates every four-vertex support of the Boolean cube for dimensions two through five. In dimension five this means all $\binom{32}{4} = 35960$ supports. It compares a classification from minors with a separate classification from XOR and coordinate patterns. The counts are exactly those in Lemma III.2.5. It also computes the coefficients of the full cube polynomial directly, without using the proposed quartic formula, on cyclic groups of orders two through six in dimensions two and three. The low-degree cancellations, torsion correction, and parity polynomial all pass.

`verify_estimates.py` checks the sharp selection extremizers with exact rational weights, evaluates the cubic certificate including its boundary and tangent cases, constructs proper phase partitions at several finite scales, and enumerates all subsets of $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z}$ for the progression bounds at lengths three and four. The norm powers in the latter check are rational; taking roots and comparing the bounds uses floating-point arithmetic with an explicit tolerance. The phase oscillation checks are also numerical. These are clearly distinguished from the exact cube and scalar checks.

Independent written reviews checked the asymmetric selection extremizers, finite progression geometry, Cauchy–Schwarz induction, quartic support classification and Bernoulli rounding, cubic minorant, and four-label counting injection. These reviews and finite computations are useful error checks; the proofs in the article, rather than the tested finite range, justify the theorems for arbitrary parameters.

V.15.2 Building the document

From the extracted directory, run

```
python3 verify_cubes.py
python3 verify_estimates.py
pdflatex -interaction=nonstopmode -halt-on-error gowers_local_refinements.tex
pdflatex -interaction=nonstopmode -halt-on-error gowers_local_refinements.tex
```

The figure is already included, so compiling the document does not require Python or a plotting library. To regenerate the figure, run `make_figure.py` in an environment with NumPy and Matplotlib. The LaTeX source uses standard packages including AMS mathematics, Palatino fonts, `microtype`, `booktabs`, and `hyperref`.

V.16 Quantitative comparisons and what they enable

Source 04, delivered Section 9.

The preceding results address different sources of loss. Their benefits should therefore be read at the interface where each estimate is used. The following table records those interfaces without combining unrelated parameters into a single claimed global improvement.

Interface	Coarse source estimate	Strengthened estimate here
Graph moment, input $\gamma B ^3$	$\gamma^7\beta^{21}p^{15}$	$\gamma^7\beta^{15}p^{15}$; arbitrary moment order and equality cases
Almost-homomorphism restriction, order eight	$(\gamma^7\beta^6\eta/4)^{2^{19}} B ^{15}$	$2^{-95}\gamma^{112}\eta^{15} B ^{15}$, under the explicit size hypothesis in Corollary II.9.5
Three-cube upper bound, $u = \ f\ _{U^3}$	$(\delta + u)^8$	Excess begins with $6\sqrt{2}\delta^4u^4$, with explicit higher terms
k -term progression counting	Error bounded by $2^k u$	Error bounded by $u \sum_{j=0}^{k-3} \delta^j$
Section 17 local coefficient	$2^{-2(k+1)^3}$	$2^{-k^2 - k \log_2 k - O(k)}$, with an explicit constant

Table 11: Comparisons with the specified local estimates in [1, 23]. The variables and hypotheses are those of the corresponding sections. The improved restriction theorem includes a large-field condition, and the phase theorem includes a characteristic condition.

The field-size requirement in the restriction theorem has a particularly simple interpretation:

$$|B| \geq \Lambda_d(\alpha\eta)^{-2d}.$$

The ambient density does not otherwise enter this sufficient condition. Two elementary decisions are responsible. A rank-two exceptional relation leaves at most $|B|^{2d-2}$ tuples, because its free coordinates can be chosen from B . These contributions are then summed with their nonnegative Fourier weights, whose total is at most one. Thus the number of possible coefficient vectors does not enter the error bound. Repeated entries are handled separately. These refinements are useful even before any attempt to optimize the kernel.

For an illustrative comparison, take $\gamma = \eta = \beta = 1/2$. The retained fraction of $|B|^{15}$ in the source display is $2^{-8388608}$, whereas the convenient bound here is 2^{-222} . The corresponding sufficient field-size condition here is $p \geq 2^{231}$. These are deliberately transparent evaluations of the proved formulas, not practical finite-size estimates. They show the change in parameter dependence and also the cost of the conservative exceptional-tuple estimate.

The cube results answer two precise local extremal questions. The best universal leading coefficient in the real centered, odd-order setting is $6\sqrt{2}$. In the setting without 2- or 3-torsion, the first possible correction after that leading term has optimal exponent $16/3$. What remains undetermined is the exact coefficient of this correction and the existence of a universal asymptotic expansion for the extremal envelope. These are local conclusions with explicit models, independent of any claim about the best quantitative bound for arithmetic-progression-free sets.

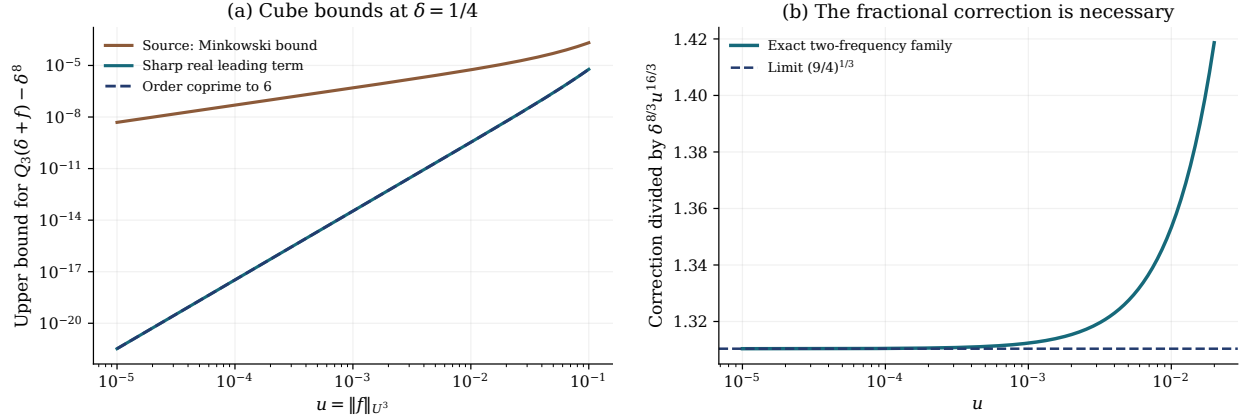


Figure 2: Proved bounds and an exact extremizing model at $\delta = 1/4$. Left: the excess above δ^8 in the source triangle-inequality bound and the two refined bounds of Theorems III.3.6 and III.3.8. The refined curves nearly coincide at small u because they have the same sharp leading term. Right: after subtracting that leading term, the explicit two-frequency family of Proposition III.3.9, divided by $\delta^{8/3} u^{16/3}$, tends to $(9/4)^{1/3}$. The figure evaluates formulas; it is not a statistical fit.

V.17 Further research questions and a proposed program

Source 04, delivered Section 10.

The questions below are proposed research targets arising from the calculations. The list distinguishes exact optimization problems from larger integration projects. It is not a claim that each formulation is already recognized as an open problem in the literature; a more extensive priority review should accompany work toward publication.

V.17.1 Extremal cube counts

Q1. Does the secondary extremal coefficient exist, and what is it? Fix $0 < \delta < 1$, and let $M_\delta(u)$ be the supremum of $Q_3(W)$ over prime cyclic groups of order greater than 16 and weights $0 \leq W \leq 1$ satisfying $\mathbb{E}W = \delta$ and $\|W - \delta\|_{U^3} = u$. Theorems III.3.8 and Proposition III.3.9 imply

$$(9/4)^{1/3} \leq \liminf_{u \downarrow 0} \frac{M_\delta(u) - \delta^8 - 6\sqrt{2}\delta^4 u^4}{\delta^{8/3} u^{16/3}} \leq \limsup_{u \downarrow 0} \frac{M_\delta(u) - \delta^8 - 6\sqrt{2}\delta^4 u^4}{\delta^{8/3} u^{16/3}} \leq 6. \quad (\text{Q1})$$

Determine whether the limit exists. Is the two-frequency value $(9/4)^{1/3}$ optimal? A useful first step is a sharper estimate of the quintic correlation in terms of the exact loss in the norm-gap inequality, instead of the separate bounds used above.

Q2. What is the correct stability statement for cube extremizers? Corollary III.3.5 controls the fourth-power Fourier mass outside one conjugate pair, equivalently a U^2 remainder. This does not alone control an L^2 remainder: many small Fourier coefficients can have little fourth-power mass but substantial second-power mass. Determine the strongest stability conclusion for bounded weights that nearly attain the leading cube bound, and identify any additional hypothesis needed for a stronger metric. For near optimal secondary behavior, test whether concentration on frequencies $\{\pm\xi, \pm 2\xi\}$ follows.

Q3. What replaces the sharp norm gap in higher dimensions? For $d \geq 4$, determine the best constant in the comparison of $\|f\|_{U^d}^{2^d}$ and $\|f\|_{U^2}^{2^d}$ for real centered functions, specifying the allowed group orders. Combine it with the exact coefficient P_d of Proposition III.3.2. The Boolean support matroid gives a finite organizing device for the next correlations, but their values and sharp norm bounds require additional harmonic analysis. Separate large-characteristic behavior from the extra identities produced by small torsion.

[write] **6 October 2026.** For $d = 3$ the comparison is Theorem III.3.4; for $d \geq 4$ this is source 03's Research question V.14.1 (which asks for the constant over odd-order groups). Merged in Section V.2, item III.1.

V.17.2 Restriction and moment estimates

Q4. Can the size threshold be substantially lowered? Theorem II.9.1 requires $|B| \geq \Lambda_d(\alpha\eta)^{-2d}$. The proof already controls exceptional relations in expectation using their Fourier weights. Can their contribution be made to reflect the additive structure of B , rather than the uniform rank-two bound? Can a kernel with a more favorable coefficient distribution reduce the required size without weakening the retained tuple count? Any proposed improvement must handle repeated entries as well as extra relations among distinct entries.

Q5. What is the true optimal retention law? The bound $\kappa_d \alpha^{2d} \eta^{2d-1} |B|^{2d-1}$ has the natural parameter powers for the single-kernel expectation budget in Proposition II.9.6. That proposition does not establish optimality among all selection procedures. Construct upper-bound examples, or find a different selection method with better powers. Also ask for useful simultaneous guarantees on $|B'|$, its ordinary additive energies, and the density of respected relations.

Q6. How far does restriction extend beyond prime cyclic groups? The interpolation theorem already applies to arbitrary finite abelian groups. The restriction theorem uses a field to turn two independent linear equations into a uniform bound on the number of solutions. Develop versions for $\mathbb{Z}/N\mathbb{Z}$, finite-field vector spaces, and maps into other groups. The appropriate hypotheses should quantify kernels of the relevant coefficient maps, rather than silently treat nonzero determinants as invertible. A weighted formulation would also allow nonuniform input measures.

Q7. Can the cross-section profile be propagated through the proof? For arrangements the gain is governed by $\sigma = \mathbb{E}_r \beta_r^3$, not just by the global density. Track this profile through subsequent restrictions and changes of variables. Determine which operations preserve useful control of σ , and whether a collection of higher cross-section moments gives a stronger interface than a single density parameter. Equality models in Section II.10 provide tests against overly optimistic exponents.

V.17.3 Patterns, counting, and integration

Q8. Determine the asymptotics of the fractional pattern counts. The proved estimates imply

$$\frac{1}{2} \leq \liminf_{k \rightarrow \infty} \frac{\log_2 F_k}{k^2} \leq \limsup_{k \rightarrow \infty} \frac{\log_2 F_k}{k^2} \leq 1, \quad (\text{Q2})$$

and the same bounds for A_k . Determine the leading constant, or improve the gap in these bounds. Relate the particular affine arrangements here to established results on threshold patterns and subset-sum arrangements before assigning novelty. Exact certified counts for $k = 4$ are a concrete computational next step; exploiting symmetries is likely to matter more than merely using a faster numerical solver.

[write] 6 October 2026. *Answered by source 01* (cross-source finding of the intake, checked at the write). The F_k here is source 01's C_k and the A_k here is source 01's A_k : the definitions are identical. Source 01's Theorem IV.2.3 gives $k^2 - k \log_2 k - O(k) \leq \log_2 C_k \leq k^2 + (\log_2 e - 1)k$. Its Theorem IV.3.1 gives $C_k \leq A_k$ and $\log_2 A_k \leq k^2 + O(k)$. Hence both limits in (Q2) exist and equal 1. The lower half uses one external input, the classical threshold-function count (Zuev; Kahn–Komlós–Szemerédi, as recorded by Baldi–Vershynin [8]), which source 01 cites and does not reprove. Source 01's Proposition IV.5.1 certifies $F_4 = C_4 = 154$, the “concrete computational next step” asked for. A_4 is not certified (an uncertified recount gives 770), and the secondary term is source 01's first question; both remain open (Section V.2, items IV.2 and IV.1).

[write] 6 October 2026; sources 06–10. Source 06 answers the same question independently: its Theorem IV.7.6 gives $\log_2 C_k = (1 + o(1))k^2$ and $\log_2 D_k = (1 + o(1))k^2$ (the same external threshold count), uniformly modulo every $M \geq 2$, and its Section IV.9 certifies $C_4 = 154$ a second time.

- Q9. Which global and local profiles are actually realizable?** The phase-extraction proof uses the uniform bound 2^k for the profiles at a fixed slope and an arrangement bound for all slopes. The slopes there come from a short discrete interval. Determine the joint distribution of profiles in that arithmetic setting, including small moduli M , and test whether an average multiplicity can replace the worst-case local multiplicity. The exact modular formula of Theorem IV.6.3 isolates one source of collisions.
- Q10. Can progression counting use more of the mixed information?** The dynamic program in Section III.4 finds the best bound within a specified telescoping family. It is not an optimality theorem for all counting arguments. Investigate whether mixed uniformity norms, additional cancellations, or information about particular pairs of factors can lower that bound. Any extension to nonadmissible slopes should display the relevant subgroup averages explicitly; the $\mathbb{Z}/2\mathbb{Z}$ example rules out simply dropping the invertibility assumptions.
- Q11. What survives complete propagation through the density increment?** Create a dependency graph for the quantitative parameters of the original proof. Substitute the improved moment, restriction, and profile estimates, and recompute every subsequent threshold with the same hypotheses. The output should be an explicit final function of k and density, with each loss traceable to a proved lemma. Compare the resulting bound with both the original argument and later work [2, 3]; an improvement in the historical proof need not improve the stronger modern theorem.
- Q12. Can the refinements be verified by a proof assistant?** Formalize the lowest-risk interfaces first: exact moment interpolation, the density-weighted telescope, and the finite support enumeration. Then formalize the selection theorem, where the main obligations are finite Fourier orthogonality, independent sampling, and the exceptional rank count. A fully checked dependency chain would clarify which quantitative gains are compatible with the later induction, and would separate a correct theorem statement from a completed formal proof. Appendix V.19 gives a proposed implementation order.

Questions 1, 4, and 8 offer relatively contained starting points: a sharp analytic constant, a finite selection threshold, and a concrete arrangement census. Questions 7 and 11 are integration projects,

for which local success must be accompanied by careful propagation of hypotheses. The distinction helps keep a proposed research program testable.

V.18 Closing perspective

Source 04, delivered Section 11.

The strongest conclusions of this investigation are explicit local theorems with complete proofs. Exact moment accounting feeds a polynomial restriction theorem; centered Fourier structure yields sharp cube coefficients and a fractional correction exponent; and simultaneous floor-pattern counting replaces a cubic exponential local loss by a quadratic one. The sharpness examples identify which aspects have already reached their natural limit and which constants remain to be determined.

The next substantial step is to turn these interfaces into a fully checked quantitative dependency chain. That would establish how much of the local improvement survives later stages of Gowers’s argument, while the extremal and pattern questions above offer independent research directions with precise targets.

V.19 A map to the Lean development

Source 04, delivered Appendix A.

This appendix identifies interfaces in the inspected snapshot [23]. It records a plan for additional work; it is not a report that new Lean declarations have been proved. A declaration defining a proposition, a theorem with a proof term, and a theorem successfully checked by the Lean kernel are different pieces of evidence.

One signature issue deserves explicit review. The inspected definitions of `lemma_9_3` and `corollary_9_4` in `Sections08_09.lean` quantify over nonzero moduli without an explicit primality typeclass. The selection theorem in this article uses primality essentially. It must therefore be formalized as a separate prime-specialized statement, or accompanied by a justified revision of the intended source interface. It is not automatically a proof of the broader existing proposition. The repository already has proofs of those statements through another route; the present claim is a stronger prime-field interface. By contrast, the exact moment inequality itself does hold over arbitrary finite abelian groups.

[write] 6 October 2026. Checked at HEAD: `lemma_9_3` and `corollary_9_4` still quantify over all $N \geq N_0$ without primality and have exact companions, as stated. The table omits one interface that was already formal at source 04’s pin: Lemma III.4.5 is `hasNatAP_of_short_modular_sequence` (note there). The other proposed strengthenings are not formalized.

For a practical implementation order, first isolate a normalized and an unnormalized Fourier convention with explicit conversion lemmas. Next prove the exact moment and telescoping statements, which need little specialized finite enumeration. Then introduce a library of finite Boolean support certificates for the cube identities. Finally add the probabilistic selection and floor-pattern arguments. Each step should compile against the pinned toolchain of the repository, and the final verification report should name the checked declarations and the commit. None of those future verification steps is represented as complete here.

Inspected interface	Proposed strengthening and principal proof obligations
<code>Proofs09Moments.lean;</code> <code>lemma_9_2_holds</code>	Keep the exact second moment before applying the higher-moment inequality. State the arbitrary-order graph version, with $ B $ in the denominator. The proof already computes an exact cardinality before weakening it.
<code>Proofs03CubeUpper.lean;</code> <code>lemma_3_10_holds</code>	Add centered cancellation and the real odd-order norm gap as separate lemmas. Prove the twelve quartic and eight quintic support classifications with explicit finite certificates or a transparent finite case split.
<code>Proofs03ProgressionAverage.lean;</code> <code>theorem_3_2_holds</code>	Generalize the required changes of variables to admissible slopes. Make multiplication automorphisms explicit, and reuse the Gowers–Cauchy–Schwarz infrastructure.
<code>Proofs03ProgressionCount.lean;</code> <code>corollary_3_3_holds</code>	Replace the full subset expansion by a finite telescoping identity. Retain density products and cancel the exact two-factor average.
<code>Proofs09Restriction.lean;</code> <code>lemma_9_3_holds</code> and <code>corollary_9_4_holds</code>	Introduce a prime-specialized version of the strengthened restriction theorem. Formalize the rank-two completion bound, the nonnegative Fourier weights, and the random-selection expectation.
<code>Sections17_18.lean</code>	Add definitions of fractional patterns and normalized profiles, followed by the hyperplane bound and the local phase-extraction theorem with explicit characteristic assumptions.

Table 12: Suggested interfaces for formalizing the proved refinements.

V.20 Reproducibility and supplementary checks

Source 04, delivered Appendix B.

The source archive contains the complete L^AT_EX manuscript, its compiled PDF, the figure source and PDF, a provenance manifest, and three independent verification programs. The programs are deliberately supplementary to the proofs. In particular, testing finitely many functions cannot prove the norm-gap theorem, and checking small groups cannot prove the restriction theorem for all sufficiently large primes.

V.20.1 Energy, cube, and coefficient checks

`verify_refinements.py` uses the Python standard library. The recorded run checked all 8 470 nonempty graphs of partial maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ for $n = 2, 3, 4, 5$, giving 59 290 exact integer instances of the energy inequality for moment orders $2, \dots, 8$. It also enumerated the three-cube support ranks over the rationals and over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, and counted parallelograms through dimension seven. These computations exhibit the different support count in characteristic two.

The kernel checks use exact rational arithmetic for the selected coefficient ratios and the central-binomial inequality, and exact integer arithmetic for the convenient powers of two in Corollary II.9.5. The complete two-frequency polynomial in Proposition III.3.9 is independently recovered with rational coefficients.

Floating-point checks are identified separately in the results file. They test the Fourier identity and norm gap on 40 real centered functions on small odd cyclic groups, the cosine examples, and 36 density-weighted progression identities, including an admissible example modulo 25. The largest recorded Fourier-identity discrepancy is less than 1.1×10^{-16} , and the progression identity

discrepancy is less than 3.4×10^{-17} . The asserted numerical tolerance is 2×10^{-11} . These are consistency checks, not rational certificates for arbitrary functions.

The supplied `verify_selection.py` checks the selection expectation bounds directly by exact rational Fourier expansion in 10 examples over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, covering 40 584 ordered additive tuples. It treats each distinct selected point only once and includes 504 distinct unrespected six-tuples. These checks use the degree-one kernel and supplement the proof of the general weighted selection lemma; the large-degree constants are checked separately.

V.20.2 Exact floor-pattern certificates

`pattern_census.py` verifies the supplied `pattern_census.json` using only rational arithmetic from the Python standard library. It obtains

$$(F_1, F_2, F_3) = (1, 2, 10), \quad (A_1, A_2, A_3) = (2, 6, 38).$$

Every branch in the hyperplane enumeration has one of two certificates: a rational point satisfying all relevant strict inequalities, or nonnegative rational weights whose combination proves that those strict inequalities are infeasible. The verifier checks both sides of every split, so it checks completeness as well as feasibility. There are 450 certified branches in the supplied record. Generating the certificates uses a numerical linear-programming solver to propose candidates, but accepting them and verifying the saved census does not use numerical tolerances.

V.20.3 Build and verification commands

From the extracted archive directory, the following commands rebuild the article and check its supplementary records:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  gowers_quantitative_refinements.tex
python3 -S verify_refinements.py --output verification_results.json
python3 -S verify_selection.py --report selection_verification.json
python3 -S pattern_census.py --verify pattern_census.json
```

The prebuilt vector figure is included, so rebuilding the article does not require Python plotting libraries. To regenerate the figure, run `python3 make_figures.py`; that optional step requires NumPy and Matplotlib. Certificate generation additionally requires SciPy. The source manifest records the inspected repository commit and the relevant Git object identifiers. The archive does not redistribute the original paper or the full repository.

V.21 Relation to the Lean development

Source 05, delivered Section 7.

V.21.1 Inspected source and a conservative integration plan

The relevant files are `Proofs02Partition.lean`, `Proofs02PhasePartition.lean`, and `Proofs02DensityIncrement.lean`; the first and third were inspected in detail in the ranges

relevant to these results. The existing density-increment file imports the phase-partition file. The repository uses `NatAP`, its carrier and properness predicate, and a finite partition relation. Its partition construction already groups residue classes into blocks with lengths at least L and less than $2L$, matching the integer mechanism used here [25].

The efficient formalization route is to keep that construction and expose a parameterized theorem, rather than duplicate it for each numerical constant. The mathematical contract should assume $2L(L-1)M \leq rs$ in the modular version and conclude a partition with cell lengths in $[L, 2L-1]$ and image diameter at most s . The proof then invokes the existing small-image-step lemma with $Q = \lfloor r/L \rfloor$.

The sharp capped result becomes a separate integer calculation with $U = \lfloor \sqrt{rs/M} \rfloor$ and $K = \lceil U/2 \rceil$. The optimality example can be formalized independently using the cardinality of a disjoint union of two-element cells. It does not require complex analysis.

For the density theorem, the main additional interfaces are the exact identity $\sum |\mathbf{1}_A - \delta| = 2\delta(1-\delta)N$, a midpoint or disk-center phase estimate, and the capacity-sensitive positive-mass lemma. The existing file already proves the zero-sum identity, density bounds, positive/negative mass equality, and nonzero Fourier-coefficient bounds by both the set and its complement [26].

V.21.2 Recommended separation of proof layers

There are four natural layers. The first is integer combinatorics: proper progressions, residue classes, lengths, and disjointness. The second is phase approximation on $\mathbb{R}/\mathbb{Z}$ or $\mathbb{ZMod} N$. The third is finite weighted real and complex inequalities. The fourth specializes to Fourier coefficients and density increments. Keeping these layers separate permits the same weighted lemmas to be used with later polynomial-phase partitions.

For a minimum-risk first contribution, formalize the sharp $1/3$ modular theorem and the strengthened Corollary 2.4 before introducing the spectral-energy material. The simultaneous box lemma is another independent milestone. The algebraic obstruction is mathematically self-contained, but its Lean implementation would additionally need algebraic-number and resultant or norm infrastructure; it need not block the upper partition bounds.

V.21.3 Verification status

No new Lean file in this package is claimed to compile, and no new theorem has been kernel-checked here. The package supplies ordinary mathematical proofs, an exact-arithmetic implementation of the finite partition constructions, and a formalization roadmap. We did not perform a whole-repository proof audit or infer the status of the complete Gowers development from the inspected modules.

The observed `main` reference during this work was

327773149bd272239f1f6c0015f76d5097512c88.

The accompanying source-audit file records the content hashes of the compared files. The partition module was also fetched explicitly at that commit. The other inspected contents are identified by their returned blob hashes; the moving branch is not treated as a permanent identifier.

[write] **6 October 2026.** Checked at HEAD: the cited blobs are correct and unchanged (291b149 for `Proofs02Partition.lean`, 85b5c3a for `Proofs02DensityIncrement.lean`, 41085a0 for the transcription). One defect of the shipped audit file `05-phase-partitions-SOURCE_AUDIT.md`: it says the commit-pinned

read of `Proofs02Partition.lean` “included lines 580–780”, but that file has 702 lines; the blob and the construction it describes (blocks of lengths at least L and less than $2L$) are correct. The audit file is shipped unchanged (README).

V.22 Reproducible computation and the limits of testing

Source 05, delivered Section 8.

The companion implementation `code/phase_partitions.py` uses Python integers and rational fractions for all partition decisions. It implements the exact scale (I.16.2), the capped $1/3$ theorem, the simultaneous ceiling condition (I.18.2), and the tolerance allocation (I.19.10). An independent checker verifies complete coverage, no repeated points, proper arithmetic progressions, one common difference, the claimed cell lengths, the number-of-cells bound, and exact circular arc widths.

The completed exact tests comprise 13,325 one-phase cases, 11,440 capped cases, 320 simultaneous cases, and 200 tolerance-allocation cases. The one-phase tests vary the interval length from 1 to 41, all slopes with denominators from 2 to 11, and five rational widths. Of the simultaneous cases, 166 have minimum length greater than one. The seeded random tests use dimensions from one to four. These checks test rounding and implementation; they are not substitutes for the proofs in Sections I.16 and I.18.

Fourier diagnostics use 80 decimal digits. There are 60 correlation-and-popularity cases and 96 spectral-energy cases. Twenty-four spectral cases have a non-singleton guaranteed minimum length, including six genuinely multifrequency cases at larger sizes. The larger examples are modular pullbacks of intervals, with their Fourier coefficients computed by a geometric-series formula independently of the partition construction. All tested inequalities passed with positive observed margins. These are numerical diagnostics, not exact certificates for transcendental inequalities.

The verification files distinguish exact structural tests from high-precision diagnostics and explicitly record that new Lean verification was not performed. To reproduce the results, run the commands in the package README. No source paper, repository file, or font file is redistributed.

V.23 Further research questions

Source 05, delivered Section 9.

The following questions arise naturally from the proved results. Unless explicitly stated otherwise, they are research directions formulated here; we do not assert that they are all previously recognized open problems.

1. Exact finite partition profiles. The universal constant $1/3$ with the fixed upper cap is settled above, but the exact best lower length for each triple (r, s, M) and each slope can be much larger. Determine an efficient characterization of the feasible pairs of minimum and maximum lengths, taking account of residue-class sizes rather than only a worst-case blocking lemma. The explicit obstruction at total size nine explains one universal constant, not the full profile.

2. Best constants in the simultaneous minimax theorem. The exponent and the symmetric dependence on ε are sharp, but the upper constant obtained from the elementary algebraic norm argument is very poor. Determine the optimal leading constants, first for one or two phases, and

the extent to which requiring a common difference across the whole partition changes them. One must specify the order of the limits in N and ε .

3. Anisotropic lower bounds. The lower construction depends on $\prod_i \varepsilon_i$. Is there a matching minimax upper bound in all highly anisotropic regimes, up to dimension-dependent constants and an additive one? A coordinate with a very loose tolerance can cease to be a genuine constraint, so a naive product formula may require a regime decomposition. Weighted Diophantine approximation is the natural language, but no such theorem is assumed here.

4. Optimal density-sensitive correlation transfer. The disk radius and the capacity inequality are separately sharp, but their simultaneous extremizers need not arise from an indicator and a linear phase. Determine the exact feasible region of block positive mass, Fourier correlation, density, and phase width. Even a finite-dimensional extremal formulation could improve the trade-off (I.17.9) and identify which constant losses are genuine.

5. Adaptive selection of frequencies and tolerances. The continuous allocation problem is solved by Proposition I.19.3, but selection of the frequency subset and optimization of the exact ceiling formula are not. Develop an algorithm that balances captured energy, coefficient imbalance, dimension, and the final integer progression length. Its correctness should be stated relative to a certified Fourier input rather than floating-point estimates alone.

6. Higher-degree and nilsequence interfaces. Apply the residual-variance inequality to valid higher-degree localization theorems. The challenge is to track density-sensitive oscillation budgets together with geometric complexity and exceptional cells. A useful target is an explicit theorem where all losses are parameterized, so that an improvement of a partition result propagates mechanically through the density argument. This would still need a separate analysis before it implied any global Szemerédi improvement.

7. Formalizing sharpness as well as existence. A Lean formalization should include the parity obstruction proving optimality of $1/3$, not just the existence theorem. This makes the quantitative API resistant to later accidental strengthening. The next independent target is the exact residual inequality (I.19.4); it isolates a reusable finite-probability fact without depending on the full higher-order proof.

8. Interval-localized spectral extraction. On a proper interval, restricted characters have a nontrivial Gram matrix. Corollary I.19.4 provides an explicit scalar-norm correction, but a large value of κ can erase the gain from several correlations. Establish useful certified conditioning bounds for specific frequency configurations, and optimize the linear combination using more of the Gram matrix than its largest eigenvalue. This would turn the abstract extension into an effective local Fourier theorem.

V.24 Conclusion

Source 05, delivered Section 10.

The finite localization step admits stronger statements than the corrected repository currently exports. The lower constant $1/3$ in the capped partition theorem is exactly optimal; the new Corollary 2.4 retains the same upper cell size with a better lower size and correlation retention; and the density increment uses the true mass $2\delta(1 - \delta)$ instead of a unit bound. Simultaneous localization has a clean, sharp exponent, and a residual-variance inequality turns an appropriate Fourier-energy hypothesis into a multiplicative increment with a density-sensitive tolerance budget.

These are local quantitative results with complete proofs and reproducible computational checks. Their place in the larger program is to provide stronger interfaces and clearer bottlenecks, not to bypass the higher-order structural work needed for a new global theorem.

V.25 An explicit potential for density-dependent iteration

Source 06, delivered Section 6.

The iteration in the proof of Gowers' Theorem 18.2 estimates every density increment by the one available at the starting density [1]. Tracking the increasing density instead gives a sharper bound. We first prove a general deterministic lemma; it separates this bookkeeping improvement from the local structural theorem that supplies the increment.

Lemma V.25.1 (Discrete potential). *Let $0 < \delta_0 \leq 1$ and $t \geq 1$. Let F be positive, nondecreasing, and continuously differentiable on $[\delta_0, 1]$. Let G be nonnegative and nonincreasing there. Suppose*

$$\delta_0 < \delta_1 < \cdots < \delta_{t-1} \leq 1, \quad \delta_{i+1} \geq \delta_i + F(\delta_i).$$

Then

$$\sum_{i=0}^{t-1} G(\delta_i) \leq G(\delta_0) + \int_{\delta_0}^1 G(u) \left(\frac{1}{F(u)} + \frac{F'(u)}{F(u)} \right) du. \quad (\text{V.25.1})$$

Proof. For $x < y$ with $y - x \geq F(x)$, put $r = F(y)/F(x) \geq 1$. Monotonicity gives

$$\begin{aligned} \int_x^y \left(\frac{1}{F(u)} + \frac{F'(u)}{F(u)} \right) du &\geq \frac{y - x}{F(y)} + \log r \\ &\geq \frac{1}{r} + \log r \geq 1. \end{aligned}$$

The last inequality follows because $r^{-1} + \log r$ is nondecreasing for $r \geq 1$ and equals one at $r = 1$. Since $G(u) \geq G(y)$ on $[x, y]$, the same integral weighted by G is at least $G(y)$. Apply this observation to the interval preceding each summand except the first, and add. $\square$

Taking $G = 1$ gives a useful count alone:

$$t \leq 1 + \int_{\delta_0}^1 \frac{du}{F(u)} + \log \frac{F(1)}{F(\delta_0)}. \quad (\text{V.25.2})$$

For $F(u) = cu^p$ with $p > 1$, this becomes

$$t \leq 1 + \frac{\delta_0^{1-p} - 1}{c(p-1)} + p \log \frac{1}{\delta_0}. \quad (\text{V.25.3})$$

The logarithmic correction makes the integral estimate valid for the discrete process without assuming infinitesimal steps.

V.25.1 A power-law density increment and length loss

Theorem V.25.2 (Explicit iteration budget). *Suppose a hereditary configuration problem has the following valid local alternative. At any density $0 < \delta \leq 1$ and length $N \geq N_* > 1$, either the desired configuration exists, or restriction to a subprogression gives density and length satisfying*

$$\delta' \geq \delta + c\delta^p, \quad N' \geq N^{a_0\delta^q}, \quad (\text{V.25.4})$$

where $c > 0$, $0 < a_0 \leq 1$, $p \geq 1$, and $q \geq 0$ are fixed. Assume the same alternative applies after each restriction, and fix an initial density $0 < \delta_0 \leq 1$. Set

$$A = \log(1/a_0), \quad L = \log(1/\delta_0).$$

For $p > 1$, put $r = p - 1$ and define

$$\begin{aligned} B(\delta_0) = A + qL + \frac{A\delta_0^{-r} - 1}{c} \frac{1}{r} \\ + \frac{q\delta_0^{-r}(rL - 1) + 1}{c} \frac{1}{r^2} + pAL + \frac{pq}{2}L^2. \end{aligned} \quad (\text{V.25.5})$$

For $p = 1$, define instead

$$B(\delta_0) = A + qL + \left(\frac{1}{c} + 1\right) \left(AL + \frac{q}{2}L^2\right). \quad (\text{V.25.6})$$

Then a sufficient initial length is

$$\log \log N \geq \log \log N_* + B(\delta_0). \quad (\text{V.25.7})$$

All logarithms in these formulas are natural.

Proof. Writing $H = \log \log N$, the length conclusion in (V.25.4) loses at most

$$G(\delta) = A + q \log(1/\delta)$$

from H at one step. Apply Lemma V.25.1 with $F(u) = cu^p$. Its integral is

$$\int_{\delta_0}^1 (A + q \log(1/u)) \left(\frac{1}{cu^p} + \frac{p}{u} \right) du. \quad (\text{V.25.8})$$

For $p > 1$, the elementary identities

$$\int_{\delta_0}^1 u^{-p} du = \frac{\delta_0^{1-p} - 1}{p - 1}$$

and

$$\int_{\delta_0}^1 u^{-p} \log(1/u) du = \frac{\delta_0^{1-p}((p-1)L - 1) + 1}{(p-1)^2}$$

evaluate (V.25.8) and give (V.25.5). Direct integration for $p = 1$ gives (V.25.6).

If no desired configuration exists, apply the local alternative repeatedly. After every finite number of restrictions, (V.25.1) and (V.25.7) leave $\log \log N_i \geq \log \log N_*$, so the local alternative remains applicable. Every step increases density by at least $c\delta_0^p > 0$. Such a sequence cannot stay at most one indefinitely, which is a contradiction. $\square$

A nonincreasing density-dependent threshold $N_*(\delta)$ causes no change: use $N_*(\delta_0)$ as the common threshold. A fixed length prefactor is also explicit. If the local length conclusion is

$$N' \geq C^{-1} N^{a_0 \delta^q}, \quad C \geq 1,$$

use the stronger working threshold

$$N_{\text{work}} = \max \left\{ N_*, \exp \left(\frac{2 \log C}{a_0 \delta_0^q} \right) \right\}. \quad (\text{V.25.9})$$

At every retained density, this implies

$$\log N' \geq a_0 \delta^q \log N - \log C \geq \frac{a_0 \delta^q}{2} \log N.$$

Thus Theorem V.25.2 applies with a_0 replaced by $a_0/2$ and N_* replaced by N_{work} .

V.25.2 Optimal leading order and the scope of the improvement

For fixed a_0, c, p, q with $p > 1$, formula (V.25.5) has the expansion

$$B(\delta) = \frac{\delta^{1-p}}{c} \left(\frac{A + q \log(1/\delta)}{p-1} - \frac{q}{(p-1)^2} \right) + O((\log(1/\delta))^2 + 1). \quad (\text{V.25.10})$$

Its leading term is optimal for the abstract local rules (V.25.4). Indeed, take the slowest allowed density sequence

$$\delta_{i+1} = \delta_i + c \delta_i^p$$

and charge exactly $G(\delta_i)$ at each step until a fixed positive target δ_* is reached, choosing $\delta_* + c \delta_*^p < 1$ so that the last step stays below 1. Since G/F is nonincreasing,

$$\int_{\delta_i}^{\delta_{i+1}} \frac{G(u)}{F(u)} du \leq \frac{G(\delta_i)}{F(\delta_i)} (\delta_{i+1} - \delta_i) = G(\delta_i).$$

Summing gives a lower bound by the main integral, up to a constant depending on the fixed target. Lemma V.25.1 exceeds that integral by only $O((\log(1/\delta))^2 + 1)$. Consequently, the leading coefficient in (V.25.10) cannot be improved using only the local recurrence and its charged length loss. Additional structural information would be necessary.

For comparison with the scale in Gowers' final iteration, suppose a valid local theorem supplies

$$\delta' \geq \delta + (\delta/2)^p, \quad N' \geq N^{(\delta/2)^p}. \quad (\text{V.25.11})$$

Then $c = a_0 = 2^{-p}$ and $q = p$. For fixed $p > 1$ and $\delta \rightarrow 0$, the leading double-logarithmic cost is

$$\frac{2^p p}{p-1} \delta^{1-p} \log(1/\delta).$$

Freezing density throughout gives a cost asymptotic to

$$2^p p \delta^{-p} \log(1/\delta).$$

Their ratio is asymptotic to $\delta/(p-1)$: the potential saves one power of the starting density in the double-logarithmic threshold. This changes neither the tower height nor the local structural input.

In particular, the repository snapshot explicitly records that its edited proof of Theorem 18.1 no longer establishes the displayed tower exponent after its revision of Corollary 16.11 [41]. Specializing (V.25.11) to that particular exponent is therefore conditional on a valid local increment theorem; Theorem V.25.2 itself has the complete proof given above.

The two refinements in these sections are complementary. The extraction theorem preserves the information already present in a partition, including the occupied negative mass and any discarded exceptional set. The iteration theorem preserves the subsequent increase in density. Both identify exactly which additional information would be required for a larger improvement.

[write] **6 October 2026; sources 06–10.** Iteration bookkeeping is also in source 03’s Proposition V.12.1 and source 10’s Propositions V.42.1 and V.42.2; the leading coefficients of the step counts are compared in the note after source 03’s proposition (the coefficient here is the smallest). The statement about Theorem 18.1 is still true at HEAD: `theorem_18_1`, `theorem_18_2` and `corollary_16_11` are open in the ledger.

V.26 Compatibility with the Lean development

Source 06, delivered Section 8.

The new mathematical statements in this report have not been added to the repository or compiled in Lean. The purpose of this section is to give an implementation map grounded in inspected definitions and proof endpoints, so that a subsequent formalization has a precise starting point. All names below belong to the namespace `LeanProofs.GowersSzemerédi` in the pinned snapshot [41].

V.26.1 Normalization and statement boundaries

The repository’s `gowersNorm d f` is unnormalized:

$$\|f\|_{U^d(G)} = \frac{\text{gowersNorm } d \ f}{N^{(d+1)/2^d}}.$$

Its predicate `UniformOfDegree f alpha d` corresponds to

$$\|f\|_{U^{d+1}}^{2^{d+1}} \leq \alpha.$$

Both the normalization and the index shift must be preserved when translating the quadratic counting criterion. The development’s `progressionAverage` is an unnormalized double sum, so the normalized discrepancy formulas here acquire a factor N^2 .

The existing endpoint `theorem_3_2_holds` lives in `Proofs03ProgressionAverage.lean`. The exact source companion `corollary_3_3_holds` is in `Proofs03ProgressionCount.lean`; it currently retains the 2^k factor. The source’s monotonicity and progression-existence companions are in `Proofs03Basic.lean` and `Proofs03ProgressionExistence.lean`. The public `gowers_cauchy_schwarz` in `Proofs03GowersCS.lean` supplies the mixed-cube inequality used in the localized phase proof.

V.26.2 A practical sequence of formalization steps

1. **Introduce a generic weighted extraction theorem.** Formalize Theorem I.21.1 using finite positive weights and cell averages, independently of cyclic groups. Derive `lemma_5_15_holds`

and its stronger replacements as special cases. The proof needs finite-sum identities, non-negativity, and pigeonhole arguments. The rational extremizers provide explicit sharpness witnesses.

2. **Replace the subset expansion in progression counting.** First add the indexed telescoping theorem without changing the existing uniformity assumptions. Then prove the two-target Cauchy–Schwarz induction and the quadratic bound. Keeping the first-order theorem separate is useful because it needs uniformity only for the later inputs, whereas the convenient homogeneous quadratic statement assumes a common higher norm bound.

3. **Expose finite carry families.**

The source defines `floorDotPattern` and `floorAffinePattern`. The two exact companions are `corollary_17_5_holds` and `corollary_17_6_holds`. Their proofs appear in `Proofs17FinitePatterns.lean`. That file splits the intercept into integer and fractional parts but does not use the exact split of every slope developed here. New finite-image definitions for $C(V)$ and $D(V)$ would make Theorems IV.7.3 and IV.7.4 natural cardinality statements. The one-sided perturbation lemma should be formalized before invoking the region count.

4. **Parameterize localized phase removal by a pattern bound.** The proof in `Proofs17LocalizedPhaseRemoval.lean` already isolates the raw inequality with a factor L^2 and a numerical lemma requiring $L^2c < 1/4$. Expose a version whose input is any valid pattern bound $L \leq K$; instantiate it with $c = 1/(8K^2)$. This permits a local replacement without rewriting the Fourier and cube algebra.
5. **Keep the Section 18 dependency conditional until its inputs are proved.** The actual theorem `section18_cyclic_density_increment_holds_of_theorem_18_1` has an explicit Theorem 18.1 hypothesis. Writing its exponent as ε , it obtains a density gain $\varepsilon/4$ and a progression size at least $(\varepsilon/4)N^\varepsilon$. The size prefactor must be retained or absorbed by a proved threshold estimate. The potential in Section V.25 is a separate, reusable way to perform the remaining iteration once a stable local theorem with explicit hypotheses is available.

The interval and affine transfer modules already document an additional quantitative issue: embedding a progression of length L into a cyclic group of order M changes ambient density by the factor L/M . A proof that repeatedly uses the new modulus cannot keep the old density without accounting for this factor. Our modular existence criterion does not remove that obligation.

V.26.3 What a completed formalization would certify

The existing audit distinguishes ordinary theorem companions from conditional implications and checks their axiom dependencies. A formalization of the proposed refinements should retain that distinction. There are three different goals: proving each auxiliary inequality, proving that its hypotheses hold at a particular use site, and proving that all constants survive the complete iteration. The first two can be made substantially more transparent by the exact interfaces above; the third remains a separate quantitative theorem.

[write] **6 October 2026; sources 06–10.** *Checked at HEAD (764740f07):* every declaration named above exists, with the roles described (`theorem_3_2_holds`, `corollary_3_3_holds`, `corollary_3_6_holds`, `gowers_cauchy_schwarz`, `corollary_17_5_holds`, `corollary_17_6_holds`, `lemma_5_15_holds`, `sectio`

n18_cyclic_density_increment_holds_of_theorem_18_1 in Proofs18QuantitativeSzemerédi.lean).
None of this source’s statements is formalized.

V.27 Further research questions and a concrete program

Source 06, delivered Section 9.

The questions below arise directly from the proved results. Unless a separate literature status is given, “question” means that the issue is not settled in this report; it does not assert that no answer has ever appeared elsewhere. This qualification is particularly relevant to small chamber counts and to inequalities obtainable by standard Cauchy–Schwarz arguments.

1. **Exact carry enumeration and a combinatorial interpretation.** Find a structural formula or recurrence for C_k and D_k , beginning with the certified values $C_4 = 154$ and $D_3 = 38$. The arrangement has normals indexed by nonempty subsets, integer thresholds, and a distinguished unit cube. Its symmetries, intersection lattice, and periodic extension are natural tools. An OEIS investigation should identify the underlying objects and definitions, not merely match a short numerical prefix. The exact certificates here provide a small benchmark for a symmetry-aware enumerator.
2. **The next term in carry entropy.** The leading coefficient in $\log_2 C_k \sim k^2$ is now determined by the threshold reduction and arrangement bound. Determine the order and coefficient of the next term, or prove a substantially smaller upper/lower gap. The current elementary upper bound is $k^2 + O(k)$; a sharper use of the threshold embedding or of the geometry internal to the unit cube may distinguish C_k from the unrestricted threshold count.
3. **Quantify information lost modulo a fixed modulus.** Although $C_{k,M}$ has the same leading entropy as C_k for every $M \geq 2$, this leaves open a large range of possible ratios $C_{k,M}/C_k$. Determine that ratio, or its logarithmic scale, for $M = 2$ and $M = 3$. The exact factorization (IV.7.6) means that any such result immediately improves modular pattern bounds without changing the integer-part argument.
4. **An exact affine orbit formula.** The nonaffine decomposition is injective because singleton carries vanish. Affine carries do not have that property. Classify precisely when two fractional affine carry maps differ by an integer linear map modulo M , and use this classification to determine $L_k(r, M)$ in the saturated range $2r \geq M$. This is a finite equivalence problem that can be explored with the certified carry families before seeking an asymptotic theorem.
5. **Avoid the square of the pattern count analytically.** The proof of Theorem IV.8.1 uses one Cauchy–Schwarz step over classes and one choice of a class, giving L^2 . Is there useful orthogonality, an energy-weighted decomposition, or additional structure among the realized classes that reduces this loss? The entropy lower bound rules out subquadratic enumeration of all classes, but it does not rule out an analytical argument that pays only for the classes carrying significant energy.
6. **Optimal density dependence in higher progression counts.** For $k = 3$, the variance factor in the quadratic discrepancy bound is sharp, as singletons show. For $k \geq 4$, determine the best coefficient depending on k and δ that can replace $\sqrt{\delta}B_k(\delta)$. One can first ask the narrower question within the family of repeated telescoping and mixed-norm Cauchy–Schwarz proofs. This would separate improvements of the method from a sharp extremal theorem about all sets.

7. **Torsion-sensitive forms of the two-target inequality.** When multiplication by a slope difference has a kernel, the variable changes in the proof cease to be measure-preserving bijections. Determine the best replacement constants in terms of those kernels and their interactions. Three-term progressions are the first test case; longer systems may require data beyond individual kernel cardinalities. This would extend the present theorem to moduli excluded by $\gcd(N, (k-1)!) = 1$.
8. **Use the mass of all dense cells.** The sharp extraction theorem quantifies the total mass above each threshold, not just the size of one selected cell. Can a branching density-increment argument exploit this mass to improve a structural or counting bound? The extremizers show that an improvement from the same scalar data alone is impossible; one must retain geometric information about the collection of high-density cells.
9. **Optimal finite corrections to the iteration potential.** For the power-law recurrence, the leading integral cost is already optimal. Determine a minimal discrete correction, or an exact dynamic-programming potential, with a useful closed form for finite starting density. A companion problem is to incorporate a density-dependent length prefactor and a nonmonotone admissibility threshold without replacing them everywhere by their worst initial values.
10. **Close the quantitative dependency chain in the inspected formalization.** The highest-impact repository task is to complete the outstanding structural estimates around Sections 13 and 16, establish the needed form of Theorem 18.1 with explicit thresholds, and only then combine extraction, transfer, and iteration. The local improvements here should be retained as parameters rather than immediately weakened to one coarse tower expression. A second route is to plan a formal treatment of modern inverse-theorem inputs, making clear which substantial theorems are being imported.
11. **Other integer-valued configuration families.** Theorem IV.7.3 already handles finite non-negative integer coefficient families V . Study which families have carry entropy comparable to the rank times the logarithm of their size, and which have additional cancellations. For signed families, first resolve the lower-dimensional floor-boundary obstruction identified after that theorem. Such a generalization could support localized arguments for configurations other than a Boolean cube.

For immediate implementation, the weighted extraction theorem and the indexed telescoping estimate are the smallest independent formalization tasks. The exact carry decomposition is the next natural target; it has an elementary proof, useful finite certificates, and a direct downstream consumer in phase removal. A global improvement requires the additional structural inputs identified above, with their actual density and scale dependence preserved.

[write] 6 October 2026; sources 06–10. These eleven questions are merged into Section V.2 with credits (items IV.2, IV.1, IV.3, IV.4, III.8, III.9, I.11, V.3, V.1 and IV.6). Question 7 is partly advanced by source 10’s Theorem III.28.2, which handles kernels for one distinguished function.

V.28 Verification and the Lean integration boundary

Source 07, delivered Section 9.

V.28.1 Exact finite checks actually performed

The supplementary `checks.py` uses only Python’s standard library. All decisions use integers or `fractions.Fraction`; the sole decimal parameter ratio is for display. The recorded run produced `verification.json`, with all checks passing.

Check	Completed instances
All maps $\phi, \psi : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$, all order-two source relations	6,561 function pairs; 419,904 relations
Equal-total word pairs in 220 unequal-fiber or punctured examples, using both the original and a second starting law	24,200 relation–law pairs
Primal profile versus exact one-variable dual	16,472 equalities
Nested-to-average envelope, including empty targets	9,059 applicable cases
Collision-to-mode inequality	1,936 cases
Sharp cyclic winding family, $2 \leq k \leq 20$, $2 \leq m \leq 5$	76 examples
General allowance at $\eta = (40k)^{-5}$, $2 \leq k \leq 1000$	999 rational endpoint checks

The randomized finite suite has the fixed seed 20261006 and includes both unstructured values and nearly affine examples. It recorded 6,058 strict relation certificates, so its tests are not confined to vacuous large-error bounds. The exhaustive singleton suite recorded 5,880 strict certificates and 157,464 violated relations, each with budget at least one. These are finite sanity checks on normalization, strictness, and source-law handling. They do not establish any universal statement by themselves; the general proofs are Sections II.14–II.19.

The inequalities for the order-five, order-nine, and relaxed order-two endpoints were also evaluated exactly. For example, the first budget at $\eta = 360^{-5}$, $\theta = 1/36$, is

$$18 \left(\frac{71}{1296} + 360^{-5} \right) = \frac{331257600001}{335923200000} < 1.$$

The original hypothesis is strict, but checking this closed endpoint gives a slightly stronger numerical verification of the corollary.

[write] **6 October 2026; sources 06–10.** Rechecked at the second write in exact arithmetic: $q(1/36) = 71/1296$, $18(71/1296 + 360^{-5}) = 331257600001/335923200000 < 1$, and $360^{-5} = (40 \cdot 9)^{-5}$, which is why Corollary II.17.6 follows from Theorem II.17.5 at $k = 9$.

V.28.2 What was inspected in the repository

The baseline is commit `ae0e895748209f0dd079c6b5ca6df6c6fc065a81`. The relevant files and roles are:

Section10.lean

Its local-model definition records the two nested agreement conditions. Its shift-regular definition includes fiber saturation, symmetry, factor-two fiber comparability, and overlap. The catalogue entry `lemma_10_9` concludes `FreimanHom 2 B' psi`.

Proofs10Shift.lean

The file contains `lemma_10_9_holds`. Its proof includes explicit handling of nonempty terminal fibers; this note does not attribute a vacuity bug to that proof. The proposed refinement replaces its two-step exceptional-set organization with averaged path marginals.

Audit.lean

Its comments distinguish an axiom-boundary audit from discharging the hypotheses of conditional results and from matching the numbered catalogue to the paper. This distinction is retained here.

The statements and proof source were read, but no Lean build or audit was run for this article. In particular, the existence of the old companion proof is not evidence that the new order-five or order-nine theorem has been formalized. The references [42, 43, 44] and the machine-readable source manifest identify the comparison precisely.

[write] 6 October 2026; sources 06–10. At HEAD (764740f07) Section10.lean, Proofs10Shift.lean and Audit.lean are unchanged since the pin ae0e89574; lemma_10_9 is at line 335 and lemma_10_9_holds at line 717. The ledger has changed since the pin (4 statements closed: lemma_16_1, lemma_16_5, lemma_16_6 and lemma_16_9), none of them in Section 10.

V.28.3 A concrete formalization route

A practical integration should first prove a finite counting version of the four-edge theorem and only then introduce arbitrary paths and optimized profiles. A proposed module decomposition is:

1. **Directional kernels.** Define internal target fibers, boundary counts, and nonnegative rational edge defects. Prove the identities (II.13.3)–(II.13.4), with positivity hypotheses attached only to denominators actually used.
2. **Transport bounds.** Prove the simple domination lemma $\sum \mu_s f \leq c(s) \sum \mu f$ for $0 \leq f$, and derive the factor-two and additive-invariance instances. The full profile optimization can be postponed; it is unnecessary for the five/nine corollaries.
3. **A shared-endpoint finite sample.** Represent paths by finite lists or `Fin`-indexed vertices, with two explicit endpoint variables and independent internal variables. Prove a weighted union bound before translating it into existence of a successful sample.
4. **The local interface theorem.** Prove Lemma II.17.1, then the factor-two order- k criterion and the order-five consequence. Keep these separate from the stronger theorem using `DomainInvariant` and a fiber lower bound.
5. **Exact constants and audits.** Derive the real-power comparison once, discharge the rational polynomial estimates, and run the namespace axiom audit on the resulting new theorem declarations. Do not relabel an unproved proposition-valued catalogue entry as a completed theorem.

Two technical design choices deserve explicit tests. First, when a path returns to its starting index, its terminal *sample* must not be identified with its initial sample; otherwise the intended pair marginals can change. Second, restricting to previously successful equations is not the same as restricting to surviving indices. Only the latter restriction is used in the marginal bound. The file `FORMALIZATION.md` in the package records these obligations separately from completed mathematics.

V.29 Further research questions and priorities

Source 07, delivered Section 10.

The following are proposed problems arising from this framework. Except for the sharp balanced threshold already settled in Theorem II.16.1, we make no claim that these are previously catalogued open problems or that the listed approaches resolve them.

Question V.29.1 (Optimal nonuniform threshold). *For fixed $k \geq 2$ and $\kappa > 1$, what is the largest universal δ -threshold forcing order- k relations when fiber sizes vary by at most κ ? How does a separate boundary bound improve it?*

The bound $1/\{2[1 + (k - 1)\kappa]\}$ is proved, but only its balanced specialization is shown sharp. A promising finite search would optimize fiber sizes and actual value assignments, rather than arbitrary independent error profiles. The latter relaxation may exaggerate what a single ϕ can realize simultaneously.

Question V.29.2 (The strongest order at the original interface). *What is the maximal guaranteed Freiman order under the precise assumptions of the recorded Lemma 10.9, and under the fuller ambient-invariance package?*

Five and nine are lower bounds on the guaranteed order, not maximality results. The exact nested envelope, the actual value of ρ , optimized word ordering, and nonuniform directional defects can all improve the sufficient certificate. Counterexamples must also satisfy the nested local model, not merely a scalar defect bound.

Question V.29.3 (Stability of winding obstructions). *If constant fibers and a violated order- k relation have total budget close to one, must their error sets resemble a small collection of cuts or carry boundaries after an appropriate structural reduction?*

The cyclic construction has a concrete wraparound boundary. Near equality in the union bound suggests that edge-failure events should be almost disjoint in a shared-endpoint experiment, but translating that observation into structural information about ϕ requires additional work.

Question V.29.4 (Efficient certificates for many relations). *Can the root-law linear programs for a large family of relations be compressed into a small reusable set of certificates, perhaps by exploiting translation symmetries or a presentation of the generated group?*

Theorem II.18.3 gives a correct finite optimization for one relation. It does not give an efficient algorithm when the number of relations is exponential in the number of directions. A useful implementation would report rational primal witnesses whenever it certifies a relation.

Question V.29.5 (Joint optimization of extraction and transport). *Can one construct the regular component and its source law together, so that the extracted domain is large and its relevant transport profiles are small, instead of first paying for coarse fiber comparability?*

Changing the source law affects the hypothesis under which local errors are controlled. The alternative-root result avoids silently changing that hypothesis, but it does not solve the joint extraction problem. A successful method would need to account explicitly for lost cardinality and for the relation between the original energy weights and the chosen law.

Question V.29.6 (Higher-dimensional coherence). *Can shared-vertex sampling on cubes or more general finite diagrams replace repeated nested exceptional-set arguments in bilinear and multilinear exactification steps?*

The two-path proof suggests a diagrammatic error budget, but additional compatibility equations can correlate the required marginals. The right problem is to specify a joint sampling law first and prove every face or edge marginal under that law, rather than assume independent tests can be glued automatically.

Question V.29.7 (Improving the earlier extraction exponent). *Which quantitative losses before the local model are necessary, and can the appearance of $\eta^{1/5}$ itself be improved while retaining a sufficiently large regular component and direction set?*

The present method removes the later square root; it does not improve the earlier fifth-root construction. This is a more consequential target than optimizing the last few units in a rounded denominator such as 75.

Question V.29.8 (An end-to-end quantitative payoff). *After substituting the transport-sensitive exactification theorem, which later bottleneck becomes dominant, and does any improvement remain in a fully tracked final bound?*

Answering this requires a complete dependency ledger through the later Bohr-set, multilinear, and density-increment steps. The initial deliverable should be a mechanically checkable parameter recurrence, followed by a comparison with the unchanged proof, before any improved global exponent is announced.

Recommended order of work. First formalize the factor-two order-five theorem and its explicit boundary handling. Next build an exact rational profile-certificate tool and search for sharp nonuniform examples. Only then undertake the full parameter propagation and higher-dimensional extensions. These tasks separate near-term checkable improvements from the substantially larger research program.

[write] **6 October 2026; sources 06–10.** The eight questions are items II.9–II.15 and V.1 of Section V.2.

V.30 Conclusion

Source 07, delivered Section 11.

The local algebraic obstruction is a closed diagram with inconsistent edge labels. Once a joint law with shared endpoints is fixed, a violated relation must consume at least one unit of total failure probability. Transport profiles express exactly how an error known under the original law can contribute along a translated path. This yields linear, rather than square-root, error propagation and preserves useful information about higher-order relations.

The concrete consequences are a sharp balanced threshold, an order-five strengthening at the recorded Lemma 10.9 interface, an order-nine conclusion when ambient invariance is retained, and larger explicit local order-two allowances. The article proves those claims and supplies finite exact-arithmetic checks. Their significance is local and structural; an improved global Szemerédi bound would require the additional end-to-end analysis identified above.

V.31 Reproduction commands

Source 07, delivered Appendix B.

From the unpacked directory, the mathematical checks and PDF can be reproduced with

```
python3 checks.py --output verification.json
latexmk -pdf -interaction=nonstopmode -halt-on-error transport_rigidity.tex
```

A Makefile provides `make check`, `make pdf`, and `make clean`. The document uses an embedded bibliography and does not need BibTeX. Recompilation requires the LaTeX packages listed in its preamble. No network access is required for either command once those packages are installed.

V.32 Integration with the supplied Lean development

Source 08, delivered Section 11.

V.32.1 The normalization boundary

The inspected definitions use $\mathbb{Z}/N\mathbb{Z}$, complex-valued functions, the unnormalized Fourier transform, and the unnormalized cube norm [45]. Translating the mathematical notation gives

$$\begin{aligned}\text{cubeCount}(d, A) &= N^{d+1} \mathcal{C}_d(\mathbf{1}_A), \\ \text{gowersNorm}(d, f) &= N^{(d+1)/2^d} \|f\|_{U^d}, \\ \text{UniformSetOfDegree}(A, \alpha, d-1) &\iff \|\mathbf{1}_A - \delta\|_{U^d}^{2^d} \leq \alpha \quad (d \geq 1).\end{aligned}$$

The last equivalence uses the successor-dimension identity for the iterated difference sum. It is already the normalization step used by the existing proof of Lemma 3.10.

For $d \geq 2$, the proposed replacement conclusion is therefore

$$\text{cubeCount}(d, A) \leq N^{d+1} \left[\delta^{2^d} + \sum_{r=4}^{2^d} \binom{2^d}{r} \delta^{2^d-r} \alpha^{r/2^d} \right]. \quad (\text{V.32.1})$$

The dimension restriction avoids both the U^1 seminorm issue and the truncated natural subtraction in the expression $d-1$ at $d=0$. Dimensions zero and one already have elementary exact counts and should remain separate cases rather than being forced through this theorem.

V.32.2 A proposed proof decomposition

No new Lean file is included as if it were verified. The following are proposed lemma boundaries, not claims that these identifiers already exist. They are intended to let the existing analytic development carry most of the proof.

Suggested lemma or module	Mathematical obligation
<code>cubeForm_subset_expansion</code>	Expand the finite product into a sum over subsets of Boolean vertices.
<code>cubeProjection_three_uniform</code>	Build the affine equivalence implementing the unit-minor argument for three distinct vertices.
<code>centeredCubeTerm_eq_zero</code>	Factor averages for subsets of cardinality one, two, or three.
<code>centeredCubeTerm_norm_le</code>	Pad omitted vertices with 1 and apply the existing cube Cauchy–Schwarz inequality.
<code>cubeCount_fourthOrder_le</code>	Combine cancellation, the cardinality strata, and the normalization factor N^{d+1} .
<code>cubeQuartic_torsion_formula</code>	Add the Fourier classification of four-vertex subsets and the two closed counting formulas.

A useful implementation order is to prove the first four statements for the current cyclic setting before generalizing the entire API to finite abelian groups. The elementary fourth-order improvement does not require Fourier analysis, classification of finite abelian groups, or any torsion assumptions. The exact quartic formula is naturally a second, separate layer.

For the indicator theorem, the function `balanced A` is real-valued inside `ℂ`. Recording that property explicitly lets parity conjugations simplify before the subset expansion. The existing cube argument uses a minus sign, while the counting convention uses a plus sign; negating all increments gives the required measure-preserving equivalence.

The proof of the analytic upper bound can be isolated from real-power arithmetic by first proving a statement in a nonnegative parameter t with a hypothesis on the normalized balanced norm. Only at the final step need one substitute $t = \alpha^{1/2^d}$. That separation reduces the chance that a bookkeeping error in roots or normalization obscures the combinatorics.

V.32.3 What was inspected and what was not verified

The two main source files used for this integration analysis were `Definitions.lean` and `Proofs03CubeUpper.lean`. Their inspected Git blob identifiers begin with `97113b7afa69` and `c017515bdd43`, respectively; complete identifiers and source URLs are in the accompanying provenance file. The latter file proves `lemma_3_10_holds` by applying Minkowski [46]. The analysis is not a repository-wide axiom audit.

No Lean executable was available in the working environment, and no Lean build or kernel verification of the new results was performed. The repository was not modified. The deliverable contains mathematical proofs, exact finite checks, and this integration plan, rather than an untested formalization presented as complete.

[write] **6 October 2026; sources 06–10.** At HEAD (`764740f07`) `lemma_3_10_holds` is still proved in `Proofs03CubeUpper.lean` (line 82) by the Minkowski bound; the statements of this source are not formalized.

V.33 Exact checks and reproducibility

Source 08, delivered Section 12.

The accompanying file `verify_cube_refinements.py` uses only Python’s standard library. All its asserted comparisons use integers or `Fraction`; it does not approximate trigonometric values or compare floating-point cube counts.

The following checks were run successfully in preparing this memorandum. First, the program enumerates every four-element subset in Boolean dimensions 2 through 6, verifying the R_d, T_d, F_d formulas on a total of 673,227 subsets. It also checks the unit-minor property on all 5,580 triples in dimensions 2 through 5.

Second, it classifies all three-cube five-element supports into 48 containing a parallelogram and eight exceptional supports, and it verifies the missing-pair multiplicities 12, 12, 4 for six-element supports.

Third, it computes the complete coefficients of $\mathcal{C}_3(\delta + \eta f)$ by direct averaging for explicit integer-valued, mean-zero functions on cyclic groups of orders 3 through 9. This verifies the vanishing of degrees one through three, the all-group quartic formula, and, in odd order, the full three-cube identity. Additional quartic checks use the groups $(\mathbb{Z}/2\mathbb{Z})^2$, $(\mathbb{Z}/2\mathbb{Z})^3$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, and $(\mathbb{Z}/3\mathbb{Z})^2$. These cover examples with several order-two characters as well as odd-order noncyclic groups.

Finally, two independent Fourier enumerations verify (III.10.3). One enumerates all eight frequencies; the other uses the four-parameter kernel formula (III.10.4). For a single cosine on $\mathbb{Z}/11\mathbb{Z}$, the same exact method gives

$$\mathcal{C}_3(\delta + \eta \cos(2\pi x/11)) = \delta^8 + \frac{3}{2}\delta^4\eta^4 + \frac{1}{32}\eta^8. \quad (\text{V.33.1})$$

The order-two quartic coefficient is checked algebraically in dimensions 2 through 7.

To rerun the tests from the extracted archive, use

```
python verify_cube_refinements.py --output verification_results.json
```

The default enumeration stops at dimension six. The optional argument `-max-d 5` shortens the quartic enumeration. Running the program rewrites only the chosen output JSON file; it makes no network requests and does not access the repository.

These checks are useful for catching coefficient, sign, and torsion mistakes. They are not proofs of statements quantified over all groups or all dimensions. Those statements are supported by the arguments in the preceding sections. The finite counterexample identity has its own explicit exhaustive certificate, whose relation to the Fourier equations is given above.

V.34 Further research questions

Source 08, delivered Section 13.

The following questions are proposed directions arising from the present results. Unless explicitly connected to a cited source, their appearance here does not assert that they are established open problems in the literature. The priority order favors tasks that can produce a concrete theorem or a checkable counterexample.

V.34.1 Optimal odd-order constants

Question V.34.1. *For $d \geq 3$, determine*

$$\kappa_d^{\text{odd}} = \sup_{\substack{G \text{ finite, odd order} \\ f: G \rightarrow \mathbb{R}, \mathbb{E}f=0, f \neq 0}} \frac{E_4(f)}{\|f\|_{U^d}^4}.$$

Is $\kappa_3^{\text{odd}} = 1/\sqrt{2}$?

Norm monotonicity gives $\kappa_d^{\text{odd}} \leq 1$. The single-cosine calculation (V.33.1) gives $\kappa_3^{\text{odd}} \geq 1/\sqrt{2}$. The best odd-order leading coefficient in a fourth-order bound purely in t is $R_d \kappa_d^{\text{odd}}$, not automatically R_d . A useful first step is a constrained Fourier optimization on odd cyclic groups, followed by a proof of any proposed inequality. Small-group optimization alone would not settle the supremum over all groups.

[write] **6 October 2026; sources 06–10.** *Answered for $d = 3$.* This κ_d^{odd} is source 03's c_d^{odd} . Source 04's Theorem III.3.4 gives $\kappa_3^{\text{odd}} = 2^{-1/2}$, attained by real parts of characters (proof in the note after that theorem), so the answer to the last sentence of the question is yes. For $d \geq 4$ it is open (Section V.2, item III.1).

Question V.34.2. *What is the largest constant c such that*

$$\mathcal{D}_3(F) \geq c \delta^4 E_4(F - \delta)$$

for every odd-order group and every $0 \leq F \leq 1$ with $\delta = \mathbb{E}F > 0$?

The results here give $2 \leq c \leq 15337/1296 < 12$. The lower endpoint follows from Theorem III.12.1; the upper endpoint follows from the explicit weight in (III.10.5). Optimizing the two-frequency family gives a slightly stronger upper bound by minimizing $12 - 2s + 6s^2 + s^4$, but there is no reason to assume this family is globally extremal. A proof separating local and global optimizers would already clarify the geometry of the cube polynomial.

[write] **6 October 2026; sources 06–10.** The failure of the constant 12 is kept on record as Remark W1; the question is item III.10 of Section V.2, merged with source 09's question on a corrected lower bound.

V.34.2 Higher coefficients and arithmetic dependence

Question V.34.3. *Compute the complete fifth- and sixth-order coefficients for general d , including their dependence on the torsion of G .*

The quartic coefficient only sees order-two characters. For larger supports, integer minors can have additional prime factors, and Fourier relations can couple several multiples of a frequency. A concrete program is to classify support matrices up to cube symmetry, compute their integer relation lattices, and produce exact character-sum profiles. The desired output is a proved formula with orbit counts, not just a table of numerical values in sample groups.

Question V.34.4. *Can the survivor polynomial $\sum_r A_{d,r}(G)z^r$ be computed substantially faster than enumerating all subsets of V_d ?*

Proposition III.14.2 reduces much of the large-prime case to column dependence. Symmetry reduction, deletion–contraction ideas, and certified integer normal forms may make dimensions four and five tractable. Small-characteristic exceptions must be retained rather than inferred from the rational calculation. It would be valuable to output compact certificates that a proof assistant can check independently of the enumeration engine.

V.34.3 Stability and explicit constructions

Question V.34.5. *Describe near-extremizers for the quartic coefficient $R_d E_4(f) + T_d E_{(2)}(f)$ relative to $\|f\|_{U^d}^4$.*

On groups with an order-two quotient, the exact extremal model supplied here is a real character. A stability theorem would quantify whether a nearly optimal ratio forces concentration on one character or on a small collection of characters. On odd groups, the correct model is not yet determined here. This question concerns a small-norm perturbation of a constant, rather than near-maximal U^d norm under an L^∞ bound; those are distinct extremal problems.

Question V.34.6. *Derandomize the indicator sharpness construction while keeping its fourth-order asymptotic coefficient.*

The rounding proof is existential and uses very large prime groups. A deterministic construction should guarantee rounding error $o(\eta^4)$ in U^d , not merely small discrepancy against individual Fourier characters. The collision estimate gives a clear quantitative target against which an explicit construction can be measured.

V.34.4 Localization, sparse weights, and formalization

Question V.34.7. *Which exact cancellations survive when cube averages are localized to intervals, progressions, or Bohr neighborhoods?*

Global three-vertex independence is the key input to the fourth-order bound. Restricting the parameter domain destroys that independence in general. A localized theorem would need either specially compatible averaging measures or explicit boundary and correlation errors. Establishing such a result is a more plausible route toward density-increment applications than directly substituting the present global estimate into an unrelated step.

Question V.34.8. *Can the inhomogeneous estimate be transferred to sparse majorized weights with an explicit error budget?*

For a relative counting theorem, the means and the majorant cannot be treated as interchangeable constants. One should first expand around the actual majorant model and identify which low-order correlations are controlled by its pseudorandomness assumptions. The finite character-kernel criterion may then organize the remaining balanced terms, but it does not supply the required majorant estimates by itself.

Question V.34.9. *Can the elementary fourth-order improvement and the quartic torsion formula be formalized as separate verified modules in the supplied development?*

The first task needs finite product expansion, affine equivalences, and the existing cube inequality. The second adds finite Fourier analysis and the classification of four vertices. A useful milestone is a theorem with the original unnormalized conclusion (V.32.1), together with tests that instantiate both odd moduli and even moduli. General finite abelian groups can be treated after the normalization and subset-expansion interfaces are stable.

V.34.5 From vanishing supports to sharper configuration bounds

Question V.34.10. *For a fixed non-cube system of linear forms, can one combine the exact full-support criterion with a true-complexity estimate to obtain the correct first nonzero power of the balanced uniformity parameter?*

This would separate two sources of quantitative loss: terms that should be zero before any norm inequality is used, and terms that are genuinely present but estimated inefficiently. The work of Gowers–Wolf and Manners [29, 14] supplies relevant norm-control theory, but the leading-power problem depends on the particular system and its arithmetic. It should not be assumed that arithmetic progressions inherit the same three cancellations as a full additive cube.

V.35 Conclusion

Source 08, delivered Section 14.

The main gain is a sharp change of scale in the error term of the selected cube-count lemma. Centering removes the first three perturbative orders, and the first surviving order is exactly controlled by parallelograms and order-two Fourier mass. The resulting exponent $4/2^d$ is optimal even for sets in odd prime cyclic groups.

The more detailed three-cube calculation reveals a second lesson: the next coefficient is signed. A positive fourth-order expansion does not imply a global lower bound with the same coefficient. The explicit two-frequency counterexample makes that obstruction quantitative.

These results leave the global Szemerédi problem unchanged, but provide a more precise local component, a finite-support classification framework, and concrete next problems with a path to computational and formal checks.

V.36 Build and source provenance

Source 08, delivered Appendix B.

The source document is self-contained LaTeX with an internal bibliography. From the extracted archive, compile it with

```
pdflatex -interaction=nonstopmode -halt-on-error fourth_order_cube_counts.tex
pdflatex -interaction=nonstopmode -halt-on-error fourth_order_cube_counts.tex
```

Two passes resolve references and the table of contents. The archive also includes `build.sh`, the exact-arithmetic verifier, its captured JSON output, a README, and `source_provenance.json`. It does not redistribute the original Gowers paper or any font files.

The original paper was checked at printed pages 483 and 485, corresponding to PDF pages 19 and 21 in the linked 124-page copy. The repository source inspection was read-only and took place on 6 October 2026. The provenance file identifies inspected blobs; it does not claim that the entire repository was frozen or audited at a single commit.

V.37 Reproducible exact checks

Source 09, delivered Section 11.

The supplied computations use exact integers and rational numbers. They are tests of the formulas and finite certificates, not a substitute for the all-dimension classification proofs.

V.37.1 Direct finite-group cube averages

The standard-library script `verification/verify_expansion.py` was run successfully on 24 specified examples. It covers cyclic groups of orders two through nine in dimension three, several cyclic groups in dimensions two and four, and product groups including $(\mathbb{Z}/2\mathbb{Z})^2$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, and $(\mathbb{Z}/3\mathbb{Z})^2$. For each case it constructs a deterministic integer-valued mean-zero function and enumerates every parameterized cube.

The coefficient of δ^{m-s} is computed directly as the s th elementary symmetric polynomial of the m evaluated function values, averaged over cube parameters. The script compares the first five coefficients with the formulas in Theorem III.21.1. This avoids both numerical roots of unity and Fourier rounding error. For the right-hand side it evaluates equivalent convolution formulas, including the quotient averages on $G/2G$ and $G/3G$ for the torsion statistics. The recorded file `results.json` contains every input and exact output.

The same script checks the Stirling formulas for $2 \leq d \leq 14$ and all $5^8 = 390625$ frequency assignments needed for the two-frequency polynomial in Appendix III.26.

V.37.2 Exhaustive small-cube lattice census

The optional SymPy script `verification/verify_lattices.py` was also run successfully with SymPy 1.14.0. It uses exact Smith normal forms for every four- and five-vertex subset in dimensions three and four. The five-vertex census is particularly informative:

Five-vertex type	$d = 3$	$d = 4$
Parallelogram plus independent fifth vertex	48	1200
Full-support real circuit (star)	8	160
Full rank, index one	0	2672
Full rank, index two	0	320
Full rank, index three	0	16
Total	56	4368

The index-two five-vertex configurations are a useful warning: their lattices are not primitive, but their moments still vanish because a character coordinate is forced to be trivial. Counting only nonprimitive supports would overcount the fifth-order contribution.

For four-vertex subsets the surviving counts are $(12, 2)$ in dimension three and $(100, 40)$ in dimension four, matching (P_d, T_d) . The same script checks the determinant, Smith invariants, and full modulo-five kernel certificate in Theorem III.23.1. Its complete output is `lattice_results.json`.

V.37.3 What was and was not verified

Both scripts completed with all assertions passing. The manuscript proofs handle arbitrary dimension and arbitrary finite abelian groups; the scripts do not exhaust those infinite families. No Lean compiler was available in the execution environment, no new theorem was checked by Lean, and no repository files were modified. The supplied formalization roadmap is therefore a proposal, not a verification certificate.

V.38 Integration with the supplied Lean development

Source 09, delivered Section 12.

V.38.1 Pinned source audit

The inspected repository checkpoint is commit

7d99e5deeb209a4671060990cb1ea3d672d57064.

The relevant directory is

Combinatorics/Ramsey/Lean/GowersSzemerédi.

The following are observations from source inspection, not claims that we rebuilt the repository:

Definitions.lean

Defines `fourier` as the unnormalized discrete Fourier transform, `balanced` as indicator minus density, `cubeForm` with parity conjugation and the argument $s - e \cdot x$, `gowersNorm` as an unnormalized cube-sum root, and `cubeCount` by parameterized additive cubes. The inspected blob is 97113b7afa6925a2dd4b76641eeaeff09597ab6a.

Sections01_03.lean

Contains the proposition-valued statement `lemma_3_10`. A proposition-valued catalogue entry alone is not a proof; here there is also a separate proof file.

Proofs03CubeUpper.lean

Proves `lemma_3_10_holds`. It imports `Proofs03Minkowski` and `Proofs03Cubes`, identifies indicator and constant norm powers, and applies `gowersNorm_add_le`. The inspected blob is c017515bdd43b3c7dd8a5d0e90747a20403d4ddc.

Proofs01_03.lean

Contains the finite Fourier identities, including Parseval and the fourth-moment/additive-quadruple identity. These are natural inputs for the exact quartic theorem.

V.38.2 Normalization and sign conventions

For $G = \mathbb{Z}/N\mathbb{Z}$, the dictionary is

Manuscript quantity	Repository quantity
$\widehat{f}(r)$	N^{-1} <code>fourier f r</code>
$\ f\ _{U^d}$	$N^{-(d+1)/2^d}$ <code>gowersNorm d f</code>
$\mathcal{C}_d(\mathbf{1}_A)$	$N^{-(d+1)}$ <code>cubeCount A</code>
$\ f_A\ _{U^d}^{2^d} \leq \alpha$	<code>UniformSetOfDegree A</code> $\alpha (d - 1)$, for $d \geq 2$

The manuscript uses $x + v \cdot h$ whereas `cubeForm` uses $s - e \cdot x$. Replacing every side variable by its negative converts between them. Real-valued functions make parity conjugation harmless, but a formal proof must still discharge the corresponding real-cast and conjugation identities. The indicator count includes degenerate parameterized cubes; it is not a count of distinct vertex sets.

V.38.3 A staged implementation plan

The most economical first formal target is Theorem III.17.2, not the entire lattice classification. It needs only a subset expansion, three-vertex uniformity, and the existing Gowers–Cauchy–Schwarz infrastructure.

A proposed first module, `Proofs03BalancedExpansion.lean`, would prove that mixed cube sums supported on one, two, or three distinct Boolean vertices vanish for a real mean-zero function. Prove explicit coordinate reindexing equivalences rather than relying on field linear algebra over $\mathbf{ZMod\ N}$. This avoids an unnecessary primality assumption. Then sum by subset cardinality and recover (III.17.2). A safe initial theorem should assume $d \geq 2$ and $\alpha \geq 0$ explicitly; this avoids accidental dimension-zero or real-power edge cases.

A second module, `Proofs03QuarticCircuits.lean`, would classify four-element Boolean subsets, count the two types, and prove their Fourier formulas. It can initially state a torsion-aware theorem for arbitrary $\mathbf{N}$; an odd-modulus corollary then removes the order-two term. Using an integer matrix or explicit coordinate-block normal form is essential. A theorem asserting independence over $\mathbf{ZMod\ N}$ merely from rational rank would be false.

A third module, `Proofs03QuinticCircuits.lean`, would implement the rank-four and rank-five cases separately. The two especially important obligations are the primitive-minor argument for a parallelogram plus a fifth vertex, and the index-two case with a forced trivial Fourier coordinate. The general character-kernel formula could be developed either first as a reusable library theorem or afterwards as a unification of the small cases.

A final `Proofs03CubeRefinement.lean` would combine these results with explicit binomial tails. The output theorem should coexist with `lemma_3_10_holds`, with the old theorem recovered as a corollary by adding back nonnegative omitted terms. There is no reason to delete a valid existing interface.

V.38.4 Recommended proof obligations and audit discipline

Useful public lemmas would include joint uniformity of at most three Boolean affine forms over an arbitrary finite abelian group; Fourier orthogonality for an integer matrix of forms; conversion of cube sums to normalized averages; counting disjoint nonempty coordinate blocks; and the binomial-tail inequality (III.16.5).

These should be genuine proved lemmas, not axioms or proposition placeholders. After implemen-

tation, build the exact pinned dependency set, audit the types of the new public theorems, inspect the axiom closure, and run the repository’s own status checks. A successful small experimental script is not a substitute for these steps, and the present package does not claim they have been carried out.

[write] **6 October 2026; sources 06–10.** At HEAD (764740f07) the inspected files still play the roles described; `lemma_3_10_holds` is at line 82 of `Proofs03CubeUpper.lean`. None of this source’s statements is formalized.

V.39 Further research questions and a prioritized program

Source 09, delivered Section 14.

The following are precise continuations of the proved results. They are proposed research targets, not assertions that each formulation is historically new or currently unsolved in the entire literature.

V.39.1 Complete the sixth-order coefficient

Research question V.39.1 (A dimension-uniform sextic expansion). Classify all six-vertex Boolean subsets that admit a full-support character tuple, and express the coefficient of δ^{m-6} in a finite basis of Fourier or higher-order statistics with closed formulas for their multiplicities as functions of d .

Theorem III.23.1 proves that an order-five term must occur. There will also be real relations and possibly several independent character parameters. A useful first computational stage is an orbit census by coordinate-pattern multisets, followed by integer-lattice certificates for each orbit. The real challenge is a finite normal-form classification that does not grow with the ambient dimension. The goal should be an exact sixth coefficient and a seventh-order remainder, not merely a longer numerical table.

V.39.2 Determine the prime-onset function

Research question V.39.2 (Support-efficient torsion). Determine $h(7)$ and $h(11)$, and improve the general bounds between (III.23.4) and $h(\ell) \leq \ell + 2$.

A determinant divisible by ℓ is necessary but not sufficient: the corresponding modulo- ℓ kernel must have full support, or balancing makes the term vanish. This extra support condition should be included from the start in a search. A practical certificate would contain the binary matrix, a nonzero determinant, its Smith invariants, and an explicit full-support kernel vector. The exact order-five example supplies a model for such a certificate.

V.39.3 Optimize constants at a fixed density and norm ratio

Research question V.39.3 (The constrained extremal envelope). For fixed d , density δ , and prescribed $(\varepsilon, \tau, \rho)$, determine or tightly bound

$$\sup\{\mathcal{C}_d(F) : 0 \leq F \leq 1, \mathbb{E}F = \delta, \|F - \delta\|_{U^d} = \varepsilon, \|F - \delta\|_{U^2} = \tau, \widehat{F - \delta}_\infty = \rho\}.$$

The coefficient A_d is optimal when all groups are allowed, but the unsigned fifth-order estimate need not be jointly sharp with it. Order-two characters maximize the universal fourth coefficient while forcing the quintic term to vanish. This incompatibility suggests that a coupled bound may improve (III.21.7). The odd-order problem is especially natural: determine the best leading coefficient relative to ε^4 , rather than relative to the exactly known Fourier energy τ^4 .

V.39.4 Stability of nearly extremal quartic corrections

Research question V.39.4 (Structure near the universal fourth-order maximum). If

$$P_d E_4(f) + T_d T_2(f) \geq (1 - \eta) A_d \|f\|_{U^d}^4,$$

what quantitative structural conclusion follows as $\eta \rightarrow 0$?

The order-two character example is an exact extremizer, but uniqueness and stability have not been proved here. One must understand both near-equality in norm monotonicity and concentration on the two-torsion spectrum. It would be useful to separate a statement valid for all real functions from stronger statements using $0 \leq \delta + f \leq 1$.

V.39.5 A useful corrected lower bound

Research question V.39.5 (Replace the false quartic lower bound). Find natural hypotheses under which M_5 is nonnegative, or bound its negative part more sharply in terms of the fourth-order statistics and density, so that a nontrivial corrected cube lower bound follows.

Nonnegative Fourier coefficients make all subset moments nonnegative, but this is a strong restriction. A less restrictive condition on the relative phases of a_χ and a_{χ^2} may be sufficient. The explicit expression (III.21.3) identifies the relevant phase interaction, and Proposition III.22.2 shows why an unconditional positivity argument cannot work.

V.39.6 Boundary-sensitive interval analogues

Research question V.39.6 (Cubes restricted to an interval). Develop a version of the expansion for cubes constrained to lie in $[N]$, with explicit boundary weights, and determine which of the first three cancellations survive exactly or approximately.

The current proof averages over a whole finite group. Conditioning on all vertices lying in an interval destroys the uniformity argument used in Lemma III.17.1. A direct transplantation would be incorrect. One possible route is to expand relative to a weighted reference measure adapted to the interval, rather than to a constant function, and explicitly retain the resulting lower-order boundary terms.

V.39.7 Algorithmic use of the measured residual

Research question V.39.7 (A residual-sensitive inverse step). Can the sixth-order certificate (III.21.11), together with an efficient estimate of the four Fourier statistics, improve a concrete inverse or counting algorithm in a useful parameter regime?

The question requires a complete cost and parameter analysis. A norm lower bound by itself does not deliver structure. A convincing result would specify the input access model, the cost of estimating $E_4, T_2, M_5, T_{3,5}$, the residual threshold, and the quantitative gain after an available inverse theorem is invoked.

V.39.8 Formalize primitive Boolean affine maps first

Research question V.39.8 (A reusable Lean library for arithmetic configurations). Build a library of exact reindexing equivalences and integer-lattice certificates for Boolean affine forms, then formalize Theorems III.17.2 and III.21.1 over arbitrary finite abelian groups.

The initial cancellation theorem is an accessible first milestone. The general character-kernel identity and a sound bridge from Smith normal form to finite-group character constraints would have much broader reuse. It is important to preserve non-prime groups in the interface: they expose the precise arithmetic phenomenon that the paper identifies.

V.39.9 Beyond diagonal cube counts

Research question V.39.9 (Colored cubes and other linear systems). For other finite systems of integer linear forms, identify the least support size at which a mean-zero expansion can contribute, and classify the first torsion-sensitive terms.

The proof mechanism is the same as Proposition III.18.1, but the first nonzero order need not be four. The heterogeneous theorem suggests applications to colored configuration counts and joint distribution tests. A useful result should state the primitive-lattice hypotheses explicitly, rather than replacing them by a characteristic-dependent rank test.

V.39.10 From a local gain to a global theorem

Research question V.39.10 (Locate an actual quantitative dependency). Identify a step in a complete progression argument whose bound depends on the local excess $\mathcal{C}_d(F) - \delta^m$ in such a way that replacing a first-order estimate by the fourth- or sixth-order estimate changes the final quantitative conclusion.

This is the necessary bridge to a stronger Szemerédi-type result. It may turn out that the paper’s dominant losses lie elsewhere, in which case the local sharpening has no effect on the final asymptotic bound. The correct research procedure is to write and audit the full dependency recurrence, not to infer a global improvement from a better constant or exponent in an auxiliary lemma.

[write] **6 October 2026; sources 06–10.** The ten questions are merged into Section V.2 (items III.2, III.16, III.17, III.12, III.10, III.13, III.18, V.2, III.15 and V.1).

V.40 Conclusion

Source 09, delivered Section 15.

Balancing contains more information than a triangle inequality retains. For full cube counts, three exact cancellations move the first error to degree four. The next two homogeneous coefficients

admit a complete arithmetic description: parallelograms and stars give ordinary Fourier statistics, while primitive failures of indices two and three give explicit torsion corrections. The multiplicities are simple Stirling-number expressions, and the residual is sixth order with a sharp exponent.

The small-order classification also explains its own limitations. The fifth-order term can be negative; the universal fourth coefficient is attained by a torsion example; and a determinant-five configuration appears one order later. These facts make the refinement more than a formal binomial improvement: they identify exactly which information controls the first deviations from a constant and which arithmetic features a correct generalization must preserve.

The resulting package provides proved local mathematics, exact computational checks, and a concrete route toward formalization. The questions above separate the immediately actionable extensions from the additional work needed for a genuinely stronger global progression bound.

V.41 Proof and artifact dependency map

Source 09, delivered Appendix C.

The logical dependence is as follows. The analytic input is Gowers–Cauchy–Schwarz, the finite Fourier inversion formula, and ordinary finite-dimensional integer linear algebra. Three-vertex primitive uniformity gives the fourth-order bound immediately. The small determinant bounds give the four- and five-vertex classifications; coordinate-block counting gives their multiplicities. Combining these with the subset expansion gives the main exact theorem. Sharpness uses explicit characters and two-frequency functions, with random rounding for indicator transfer. The order-five obstruction has its own determinant and kernel certificate.

The numerical files are independent cross-checks of those mathematical steps. The optional finite table in Appendix III.26 is the one additional exact identity supported by a full explicit enumeration. The source audit and formalization plan do not enter any mathematical proof and do not assert a successful Lean build.

V.42 What the local gains do to a quantitative iteration

Source 10, delivered Section 10.

A better local lemma is useful only after its hypotheses and losses have been propagated through the later argument. We record two exact bookkeeping statements to make that distinction concrete.

Proposition V.42.1 (A reciprocal-density bound on the number of steps). *Suppose $c > 0$, $a \geq 1$, and $0 < \delta_0, \dots, \delta_t \leq 1$ satisfy $\delta_{i+1} \geq \delta_i + c\delta_i^a$ for $0 \leq i < t$. If $a = 1$, then*

$$t \leq \frac{\log(1/\delta_0)}{\log(1+c)}.$$

If $a > 1$, then

$$t \leq \frac{\delta_0^{-(a-1)} - 1}{1 - (1+c)^{-(a-1)}}. \tag{V.42.1}$$

In particular, if $\delta_{i+1} \geq \delta_i + \delta_i^2/320$, then $t \leq 321(\delta_0^{-1} - 1)$.

Proof. For $a = 1$, iteration gives $\delta_t \geq (1 + c)^t \delta_0$. For $a > 1$, put $q = a - 1$ and $u = c\delta_i^q \in (0, c]$. Then

$$\delta_i^{-q} - \delta_{i+1}^{-q} \geq c \frac{1 - (1 + u)^{-q}}{u}.$$

The function in the fraction is nonincreasing in u , since

$$\frac{1 - (1 + u)^{-q}}{u} = q \int_0^1 (1 + su)^{-q-1} ds.$$

Thus each decrement is at least $1 - (1 + c)^{-q}$. Sum over i and use $\delta_t^{-q} \geq 1$. For $a = 2$, the denominator is $c/(1 + c)$; substitution of $c = 1/320$ gives the stated specialization. $\square$

The last estimate sharpens the $640\delta_0^{-1}$ dyadic count displayed in the source's proof of Theorem 2.6, while keeping precisely the same increment hypothesis. It is an elementary refinement of that bookkeeping, not a new form of Roth's theorem. If an argument tries to force a contradiction by performing more steps, it must allow $\lfloor 321(\delta_0^{-1} - 1) \rfloor + 1$ attempted increments, and must verify that the size assumptions remain available at every one.

[write] **6 October 2026; sources 06–10.** The transcription's count $640\delta_0^{-1}$ is in the proof of Theorem 2.6 (exact companion `theorem_2_6_holds`, `Proofs02Roth.lean`). The step counts of source 03's Proposition V.12.1 and source 06's Lemma V.25.1 are compared with this one in the note after source 03's proposition; source 03's (V.12.2) is the next proposition with constant parameters.

Proposition V.42.2 (Exact propagation of length losses). *Suppose $N_{i+1} \geq b_i N_i^{\vartheta_i}$, where $N_i > 0$, $b_i > 0$, and $0 < \vartheta_i \leq 1$. Let $P_t = \prod_{i=0}^{t-1} \vartheta_i$, with $P_0 = 1$. Then*

$$\log N_t \geq P_t \log N_0 + \sum_{j=0}^{t-1} \left(\prod_{i=j+1}^{t-1} \vartheta_i \right) \log b_j.$$

For positive target thresholds $M_0, \dots, M_T$, a sufficient condition for $N_t \geq M_t$ for every $0 \leq t \leq T$ is

$$\log N_0 \geq \max_{0 \leq t \leq T} \frac{1}{P_t} \left[\log M_t - \sum_{j=0}^{t-1} \left(\prod_{i=j+1}^{t-1} \vartheta_i \right) \log b_j \right].$$

For constant $\vartheta \in (0, 1)$, constant $b \in (0, 1]$, and $M \geq 1$, the simpler sufficient condition is

$$\log N_0 \geq \vartheta^{-T} \left(\log M + \frac{\log(1/b)}{1 - \vartheta} \right) - \frac{\log(1/b)}{1 - \vartheta}.$$

Proof. Take logarithms and repeatedly substitute the preceding inequality. This yields the first formula by induction. Solve each of the finitely many resulting lower bounds for $\log N_0$ and take their maximum. For constant parameters the geometric sum evaluates explicitly; its lower bound for $\log N_t$ decreases with t under the displayed condition, so it is enough to check $t = T$. $\square$

For Theorem I.24.2, if the continuous square-root expression is at least two, its floor is at least half that expression. One may therefore take

$$\vartheta_i = \frac{1}{2}, \quad b_i = \frac{1}{2} \sqrt{\frac{\lambda_{\gamma,i}}{2\pi}},$$

provided the appropriate correlation hypothesis is available at stage i . This gives a direct route for comparing choices of the retained correlation fraction γ . It also shows why a claimed improvement in a final tower exponent cannot be inferred from a local constant alone: one must supply the entire sequence of b_i, ϑ_i, M_i and the number of permitted density increments.

V.42.1 The particular limitation in the supplied development

At the pinned snapshot, the edited text places an explicit warning before Theorem 18.1: after replacing an estimate in Corollary 16.11 by the bound supplied by its argument, the displayed assembly no longer establishes the intended quantitative exponent. The source ledger also leaves Corollary 16.11 and Theorems 18.1–18.2 open [48]. These are statements about the repository’s present proof chain. They do not negate the published theorem of Gowers or the subsequent quantitative literature.

The results of this report do not discharge those later dependencies. They can replace specific finite lemmas and reduce some input losses, but a new global bound for $r_k(N)$ would require a separate completed argument through the multilinear extraction, local inverse step, interval transfer, and iteration. The precise propagation formula above identifies what that future argument must certify.

V.43 A map to the Lean development and reproducible checks

Source 10, delivered Section 11.

V.43.1 What has actually been verified

The general results above have mathematical proofs in the article. Independent review checked the noninvertible reindexing, mixed-norm induction, strict tail threshold, exceptional-set denominator, integer partition bounds, diagonal correction in the random selection, and the two representation-counting injections. That review corrected an asymptotic wording issue: when density varies, the length comparison is multiplicative, as stated in Section I.24.

[write] **6 October 2026; sources 06–10.** The independent review mentioned here is not part of the delivery and is not shipped; nothing in this report rests on it (as for source 03’s reviews). The intake reran the four diagnostic scripts on a copy (all exit 0, outputs identical).

The package also contains independent finite diagnostics. Their role is to expose errors in formulas, normalization, and boundary cases; they do not establish a theorem for all finite groups. The executed checks include the following.

Family	Cases	What is checked
Density profiles	1,230	Exact cap and variance inequalities.
Density tails	4,098	The strict-threshold mass estimate.
Exceptional sets	2,532	The retained-cell density guarantee.
Sharp tail examples	45	Exact equality for three-point profiles.
Phase partitions	39,648	Coverage, integer lengths, circle approximation, and 1,408,790 individual cells.
Prime-codomain fibres	23,538	Exhaustive small-field function restrictions, the size guarantee, and the Freiman property.
Weighted representations	41	Exact codegree and representation counts.
Mixed-norm averages	432	Arbitrary complex functions, both AP and general coefficient systems.
Heterogeneous AP counts	210	The weighted telescoping bound.
Torsion sharpness	5	Four characteristic-two dimensions and the exact characteristic-three cube value 49/81.
Cubic norm	125	The determinant identity and unique zero.
Polynomial coloring	390,000	Every ordered nontrivial 4-AP in $\mathbb{F}_5^4$.

The rational profile, fibre, determinant, and explicit coloring checks use exact arithmetic. In the phase script the coverage, circle-step, and half-angle checks are exact rational calculations; complex distances are checked numerically with tolerance 10^{-12} . The arbitrary complex Gowers calculations use NumPy and tolerance 10^{-9} , with random seed 20261006. Their cube powers are computed from the defining sums, independently of the induction used in the proof. The torsion sharp examples are also verified exactly.

The file `verification_results.json` records the actual script outputs and their hashes. The file `source_manifest.json` records the pinned source identities. The README supplies the commands needed to rebuild the PDF and rerun the diagnostics.

V.43.2 Suitable formalization units

The current development defines `UniformOfDegree`, `gowersNorm`, `NatAP`, `ModAP`, `FreimanHom`, and the relevant catalogue statements [48]. Its existing modules suggest the following interfaces. This table is a proposed integration map, not a list of newly compiled declarations.

Mathematical unit	Existing source location	Main obligation
Partition cap and tail	<code>Proofs02</code> <code>DensityIncrement.lean</code>	Finite weighted sums; positivity of $2\delta - S$; equality cases.
Phase partition	<code>NatAP</code> and the Section 2 partition modules	Circle approximation, injective integer progression coordinates, and cardinalities with floors retained.
Kernel average identity	Generic finite-group infrastructure, with a cyclic specialization	Exact equal-size fibres of the multiplication endomorphism.
Mixed-norm induction	<code>Proofs03</code> <code>ProgressionAverage.lean</code> and the cube infrastructure	Hölder, derivative moments, and the precise invertible-neighbor hypothesis.
Weighted telescoping	<code>Proofs03</code> <code>ProgressionCount.lean</code>	Vanishing of the first two terms and a separate count of repeated progressions.
Weighted energy extraction	<code>Proofs07BalogSzemerédi.lean</code>	Finite random sampling or equivalent counting, diagonal correction, and injectivity of the tuple encodings.

Mathematical unit	Existing source location	Main obligation
Obstruction fibre and torsion budget	<code>Proofs07</code>	The graph projection injection; exact bad-step fibres; translation averaging.
Polynomial obstruction	<code>FreimanRestriction.lean</code> A new finite-field polynomial module	Coefficient homogeneity, a Vandermonde determinant, Chevalley–Warning, and the norm construction.

For the mixed-norm theorem, it may be easier first to formalize its equivalent version after raising to the power 2^{k-1} :

$$|\Lambda_{\mathbf{a}}(f_1, \dots, f_k)|^{2^{k-1}} \leq K \|f_k\|_{U^{k-1}}^{2^{k-1}} \left(\prod_{i=1}^{k-2} (\mathbb{E}|f_i|^{2^i})^{2^{k-1-i}} \right) \cdot (\mathbb{E}|f_{k-1}|^{2^{k-2}})^2.$$

This formulation avoids fractional powers in the inductive algebra. Taking nonnegative roots can be a final wrapper.

The probabilistic language in Theorem II.20.1 also has an entirely finite interpretation. For a fixed sampled $v \in A$, sample independent integers $u_a \in \{1, \dots, m\}$ and retain a when $u_a \leq r_A(v - a)$. This exactly realizes the required rational membership probabilities. The proof can therefore be phrased using finite sums over $A \times \{1, \dots, m\}^A$, without introducing a general probability-space interface.

The most important source-to-theorem checks remain mathematical: the normalization must match, positivity must justify each reindexing bound, the progression must have the claimed distinctness, and the new theorem’s assumptions must actually be available at its call sites. A successful axiom audit alone would not verify those comparisons.

V.44 Further research questions and proposed next steps

Source 10, delivered Section 12.

The questions below separate proposed refinements of the present theorems from established questions in the wider literature. A question introduced here is not asserted to be previously unknown or to have been recognized as open by the community.

V.44.1 Finite problems closest to the proved results

1. **Optimal torsion constants beyond the sharp families.** For a fixed finite abelian G , coefficient pattern, and target having at least one invertible coefficient difference, determine the least constant C in $|\Lambda_{\mathbf{a}}(f)| \leq C \|f_j\|_{U^{k-1}}$ for 1-bounded functions. Theorem III.28.2 gives an explicit upper bound, and Proposition III.28.4 settles particular families. The next small case is the exact constant for four-term patterns on elementary abelian 3-groups. Small finite optimizations can suggest extremizers, but an analytic extremal proof is the target.
2. **Use restricted direction averages before losing them.** The fibre identity retains the more informative quantity $\mathbb{E}_h \mathbf{1}_{cG}(h) V(h)$. Determine when the images cG arising at successive Cauchy–Schwarz steps have enough compatibility to improve the product of kernel cardinalities. An exact comparison for nested images would already strengthen the generic bound.

3. **Optimize the geometric part of the phase tradeoff.** The density cap is sharp, but the passage from one approximate period to blocks of lengths L through $2L - 1$ may leave room in the constants. Seek the best universal minimum block length for a prescribed phase disc radius, with all integer rounding retained. A useful first goal is a matching obstruction or a better construction at fixed η/δ as $\delta \rightarrow 0$.
4. **Simultaneous correlation data.** If several approximately constant characters or polynomial phases are available on the same partition, their correlations can force a larger conditional second moment than one correlation alone. Determine the best Gram-matrix bound for that moment and combine it with (I.23.3). The geometry must supply the common partition; assuming it for free would conceal the main cost.
5. **A sharp vertical-fibre restriction theorem.** How close is $|B|/[k(h_k - 1) + 1]$ to the best bound determined only by the defect fibre? Can the factor k be reduced under additional additive structure of H_k ? The exact case $H_k = \{0\}$ is settled, but the extremal nonzero-fibre case remains a natural proposed problem.
6. **Exploit overlaps in the bad-step union.** Theorem II.21.3 uses a union bound over $E_j = \{d \neq 0 : jd \in F\}$. Exact intersections between these sets can be substantial. A formula adapted to a cyclic prime-power codomain, using valuations and the distribution of defects in jV , is a concrete route to a better general-codomain retention bound.
7. **Optimize the graph-energy input and its propagation.** The explicit proof here retains a proportion $\gamma/4$ before Freiman extraction; the published Reiher–Schoen theorem retains almost $\sqrt{\gamma}$ [12]. Propagate the latter, with the exact vertical fibre when available, through the later Bohr-neighborhood and local-linearity stages of the supplied proof. The desired output is a complete new chain of numerical inequalities, not an isolated replacement of the exponent 1164.

V.44.2 Polynomial colorings after the exact low-degree answer

8. **Higher degrees.** Theorem III.30.3 completely answers the specified coordinate problem for $D_\ell \leq k$. Above that range, the higher-degree obstruction proved in Section III.30 gives necessary dimension bounds without matching constructions. For $k = 4$, a scalar polynomial of degree five or six has the resulting upper dimension bound eight. Determine the largest attainable dimension, starting with dimension five. This is a proposed finite extremal problem, not a claimed established open conjecture.
9. **Stability at the dimension threshold.** At $d = \sum_\ell \lceil D_\ell/2 \rceil^2$, must every avoiding polynomial have some structural feature resembling an anisotropic odd-degree decomposition? Full equivalence to the norm construction may be too strong, but a classification in the first cubic four-dimensional case would test the idea. Coordinate changes and linear changes of the color coordinates should be factored out before comparing examples.
10. **Actual colors versus declared coordinates.** The exact theorem controls the number of polynomial coordinates. Can a map with more coordinates but an unusually small image outperform the full-palette exponent? A useful result would relate avoidance, fiber geometry, and actual image cardinality. The theorem proved here does not supply such a relation.

11. **Unrestricted symmetric-progression colorings.** Deng, Tidor, and Zhao ask whether $N^{o(1)}$ colors suffice to avoid nontrivial symmetrically colored four-term progressions [37]. Shi and Dong’s $O(N^{1/4})$ construction improves the known polynomial exponent [33]. The dimension obstruction here explains a limit of one bounded-degree polynomial-coordinate model. It leaves unrestricted colorings, higher-degree constructions, and more economical treatment of digit information as meaningful directions for further work.

V.44.3 A practical order of work for this repository

The first implementation target should be the optimal partition cap, tail bound, and kernel-fibre averaging identity: they are small, independent facts with clear finite-sum statements. Next come the mixed-norm progression theorem and weighted counting wrapper, where normalization and torsion hypotheses are the main integration risks. The obstruction-fibre injection is another compact structural target.

The weighted energy proof is a larger, but still self-contained, formalization project. Its exact finite sampling and representation injections make its dependencies transparent. The polynomial dimension theorem then forms a separate finite-field development with a complete mathematical endpoint and a useful explicit example.

Finally, any effort to improve the headline Szemerédi estimate should return to the unresolved quantitative dependencies in the source ledger, especially the transition through Section 16. Once those hypotheses and bounds are discharged, the local improvements in this report can be inserted into an audited parameter calculation using Propositions V.42.1 and V.42.2.

[write] **6 October 2026; sources 06–10.** The eleven numbered questions, and the two question paragraphs at the ends of Sections III.29 and II.21, are merged into Section V.2 with credits. The intake dossier counted ten numbered questions for this source; there are eleven (seven in its first subsection, four in its second).

V.45 External inputs, provenance, and reproducibility

Source 10, delivered Appendix A.

The new quantitative deductions rely on elementary finite algebra and analysis except where the following established inputs are explicitly used: Fourier orthogonality and Parseval, finite-field extension norms, and the Plünnecke–Ruzsa sum-difference inequality. The needed Chevalley–Warning statement is proved in the article. The Reiher–Schoen result is used only for the separately labeled modern-input alternative; the explicit $2^{-209}\gamma^{103}$ corollary uses the energy theorem proved here.

The source comparison was made against the pinned repository files, including the paper transcription, `Definitions.lean`, the numbered Sections 1–3 and 6–7, the relevant density/counting modules, the Section 18 quantitative wrapper, the status ledger, and the axiom audit. The manifest gives exact paths and blob identifiers for these materials. The bibliography points to the original publication separately from the edited repository version.

The main $\text{\LaTeX}$ file contains the complete article and bibliography. Its only external illustration is `figures/phase_frontier.pdf`; the script `verification/make_figure.py` regenerates it and the numerical examples. The plotted points evaluate theorem guarantees and do not represent observed extremal sets. The four mathematical diagnostic scripts can be run separately or through `verification/run_all.py`. None requires network access.

References

Merged from the five sources' bibliographies. A work cited by several sources appears once, with each source's annotations; the repository entries are kept per source, because each records that source's pin.

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- [14] F. Manners, *True complexity and iterated Cauchy–Schwarz*, 2021. <https://arxiv.org/abs/2109.05731> (source 04).

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- [16] [source 01] *Definitions.lean*, in the snapshot [15], under *Combinatorics/Ramsey/Lean/GowersSzemerédi/*. Inspected definitions: partition, density, balanced function, unnormalized Fourier transform, natural progression, and average cell size.
- [17] [source 01] *Proofs02DensityIncrement.lean*, in the snapshot [15]. Inspected header and preparatory density and positive-mass lemmas.
- [18] [source 01] *Proofs17FinitePatterns.lean*, in the snapshot [15]. Inspected header and finite-dimensional sign-family infrastructure.
- [19] [source 01] *Sections17_18.lean*, in the snapshot [15]. Inspected floor-pattern specifications, affine patterns, partition uniformity, and the editorial dependency warning attached to Theorem 18.1.
- [20] [source 02] V. Reshetnikov, *ProveIt repository*, inspected October 6, 2026. Paper transcription at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Combinatorics/Ramsey/Papers/sz-thm-gowers-proof>; Lean development at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Combinatorics/Ramsey/Lean/GowersSzemerédi>. Specific inspected blobs are listed in source 02’s Appendix A (Section V.11). These links follow the moving branch *main*; the blob identifiers are the permanent record.
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- [25] [source 05] V. Reshetnikov, ProveIt repository, *Proofs02Partition.lean*, in *Combinatorics/Ramsey/Lean/GowersSzemerédi*. Returned blob identifier 291b1494ab5af839456dcac35f4bb447d4c90844. [Commit-pinned partition module](#).
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Added in the second write (sources 06–10, 6 October 2026). Works above that sources 06–10 also cite: [1] (06, 07, 08, 09, 10; source 07 adds “Section 10, particularly Lemmas 10.6, 10.8, and 10.9”, source 08 “in particular, Lemmas 3.7–3.10” and the visual check of printed pages 483 and 485); [2] (all five); [3] (06); [9] (06, with the MathNet record <https://www.mathnet.ru/eng/dan7111>); [8] (06, as the journal version: SIAM J. Math. Data Sci. **1** (2019), 699–729, [doi:10.1137/19M1257792](https://doi.org/10.1137/19M1257792)); [12] (10, Theorem 1.2); [6] (10); [14] (08). The entries below are new.

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- [41] [source 06] V. Reshetnikov, *ProveIt: Gowers–Szemerédi paper and Lean development*, repository snapshot 7d99e5deeb209a4671060990cb1ea3d672d57064, inspected 6 October 2026. [Edited paper](#); [Lean sources](#); [formalization checkpoint](#); [statement ledger](#).
- [42] [source 07] V. Reshetnikov, ProveIt repository, *GowersSzemerédi/Section10.lean*, at commit ae0e895748209f0dd079c6b5ca6df6c6fc065a81, inspected 6 October 2026. Source blob 302e8a223f56dcdabf29f30ca3c80ac62a7adbf. [Pinned statement source](#). The repository’s paper transcription is under `Combinatorics/Ramsey/Papers/sz-thm-gowers-proof/`.

- [43] [source 07] V. Reshetnikov, *ProveIt* repository, *GowersSzemerédi/Proofs10Shift.lean*, at the same pinned commit. Source blob `ab399d43dce38f7c8387cd011e2df3915a75ee19`. [Pinned proof source](#), including `lemma_10_9_holds` and its nonvacuous two-step preparation.
- [44] [source 07] V. Reshetnikov, *ProveIt* repository, *GowersSzemerédi/Audit.lean*, inspected 6 October 2026. Source blob `10b6a30720f41111569bc147d693d7f86508dab0`. [Audit source](#). Its distinction between an axiom audit and completion of conditional mathematical inputs is relevant to the status of source 07.
- [45] [source 08] V. Reshetnikov, *ProveIt* repository, *Definitions.lean*, GowersSzemerédi development, *Combinatorics/Ramsey/Lean*. Read-only source inspection, 6 October 2026. [Repository source](#) (a link to the moving branch `main`; the blob identifier in source 08’s `source_provenance.json` is the permanent record).
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- [47] [source 09] V. Reshetnikov, *ProveIt*, GitHub repository, *Combinatorics/Ramsey* source paper and Lean development. Inspected checkpoint `7d99e5deeb209a4671060990cb1ea3d672d57064`, 6 October 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Source 09’s source audit (shipped as `09-cubical-expansions-SOURCE_AUDIT.md`) records the relevant paths and blob identifiers.
- [48] [source 10] V. Reshetnikov, *ProveIt*, repository snapshot `7d99e5deeb209a4671060990cb1ea3d672d57064`, 6 October 2026. [Edited paper](#); [Lean development](#); [source status ledger](#).